

Lexicographic Orders of Well-Orderings of the Surreal Numbers

Set approximations, the singular birthday transition, set-length words,
class orders, and higher-order structure

Research report merged from ten independent manuscripts
prepared for Vladimir Reshetnikov (sources 08, 09, 11 and 12 of batch 80;
sources 13–18 of batch 81)

Sources dated 2 October 2026 (source 11: 3 October 2026, UTC) and 3 October 2026
(sources 13–18); merged 2–3 October 2026

Abstract

The phrase *all well-orders of the surreal numbers* has several inequivalent meanings. Well-orders of a fixed birthday fragment form a set; well-orders of arbitrary set-sized subsets form an ordinary class; well-orders of the whole surreal line are themselves classes, and their totality belongs to a higher logical level. We develop these settings and distinguish set-like from unrestricted class well-orders.

For an infinite set-sized alphabet X , the existing ProveIt theory gives $|\mathcal{W}(X)| = 2^{|X|}$ and the exact ordinal embedding spectrum $\beta < (|X|)^+$. We apply it to surreal birthday fragments. For arbitrary set-length surreal words, we give an explicit sign-sequence embedding into **No**, with birthday bounds, and identify the one-sided cuts created by limit-length words.

For the whole surreal line, we define lexicographic comparison directly on labeled class well-orders, without assuming ordinal enumerations or a class of class order types. Every such well-order has a canonical initial segment of type Ord; its remaining tail determines a convex fiber. In the set-like fragment, bounded-support permutations of one global enumeration form an ordinary class isomorphic to **No**, and interpolate every uniformly set-indexed strict cut of the entire fragment. The fragment contains every class linear order, admits binary class-word coding, and cannot have an exhaustive uniform class of set-codes. The unrestricted order has adjacent pairs, all arising from finite residual alphabets. An omitted-range construction exhibits a proper-class cut filled by a non-set-like well-order. In a full inaccessible-universe interpretation, the unrestricted order does not embed into its set-like fragment.

The proofs use **ZF**, **ZFC**, or **GBC**, as specified, with a further uniform nonembedding theorem in **KM** + **GC**. We give universe-relative and proof-assistant formulations, and propose further research on the higher-order structure.

[merge] This report merges four manuscripts written independently on 2 October 2026 on the same question and delivered together in batch 80: source 11 (the base, whose abstract is printed above), and sources 12, 09 and 08. They prove the same core up to four times. Each shared theorem is printed once; the other sources' versions are recorded in numbered notes, and their genuinely different arguments are kept as marked routes. Source 12 adds an exact dichotomy at birthday cutoffs: for an infinite cardinal κ and $\mu = 2^{<\kappa}$, the minimal-length enumeration order $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ is equimorphic with the binary cube $\mathbb{B}_{\kappa \cdot \kappa}$ (ordinal product) exactly when κ is a singular strong limit, and with $\mathbb{B}_\mu$ otherwise, with $\kappa = \beth_\omega$ as an unconditional example. It also adds a second direct comparison of all class well-orders, by their largest common labelled prefix, and an exact external identification of both global orders in a full-class model at an inaccessible cardinal. Source 09 adds bounded reflection of set-indexed families, an explicit unfilled (Ord, Ord)

configuration, and the embeddability in **No** of every uniformly class-coded family. Source 08 adds a separator of exactly prescribed birthday, the failure of finite support, and relation-code lexicography of all class well-orders. Source 11's choiceless proof answers source 12's question on the weak-choice strength; the direct comparisons of sources 11 and 12 answer source 08's question on comparing non-set-like class well-orders; and source 12's dichotomy answers in part a question of the repository's report on the reals. The whole set-sized general-alphabet layer re-proves that report and is printed with pointers to it, never as new.

[merge] (3 October 2026.) Six further manuscripts of batch 81, sources 13–18, continue this report and are merged into it as Parts VI–IX. They compute the density, cellularity, characters, gap spectrum and gap multiplicities of the set-sized strata at birthday cutoffs and the coding lengths of the strata $\mu \cdot \gamma + n$; classify the termination conventions of word orders; classify every proper class cut of the set-coded core and recognize the cuts of global enumerations, which answers source 11's question on represented cuts, with automorphism orbits, class-extension absoluteness and support thresholds; decompose every class well-order canonically into Ord-blocks, with every class well-order as a block-index order and explicit condensation histories; settle source 08's question on full virtual products; bound the comprehension needed for the uniform nonembedding by Σ_1^1 ; and classify the full order of every countable ω -standard model as a factorial shuffle. Results proved by several of them are printed once, with every proof credited; their re-proofs of this report are notes.

Status. This is a mathematical research report with proofs, not a claim of machine verification or bibliographic priority. Existing general set-alphabet results are credited to the repository. The further constructions are proved here and should be evaluated by their arguments. The repository audit was targeted; no Lean build or remote change was made.

[merge] **Status of the merged report.** AI-assisted and unrefereed; not formalized. Sources 12 and 09 call themselves AI-assisted in their title pages; sources 08 and 11 do not use the word (author lines: 08 and 09 “Research manuscript prepared for Vladimir Reshetnikov”, 11 “Research report”, 12 “Research report prepared for Vladimir Reshetnikov”). None of the four claims historical priority, peer review or machine verification. Their finite checks verify finite instances only, never a transfinite or class-theoretic statement. No statement of this report is formalized; the formal status of the inputs it uses is in [section 1.5](#).

[merge] **Batch 81 (3 October 2026).** Sources 13–18 are research manuscripts with written proofs. Sources 13, 15, 17 and 18 call themselves AI-assisted, source 14 unrefereed, and source 16 says that its checks are not external peer review. None claims priority, peer review or machine verification, and none of their statements is formalized ([sections 1.4 and 1.8](#)).

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1 About this report

1.1 Four manuscripts, one report

[merge] (Added 3 October 2026: six further manuscripts of batch 81 are merged as sources 13–18 and described in [section 1.8](#). This subsection describes the four batch-80 sources.) [merge] All four manuscripts arrived in commit 4e270aa46 and were placed in ccc046989 (batch 80, cluster K3). Their batch-80 manuscript numbers are their source numbers here.

Source	Manuscript (archive)	Pin and date line	Labels and files
11 (base)	<i>Lexicographic Orders of Well-Orderings of the Surreal Numbers: Set Approximations, Class Orders, and Higher-Order Structure</i> ; 2,807 lines, 35-page PDF (surreal_well_orders (1).zip)	ae7c1e6aa (2 October 2026, 18:03:53 PDT = 3 October 2026, 01:03:53 UTC); dated “3 October 2026” (the UTC date)	prefix swo::; files 11-raw-orders-*
12	<i>Well-Orders of the Surreal Numbers and Their Lexicographic Geometry: Class-sized common prefixes, exact embedding spectra, a singular-cardinal transition, and size-safe formalization</i> ; 1,689 lines, 24-page PDF (surreal_well_orders.zip)	63bb8b1d9 (2 October 2026, 18:01:32 PDT); dated 2 October 2026	prefix swo:st::; files 12-singular-*
09	<i>Well-orders of the Surreal Numbers: Lexicographic universality, a surreal dense core, and the boundary between sets and classes</i> ; 1,677 lines, 24-page PDF (Surreal_Well_Orders_Research.zip)	63bb8b1d9; dated 2 October 2026	prefix swo:ec::; files 09-core-*
08	<i>Lexicographic Orders of Surreal Well-Orderings: Set-sized spectra, global enumeration orders, and a surreal-sized dense skeleton</i> ; 2,022 lines, 26-page PDF (Surreal_Well_Orders_Research (1).zip)	ae7c1e6aa; dated 2 October 2026	prefix swo:sk::; files 08-skeleton-*

The two pairs of archives sharing a name (08 and 09, 11 and 12) are not editions of one another. Every pair of the four texts has word 8-gram overlap of 0.6–1.1 % (preamble boilerplate), no member file is shared, the pins differ within each pair, and the result sets differ. They are four independent manuscripts on one spine; none supersedes another, and all four are merged (union).

Source 11 is the base. Its global-choice equivalence is proved over GB without set choice, the weakest base theory of the four ([theorem 13.1](#)); it covers set alphabets, set-length words, and set-like and unrestricted class well-orders; and it alone attributes the general set-alphabet theorems to the repository’s report on the reals, `SetTheory/Cardinals/docs/reports/ordinals-and-order-types/lexicographic-well-orderings-of-reals/` (cited below as the reals report, labels lwo:).

1.2 How the four texts are printed

[merge] Source 11’s text is printed in full and in its own order. The other sources’ material is inserted where it belongs.

- Their introductions ([section 4](#)) and foundational sections are printed whole, as sections headed “Source N ”.

- A result that a printed result already proves by the same argument is recorded in a numbered *Note*: the note gives the source’s statement, its number in that source, and what differs (hypotheses, strength, extra remarks). Prose of the source that makes a point not made elsewhere is kept in the note.
- A genuinely different proof is printed in full as a marked route (“Source 12’s route”, “second proof”). A result proved only by one source is printed in full where it belongs, with its source named in the heading or the first sentence.
- Every label of every source survives, with the source’s prefix (**swo:** for 11, **swo:st:** for 12, **swo:ec:** for 09, **swo:sk:** for 08). Numbered statements without a label in their source carry **swo:sub:nnumber**, for example **swo:st:n10.4** for source 12’s Corollary 10.4. [Section D](#) maps every numbered statement of the four sources to its place here.
- Text written for the merge is marked **[merge]**. Dated notes of 2 October 2026 record where one source answers another ([section 1.7](#)).

Statements and sections are numbered by their place in this report; source numbers appear in headings, notes and [section D](#). Apart from the renamings of [section 1.3](#) and the corrections listed in [section 1.6](#), the sources’ text is verbatim. Statements of the reals report and of other repository reports are cited by label name, as the sources do; those reports are not reproduced. The same rules govern sources 13–18 ([section 1.8](#)); their introductions are in [section 4.5](#).

1.3 One notation for four sources

[merge] The sources’ symbols collide. The following convention is fixed before any of them is used; every renaming is listed, and the tempting false reading is printed where confusion is likely.

Here	Meaning, and the sources’ symbols
GB, GBC	GB has a ZF set part, elementary class comprehension and class replacement, and <i>no</i> choice principle, as in the foundations report (found:sub:gbconvention) and the collection’s notation guide; GBC = GB + GC. Sources 09 and 12 analyse the necessity of choice “over GB plus set-level AC” (09, proposition 13.7 ; 12, theorem 13.2); that theory is written GB + AC in the merge text. Source 11 works over GB itself. <i>False reading</i> : the letters GB in sources 09 and 12 do not include set choice here.
bd(x)	Birthday, the length of the sign sequence (08, 09; the foundations and birthday-cutoff reports). Source 11’s $\ell(x)$ and source 12’s $b(s)$ are renamed bd(x), bd(s). In sources 08 and 09 the letter b is the baseline enumeration, never a birthday.
$\mathbf{No}_{<\kappa}$	$\{x \in \mathbf{No} : \text{bd}(x) < \kappa\}$, the birthday cutoff (foundations report $\mathbf{No}_{<\alpha}$). Renamed: source 08’s X_λ , X_κ ; source 09’s X_θ (its $\kappa_\theta = \mathbf{No}_{<\theta} $); source 11’s X_κ ; source 12’s S_κ . The reals report writes $S_{<\theta}$. <i>False reading</i> : source 12’s S_κ is $\mathbf{No}_{<\kappa} = S_{<\kappa}$, not $S_{<\kappa^+}$.
$\ell_2(L)$	Binary coding length $\min\{\gamma : L \hookrightarrow \mathbb{B}_\gamma\}$, as in the reals report (lwo:def:binary-length). Source 12 prints the same symbol ℓ_2 and calls it the “binary coding rank” (its macro was named \rk); its prose is kept. rank(x) is the von Neumann rank of a set (source 11’s \rk), used only in the proof of theorem 13.1 . <i>False reading</i> : the “rank” of source 12 is ℓ_2 , not rank.
$\mathcal{W}(X)$, $\mathcal{W}_\alpha(X)$	All exhaustive well-order enumerations of a set X (lengths below $ X ^+$), and those of length exactly α , compared at the first unequal entry in the given order of X (reals report’s $\mathcal{W}$, $\mathcal{W}_\lambda$). The minimal slice is $\mathcal{W}_{ X }(X)$. Source 11’s $\mathcal{P}_\lambda(X)$ is renamed $\mathcal{W}_\lambda(X)$; source 12’s script $\mathscr{W}$ is printed $\mathcal{W}$.

Here	Meaning, and the sources' symbols
$\mathcal{I}(X), \mathcal{I}, \mathcal{T}$	$\mathcal{I}(X)$: injective words of length below $ X ^+$ over a set X (source 11). $\mathcal{I} = \mathcal{I}(\mathbf{No})$: the class of all injective set-length surreal words, proper prefixes first (source 11's $\mathcal{I}$; source 08's $\mathcal{I}(\mathbf{No})$; source 09's $\mathcal{T}$, renamed). $\mathcal{T}$: all set-length words, not necessarily injective (source 11's $\mathcal{T}$). <i>False reading</i> : source 09's $\mathcal{T}$ was the injective class. The reals report's $\mathcal{I}$ (injections $\kappa \rightarrow \mathbb{R}$) is unrelated.
$\mathfrak{W}_{sl}(\mathbf{No}), \mathfrak{W}_{all}(\mathbf{No})$	Set-like global well-orders of $\mathbf{No}$, equivalently class bijections $\text{Ord} \rightarrow \mathbf{No}$; and all class well-orders of $\mathbf{No}$. Both are predicates on classes ("metanotation"), never classes of classes. Source 12's $\mathcal{E}(\mathbf{No})$, written $\text{SL}(\mathbf{No})$ in its macros, and its $\mathcal{W}_{cl}(\mathbf{No})$; source 09's $\mathcal{E}$; source 08's "virtual" $\mathcal{E}$ are renamed. $\mathfrak{J}$ is source 11's collection of class injections $\text{Ord} \rightarrow \mathbf{No}$.
$\mathbb{B}_\gamma, \mathbb{B}_{\text{Ord}}$	$\mathbb{B}_\gamma = (\{0, 1\}^\gamma, <_{\text{lex}})$ (reals report, 08, 12). $\mathbb{B}_{\text{Ord}}$: class binary words on Ord compared at their least disagreement (source 08's $\mathbb{B}_{\text{Ord}}$; source 12's $\mathfrak{B}$, renamed; source 11 writes $\{0, 1\}^{\text{Ord}}$).
$\simeq_e, \hookrightarrow$	Equiporphism (embeddings both ways; not isomorphism) and order embedding. Source 11's $\approx_{\text{emb}}$ is printed $\simeq_e$.
$\mathcal{P}_{bd}(e_0), \mathcal{P}_s(b)$	The <i>core</i> : enumerations that agree with a fixed baseline enumeration outside a set of positions. Source 11 codes them by normalized permutations $\sigma : \lambda \rightarrow \lambda$ ($\mathcal{P}_{bd}(e_0)$, definition 19.3); sources 09 and 08 by fixed-point-free bijections $\sigma : S \rightarrow S$ of a set S of ordinals (source 09's $\mathcal{C}_b$, source 08's $\mathcal{P}_b$, both written $\mathcal{P}_s(b)$ here; source 08's code carrier $\mathcal{P}$ is $\mathcal{P}_s$, and its power-set $\mathcal{P}(V_\kappa)$ keeps the letter $\mathcal{P}$). The two codings have the same evaluations: restrict the identity extension of a support code to $\lambda = \sup\{\xi + 1 : \xi \in S\}$, and conversely discard fixed points. With the same baseline, every order statement about one is a statement about the other.
e_0, B, b	The fixed baseline class bijection $\text{Ord} \rightarrow \mathbf{No}$ of sources 11, 12, and 09 and 08 respectively.
$d(F, G), \Delta(E, F), \delta(e, f)$	The least ordinal at which two class functions on Ord differ (sources 11, 09, 08).
$\eta_{\text{Ord}}, \eta_\kappa$	The set-cut property, and its version for cuts of fewer than κ elements, empty sides allowed (11, definition 5.1 ; 08, definition 5.8); source 09 writes "set-cut property". The subscript is a cardinal bound, not an $\aleph$ -index.
Termination	Sign sequences compare with $- < \text{termination} < +$ (numerical order); words compare with proper prefixes first. Exhaustive enumerations of one set are prefix-free, so no convention is needed there.
(S08. n)	Source 08's displayed equations (1)–(13) keep their numbers as (S08.1)–(S08.13). (S08.10), an inequality inside a proof recorded in note 19.13 , is described there, not displayed.

1.3.1 Addendum of 3 October 2026: notation for sources 13–18

[merge] The six batch-81 sources ([section 1.8](#)) reuse the symbols above with other meanings, and several reuse this report's macro names for other objects. The convention below is fixed before any of their material is printed; it applies to Parts VI–IX and changes nothing in Parts I–V. Every renaming is listed; the tempting false reading is printed where confusion is likely.

Here	Meaning, and the batch-81 sources' symbols
κ, μ, r, q	κ is an infinite cardinal used as a birthday cutoff, $\mu = 2^{<\kappa} = \mathbf{No}_{<\kappa} $, $r = \text{cf } \kappa$ and $q = \text{cf } \mu$, as in source 17. Renamed: source 18's ν, τ (printed r, q ; in source 18's general principle, ν stays the point character of an abstract order D); source 14's $\tau = \text{cf } \kappa$ (printed r); source 15's $\delta = \text{cf } \kappa$ in its saturation section (printed r). Source 16 writes κ for the cardinality of a general alphabet and θ for the cutoff; in its text these are printed μ and κ . Sources 13–18's X_κ and source 16's X_θ are printed $\mathbf{No}_{<\kappa}$. <i>False reading:</i> r is a cofinality, never a coding length (next row).
$X, X = \mu$	A general alphabet: an infinite linearly ordered set, of cardinality μ (source 18's convention; source 16's κ).
$\rho_\kappa, \mathbb{B}_\sigma \times_{\text{lex}} m$	Source 15's r_κ ($\kappa \cdot \kappa$ at a singular strong-limit cutoff, μ otherwise; by theorem 8.3 it is $\ell_2(\mathcal{W}_\mu(\mathbf{No}_{<\kappa}))$), renamed to keep $r = \text{cf } \kappa$; and source 15's $D_{\sigma,m} = \mathbb{B}_\sigma \times_{\text{lex}} \{0, \dots, m-1\}$, the binary cube with a finite chain of m points as last coordinate, renamed because source 15's core is $\mathcal{D}$.
d, cell, w	Source 18's letter ρ for a regular cardinal is printed θ , as in source 17. Density, cellularity (a supremum, not necessarily attained) and weight of an order topology, as in the reals report; sources 15–18 write d, c, w . Cellularity is never written c . Source 15's $d(a, b)$ (a first-difference coordinate) and source 18's $d_* = \mu^{<\mu}$ keep their letters.
ci, characters	ci is coinitality. The left and right characters of a point x are $\text{cf}(L_{<x})$ and $\text{ci}(L_{>x})$ (sources 17, 18; source 15 writes $\text{cf}^-(f)$, $\text{ci}^+(f)$). A point character, the global cofinality of the order, and the character of a gap are three different invariants (source 18's warning).
GapChar(L), Reg	The set of characters ($\text{cf } A, \text{ci } B$) of the gaps (A, B) of L , a gap having nonempty sides with no endpoints (principal cuts and the external ends are excluded). Source 17's $\Sigma(L)$ and source 18's $\text{GapSpec}(L)$ are printed with the reals report's symbol GapChar ; the report surreal-fields-across-universes writes GapSpec . Reg is the class of infinite regular cardinals (source 17's Reg_∞ ; the reals report writes $\text{Reg}(<\kappa)$ for those below κ).
η_ν	Source 17: for regular ν , every cut $A < B$ with $ A , B < \nu$ is filled; sources 15 and 18: every cut with $ A + B < \nu$ (resp. $ A \cup B < \nu$) is filled, for any infinite cardinal ν . Empty sides are allowed throughout. For regular ν the three definitions agree with this report's η_κ (section 1.3), and every threshold statement printed concerns the regular cardinal r .
$P, [p], C_p$; straddling	$P = \mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ is the minimal slice (sources 17, 18; source 15 writes L). For an injective prefix p , $[p]$ (source 17) and C_p (sources 16, 18, and source 11 in Part IV) are the same cylinder. A cylinder that meets both sides of a cut <i>straddles</i> it; this is the word of sources 16, 17 and 18 and is used in all batch-81 Parts. Sources 13 and 14 say that a prefix <i>crosses</i> a cut, source 15 that it is <i>persistent</i> ; these words are explained where printed.
$\mathcal{T}, \mathcal{I}, \mathcal{T}^{\text{inc}}$	All set-length words, injective set-length words, numerically increasing set-length words. Source 16's $\mathcal{T}, \mathcal{T}^{\text{inj}}, \mathcal{T}^{\text{inc}}$, whose macro <code>\Words</code> names <i>all</i> words while this report's <code>\Words</code> names the injective ones; source 13's $\mathcal{T}, \mathcal{I}$ agree with this report. Source 18's centered classes $\mathcal{C}, \mathcal{C}_{\text{inj}}, \mathcal{C}_{\text{inc}}$ are printed $\mathcal{T}_c, \mathcal{I}_c, \mathcal{T}_c^{\text{inc}}$ (its $\mathcal{C}$ would clash with $\mathcal{C}_b$). Source 16's termination types $T_{\min}, T_\kappa$ are printed $\mathbf{T}_{\min}, \mathbf{T}_\kappa$. <i>False reading:</i> source 16's $\mathcal{T}$ is not the injective class.

Here	Meaning, and the batch-81 sources' symbols
$\mathfrak{W}_{\text{sl}}(\mathbf{No}), \mathfrak{W}_{\text{all}}(\mathbf{No})$	As in section 1.3 . Renamed: $\mathfrak{E}$ (sources 13, 14, 16, 17) for set-like global orders (source 15 already writes $\mathfrak{W}_{\text{sl}}$); source 14's $\mathfrak{W}$ (macro <code>\Hwell</code>), source 16's $\mathfrak{H}$ and source 17's $\mathfrak{W}$ (macro <code>\HW0</code>) for all class well-orders (sources 13 and 15 already write $\mathfrak{W}_{\text{all}}$). Source 18's macros <code>\Hsl</code> , <code>\Hall</code> print script $\mathcal{H}_{\text{sl}}$, $\mathcal{H}_{\text{all}}$ there and $\mathfrak{W}_{\text{sl}}$, $\mathfrak{W}_{\text{all}}$ here.
$\mathfrak{J}$	Class injections $\text{Ord} \rightarrow \mathbf{No}$, as source 11's $\mathfrak{J}$ (sources 13, 14, 17 write $\mathfrak{J}$ with the macro <code>\J</code> ; source 16 writes $\mathfrak{J}$ with the macro <code>\I</code> ; source 15's $\mathfrak{J}$). <i>False reading</i> : source 16's <code>\I</code> is not this report's $\mathcal{I} = \mathcal{I}$, the class of injective set-length words.
$\mathcal{P}_{\text{bd}}(b), \mathcal{P}_{\text{s}}(b)$	The core, as in section 1.3 , in the source's own coding: normalized permutations $\mathcal{P}_{\text{bd}}(b)$ for sources 15 ($\mathcal{D}$) and 16 ($\mathcal{P}_b$); fixed-point-free support codes $\mathcal{P}_{\text{s}}(b)$ for sources 13 ($\mathcal{D}$), 14 ($\mathcal{P}_b$), 17 ($\mathcal{C}_b$) and 18 ($\mathcal{C}_b$, macro <code>\Pbd</code>). <i>False reading</i> : source 18's <code>\Pbd</code> is the support coding, not this report's $\mathcal{P}_{\text{bd}}(e_0)$; the two codings have the same evaluations.
$\mathcal{P}_{<\kappa}(b)$	Source 14's support-cardinality layer: codes whose support has cardinality less than κ . <i>False reading</i> : it is not $\mathcal{P}_{\text{bd}}$, whose subscript means bounded (set-sized) support; source 14 itself warns that a cardinality filtration and an ordinal-height filtration must not share the word "bounded".
$\simeq_e, <_{\text{lex}}$	Equimorphism is $\simeq_e$ for all six sources (source 16's $\simeq_{\text{emb}}$; source 18's macro <code>\equim</code> prints $\rightleftharpoons$ there and $\simeq_e$ here). Source 14's macro <code>\lex</code> prints the relation $<_{\text{lex}}$; this report's <code>\lex</code> is the subscript lex , so source 14's relation is printed $<_{\text{lex}}$. Source 18's bold $\mathbf{B}_\gamma$ is printed $\mathbb{B}_\gamma$.
Cuts of the core	A normalized class cut (L, U) has a lower side L without greatest element; the cut of an injective branch F is written L_F (source 16's L_F , source 17's L_f ; source 14's $D_F, D_p^\pm, D_{p,H}$ and source 13's (L, U) are explained where printed). In Part VIII, source 13's code cylinder $\mathcal{D}_p$ is printed $\mathcal{P}_{\text{s}}(b)_p$, and source 17's $\mathcal{C}_p$ and source 18's $\mathcal{C}_b(p)$ are printed $\mathcal{C}_p$; in source 14's support section its decreasing sequence (r_i) is printed (u_i) and the regular cardinal of its definition of η -orders θ , since $r = \text{cf } \kappa$ there.
$\varepsilon, \text{end}, \dagger$	A formal termination symbol adjoined to the alphabet of a word order; it is never a surreal number. Source 16's ε sits at a cut $L_\varepsilon < \varepsilon < R_\varepsilon$ of $\mathbf{No}$, and its $0^-, 0^+$ sit immediately below and above zero; source 14 writes <code>end</code> ; source 18's centered $\dagger$ takes the place of zero, which is then not a letter, and its $\dagger_-, \dagger_+$ sit immediately below and above an available zero. Source 13's central termination pads a word with zeros, zero again not being a letter. Source 18 writes $\ell(p)$ for the length of a word p and $\mathcal{J}$ for the increasing words, printed $\mathcal{T}^{\text{inc}}$. <i>False reading</i> : the Conway sign order $- < \text{end} < +$ is not prefix-first dictionary order.
$(\text{SNN}.n)$, labels	Displayed equations of source NN keep their numbers as $(\text{SNN}.n)$, as for source 08. Labels: <code>swo:fs</code> : (13, replacing its <code>swf</code> :), <code>swo:tc</code> : (14), <code>swo:cb</code> : (15, keeping its internal <code>cce</code> : and <code>new</code> :, since dropping them would merge two of its labels), <code>swo:ds</code> : (16), <code>swo:gs</code> : (17), <code>swo:sh</code> : (18, keeping its internal <code>wv</code> :); unlabelled numbered statements get <code>swo:sub:nnumber</code> ; labels written for the merge in Part VI begin <code>swo:vi</code> :, in Part VII <code>swo:vii</code> :, in Part VIII <code>swo:viii</code> :, in Part IX <code>swo:ix</code> :. Two labels of source 13 printed in Part VI carry its old prefix as well, <code>swo:fs:swf:sec:fixed</code> and <code>swo:fs:swf:thm:fixed</code> ; like every label they are not renamed.

Here	Meaning, and the batch-81 sources' symbols
Blocks, ranks, κ in Part IX	Source 15's B_W , C_b (least representatives, block of b), source 16's J_R , α_R and source 18's B_R , θ_R (representatives of the proper blocks, type of the final set block) are kept; $\rho_R(x)$ (sources 16 and 18) is a rank inside a block, not ρ_κ . <i>False readings:</i> in source 15's block section b is not the baseline and C_b is not a cylinder; source 14's J_R is an initial branch, not source 16's J_R . In Part IX's sections on larger universes κ is strongly inaccessible (source 14's Ω , sources 17's and 18's λ). Source 18's plain H_{sl} , H_{all} are printed $\mathfrak{W}_{sl}$, $\mathfrak{W}_{all}$.

1.4 What the report claims, and what it does not

[merge] The report claims the written proofs printed below, each in the foundation its statement names (ZF, ZFC, GB, GB + AC, GBC, KM + GC, or ZFC with a strongly inaccessible cardinal). The results new against the repository are: the singular transition at birthday cutoffs ([theorem 8.3](#)); the choiceless global-choice equivalence ([theorem 13.1](#)); the direct comparison of all class well-orders ([theorem 14.2](#), [theorem 14.7](#)), its canonical initial part and convex fibres, and its adjacency criterion ([theorems 15.1 and 16.2](#) and [proposition 15.3](#)); the core of bounded-support perturbations and its interpolation theorem ([theorem 19.10](#) and [corollary 19.15](#)); binary-class equimorphism and the diagonal non-coding theorems ([theorems 19.32, 19.37 and 19.38](#)); omitted ranges ([theorems 19.42 and 19.43](#) and [corollary 19.46](#)); bounded reflection ([theorem 19.30](#)); the failure of finite support ([proposition 19.31](#)); relation-code lexicography ([proposition 17.1](#)); explicit word coding and the word-cut classification ([theorems 9.1 and 11.1](#)); and the universe models and non-embedding theorems ([theorems 20.1, 20.2, 21.1 and 22.1](#)). None of these is claimed to be historically first.

It does *not* claim the following; every limitation stated by any source is kept.

- *No novelty for re-proofs.* The set-sized general-alphabet layer (prefix-freeness, cardinality $2^{|X|}$, universality, no $|X|^+$ -chain, ordinal and cube spectra, coding length $|X|^+$, extrema, adjacency, $n!$ blocks, strict cubes) re-proves the reals report for every infinite linear alphabet and is printed with pointers to it ([section 6.1](#)). Universality and homogeneity of **No** are classical (Ehrlich [15]; Hamkins [20]); all four sources say so. The coding of sets behind the choice equivalence is, in source 09's words, a standard well-founded-coding argument; the connection is discussed by Hamkins [18, 19].
- *Source 11* (its Status paragraph, its delivery README's "Scope and verification" and its audit): no machine verification, no Lean build, no new Lean file; no bibliographic priority for constructions proved in the report; research questions are proposed directions, not a certified list of published open problems; the repository audit was targeted and repository-wide correctness was not assessed; the finite checks do not verify transfinite proofs, cardinal arithmetic or class comprehension. Its README states: "The mathematical proofs underwent a separate proof review." That is the delivery's own statement; no such review is recorded in the repository, and this report does not rely on it.
- *Source 12* (title page, its Section 1.2, and 12-singular-RESEARCH_STATUS.md, "What is not claimed"): AI-assisted, unrefereed; historical priority not claimed; no previously catalogued open problem is claimed settled, nor that its exact formulations are new; no new Lean or Rocq verification; the Python checks are not substitutes for the proofs; the bounded theorem is not a construction of a full class model at a singular cardinal; the full collection of class well-orders is not treated as a set or ordinary class at the same foundational level; no universal semantic decoder for formula codes is assumed; its nine research questions are proposed continuations;

its source search was not sufficient to establish originality.

- *Source 09* (title page, its Appendix A, and `09-core-RESEARCH_STATUS.md`, “Important boundaries”): AI-assisted, not peer-reviewed or Lean-verified, no historical priority; classical ingredients and repository results are distinguished from its constructions; no class whose members are proper classes is formed; the global-enumeration representation covers set-like class well-orders only, and no complete classification of the relation-code comparison is claimed; the core realizes uniform set-sized cuts, not all proper-class cuts (an explicit counterexample is supplied); for fixed alphabets the absence of κ^+ and its reverse is not claimed to be a sufficient embeddability criterion; automorphisms and higher-level cuts are not classified; its research directions are proposals; literature and repository searches were targeted, not exhaustive.
- *Source 08* (its Status box and delivery README): explicit derivations, not a claim of historical priority; not independently refereed and not checked in Lean; no new Lean formalization; repository declarations are separated from proposed extensions; repository sources were read but not rebuilt; the finite tests do not establish any transfinite or class-theoretic theorem; every use of “class” is relative to the stated foundational model.
- *Proposed Lean modules are proposals.* The module names in the Lean plans of [sections 20 and 24 to 26](#) do not exist in the repository; the existing declarations they cite do ([section 1.5](#)).
- *Review status.* The repository’s Hilbert’s-tenth research tree holds a written-proof review of the four archives (`Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_batch80_surreal.md`, commit `c338541e3`). It found their principal arguments sound under the stated foundations, with one scoped correction to source 08, which is applied here ([section 5.6](#)), and it reran the four finite checks. A second note there authenticates the placement (`review_surreal_placement_ccc046989.md`, commit `3aa123856`). That review is pinned to the delivered archives and does not cover the text written for this merge; it is not an entry of the surreal collection’s review record, where this report’s review is pending. ([merge] Added 3 October 2026: the same tree has since reviewed the merged text as well. Its synthesis review (`review_surreal_synthesis_0be9b9134.md`, commit `280583d2f`) read every passage written for the merge and found no mathematical defect under the stated hypotheses, but two overstatements, both corrected here in place: the choice recursion of [theorem 13.1](#) had been called a “coherent” sequence of well-orders in [remark 13.4](#) and [remark 61.15](#), and the README had understated the largest delivered file. Its preservation audit (`review_surreal_transfer_0be9b9134.md`, commit `afb706742`) accounts for all 166 formal statements of the four sources (124 printed with normalized-exact bodies, 41 as duplicate-result notes, one question inside an answered remark). Neither is an entry of the collection’s review record either.)

[merge] *Addendum of 3 October 2026: sources 13–18.* The batch-81 addition claims the written proofs printed in Parts VI–IX, each in the foundation its statement names. Its results that are new against the repository are those called new in [sections 33, 39, 50 and 60](#); its re-proofs of this report, of the reals report and of the report `surreal-fields-across-universes` are notes. None is claimed to be historically first, and results proved by several batch-81 sources are credited to all of them. Every limitation stated by the six sources is kept:

- *Source 13* (`13-products-RESEARCH_STATUS.md`): AI-assisted and unrefereed; no peer review or priority claimed; a targeted audit of this report at `0ecd158fd`, no Lean build, no new Lean proof. Its six “critical scope conditions”: cut reconstruction concerns normalized cuts and all set-ordinal levels, exhaustion being a separate coverage condition; the classification is relative to a supplied coded core and prefix system, which the pure order does not canonically

recover; the product nonembedding is proved for arbitrary external maps in the full inaccessible universe and for fixed-formula uniform maps in $\text{KM} + \text{GC}$, and is not advertised as a theorem of bare GBC about a third sort; externally countable models admit external embeddings that need not be internally uniform; central termination reserves zero as a marker; ordinary classes of set codes, class relations and a higher collection of all class relations are never identified. Its finite checks do not verify ordinal limits, class comprehension, global choice or transfinite recursion.

- *Source 14* (`14-trichotomy-RESEARCH_STATUS.md`): GBC with a fixed baseline; the inaccessible model is an illustration, not a hypothesis; historical priority not established; no Lean theorem compiled. Its scope cautions: the trichotomy uses named prefix cylinders; the orbit theorem is a coarser classification; equimorphism is not isomorphism; a principal normalized cut has its point on the upper side; cardinal support bounds are not ordinal location bounds; the class-extension theorem fixes the sets and the baseline; raw trace fibres forget tails, and singleton fibres can be non-set-like; a coherent injective branch can omit labels; a virtual quotient is not a collection of class equivalence classes. Its status file says that the inspected Lean simplicity theorem “is NOT an independent arbitrary-small-cut existence theorem”; that is true of `SignSequenceSimplicity.lean`, while the existence theorem is `small_cut_fillers` in `SignSequenceCut.lean` (section 1.5).
- *Source 15* (its delivery `README`, not shipped): AI-assisted and unrefereed; its extensions “received independent mathematical review within the task”, with no claim of priority or formal certification; the fixed-type classification covers $\mu \cdot \gamma + n$ with finite n only.
- *Source 16* (`16-strata-SOURCE_STATUS.txt`): a targeted audit at 491ed8411; “The proofs underwent independent mathematical checks during preparation”, the delivery’s own statement, which “is not external peer review, a proof of historical novelty, or verification by a proof assistant”; no Lean build; its Σ_1^1 theorem is a sufficient bound, with no reversal.
- *Source 17* (`17-gaps-research_status.md`): AI-assisted, no referee review, no compiled Lean; proposed module names only; a targeted audit; no priority. Its unresolved scope: no isomorphism classification of the minimal slice; nonminimal types and noncardinal cutoffs not classified; the automorphism group, arbitrary baseline changes, higher residual blocks and their recursion strength open; the condensed unrestricted order not identified with the set-like order or with **No**; the class-extension theorem not a reconstruction of the class realization. Its reserve formula minimizes the number of extra moved positions, not an ordinal support bound or a birthday.
- *Source 18* (its delivery `README`, not shipped): “conventional proofs and independent AI-assisted proof review”; no literature priority, peer review or machine-checked formalization; no Lean build or imported-axiom audit; its status table is printed whole (section 63.8).

The repository’s Hilbert’s-tenth research tree, which reviewed the batch-80 sources, holds no review of sources 13–18 at the time of this write.

1.5 Formal status

[merge] No statement of this report is formalized, and no `swo:` label has a Lean mapping in the collection’s formalization ledger (`Algebra/SurrealNumbers/docs/FORMALIZATION.md`). Placement in the surreal collection, beside the Lean foundations of `Algebra/SurrealNumbers/Surreal/Foundations/`, confers no formal status.

Four inputs that the sources use *are* proved in Lean, in the universe-relative form recorded in

that ledger, where a “set” of surreals is a family indexed by a `Small.{u}` type and the sign carrier lives in `Type (u+1)`:

- properness and bounded birthdays (found:prop:proper; [proposition 5.4](#), first three clauses): `SignSequence`, `birthdays_bounded`, `small_bounded`, `not_small` in `SignSequence.lean`;
- filling of small cuts (found:sub:cutdata): `small_cut_fillers` in `SignSequenceCut.lean` and `exists_separator_of_small_sets` in `SignSequenceCutOperation.lean`;
- a separator of exactly a prescribed birthday above all option birthdays, [theorem 5.6](#): `exists_cut_separator_of_birthday_lt` in `SignSequenceCut.lean`, by the same route as source 08’s proof (the complete lexicographic lattice of Boolean words of one length, which is [lemma 5.5](#), there taken from Mathlib);
- the inconsistency of unrestricted cuts (found:prop:allcuts): `noUnrestrictedCuts` in `Size0bstructions.lean`.

The other declarations the sources cite exist at the current revision: `ordinalOrderEmbedding`, `small_strict_upper_bounds`, `simpler_wellFounded`, `existsUnique_simplest_separator`, `cut_realizes`, `cut_isPrefix`, `birthday_cut_le`, `birthday_cut_le_iSup`, `cut_congr`, `SmallCutData`, `HasSmallCutFillers`, `not_small_of_small_cut_fillers`, `lt_iff`, and the others in their tables. None of them states a result of this report.

[merge] (3 October 2026.) The same holds for sources 13–18: none ships Lean code, no statement of theirs is formalized, the declarations they cite exist at the current revision (including `le_iff_options_lt` and `lt_iff_exists_option` in `SignSequenceComparison.lean`, cited by source 18), and the modules they propose do not exist ([note 59.1](#)).

1.6 Where the merge had to choose

[merge]

- *Base and order.* Source 11, for the reasons in [section 1.1](#); its section order is kept, and member material follows the base result it duplicates or extends.
- *Strongest statement printed as the statement.* The choice equivalence is source 11’s (over GB; sources 12 and 09 assume set choice, and source 09 treats set-like orders only). The diagonal theorem is printed twice: source 11’s ([theorem 19.37](#), rows of set codes) and source 12’s ([theorem 19.38](#), every uniform family of class well-orders, with a set-like witness); sources 09’s and 08’s are notes (source 08’s indexes by Ord only). The birthday-cutoff statement is source 12’s dichotomy; source 08’s regular case is a note.
- *Different routes kept in full:* source 09’s stabilization proof of the ordinal spectrum; source 12’s route to the set-sized spectra through binary coding length; the second and third proofs that a set can be avoided ([lemmas 5.3](#) and [5.10](#)); source 12’s and 09’s proofs of the choice equivalence over GB + AC; source 12’s largest common labelled prefix; source 12’s residual decomposition and adjacency criterion; source 12’s concatenation proof of the binary-class embedding; source 12’s prefix extension through the complement of the prefix; source 09’s support coding of the core; source 09’s explicit unfilled configuration; source 09’s general universality and homogeneity theorems; source 12’s birthday-bounded universality.
- *Kept whole:* each member’s introduction, foundational section, universe and model sections, Lean plan, research questions, conclusion, ledger and reproducibility record.
- *Corrections to source text.* The renamings of [section 1.3](#); one mistyped reference in source 11 (a proposition cited as “Theorem”, in [section 12](#)); source 08’s predecessor citation ([section 4.3](#)

and the note after it); source 12’s question on the weak-choice strength, printed as an answered remark (remark 61.15); source 08’s question 11 annotated as answered; one sentence of source 08 on well-foundedness replaced by the correction of the Hilbert’s-tenth review (section 5.6 and the note before it). Statements that name delivery files (build scripts, `finite_checks.py`, `surreal_well_orders.tex`) are kept and annotated with the shipped names (section B).

1.7 Where one source answers another

[merge] Dated notes, 2 October 2026.

- (a) *Source 12’s question on the weak-choice strength* (remark 61.15, and its remark 13.3): answered by source 11. Over GB without set choice, a class well-order of **No** exists exactly when global choice holds (theorem 13.1); see remark 13.4. The same theorem shows that the set-choice hypothesis of source 09’s proposition 13.7 is unnecessary. Source 09’s direction on the exact axiomatic strength of *other* universality and homogeneity assertions (section 61.3) is unaffected.
- (b) *Source 08’s question 11, comparison of non-set-like class well-orders* (section 61.4): answered by sources 11 and 12. Theorems 14.2 and 14.7 compare all class well-orders of a labelled class directly, with elementary comprehension, without ETR and without comparing abstract class order types; this also supplies the single comparison that source 08’s section 17 says does not follow by aligning domains (see the note after it). Source 09’s relation-code direction concerns a different comparison and stays open.
- (c) *Source 08’s question 3, singular birthdays and code lengths*: answered in part by source 12’s theorem 8.3: at a singular cutoff the least binary coding length of the fixed-length enumeration order is $\kappa \cdot \kappa$ when κ is a strong limit and $2^{<\kappa}$ otherwise. Which cuts fail at singular cutoffs stays open.
- (d) *Source 08’s question 2, invariants at regular cutoffs*: partly answered by source 11’s proposition 12.1 (the minimal slice is η_κ , with every global and local cofinality and coinitality κ); density, cellularity and the cut cofinalities beyond these stay open.
- (e) *The reals report’s question “Other ground orders”* (its Section “Further research questions”, `lwo:sec:research`, which names “set-sized suborders of surreal numbers”): answered in part by theorem 8.3 for $X = \mathbf{No}_{<\kappa}$ and the minimal stratum ($n = 0$): $\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \simeq_e \mathbb{B}_\mu$, except at singular strong-limit κ , where it is $\simeq_e \mathbb{B}_{\kappa \cdot \kappa}$ and not $\simeq_e \mathbb{B}_\kappa$. It stays open for $n > 0$, for longer strata, and for other alphabets (source 12’s research question 61.11, source 08’s question 4).
- (f) *Hamkins’s question whether universality of the surreal line is a weak global choice principle* [21]: not settled here, as source 12 says after its remark 61.15.

[merge] Dated note, 3 October 2026 (batch 81). Items (c) and (d) are completed and item (e) is extended by Part VI: the cut spectrum at singular cutoffs, the density, cellularity and cut cofinalities at regular cutoffs, and the longer strata $\mu \cdot \gamma + n$ of **No**_{<κ} (section 33).

1.8 The batch-81 addition: six further manuscripts

[merge] (3 October 2026.) Six manuscripts of batch 81, all dated 3 October 2026 and all continuing this report, are merged into it as one addition, sources 13–18. Sources 13–17 arrived in commit 9a4ce14e9 and were placed in f0cd7032d (batch 81L); source 18 arrived in bd599ac06 and was placed in c4720e1b2 (batch 81L2). Their local numbers continue this report’s sequence; the batch-81 manuscript number is in the second column.

Src.	B81	Manuscript (archive), pages	Pin	Labels, files
13	01	<i>Lexicographic Orders of Surreal Well-Orderings: Further Structure</i> ; 33 pp. (Surreal_Well_Orders_Further_Study.zip)	0ecd158fd	swo:fs;; 13-products-*
14	02	<i>Lexicographic Well-Orders of the Surreals II: cut classification, support thresholds, and class-model absoluteness</i> ; 33 pp. (lexicographic_surreal_well_orders_II.zip)	0ecd158fd	swo:tc;; 14-trichotomy-*
15	03	<i>Lexicographic Orders of Surreal Well-Orders: interpretations, canonical blocks, represented cuts, and fixed-type spectra</i> ; 37 pp. (surreal_lexicographic_orders.zip)	0ecd158fd	swo:cb;; 15-blocks-*
16	04	<i>Lexicographic Orders of Surreal Well-Orders: Cuts, Canonical Blocks, and the Limits of Set-Sized Coding</i> ; 33 pp. (surreal_well_orders_further.zip)	491ed8411	swo:ds;; 16-strata-*
17	05	<i>Lexicographic Well-Orders of the Surreal Numbers, II: exact cut spectra, branch recognition, optimal support reserves, and automorphism-extension obstructions</i> ; 29 pp. (surreal_well_orders_further_study(1).zip)	0ecd158fd	swo:gs;; 17-gaps-*
18	06	<i>Lexicographic Well-Orders of the Surreal Numbers: Cuts, canonical class blocks, and changes of interpretation</i> ; 53 pp. (surreal_lexicographic_orders_further_study.zip)	83befe707	swo:sh;; 18-shuffle-*

The six are independent texts, not editions of one another. Every pair shares 0.6–1.4% of word 8-grams (preamble, this report’s title and address, bibliography), no member file is shared, and the theorem sets differ. Sources 14 and 17 share only the inner directory name `surreal_well_orders_II` and a “II” in the title; source 18’s archive name extends source 15’s, but its pin, structure and results differ. None supersedes another; all six are merged (union). Sources 13–17 read this report at commit `0be9b9134`, before the corrections of `0f52c0dca`; source 18 read it at `83befe707`, after them, and before sources 13–17 arrived, so it never saw them.

They overlap heavily with each other and with this report. The represented-cut theorem that answers source 11’s [research question 61.1](#) is proved in all six by one argument; all six re-prove the core layer of Part IV (fresh separators, completion, interpolation, the core isomorphic to **No**, the diagonal theorem, direct comparison); and the set-sized layer re-proves the reals report. What is new is a second wave of results on questions of this report: the topology, cut spectra and coding lengths of the set-sized strata at birthday cutoffs; the termination classification of word orders; the trichotomy, characters, automorphism orbits, absoluteness and support thresholds for cuts of the set-coded core; the canonical block decomposition of class well-orders with explicit condensation histories; product (non)absorption; a Σ_1^1 comprehension bound; and the countable models.

The addition is printed by theme, not by source, because the sources overlap: Part VI ([part VI](#)) prints the set-sized strata; Part VII ([part VII](#)) prints the six sources’ maps of interpretations, the word orders, the termination conventions and the other comparison orders; Part VIII ([part VIII](#)) prints the class cuts of the core: their classification and recognition, automorphisms, class extensions and support; Part IX ([part IX](#)) prints the unrestricted class well-orders: the canonical block decomposition and condensation histories, finite components and trace fibres, products, a Σ_1^1 comprehension bound, larger universes and countable models, and the sources’ formalization plans. The sources’ introductions are printed in [section 4.5](#), their questions in [section 61.5](#), and their

conclusions, ledgers, reproducibility records, title pages and statement crosswalks after those of the batch-80 sources. The rules of [section 1.2](#) apply unchanged: each shared result is printed once, with every source that proves it named; the other versions are numbered notes, genuinely different proofs are marked routes, and every label of every source survives with the prefixes of [section 1.3.1](#). Re-proofs of results of this report or of other repository reports are printed as notes with pointers, never with a novelty claim.

Status. Sources 13, 15 and 17 call themselves AI-assisted and unrefereed, source 14 unrefereed ([merge] corrected 3 October 2026: source 14 does not call itself AI-assisted); source 18 calls itself “an AI-assisted research manuscript with mathematical proof review”, and its validation “consists of conventional proofs and independent AI-assisted proof review”; source 16 says its checks are “not external peer review”. None claims historical priority, peer review or a machine-checked formalization, and none is formalized here. The status files of sources 13, 14, 16 and 17 are shipped in this directory under their prefixed names.

Where the last write of the addition had to choose (3 October 2026). The six introductions are printed in [section 4](#), with the batch-80 introductions; the foundations sections, the universe sections and the Lean plans of sources 13–18 are printed at the end of Part IX rather than in Part V, so that no existing section is renumbered. Corrections made to source text: “In GBC” added to source 15’s [proposition 51.10](#); source 13’s “new point” about split products annotated as already contained in source 11’s proofs ([note 55.6](#)); source 15’s sentence that adding classes can destroy internal well-foundedness withdrawn for GBC realizations ([note 57.5](#)); source 17’s question table and source 14’s “Four main results” annotated ([section 4.5](#)); citations of this report replaced by cross-references, with the sources’ numbers “at the pin”. Source equations are tagged ($SNN.n$).

2 The objects being studied

There are two orders in this problem. Each object under consideration is itself a well-order of some collection of surreal numbers. We then compare these objects lexicographically using the *numerical* order on their surreal labels. This second order need not be a well-order. Indeed, the orders below contain abundant descending sequences.

Write $L \hookrightarrow M$ for an order embedding and $L \simeq_e M$ for embeddings in both directions. Equimorphism does not imply isomorphism.

Object	Elements	Logical size and interpretation
$\mathcal{W}(X)$	Ordinal-indexed bijections onto a fixed set $X \subseteq \mathbf{No}$	A set; ordinary ZFC order theory
$\mathcal{I}(\mathbf{No})$	Injective set-length ordinal words in $\mathbf{No}$	A class of sets; well-orders of arbitrary set-sized subsets
$\mathfrak{W}_{\text{sl}}(\mathbf{No})$	Set-like class well-orders of all $\mathbf{No}$, equivalently class bijections $\text{Ord} \rightarrow \mathbf{No}$	A higher-order totality; use class variables or uniform families
$\mathfrak{W}_{\text{all}}(\mathbf{No})$	All class well-orders of all $\mathbf{No}$, including non-set-like ones	A higher-order totality; direct labeled comparison is needed

The fraktur notation is deliberate. It does not declare a class with proper classes as members. Statements about $\mathfrak{W}_{\text{sl}}$ and $\mathfrak{W}_{\text{all}}$ are formulas with class variables, or statements relative to a specified larger universe. An embedding of an ordinary class order into one of these totalities is a uniform family: one class relation encodes the object assigned to each set-valued index.

Principal answers. For a fixed infinite alphabet X , the exact ordinal ceiling is $(|X|)^+$. Passing to all set-length surreal words removes that ceiling: there is an explicit embedding of the entire class into **No**, and the singleton words embed **No** in the other direction. The two orders are nevertheless not isomorphic, because limit-length words have set-sized left cofinality.

Under global choice, the set-like global fragment contains a uniformly indexed copy of every class linear order. This includes every set ordinal, Ord, reverse Ord, Ord + Ord, and the whole numerical surreal order. Yet the entire fragment cannot be exhausted by one uniform class of set-codes. Class universality of a subfamily is compatible with a larger totality of class objects.

The unrestricted global order is different again. It has adjacent pairs and convex residual fibers. Every raw well-order begins with a canonical Ord-sequence; if that sequence omits labels, those labels form a later well-ordered tail. This decomposition permits a precise study of *all* well-orders without treating them all as set-like. It also supplies a nonembedding obstruction: the unrestricted order has too many disjoint two-point fibers to embed into the set-like order in the full containing-universe model. A related uniform definability obstruction is provable in KM + GC.

3 Relationship to ProveIt and the literature

We inspected the repository at the immutable revision

ae7c1e6aae3dc58c9c638b35646865ad6c10387d.

The closest precursor is the report on lexicographic well-orderings of the reals [1], in the subdirectory `lexicographic-well-orderings-of-reals/` of the repository's ordinal-order reports. Its general alphabet theorems already establish the cardinality, universality, and ordinal cutoff for every infinite linearly ordered set X . Their application to birthday fragments is a transfer of existing results, not a new generic alphabet theorem.

That report also proves, with $\mathfrak{c} = 2^{\aleph_0}$,

$$\mathbf{No}_{<\theta} \hookrightarrow \mathcal{W}(\mathbb{R}) \quad (\theta < \mathfrak{c}^+), \quad \mathbf{No}_{<\mathfrak{c}^+} \not\hookrightarrow \mathcal{W}(\mathbb{R}).$$

The negative assertion follows from the all-plus chain of length $\mathfrak{c}^+$. It warns against taking an incoherent union of stagewise embeddings.

The repository's foundational report [3] supplies the relevant size discipline: individual surreals are set-coded, bounded fragments are sets, arbitrary class cuts need not be fillable, and quantifying over class functions does not create a class of those functions. Its local set-valued recursion arguments are directly useful. A nearby vector-space report uses the same distinction to avoid an unnecessary ETR hypothesis [5].

The sign-sequence presentation is standard [12, 14]. Absolute universality and homogeneity of the surreal order are part of the established theory discussed by Ehrlich [15]. We supply the short recursion arguments needed here.

The Kanovei–Shelah connection. There is a methodological connection, but the objects are different. Kanovei and Shelah's actual index consists of maps

$$a : \mathfrak{c} \longrightarrow \mathcal{P}(\mathbb{N})$$

whose ranges are ultrafilters. They compare these maps lexicographically using the binary order on $\mathcal{P}(\mathbb{N})$ [16, Section 1]. The maps need not be injective and do not enumerate the entire real line. Their construction illustrates how a definable linear order on many codes can replace a definably selected well-order. The present problem shares that idea while introducing a different level of class-size complexity.

Contribution and attribution. The central developments here are explicit surreal-word compression, direct comparison of unrestricted class well-orders, their canonical initial-Ord decomposition, the bounded-support interpolation theorem, and the relation between omitted ranges and non-set-like tails. No claim about real-specific countability, separability, or gap spectra is transferred to surreal alphabets without proof.

4 The other sources' introductions

[merge] Each of the other manuscripts opens with its own statement of the problem, of its relation to ProveIt, and of the Kanovei–Shelah connection. These introductions are printed whole, one subsection each, followed by notes where the merge corrects or qualifies them. Their summaries of results are not repeated claims: every result they announce is printed, once, in the body.

4.1 Source 12: scope, predecessors, and the results at a glance

The phrase “well-orders of surreal numbers” hides several distinct objects. The usual numerical order of **No** is not a well-order. One may instead put an unrelated well-order on the entire surreal carrier, or consider numerical well-orders on subclasses, or work with an ordinary set of bounded-birthday surreals. This article treats all three viewpoints, but the principal object is the lexicographic comparison of *exhaustive well-orders of a fixed carrier*.

The supplied ProveIt repository contains a substantial predecessor, *Lexicographic Orders of the Well-Orderings of the Reals*, merging three reports and distinguishing the full enumeration order from its minimal-length slice [1]. The inspected README explicitly marks that report as unrefereed and not formalized. Its real-number spectra, binary coding, finite-tail geometry, and warnings about isomorphism versus equimorphism motivate the present work. The repository also contains a Lean development of simplest elements of convex sets of sign sequences [10]. These are useful foundations, not a formal verification of the new theorems below.

We pinned the audit to repository commit

63bb8b1d99359e7188abf0a33fb152e121858150.

The audit was targeted, not an exhaustive review of the repository. All essential arguments used here are written out, so no unreviewed research claim from the repository is required as a black box.

4.1.1 Three different orders

For an infinite set-sized linear order X of cardinality μ , define

$$\mathcal{W}_\alpha(X) = \{e : \alpha \longrightarrow X : e \text{ is bijective}\}, \quad \mathcal{W}(X) = \bigcup_{\mu \leq \alpha < \mu^+} \mathcal{W}_\alpha(X).$$

They are ordered by their first unequal entry, using the given order of X , not the well-order being represented.

For the proper class **No**, let $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ denote, as a *predicate on classes*, the class bijections $E : \text{Ord} \rightarrow \mathbf{No}$. These represent exactly the set-like global well-orders. Let $\mathfrak{W}_{\text{all}}(\mathbf{No})$ denote the predicate of being an arbitrary class well-order relation on **No**. Neither displayed notation asserts the existence of a class having these class-sized objects as members.

The comparison on $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ uses a least ordinal of disagreement. The comparison on $\mathfrak{W}_{\text{all}}(\mathbf{No})$ uses a largest common *labelled* initial segment. The latter can be a proper class, and this is essential rather than a cosmetic extension.

4.1.2 What is proved, and what is not being claimed

The principal developments are the elementary common-prefix theorem ([theorem 14.7](#)), the global-choice equivalence ([theorem 13.2](#)), the non-enumerability theorem ([theorem 19.38](#)), and the singular transition ([theorem 8.3](#)). The full and minimal bounded spectra are proved in [sections 6.2](#) and [8](#). The difference between the dense set-like order and finite-tail geometry in the full order is proved in [sections 16.1](#) and [19.2](#).

These are mathematical claims with supplied proofs. We do not claim that they settle a previously catalogued open problem, that their exact formulations are historically new, or that they have been independently refereed. The proposed research questions at the end are precisely that: questions suggested by the results, not an assertion of a complete literature census.

4.1.3 The Kanovei–Shelah connection

Kanovei and Shelah order maps $a : \mathfrak{c} \rightarrow \mathcal{P}(\mathbb{N})$ whose ranges are ultrafilters, using a fixed lexicographic order on $\mathcal{P}(\mathbb{N})$ [[16](#), §1]. Their maps need not be bijections, and their index is not the literal collection of all well-orders of $\mathbb{R}$, still less all class well-orders of **No**. The useful common idea is to compare representations at their first difference instead of choosing one canonical representative. Their index construction remains set-sized. Moving to all global surreal well-orders creates an additional size level that their construction does not remove.

[merge] *On source 12's introduction.* Source 12 read the reals report's README and the Lean file `SignSequenceSimplicity.lean`, not the reals report's article; it therefore re-proved the set-sized patterns it needed. They are printed here as a marked route with pointers to the reals report ([sections 6.1](#) and [6.2](#)). The reals report's article has not changed since source 12's pin. Source 12's principal developments are printed as [theorem 14.7](#) (a second route to [theorem 14.2](#)), [theorem 13.2](#) (a second route to [theorem 13.1](#), over $\text{GB} + \text{AC}$), [theorem 19.38](#) (printed beside [theorem 19.37](#)) and [theorem 8.3](#) (new).

4.2 Source 09: the question and the principal conclusions

A surreal number is a set-sized object, but the totality **No** of surreal numbers is a proper class. Consequently, taking “all well-orders” changes logical level when one passes from $\mathbb{R}$ to **No**. The main task is not merely to replace the real alphabet by a larger alphabet. It is to specify which objects are being ordered and which collections of those objects are available in the foundational theory.

Throughout, the word *embedding* means an order-preserving and order-reflecting injection. It need not preserve cuts, suprema, the order topology, birthdays, simplicity, or field operations. An order is *equimorphic* with another when each embeds into the other; this is weaker than isomorphism.

4.2.1 Three different carriers

For a set $X \subseteq \mathbf{No}$, let $\mathcal{W}(X)$ consist of all bijections $e : \alpha \rightarrow X$, with α an ordinal. These are exactly the exhaustive enumerations associated with well-orders on the fixed labeled set X . For different e, f , comparison takes place at the first unequal value. Because both are exhaustive, neither can be a proper initial segment of the other.

Let $\mathcal{I}$ instead contain every set-length injective word $p : \alpha \rightarrow \mathbf{No}$, allowing its range to vary. Here proper-prefix pairs do occur, so an end-of-word convention is necessary. We use the convention that a proper prefix comes first. This is ordinary dictionary order, *not* the convention used to compare surreal sign sequences.

Finally, let $\mathfrak{W}_{\text{sl}}$ be informal notation for the global class bijections $E : \text{Ord} \rightarrow \mathbf{No}$. Each represents a set-like global well-order. In GBC, $\mathfrak{W}_{\text{sl}}$ is notation for a condition on class variables, not a class whose members are these class functions. A precise, ordinary class substitute is the set-coded core $\mathcal{P}_s(b)$ of eventual perturbations of a fixed $b : \text{Ord} \rightarrow \mathbf{No}$.

Main global conclusion. In GBC, for every fixed global enumeration b ,

$$(\mathcal{P}_s(b), <_{\text{lex}}) \cong (\mathbf{No}, <).$$

The core is order-dense among all global enumerations and even supplies interpolants for every uniformly set-indexed separated cut involving arbitrary global enumerations. Nevertheless, there is no uniform class-indexed enumeration of *all* global enumerations, and certain proper-class-sized separated families have no interpolant at all.

4.2.2 How this extends the repository’s real-line investigation

The inspected ProveIt snapshot contains a substantial merged report on lexicographic well-orderings of the reals [1]. That report studies ordinal spectra, binary coding, finite tails, local characters, and gaps. Its set-sized perspective is useful here, but its continuum bounds cannot simply be applied to the entire surreal class. We reprove the set-sized arguments needed below for an arbitrary linearly ordered set, without importing unverified transfinite claims from that report.

The genuinely global development is organized around four concrete constructions: bounded-support extension of an injective prefix; set-indexed lexicographic interpolation; an eventually-equal surreal core; and diagonalization at independent pairs of positions. These constructions are the main contribution of this manuscript. Whether particular formulations have appeared previously would require a more exhaustive literature search; they are not presented as established historical firsts.

Kanovei and Shelah’s nonstandard-model construction provides motivation for lexicographically ordering redundant codes. Their actual index is not the collection of all real well-orders: it consists of maps from the continuum cardinal to $\mathcal{P}(\mathbb{N})$ whose ranges are ultrafilters, ordered at their first disagreement [16, §1]. This distinction remains essential in any proposed surreal analogue.

[merge] *On source 09’s introduction.* Source 09 read the reals report’s README and the Lean files `SignSequence.lean` and `SignSequenceCutOperation.lean`. Its set-sized arguments, which it “reproves ... without importing unverified transfinite claims”, are printed as notes and one marked route (section 6.3). The four constructions it calls its main contribution were found independently by the other sources on the same day: bounded-support prefix completion also by sources 11 and 08 (lemma 19.4 and note 19.8); set-indexed interpolation by sources 11, 12 and 08 (theorem 19.10 and notes 19.11 and 19.13); the eventually-equal core by sources 11 and 08 (corollary 19.15 and note 19.17); and the pair-switch diagonalization by sources 11, 12 and 08 (theorems 19.37 and 19.38 and note 19.41). No priority among the sources is implied by the order of printing. The box’s last sentence is theorem 19.42 and theorem 19.43.

4.3 Source 08: the question, disambiguated

The phrase “well-orders of surreal numbers” has two independent ambiguities. First, is the carrier a set of surreal numbers or the whole proper class? Second, is a well-order required to be the restriction of the numerical order, or is it a new ordering of the labelled carrier? A third choice enters when the well-orders themselves are compared: which data are read lexicographically?

This paper uses the following principal convention. An arbitrary well-order on a labelled carrier is represented by its exhaustive increasing enumeration *in that well-order*. Two enumerations are then compared at their first unequal entry, using the *numerical* order of their surreal values. The enumerations themselves need not be numerically increasing. This distinction is indispensable.

For example, the two lists $(0, 1, -1)$ and $(0, -1, 1)$ represent two well-orders of the same three-element set. The second list is smaller in our lexicographic order, since $-1 < 1$. Both underlying relations have ordinal type 3. Passing to ordinal isomorphism types would identify them and destroy the object being studied.

4.3.1 Five different carriers

Object	Elements or parameters	Foundational size
$\mathcal{W}(X)$	Exhaustive enumerations of one fixed set X	A set in ZFC
$\mathcal{I}(\mathbf{No})$	Injective ordinal-length, set-sized words in $\mathbf{No}$	A class of sets
$\mathfrak{M}_{sl}$	Class bijections $\text{Ord} \rightarrow \mathbf{No}$	A virtual collection of classes
$\mathcal{P}_s(b)$	Set codes for set-supported changes of a fixed b	A class of sets
Arbitrary class well-orders	Class relations, possibly not set-like	A virtual collection; another comparison rule is needed

A *virtual collection* is notation for a condition on class variables, not a new object whose members are proper classes. Thus “ $e \in \mathfrak{M}_{sl}$ ” abbreviates “ e is a class bijection from Ord onto $\mathbf{No}$.” All statements involving several such parameters can be translated into the two-sorted language of class set theory. Whenever a family of classes is used, it is represented by one uniform class relation. This convention is made precise in [section 19.1](#).

4.3.2 Main conclusions

The central distinction is between a very large ambient order and its class-sized dense skeleton. Fix a global enumeration $b : \text{Ord} \rightarrow \mathbf{No}$. Write $\mathcal{P}_s(b)$ for canonical set codes for permutations σ of Ord that equal the identity outside a set, ordered by the numerical lexicographic comparison of $b \circ \sigma$. We prove

$$(\mathcal{P}_s(b), <_b) \cong (\mathbf{No}, <). \quad (\text{S08.1})$$

Moreover, if uniformly given set-sized families L, R of arbitrary global enumerations satisfy $L <_{\text{lex}} R$, then some $b \circ \sigma$ strictly separates them. The separator is in the same fixed skeleton regardless of the supports, or lack of supports, of the input enumerations.

This yields a useful answer to the embedding question. Every *set* linear order, every ordinal, and indeed every *class linear order whose elements are sets* embeds into $\mathcal{P}_s(b)$. In particular, Ord , $\text{Ord} + 1$, $\text{Ord} + \text{Ord}$, their reversals, $\mathbf{No}$, and each set-indexed lexicographic power of $\mathbf{No}$ embed. This does not say that the full virtual order $\mathfrak{M}_{sl}$ is itself a class order or is isomorphic to $\mathbf{No}$.

At a fixed set-sized carrier X , the answer is different. If $|X| = \kappa$ is infinite, then

$$|\mathcal{W}(X)| = 2^\kappa, \quad \beta \hookrightarrow \mathcal{W}(X) \iff \beta < \kappa^+. \quad (\text{S08.2})$$

Thus making the carrier larger changes the allowed ordinal spectrum; there is no legitimate substitution of a “cardinal $|\mathbf{No}|$ ” into [\(S08.2\)](#).

4.3.3 Relationship to existing work

The set-sized part extends the exhaustive-enumeration viewpoint of the prior working manuscript on the continuum [2]. Its results are reproved here in a form that does not assume separability of the alphabet. The standard surreal background is the sign-sequence and cut-filling viewpoint associated with Conway, Gonshor, and Alling [11, 12, 13]; Ehrlich’s work places the surreal continuum in a broader universality framework [15].

The immediate formalization base is the inspected ProveIt sign-sequence construction and its small-cut theorem [7, 8]. We do not need the repository’s field arithmetic or stronger research claims. The resemblance to the Kanovei–Shelah definable hyperreal construction is real but limited: their actual index consists of fixed-length maps whose ranges are ultrafilters, not literally all well-orders of the reals [16]. We return to this distinction in [section 4.4](#).

[merge] *On source 08’s predecessor.* The “prior working manuscript on the continuum” that source 08 cites [2] is batch-76 manuscript 02, *Lexicographic Orders of Well-Orderings of the Continuum*, which is now the base source 02 of the repository’s reals report (placed in 2a04b60f2, written in 718bfa3b5 and 603f6ed31). Source 08 did not read that report. Its set-sized results re-prove the reals report’s general-alphabet theorems, which hold for every infinite linear alphabet, separable or not; they are recorded with pointers in [sections 6.1](#) and [6.4](#). The display (S08.1) is [note 19.17](#), and (S08.2) is [notes 6.21](#) and [6.23](#).

4.4 Source 08: the Kanovei–Shelah connection, carefully stated

Kanovei and Shelah’s definable nonstandard model uses an index A of maps

$$a : \mathfrak{c} \longrightarrow \mathcal{P}(\mathbb{N})$$

whose ranges are ultrafilters. They compare these maps lexicographically using the binary characteristic-function order on subsets of $\mathbb{N}$. Repetitions are allowed; the index is not a well-order [16]. It is therefore not literally the order of all bijective well-order enumerations of $\mathbb{R}$.

The relevant common idea is that ordinal-indexed codes can be linearly compared at their first unequal value even when no definable choice of one representative from each coded object is supplied. Our paired codes, terminal codes, and virtual relations explore that order-theoretic idea in the surreal setting. They do not provide a class ultrapower or a transfer theorem.

An extension to a class-sized ultrapower would need separate decisions about admissible predicates, function families, equivalence-class coding, truth or satisfaction, and the exact size of saturation. Neither a dense index order nor a surreal-sized skeleton settles these questions. In particular, the set-cut interpolation theorem is an order-theoretic theorem, not a claim of saturation for every language or every class-sized family of types.

[merge] The Kanovei–Shelah distinction is made in the same terms by all four sources (source 11 in [section 3](#), source 12 in [section 4.1](#), source 09 in [section 4.2](#), source 08 above) and by the reals report itself, which analyses the actual index (`lwo:eq:KS`, `lwo:sec:KS-topology`, `lwo:rk:sec:KS`). The uniform constructions it suggests are research questions here ([research question 61.10](#); source 09’s and source 08’s last directions).

4.5 The introductions of sources 13–18

[merge] (3 October 2026.) The opening sections of the six batch-81 sources are printed whole in the following subsections, in source order, followed by the Kanovei–Shelah remarks of sources 14 and 17; the other four batch-81 sources discuss the Kanovei–Shelah index in their maps of interpretations

(Part VII, [section 34](#)). Every result they announce is printed once in Parts VI–IX. Their equations are tagged (SNN.*n*) as elsewhere. Corrections and qualifications:

- *Stale descriptions of this report.* All six describe this report at their pins, as merged from four manuscripts; sources 13 and 15 (and source 14 in its audit section) give its length as 115 pages. It now merges ten manuscripts in nine Parts. Their citations of this report are printed as cross-references, with section and theorem numbers “at the pin” where the source gives them.
- *One theorem announced six times.* Each source presents the represented-cut criterion among its main results; it is one theorem, proved independently in all six ([note 46.1](#)). Source 16’s “all ordinal-class blocks at once” and source 18’s ([S18.3](#)) are likewise one theorem with source 15’s ([note 52.1](#)), and source 18’s density display ([S18.2](#)) is source 16’s theorem ([theorem 27.2](#)). Source 18, which could not know sources 13–17, calls these results “advances” over the repository in its table; relative to the merged report they are independent proofs of results of the sibling sources, and only the items listed as new in [sections 33, 39, 50 and 60](#) carry that word here.
- *Source 14’s “Four main results”* lists five; the heading is kept.
- *Source 17’s question table* maps its reserve theorem to source 11’s [research question 61.6](#), which asks for birthday bounds of word embeddings; the reserve theorem bears on source 09’s direction 6 and source 08’s question 13 instead ([note 49.11](#)). The table also omits source 08’s questions 2, 3, 6 and 12, on which its results bear ([sections 33 and 50](#)).
- *Parameters.* Source 16’s overview writes κ for the size of an alphabet and source 18’s writes ν, τ, ρ ; they are printed μ and r, q, θ as in [section 1.3.1](#). Source 18’s display defining $X_\kappa = \mathbf{No}_{<\kappa}$ is omitted with its symbol.

4.6 Source 13: scope, provenance, and the main advances

The phrase “the lexicographic ordering of all well-orders of surreal numbers” does not determine a single object. There are three independent choices: what is being well-ordered, how that well-order is represented, and in what universe the collection of representations is allowed to exist. A fourth choice appears when representations can have different lengths: the position of a termination symbol.

The ProveIt repository already contains a substantial report on these issues, assembled from four independent manuscripts. Its maintained `article.pdf` is described as 115 pages, and the report explicitly distinguishes written proofs from machine-checked results (this report and its README at the pin; [\[merge\]](#) the page count and the number of sources have grown with the batch-81 addition, and the report is still not formalized). We use the snapshot `0ecd158fd163bc4aca0ee40ccf6479ca6cf5f2a8`. The earlier report supplies the fixed-alphabet theory, the surreal-sized dense core, the direct comparison of raw class well-orders, and several nonembedding and coding arguments. These are credited below and, where needed for the new proofs, re-established in a compact form.

The present investigation addresses four of that report’s proposed continuations rather than relabeling its existing results as discoveries. The term “new” below means an extension developed in this manuscript relative to the inspected material; it is not a bibliographic priority claim.

Main conclusions. A normalized cut comes from an injective Ord-word exactly when it crosses a unique prefix cylinder at every set-ordinal length. It comes from a global enumeration exactly when these prefixes eventually include every label. Nonbranch cuts stop either at a successor step, where they separate whole next-letter cylinders, or at a limit step, where they lie on the boundary of one cylinder.

For the virtual order $\mathfrak{W}_{\text{sl}}$ of all class bijections $\text{Ord} \rightarrow \mathbf{No}$,

$$\mathfrak{W}_{\text{sl}} \cong \mathbf{No} \times_{\text{lex}} \mathfrak{W}_{\text{sl}},$$

but $\mathfrak{W}_{\text{sl}} \times_{\text{lex}} 2 \not\rightarrow \mathfrak{W}_{\text{sl}}$ in the full inaccessible model and in the specified uniform $\text{KM} + \text{GC}$ sense. This is not a contradiction: $\mathbf{No}$ is an ordinary class of sets, whereas the full $\mathfrak{W}_{\text{sl}}$ is a domain of class objects.

An order automorphism can send a prefix cylinder to a non-cylinder. Consequently the numerical linear order on enumerations does not, by itself, canonically specify the prefix tree. Central termination, on the other hand, repairs the small one-sided characters of ordinary dictionary words and produces an order isomorphic to $\mathbf{No}$.

4.6.1 What is proved, and what remains conditional

The cut-reconstruction, product-absorption, recoding, and central-word results are proved in GBC. The product obstruction is proved in two separate interpretations: ambient ZFC with a strongly inaccessible universe, and a fixed-formula uniformity schema in $\text{KM} + \text{GC}$. The latter proof identifies its impredicative comprehension step. We do not claim that the same argument works over bare GBC.

This separation also prevents a false “absolute” assertion about the full global order. In an externally countable model, the available global enumerations form an externally countable dense order without endpoints, hence a copy of the rationals. Arbitrary external maps and internally uniform constructions therefore answer different embedding questions.

4.6.2 Reading paths

Section 34.1–32.2 survey the interpretations and size constraints. Section 42.2–46.4 establish the dense-core and cut machinery. Section 55–36.4 contain the other main extensions. Sections 57.3 and 59.1 specify foundational and proof-assistant interfaces. The final research agenda distinguishes questions resolved here from stronger problems still to be investigated.

4.7 Source 14: the continuation and its main conclusions

The phrase “the lexicographic ordering of all well-orders of surreal numbers” does not specify one mathematical object. It leaves open what is being well-ordered, how it is represented, how termination is compared, and which logical universe supplies the word “all.” These choices are not cosmetic: they change cardinality, adjacency, completeness, and even the logical sort of the objects under study.

The present article continues the merged ProveIt report (this report and its README), inspected at commit 0ecd158fd163bc4aca0ee40ccf6479ca6cf5f2a8. That report combines four earlier manuscripts and already proves a large body of results. In particular, its Section 19 ([merge] Part IV here) supplies the dense set-supported skeleton, and its research agenda explicitly asks which cuts of that core represent exhaustive branches. Repeating the skeleton isomorphism or the set-sized ordinal spectrum would not answer that question.

Our main ambient theory is $\text{GBC} = \text{GB} + \text{GC}$, with Power Set and ordinary set Foundation. The letters $\mathfrak{W}_{\text{sl}}$, $\mathfrak{J}$, and $\mathfrak{W}_{\text{all}}$ denote predicates on classes, not classes whose members are proper classes. Fix a class bijection

$$b : \text{Ord} \longrightarrow \mathbf{No}.$$

The ordinary class $\mathcal{P}_s(b)$ consists of set codes for permutations of $\mathbf{Ord}$ with set support, evaluated against b . The virtual order $\mathfrak{W}_{sl}$ consists of all admitted class bijections $\mathbf{Ord} \rightarrow \mathbf{No}$.

Four main results. ([merge] Five are listed below.)

Cut classification and recognition. Every proper normalized class cut of $\mathcal{P}_s(b)$ is uniquely a cylinder edge, a numerical gap after a set prefix, or an $\mathbf{Ord}$ -long injective branch. Such a cut is the trace of an element of $\mathfrak{W}_{sl}$ exactly when its crossing prefixes occur at every ordinal length and together contain every surreal (Theorem 43.3 and 45.1).

No newly filled old cuts. For two GBC class realizations with the same sets and the same b , an old skeleton cut is realized by a global enumeration in the larger realization if and only if it already was in the smaller one (Theorem 48.1).

Exact support threshold. For every infinite set cardinal κ ,

$$\mathcal{P}_{<\kappa}(b) \text{ is } \eta_{cf \kappa} \text{ and is not } \eta_{(cf \kappa)^+}.$$

A witnessing omitted configuration has a minimum possible support cost of exactly κ in the full core (Theorem 49.3).

What unrestricted tails hide. Two labelled class well-orders have the same trace on $\mathcal{P}_s(b)$ exactly when they have the same canonical initial $\mathbf{Ord}$ -enumeration. Above a fixed injective branch F , the fiber is precisely the family of well-orders of $\mathbf{No} \setminus \text{ran } F$ (Theorem 54.6).

Bare-order cut orbits. Character pairs completely determine the automorphism orbits of normalized cuts, yet exhaustive representability is not preserved by those automorphisms (Theorem 47.2).

The results are complementary. The first determines the missing boundary points; the second studies dependence on the admitted classes; the third measures the amount of permutation support needed for interpolation; and the fourth distinguishes the full relation order from its set-like part. No one of these conclusions identifies the full collection of class well-orders with an ordinary surreal-sized class.

4.8 Source 15: scope, notation, and the audit boundary

The useful starting point is to specify what the entries of the proposed order actually are. A well-order relation on a fixed set, an enumeration of that set, an isomorphism type, a relation table, and a naturally increasing subclass of $\mathbf{No}$ are different kinds of data. Some determine one another, but their natural-looking lexicographic comparisons need not agree.

The ProveIt report *Lexicographic Orders of Well-Orderings of the Surreal Numbers* combines four earlier manuscripts into a 115-page document ([merge] at that revision). We inspected its source, guide, the related real-alphabet report, and the sign-sequence foundation at revision 0ecd158fd163bc4aca0ee40ccf6479ca6cf5f2a8. The surreal report's source blob is f20837b76f8d981cb1dce3da79f633e280edad41; the same source was present at the earlier search-index revision 491ed8411cccf7bd57916eb7666bbba762e92f96. This article uses the newer revision as its audit boundary (this report; [1, 7]).

The prior report already establishes general set-sized cardinal and ordinal spectra, direct comparison of labelled class well-orders without abstract class ordinals, the global-choice equivalence, a dense bounded-support core isomorphic to $\mathbf{No}$, an omitted-range construction, and the singular minimal-slice dichotomy. We preserve those results as background. The main continuation consists of the block decomposition in Section 52.1, the cut classification in Section 46.2, and the longer

fixed-type calculation in [Section 31](#). The interpretation obstructions in [Section 38](#) explain why these results cannot simply be transferred to every other coding.

Foundational convention. GB means Gödel–Bernays set theory with Power Set and with no choice axiom built into the notation. $\text{GBC} = \text{GB} + \text{GC}$, where global choice provides a class choice function on nonempty sets. Set-sized results are in ZFC unless stated otherwise. A class well-order is a class strict linear order in which every nonempty available subclass has a least element. This explicit convention avoids silently confusing internal class well-foundedness with well-foundedness in an external metatheory. The standard equivalent formulations under global choice are discussed in [Section 57.4](#).

Order notation. An embedding preserves and reflects strict order. $L \simeq_e M$ means that there are embeddings in both directions; it does not mean that L and M are isomorphic. The product $L \times_{\text{lex}} M$ compares the L coordinate first. Ordinary ordinal multiplication keeps its usual convention: $\alpha \cdot \beta$ is β consecutive copies of α . We write $\mathbb{B}_\theta = (2^\theta, <_{\text{lex}})$, where $0 < 1$. The binary coding length $\ell_2(L)$ is the least ordinal θ for which $L \hookrightarrow \mathbb{B}_\theta$, when it exists. Birthday is written $\text{bd}(x)$.

An essential size distinction. Symbols such as $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ and $\mathfrak{W}_{\text{all}}(\mathbf{No})$ name predicates on class variables, with a comparison formula. They are not assertions that an ordinary GB class has other proper classes as its elements. A statement about a uniform family means one class relation whose rows are indexed by sets, possibly by a proper class of set indices. A larger universe may instead make the whole collection into an external set. The two readings are related but must be kept explicit.

4.9 Source 16: main conclusions and relationship to the earlier report

There is a coherent and quite rich theory here, but it is essential to say which objects are being compared. A well-order on surreal labels need not agree with their numerical order. An ordinal-indexed enumeration of the whole surreal class is necessarily a class function. An arbitrary class well-order need not possess such an enumeration. Finally, even when each well-order is a legitimate class, a collection having those classes as its elements is not an ordinary class of the same two-sorted theory.

The ProveIt manuscript (this report, read at revision 491ed8411) already establishes the basic distinctions, exact ordinal embedding bounds for set alphabets, the surreal-sized bounded-support core, direct comparison of non-set-like orders, and a full-universe nonembedding result. Its merged source contains four earlier manuscripts and explicitly leaves several sharper questions. The present article develops those questions rather than treating the earlier conclusions as new. The principal additions are as follows.

1. **Topology at every enumeration length.** For an infinite linearly ordered set X of cardinality μ and an ordinal α with $|\alpha| = \mu$,

$$\text{d}(\mathcal{W}_\alpha(X)) = \text{cell}(\mathcal{W}_\alpha(X)) = \begin{cases} \mu^{<\mu}, & \alpha \in \{\mu, \mu + 1, \mu + 2\}, \\ 2^\mu, & \text{otherwise.} \end{cases}$$

For the full order, both invariants equal 2^μ . Every infinite stratum, and the full order, has a countable witness to failure of conditional Dedekind completeness.

2. **Regular-stage gap spectrum.** If X is an η_μ order of size μ , for regular infinite μ , the unfilled two-sided cuts of $\mathcal{W}_\mu(X)$ have exactly the characters (μ, λ) and (λ, μ) , where λ ranges over infinite regular cardinals at most μ .
3. **Represented cuts.** A lower cut of the bounded-support core represents an injective Ord-branch exactly when it is straddled by a prefix cylinder of every set ordinal length. The branch is exhaustive exactly when those prefixes collectively contain every label. A failed test has a first successor or limit obstruction; arbitrary non-set-like well-orders do not remove either obstruction.
4. **All ordinal-class blocks at once.** Every class well-order R admits the canonical decomposition

$$(X, R) \cong \left(\sum_{j \in J_R} \text{Ord} \right) + \alpha_R,$$

where the summands are precisely the equivalence classes for set-sized closed intervals, and α_R is the optional final set block. The block index can itself be any class well-order. The construction is elementary in GB, and directly defines the entire family of residual tails.

5. **Word conventions and formal strength.** For the shorter-prefix-first convention, arbitrary, injective, and numerically increasing set-length words are isomorphic in GBC by maps preserving lengths and prefixes. Other placements of a termination symbol can instead make the whole unrestricted or injective word order isomorphic to **No**. For elementary relationally specified transformations, a uniform nonembedding theorem needs only Σ_1^1 class comprehension over GBC, rather than all of KM.

These are extensions relative to the inspected report, not announcements that previously catalogued problems in the literature have been solved. The proofs below are intended to be independently readable. When a longer existing theorem is only summarized, its provenance and exact scope are stated. We separate mathematical proof, finite diagnostic checks, and formal proof throughout.

4.10 Source 17: relation to the repository and statement of the advance

4.10.1 Pinned source and scope

The main input is the merged ProveIt report *Lexicographic Orders of Well-Orderings of the Surreal Numbers* (this report), at repository revision

0ecd158fd163bc4aca0ee40ccf6479ca6cf5f2a8.

The relevant path is

Algebra/SurrealNumbers/docs/foundations-and-computation/surreal-well-orders/article.tex.

Its blob identifier is f20837b76f8d981cb1dce3da79f633e280edad41. The accompanying README describes a union of four independent manuscripts, not four successive editions. It also records the unrefereed and unformalized status of that report. We use the merged text rather than silently selecting one of the earlier conventions.

The source already proves the basic set-alphabet embedding results, the direct comparison of unrestricted labelled class well-orders, bounded-support interpolation, the existence of a surreal-sized dense core, a diagonal no-coding theorem, and a singular-cardinal transition in binary coding length. We do not present these as new. In particular, binary equimorphism is not an order-isomorphism classification, and it does not by itself determine local characters or the cut spectrum.

4.10.2 Questions addressed

The following correspondence uses the source's stable LaTeX labels, so it does not depend on pagination.

Source question or topic	Result here	Limitation
<code>swo:n16.7</code> ; <code>swo:st:n14.3</code>	Exact point characters, cut spectrum, density, cellularity, weight, saturation of the minimal slice	Not an isomorphism classification; not every nonminimal type
<code>swo:n16.1</code> ; source 09, direction 3	Cylinder completeness and coverage characterize represented cuts	Uses the labelled prefix structure, not just the bare core order
<code>swo:n16.6</code> ; source 09, direction 6	Optimal additional-support cardinal for prefix completion	Not an optimal birthday bound for every word code
<code>swo:n16.4</code> ; source 09, direction 4	Exact automorphism extension criterion and a nonextendible core automorphism	No classification of the entire automorphism group
<code>swo:n16.8</code> ; <code>swo:st:n14.8</code>	Same-set class extensions are detected by newly represented cuts	Does not reconstruct the entire class realization up to isomorphism
<code>swo:st:n14.7</code>	Maximal finite adjacency blocks and density of their quotient	No full quotient isomorphism classification

These are extensions of specific research questions posed in the repository, not assertions that the questions have an established history as major open problems. Every result used from the source is either restated with a proof or clearly identified as background.

4.11 Source 18: main results and the scope of the continuation

The current **ProveIt** report on surreal well-orders (this report, read at revision 83befe707) merges four earlier manuscripts. It already distinguishes set and class interpretations, proves the set-sized ordinal spectrum, analyzes binary coding at singular birthdays, compares arbitrary class well-orders directly, characterizes adjacency, and constructs a dense bounded-support core in the set-like global order. Repeating these conclusions would not answer the request for further study. We use them as a starting point, prove the elementary tools needed below, and address the outstanding questions on cuts, block decompositions, termination conventions, and model dependence.

Here is the principal bounded result. Throughout the article, κ is an infinite initial cardinal and

$$\mu = |\mathbf{No}_{<\kappa}| = 2^{<\kappa}, \quad r = \text{cf } \kappa, \quad q = \text{cf } \mu.$$

Let $P = \mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ consist of all bijections $\mu \rightarrow \mathbf{No}_{<\kappa}$, ordered by their first unequal values. We prove

$$|P| = 2^\mu, \quad d(P) = \text{cell}(P) = \mu^{<\mu}, \quad \text{cf}(P) = \text{ci}(P) = r,$$

and every point has left and right character q . The order fills all cuts specified by fewer than r points. Its proper Dedekind gaps have exactly the character pairs

$$\{(r, r), (q, q)\} \cup \{(\theta, r), (r, \theta) : \theta < \mu \text{ is an infinite regular cardinal}\}. \quad (\text{S18.1})$$

This is a gap spectrum, not an isomorphism classification. Orders with the same spectrum can still have different global arrangements of cuts. There are exactly 2^μ proper gaps of character (q, q) , and exactly $\mu^{<\mu}$ of every other character in the displayed spectrum. Thus the Dedekind completion has cardinality 2^μ .

For every infinite ordered alphabet X of size μ , a separate fixed-type result gives

$$d(\mathcal{W}_\alpha(X)) = \text{cell}(\mathcal{W}_\alpha(X)) = \begin{cases} \mu^{<\mu}, & \alpha = \mu, \mu + 1, \mu + 2, \\ 2^\mu, & \mu + 3 \leq \alpha < \mu^+. \end{cases} \quad (\text{S18.2})$$

The third terminal position is the first that supplies an interior point of a finite permutation fiber. This is an order-topological transition, and its numerical values need not differ when $\mu^{<\mu} = 2^\mu$.

On the class side, let R be any supplied well-order of a proper class. Declare two labels equivalent when the closed interval between them is a set. Every block has an actual least label, which represents it without making a class an element. In GB this gives

$$R \cong (B_R \times \text{Ord})_{\text{lex}} + \theta_R, \quad (\text{S18.3})$$

with all proper blocks of type Ord and at most one final set block of type θ_R . The block index B_R can be any class well-order; it can even be isomorphic to R . Thus simultaneous decomposition is elementary, while repeated quotienting need not lower isomorphism type.

A class cut of the bounded-support core comes from a raw global well-order precisely when, at every ordinal length, one prefix cylinder meets both sides. These prefixes uniquely reconstruct an injective Ord -word by elementary comprehension. The cut has a set-like representative exactly when that word is onto. A cut that would require the first two entries to be the same surreal is represented by no raw well-order. Allowing arbitrary class well-orders therefore does not fill all class cuts, even though it fills the omitted-range branch cuts.

A further theorem concerns an external, countable model of the class theory. If the model is ω -standard, the full admitted order $\mathfrak{W}_{\text{all}}$ is a dense shuffle of finite chains of sizes

$$1, 2, 6, 24, 120, \dots$$

Its finite-condensation quotient is the rational order with every factorial size appearing densely as a color. The admitted set-like order $\mathfrak{W}_{\text{sl}}$ is externally isomorphic to $\mathbb{Q}$, but it is not definable in the pure full order, even with finitely many parameters. These are external statements; they do not supply an internally coded embedding of all global orders into a class of sets.

The final new group of results concerns set-length words. Ordinary dictionary comparison puts a proper prefix first. Under GBC, arbitrary, injective, and strictly numerically increasing words with this convention are isomorphic as prefix trees and as linear orders. Centered termination over $\mathbf{No} \setminus \{0\}$ instead makes arbitrary and injective words isomorphic to $\mathbf{No}$. Increasing words retain a precisely identified limit-prefix obstruction. Keeping the actual digit 0 immediately above a distinct terminator creates adjacent pairs. Hence “lexicographic” must include a termination rule.

4.11.1 What is established here

Topic	Repository starting point	Advance in this article
Birthday cutoffs	Coding rank and selected regular-stage results	Exact density, cellularity, point characters, saturation, and proper-gap spectrum with multiplicities
Fixed ordinal types	Finite-tail geometry; longer strata proposed	Exact density/cellularity transition at $\mu + 3$
Class blocks	Initial set-like part; iteration strength proposed	Elementary simultaneous decomposition, full quotient spectrum, self-similarity
Class cuts	Dense core and omitted-range cuts	Exact prefix criterion, elementary reconstruction, and first-failure classification
Countable models	Internal/external distinction	Factorial-shuffle classification and pure-order nondefinability
Word conventions	Prefix-first words and their limit obstructions	Tree isomorphisms; centered conventions and exact increasing-word obstruction

“Advance” means an extension of the inspected repository text. We make no assertion that these formulations have never appeared elsewhere, and do not describe the repository’s proposed directions as a certified list of open problems in the published literature. The proofs below are independent of the repository’s claimed Lean theorems except where a formalization reuse point is explicitly discussed.

4.12 Source 14: the precise Kanovei–Shelah connection

Kanovei and Shelah’s definable nonstandard-model construction uses a lexicographically ordered family of maps from the continuum cardinal to $\mathcal{P}(\mathbb{N})$ whose ranges are ultrafilters; the maps may repeat values [16]. That index set is not literally the set of all well-orders of $\mathbb{R}$. Its relevance is methodological: a linear order on an entire family of presentations can organize a construction without selecting one privileged presentation. Importing that method into a surreal setting must still state the indexing sort, support restriction, and evaluation relation. The diagonal theorem prevents replacing all global enumerations by one exhaustive ordinary class of set presentations.

4.13 Source 17: the Kanovei–Shelah connection, with its size boundary

The definable nonstandard-real construction is a useful motivation for ordering many presentations rather than selecting one preferred presentation. Its actual index should be stated accurately. In Section 1 of Kanovei–Shelah [16], the index consists of maps from the continuum cardinal into $\mathcal{P}(\mathbb{N})$ whose ranges are ultrafilters; the maps need not be injective. The authors order this *set* of maps by first difference, using the binary lexicographic order of subsets of $\mathbb{N}$, and then use finite-support functions. It is therefore not literally the order of all well-order relations on $\mathbb{R}$.

The methodological analogy remains valuable: a specified linear order of codes can organize a construction without a definable choice of one special code. In the present setting, $\mathcal{W}_\mu(X)$ is a legitimate set index, and a uniform subfamily of global enumerations can be indexed by an ordinary class of set codes. Neither observation makes the full higher-order predicate $\mathfrak{W}_{\text{sl}}$ into a set index. Any proposed surreal analogue must state where its functions, supports, evaluation maps, and

ultrafilters live. The diagonal obstruction of [Theorem 42.16](#) rules out one tempting shortcut, while leaving many bounded or explicitly uniform constructions available.

[merge] The Kanovei–Shelah distinction is thus made by all ten sources, in the same terms ([section 4.4](#) and the note after it).

5 Surreal order facts and foundational conventions

5.1 Signs, birthdays, and set cuts

A surreal number is represented by a sign sequence

$$s : \alpha \longrightarrow \{-, +\}$$

of set-ordinal length α , its birthday $\text{bd}(s)$. Numerical comparison at the first disagreement uses

$$- < \text{termination} < +.$$

This differs from dictionary order with every proper prefix smaller than every extension. A sign sequence can be on either side of an extension, depending on its next sign.

For an ordinal θ , let

$$\mathbf{No}_{<\theta} = \bigcup_{\alpha < \theta} \{-, +\}^\alpha.$$

It is a set. The full $\mathbf{No}$ is proper-class-sized, because all-plus sequences embed all ordinals. Every set of surreals has bounded birthdays by Replacement.

Definition 5.1. A class linear order L has the *set-cut property*, called η_{Ord} here, if for all sets $A, B \subseteq L$ with $A < B$ there is x with $A < x < B$. Empty sides are allowed. For a cardinal κ , the set-sized notation η_κ requires the same property when $|A \cup B| < \kappa$.

The surreal cut construction supplies this property for $\mathbf{No}$ [[14](#), Section 2]. A separator can be chosen canonically as the unique simplest separator. This statement concerns *sets* of options; applying it to the whole class against the empty right side would be inconsistent.

Lemma 5.2 (Deleting a set). *If $S \subseteq \mathbf{No}$ is a set, then $\mathbf{No} \setminus S$ has the set-cut property. Every nonempty open interval of $\mathbf{No}$ is a proper class.*

Proof. Let $A < B$ be sets in $\mathbf{No} \setminus S$. First choose x with $A < x < B$, and put

$$S_- = \{s \in S : s \leq x\}, \quad S_+ = \{s \in S : x < s\}.$$

Then $A \cup S_- < B \cup S_+$. A separator of this new set cut avoids both parts of S . The argument allows empty sides. If an interval (a, b) were a set S , applying the result to $\{a\} < \{b\}$ in $\mathbf{No} \setminus S$ would be impossible. $\square$

Choosing both separators by simplicity makes the deletion argument uniform in the set parameters without global choice.

[merge] *Third route to the first clause of [lemma 5.2](#) (source 12, Lemma 6.2).* Source 12 avoids a forbidden set by using that open intervals of $\mathbf{No}$ are proper classes.

Lemma 5.3 (Avoiding a forbidden set). *If $A < B$ are sets of surreals and D is a set, then some separator z of A, B lies outside D .*

Proof. Choose u between A and B , and then v between $\{u\}$ and B . Every point of (u, v) separates A, B , and that interval is a proper class. It cannot be contained in D . With a fixed global well-order, choose the first acceptable point, making the operation uniform. $\square$

[merge] Source 08 opens with its own toolkit, printed next. Its [proposition 5.4](#) restates `found:prop:proper` (Lean-proved in universe-relative form; [section 1.5](#)) and adds that neither the numerical order nor “birthday first” well-orders **No**, which source 12 also proves ([section 13.1](#)). Its [theorem 5.6](#) sharpens the birthday bound of the cut theorem to a separator of *exactly* any prescribed birthday above the option birthdays; this is proved in Lean as `exists_cut_separator_of_birthday_lt`, by the same complete-lattice argument ([lemma 5.5](#)). Its [corollary 5.7](#) contains the first clause of [lemma 5.2](#) with a prescribed birthday, and its [definition 5.8](#) is [definition 5.1](#).

5.2 Source 08: surreal order and size, the minimal toolkit

We take a surreal number to be a sign sequence

$$s : \alpha \longrightarrow \{-, +\}, \quad \alpha \in \text{Ord}.$$

Its birthday is $\text{bd}(s) = \alpha$. To compare two sequences, extend them by a termination symbol 0 and compare at the first disagreement using $- < 0 < +$. This is the numerical order. The simplicity relation is proper initial-segment extension, and is not the numerical order.

Proposition 5.4. *Every birthday-bounded collection of surreals is a set, every set of surreals has bounded birthdays, and **No** is a proper class. Neither the numerical order nor “birthday first, then numerical order” is a well-order of **No**.*

Proof. For a fixed ordinal λ , the sign sequences of length below λ form the set $\bigcup_{\alpha < \lambda} \{-, +\}^\alpha$. Replacement bounds the birthdays of any given set. Conversely, the all-plus sequences give a distinct surreal for every ordinal, so a set containing all surreals would bound all ordinals. The negative integers already descend numerically. For the second claim, let s_n have birthday ω , with n initial minus signs followed by plus signs forever. At coordinate n one has $s_{n+1}(n) = - < + = s_n(n)$, so $s_0 > s_1 > s_2 > \dots$ within a single birthday level. $\square$

5.2.1 A birthday-controlled separator theorem

We record a proof of the exact order-theoretic fact used later. It also explains the repository’s bounded-word implementation [8].

Lemma 5.5. *For every set ordinal β , the binary lexicographic order $\mathbb{B}_\beta = (\{0, 1\}^\beta, <_{\text{lex}})$ is a complete lattice.*

Proof. The least and greatest elements are the constant 0 and 1 words. Given a set A of words, form w recursively. At coordinate ξ , put $w(\xi) = 1$ exactly when some $a \in A$ agrees with the already constructed prefix $w \upharpoonright \xi$ and has $a(\xi) = 1$; otherwise put 0. If some $a \in A$ exceeded w , its first disagreement would contradict the rule for a 0 coordinate. If $v < w$, then at their first disagreement ξ we have $v(\xi) = 0$, $w(\xi) = 1$, and a witness $a \in A$ agreeing with w through that 1 coordinate. Thus $v < a$ and v is not an upper bound. This proves that $w = \sup A$, including $A = \emptyset$. Infima follow by reversing all bits. $\square$

Theorem 5.6 (Birthday-controlled cut filling). *Let A, B be sets of surreals with $A < B$, meaning $a < b$ for all $a \in A$, $b \in B$. If β is an ordinal strictly greater than every birthday in $A \cup B$, then a surreal z of birthday exactly β satisfies $A < z < B$. Either side may be empty.*

Proof. For a shorter word x , let $u_\beta(x)$ be the word of length β obtained by appending a plus to x and then filling all remaining coordinates with minuses. It is the least full-length word strictly above x . Dually, let $v_\beta(x)$ be x followed by a minus and then pluses; it is the greatest full-length word strictly below x . These assertions follow by inspecting the first differing sign: an extension above x must choose plus when x terminates, and the least possible continuation then uses only minuses.

If $x < y$ and both have length below β , then $u_\beta(x) \leq v_\beta(y)$. Indeed $u_\beta(x) \leq y$ by the same first-disagreement check; equality is impossible because their birthdays differ. Applying the dual greatest-word property gives the claim. By lemma 5.5, full-length sign words have a supremum, and

$$z = \sup\{u_\beta(a) : a \in A\}$$

satisfies $a < u_\beta(a) \leq z$ for every $a \in A$ and $z \leq v_\beta(b) < b$ for every $b \in B$. All words in this complete lattice have birthday exactly β . If A is empty, its supremum is the all-minus word. The vacuous case $\beta = 0$ causes no problem. $\square$

Corollary 5.7 (Fresh separators). *If $A < B$ are sets and S is a set of forbidden surreals, a separator $z \notin S$ exists. It can be required to have any specified birthday strictly exceeding all birthdays in $A \cup B \cup S$.*

Definition 5.8. A class linear order is an η_{Ord} -order if every pair of sets $A < B$ has a strict separator, including empty sides. For an infinite cardinal κ , an η_κ -order satisfies the same condition when $|A \cup B| < \kappa$.

The notation η_κ here uses a cardinal bound directly; it should not be confused with conventions in which the subscript is the index α in $\aleph_\alpha$. By theorem 5.6, **No** is an η_{Ord} -order. Every interval is a proper class: if (a, b) were a set, a separator of $\{a\} \cup (a, b)$ and $\{b\}$ would be a new member of the same interval above all its members. No set is cofinal or coinital in **No**.

5.3 Which recursion principle is used?

Here GB has a ZF set part, elementary class comprehension, and class replacement; choice is stated separately. We use $\text{GBC} = \text{GB} + \text{GC}$, which includes set choice. Cardinality calculations and cardinal-valued cofinality statements throughout the article are read in ZFC, or in the set part of GBC. The explicitly definable sign code and the direct set-cut constructions retain their stated choice-free strength.

Ordinary recursion along Ord, with a set output at every stage and a rule using the set of earlier values and fixed class parameters, is available in GB. On each set ordinal perform the usual recursion with those parameters. Uniqueness makes the local solutions agree; elementary comprehension assembles their graph.

This is different from arbitrary elementary transfinite recursion, ETR, whose stages may themselves be classes. The latter supports iterated truth predicates and is stronger than GBC [22]. The prefix searches, partial permutations, and back-and-forth maps used here have set-sized stages.

[merge] The other sources fix their foundational conventions in their own sections, printed whole next. They agree with source 11's conventions, with one difference of base theory: when it analyses choice, source 12 works over $\text{GB} + \text{AC}$ (section 1.3).

5.4 Source 12: the surreal carrier and the foundational conventions

5.4.1 Sign sequences

We use the sign-sequence presentation

$$\mathbf{No} = \bigcup_{\alpha \in \text{Ord}} \{-, +\}^\alpha.$$

Each individual sequence is a set. Its length $\text{bd}(s) = \text{lh}(s)$ is its birthday. At the first difference, signs and termination are compared by

$$- < \text{end} < +.$$

This is the usual numerical order, not the ordinary “shorter comes first” dictionary convention. Simplicity is the separate relation of being a proper sign prefix. The standard surreal construction and its simplicity structure are treated by Gonshor and Ehrlich [12, 15].

We shall use the following standard feature, also obtainable directly from the construction: whenever A, B are *sets* of surreals with $A < B$, there is a surreal z with $A < z < B$. There is a unique such separator of minimum birthday. Moreover, if every member of $A \cup B$ has birthday less than an ordinal γ , a separator can be constructed with birthday at most γ . This is interpolation, not Dedekind completeness: no least upper bound is being asserted.

Every set of surreals has a birthday bound, by Replacement. Every nonempty numerical interval contains a proper class of surreals. To see the latter without arithmetic, start with $a < b$ and recursively choose x_α between $\{a\} \cup \{x_\xi : \xi < \alpha\}$ and $\{b\}$. The previous values form a set at each stage. Thus one obtains distinct points for all ordinals α .

5.4.2 Well-orders and set-likeness

A class well-order on a class X is a strict linear class relation R for which every nonempty subclass has an R -least element. In GBC this agrees with the usual formulations using nonempty set subsets or absence of descending ω -sequences; the relationship is discussed in [27, §3]. We use strict relations throughout.

It is *set-like* if

$$\text{Pred}_R(x) = \{y \in X : yRx\}$$

is a set for every x . A proper-class set-like well-order is isomorphic to Ord : map x to the ordinal type of its set of predecessors. The range is an initial segment of Ord , and cannot be a set ordinal when X is proper. This construction uses only set-sized order types.

Not all class well-orders are set-like. The ordered sums $\text{Ord} + 1$, $\text{Ord} + \text{Ord}$, and $\text{Ord} + \text{Ord} + 1$ are examples. Their notation means ordered sums of class orders, not new von Neumann ordinals.

5.4.3 Foundational levels

Our set-sized results are theorems of ZFC. Our global results are in GBC, meaning Gödel–Bernays set theory with global choice. When analyzing the necessity of choice we work over GB plus ordinary, set-level AC. The full collection of global well-orders is handled by formulas with class variables, not by pretending that a proper class is an element of another GBC class.

Kelley–Morse theory allows stronger comprehension, but its classes still have sets, not proper classes, as members. Moving from GBC to KM does not by itself create a powerclass object of all subclasses of $\mathbf{No}$. One can instead use a larger set-theoretic universe, a set model with a specified collection of classes, or an explicitly higher-order theory. These are distinct interpretations and will be kept distinct.

5.4.4 Why an ordinal-length recursion here need not invoke ETR

A recursion on Ord that produces one *set* at each stage and uses the set of previous values is available by the ordinary recursion theorem with class parameters in GB. In contrast, ETR permits a genuinely class-valued recursive stage, represented by a class of pairs of a stage and an arbitrary set. Even a truth hierarchy brings stronger principles into play [27, §2]. Our forth constructions, block lengths, and canonical enumeration of a class with set-sized initial segments are set-valued recursions. We do not invoke a universal satisfaction predicate or unrestricted class-valued recursion.

[merge] Like source 12, source 09 works over $\text{GB} + \text{AC}$ when it analyses choice. Its lemma 5.10 is a second route to the first clause of lemma 5.2: it adds the forbidden points inside the cut to the left side, where source 11 first splits the forbidden set at a separator. Its bound (1) is hset:eq:cut-bound of the birthday-cutoff report [4], and its definition 5.9 is definition 5.1.

5.5 Source 09: foundational conventions and the surreal order

5.5.1 Sets, classes, and the meaning of a family

All set-sized results are in ZFC. Global results are in GBC, here meaning Gödel–Bernays class theory with global choice. Classes have sets as members. A proper class cannot be an element of another class. An expression such as “the set of class functions E_i ” is therefore replaced by a *uniform family*: a class relation

$$F \subseteq I \times \text{Ord} \times \mathbf{No}$$

whose i th section is the graph of E_i . The index I may be a set or a class of sets. If I is a set, this is a uniformly set-indexed family, not a literal set of proper classes.

This distinction does not prevent quantification over individual class variables. For example, “for every two class bijections E, F , there is a least ordinal where they differ” is a legitimate GBC assertion. What is unavailable is a class having all such bijections as members. Rangel and Mariano explicitly distinguish the set-subsets of a class from a nonexistent class of all its subclasses [17, §1.1.1].

Global choice will be used in its standard consequence that a proper class can be enumerated by Ord with set-sized initial segments. One way to obtain this consequence is to well-order each rank layer using a global choice function, concatenate the layers, and collapse the resulting set-like well-order. Restricting such an order to any proper subclass again gives order type Ord .

5.5.2 Sign sequences and the cut property

Use the sign-sequence realization

$$\mathbf{No} = \bigcup_{\alpha \in \text{Ord}} \{-, +\}^\alpha.$$

Each displayed element is a set function. Its domain is its *birthday*, denoted $\text{bd}(x)$. Numerical order compares the first different sign using

$$- < \text{termination} < +.$$

It is different both from ordering by birthday and from the dictionary order on arbitrary surreal-valued words used later. The all-plus word of length α represents the ordinal α .

The standard simplicity/cut theorem says that, for sets $A, B \subseteq \mathbf{No}$ with $A < B$, there is a unique simplest surreal strictly between them. Here $A < B$ means $a < b$ for every $a \in A, b \in B$, with empty sides allowed. We write this simplest separator as $\{A \mid B\}$. A useful birthday bound is

$$\text{bd}\{A \mid B\} \leq \sup\{\text{bd}(x) + 1 : x \in A \cup B\}. \quad (1)$$

These are standard sign-sequence facts [12]; the inspected repository also contains explicit small-cut and birthday-bound statements on its sign carrier [9].

In particular, every set of surreal numbers has strict upper and lower bounds. No set is cofinal or cointial in **No**, and no set is the whole of **No**. These assertions do not authorize a cut constructor with arbitrary proper-class option families.

Definition 5.9. A class linear order D has the *set-cut property* if every pair of sets $A, B \subseteq D$ satisfying $A < B$ has an element $d \in D$ with $A < d < B$. Empty sides are allowed.

The surreal order has the set-cut property. In terminology used in the literature this is an absolute or class analogue of an η -order. We use the explicit definition rather than relying on potentially ambiguous subscripts such as η_{Ord} .

Lemma 5.10 (Avoiding a prescribed set). *If D has the set-cut property and $U \subseteq D$ is a set, then $D \setminus U$ also has the set-cut property.*

Proof. Let $A, B \subseteq D \setminus U$ be sets with $A < B$, and put

$$U_0 = \{u \in U : A < u < B\}.$$

Then $A \cup U_0 < B$. Choose x with $A \cup U_0 < x < B$. This x is in the original open cut. If it belonged to U , it would belong to U_0 , giving $x < x$. Thus $x \notin U$. $\square$

5.5.3 The precise recursion strength used here

We repeatedly construct a sequence of *sets* indexed by ordinals, where the next value is a uniquely specified set obtained from the set-sized previous history and fixed class parameters. This recursion is available in GBC: set recursion with class parameters constructs all set-length approximations, uniqueness makes them compatible, and elementary class comprehension joins them.

This is not the principle ETR, whose stages can themselves be proper classes. The latter is genuinely stronger than GBC and must not be silently invoked by saying “perform class recursion” [24, §2]. Every global recursion in the proofs below is explicitly of the former, set-valued kind.

[merge] Source 08’s foundational section, printed next except for its Theorem 8.2, which is recorded in [note 5.12](#). One sentence is corrected, as proposed by the review of the four archives in the repository’s Hilbert’s-tenth research tree (`Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_batch80_surreal.md`, commit `c338541e3`, with the patch `surreal_research1_wellfounded_scope.patch` there). Source 08 wrote: “For non-set-like relations, the model’s class collection can matter to the well-order assertion.” Under GBC, internal well-foundedness of a fixed class relation is equivalent to the absence of a set-coded descending ω -sequence ([27, Section 3]; source 11 makes the same distinction in [section 20](#)), so two class expansions with the same sets agree about it for a relation they share; what can change is which relations are available, and internal well-foundedness differs from external well-foundedness. The printed sentence is the patch’s text. No statement of source 08 depends on the original sentence.

5.6 Source 08: class foundations and set-like well-orders

From this section onward, the default theory is GBC: Gödel–Bernays class set theory with Global Choice. Sets are the objects that may be members; classes are collections of sets. A class function maps set arguments to set values. Its image of a set is a set by class replacement. Elementary class comprehension allows formulas with class parameters but only set quantifiers in the defining formula.

5.6.1 Why Global Choice is named explicitly

Global Choice supplies a set-like global well-order of the universe. Restricting it to any class gives a set-like well-order, and every proper class can then be enumerated by Ord . We use this to choose one baseline $b : \text{Ord} \rightarrow \mathbf{No}$, to enumerate class carriers during back-and-forth, and to select witnesses uniformly. Ordinary set Choice alone does not justify these global choices.

Many local results need less. Pairwise lexicographic comparison, the extension of one specified set prefix relative to a given b , and the set-cut argument use only class replacement, elementary comprehension, set Choice, and the displayed class parameters. Keeping b as an explicit parameter is therefore useful in weaker foundations.

Definition 5.11. A class well-order $\prec$ is *set-like* when, for every element x , its predecessor class $\{y : y \prec x\}$ is a set. By a class well-order here we mean a linear class relation such that every nonempty subclass available in the model has a least element.

For a set-like relation, it suffices to require well-foundedness on sets: a nonempty subclass has a witness x , and one may search the set of predecessors of x together with x for its least member. Over GBC, well-foundedness of any fixed class relation is equivalent to the absence of a set-coded descending ω -sequence. Keeping the sets and the relation fixed while enlarging the class collection therefore does not change its internal well-foundedness. What may change is which class relations are available; internal and external well-foundedness must also be distinguished. [merge] (This replaces one sentence of source 08; see the note before this section.)

Note 5.12 (Source 08, Theorem 8.2). *Every set-like well-order of a proper class has a unique order isomorphism from Ord . Conversely, every class bijection $e : \text{Ord} \rightarrow \mathbf{No}$ induces a set-like well-order of $\mathbf{No}$.* This is the last paragraph of the proof of [theorem 13.1](#) and source 09’s [proposition 13.6](#), proved by the same rank map: the predecessors of x form a well-ordered set of ordinal type $r(x)$; the range of r is downward closed, and it cannot be a set ordinal, since the inverse of r would then map a set ordinal onto a proper class, contradicting class replacement. Conversely the predecessors of $e(\alpha)$ form the set $e[\alpha]$.

Thus all set-like proper-class well-orders have the same abstract order type. The interesting lexicographic structure lies in their labels, not in distinct ordinal types. It is incorrect either to introduce a set ordinal equal to “the order type of all ordinals” or to quotient away the enumerations and expect the lexicographic information to survive.

5.6.2 The recursion principle actually used

All transfinite constructions below have *set-valued* stages and set-sized histories at each ordinal stage. In GBC, fix the class parameters and a rule assigning a unique set to each set history. The ordinary set recursion theorem with those parameters constructs a solution through each set ordinal. Uniqueness makes these initial solutions compatible, and elementary comprehension forms their union as a class. This is the set-valued recursion needed here.

It is not the principle of elementary transfinite recursion with *class-valued* stages. For example, constructing a new truth class at each stage is a substantially different task. The distinctions among GBC, GBC + ETR, and KM are genuine; the class-recursion and class forcing literature documents their strength [23, 25]. No equivalence with those stronger principles is claimed for the constructions in this paper.

5.7 Universality and uniqueness of the saturated class order

Proposition 5.13. *In ZFC, every set linear order embeds into $\mathbf{No}$. In GBC, every class linear order embeds uniformly into $\mathbf{No}$. Any two proper-class η_{Ord} orders are isomorphic in GBC.*

Proof. Enumerate the source in a cardinal-length well-order, or in type Ord for a proper class, using global choice. At stage α , the earlier images form a set. Separate the images of earlier points below the new point from those above it, and choose a separator. This extends an order embedding. Each partial map is a set.

For uniqueness, globally enumerate both class orders and alternately insert the first unused point from each enumeration. The appropriate set cut in the other order is fillable. Global choice selects a separator when none is distinguished. The union of the partial isomorphisms is a surjective class order isomorphism. $\square$

The same construction, started from a specified set-sized partial isomorphism, gives set homogeneity. These are the order-theoretic universality and homogeneity features of $\mathbf{No}$ [15]. The assertion that every class order embeds is made with global choice; it is not used as an unproved converse to global choice.

[merge] *Source 12's version (its Lemmas 6.5 and 6.6).* It exposes the size of the forth recursion, with a birthday bound for set orders, and proves the class isomorphism $\mathbf{No} \setminus A \cong \mathbf{No}$ that source 11 uses in [proposition 19.25](#).

Lemma 5.14 (Surreal universality, with the size of the recursion exposed). *In GBC, every class linear order admits a class order-embedding into $\mathbf{No}$. For a set order enumerated in type λ , one can arrange that the image of its ξ th enumerated element has birthday at most ξ , for each $\xi < \lambda$.*

Proof. Use a global-choice enumeration of the carrier, either by an ordinal or by Ord. When the next element is considered, its previously embedded predecessors and successors form two sets $A < B$. Send it to the simplest surreal separator. It is distinct from all previous images and preserves order. Every stage has set-sized history. Induction gives the birthday bound, because the earlier birthdays are less than the current stage. This is the standard forth argument behind surreal universality [15, 20]. $\square$

Lemma 5.15 (Deleting a set does not change the numerical class order). *In GBC, $\mathbf{No} \setminus A$ is order-isomorphic to $\mathbf{No}$ for every set A . Every nonempty open numerical interval is also order-isomorphic to $\mathbf{No}$.*

Proof. Each target has interpolation for arbitrary separated sets, by [lemma 5.3](#), adding interval bounds when needed. Each is a proper class and has an Ord-enumeration by global choice. Carry out a back-and-forth between the two enumerations. At each ordinal stage, extend the partial isomorphism to the next unused domain point and then to the next unused target point. The previous partial map is a set, and the required cut can be filled while avoiding its set-sized range. At the end the class map is onto and order-preserving. $\square$

[merge] *Source 09's route.* Source 09 proves universality into any class order with the set-cut property, uniqueness together with set-homogeneity, and the isomorphisms $\mathbf{No} \setminus U \cong \mathbf{No}$ and $(a, b) \cong \mathbf{No}$; its proofs are the forth and back-and-forth recursions of [proposition 5.13](#), stated for a general target.

5.8 Source 09: the classical universality mechanism, with proofs

The following facts are classical features of the surreal order; we include the arguments because the same mechanism will identify our lexicographic core. The class-universal and set-homogeneous viewpoint appears, for example, in [17].

Theorem 5.16 (Class universality). *In GBC, every class linear order embeds into $\mathbf{No}$. More generally, every class linear order embeds into any class linear order with the set-cut property.*

Proof. For a set-sized domain, choose a well-ordering and enumerate it by an ordinal. For a proper-class domain L , use global choice to fix a class bijection $a : \text{Ord} \rightarrow L$. At stage α , assume the values $h(a(\xi))$ for $\xi < \alpha$ have been selected. Form the two sets

$$A_\alpha = \{h(a(\xi)) : \xi < \alpha, a(\xi) <_L a(\alpha)\}, \quad B_\alpha = \{h(a(\xi)) : \xi < \alpha, a(\alpha) <_L a(\xi)\}.$$

Earlier choices preserve order, so $A_\alpha < B_\alpha$. Choose a separator in the target, using its set-cut property and a fixed global well-order for a definite choice. The new value is different from every earlier value and has exactly the required comparisons. At each stage the partial map is a set. The resulting class function is the embedding. $\square$

Theorem 5.17 (Uniqueness and set-homogeneity). *In GBC, any two class linear orders with the set-cut property are isomorphic. Moreover, an order isomorphism between set-sized suborders of these orders extends to a class order isomorphism.*

Proof. Neither order is a set: applying the set-cut property with the entire carrier on the left would otherwise be contradictory. Fix Ord-length enumerations of both carriers. Start with the given set partial isomorphism. At each ordinal stage, first put the least not-yet-mapped element of the first enumeration into the domain. Its required image is constrained by two sets of already mapped elements; the set-cut property supplies an image. Next put the least omitted element of the second enumeration into the range by the symmetric construction. Take unions at limit stages. Every initial partial map is a set, so the recursion is of the permitted kind. The final map is total and onto. $\square$

Corollary 5.18. *For every set $U \subseteq \mathbf{No}$, $\mathbf{No} \setminus U \cong \mathbf{No}$. Every nonempty open interval of $\mathbf{No}$ is likewise isomorphic to $\mathbf{No}$.*

Proof. Use lemma 5.10 and theorem 5.17. For an interval (a, b) , add a to every lower constraint set and b to every upper constraint set. $\square$

These isomorphisms concern numerical order alone and need not preserve simplicity or arithmetic.

[merge] Source 08's Section 12 ("The skeleton is a surreal line") contains two further statements of the classical mechanism and the skeleton theorem (note 19.17).

Note 5.19 (Source 08, Theorem 12.1, "Uniqueness of a class saturated line"). *In GBC, any two proper-class η_{Ord} -orders whose elements are sets are order-isomorphic.* This is the last clause of proposition 5.13 and theorem 5.17, by the same back-and-forth. Source 08 spells out that the separator chosen in the forward step cannot be a previous image, that a fixed set-like global well-order chooses each separator uniformly among sets, and that the least-unmatched rule handles every element: after enough stages all elements of a given bounded index have been included.

Note 5.20 (Source 08, Theorem 12.3, “Universality for class linear orders”). *Every class linear order L with set-valued elements admits a class order embedding into $\mathbf{No}$, and hence into $\mathcal{P}_s(b)$.* This is the second clause of [proposition 5.13](#) and [theorem 5.16](#), composed with [note 19.17](#). Source 08 adds: “The input order need not be set-like. A point with a proper class of predecessors is harmless because only the previously *processed* elements, rather than all its predecessors, enter a stage of the construction. This explains why $\text{Ord} + 1$ and more complicated non-set-like class well-orders can embed into the skeleton even though each *enumeration represented by* a skeleton point induces a set-like well-order on $\mathbf{No}$.”

Part I

Sets of surreals

6 Well-order codes on a fixed set alphabet

Let X be an infinite linearly ordered set, and put $\lambda = |X|$. We work in ZFC throughout this section. A well-order on the whole carrier X has a unique increasing enumeration $f : \alpha \rightarrow X$, where increasing refers to that well-order, not to the native order of X . Its domain satisfies $|\alpha| = \lambda$, and hence $\alpha < \lambda^+$. Write $\mathcal{W}(X)$ for these bijective enumeration codes, and introduce two related sets:

$$\mathcal{I}(X) = \bigcup_{\alpha < \lambda^+} \{f : \alpha \hookrightarrow X\}, \quad \mathcal{W}_\lambda(X) = \{f : \lambda \rightarrow X : f \text{ is bijective}\}.$$

Thus $\mathcal{I}(X)$ codes well-orders of arbitrary subsets of X , including the empty subset, whereas $\mathcal{W}(X)$ always uses the whole carrier. All three sets inherit the lexicographic order defined earlier. A proper prefix is smaller than its extension in $\mathcal{I}(X)$. Two distinct members of $\mathcal{W}(X)$ cannot be proper prefixes of one another: the shorter enumeration would already have used every element of X .

The repository’s real-well-ordering report already proves the exhaustive and cardinal-length statements for arbitrary infinite alphabets [1]. We record those results together with the injective-word variant, and include a common proof to keep the size distinctions explicit. Neither density nor completeness of the native order is needed.

Theorem 6.1 (Cardinality and exact ordinal spectrum). *Each of $\mathcal{W}_\lambda(X)$, $\mathcal{W}(X)$, and $\mathcal{I}(X)$ has cardinality 2^λ , contains a copy of every linear order of cardinality at most λ , and contains neither λ^+ nor $(\lambda^+)^*$. Consequently the ordinals embeddable in any of them are exactly the ordinals $\alpha < \lambda^+$.*

Proof. There are at most λ^+ permitted domains. For each domain there are at most $\lambda^\lambda = 2^\lambda$ functions, so $|\mathcal{I}(X)| \leq \lambda^+ \cdot 2^\lambda = 2^\lambda$. For the reverse inequality, partition X into pairs $\{a_\xi, b_\xi\}$, $\xi < \lambda$, with $a_\xi < b_\xi$. The lexicographic order on $\lambda \times \{0, 1\}$, taking ξ first, has order type $2 \cdot \lambda = \lambda$. For $\varepsilon \in 2^\lambda$, list the ξ -th pair as (a_ξ, b_ξ) if $\varepsilon(\xi) = 0$, and as (b_ξ, a_ξ) otherwise. The resulting map

$$2^\lambda \rightarrow \mathcal{W}_\lambda(X)$$

is an order embedding when 2^λ has lexicographic order. This proves all three cardinality assertions.

Let L be any linear order of cardinality at most λ , and fix an injection $\iota : L \hookrightarrow \lambda$. Associate to $t \in L$ the binary function c_t whose value at $\iota(s)$ is 1 precisely when $s < t$, and whose remaining coordinates are 0. If $t < u$, then $c_t \leq c_u$ coordinatewise, and the inequality is strict at $\iota(t)$. Their first difference therefore has the correct direction. The preceding pair construction embeds L into $\mathcal{W}_\lambda(X)$.

It remains to exclude longer monotone sequences. Suppose that $(f_\xi)_{\xi < \lambda^+}$ is strictly increasing or strictly decreasing in $\mathcal{I}(X)$. We recursively construct eventual common prefixes p_α of length α , for $\alpha < \lambda^+$. At a limit stage, take the union of the earlier prefixes. Regularity of λ^+ ensures that a tail of the original sequence extends that union. At a successor step, at most one member of this tail equals p_α ; discard it and all earlier indices. The values at coordinate α are now monotone. A monotone sequence indexed by a tail of λ^+ , taking at most λ values, is eventually constant: the first occurrence indices of all its values have a supremum below λ^+ . Append this eventual value to p_α . Every constructed prefix is injective because it occurs in members of the original sequence. The union of the prefixes through λ^+ would therefore be an injection $\lambda^+ \hookrightarrow X$, a contradiction. Finally, an ordinal at least λ^+ contains λ^+ , whereas every smaller ordinal has cardinality at most λ . $\square$

The distinction between the models also affects their elementary order topology. The order $\mathcal{W}_\lambda(X)$ is dense. Indeed, let $f <_{\text{lex}} g$, with first difference at ξ . If the tail of f after ξ has an ascent, interchanging the two entries produces a point strictly between f and g . If the tail of g has a descent, the corresponding interchange works in the other direction. If neither exists, the first remaining tail is strictly decreasing and the second strictly increasing. Their common set of entries, after removing the two entries at ξ , is infinite and would be both well-ordered and reverse-well-ordered. This is impossible.

By contrast, $\mathcal{W}(X)$ has adjacent pairs. Choose $a < b$ in X , enumerate $X \setminus \{a, b\}$ as a prefix p , and compare $p \frown \langle a, b \rangle$ with $p \frown \langle b, a \rangle$. Any intervening enumeration would have to share p , after which only these two orders remain. Thus a cardinal-length convention changes the resulting linear order even though it does not change its cardinality or ordinal embedding spectrum.

Proposition 6.2 (Exactly where jumps occur). *Suppose $f <_{\text{lex}} g$ in $\mathcal{W}(X)$, with common prefix p up to their first difference. Put $Y = X \setminus \text{ran}(p)$, $a = f(\text{dom}(p))$, and $b = g(\text{dom}(p))$, so $a < b$. The pair is adjacent precisely when Y is finite, a, b are consecutive in its native order, the tail of f after a lists $Y \setminus \{a\}$ decreasingly, and the tail of g after b lists $Y \setminus \{b\}$ increasingly.*

Proof. If an unused element lies strictly between a and b , start an enumeration with that element after p and complete it; this lies between f and g . Likewise, an ascent after a , or a descent after b , can be interchanged to create an intervening enumeration. Adjacency thus forces the two remaining tails to be decreasing and increasing, respectively. Their common native suborder $Y \setminus \{a, b\}$ is then both well-ordered and reverse-well-ordered. It must be finite: an infinite well-order contains its first ω elements, a subset with no greatest element. This proves necessity. Conversely, an intervening enumeration must share p . There is no possible next entry strictly between a and b . Among enumerations beginning with a , the displayed decreasing continuation is the largest; among those beginning with b , the increasing continuation is the smallest. Neither branch admits an intervening point. $\square$

This description also identifies the finite components generated by adjacency. Write the domain of an enumeration uniquely as $\theta = \delta + n$, with δ a limit ordinal and $n < \omega$. Its component consists of the $n!$ permutations of the final n entries, keeping the prefix of length δ fixed. In particular, words of limit length have no immediate neighbor. A jump cannot cross from one such component to another, because its common-prefix remainder would be infinite.

Endpoints can be classified just as concretely. The least element of $\mathcal{W}(X)$, if one exists, must always choose the least unused native element: otherwise interchanging that entry with a later smaller one decreases the enumeration. Hence a least element exists exactly when the native order on X is a well-order, and is its increasing enumeration. The greatest-element statement is the reverse assertion. For $\mathcal{W}_\lambda(X)$, the same argument shows that a least element exists exactly when the native

order type is the initial ordinal λ , not merely an arbitrary ordinal of cardinality λ . For example, an alphabet of native type $\lambda + 1$ has a least full enumeration, but no least λ -length permutation.

6.1 Pointers to the reals report

[merge] The theorems of this section hold for every infinite linearly ordered set and are theorems of the repository’s reals report, which proves them in that generality. Source 11 says so ([section 3](#)); its repository audit (`11-raw-orders-repository_audit.md`) adds that their application to birthday fragments “should be attributed as applications of existing repository results, not marketed as new general theorems”. All four sources re-prove them: source 08 from batch-76 manuscript 02 (now the reals report’s base source), sources 09 and 12 deliberately, and source 11 to keep the size distinctions explicit. They are printed once, here, with these pointers; no novelty is claimed for any of them.

Statement	Here	Reals report
Prefix-freeness of exhaustive enumerations	before theorem 6.1 ; lemma 6.3 ; note 6.11 ; note 6.18	<code>lwo:lem:prefix-free</code>
Cardinality 2^λ ; pair swaps embed $\mathbb{B}_\lambda$ in $\mathcal{W}_\lambda(X)$ and $\mathbb{B}_\theta$ in $\mathcal{W}(X)$ for $\theta < \lambda^+$	theorem 6.1 ; lemma 6.6 and theorem 6.8 ; note 6.12 ; notes 6.19 and 6.21	<code>lwo:prop:cardinality</code> , <code>lwo:lem:all-small-cubes</code>
Every order of cardinality $\leq \lambda$ embeds (cut coding), even in every fixed-type slice	theorem 6.1 ; lemma 6.5 ; note 6.12 ; note 6.20	<code>lwo:lem:cut-code</code> , <code>lwo:thm:ordinal-spectrum(a)</code> , <code>lwo:sp:thm:universality</code>
No monotone λ^+ -sequence; ordinal spectrum $\beta < \lambda^+$	theorem 6.1 ; lemma 6.7 ; lemma 6.13 and theorem 6.14 ; notes 6.22 and 6.23	<code>lwo:lem:eventual</code> , <code>lwo:thm:no-long</code> , <code>lwo:thm:ordinal-spectrum</code>
Strict hierarchy of binary cubes	lemma 6.4	<code>lwo:thm:cube-strict</code>
Cube spectrum $\theta < \lambda^+$ and coding length $\ell_2(\mathcal{W}(X)) = \lambda^+$	theorem 6.8 ; notes 6.23 and 6.24	<code>lwo:thm:full-length</code>
Least and greatest enumerations	paragraph after proposition 6.2 ; note 6.16	<code>lwo:lem:extrema</code>
Adjacent pairs; finite blocks of $n!$ points	proposition 6.2 and the paragraph after it; note 6.17 ; notes 6.25 and 6.26	<code>lwo:thm:adjacency</code> , <code>lwo:thm:blocks</code>
Absence of λ^+ is not a sufficient embeddability criterion	remark 6.9 ; remark 6.15	the reals report’s non-claims

The arguments are the reals report’s: pair swaps, characteristic-function cut codes, stabilization of a common prefix along a long monotone sequence, and cylinder fusion for strict cubes (the reals report credits the strict cube hierarchy to Corollary `cor:strict` of the point-separating game report and prints its fusion proof as a second route). Two of the sources’ routes are printed in full: source 12’s Section 9, because the singular transition of [section 8](#) uses its lemmas, and source 09’s stabilization lemma for a regular cardinal ([lemma 6.13](#)). Source 08’s statements are recorded in notes ([section 6.4](#)).

6.2 Source 12’s route: a general set-sized spectrum theorem

We temporarily forget the surreal structure. Let X be any infinite linearly ordered set, and write $|X| = \mu$, an initial cardinal. The spectra below depend only on the cardinality, not on density,

separability, or field operations of X .

Lemma 6.3 (Prefix-freeness of exhaustive enumerations). *Two distinct exhaustive injective ordinal enumerations of X are never proper prefixes of one another. Hence first-difference lexicography is unambiguous on $\mathcal{W}(X)$.*

Proof. An exhaustive prefix has already used every element of X . An injective proper extension would have to use a new element, which is impossible. $\square$

For any ordinal γ , let $\mathbb{B}_\gamma = (2^\gamma, <_{\text{lex}})$. For linear orders L, M , write $L \hookrightarrow M$ for an order-embedding and $L \simeq_e M$ for embeddings in both directions. Define

$$\ell_2(L) = \min\{\gamma : L \hookrightarrow \mathbb{B}_\gamma\}$$

when such a binary coding ordinal exists. This is an ordinal-valued rank, not a cardinal invariant.

Lemma 6.4 (Strict binary hierarchy). *For ordinals α, β ,*

$$\mathbb{B}_\alpha \hookrightarrow \mathbb{B}_\beta \iff \alpha \leq \beta.$$

In particular $\ell_2(\mathbb{B}_\alpha) = \alpha$.

Proof. Zero-padding proves the forward construction when $\alpha \leq \beta$. It suffices to rule out an embedding $f : \mathbb{B}_{\beta+1} \rightarrow \mathbb{B}_\beta$.

Recursively construct an input prefix s_ξ of length ξ and an output prefix t_ξ of length ξ , so that every input extending s_ξ is sent to an output extending t_ξ . At a successor stage, split the input cylinder into its two next-bit halves and split the output cylinder likewise. The two input halves have strictly ordered images. They cannot both have images meeting both output halves: an image point in the upper output half from the lower input half would exceed an image point in the lower output half from the upper input half. Thus one input half has image contained in a single output half. Choose those halves to continue.

At a limit take unions; the corresponding cylinders are nonempty and the containment persists. At stage β , two inputs remain, differing in their last bit, but their images have the same full β -long output. This contradicts injectivity. For $\alpha > \beta$, padding embeds $\mathbb{B}_{\beta+1}$ into $\mathbb{B}_\alpha$, giving the general impossibility. $\square$

Lemma 6.5 (Characteristic-cut coding). *Every linear order of cardinality at most μ embeds in $\mathbb{B}_\mu$.*

Proof. Enumerate its carrier by distinct coordinates d_ξ , padding unused coordinates if necessary. Send x to the bits $\mathbf{1}[d_\xi < x]$. For $x < y$, every differing coordinate changes from 0 to 1, and the coordinate representing x does change. Thus the least difference gives the correct order. $\square$

Lemma 6.6 (Pair orientation coding). $\mathbb{B}_\mu \hookrightarrow \mathcal{W}_\mu(X)$. *More generally, $\mathbb{B}_\gamma \hookrightarrow \mathcal{W}(X)$ for every $\gamma < \mu^+$.*

Proof. Partition X into μ unordered pairs, sort each pair by X 's order, and orient the pair according to its bit. The ordinal sum of μ two-element blocks has type $2 \cdot \mu = \mu$.

For a general $\gamma < \mu^+$, choose $|\gamma|$ disjoint pairs and retain a disjoint reserve of size μ . Such a choice is possible even when $|\gamma| = \mu$. List the pairs in type γ , then well-order all remaining labels in type μ . The resulting enumeration has length $2 \cdot \gamma + \mu < \mu^+$. Its comparison is decided by the first oriented pair at which the inputs differ. $\square$

Lemma 6.7 (No μ^+ -long monotone chain). *Neither μ^+ nor its reverse embeds into $\mathcal{W}(X)$.*

Proof. Suppose $(e_i)_{i < \mu^+}$ is strictly increasing. We recursively stabilize its entries on final segments of the index set. Assume that a final segment agrees on a prefix p of length $\delta < \mu^+$. No enumeration in a further infinite final segment can end there, by [lemma 6.3](#). Their δ th entries form a nondecreasing μ^+ -sequence in a set of size μ .

Such a sequence is eventually constant: choose one first occurrence for each value in its range; their at most μ indices have supremum below μ^+ by regularity. Past that supremum no larger value can first appear. At a limit stage $\delta < \mu^+$, take the supremum of the earlier stabilization indices, still below μ^+ , and the union of the earlier prefixes.

This would produce an injective sequence of labels from X of length μ^+ : every set-sized earlier part is shared by a final segment of injective enumerations. That contradicts $|X| = \mu$. Replace nondecreasing by nonincreasing for the reverse order. $\square$

Theorem 6.8 (Cardinality and exact spectra). *For every infinite linearly ordered set X of cardinality μ ,*

$$\begin{aligned} |\mathcal{W}(X)| &= |\mathcal{W}_\mu(X)| = 2^\mu, \\ \beta &\hookrightarrow \mathcal{W}(X) \iff \beta < \mu^+, \\ \beta &\hookrightarrow \mathcal{W}_\mu(X) \iff \beta < \mu^+, \\ \mathbb{B}_\gamma &\hookrightarrow \mathcal{W}(X) \iff \gamma < \mu^+, \\ \ell_2(\mathcal{W}(X)) &= \mu^+. \end{aligned}$$

The ordinal statements also hold for reversed ordinals, and every linear order of cardinality at most μ embeds even into $\mathcal{W}_\mu(X)$.

Proof. Pair coding gives 2^μ elements, while all well-orders are relations on X , giving the upper bound $2^{|X \times X|} = 2^\mu$. The lower ordinal bounds and universality follow from [lemmas 6.5](#) and [6.6](#); [lemma 6.7](#) gives the upper bounds. The same characteristic coding applies to reversed ordinals.

For the cube spectrum use [lemma 6.6](#). If $\gamma \geq \mu^+$, $\mathbb{B}_\gamma$ contains an increasing chain of type μ^+ : use the sequences that are 1 before a moving cut and 0 afterward. Thus it cannot embed into $\mathcal{W}(X)$.

Embed X into $\mathbb{B}_\mu$ using [lemma 6.5](#). Encode an exhaustive enumeration of length $\alpha < \mu^+$ by concatenating its μ -long entry codes, then pad with zeroes to length μ^+ . The unpadded length $\mu \cdot \alpha$ has cardinality at most μ , so is less than μ^+ . Distinct enumerations disagree before either ends, and their encoded first disagreement preserves the order. This proves $\mathcal{W}(X) \hookrightarrow \mathbb{B}_{\mu^+}$.

If it embedded in $\mathbb{B}_\rho$ for $\rho < \mu^+$, then the copy of $\mathbb{B}_{\rho+1}$ supplied by [lemma 6.6](#) would contradict [lemma 6.4](#). Hence the rank is exactly μ^+ . $\square$

Remark 6.9 (What the rank does not say). Although $\ell_2(\mathcal{W}(X)) = \mu^+$, the cube $\mathbb{B}_{\mu^+}$ does *not* embed back into $\mathcal{W}(X)$. An order can first admit a binary encoding at an ordinal without containing the full cube of that ordinal. Nor is the absence of μ^+ and its reverse claimed sufficient for an arbitrary order to embed in $\mathcal{W}(X)$.

6.3 Source 09's route

[merge] Source 09's Section 4 works for any linearly ordered set. Its first statements are recorded in notes; its stabilization lemma and its proof of the ordinal spectrum are a second route and are printed in full.

Note 6.10 (Source 09, Definition 4.1). $\mathcal{W}_\alpha(X) = \text{Bij}(\alpha, X)$ and $\mathcal{W}(X)$ is the union over all ordinals α ; if $|X| = \kappa$ is infinite, the possible lengths are exactly $\kappa \leq \alpha < \kappa^+$. This is the $\mathcal{W}(X)$ of [section 6](#) and of the reals report.

Note 6.11 (Source 09, Lemma 4.2, “Exhaustiveness removes the prefix ambiguity”). *Distinct members of $\mathcal{W}(X)$ have a first unequal coordinate strictly below both of their lengths, so $e <_{\text{lex}} f$ at that coordinate is a strict linear order (for transitivity compare the two first-disagreement ordinals of $e < f$ and $f < g$). This is `lwo:lem:prefix-free` and [lemma 6.3](#), by the same argument.*

Note 6.12 (Source 09, Theorem 4.3, “Cardinality and universality”). *For every infinite linearly ordered set X of cardinality κ , $|\mathcal{W}_\kappa(X)| = |\mathcal{W}(X)| = 2^\kappa$; the lexicographic cube 2^κ embeds into $\mathcal{W}_\kappa(X)$; consequently every linear order of size at most κ embeds into $\mathcal{W}_\kappa(X)$ and hence into $\mathcal{W}(X)$. This is [theorem 6.1](#), `lwo:prop:cardinality` and `lwo:thm:ordinal-spectrum(a)`, with the same pair-orientation and characteristic-function codes. Source 09 stresses that $2 \cdot \kappa = \kappa$ is ordinal multiplication.*

Second route (source 09): eventual stabilization at a regular cardinal.

Lemma 6.13 (Eventual stabilization). *Let μ be a regular infinite cardinal and let X have cardinality less than μ . Every nondecreasing map $f : \mu \rightarrow X$ is eventually constant.*

Proof. Some fiber must be unbounded: otherwise the union of fewer than μ bounded fibers would be bounded. Once a value in such a fiber is reached, monotonicity and the unboundedness of the fiber prevent any subsequent strict increase. $\square$

Theorem 6.14 (Exact ordinal spectrum). *For $|X| = \kappa$ infinite and any ordinal β ,*

$$\beta \hookrightarrow \mathcal{W}(X) \iff \beta < \kappa^+.$$

The same equivalence holds for β^ , the reverse order of β .*

Proof. The positive direction follows from [note 6.12](#). For the negative direction put $\mu = \kappa^+$ and suppose that $(e_\xi)_{\xi < \mu}$ is strictly increasing in $\mathcal{W}(X)$. We construct an injective common eventual prefix of length μ , which is impossible in the alphabet X .

Assume a prefix p of length $\delta < \mu$ is shared by all e_ξ beyond some index $t_\delta < \mu$. At limits this holds by taking the supremum of the earlier cutoffs; regularity ensures that the new cutoff remains below μ . If p exhausts X , every enumeration on this tail is exactly p , contrary to strict increase. Otherwise each tail enumeration has an entry at coordinate δ . As the words share all earlier coordinates, these entries are nondecreasing in ξ . By [lemma 6.13](#) they eventually have one common value. Add it to p . Injectivity of the enumerations makes this a new value, not used earlier in p .

Continuing for every $\delta < \mu$ gives μ distinct elements of X , a contradiction. An embedding of any $\beta \geq \mu$ would restrict to one of μ . Reversing the alphabet proves the negative statement for μ^* and its longer extensions. $\square$

Remark 6.15. The absence of κ^+ and its reverse is not asserted to characterize all linear orders embeddable in $\mathcal{W}(X)$. Indeed $\mathcal{W}(X)$ already contains the binary cube of cardinality 2^κ , so “size at most κ ” is a universality guarantee, not a necessary cardinality condition for an individual order to embed.

Note 6.16 (Source 09, Proposition 4.7). *$\mathcal{W}(X)$ has a least element exactly when the alphabet order on X is a well-order, and a greatest element exactly when its reverse is.* This is the paragraph on endpoints after [proposition 6.2](#) and `lwo:lem:extrema`, by the same argument (a least enumeration must take the least unused element at every position).

Note 6.17 (Source 09, Proposition 4.8, “Finite residuals are exactly the adjacency mechanism”). *If $e < f$ in $\mathcal{W}(X)$ have common prefix p before their first disagreement and $R = X \setminus \text{ran}(p)$, they are adjacent precisely when R is finite and their remaining words are consecutive permutations of R in finite lexicographic order.* This is [proposition 6.2](#) and `lwo:thm:adjacency`, by the same argument through [note 6.16](#).

Even for a dense alphabet X , the full space $\mathcal{W}(X)$ can therefore have adjacent points. This finite-tail phenomenon disappears for global enumerations: deleting a set prefix from **No** never leaves a finite remainder.

6.4 Source 08’s set-sized sections

[merge] Source 08’s Sections 3 (“All well-orderings of a fixed set”), 4 (“The exact ordinal and binary-cube spectra”) and 5 (“Adjacent pairs and finite tails”) re-prove the reals report’s general theorems by the same arguments. They are recorded here with source 08’s own remarks. Throughout, $(X, <)$ is an infinite linearly ordered set of cardinality κ , its initial ordinal.

Note 6.18 (Source 08, display (3) and Lemma 3.1, “No prefix ambiguity”). Source 08 defines

$$\mathcal{W}_\alpha(X) = \{e : \alpha \rightarrow X : e \text{ is a bijection}\}, \quad \mathcal{W}(X) = \bigcup_{\kappa \leq \alpha < \kappa^+} \mathcal{W}_\alpha(X) \quad (\text{S08.3})$$

and proves: *two distinct members e, f of $\mathcal{W}(X)$ disagree before either one ends; consequently*

$$e <_{\text{lex}} f \iff e(\delta) < f(\delta), \quad \delta = \min\{\xi < \min(\text{dom } e, \text{dom } f) : e(\xi) \neq f(\xi)\} \quad (\text{S08.4})$$

defines a linear order without an end-of-word convention. This is `lwo:lem:prefix-free`. Source 08 adds: “Sorting by length *before* reading entries gives a different order. For instance, choosing one word of each possible length would produce a chain of type κ^+ in that alternative.” [Note 6.22](#) shows this is impossible in [\(S08.4\)](#).

Note 6.19 (Source 08, Lemma 3.2, “Pair coding”). *For every ordinal $\theta < \kappa^+$ there is an order embedding $\mathbb{B}_\theta \hookrightarrow \mathcal{W}(X)$; in particular $\mathbb{B}_\kappa \hookrightarrow \mathcal{W}_\kappa(X)$.* This is `lwo:lem:all-small-cubes`, `lwo:prop:cardinality` and [lemma 6.6](#), with the same blocks (a_ξ, c_ξ) and (c_ξ, a_ξ) followed by a fixed enumeration of the unused elements.

Note 6.20 (Source 08, Lemma 3.3). *Every linear order L of cardinality at most κ embeds into $\mathbb{B}_\kappa$, by the bits $\mathbf{1}[d < x]$ at distinct coordinates.* This is `lwo:lem:cut-code` and [lemma 6.5](#).

Note 6.21 (Source 08, Theorem 3.4). *$\mathcal{W}(X)$ has cardinality 2^κ and contains every linear order of cardinality at most κ ; the embedding can already be made into $\mathcal{W}_\kappa(X)$.* This is `lwo:prop:cardinality` and `lwo:thm:ordinal-spectrum(a)`. Source 08 notes that neither the continuum hypothesis nor any regularity of κ occurs, so the theorem applies to arbitrary birthday cutoffs and to numerically scattered as well as dense sets of surreals.

Note 6.22 (Source 08, Theorem 4.1, “Prefix stabilization”). *There is no strictly increasing or strictly decreasing sequence of length κ^+ in $\mathcal{W}(X)$.* This is `lwo:thm:no-long`; the proof is the stabilization of [lemma 6.13](#) and [theorem 6.14](#): a monotone κ^+ -sequence with at most κ values has an unbounded fibre in the regular cardinal κ^+ and is constant from its first occurrence on.

Note 6.23 (Source 08, Corollary 4.2, “Exact spectra”). *For every ordinal β and set ordinal θ : $\beta \hookrightarrow \mathcal{W}(X)$, $\beta^{\text{op}} \hookrightarrow \mathcal{W}(X)$ and $\mathbb{B}_\theta \hookrightarrow \mathcal{W}(X)$ each hold exactly when the ordinal is below κ^+ .* This is `lwo:thm:ordinal-spectrum` and the cube clause of `lwo:thm:full-length`, as in [theorem 6.8](#); the chain $f_\eta = \mathbf{1}[\xi < \eta]$ of type κ^+ in $\mathbb{B}_{\kappa^+}$ excludes the larger cubes.

Note 6.24 (Source 08, Proposition 4.3, “A uniform binary upper coding”). *There is an order embedding $\mathcal{W}(X) \hookrightarrow \mathbb{B}_{\kappa^+}$, by concatenating cut codes of the entries and padding with zeros. This is the upper bound of `lwo:thm:full-length`, as in [theorem 6.8](#). Source 08’s remark: these results locate $\mathcal{W}(X)$ between a long hierarchy of binary cubes without identifying it with one; the absence of a κ^+ -chain is compatible with cardinality 2^κ and does not mean that all suborders are small.*

Note 6.25 (Source 08, Theorem 5.1, “Adjacency classification”). *Let $e < f$ in $\mathcal{W}(X)$ first disagree at δ , with common prefix p , remaining alphabet $Y = X \setminus \text{ran } p$, $a = e(\delta)$ and $c = f(\delta)$. They are adjacent exactly when (i) Y is finite, (ii) $a < c$ are consecutive in Y , and (iii) after a the word e lists $Y \setminus \{a\}$ decreasingly and after c the word f lists $Y \setminus \{c\}$ increasingly. This is [proposition 6.2](#) and `lwo:thm:adjacency`, by the same argument.*

Note 6.26 (Source 08, Corollary 5.2, “Finite condensation blocks”). *If the length of e is $\lambda + n$ with λ zero or a limit and $n < \omega$, its finite-condensation block consists of the words with the same prefix of length λ and all $n!$ permutations of the remaining n values; a limit-length word has a singleton block. This is the paragraph after [proposition 6.2](#) and `lwo:thm:blocks`. Source 08 adds: “The phrase ‘singleton block’ does not assert that the point is isolated: it means that it has no finite chain of jumps to another point.”*

7 Birthday cutoffs and their set-sized orders

Use the sign presentation of surreal numbers: $x \in \mathbf{No}$ is a function from an ordinal $\text{bd}(x)$ to $\{-, +\}$. At the first difference, the comparison is $- < \text{termination} < +$. This presentation and the fundamental set-cut theorem are treated in Gonshor and in the modern account of Bagayoko and van der Hoeven [12, 14]. For an infinite cardinal κ , define

$$\mathbf{No}_{<\kappa} = \{x \in \mathbf{No} : \text{bd}(x) < \kappa\}.$$

This is a set. Counting sign sequences gives

$$|\mathbf{No}_{<\kappa}| = \sum_{\alpha < \kappa} 2^{|\alpha|} = 2^{<\kappa}.$$

Here $2^{<\kappa}$ denotes the supremum of 2^μ over cardinals $\mu < \kappa$; it is at least κ .

Corollary 7.1. *Put $\lambda = 2^{<\kappa}$. Then $\mathcal{W}(\mathbf{No}_{<\kappa})$ has cardinality 2^λ , is universal for linear orders of cardinality at most λ , and admits exactly the ordinals below λ^+ . The same conclusions hold for its cardinal-length permutation suborder and for the injective-word order $\mathcal{I}(\mathbf{No}_{<\kappa})$.*

The parameter controlling this spectrum is the cardinality λ of the alphabet. It need not equal the birthday cutoff κ . For $\kappa = \omega$, the alphabet consists of the dyadic rationals. Its well-ordering order has cardinality $2^{\aleph_0}$, contains every countable linear order, and does not contain ω_1 . For $\kappa = \omega_1$, the alphabet consists of the surreals with countable sign expansions and has cardinality $\mathfrak{c} = 2^{\aleph_0}$. Its well-ordering order consequently has cardinality $2^\mathfrak{c}$ and ordinal spectrum $\alpha < \mathfrak{c}^+$, without any use of the continuum hypothesis. When $2^{<\kappa} = \kappa$, the spectrum simplifies to $\alpha < \kappa^+$.

Lemma 7.2. *If κ is regular infinite, $\mathbf{No}_{<\kappa}$ fills every cut $A < B$ with $|A \cup B| < \kappa$. Such a cut can still be filled after excluding any prescribed subset of cardinality less than κ .*

Proof. The birthdays of the members of $A \cup B$ have supremum below κ , by regularity. The simplest surreal between them has birthday at most the supremum of their birthdays plus one, and hence still belongs to $\mathbf{No}_{<\kappa}$. For the exclusion statement, let S be the forbidden set, and let S_0 consist of its members that already lie strictly between A and B . Fill the cut $A \cup S_0 < B$. Its filler lies in the original interval but strictly above every forbidden point in that interval. $\square$

We call this property η_κ , always allowing either side of a cut to be empty. The cardinal parameter in this notation is explicit; conventions indexing eta orders by ordinal subscripts vary in the literature. This elementary form of saturation is all that the stage arguments below require.

Regularity is necessary for this particular cutoff conclusion. If κ is singular, a cofinal sequence of ordinals below κ , of length $\text{cf}(\kappa) < \kappa$, is unbounded in $\mathbf{No}_{<\kappa}$. Indeed, a surreal of birthday β is at most the ordinal β , the all-positive sign expansion of that length. Every candidate upper bound is therefore exceeded by some member of the cofinal sequence. The cut consisting of this sequence on the left and the empty set on the right has no filler in $\mathbf{No}_{<\kappa}$. This failure concerns the bounded stage; it does not weaken the set-cut theorem for the full surreal class.

[merge] Source 09's Section 5 adds the stage sizes for every ordinal θ , not only for cardinals, the density and cofinality of $\mathbf{No}_{<\theta}$, and a table. The cardinal clause of its [proposition 7.3](#) is source 11's count above, and (2) is [corollary 7.1](#).

7.1 Source 09: birthday-bounded versions of the problem

Set

$$\mathbf{No}_{<\theta} = \{x \in \mathbf{No} : \text{bd}(x) < \theta\}, \quad \kappa_\theta = |\mathbf{No}_{<\theta}|.$$

Each $\mathbf{No}_{<\theta}$ is a set, and every set of surreal numbers is contained in one such $\mathbf{No}_{<\theta}$. The latter assertion follows by taking the supremum of successor birthdays.

Proposition 7.3. *For every ordinal θ ,*

$$|\mathbf{No}_{<\theta}| = \sum_{\alpha < \theta} 2^{|\alpha|}.$$

In particular, for an infinite cardinal τ , $|\mathbf{No}_{<\tau}| = 2^{<\tau} := \sup_{\lambda < \tau} 2^\lambda$. If θ is an infinite limit ordinal, $\mathbf{No}_{<\theta}$ is dense without endpoints and has cofinality and coinitality $\text{cf}(\theta)$.

Proof. There are exactly $2^{|\alpha|}$ sign words of length α , and words of different lengths are distinct. For an infinite cardinal τ , the supremum of these cardinalities is at least τ , so multiplying it by τ does not enlarge it.

For $x < y$ in an infinite limit stage, the simplest cut separator has birthday at most $\max(\text{bd}(x), \text{bd}(y)) + 1 < \theta$. Larger positive and negative ordinals of birthday below θ rule out endpoints. The ordinals below θ are cofinal, and their negatives are coinital. Conversely, a subset whose birthdays are bounded below θ is bounded in $\mathbf{No}_{<\theta}$ by an ordinal below θ . A cofinal subset must therefore have birthdays cofinal in θ , establishing the lower cofinality bound. $\square$

The fixed-set results now give the exact ordinal spectrum

$$\beta \hookrightarrow \mathcal{W}(\mathbf{No}_{<\theta}) \iff \beta < \kappa_\theta^+ \quad (\kappa_\theta \text{ infinite}). \quad (2)$$

This is a cardinal bound attached to the entire alphabet, not merely to the cofinality of its numerical order.

Stage	Alphabet size	Size of its well-ordering space	Ordinal spectrum
$\mathbf{No}_{<\omega}$	$\aleph_0$	$2^{\aleph_0}$	$\beta < \omega_1$
$\mathbf{No}_{<\omega_1}$	$2^{\aleph_0} = \mathfrak{c}$	$2^{\mathfrak{c}}$	$\beta < \mathfrak{c}^+$
$\mathbf{No}_{<\tau}$, τ strongly inaccessible	τ	2^τ	$\beta < \tau^+$

The second row does not identify $\mathfrak{c}^+$ with ω_2 without an additional continuum hypothesis. At finite stages, $|\mathbf{No}_{<n}| = 2^n - 1$, so the number of well-orders is $(2^n - 1)!$.

[merge] Source 08's Section 6 begins with the same count, as (S08.5), and then treats the regular case; its statements are recorded in the notes after this extract.

7.2 Source 08: birthday cutoffs and the regular-cardinal model

For an ordinal λ , write

$$\mathbf{No}_{<\lambda} = \{x \in \mathbf{No} : \text{bd}(x) < \lambda\}.$$

Its exact cardinality is

$$|\mathbf{No}_{<\lambda}| = \sum_{\alpha < \lambda} 2^{|\alpha|}. \quad (\text{S08.5})$$

This follows directly from the disjoint birthday levels in the sign-sequence representation. If this cardinal is infinite, all the results of section 6.4 apply to $\mathcal{W}(\mathbf{No}_{<\lambda})$. No algebraic closure property of the cutoff is needed.

A particularly transparent case occurs at a regular infinite cardinal κ satisfying $2^{<\kappa} = \kappa$. This includes $\kappa = \omega$ and every strongly inaccessible cardinal; it also includes successor cardinals under appropriate cardinal arithmetic hypotheses.

Note 7.4 (Source 08, Proposition 6.1). *If κ is regular infinite and $2^{<\kappa} = \kappa$, then $|\mathbf{No}_{<\kappa}| = \kappa$ and the numerical order of $\mathbf{No}_{<\kappa}$ is an η_κ -order.* The second clause is lemma 7.2 (which needs only regularity), proved by the same birthday bound; it is the classical saturation of $\mathbf{No}_{<\kappa}$ recorded in the birthday-cutoff report (`hset:rem:saturation`, through `hset:eq:cut-bound`) [4].

Note 7.5 (Source 08, Lemma 6.2). *Replace each minus by 00 and each plus by 11 and append the terminal block 01; the resulting binary word $d(s)$ of length $2 \cdot \text{bd}(s) + 2$ is prefix-free and $s < t$ exactly when $d(s) <_{\text{lex}} d(t)$.* This is source 12's lemma 19.33, by the same comparison $00 < 01 < 11$.

Note 7.6 (Source 08, Theorem 6.3, “The initial-length cutoff order”). *If κ is regular infinite and $2^{<\kappa} = \kappa$, then*

$$\mathcal{W}_\kappa(\mathbf{No}_{<\kappa}) \simeq_e \mathbb{B}_\kappa, \quad |\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})| = 2^\kappa; \quad (\text{S08.6})$$

moreover $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$ is an η_κ -order, and its ordinal and reverse-ordinal spectra consist exactly of the ordinals below κ^+ . The equimorphism is case 2 of source 12's theorem 8.3, proved the same way (pair coding one way, the unpadding code note 7.5 with partial sums below κ by regularity the other way). The η_κ clause is source 11's proposition 12.1 for $X = \mathbf{No}_{<\kappa}$; source 08's set-sized cut argument is the same common-coordinate search, with fresh values from theorem 5.6 and a completion of length κ , and it refers to note 19.13 for the detailed invariant. The spectra are theorem 6.1.

Note 7.7 (Source 08, Remark 6.4). Equimorphism in (S08.6) is not isomorphism: the binary cube has endpoints and adjacent pairs, whereas $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$ is dense without endpoints. Order embeddings need not preserve jumps or be continuous for the order topologies.

The regularity assumption has specific jobs: it bounds option birthdays and first-disagreement coordinates, and controls partial sums of code lengths. At a singular cutoff these steps may fail. The general cardinality and ordinal-spectrum theorems remain valid, but the η_κ and fixed-length equimorphism conclusions are not asserted without their hypotheses.

[merge] The singular case that this paragraph leaves open is settled for the coding length by source 12's dichotomy (section 8): at a singular strong limit the fixed-length order is $\simeq_e \mathbb{B}_{\kappa \cdot \kappa}$, not $\mathbb{B}_\kappa$, and it is $\simeq_e \mathbb{B}_{2^{<\kappa}}$ when $2^{<\kappa} > \kappa$ (section 1.7, item (c)). Source 11's remark at the end of section 7 gives the failure of η_κ at a singular cutoff.

[merge] The next section is source 12’s Section 10. Its dichotomy is new against the repository; it answers in part the reals report’s question “Other ground orders” (section 1.7). It uses source 12’s self-delimiting code (lemma 19.33, printed in part IV) and its lemmas of section 6.2. Here S_κ of source 12 is printed $\mathbf{No}_{<\kappa}$, and its “binary coding rank” is the binary coding length ℓ_2 of the reals report.

8 Birthday cutoffs and the singular transition (source 12)

Let κ be an infinite initial cardinal and put

$$\mathbf{No}_{<\kappa} = \{s \in \mathbf{No} : \text{bd}(s) < \kappa\}, \quad \mu = |\mathbf{No}_{<\kappa}| = 2^{<\kappa} = \sup_{\lambda < \kappa} 2^{|\lambda|}.$$

The cardinality identity follows by counting all levels. The supremum is at least κ , and summing κ levels of size at most that supremum does not increase it. The reverse inequality is supplied by the levels themselves. We do not assume $\mathbf{No}_{<\kappa}$ is a field; none of the following proofs uses arithmetic closure.

8.1 A scheduler that preserves the first varying coordinate

Lemma 8.1 (Row scheduler). *Suppose a set order X of infinite cardinality κ has disjoint subsets $(X_i)_{i < \kappa}$, each containing a copy of X and each of cardinality κ . Suppose their complement is either empty or has cardinality κ . Then*

$$X_{\text{lex}}^\kappa \hookrightarrow \mathcal{W}_\kappa(X).$$

Proof. Fix embeddings $\phi_i : X \rightarrow X_i$ and a baseline enumeration of each X_i in length κ . For $x \in X$, transpose the label $\phi_i(x)$ with the first label of that baseline, giving an enumeration $r_{i,x}$ of X_i whose first value is $\phi_i(x)$. Its dependence on x is fixed and uniform. If the complement is nonempty, add a fixed filler row indexed before all variable rows. The row index order still has type $1 + \kappa = \kappa$.

Well-order row-stage pairs $(i, j) \in \kappa \times \kappa$ first by $\max(i, j)$ and then lexicographically within a level. This schedule has type κ : the predecessors of any pair have cardinality less than κ , since both coordinates there are bounded below κ , while the entire set of pairs has size κ . The first appearance of row i is $(i, 0)$, and these first appearances occur in increasing row order.

For an input $(x_i)_{i < \kappa}$, output the j th entry of r_{i,x_i} when pair (i, j) is scheduled. The output is an exhaustive bijection in type κ . If two inputs first differ in row i , every earlier row agrees, no later row has appeared before $(i, 0)$, and the first output discrepancy is $\phi_i(x_i)$ versus $\phi_i(y_i)$. Its sign is exactly the sign of the input comparison. $\square$

Lemma 8.2 (Surreal cone decomposition). *If $|\mathbf{No}_{<\kappa}| = \kappa$, then*

$$\mathcal{W}_\kappa(\mathbf{No}_{<\kappa}) \simeq_e (\mathbf{No}_{<\kappa})_{\text{lex}}^\kappa.$$

Proof. One embedding is the inclusion of bijective sequences among all sequences. For the other, partition the sign sequences with a minus into the cones with prefixes

$$p_i = +^i \frown -, \quad i < \kappa.$$

They are disjoint, and prefixing by p_i gives a numerical order isomorphism from $\mathbf{No}_{<\kappa}$ onto that cone. For $i, \alpha < \kappa$, $i + 1 + \alpha < \kappa$; this uses that κ is an initial cardinal and is valid even when it is singular. Removing the initial prefix also leaves a sign sequence of length less than κ . The complement consists of all-plus sequences, one for each length below κ , so has cardinality κ . Apply lemma 8.1. $\square$

Theorem 8.3 (Exact minimal-slice dichotomy). *For κ and $\mu = 2^{<\kappa}$ as above,*

$$\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \simeq_e \begin{cases} \mathbb{B}_{\kappa \cdot \kappa}, & \mu = \kappa \text{ and } \text{cf } \kappa < \kappa, \\ \mathbb{B}_\mu, & \text{otherwise.} \end{cases}$$

Consequently its binary coding rank is respectively $\kappa \cdot \kappa$ or μ . The exceptional case is exactly the case of a singular strong-limit cardinal κ .

Proof. We separate the three mechanisms.

Case 1: $\mu > \kappa$. By lemma 19.33, each member of $\mathbf{No}_{<\kappa}$ has a prefix-free numerical binary code of length less than κ . Pad each such code to length κ . Concatenating the entry codes of a μ -long enumeration produces a binary word of length $\kappa \cdot \mu = \mu$. The equality holds because μ is an initial cardinal strictly larger than κ : every proper partial product $\kappa \cdot \xi$, $\xi < \mu$, has cardinality below μ . Thus $\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \hookrightarrow \mathbb{B}_\mu$. Pair coding gives the reverse embedding.

Case 2: $\mu = \kappa$ and κ is regular. Use the unpadding, self-delimiting code $c(s)$. In a sequence of κ entries, the sum of code lengths before any stage $\xi < \kappa$ is less than κ , by regularity. The full sum is κ , since every length is positive. Consequently concatenation embeds even $(\mathbf{No}_{<\kappa})_{\text{lex}}^\kappa$ into $\mathbb{B}_\kappa$. Pair coding again supplies the reverse embedding for $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$.

Case 3: $\mu = \kappa$ and $\delta = \text{cf } \kappa < \kappa$. The padded single-entry code embeds $\mathbf{No}_{<\kappa}$ into $\mathbb{B}_\kappa$, whence

$$\mathcal{W}_\kappa(\mathbf{No}_{<\kappa}) \hookrightarrow (\mathbf{No}_{<\kappa})_{\text{lex}}^\kappa \hookrightarrow \mathbb{B}_{\kappa \cdot \kappa}.$$

Choose increasing nonzero ordinals $(\lambda_j)_{j < \delta}$ cofinal in κ . Their ordinal sum is κ . As $\delta \cdot \kappa = \kappa$, partition the κ input coordinates into κ consecutive blocks of length δ . In each block, restrict its j th coordinate to sign sequences of exactly length λ_j . The lexicographic product of those coordinates is the binary cube of the sum of their lengths. Thus, over all the blocks, this restricted product is

$$\mathbb{B}_{\sum_{\xi < \kappa} \sum_{j < \delta} \lambda_j} = \mathbb{B}_{\kappa \cdot \kappa}.$$

It embeds into $(\mathbf{No}_{<\kappa})_{\text{lex}}^\kappa$, which embeds into $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$ by lemma 8.2. This gives the claimed equimorphism. The rank assertions follow from lemma 6.4.

Finally, a singular strong limit satisfies $2^{<\kappa} = \kappa$. Conversely, suppose κ is singular and $2^{<\kappa} = \kappa$. If $2^\lambda = \kappa$ for some $\lambda < \kappa$, choose $\rho < \kappa$ with $\rho \geq \lambda, \text{cf } \kappa$. Then $\kappa \leq 2^\rho \leq 2^{<\kappa} = \kappa$, so $2^\rho = \kappa$, contradicting König's inequality $\text{cf}(2^\rho) > \rho$. Thus $2^\lambda < \kappa$ for every $\lambda < \kappa$. $\square$

Corollary 8.4 (A complete cube test for the minimal slice). *Let $\rho = \kappa \cdot \kappa$ in the exceptional case of theorem 8.3, and let $\rho = \mu$ otherwise. Then*

$$\mathbb{B}_\gamma \hookrightarrow \mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \iff \gamma \leq \rho.$$

In every case its ordinal embedding spectrum is still precisely $\{\beta : \beta < \mu^+\}$.

Example 8.5 (An unconditional singular example). Let $\kappa = \beth_\omega = \sup_{n < \omega} \beth_n$. It is a singular strong-limit cardinal of cofinality ω , independently of GCH. Thus

$$\ell_2(\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})) = \kappa \cdot \kappa > \kappa, \quad \ell_2(\mathcal{W}(\mathbf{No}_{<\kappa})) = \kappa^+.$$

All three displayed ordinals must be distinguished. The first two have the same cardinality, but are different coding lengths; the successor cardinal is larger still.

Example 8.6 (Regular cutoffs). For $\kappa = \omega$, $\mathbf{No}_{<\kappa}$ is the order of finite-birthday surreals, and the minimal slice is equimorphic with $\mathbb{B}_\omega$. For $\kappa = \omega_1$, its cardinality is $2^{\aleph_0}$, and the minimal slice is equimorphic with $\mathbb{B}_{2^{\aleph_0}}$, whether or not CH holds. At a strongly inaccessible κ , it is equimorphic with $\mathbb{B}_\kappa$.

8.2 Why the singular phenomenon is genuinely ordinal

A set of κ coordinates does not license arbitrary reordering of those coordinates in a lexicographic product. In the singular case, a short cofinal block of coordinate lengths can already consume ordinal length κ ; repeating that event κ times consumes $\kappa \cdot \kappa$. Replacing this ordinal sum by its cardinality would incorrectly collapse the rank back to κ . The strict-cube lemma shows that the lost information is mathematically detectable.

The row scheduler addresses a separate possible error: an arbitrary bijection $\kappa \times \kappa \rightarrow \kappa$ need not preserve the first varying row. Bounding by the maximum coordinate ensures that all earlier observable output is independent of later varying rows. This is why the product-to-permutation embedding is valid even at singular κ .

[merge] The last sentence of [corollary 8.4](#) is [corollary 7.1](#) for the minimal slice. The dichotomy computes binary coding length only; source 11's question on the exact cut spectrum and point characters of $\mathcal{W}_\lambda(\mathbf{No}_{<\kappa})$ without $2^{<\kappa} = \kappa$ ([research question 61.7](#)) stays open.

[merge] Dated note, 3 October 2026 (batch 81): that question is now answered by sources 17, 15 and 18 in Part VI: point characters (q, q) with $q = \text{cf } \mu$, the exact gap spectrum and its multiplicities, and the saturation threshold $\eta_{\text{cf } \kappa}$ of $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$, for every infinite cardinal κ ([sections 30](#) and [33](#)). Source 15 extends the dichotomy itself to the strata $\mu \cdot \gamma + n$ ([theorem 31.2](#)).

Part II

Set-length words

9 Set-length surreal words embed explicitly in the surreal line

Let $\mathcal{T}$ be the class of all set-length words $f : \alpha \rightarrow \mathbf{No}$, and let $\mathcal{I}$ be its injective subclass. The latter codes well-orders of set-sized subsets of $\mathbf{No}$. Both are ordered lexicographically with proper prefixes smaller. Their members are sets, so these are ordinary definable classes in $\mathbf{GB}$, not collections whose members are proper classes. They should not be confused with all global well-orders of the entire surreal class.

In particular, a word cannot exhaust $\mathbf{No}$: its range is a set by Replacement. Passing from fixed alphabets to $\mathcal{I}$ consequently changes which carrier is being well-ordered from one point of the lexicographic order to another. This is a mathematically natural extension, but it answers a different question from ordering all class well-orders on a single proper-class carrier. Making this distinction before forming any collection of codes is what prevents a hidden ascent from classes to collections of classes.

The following construction uses neither a global enumeration of $\mathbf{No}$ nor Global Choice. It uses only ordinal concatenation of sign expansions, whose general properties are discussed in [\[14, Section 3.2\]](#).

Theorem 9.1 (Explicit sign coding). *There is an explicitly defined order embedding $\mathcal{E} : \mathcal{T} \rightarrow \mathbf{No}$. It maps the empty word to 0, maps every nonempty word to a positive surreal, and restricts to an order embedding of $\mathcal{I}$. Singleton words give embeddings in the opposite direction.*

Proof. For a sign expansion s , let $D(s)$ duplicate each sign: replace $-$ by $(-, -)$, and $+$ by $(+, +)$. For $x \in \mathbf{No}$, put

$$C(x) = (+) \frown D(s_x) \frown (+, -).$$

The length here is the ordinal $1 + 2 \cdot \text{bd}(x) + 2$. If s_x, s_y first differ at a sign, their doubled blocks have the same comparison. If s_x is a proper prefix of s_y , the terminal block $(+, -)$ of $C(x)$ is compared with the next doubled block of $C(y)$. We have

$$(-, -) < (+, -) < (+, +).$$

This is precisely the comparison of a missing sign with a negative or positive continuation. Thus C preserves the surreal order. Moreover, two distinct values of C cannot be prefixes of one another: the same first differing block supplies an actual differing sign before either code ends.

For $f : \alpha \rightarrow \mathbf{No}$, concatenate these blocks in domain order:

$$\mathcal{E}(f) = \bigcap_{i < \alpha} C(f(i)).$$

Replacement and ordinal summation make this a set-length sign expansion. If two words first differ at i , their preceding blocks coincide and their blocks at i have a first differing sign. Prefix freedom prevents that difference from being displaced into a subsequent block. The comparison therefore agrees with the comparison of the original entries. If f is a proper prefix of g , the expansion $\mathcal{E}(g)$ continues $\mathcal{E}(f)$ with the initial $+$ of the next block, so $\mathcal{E}(f) < \mathcal{E}(g)$. This also proves the assertions about empty and nonempty words. Finally, $x \mapsto \langle x \rangle$ embeds $\mathbf{No}$ into both word classes. $\square$

Accordingly, $\mathcal{T}$, $\mathcal{I}$, and $\mathbf{No}$ are pairwise equimorphic as class linear orders. This conclusion asserts embeddings both ways, not isomorphism. Section 10 exhibits an obstruction to isomorphism that persists after the empty word is removed.

Proposition 9.2 (Birthday bounds). *The coding has the following bounds.*

1. If κ is regular infinite, $\alpha < \kappa$, and $f(i) \in \mathbf{No}_{<\kappa}$ for every $i < \alpha$, then $\mathcal{E}(f) \in \mathbf{No}_{<\kappa}$.
2. For any infinite cardinal κ , put $\lambda = 2^{<\kappa}$. If f is an arbitrary injective word over $\mathbf{No}_{<\kappa}$, then $\mathcal{E}(f) \in \mathbf{No}_{<\lambda^+}$.

Proof. The birthday of the image is

$$\text{bd}(\mathcal{E}(f)) = \sum_{i < \alpha} (1 + 2 \cdot \text{bd}(f(i)) + 2),$$

with ordinal addition and multiplication. For the first assertion, this sums fewer than κ ordinals below κ , and regularity keeps the sum below κ . For the second, injectivity gives $|\alpha| \leq \lambda$; each block has cardinality at most λ . The whole sign expansion has cardinality at most λ , so its ordinal length is below λ^+ . $\square$

The first bound need not hold at a singular cutoff. Choose distinct ordinals $\gamma_i < \kappa$, $i < \text{cf}(\kappa)$, cofinal in κ , and regard them as surreal entries. The word has length less than κ , but the lengths of its code blocks are unbounded in κ , so its encoded birthday is at least κ . Also, injectivity is essential to the unrestricted second bound: constant words over a one-element alphabet can have every ordinal length.

10 Local characters and set suprema of word orders

In this section $\mathcal{W}$ denotes either $\mathcal{T}$ or $\mathcal{I}$. Deleting the set of entries of an existing prefix causes no difficulty when choosing a new coordinate: apply Lemma 5.2. We say that a segment has no set cofinal subset rather than assigning it a cardinal cofinality that does not exist.

Theorem 10.1 (Exact local character). *Let $f \in \mathcal{W}$ have domain α .*

1. *If α is a nonzero limit ordinal, then $\text{cf}(\mathcal{W}_{<f}) = \text{cf}(\alpha)$.*
2. *If α is a successor ordinal, then $\mathcal{W}_{<f}$ has no set cofinal subset.*
3. *For every α , the segment $\mathcal{W}_{>f}$ has no set coinitial subset.*

The empty word is the minimum of $\mathcal{W}$.

Proof. Suppose first that α is a nonzero limit. Its proper prefixes $f \upharpoonright \beta$, $\beta < \alpha$, are cofinal below f . A word $g < f$ is either a proper prefix, or differs from f first at some coordinate ξ ; in the latter case it lies below every prefix long enough to include coordinate ξ . A cofinal sequence of indices in α therefore supplies a cofinal subset of size $\text{cf}(\alpha)$. Conversely, to each $g < f$ assign a prefix length strictly large enough that the corresponding prefix exceeds g . Fewer than $\text{cf}(\alpha)$ such lengths are bounded below α , so the corresponding words cannot be cofinal.

Next let $\alpha = \beta + 1$, write $p = f \upharpoonright \beta$, and take a set $A < f$. For each member a properly extending p , its next coordinate satisfies $a(\beta) < f(\beta)$. Choose an unused surreal d strictly above all these coordinates and strictly below $f(\beta)$. Then $h = p \frown \langle d \rangle$ is below f . It exceeds every member of A extending p ; the remaining members are at most p , so they also lie below h . Thus A is not cofinal.

Finally, take a set $B > f$. Collect $b(\alpha)$ for those $b \in B$ properly extending f , and choose an unused d below all these coordinates. The word $h = f \frown \langle d \rangle$ lies above f . It lies below the members of B extending f . Every other member of B first differs earlier with a larger coordinate, and consequently exceeds every extension of f , including h . Hence $f < h < B$. $\square$

Corollary 10.2. *Neither $\mathcal{T}$ nor $\mathcal{I}$, with or without its empty word, is order-isomorphic to $\mathbf{No}$. Each has nontrivial set suprema, but a set with no minimum cannot have an infimum in either word order.*

Proof. Take an injective word f of length ω . Its nonempty finite prefixes increase to f , and f is their supremum by the preceding theorem. No surreal can be the supremum of a set lying strictly below it: the surreal set-cut property would insert a smaller upper bound. This distinguishes the orders even after deleting their minimum. If f were the infimum of a set B not containing f , part (3) would produce h with $f < h < B$, contradicting the definition of infimum. $\square$

These local obstructions coexist with very rich intervals. Given $f <_{\text{lex}} g$ with a first differing coordinate, vary that coordinate through a surreal interval strictly between the two values and disjoint from the set of preceding entries. If f is a proper prefix of g , vary a new coordinate below the next entry of g instead. The surreal set-cut property provides such an interval after excluding the preceding entries. Every surreal interval (u, v) is order-isomorphic to $\mathbf{No}$, for example by

$$t \mapsto u + \frac{v - u}{2} \left(1 + \frac{t}{1 + |t|} \right).$$

Thus every open word interval contains a copy of $\mathbf{No}$, and Theorem 9.1 then places a copy of the entire word order inside it. Interval universality therefore does not imply that set cuts are filled.

11 The complete obstruction to filling a set cut

The preceding prefix example is the only nontrivial obstruction. Here and below, $A < B$ means that every member of A is strictly below every member of B ; either set may be empty.

Theorem 11.1 (Word-cut classification). *A set cut $A < B$ in $\mathcal{W}$ has no interpolant if and only if one of the following holds:*

1. *B contains the empty word;*
2. *B has a minimum b of nonzero limit length, and A is cofinal in $\mathcal{W}_{<b}$.*

If neither obstruction occurs, an interpolant can be chosen to have nonzero successor length.

Proof. Either displayed obstruction plainly prevents an interpolant. For the converse, perform the following common-prefix search. Start with p empty. Maintain these invariants: members of A not extending p lie below p ; members of B not extending p lie above every extension of p . Here extending allows equality.

If $p \in B$, stop. Otherwise, with $\delta = \text{dom}(p)$, form the sets of next coordinates

$$A(p) = \{a(\delta) : a \in A, p \text{ is a proper prefix of } a\}, \quad B(p) = \{b(\delta) : b \in B, p \text{ is a proper prefix of } b\}.$$

Every member of $A(p)$ is at most every member of $B(p)$, since the corresponding words satisfy $a < b$. If these coordinate sets are disjoint, all these inequalities are strict. Choose a surreal c strictly between them, outside $\text{ran}(p)$. Then $p \frown \langle c \rangle$ interpolates the original cut by the invariants, and the construction is complete.

If the two coordinate sets meet, their intersection consists of a unique element c : two distinct common elements would violate the preceding inequalities. Replace p by $p \frown \langle c \rangle$. The invariants persist because the discarded A -branches have smaller next coordinates and the discarded B -branches have larger ones. In the injective-word case, c is unused, since it is witnessed by an injective word extending the previous prefix. At a limit stage take the union of the prefixes. It is a set word, is injective when required, and satisfies the same invariants. A member of B cannot have become a proper prefix of this union: the search would already have stopped upon reaching that member.

This procedure requires only set recursion. Choose an ordinal strictly above the domains of all words in $A \cup B$. The search cannot continue taking common next coordinates up to that length, because there would be no words supplying those coordinates. It therefore terminates. The remaining possibility is termination at $p \in B$. The invariants show that $p = \min B$. If its length is zero, the first obstruction holds.

Termination cannot occur at a successor length $\delta + 1$. The last appended coordinate was chosen from the intersection, so some member of A extends this successor prefix. Such a word is at least p , contrary to $A < B$. Hence a nonzero terminal length α is a limit. For every $\beta < \alpha$, a member of A witnessed the common coordinate at stage β ; it extends $p \upharpoonright (\beta + 1)$ and lies below p . These witnesses exceed prefixes of arbitrarily large length. Since the proper prefixes are cofinal below p , A is cofinal below p . This is exactly the second obstruction. $\square$

Corollary 11.2 (Successor words). *The suborder of $\mathcal{W}$ consisting of words of nonzero successor length fills every set cut. In GBC, it is order-isomorphic to $\mathbf{No}$.*

Proof. Neither obstruction in Theorem 11.1 can occur when the upper set consists of successor words. The construction returns another successor word, proving the set-cut property. Under Global Choice, enumerate each of the two proper classes by a set-like class well-order and perform back-and-forth. At every ordinal stage the previously matched elements form a set; their images specify a set cut, and an unused filler extends the partial order isomorphism. The union is an isomorphism. This is the class analogue of the usual eta-order back-and-forth, with its choice assumption made explicit; compare the class framework in [15]. $\square$

The distinction is precise. Deleting the empty word alone does not remove the prefix-supremum obstruction. Restricting to successor lengths does remove it. The explicit embeddings of Theorem 9.1 require no Global Choice, while the back-and-forth identification just given uses it.

[merge] Source 09’s class of set-length injective words is $\mathcal{I}$ (its $\mathcal{T}$). Its [proposition 11.3](#) obtains the equimorphism with **No** from class universality in GBC; source 11’s [theorem 9.1](#) gives explicit embeddings in ZF, and [corollary 10.2](#) the non-isomorphism.

11.1 Source 09: allowing the underlying set to vary

11.1.1 The set-word class

Every well-order on a set of surreals has a unique injective exhaustive enumeration of its range. Thus all such labeled set well-orders are represented by the genuine proper class

$$\mathcal{I} = \{p : p \text{ is an injective function } \alpha \rightarrow \mathbf{No} \text{ for some ordinal } \alpha\}.$$

Every member of $\mathcal{I}$ is a set. The collection is proper because it contains every singleton word (x) with $x \in \mathbf{No}$.

For $p, q \in \mathcal{I}$, compare at the first unequal entry when one exists; otherwise the proper prefix is smaller. This is a strict linear order. One can verify transitivity by padding words with a formal termination symbol below every surreal, but this symbol is *not itself a surreal number*. It is simply part of the definition of word order.

Proposition 11.3. *The dictionary order on $\mathcal{I}$ is dense, has the empty word as its least element, and has no greatest element. In GBC it is equimorphic with **No**, but it is not isomorphic to **No**, even after removing the empty word.*

Proof. If $p < q$ differ at coordinate δ , their common prefix has set range. By [lemma 5.10](#), choose a fresh surreal between the two entries at δ and append it to this prefix. The resulting word lies strictly between p and q . If p is a proper prefix of q , append to p a fresh value smaller than $q(\text{dom } p)$; this also gives an intermediate word. Every word has a proper injective extension, so there is no greatest word.

The singleton map $x \mapsto (x)$ embeds **No** into $\mathcal{I}$. Conversely, $\mathcal{I}$ is a class linear order and embeds into **No** by [theorem 5.16](#). The final nonisomorphism assertion follows from the next proposition. $\square$

Note 11.4 (Source 09, Proposition 6.2, “Limit prefixes create small one-sided character”). *If $p \in \mathcal{I}$ has nonzero limit length λ , its proper prefixes are cofinal in the open initial segment below p , the cofinality of that segment is exactly $\text{cf}(\lambda)$, and $p = \sup_{\mathcal{T}}\{p \upharpoonright \xi : \xi < \lambda\}$.* This is part (1) of [theorem 10.1](#) and the first step of the proof of [corollary 10.2](#), by the same argument.

In **No**, the set-cut property forbids a set cofinal immediately below any fixed point. Thus the difference in [note 11.4](#) is intrinsic, not just an artifact of the empty word. An embedding into **No** does not contradict this: the image need not preserve suprema in the ambient order.

11.1.2 Labeled objects versus order types

There is only one increasing exhaustive enumeration of a set already well-ordered by the numerical surreal order. In contrast, $\mathcal{W}(X)$ allows all well-orders on the labeled set X . If one instead quotients all set well-orders by abstract order isomorphism, only their ordinal types remain. Lexicographic comparison of their numerical entries does not descend to that quotient: two distinct two-letter words can have identical abstract order type and different positions in dictionary order. The distinctions between labels, inherited order, and order type must be retained.

[merge] Source 08’s Section 7 (“Varying set-sized subsets: a different class order”) uses the same class $\mathcal{I}(\mathbf{No}) = \mathcal{I}$ with proper prefixes first.

Note 11.5 (Source 08, Proposition 7.1). *The prefix-small lexicographic order on $\mathcal{I}(\mathbf{No})$ is linear; its empty word is a minimum, its nonempty part is dense without endpoints, and it contains a copy of $\mathbf{No}$ through one-entry words. Nevertheless its nonempty part is not an η_{Ord} -order: for an injective word p of length ω , the set of its nonempty finite prefixes and $\{p\}$ have no separator. This is [corollary 10.2](#) with the obstruction (2) of [theorem 11.1](#), by the same argument. Its last paragraph is printed next.*

More generally, for a word p of nonzero limit length λ , its proper prefixes are cofinal below it, and the left cofinality at p is $\text{cf}(\lambda)$. For the lower bound, a family of fewer than $\text{cf}(\lambda)$ words below p has its first disagreement or termination coordinates bounded below λ , and so is bounded by a proper prefix. Thus even very large numerical alphabets do not erase cofinalities created by termination conventions.

If “well-orders of subsets” is instead required to mean *numerically increasing* enumerations, it defines a further subclass of $\mathcal{I}(\mathbf{No})$. The singleton embedding of $\mathbf{No}$ still works, and the limit-prefix obstruction persists using an increasing p . We do not identify that subclass with the arbitrary-enumeration class. Likewise a numerically well-ordered *proper subclass* of $\mathbf{No}$ need not be set-like: a bounded increasing copy of Ord followed by its upper endpoint has type $\text{Ord} + 1$. Such copies also follow from the universality theorem in [section 5.8](#).

12 Saturation at a regular stage

For regular infinite κ , restrict to successor-length injective words of length below κ , with entries in $\mathbf{No}_{<\kappa}$. This order is η_κ . Indeed, a cut with fewer than κ words has all its word lengths bounded below κ . The common-prefix search in the preceding proof consequently stops below κ , and [Lemma 7.2](#) supplies the last coordinate while excluding the fewer than κ entries already used. [Proposition 9.2](#) embeds this order into $\mathbf{No}_{<\kappa}$; singleton words embed $\mathbf{No}_{<\kappa}$ back. When $2^{<\kappa} = \kappa$, both orders have cardinality κ , and the ordinary back-and-forth of length κ makes them isomorphic.

A related saturation statement applies to full permutations.

Proposition 12.1. *Let X be an η_κ -order of cardinality κ , where κ is regular infinite. Then $\mathcal{W}_\kappa(X)$ is η_κ . Its global and local cofinalities and coinitalities are all exactly κ .*

Proof. For a cut $A < B$ of total cardinality below κ , apply the common-coordinate search to its permutations. If both sides are nonempty, the first-disagreement indices of the pairs in $A \times B$ have supremum below κ . The search cannot continue sharing a coordinate past this bound; if either active side becomes empty, it stops sooner. The common prefix therefore has length below κ . The η_κ property chooses an unused coordinate strictly between the active coordinate sets. Complete the resulting prefix to a κ -permutation: its remaining entries and remaining positions both have cardinality κ , and its remaining interval of positions has order type κ . The completed permutation fills the cut.

Saturation excludes cofinal or coinital subsets of size below κ , both globally and on either side of a point. Conversely, [Theorem 6.1](#) excludes monotone sequences of length κ^+ . If one of these segments had no cofinal subset of size at most κ , recursively choosing a point above all earlier choices would produce such an increasing sequence. The decreasing argument is identical. Thus each of the stated characters equals κ . $\square$

In particular this proposition applies to $\mathbf{No}_{<\kappa}$ when $2^{<\kappa} = \kappa$. The resulting permutation order has cardinality 2^κ , so it is substantially larger than its alphabet while retaining the same local characters. It is not η_{κ^+} : that stronger cut property would construct an increasing κ^+ -sequence, contrary to Theorem 6.1.

Part III

Global choice and all class well-orders

13 Global choice and well-ordering the surreal class

The distinction between a set of surreal numbers and the entire surreal class changes the existence question. Here GB has the sets of ZF, without a choice axiom, and GC asserts the existence of a class function choosing an element of every nonempty set. A class well-order is a strict linear class relation for which every nonempty available subclass has a least element. Its underlying objects are still sets. The relation itself can be a proper class, and its predecessor classes need not be sets. These are the size distinctions emphasized in the repository's foundations discussion [3].

Throughout this section, $\mathfrak{W}_{\text{all}}$, $\mathfrak{W}_{\text{sl}}$, and $\mathcal{W}(X)$ when X is a proper class are metanotation for the indicated collections of class relations. We quantify over their members as class variables; we do not assert that those collections are GB classes. When X is a set, $\mathcal{W}(X)$ retains its ordinary set meaning from the preceding section. All constructions below are uniform formulas in the stated class parameters. In particular, restricting such a construction to an already given uniformly indexed family produces an ordinary class relation on the indexing domain.

Concretely, a family indexed by a class I is supplied by one class $\mathcal{R} \subseteq I \times \mathbf{No} \times \mathbf{No}$, whose slice at i defines the relation R_i . Writing this family does not make R_i a set or a member of I : the index i is a set, and the slice is a class defined with that parameter. This distinction permits uniform families of proper class relations while preserving the usual GB typing rules. No exhaustive indexing family is assumed in this section.

Theorem 13.1 (Existence and global choice). *Over GB, the following statements are equivalent:*

1. GC holds;
2. there is a class well-order of $\mathbf{No}$;
3. there is a set-like class well-order of $\mathbf{No}$;
4. there is a class bijection $\text{Ord} \rightarrow \mathbf{No}$.

Here set-like means that the predecessors of every element form a set.

Proof. We give the implication from an arbitrary well-order to global choice in detail. It also explains why set choice need not be assumed in the base theory. Let W be a class well-order of the sign-sequence presentation of $\mathbf{No}$. By ordinary set-valued transfinite recursion, with the fixed class parameter W , construct a set relation w_α well-ordering V_α , for every ordinal α .

Set $w_0 = \emptyset$. Suppose w_β has been constructed. There is a unique ordinal θ and a unique order isomorphism

$$e : \theta \longrightarrow (V_\beta, w_\beta).$$

For $A \subseteq V_\beta$, let s_A be the sign sequence of length θ whose value at ξ is $+$ exactly when $e(\xi) \in A$. The map $A \mapsto s_A$ is a bijection between $\mathcal{P}(V_\beta)$ and the set of all sign sequences of length θ . Define

$$A \, w_{\beta+1} \, B \iff s_A \, W \, s_B.$$

This is a set relation by separation with a class parameter, and it well-orders $V_{\beta+1} = \mathcal{P}(V_\beta)$. No selection of an enumeration is involved: the enumeration of an already specified set well-order is unique.

At a nonzero limit λ , order V_λ first by rank, and within rank γ by the restriction of $w_{\gamma+1}$:

$$x w_\lambda y \iff \text{rank}(x) < \text{rank}(y) \text{ or } (\text{rank}(x) = \text{rank}(y) = \gamma \text{ and } x w_{\gamma+1} y).$$

Since λ is a limit, every required $\gamma + 1$ is below λ . A nonempty subset of V_λ has a least occurring rank; the specified well-order on that rank then supplies its least member. Thus the recursion continues. Its stages are sets, even though their union over all ordinals is a class. This is precisely the ordinary GB recursion theorem, rather than a recursion whose individual values are proper classes.

For a nonempty set A , every element of A belongs to $V_{\text{rank}(A)}$. Choose the $w_{\text{rank}(A)}$ -least element of A . The recursion and this prescription define a single class choice function, establishing GC.

Conversely, global choice implies choice for sets. For each ordinal α , the set of well-orders of the birthday layer $S_\alpha = \{-, +\}^\alpha$ is nonempty. Apply the global choice function to this set, and order **No** first by birthday and then by the selected order within the birthday layer. This is a class well-order. Every predecessor class is contained in $\bigcup_{\beta \leq \alpha} S_\beta$ for some α , hence is a set. This proves the set-like version directly.

Finally, for any set-like well-order R of the proper class **No**, assign to x the ordinal order type of its predecessor set. These ranks form an initial subclass of Ord. That subclass cannot be a set: otherwise the inverse rank map, by class replacement on a set domain, would make **No** a set. It is therefore all of Ord, and the inverse rank map is the required bijection. Conversely, a class bijection from Ord transports its well-order to a set-like well-order of **No**. $\square$

The connection between surreal enumerations and global choice is discussed by Hamkins [18]; the general equivalences for global choice are collected in [19]. The explicit rank recursion above proves the assertion over the stated choiceless GB base. It does not assert that a global choice class is definable without parameters in the first-order language of set theory. Definability is an additional requirement.

[merge] *Second proof over GB + AC (source 12)*. Source 12 proves the equivalence by coding transitive closures on ordinals; set choice is used only to code each $\text{TC}(\{x\})$ on an ordinal. Its equivalence includes non-set-like well-orders, as source 11's does. Its subsection on birthday order follows its theorem.

13.1 Source 12's route over GB with set choice

Theorem 13.2 (Choice equivalence). *Over GB plus set-level AC, the following are equivalent:*

- (i) *global choice;*
- (ii) *a class bijection $\text{Ord} \rightarrow \mathbf{No}$;*
- (iii) *a set-like class well-order of $\mathbf{No}$;*
- (iv) *a class well-order of $\mathbf{No}$, not necessarily set-like.*

Proof. Global choice supplies a set-like global well-order of the universe, and hence of the proper class **No**. A proper set-like well-order has type Ord, giving (i) $\Rightarrow$ (iii) $\Rightarrow$ (ii). A bijection in (ii) pulls back ordinal order, giving (iii), and (iii) $\Rightarrow$ (iv) is immediate.

We prove the substantive reverse implication by coding sets. Suppose R well-orders **No**. Let x be a nonempty set and let $T_x = \text{TC}(\{x\})$; it contains x and all its elements. By set choice, there is

a least initial ordinal λ equinumerous with T_x . There is a fixed definable ordinal τ_λ large enough to code a binary relation on λ , together with one distinguished index, by a binary sequence of length τ_λ . For example, concatenate a $\lambda \cdot \lambda$ -long relation table and a λ -long singleton marker.

Consider the *set* of codes for well-founded extensional relations on λ whose Mostowski collapse has range T_x , and whose marked index collapses to x . This set is nonempty. Convert 0, 1 to $-$, $+$, view the codes as surreals, and take its R -least member. Decode it, and choose the element of x represented by the least ordinal index that belongs to the marked node in the coded membership relation.

The result is an element of x , uniquely determined by x and the class parameter R . Elementary class comprehension gives the graph of this choice operation on all nonempty sets. It is a global choice function. Notice that only a set of codes for each x was minimized: set-likeness of R was not needed. $\square$

Remark 13.3 (The explicit set-choice hypothesis). The converse proof uses set choice to code T_x on an ordinal. We do not claim the same equivalence over GB with no set-level choice. A purported stronger choice equivalence must address that missing coding step.

13.1.1 Birthday is not a canonical global well-order

Ordering first by birthday and then by numerical value does not work. For $n < \omega$, let s_n be the length- ω sign sequence consisting of n minuses followed by pluses forever. Then

$$\text{bd}(s_n) = \omega, \quad s_{n+1} < s_n.$$

The whole descending sequence lies in one birthday level. Simplicity itself is well-founded but is only a partial order; two incompatible sign sequences need not be comparable by simplicity.

With global choice one may choose a well-order on every birthday level and concatenate them in birthday order. The result is set-like because all earlier levels, and every initial part of the current level, form a set. The noncanonical choices have not disappeared: they are exactly the kind of information accounted for by [theorem 13.2](#).

Remark 13.4 ([[merge](#)]; source 12's question answered). Dated note, 2 October 2026. Source 12's [remark 13.3](#) declines the choiceless case, and its research question on the exact weak-choice strength ([remark 61.15](#)) asks which principle is equivalent to a class well-order of the sign-sequence surreals over GB without set choice. Source 11's [theorem 13.1](#) answers it: global choice itself. The “missing coding step” that [remark 13.3](#) names is avoided, not supplied: the well-orders w_α of V_α are built by set-valued recursion from the given well-order of the sign sequences of length $\theta = \text{otp}(V_\beta, w_\beta)$, a subset $A \subseteq V_\beta$ being coded by the sign sequence s_A of length exactly θ , so no ordinal coding of a transitive closure has to be chosen. The proof thus produces a uniformly defined sequence of well-orders of the sets V_α , in particular of the power sets of ordinals, which is the comparison the question suggests. The same mechanism, at the level of sets, is the ordinal-graph coding of the birthday-cutoff report (`hset:thm:choice`, whose range over ZF is the class of sets with well-orderable transitive closure) [[4](#)]; that theorem concerns a different statement and is unchanged. Hamkins's question whether surreal universality implies global choice [[21](#)] is not affected. (Corrected in place on 3 October 2026, following the Hilbert's-tenth programme's review [Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_surreal_synthesis_0be9b9134.md](#) (commit 280583d2f): this note had called the sequence “coherent”. The orders w_α are uniformly defined, but a later one need not restrict to an earlier one. If W puts $-$ before $+$ among the sign sequences of length one and $+-$ before $--$ among those of length two, then w_2 puts $\emptyset$ before $\{\emptyset\}$ (codes $-$ and $+$), while w_3 puts $\{\emptyset\}$ before $\emptyset$ (codes $+-$ and $--$). The proof of [theorem 13.1](#) needs only the

uniform definition: it selects the $w_{\text{rank}(A)}$ -least element of each nonempty set A . Ordering all sets by rank and then, within rank γ , by $w_{\gamma+1}$, as at the limit stages, does give a class well-order whose restrictions to the sets V_α cohere; the proof does not use it.)

[merge] *Third proof over GB + AC, for set-like orders (source 09).* Source 09’s [proposition 13.6](#) is the last paragraph of the proof of [theorem 13.1](#) (and source 08’s [note 5.12](#)); its [proposition 13.7](#) recovers global choice from a class bijection $\text{Ord} \rightarrow \mathbf{No}$ through a definable class surjection of $\mathbf{No}$ onto V , using set choice only to prove surjectivity. By [theorem 13.1](#) that hypothesis is unnecessary ([remark 13.4](#)). Its last subsection leads to source 08’s relation codes ([section 17](#)).

13.2 Source 09: global well-orders and global enumerations

Definition 13.5. A class well-order $<$ on a class D is *set-like* when the predecessors of every $d \in D$ form a set. We take “class well-order” to mean that every nonempty subclass has a least element.

Proposition 13.6. *Set-like class well-orders on $\mathbf{No}$ correspond bijectively to class bijections $E : \text{Ord} \rightarrow \mathbf{No}$.*

Proof. Given E , transfer the ordinal order to its values. Every predecessor set is the image of a set ordinal, so this order is set-like and is a class well-order.

Conversely, for a set-like well-order, assign to x the ordinal order type of its set of predecessors. This assignment is an order isomorphism onto a downward-closed class of ordinals. Its range cannot be a set: otherwise its inverse, a class function on a set, would have set range $\mathbf{No}$. A proper downward-closed class of ordinals is all of Ord . Invert the assignment. $\square$

Global choice supplies at least one such bijection; fix one and call it b . None of these enumerations is increasing in the numerical order on $\mathbf{No}$. The numerical order is dense and has descending sequences, so it is not a well-order.

Proposition 13.7 (The baseline already entails global choice). *Over GB with ordinary choice for sets, the following are equivalent: global choice; a class bijection $b : \text{Ord} \rightarrow \mathbf{No}$; and a set-like class well-order of $\mathbf{No}$.*

Proof. Global choice gives the bijection as explained in [section 5.5](#), and [proposition 13.6](#) identifies the last two conditions without using global choice. It remains to recover global choice from b .

There is an elementary definable class surjection $d : \mathbf{No} \rightarrow V$, using only ordinary set choice to prove surjectivity. Here is the coding argument. A set code consists of a nonzero ordinal γ , a well-founded extensional relation $R \subseteq \gamma \times \gamma$, and a distinguished root $r < \gamma$. Its decoded value is the image of r under the Mostowski collapse of (γ, R) . Every set x has such a code: well-order a transitive set containing x , transport its membership relation to an ordinal, and mark the index corresponding to x .

These triples have a uniform injective encoding by sign sequences. Write γ plus signs, then one minus sign as a header. Append a binary word of length $\gamma \cdot \gamma$ recording R , with (u, v) assigned to position $\gamma \cdot v + u$, and then a word of length γ whose unique plus sign marks r . The first minus sign recovers γ ; the prescribed block lengths recover R and r . Decode valid codes by their set Mostowski collapses and send invalid codes to $\emptyset$. Everything in this decoder concerns set-sized relations, so it defines a class function by elementary comprehension, not a class truth predicate.

The composite $d \circ b : \text{Ord} \rightarrow V$ is a class surjection. Assign to each set x the least ordinal α with $d(b(\alpha)) = x$. These indices define a set-like global well-order of V : predecessors of x occur among the set of decoded values before its index. Taking the least member of each nonempty set in that well-order gives a global choice function. $\square$

Thus existence of the global baseline is not a choice-free substitute for global choice. The equivalence is a standard well-founded-coding argument, included to make the axiomatic cost explicit. It says nothing similar about merely well-ordering one birthday-bounded set.

13.2.1 Why non-set-like global well-orders are different

Not every class well-order is set-like. For example, $\text{Ord} + 1$ and $\text{Ord} + \text{Ord}$ are class well-orders; the final point in the first example, and every point in the second block of the second example, has a proper class of predecessors. Global choice permits transfer of these orders to the underlying class **No**.

Such orders have no exhaustive enumeration by ordinal positions alone. They are deliberately excluded from $\mathfrak{W}_{\text{sl}}$. Even two presentations of orders with a full first block of type Ord can agree throughout that block and differ later; there is then no ordinal position at which their full presentations first differ.

There is a different, fully explicit way to compare all class well-order *relations*, including non-set-like ones: fix a class bijection $c : \text{Ord} \rightarrow \mathbf{No} \times \mathbf{No}$ and compare the binary characteristic classes of the relations at their first differing c -coordinate. This is a legitimate comparison formula on class variables. However, it depends on the auxiliary coordinate enumeration and is not the exhaustive-enumeration lexicographic order analyzed here. Its collection of objects remains of the same higher logical level.

[merge] The relation-code comparison of source 09's last paragraph is source 08's [proposition 17.1](#), which computes its equimorphism type. Source 09's research direction on it ([section 61.3](#)) asks for more: a dense set-coded core and the representation of its cuts.

14 Lexicographic comparison without class ordinals

The set-sized enumeration method studied for the real numbers [1] suggests comparing the first unequal labels. Arbitrary class well-orders require a formulation that also permits a proper class of common earlier labels. No ordinal enumeration of that entire common prefix is needed.

For a strict class well-order R on a fixed class X , write

$$P_R(x) = \{y \in X : yRx\}.$$

A common labeled prefix of R, S is a subclass initial in both orders on which their restrictions agree. The adjective labeled matters: agreement here is equality of the actual elements and relations, rather than the existence of an abstract order isomorphism.

Lemma 14.1 (Unique first disagreement). *If $R \neq S$ are class well-orders of the same class X , there are unique distinct $a, b \in X$ such that*

$$P_R(a) = P_S(b) = C, \quad R \upharpoonright C = S \upharpoonright C.$$

The subclass C is their maximal common labeled prefix.

Proof. The class $D = \{x : P_R(x) \neq P_S(x)\}$ is nonempty, since predecessor classes determine a strict order. Let a be its R -least element and put $C = P_R(a)$. For every $y \in C$, $P_R(y) = P_S(y)$. In particular, ySa : otherwise aSy , which would imply aRy , contradicting yRa . Thus $C \subseteq P_S(a)$. Also, if zSy with $y \in C$, then zRy , so $z \in C$. Therefore C is S -initial as well as R -initial, and the restrictions of the orders to it agree.

The inclusion $C \subseteq P_S(a)$ is strict because $a \in D$. Let b be the S -least element outside C . The strict inclusion shows that bSa , hence $b \neq a$. Initiality and the minimality of b give $P_S(b) = C$.

For uniqueness, suppose a', b', C' satisfy the displayed conditions with $a' \neq b'$. Every element of C' has identical predecessor classes in the two orders. But $b'Sa'$, so the predecessor classes of a' differ. Consequently a' is exactly the R -least element of D ; hence $a' = a$, $C' = C$, and $b' = b$. Any common prefix properly extending C would have to include incompatible next labels, so C is maximal. $\square$

Fix now a native linear order $<_X$ on the labels. For distinct R, S , define $R <_{\text{lex}} S$ when the labels supplied by Lemma 14.1 satisfy $a <_X b$.

Theorem 14.2 (Raw lexicographic order). *This rule is a strict linear order on the class well-orders of X , understood as a definable relation on class variables. In particular it defines $\mathfrak{W}_{\text{all}}$ for $X = \mathbf{No}$ with its native numerical order.*

Proof. Existence and uniqueness of first disagreement give totality and asymmetry; the relation is irreflexive by definition. To verify transitivity, suppose $R <_{\text{lex}} S <_{\text{lex}} T$. Write (a, b) for the first unequal labels of R, S , and (c, d) for those of S, T . Their common prefixes are $C = P_S(b)$ and $D = P_S(c)$, respectively, and $a <_X b$, $c <_X d$.

If bSc , then $C \subsetneq D$. The orders S, T agree through the label b , so the first unequal labels of R, T are a, b , with common prefix C . Thus $R <_{\text{lex}} T$. If cSb , then $D \subsetneq C$; now R, S agree through c , and the first unequal labels of R, T are c, d , again giving the conclusion. If $b = c$, the prefixes coincide and the next labels of R, T are a, d . Since $a <_X b = c <_X d$, the conclusion follows in this case as well. $\square$

This proof compares labeled relations directly. It does not compare arbitrary class well-orders up to isomorphism, and makes no appeal to a collection of class ordinal types. The defining conditions use only set quantifiers with the given class relations as parameters: equality of predecessor classes means a universal statement about their set elements.

For example, the predicate that a, b are the first unequal labels can be written without quantifying over a common-prefix class:

$$\begin{aligned} a \neq b \quad \wedge \quad & \forall z (zRa \leftrightarrow zSb) \\ & \wedge \quad \forall x \forall y ((xRa \wedge yRa) \longrightarrow (xRy \leftrightarrow xSy)). \end{aligned}$$

Adding $a <_X b$ and existentially quantifying the two labels gives the lexicographic predicate. Consequently its restriction to the indexed family $\mathcal{R}$ described above is a GB class of pairs of indices. If the indexing contains duplicate orders, this relation orders the distinct relations, and becomes a strict linear order on indices whenever the indexing is injective. Passing instead to abstract isomorphism types would discard exactly the label information on which the construction depends.

Proposition 14.3 (Convex prefix cones). *Fix a well-ordered subclass (C, P) of X . The collection of all class well-orders of X extending (C, P) as an initial labeled segment is convex in the raw lexicographic order.*

Proof. Suppose R, S extend the prefix and $R <_{\text{lex}} T <_{\text{lex}} S$. Let D be the first-disagreement prefix of R, T . The subclasses C, D are initial in R , and therefore inclusion-comparable. If $C \subseteq D$, then T extends (C, P) , as required. If $D \subsetneq C$, the next R -label lies in C ; it is also the next S -label after D . Accordingly, T compares with R and S in the same direction, contradicting the strict inequalities. This reasoning permits C and D to be proper classes. $\square$

[merge] *Second route (source 12): the largest common labelled prefix.* Source 12's Section 3 proves lemma 14.1 and theorem 14.2 differently. Instead of the class D of points whose predecessor classes differ, it defines the largest common labelled prefix $C(R, S)$ by an elementary formula and proves directly that it is initial in both orders and contains every common labelled initial segment. Source 11's shipped check compares its predecessor rule with finite lexicographic order on all 266,272 pairs of permutations of at most six points (`data/11-raw-orders-finite_checks_results.txt`); source 12's checks the common-prefix formula on 15,017 finite pairs (`data/12-singular-finite_checks.json`).

14.1 Source 12's route: the largest common labelled prefix

The central observation is that comparing two *labelled* well-orders is easier than comparing arbitrary abstract class order types. Fix a class X equipped with an ambient strict linear order $<$, and let R, S be class well-orders of the same carrier X .

Definition 14.4 (Common labelled prefix). Define $C(R, S) \subseteq X$ by putting $x \in C(R, S)$ if

$$\begin{aligned} \text{Pred}_R(x) &= \text{Pred}_S(x), \\ \forall u, v \in \text{Pred}_R(x), \quad uRv &\iff uSv. \end{aligned}$$

Equality of the predecessor classes is expanded as a quantification over sets $u \in X$. Thus this is an elementary formula with class parameters R, S ; it is not a quantification over all possible common prefixes.

Lemma 14.5 (Maximal-prefix lemma). *$C(R, S)$ is an initial segment of both well-orders, their restrictions to it agree, and it contains every common labelled initial segment. If $R \neq S$, it is proper, and the least elements*

$$a = \min_R(X \setminus C(R, S)), \quad b = \min_S(X \setminus C(R, S))$$

exist and are distinct.

Proof. Suppose $x \in C(R, S)$ and yRx . Equality of the predecessor classes puts y before x in S as well. For zRy , transitivity gives zRx ; agreement of R, S below x then gives zSy . Conversely, zSy implies zSx , hence zRx , and agreement below x yields zRy . The orders agree below y because they agree below x . Therefore $y \in C(R, S)$. The symmetric argument proves initiality in S .

If $u, v \in C(R, S)$ and uRv , equality of the predecessor classes of v gives uSv . Thus the restrictions agree. If D is any common labelled initial segment and $x \in D$, both predecessor classes of x lie in D and agree, as do the restricted relations. Hence $D \subseteq C(R, S)$. If $C(R, S) = X$, equality of every predecessor class implies $R = S$.

For $R \neq S$, class well-foundedness supplies a, b . By initiality and minimality, $\text{Pred}_R(a) = C(R, S)$ and $\text{Pred}_S(b) = C(R, S)$. If $a = b$, these predecessor classes agree and the two relations agree on them. The defining formula would put a in $C(R, S)$, a contradiction. $\square$

Definition 14.6 (Full lexicographic comparison). For distinct R, S , put $R <_{\text{lex}} S$ precisely when $a < b$, with a, b from lemma 14.5. Identical relations are equal in this order.

Theorem 14.7 (Full class lexicographic order). *In GBC, the preceding comparison is a strict linear ordering of the predicate of all class well-orders of a fixed linearly ordered class X . Its definition uses only elementary class comprehension. In particular, ETR and the comparability of abstract class well-order types are not required.*

Proof. Irreflexivity and trichotomy follow from the maximal-prefix lemma and trichotomy in X . For transitivity suppose $R <_{\text{lex}} S <_{\text{lex}} T$. The classes $C(R, S)$ and $C(S, T)$ are initial segments of the same well-order S , so they are comparable by inclusion.

If $C(R, S)$ is a proper subset of $C(S, T)$, the first S -element b after $C(R, S)$ belongs to $C(S, T)$. Thus T agrees with S through that element. The first disagreement of R, T is exactly the earlier R, S disagreement, with the same two labels $a < b$. Consequently $R <_{\text{lex}} T$. The opposite proper-inclusion case uses the earlier S, T disagreement and is identical in form.

If the two common prefixes are equal to C , let a, b, c be the first labels after C in R, S, T . The hypotheses give $a < b < c$. Hence R, T also first disagree after C , and $R <_{\text{lex}} T$. Every construction of C , and every assertion about its next elements, expands into set quantifiers with the given class relations as parameters. $\square$

Remark 14.8 (Why this does not prove class order-type comparability). In general one asks whether two arbitrary class well-orders admit an initial-segment isomorphism. ETR is a sufficient hypothesis for that principle [27, Theorem 6]. Here no such isomorphism is constructed. Only their *identical labelled* portions are compared. As soon as the labels disagree, the fixed ambient order decides the answer. Confusing these two tasks would impose an unnecessary stronger axiom.

Example 14.9 (A disagreement after every ordinary ordinal). Fix $a < b$ in $\mathbf{No}$. Well-order $\mathbf{No} \setminus \{a, b\}$ in type Ord. Append first a, b , and in a second order append b, a . The orders agree throughout their initial Ord-block and differ only after it. No ordinary ordinal indexes that first disagreement, but [theorem 14.7](#) compares them immediately: their common labelled prefix is $\mathbf{No} \setminus \{a, b\}$.

Corollary 14.10 (Agreement with enumeration lexicography). *For set-sized exhaustive ordinal enumerations, and for class bijections $\text{Ord} \rightarrow X$, the common-prefix definition agrees with first-difference lexicography.*

Proof. Before the first unequal entry, exactly the same labels occur in exactly the same order. Their ranges form the common labelled initial segment. No later label can belong to that segment by initiality and the distinct next labels. In the Ord case the disagreement is a nonempty class of ordinals; restricting it below any one witness gives its least member. $\square$

[merge] Source 12's [remark 14.8](#) is source 11's remark after [theorem 14.2](#) and in [section 20](#): comparability of abstract class well-orders by initial-segment isomorphisms follows from ETR [27], and neither labelled comparison needs it. Its [example 14.9](#) is source 11's [example 16.3](#), and its [corollary 14.10](#) is the corresponding sentence after [definition 19.1](#).

15 The canonical initial set-like part

Every class well-order of $\mathbf{No}$ has a canonical initial segment of type Ord, even when its entire order is not set-like. This gives a structural description of all well-orders using ordinary class functions.

Theorem 15.1 (Canonical initial part). *For $R \in \mathfrak{W}_{\text{all}}$, let*

$$I_R = \{x \in \mathbf{No} : P_R(x) \text{ is a set}\}.$$

Then I_R is a proper initial subclass of the domain or is the entire domain; in either case it is a proper class. Its induced order is set-like and has a unique order enumeration

$$F_R : \text{Ord} \longrightarrow I_R.$$

The order R is the concatenation of this enumeration with its induced well-order on $\mathbf{No} \setminus I_R$.

Proof. If $x \in I_R$ and yRx , then $P_R(y) \subseteq P_R(x)$, so separation makes $P_R(y)$ a set. Thus I_R is initial. Suppose it were a set. Since **No** is proper, there would be an R -least element x outside it. Every predecessor of x would then belong to I_R , and initiality would place every member of I_R before x . Hence $P_R(x) = I_R$, a contradiction.

The induced order on I_R has the same predecessor sets as R . The ordinal-rank map therefore identifies it with an initial subclass of **Ord**. As in Theorem 13.1, a set-sized range would make I_R a set by replacement. Its range is consequently all of **Ord**. The inverse map is unique, and initiality gives the asserted concatenation. $\square$

Example 15.2 (Two proper blocks). Under GBC, fix $e : \text{Ord} \rightarrow \mathbf{No}$ bijective. The disjoint classes $A = \{e(2\alpha) : \alpha \in \text{Ord}\}$ and $B = \{e(2\alpha + 1) : \alpha \in \text{Ord}\}$ partition **No**, where the product is ordinal multiplication $2 \cdot \alpha$. Order each block by its displayed ordinal index and place all of A before all of B . The resulting well-order has class type $\text{Ord} + \text{Ord}$. Its canonical initial part is exactly A : a point of B has the whole proper class A among its predecessors. Hence an **Ord**-enumeration of the canonical part is not, in general, an enumeration of the entire well-order. Both blocks consist of ordinary surreal labels; their size affects their positions in the chosen well-order, not the status of individual surreal numbers as sets.

Let $\mathfrak{J}$ denote the metacollection of injective class functions $F : \text{Ord} \rightarrow \mathbf{No}$, with lexicographic comparison at their first ordinal of disagreement. Such a first disagreement exists for distinct functions: after choosing any witness, minimize within the set of earlier ordinals. Put $Y_F = \mathbf{No} \setminus \text{ran}(F)$.

Proposition 15.3 (Convex fibers). *Assume GBC. Every $F \in \mathfrak{J}$ occurs as F_R for some $R \in \mathfrak{W}_{\text{all}}$. The fiber over F is a convex copy of $\mathcal{W}(Y_F)$. Distinct fibers are ordered by the first unequal values of their functions F .*

Proof. The range of an injection from **Ord** is proper, since a set could not contain an injective image of all ordinals. Choose a well-order Q of Y_F , using the restriction of any fixed global well-order, and put the sequence F before Q . Each $F(\alpha)$ has exactly the set $F''\alpha$ of predecessors. Each residual point, however, has the proper class $\text{ran}(F)$ among its predecessors. Thus the resulting order has canonical function exactly F .

All orders with that function extend the same initial labeled order on $\text{ran}(F)$, and conversely every extension has that canonical function. Proposition 14.3 proves convexity. Restriction to Y_F and concatenation with F are inverse operations; first disagreement inside the residual orders proves that they preserve lexicographic comparison. Finally, if $F \neq G$, their first ordinal disagreement occurs before either residual order begins. It therefore determines the comparison of every member of one fiber with every member of the other. $\square$

We may summarize this result as the lexicographic-sum description

$$\mathfrak{W}_{\text{all}} \cong \sum_{F \in (\mathfrak{J}, <_{\text{lex}})} \mathcal{W}(Y_F).$$

This is uniform metanotation for the preceding theorem, not the assertion that its indexing hyperclass or all its fibers have been collected into a GB class. The set-like fragment $\mathfrak{W}_{\text{sl}}$ consists exactly of the fibers with $Y_F = \emptyset$, or equivalently the exhaustive enumerations. For finite Y_F of size m , its fiber is a convex block of $m!$ orders, with the usual lexicographic order on permutations.

Remark 15.4 (Singleton fibers and iteration). A singleton fiber need not be set-like. If F omits exactly one surreal z , its unique completion puts z after the entire **Ord**-sequence; it has class order

type $\text{Ord} + 1$. Conversely, every exhaustive F gives a singleton set-like fiber. Extracting the initial part once also does not justify transfinite iteration of this operation through class-sized tails. Such an iteration would require a separate specification of its indexing order, its objects, and its recursion principle; none is asserted here.

16 All jumps arise from finite residual permutations

The full order has jumps despite the extreme density of the native surreal line. A proper common prefix can already have used every label between the two labels at which comparison is decided.

Lemma 16.1 (Extreme enumerations). *For a class Z with a native linear order, a class well-order is lexicographically greatest among all well-orders of Z exactly when it is the native reverse order. It is lexicographically least exactly when it is the native forward order.*

Proof. If a well-order places y before z with $y < z$, transpose these two labels throughout the order. The result is another class well-order. Its first changed label is z in place of y , so it is lexicographically greater. A greatest order must therefore be strictly decreasing in the native order, which determines the entire relation. Conversely, in the reverse native order the next label at any first disagreement is the greatest remaining label, so every different well-order is smaller. The least-order assertion follows by reversing the native order. The argument includes empty and singleton domains. $\square$

The lemma is an existence criterion as well as a characterization: a greatest enumeration exists only when the native reverse order itself is well-founded. It is therefore not permissible to use “list the remaining labels in decreasing order” as an unconditional completion rule. For example, on a copy of the natural numbers with its usual native order, the increasing enumeration is least, but there is no greatest well-order. Finite residual domains have both extrema, and the adjacency theorem shows that precisely such domains can support a jump after a common prefix.

Theorem 16.2 (Exact adjacency criterion). *Let $R <_{\text{lex}} S$ be class well-orders of X . Let C be their first-disagreement prefix, let $a < b$ be their respective next labels, and put $Y = X \setminus C$. They are adjacent if and only if all three conditions hold:*

1. *there is no $y \in Y$ with $a < y < b$;*
2. *the R -tail on $Y \setminus \{a\}$ is the native reverse order;*
3. *the S -tail on $Y \setminus \{b\}$ is the native forward order.*

These conditions force Y to be finite. Thus adjacency is exactly ordinary finite permutation adjacency after a possibly proper class common prefix.

Proof. Suppose first that the pair is adjacent. A label $y \in Y$ strictly between a, b could be placed immediately after C , followed by the restriction of R to the other remaining labels. This would give a well-order strictly between R, S ; thus condition 1 holds.

Any lexicographically greater replacement of the R -tail after a would also give an intermediate order: its next label after C would still be $a < b$. Hence this tail is greatest, and Lemma 16.1 gives condition 2. A lexicographically smaller replacement of the S -tail after b similarly gives condition 3. These modifications use restrictions and finite transpositions of existing class relations; they require no new choice principle.

The overlap $Z = Y \setminus \{a, b\}$ is now well-ordered both forward and backward in the native order. If Z were not finite, repeatedly take its least element after deleting the finitely many already selected. Ordinary recursion on ω and replacement produce a set of strictly increasing selected elements.

That set has no greatest member, contradicting reverse well-ordering. Thus Z , and hence Y , is finite. This argument does not assume that Z was a set in advance.

Conversely, suppose the three conditions hold and let $R <_{\text{lex}} T <_{\text{lex}} S$. Convexity of the prefix cone forces T to extend C . Let $t \in Y$ be its next label. A label below a makes $T < R$, and a label above b makes $T > S$; condition 1 excludes intermediate labels. If $t = a$, its remaining tail cannot exceed the greatest tail of R , so $T \leq R$. If $t = b$, its tail cannot precede the least tail of S , so $T \geq S$. Every possibility contradicts strict betweenness. $\square$

Example 16.3 (An adjacent pair of type $\text{Ord} + 2$). Assume GBC, choose surreal labels $a < b$, and give $C = \mathbf{No} \setminus \{a, b\}$ a set-like well-order P . Consider

$$R = P + (a, b), \quad S = P + (b, a).$$

Each has class order type $\text{Ord} + 2$, and R, S are adjacent. They have the same canonical initial enumeration, and constitute its entire two-point fiber. The notation $\text{Ord} + 2$ describes a class concatenation; it is not a set ordinal.

For $X = \mathbf{No}$, every adjacent pair has a cofinite, hence proper class, common prefix. Neither endpoint can be set-like. Moreover, if two set-like orders are given, their common prefix is a set. The deletion property of the surreal line leaves an unused label strictly between their next labels, and that set prefix can be completed to a set-like well-order. For completeness, this extension does not require an additional recursion principle. Retain the prescribed set prefix, put the new intermediate label next, and then append the restriction of a fixed set-like global well-order of $\mathbf{No}$ to the unused labels. A predecessor class in the appended part is contained in the union of the prescribed set prefix and a predecessor set of the fixed order, together with the one inserted label. It is therefore a set. Every label occurs once, and the concatenation is a well-order, so the completion belongs to $\mathfrak{W}_{\text{sl}}$. Thus the induced order $\mathfrak{W}_{\text{sl}}$ is dense, while $\mathfrak{W}_{\text{all}}$ has the finite terminal blocks classified above. Allowing all well-orders changes the local order structure in a concrete and unavoidable way.

[merge] *Second route (source 12): residual decomposition and adjacency.* Source 12's Section 8 proves [theorem 16.2](#) in the form “the residual class is finite and the residual permutations are consecutive”; the two criteria are equivalent, since a tail that is both well-ordered and reverse well-ordered is finite. Its [proposition 16.5](#) is the description of one prefix cone given by [propositions 14.3](#) and [15.3](#). Its [corollary 16.7](#) adds that interior points of the finite blocks with $n \geq 3$ are isolated in the order topology and that the set-like suborder is not dense in the full order, and its last paragraph adds the order-reversing involution induced by negation.

16.1 Source 12's route: unrestricted class well-orders and finite-tail geometry

The full order $\mathfrak{W}_{\text{all}}(\mathbf{No})$ contains $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ as an ordered subcollection, but it is not dense. The effect is not a failure of density of the surreal alphabet; it is exhaustion of almost all labels by a proper-class prefix.

Definition 16.4 (A prefix cylinder in the full order). Let $C \subsetneq X$ carry a fixed well-order P . Consider all class well-orders of X that have (C, P) as a labelled initial segment. Write this cylinder as $[P]_X$ and put $D = X \setminus C$.

Proposition 16.5 (Residual-order decomposition). *$[P]_X$ is convex and is order-isomorphic to the lexicographic order of all well-orders of the residual carrier D , with its inherited ambient order. If D is a set, this is the ordinary set order $\mathcal{W}(D)$, with the same notation extended to finite carriers. If $|D| = n < \omega$, the cylinder has exactly $n!$ elements.*

Proof. Restriction to D and ordered concatenation with P are inverse operations. Concatenation of two class well-orders is a class well-order: a nonempty subclass has its minimum in the first part if it meets that part, and otherwise in the second. The maximal common prefix after P is exactly the common prefix of the tails, so comparisons are preserved.

If a third order differs before completing P , its first discrepancy with every extension of P is the same and places it on the same side of all of them. This proves convexity. The last assertion counts permutations of a finite carrier. $\square$

Theorem 16.6 (Exact adjacency criterion). *Let $R <_{\text{lex}} S$ be two class well-orders of X , and let $C = C(R, S)$, $D = X \setminus C$. They are adjacent in the full order if and only if D is finite and their residual permutations are consecutive in the ordinary finite lexicographic order.*

Proof. Let $a < b$ be their first residual labels. If some $d \in D$ satisfies $a < d < b$, begin the residual order with d and well-order the remaining labels by a restriction of R . This gives an order strictly between. So an adjacent pair can have no such d .

If the tail of R after a has two labels u before v with $u < v$ in the ambient order, interchange the labels u, v at their two positions. The new well-order is larger than R , and still smaller than S because its first residual label remains a . Thus adjacency forces the R tail to be numerically decreasing. Dually, the tail of S after b must be numerically increasing.

On $D \setminus \{a, b\}$, both the ambient numerical order and its reverse are therefore well-orders. Such a carrier is finite: an infinite well-order contains an increasing ω -sequence, contradicting well-foundedness of its reverse. Hence D is finite.

Conversely, if D is finite, convexity in [proposition 16.5](#) says that every order between R, S must be another permutation of D after C . There is none exactly when the two permutations are consecutive. $\square$

Corollary 16.7 (Proper-class prefixes are responsible). *Adjacent pairs in $\mathfrak{W}_{\text{all}}(\mathbf{No})$ share a proper-class prefix. There are convex blocks of every factorial size $n!$. For $n \geq 3$, interior elements of such blocks are isolated in the order topology. The set-like suborder is not dense in the full order.*

Proof. A set prefix cannot exhaust all but finitely many surreals. Conversely, well-order $\mathbf{No} \setminus D$ in type Ord for any finite D , and append all its permutations. For $|D| \geq 3$, choose two nonconsecutive permutations in the resulting finite convex block. Their nonempty open interval contains only orders with a proper-class predecessor segment, so it contains no set-like order. $\square$

The full order still has no endpoints and no set-indexed cofinal or cointial family, by the first-entry argument of [note 19.28](#). It contains every class linear order, by [note 19.21](#). But neither global density nor interpolation for all set-indexed cuts survives: an adjacent pair already gives a two-sided cut that cannot be filled.

Pointwise surreal negation gives an order-reversing involution on both global orders: transport a well-order by the bijection $x \mapsto -x$. This preserves enumeration type and set-likeness while reversing the ambient comparison of the first unequal labels.

[merge] Source 08's Section 14 proposes two alternatives to the labelled comparison of [section 14](#): a fixed larger index order, and relation-code lexicography.

17 Non-set-like well-orders and alternative coding (source 08)

The restriction to set-like well-orders is substantive, not cosmetic. Choose distinct surreals $a < c$, and enumerate $\mathbf{No} \setminus \{a, c\}$ by Ord. Append a, c at the end in one well-order and c, a in another.

These are legitimate class well-orders of type $\text{Ord} + 2$, but they are not set-like: the final points have a proper class of predecessors.

The two enumerations agree at every *set ordinal* position. Their first disagreement occurs only after a proper-class initial segment, at the first appended point. Thus (S08.7), whose index is Ord , does not compare them. Treating “ Ord ” as one more set ordinal would conceal precisely the error.

17.1 A fixed larger index order

One can fix a class well-order J , such as $\text{Ord} + 2$, and compare class bijections $J \rightarrow \mathbf{No}$ at their first difference, using the order on J and the numerical order on $\mathbf{No}$. For the indicated concrete J this is entirely rigorous. For a general J , the existence of least members for the disagreement subclasses must be part of the specified class well-order property.

What does not follow is a canonical single comparison for all bijections having all possible non-set-like class well-order domains. Unlike set ordinals, the domains do not automatically come with a common set-ordinal coordinate system. A comparison theorem for all such domains, or a recursive alignment of their proper-class initial segments, needs an explicit formulation and an axiomatic audit. We do not assume an unproved class analogue of every set well-order theorem.

17.2 Relation-code lexicography works, but changes the question

There is an alternative that does not align enumeration domains. Fix a class bijection

$$q : \text{Ord} \longrightarrow \mathbf{No} \times \mathbf{No}.$$

Encode each strict class relation R on $\mathbf{No}$ by $\chi_R(\xi) = 1$ iff $q(\xi) \in R$. Compare these characteristic functions lexicographically. This gives a comparison of *all* class well-order relations, set-like or not. It depends on q , and is not generally the numerical lexicographic comparison of their exhaustive enumerations.

Proposition 17.1. *For every fixed q , the virtual order of all class well-order relations under relation-code comparison is equimorphic with $\mathbb{B}_{\text{Ord}}$. The lower embedding can be chosen to use only set-like well-order relations.*

Proof. The upper embedding is the characteristic-function code itself. For the lower embedding, partition $\mathbf{No}$ into disjoint two-element blocks using a baseline b . For each block $\{a, c\}$, let m be the smaller of the two positions where q lists (a, c) and (c, a) . The positions m are distinct and form a proper subclass of Ord ; they have a unique increasing enumeration by Ord . Order the blocks by this enumeration.

Within each block, choose its orientation so that the relation value at its first listed directed pair is the prescribed input bit. Order all elements in earlier blocks before all elements in later blocks. This is a set-like well-order, consisting of an Ord -sequence of two-element blocks. Reversing a block changes only the two directed relations internal to that block, not any relations between blocks. At the first differing input bit, its associated m is therefore the first changed coordinate of the characteristic function. The code preserves order. All selections and enumerations are uniform relative to q, b . $\square$

This supplies two coherent meanings of “lexicographic ordering of all well-orders”: numerical enumeration comparison on the set-like part, and relation-code comparison on all class well-orders. They have uniform embeddings into one another through the binary class order, but their definitions, canonicity properties, and possible topological invariants should not be conflated.

[merge] Dated note, 2 October 2026. Source 08 says (its Section 14.1) that a canonical single comparison for all bijections from all non-set-like class well-order domains does not follow without an explicit alignment of proper-class initial segments, and asks (its question 11) for a comparison principle for arbitrary class well-orders and its recursion requirements. Sources 11 and 12 supply one without aligning domains: compare the two relations on their largest common labelled prefix, then compare the next labels ([theorems 14.2](#) and [14.7](#)); its definition uses elementary comprehension only, not ETR and not comparability of abstract class order types. For the pair of type $\text{Ord} + 2$ displayed at the start of this section, the comparison is decided at the first appended point ([example 16.3](#), [example 14.9](#)). Source 08’s warning stands for the alignment of abstract domains, and relation-code lexicography remains a different order ([section 18](#) makes the same point on three points).

18 Numerically well-ordered subclasses and other interpretations (source 12)

The positive surreal ordinals form a numerically increasing class copy of Ord , but this is not a well-order of all of $\mathbf{No}$. More generally, [lemma 5.14](#) implies that every class well-order is isomorphic to a numerically well-ordered subclass of $\mathbf{No}$. That subclass need not be set-like in its inherited numerical order.

For an explicit example, the class

$$\left\{ -\frac{1}{\alpha + 1} : \alpha \in \text{Ord} \right\} \cup \{0\}$$

has numerical order type $\text{Ord} + 1$. The arithmetic here is surreal arithmetic on the embedded ordinal values; the reciprocal reverses positive order and the minus sign reverses it back. The final point has a proper class of predecessors. Putting a positive ordinal block after this negative block gives a numerical copy of $\text{Ord} + \text{Ord}$.

For a set $A \subseteq \mathbf{No}$, numerical well-ordering has the usual ordinal order type. Its birthday bound places it inside some $\mathbf{No}_{<\kappa}$. The class of all such set-sized well-ordered subsets is a legitimate class of sets. Its members are not exhaustive enumerations of a fixed carrier, however, so one must specify what happens when one enumeration is a prefix of another. The “shorter first” convention and the surreal $- < \mathbf{end} < +$ convention are different choices; neither can be silently borrowed from the prefix-free exhaustive setting.

Finally, ordering binary relation tables rather than enumerations is another legitimate but different construction. If one fixes a set-like well-order of $\mathbf{No} \times \mathbf{No}$, compares the characteristic functions of relations at the first unequal table entry, and restricts to well-orders, one gets a table-dependent order. A first changed relation bit is not generally the first changed enumerated label. Even on three points the two lexicographic orders can differ. The present article uses labelled initial segments, not relation-table lexicography.

[merge] Source 12’s shipped check records a three-point example on which relation-table lexicography and enumeration lexicography differ (`data/12-singular-finite_checks.json`, key `relation_table_vs_enumeration_example`).

Part IV

Set-like global well-orders

19 The lexicographic structure of global enumerations

The set-like case has a particularly strong structure theorem. Although the full collection of global enumerations is a higher-order object, one ordinary class of set codes already meets every cut specified by set-indexed families. That class, with its induced order, is isomorphic to the surreal line. The distinction between a set-indexed family and an ordinal-indexed class family is essential: an explicit omitted-range construction shows exactly where the interpolation theorem stops.

Throughout this section we work in GBC and fix a class bijection

$$e_0 : \text{Ord} \longrightarrow \mathbf{No}.$$

The existence of such a baseline is supplied by global choice; the relevant equivalences are discussed in [19]. We use the surreal set-cut property, its preservation under deleting a set (Lemma 5.2), and the universality and uniqueness statements of Proposition 5.13. These are order-theoretic properties; the field operations on $\mathbf{No}$ play no role in the main construction.

19.1 Class parameters and uniform families

Definition 19.1. For this section, $\mathfrak{J}$ denotes the collection of class injections $F : \text{Ord} \rightarrow \mathbf{No}$, and $\mathfrak{W}_{\text{sl}}$ denotes the subcollection of class bijections. For distinct $F, G \in \mathfrak{J}$, put

$$d(F, G) = \min\{\alpha \in \text{Ord} : F(\alpha) \neq G(\alpha)\}, \quad F <_{\text{lex}} G \iff F(d(F, G)) < G(d(F, G)).$$

A *uniformly set-indexed family* $(F_i)_{i \in I}$ is given by a set I and one class relation whose i th row is the graph of F_i .

These notations are interpreted relationally in two-sorted class theory. An individual F is a class, not a possible element of a GB class. Thus “for all $F \in \mathfrak{W}_{\text{sl}}$ ” abbreviates quantification over class functions with the stated property. Likewise, a uniformly indexed family is a single class of triples (i, α, x) , not a literal set containing proper classes. The first-difference definition is legitimate: choose any coordinate of disagreement and minimize within the set of preceding ordinals. The usual first-difference proof gives a strict linear comparison on these objects.

Uniformity matters whenever a set bound is taken. For example, if two uniformly set-indexed families have different rows across every pair, Class Replacement makes their cross first-difference ordinals a set. No principle of choosing an arbitrary family of classes is being assumed. Finite lists of given class parameters are automatically uniform, and the families of set-coded objects constructed below have a fixed evaluation formula, so their set-indexed subfamilies are uniform as well.

[merge] Sources 09 and 08 define the same comparison of global enumerations (source 09 in section 19.3; source 08 in its Section 9, “The virtual order of global enumerations”).

Note 19.2 (Source 08, display (7) and Proposition 9.1). For two different global enumerations e, f ,

$$\delta(e, f) = \min\{\alpha \in \text{Ord} : e(\alpha) \neq f(\alpha)\}, \quad e <_{\text{lex}} f \iff e(\delta(e, f)) < f(\delta(e, f)), \quad (\text{S08.7})$$

and *this comparison satisfies trichotomy and transitivity; it defines a linear comparison in the two-sorted sense*. This is definition 19.1 and the paragraph after it, by the same argument. Source 08’s Section 9.1 makes source 11’s point about uniform families: a family indexed by J is one class

relation $T \subseteq J \times \text{Ord} \times \mathbf{No}$; for a set J , class replacement bounds all cross first-disagreement coordinates; “merely saying ‘choose one class for every member of a set’ would not supply the uniform relation, and could hide an additional class-choice principle”; with repeated rows one works with the comparison preorder or an injective subfamily.

19.2 A class of bounded-support codes

Definition 19.3. A *normalized bounded-support code* is a set permutation $\sigma : \lambda \rightarrow \lambda$ of an ordinal λ satisfying

$$\lambda = \sup\{\xi + 1 : \xi < \lambda \text{ and } \sigma(\xi) \neq \xi\}.$$

The empty permutation is allowed. Extend σ to a class permutation $\hat{\sigma}$ of Ord by the identity outside λ , and evaluate the code as

$$E_\sigma = e_0 \circ \hat{\sigma}.$$

Write $\mathcal{P}_{\text{bd}} = \mathcal{P}_{\text{bd}}(e_0)$ for the ordinary class of these codes, ordered by the lexicographic comparison of their evaluations.

Normalization removes the otherwise harmless duplication obtained by adding fixed points to the end of a code. Every class permutation with bounded support has exactly one normalized code: restrict it to the supremum of one plus its moved coordinates. Its support is invariant under the permutation, so this restriction is a permutation of the indicated ordinal. Consequently distinct normalized codes evaluate to distinct enumerations, and the induced order on $\mathcal{P}_{\text{bd}}$ is a genuine class linear order. The class is proper, since the transpositions of 0 with arbitrary nonzero ordinals have distinct codes.

Here and below, identifying a code with its evaluated enumeration is a notational convenience. It does not convert the evaluated proper-class graph into a set. In particular, the assertion that $\mathcal{P}_{\text{bd}}$ is an ordinary class refers to the codes, while assertions comparing its members with arbitrary members of $\mathfrak{W}_{\text{sl}}$ use the common evaluation relation.

Lemma 19.4 (Completion of an arbitrary set prefix). *Every injective set function $p : \delta \rightarrow \mathbf{No}$, with δ an ordinal, is the initial segment of an enumeration represented in $\mathcal{P}_{\text{bd}}$. The completion can be chosen uniformly in p .*

Proof. Let $q = e_0^{-1} \circ p$. This is a set injection from δ into Ord . Choose an infinite initial cardinal γ strictly larger than δ and than $\sup \text{ran}(q)$. Both complements

$$\gamma \setminus \delta \quad \text{and} \quad \gamma \setminus \text{ran}(q)$$

have cardinality γ , since fewer than γ elements were removed. A bijection between the complements therefore extends q to a permutation of γ . Extend that permutation by the identity above γ , and normalize its code. Its evaluation agrees with p at every coordinate below δ . To make the construction uniform, fix the least permissible γ and then the least complement bijection under a fixed global well-order. These choices select sets; they do not require a choice principle for families of proper classes. $\square$

The lemma is stronger than merely extending p to some global well-order. Even an arbitrarily long set-sized prefix can be realized by changing the baseline only on a bounded set of coordinates. The bound need not be small: it must contain the baseline indices of all values prescribed by p .

Source 12’s prefix extension. [merge] Source 12’s Section 6 (“Set-like global well-orders: interpolation and universality”) proves the results of this section by the same arguments, except for its prefix extension, printed next, which is a different construction: it appends the restriction of the baseline order to the complement of the prefix, so its completion need not have bounded support. Source 12’s interpolation theorem, local characters and local universality are recorded in notes below; its avoidance lemma and its universality lemmas are printed in [section 5](#).

We now restrict to $\mathfrak{W}_{\text{sl}}(\mathbf{No})$, the class-bijection predicate. All statements about set- or class-indexed collections of its members mean uniform families, unless a larger ambient universe is explicitly specified.

Lemma 19.5 (Extending a set prefix). *Every injective set sequence $p : \alpha \rightarrow \mathbf{No}$ extends to a class bijection $\text{Ord} \rightarrow \mathbf{No}$. The extension can be chosen uniformly from p and one fixed baseline bijection B .*

Proof. Let $A = \text{ran}(p)$. Restrict the baseline order to $\mathbf{No} \setminus A$. It is a proper-class set-like well-order, so it has its canonical increasing enumeration in type Ord . Append that enumeration to p . The class order $\alpha + \text{Ord}$ is canonically isomorphic to Ord . The construction is set-valued recursion with B as a parameter. $\square$

[merge] *Source 09’s coding of the core.* Source 09 codes the core by fixed-point-free support permutations ([section 1.3](#)). Its prefix completion [lemma 19.6](#) has the proof of [lemma 19.4](#), and its [remark 19.7](#) warns why a larger cardinal is needed.

19.3 Source 09’s coding: canonical support codes and prefix completion

19.3.1 Lexicographic comparison of global enumerations

If $E, F : \text{Ord} \rightarrow \mathbf{No}$ are distinct, choose an ordinal γ where they differ. Their disagreement class intersected with $\gamma + 1$ is a nonempty set and has a least member. Thus

$$\Delta(E, F) = \min\{\alpha : E(\alpha) \neq F(\alpha)\}$$

is a set ordinal. Define

$$E <_{\text{lex}} F \iff E(\Delta(E, F)) < F(\Delta(E, F)).$$

The same first-disagreement argument as before proves trichotomy and transitivity. This proof quantifies over individual classes, not over a new class whose members would be enumerations.

19.3.2 Canonical set codes for eventual perturbations

A *support code* is a set function $\sigma : S \rightarrow S$, where $S \subseteq \text{Ord}$ is a set, σ is bijective, and $\sigma(\alpha) \neq \alpha$ for every $\alpha \in S$. The empty code is allowed. Extend it by the identity outside S :

$$\hat{\sigma}(\alpha) = \begin{cases} \sigma(\alpha), & \alpha \in S, \\ \alpha, & \alpha \notin S. \end{cases}$$

The code determines a global enumeration

$$E_{\sigma} = b \circ \hat{\sigma}. \tag{3}$$

Let $\mathcal{P}_s(b)$ be the class of these codes, ordered by the corresponding E_σ . Because fixed points are omitted, distinct codes yield distinct enumerations. The carrier of $\mathcal{P}_s(b)$ genuinely consists of sets.

A class enumeration E occurs this way exactly when it agrees with b at every sufficiently large ordinal. Indeed any set of ordinals is bounded. Conversely, if E equals b beyond a bound, then $b^{-1} \circ E$ has set support, and that support is invariant under the permutation and its inverse. This gives its unique support code.

Lemma 19.6 (Bounded-support prefix completion). *Every injective set word $p : \alpha \rightarrow \mathbf{No}$ is a prefix of some E_σ with $\sigma \in \mathcal{P}_s(b)$. More precisely, choose any infinite initial cardinal ν strictly above α and every ordinal in $b^{-1}[\text{ran } p]$. Then σ can be chosen with support contained in ν .*

Proof. Let $r = b^{-1} \circ p : \alpha \rightarrow \nu$. Since $\alpha < \nu$, both $\nu \setminus \alpha$ and $\nu \setminus \text{ran } r$ have cardinality ν . Choose a bijection between them. Together with r , this gives a permutation $\pi : \nu \rightarrow \nu$. Extend π by the identity outside ν and remove fixed points from its set code. Then $b \circ \pi \upharpoonright \alpha = p$. $\square$

Remark 19.7 (Why extra unused positions matter). A partial injection need not extend to a permutation on the union of its domain and range. For example, $n \mapsto n + 1$ on ω cannot extend to a permutation of ω while retaining every prescribed value. The proof avoids this defect by moving to a larger cardinal ν , where both complements have the same cardinality. Merely saying “complete the partial permutation” would leave a genuine gap.

For each ordinal γ , permutations of γ extended by the identity form a set-sized suborder of the core. Increasing γ gives compatible embeddings. Therefore

$$\mathcal{P}_s(b) = \bigcup_{\gamma \in \text{Ord}} \{\text{support codes of permutations of } \gamma\}.$$

Unlike naive unions of $\mathcal{W}(\mathbf{No}_{<\theta})$, this is a coherent direct system: the same global enumeration is retained when the support bound is increased.

[merge] Source 08’s Section 10 (“Set-supported permutations and prefix extension”) uses source 09’s support codes, writes $\mathcal{P}_s$ for their class and $\mathcal{P}_s(b)$ for the class ordered by

$$s <_b t \iff b \circ \widehat{s} <_{\text{lex}} b \circ \widehat{t}, \tag{S08.8}$$

an ordinary class relation since codes are sets, and calls $O_b = \{b \circ \widehat{s} : s \in \mathcal{P}_s\}$ the virtual orbit of b .

Note 19.8 (Source 08, Lemma 10.1, “Prefix extension inside one orbit”). *Every injective set word $p : \alpha \rightarrow \mathbf{No}$ is an initial segment of $b \circ \widehat{s}$ for some $s \in \mathcal{P}_s$.* This is lemma 19.4 and lemma 19.6, by the same proof through an initial cardinal θ above α and $\sup \text{ran}(b^{-1} \circ p)$, where both complements have cardinality θ . Source 08 notes that set choice supplies the complement bijection; that a support below any such θ suffices; that an arbitrary partial permutation of a fixed infinite set need not extend to a permutation of that set; and that under global choice the extension can be selected uniformly as the least valid code in a fixed set-like global well-order, “a choice of *sets*, not an invocation of choice among arbitrary classes”.

Note 19.9 (Source 08, Corollary 10.2). *There is a uniform order embedding $j_b : \mathbf{No} \hookrightarrow \mathcal{P}_s(b)$ for which $b \circ \widehat{j_b(x)}$ has first entry x ; its image is cofinal and cointial in $\mathfrak{W}_{\text{sl}}$ in the uniform-family sense.* The code is the transposition of 0 and $b^{-1}(x)$. This is proposition 19.20 (source 11’s E_x) and the first part of source 12’s note 19.28. Source 08: the images of the positive and of the negative ordinals are cofinal and cointial, so there are uniform class-indexed cofinal families although there are no set-indexed ones; “cofinality must not be confused with exhaustive parametrization”.

19.4 Interpolation of every uniformly set-indexed cut

Theorem 19.10 (Prefix interpolation). *Let $(A_i)_{i \in I}$ and $(B_j)_{j \in J}$ be uniformly set-indexed families in $\mathfrak{J}$, with*

$$A_i <_{\text{lex}} B_j \quad (i \in I, j \in J).$$

Either index set may be empty. There is an injective set prefix p such that every member of $\mathfrak{J}$ extending p is strictly above every A_i and strictly below every B_j . In particular, a member of $\mathcal{P}_{\text{bd}}$ lies between the two families.

Proof. If both index sets are nonempty, Class Replacement gives a set of ordinals $d(A_i, B_j)$. Choose β so that

$$d(A_i, B_j) < \beta \quad (i \in I, j \in J).$$

If one side is empty, put $\beta = 0$. We build an injective prefix p , keeping track of the rows that still agree with it. Call these rows *active*. The invariant is that every discarded A_i is below every extension of p , and every discarded B_j is above every extension of p .

Suppose p has length α and the construction has not stopped. Let

$$L = \{A_i(\alpha) : A_i \text{ is active}\}, \quad R = \{B_j(\alpha) : B_j \text{ is active}\}.$$

These are sets. Since active cross pairs agree before α , the hypothesis implies $\ell \leq r$ whenever $\ell \in L$ and $r \in R$. Thus $L \cap R$ has at most one member: two distinct common values, taken in reverse order as a left and a right value, would contradict this inequality.

If $L \cap R$ is empty, all cross inequalities are strict. By Lemma 5.2, choose

$$z \in \mathbf{No} \setminus \text{ran}(p), \quad L < z < R.$$

This notation includes an empty left or right side; if both are empty, choose any fresh value. Append z and stop. Every previously active left row is now strictly below every extension of the new prefix, and every previously active right row is strictly above it. The invariant handles the already discarded rows, proving the desired conclusion.

If $L \cap R = \{z\}$, append z and retain as active exactly the rows whose next value is z . The new value is fresh because each surviving row is injective. A discarded left value is strictly below z , since $z \in R$; a discarded right value is strictly above z , since $z \in L$. Hence the invariant is preserved. At a limit stage take the union of the preceding prefixes; its length is a set ordinal and it remains injective.

The procedure cannot continue with both sides active through a prefix of length β . Such active rows would agree below β , contrary to the bound on every cross first difference. Therefore it stops no later than the next step, giving a set prefix. Lemma 19.4 supplies a bounded-support completion. Notice that surjectivity of the input rows was never used; this is why the theorem was stated for $\mathfrak{J}$. $\square$

All recursion histories in this proof are sets. Even when the input rows are proper classes, their restrictions to a set ordinal are sets, and their next values form sets because the index sets are sets. The proof therefore uses ordinary set-valued transfinite recursion with class parameters, not ETR. The same observation applies to the back-and-forth argument below.

Note 19.11 (Source 12, Theorem 6.3, “Set-indexed interpolation”). *If I, J are sets and $(E_i)_{i \in I}$, $(F_j)_{j \in J}$ are uniform families in $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ with $E_i <_{\text{lex}} F_j$ for all i, j , there is $G \in \mathfrak{W}_{\text{sl}}(\mathbf{No})$ with $E_i <_{\text{lex}} G <_{\text{lex}} F_j$; empty sides are allowed, so $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ is dense without endpoints in the indexed-family sense. This is theorem 19.10, by the same forced-prefix construction; source 12 completes the prefix by lemma 19.5, so its interpolant need not lie in the bounded-support core.*

[merge] Sources 09 and 08 prove the same theorem with their support codes (source 09’s Section 9, “The interpolation theorem”, and source 08’s Section 11, “The set-cut interpolation theorem”).

Note 19.12 (Source 09, Theorem 9.1, “Uniform set-indexed interpolation”). *For uniformly set-indexed families $(E_i) <_{\text{lex}} (F_j)$ of global enumerations (I, J possibly empty) and every fixed baseline b , there is $\sigma \in \mathcal{P}_s(b)$ with $E_i <_{\text{lex}} E_\sigma <_{\text{lex}} F_j$.* This is [theorem 19.10](#), by the same construction; source 09 bounds the forced prefix by $\lambda + 1$, $\lambda = \sup\{\Delta(E_i, F_j) + 1\}$, and notes that the theorem “permits constraints from outside the chosen core, and the interpolant can be required to come from *any* specified baseline’s eventual-agreement class”.

Note 19.13 (Source 08, Theorem 11.1, “Orbit interpolation”). *For fixed b and uniform families $(l_i)_{i \in I}, (r_j)_{j \in J}$ of global enumerations, I, J sets, with $l_i <_{\text{lex}} r_j$, there is $s \in \mathcal{P}_s$ with*

$$l_i <_{\text{lex}} b \circ \hat{s} <_{\text{lex}} r_j \quad (i \in I, j \in J); \quad (\text{S08.9})$$

either family may be empty. This is [theorem 19.10](#), by the same active-row invariant (its display (S08.10) is the inequality $a \leq b$ between active next entries). Source 08 stresses that the separator lies in one fixed orbit “regardless of the supports, or lack of supports, of the input enumerations”.

Note 19.14 (Source 08, Corollary 11.2). *The virtual order $\mathfrak{W}_{\text{sl}}$ is dense without endpoints and has no uniformly set-indexed cofinal or coinital family; every orbit O_b is dense in $\mathfrak{W}_{\text{sl}}$ and realizes every uniformly set-sized separated cut; the class order $\mathcal{P}_s(b)$ is an η_{Ord} -order.* This is [corollary 19.15](#) and [proposition 19.27](#), by specializing the interpolation theorem.

Corollary 19.15. *The class order $\mathcal{P}_{\text{bd}}$ is an η_{Ord} order and is order-isomorphic to $(\mathbf{No}, <)$. Moreover, it meets every strict cut in $\mathfrak{W}_{\text{sl}}$ specified by uniformly set-indexed families.*

Proof. A set of codes is uniformly evaluable, so Theorem [19.10](#) applies to every separated pair of sets in $\mathcal{P}_{\text{bd}}$. Empty sides give the absence of endpoints, and singleton sides give density. We already know that $\mathcal{P}_{\text{bd}}$ is a proper class. Proposition [5.13](#) therefore identifies it with $\mathbf{No}$.

Explicitly, global choice enumerates each of the two classes by ordinals. Alternately extend a set-sized partial isomorphism to cover the next element of one enumeration and then the next element of the other. The lower and upper images at either step are sets, so the cut property supplies the new point. The union of this ordinary set-valued recursion is an isomorphism. The last assertion is the direct specialization of Theorem [19.10](#) to bijective input rows. $\square$

This is an isomorphism of the class of bounded-support codes with the surreal line. It is not an isomorphism between $\mathbf{No}$ and the full higher-order collection $\mathfrak{W}_{\text{sl}}$. Different baselines produce classes with the same abstract order type, but the argument does not single out a canonical isomorphism.

Note 19.16 (Source 09, Corollary 9.2, “The surreal core theorem”). *In GBC, $\mathcal{P}_s(b) \cong \mathbf{No}$. In particular $\mathcal{P}_s(b)$ is dense without endpoints, every set-sized partial order isomorphism between its suborders extends to an automorphism, and every class linear order embeds into it.* This is [corollary 19.15](#) (with the coding of [section 1.3](#)), with homogeneity from [theorem 5.17](#). Source 09 adds that no set family is cofinal or coinital among all global enumerations, or immediately on either side of a fixed global enumeration, while proper-class cofinal and coinital families exist, with first entries running through cofinal positive and coinital negative ordinals.

Note 19.17 (Source 08, Theorem 12.2, “Surreal-sized skeleton”). *For every fixed global enumeration b , $\mathcal{P}_s(b) \cong \mathbf{No}$; the code order is a proper class, is dense in $\mathfrak{W}_{\text{sl}}$ through its represented enumerations, and realizes every uniform set cut of $\mathfrak{W}_{\text{sl}}$.* This is [corollary 19.15](#), by the same uniqueness theorem ([note 5.19](#)); properness comes from [note 19.9](#). Source 08 adds: “The isomorphism is an isomorphism of *linear orders*, not of fields, of simplicity trees, of birthday filtrations, or of permutation groups.

No canonical isomorphism independent of Global Choice and b is asserted. One could transport a field structure along a selected isomorphism, but it would be additional, choice-dependent structure, not an arithmetic operation naturally defined by lexicographic comparison of enumerations.”

19.4.1 Eventual agreement and interleaved orbits (source 08)

Two enumerations e, f differ on a set of coordinates exactly when $e = b \circ \hat{s}$, $f = b \circ \hat{t}$ for some common orbit, equivalently when $e^{-1} \circ f$ has set support. Any set of ordinal coordinates is bounded, so this is also eventual equality: there is an ordinal α such that $e(\xi) = f(\xi)$ for every $\xi \geq \alpha$. The equivalence between the support conditions follows from injectivity of the baseline enumeration and class replacement.

Distinct orbits are disjoint as collections of enumerations, but every one is dense in $\mathfrak{W}_{\text{sl}}$. Hence these equivalence classes are not convex, and the quotient by eventual equality does not inherit a linear order by simply comparing representatives. This is a concrete obstruction, not merely a size warning.

Corollary 19.18 (The class of realizations of a set cut). *Under the hypotheses of Theorem 19.10, the ordinary class*

$$K = \{p \in \mathcal{P}_{\text{bd}} : (\forall i \in I \ A_i <_{\text{lex}} p) \text{ and } (\forall j \in J \ p <_{\text{lex}} B_j)\}$$

is itself order-isomorphic to $\mathbf{No}$.

Proof. It is an ordinary class by comprehension with the two uniform input relations as parameters. Suppose $C, D \subseteq K$ are sets with $C < D$. Add the evaluated codes of C to the lower input family and those of D to the upper input family. All cross inequalities remain strict: members of K already lie between the original sides. Theorem 19.10 supplies a member of K between C and D . Thus K has the set-cut property, including empty sides. It cannot be a set, since applying that property with lower side all of K would give a member above itself. The uniqueness statement of Proposition 5.13 now gives the isomorphism. $\square$

Thus the interpolation theorem produces a full proper-class order of available choices, even when the input constraints are numerous and occur at arbitrarily complicated set-ordinal coordinates. In particular, no set of codes can be order-dense in $\mathfrak{W}_{\text{sl}}$. Given such a set, choose a point strictly above all its evaluations, and then a second, larger point. The open interval between the two contains no code from the proposed dense set. The same argument applies within K : a set cannot be dense even in the class of realizations of one specified set cut. The ordinary class $\mathcal{P}_{\text{bd}}$ is therefore a natural scale for density here, while an exhaustive class of set codes will be ruled out by a separate diagonal argument.

Note 19.19 (Source 09, Corollary 9.3, “Dense copies in every interval”). *For global enumerations $E < F$, the class $\{\sigma \in \mathcal{P}_s(b) : E < E_\sigma < F\}$ is isomorphic to $\mathbf{No}$; the core meets every open interval of the global order and supplies a copy of every class linear order inside it.* This is [corollary 19.18](#) with $I = \{E\}$ and $J = \{F\}$, by the same argument.

19.5 Universality, cylinders, and local cofinality

Proposition 19.20. *The surreal line embeds explicitly into $\mathcal{P}_{\text{bd}}$. Consequently every ordinary class linear order embeds into $\mathcal{P}_{\text{bd}}$ and hence into $\mathfrak{W}_{\text{sl}}$, in the uniform evaluation sense.*

Proof. For $x \in \mathbf{No}$, transpose the values $e_0(0)$ and x in e_0 , and denote the resulting enumeration by E_x . Its coordinate permutation has support contained in $\{0, e_0^{-1}(x)\}$, so it has a code in $\mathcal{P}_{\text{bd}}$, and

$$E_x(0) = x, \quad x < y \iff E_x <_{\text{lex}} E_y.$$

For the general assertion, compose this embedding with the class-order universality of $\mathbf{No}$ from Proposition 5.13. $\square$

Thus all set ordinals, the class order Ord , its reverse, and class orders such as $\text{Ord} + \text{Ord}$ occur as suborders of the lexicographic comparison. This statement concerns the order *between* enumerations. Each individual enumeration in $\mathfrak{W}_{\text{sl}}$ still presents a well-order of type Ord ; its lexicographic environment need not be well-ordered.

Note 19.21 (Source 12, Theorem 6.7, “Local class universality and cylinders”). *Every class linear order has a uniform order-embedding into $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ and into every nonempty open interval of $\mathfrak{W}_{\text{sl}}(\mathbf{No})$; for a set-sized injective prefix p , the cylinder $[p]$ is convex and isomorphic to $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ by uniform transformations.* This is proposition 19.20 and proposition 19.25, by the same constructions ($x \mapsto E_x$ composed with lemma 5.14; a cylinder through $\mathbf{No} \setminus \text{ran}(p) \cong \mathbf{No}$, lemma 5.15). Source 12 lists the examples “all set ordinals, their reverses, $\text{Ord} + 1$, $\text{Ord} + \text{Ord}$, all set-sized lexicographic powers of $\mathbf{No}$ formed as class orders” and adds that “every class linear order” still means an order whose elements are sets; it does not include an unrestricted higher collection such as all class well-orders themselves.

Note 19.22 (Source 09, Theorem 10.2). *Every set ordinal, every reverse set ordinal and every class linear order embeds as a uniformly class-indexed family of global enumerations; all these embeddings can be taken in $\mathcal{P}_s(b)$, or within any specified nonempty global open interval.* This is proposition 19.20 with corollary 19.18. Source 09 lists Ord , Ord^* , $\text{Ord} + 1$, $\text{Ord} + \text{Ord}$, set-sized lexicographic powers of $\mathbf{No}$ and every class linear order on a class of set codes, and stresses that “the class order *being embedded* need not be set-like, although each well-order represented by a target global enumeration is set-like. These are two entirely different orders.” Its direct embedding $x \mapsto T_x \circ b$ (exchange $b(0)$ and x) is source 11’s E_x ; it “does not by itself give the surjectivity of the isomorphism”.

Note 19.23 (Source 08, Corollary 12.4). *The skeleton contains copies of every ordinal and its reverse, every class well-order, $\text{Ord} + 1$, $\text{Ord} + \text{Ord}$, $\mathbf{No} \times_{\text{lex}} \mathbf{No}$, and $(\mathbf{No}^\alpha, <_{\text{lex}})$ for every set ordinal α ; every injectively parametrized uniform class family of global enumerations, with its induced lexicographic order, also embeds into $\mathcal{P}_s(b)$.* The first part is proposition 19.20; the last clause is source 09’s proposition 19.24, printed next. Source 08: this concerns a uniform class-indexed family, not all class enumerations simultaneously, and replacing the set ordinal α by Ord in $\mathbf{No}^\alpha$ changes the size level of the elements and is not covered.

Proposition 19.24 (Representable suborders do not exceed surreal universality). *Every uniformly class-indexed family of global enumerations, modulo equal rows, has a class-coded linear order that embeds into $\mathbf{No}$.*

Proof. For a family with class index I , choose the least representative of each equal-row class in a fixed set-like global well-order of I . Equality of rows is the first-order condition $\forall \alpha E_i(\alpha) = E_j(\alpha)$, with the family as a class parameter. The representatives form a class of sets. Their first-disagreement order is a class linear order, to which theorem 5.16 applies. $\square$

This does not say that the full higher-level totality $\mathfrak{W}_{\text{sl}}$ embeds into $\mathbf{No}$; it says that every *uniformly class-coded portion* does.

Proposition 19.25 (Prefix cylinders). *For an injective set prefix p , let*

$$C_p = \{E \in \mathfrak{W}_{\text{sl}} : E \upharpoonright \text{dom}(p) = p\}.$$

The cylinder C_p is convex and order-isomorphic to $\mathfrak{W}_{\text{sl}}$, with the isomorphism interpreted as a uniform transformation of class parameters. Every nonempty open interval in $\mathfrak{W}_{\text{sl}}$, and every interval specified by a uniformly set-indexed strict cut, contains a whole such cylinder.

Proof. An enumeration that first departs from p at a coordinate of p lies on the same side of all its extensions. This proves convexity. Put $S = \text{ran}(p)$ and $\delta = \text{dom}(p)$. By Lemma 5.2 and Proposition 5.13, fix an order-isomorphism $h : \mathbf{No} \rightarrow \mathbf{No} \setminus S$. The ordinal tail $[\delta, \text{Ord})$ has order type Ord , with its canonical increasing parametrization $\xi \mapsto \delta + \xi$. Given $E \in \mathfrak{W}_{\text{sl}}$, prepend p to the enumeration $h \circ E$ on this tail. The result is a bijection onto $\mathbf{No}$, since its prefix covers S and its tail covers exactly the complement. Conversely, restrict an element of C_p to its tail and apply h^{-1} . These transformations are inverse and preserve first-difference comparisons. The final assertion follows from Theorem 19.10. $\square$

In particular, every such interval contains a copy of the entire comparison structure $\mathfrak{W}_{\text{sl}}$, not merely a copy of an arbitrarily large set order. This self-similarity does not invoke a class whose members are the cylinders; for each set prefix the displayed construction gives the required transformation, using the indicated class parameters.

Note 19.26 (Source 09, Proposition 10.4, “Cylinder self-similarity”). *Each cylinder $[p]$ (the global enumerations extending an injective set word p ; convex, and nonempty by lemma 19.6) is isomorphic to the full global comparison order by an explicit invertible transformation of class enumerations preserving lexicographic order and having exactly $[p]$ as its image. This is proposition 19.25, by the same construction $H_E \upharpoonright \alpha = p$, $H_E(\alpha + \xi) = h(E(\xi))$ with $h : \mathbf{No} \cong \mathbf{No} \setminus \text{ran } p$ (corollary 5.18). Source 09’s paragraph on two baselines follows; source 08 makes the same observation in its subsection printed after corollary 19.15.*

For two baselines b, c , their eventual-agreement collections are either identical or disjoint. They are identical exactly when b, c agree beyond some set ordinal. Otherwise no global enumeration can be eventually equal to both. Nevertheless, *each* such collection has a genuine set-coded presentation isomorphic to $\mathbf{No}$ and is dense in the entire global comparison order. Eventual agreement is therefore not a convex relation, and its quotient does not inherit a lexicographic linear order by comparing arbitrary representatives.

Proposition 19.27 (Approximants at every point). *For each $F \in \mathfrak{J}$ there are uniformly ordinal-indexed families (a_α) and (b_α) of codes in $\mathcal{P}_{\text{bd}}$ such that, whenever $\alpha < \beta$,*

$$a_\alpha <_{\text{lex}} a_\beta <_{\text{lex}} F <_{\text{lex}} b_\beta <_{\text{lex}} b_\alpha.$$

The lower family is cofinal among the global enumerations below F , and the upper family is coinital among those above F . Neither side admits a uniformly set-indexed cofinal or coinital family, respectively.

Proof. Choose a fresh value below $F(\alpha)$, append it to $F \upharpoonright \alpha$, and use Lemma 19.4 to obtain a_α . Choose a fresh value above $F(\alpha)$ to obtain b_α . Lemma 5.2 supplies the fresh values, and the fixed global well-order makes all choices uniform. For $\alpha < \beta$, the comparison between a_α and a_β is decided at α , giving the asserted inequality; the other comparisons are identical arguments.

If $E <_{\text{lex}} F$, choose $\alpha > d(E, F)$. Then a_α agrees with F at the decisive coordinate, so $E <_{\text{lex}} a_\alpha <_{\text{lex}} F$. This proves cofinality, and the upper assertion is symmetric. Finally, given any

uniformly set-indexed family below F , Theorem 19.10 applied with upper side $\{F\}$ supplies a larger lower approximant in $\mathcal{P}_{\text{bd}}$. Hence the original family was not cofinal. The dual argument proves the coinitality assertion. $\square$

For $F \in \mathfrak{W}_{\text{sl}}$, these are the local cofinality and coinitality statements at an actual point of the order. It is appropriate to say that ordinal-length class families suffice and set-sized families do not; assigning an ordinary set cardinal to these proper-class cofinalities would obscure the distinction.

Note 19.28 (Source 12, Proposition 6.4, “Cofinality and local character”). *No set-indexed uniform family is cofinal or coinital in $\mathfrak{W}_{\text{sl}}(\mathbf{No})$; there are increasing cofinal and decreasing coinital families indexed by Ord; at every point neither side has a set-indexed cofinal, respectively coinital, family, but both have Ord-indexed ones.* This is proposition 19.27 together with the remark that first entries of a set-indexed family are bounded, by the same argument; the cofinal family is $(E_\alpha)_{\alpha \in \text{Ord}}$ for the positive ordinals α .

Corollary 19.29 (Set saturation is not Dedekind completeness). *A nonempty uniformly set-indexed family in $\mathfrak{W}_{\text{sl}}$ with no greatest member has no supremum in $\mathfrak{W}_{\text{sl}}$. Dually, a family with no least member has no infimum. Both statements also hold in the ordinary class order $\mathcal{P}_{\text{bd}}$.*

Proof. Suppose that E is the supremum of a family (A_i) with no greatest member. Every A_i is then strictly below E . By Theorem 19.10, there is a bounded-support enumeration p with $A_i < p < E$ for every i . This makes p a smaller upper bound, a contradiction. The infimum argument reverses the inequalities, and the same bounded-support interpolant proves the assertions internal to $\mathcal{P}_{\text{bd}}$. $\square$

For example, take the enumerations E_n from the transposition construction, so that $E_n(0) = n$ for $n \in \mathbb{N}$. They form an increasing sequence of bounded-support codes. The enumeration whose first value is the surreal ordinal ω is an upper bound, but there is no least upper bound in either $\mathcal{P}_{\text{bd}}$ or $\mathfrak{W}_{\text{sl}}$. The obstruction is not a failure to find bounds: Theorem 19.10, with one input side empty, bounds every uniformly set-indexed family above and below. Rather, whenever a proposed bound is strictly beyond every member, there remains room to move the bound closer. This distinction will also matter when the full order of Dedekind cuts is embedded below. An order embedding preserves comparisons, but does not automatically preserve the role of a point as the least upper bound of a specified family.

[merge] Source 08’s Section 11.2, “Saturation is not Dedekind completeness”, proves corollary 19.29 by the same argument (a strictly smaller upper bound from note 19.13) and adds that the same assertion holds in $\mathcal{P}_s(b)$ and in $\mathbf{No}$: “Set-cut saturation supplies abundant *strict* separators; it is not the Dedekind completeness familiar from the real line.”

[merge] Bounded reflection is proved only by source 09.

19.6 Source 09: bounded reflection

19.6.1 Every set family reflects into one bounded permutation stage

Theorem 19.30 (Bounded reflection). *Let $(E_i)_{i \in I}$ be a uniformly set-indexed family of pairwise distinct global enumerations. For any baseline b , there are a set ordinal λ , an infinite initial cardinal ν , and support codes $\sigma_i \subseteq \nu \times \nu$ such that*

$$E_{\sigma_i} \upharpoonright \lambda = E_i \upharpoonright \lambda, \quad E_i < E_j \iff E_{\sigma_i} < E_{\sigma_j}.$$

The family order also embeds into $\mathcal{W}_\kappa(\mathbf{No}_{<\theta})$ for some birthday bound θ , where $\kappa = |\mathbf{No}_{<\theta}|$.

Proof. Choose $\lambda \geq 1$ strictly greater than every first-disagreement ordinal of a pair of distinct rows. This is possible by Replacement because $I \times I$ is a set. The set

$$U = \bigcup_{i \in I} \text{ran}(E_i \upharpoonright \lambda)$$

is a set of surreals. Choose one infinite initial cardinal ν strictly above λ and every ordinal in $b^{-1}[U]$. Apply lemma 19.6 to every row prefix at this same bound. All comparisons were resolved before λ , so they are preserved. Choice for this set family of completions suffices.

For the birthday version, take θ large enough that $U \subseteq \mathbf{No}_{<\theta}$ and its cardinality κ is strictly larger than the ordinal λ . Each prefix then removes fewer than κ letters from $\mathbf{No}_{<\theta}$. Enumerate the remainder in type κ and append it. The total ordinal length is $\lambda + \kappa = \kappa$. The resulting exhaustive words preserve all first disagreements. $\square$

Thus no counterexample involving a *uniform set family* needs unboundedly many coordinates in an essential way. But the bound depends on the family, and there is no single set bound working for every family. For each κ , the global core contains κ^+ , whereas theorem 6.14 rules it out of any fixed alphabet of size κ .

[merge] The failure of finite support is proved only by source 08; it shows why the core must allow set support.

19.7 Why finite support is insufficient (source 08)

Let F_b be the suborder of $\mathcal{P}_s(b)$ consisting of finite support codes. The explicit transpositions in note 19.9 show that F_b already contains a copy of $\mathbf{No}$ and hence every class linear order. Nevertheless it does not have the saturation or ambient density of the full set-supported skeleton.

Proposition 19.31. *The order F_b is not an η_{Ord} -order, and its represented enumerations are not dense in $\mathfrak{W}_{\text{sl}}$.*

Proof. Put $p(n) = b(n+1)$ for $n < \omega$. This injective prefix differs from b at every natural-number coordinate. Every finite injective prefix extends to a finite perturbation of b : after applying b^{-1} , its partial injection extends to a permutation of the finite union of its domain and range, because the two remaining finite complements have equal size.

For each n , choose fresh values $a_n < p(n) < c_n$ outside $\text{ran}(p \upharpoonright n)$, and extend $(p \upharpoonright n) \cap (a_n)$ and $(p \upharpoonright n) \cap (c_n)$ to finite perturbations l_n, r_n of b . These choices can be made uniformly. For all n, m one has $l_n < r_m$: at the earlier of their altered coordinates one sees either $a_n < p(n)$ or $p(m) < c_m$, and when $n = m$ one sees $a_n < c_n$.

A separator of these two countable families must agree with p at every finite coordinate. Otherwise at its first disagreement k , the rows l_m, r_m for $m > k$ both agree with p there, forcing the candidate below a left row or above a right row. Such a separator moves every natural-number coordinate of b and cannot have finite support. This proves failure of set-cut saturation.

Finally choose $u < v$ in $\mathfrak{W}_{\text{sl}}$ with common prefix p of length ω and distinct fresh entries immediately afterward, using note 19.8. Every enumeration between them must share that prefix, so none is a finite perturbation of b . Hence F_b is not dense in $\mathfrak{W}_{\text{sl}}$. $\square$

Thus containing every ordinary class order is weaker than being a surreal line or being a dense skeleton of the virtual ambient order. Set support, not merely finite support, is what permits arbitrary set-length prefixes to be imposed.

19.8 Binary class words, cuts, and diagonalization

Theorem 19.32. *The lexicographic orders $\mathfrak{W}_{\text{sl}}$ and $\{0,1\}^{\text{Ord}}$ are order-bi-embeddable, where a binary word of length Ord means a class characteristic function. Both embeddings have uniform relational definitions in GBC.*

Proof. Partition the coordinate ordinals into successive pairs $2 \cdot \xi, 2 \cdot \xi + 1$. Here the multiplication is ordinal multiplication with 2 on the left; every ordinal has a unique such pair position. Sort the corresponding baseline values as

$$u_\xi = \min\{e_0(2 \cdot \xi), e_0(2 \cdot \xi + 1)\}, \quad v_\xi = \max\{e_0(2 \cdot \xi), e_0(2 \cdot \xi + 1)\}.$$

For a class $D \subseteq \text{Ord}$, put these two values in the order (u_ξ, v_ξ) if $\xi \notin D$, and in the order (v_ξ, u_ξ) if $\xi \in D$. Denote the resulting enumeration by T_D . Each pair retains exactly its original two values, so T_D is a bijection onto $\mathbf{No}$. At the least differing bit of two characteristic functions, 0 chooses the smaller first value and 1 the larger one. Thus $D \mapsto T_D$ preserves and reflects lexicographic comparison.

For the converse, fix an ordinal enumeration $(p_\xi)_{\xi \in \text{Ord}}$ of the set codes in $\mathcal{P}_{\text{bd}}$. Associate with $E \in \mathfrak{W}_{\text{sl}}$ the class

$$D_E = \{\xi \in \text{Ord} : p_\xi <_{\text{lex}} E\}.$$

If $E <_{\text{lex}} F$, then $D_E \subseteq D_F$, and the inclusion is strict: Corollary 19.15 supplies a code strictly between E and F . At the least ordinal belonging to $D_F \setminus D_E$, the characteristic functions therefore have values 0 and 1, respectively. This proves the second order embedding. Both constructions use elementary class comprehension with the input word or enumeration as a class parameter. $\square$

Bi-embeddability must not be replaced by isomorphism. Binary class words have a least and a greatest word and have adjacent pairs: after a set prefix, the word with next bit 0 and all subsequent bits 1 immediately precedes the word with next bit 1 and all subsequent bits 0. By contrast, $\mathfrak{W}_{\text{sl}}$ is dense and has no endpoints.

[merge] *Second route (source 12).* Source 12's Section 7 embeds $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ into binary class words by concatenating self-delimiting codes of the entries, instead of source 11's characteristic function of a dense class; the other embedding is source 11's pair swap.

19.9 Source 12's route to the binary-class representation

Let $\mathbb{B}_{\text{Ord}}$ denote the predicate of class subsets of Ord, regarded as binary functions on Ord and compared at their least disagreement. This is again notation for a predicate and relation on classes, not a powerclass object.

Lemma 19.33 (A self-delimiting numerical code). *Replace every minus in a sign sequence by 00, every plus by 11, and append the terminal block 01. The resulting binary ordinal word $c(s)$ has length $2 \cdot \text{bd}(s) + 2$. Distinct codewords are prefix-incomparable, and their first-difference comparison agrees with numerical comparison of the surreals.*

Proof. The three two-bit blocks have order $00 < 01 < 11$, exactly matching $- < \text{end} < +$. At a sign disagreement the first different block decides. When one sign sequence ends, its terminal block differs from the next sign block of the other. Hence neither complete codeword can be a prefix of another, and the numerical comparison is preserved. $\square$

Theorem 19.34 (Binary-class equimorphism). *In the sense of uniform class transformations,*

$$\mathfrak{W}_{\text{sl}}(\mathbf{No}) \simeq_c \mathbb{B}_{\text{Ord}}.$$

Proof. For $E : \text{Ord} \rightarrow \mathbf{No}$, concatenate $c(E(\alpha))$ over all ordinals. The cumulative length before each stage is a set ordinal. All blocks are nonempty, so the cumulative lengths are unbounded in Ord and the result is a binary class on all of Ord . If E, F first disagree at α , their earlier block lengths and bits agree, and the code of that entry decides the same inequality by [lemma 19.33](#).

For the converse, use the disjoint pairs $a_\alpha < b_\alpha$ in the proof of [theorem 19.38](#). For a binary class A , list the pair in ascending order if its bit is 0, and descending order if its bit is 1. Put pair blocks in ordinal order. The first different bit gives the first different pair orientation and the correct lexical inequality. All labels occur exactly once. $\square$

This is not an isomorphism assertion. The binary-class order has least and greatest elements and adjacent pairs, such as a zero followed by all ones versus a one followed by all zeros. By [note 19.11](#), $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ has no endpoints or adjacent pairs. The distinction between equimorphism and isomorphism is already visible at this level.

[merge] Source 08’s Section 13 (“Binary class sequences and the size obstruction”) defines an embedding between virtual collections as a *uniform operator*: “a formula, with displayed class parameters, assigning a unique output class to every input class, with class existence proved in the base theory. It is not a new class whose members are ordered pairs of proper classes.”

Note 19.35 (Source 08, Theorem 13.1, “Virtual binary equimorphism”). *Relative to a baseline $b : \text{Ord} \rightarrow \mathbf{No}$, there are uniform order embeddings*

$$\mathbb{B}_{\text{Ord}} \hookrightarrow \mathfrak{W}_{\text{sl}} \hookrightarrow \mathbb{B}_{\text{Ord}}; \quad (\text{S08.11})$$

thus these virtual orders are equimorphic in the uniform-operator sense; they are not order-isomorphic. This is [theorem 19.32](#). The first embedding is source 11’s pair swap; the second is source 12’s concatenation ([theorem 19.34](#)), with the code of [note 7.5](#) and block endpoints $D_0 = 0$, $D_{\xi+1} = D_\xi + (2 \cdot \text{bd}(e(\xi)) + 2)$, $D_\lambda = \sup_{\xi < \lambda} D_\xi$ defined by set-valued recursion. Source 08 adds that the jump “ $p \frown 0$ followed by constant 1” below “ $p \frown 1$ followed by constant 0” in $\mathbb{B}_{\text{Ord}}$ ceases to be adjacent in the image, because $\mathfrak{W}_{\text{sl}}$ is dense: “equimorphism does not preserve order topology”.

Corollary 19.36. *The higher-order order of all class initial segments of $\mathbf{No}$, ordered by inclusion, embeds into $\mathfrak{W}_{\text{sl}}$. In particular, this holds for any specified convention of Dedekind cuts. Conversely, each $E \in \mathfrak{W}_{\text{sl}}$ determines the initial segment $\{p \in \mathcal{P}_{\text{bd}} : p <_{\text{lex}} E\}$, and these initial segments separate distinct enumerations.*

Proof. For a class initial segment $C \subseteq \mathbf{No}$, form $e_0^{-1}[C] \subseteq \text{Ord}$ and apply the pair-swap embedding of Theorem [19.32](#). Proper inclusion of initial segments gives proper inclusion of their inverse images, which implies the correct lex comparison of their characteristic functions. The converse is the dense-class construction in the second half of that theorem. $\square$

This corollary does not form a GB class of all Dedekind cuts of $\mathbf{No}$. Each individual cut is an allowed class parameter; the rule associates an enumeration to it and compares any two such results. An encompassing object belongs to a higher type or to an explicitly larger ambient universe. Nor does the corollary assert that the natural inclusion of $\mathcal{P}_{\text{bd}}$ in $\mathfrak{W}_{\text{sl}}$ is a Dedekind completion: an order embedding of cuts need not preserve their intended suprema or their original positions relative to the embedded line.

There is a useful description of the natural cuts produced by an enumeration outside the bounded-support family. If $E \in \mathfrak{W}_{\text{sl}}$ has no code in $\mathcal{P}_{\text{bd}}$, then

$$L_E = \{p \in \mathcal{P}_{\text{bd}} : p <_{\text{lex}} E\}, \quad U_E = \{p \in \mathcal{P}_{\text{bd}} : E <_{\text{lex}} p\}$$

partition $\mathcal{P}_{\text{bd}}$ into two nonempty classes. Proposition 19.27 shows that L_E has no set-sized cofinal subset and U_E has no set-sized coinital subset, while ordinal-indexed class families suffice on both sides. This cut is unfilled in $\mathcal{P}_{\text{bd}}$: a putative filling code and E would be separated by another bounded-support code. Transporting the cut through an isomorphism $\mathcal{P}_{\text{bd}} \cong \mathbf{No}$ therefore exhibits a surreal class cut with neither side controlled by a cofinal or coinital set. This is a description of individual additional enumerations relative to the fixed dense class, rather than a classification of all surreal class cuts.

Theorem 19.37 (No exhaustive uniform class of codes). *There is no ordinary class of set codes with a single class evaluation relation whose evaluated rows include every member of $\mathfrak{W}_{\text{sl}}$.*

Proof. If such a coding existed, global choice would enumerate its codes by ordinals, with repetitions if the class of codes were a set. Write the resulting row relation as $\text{Eval}(\xi, \alpha, x)$. Use the sorted pairs $u_\xi < v_\xi$ from Theorem 19.32, and define the class

$$D = \{\xi : \text{Eval}(\xi, 2 \cdot \xi, u_\xi)\}.$$

The associated pair-swap enumeration T_D differs from row ξ on the pair $(2 \cdot \xi, u_\xi)$. If that pair belongs to the old row, it does not belong to the graph of T_D , whose value there is v_ξ . If it does not belong to the old row, it belongs to the new graph, whose value there is u_ξ . Thus T_D is a member of $\mathfrak{W}_{\text{sl}}$ omitted by the proposed exhaustive list. The argument even permits invalid or nonfunctional rows in the evaluator. $\square$

The theorem explains why class universality of $\mathbf{No}$ cannot simply be applied to the full collection $\mathfrak{W}_{\text{sl}}$. It also pinpoints the limitation of coding only bounded modifications, or any other one uniformly evaluable family of set codes: such a family may be order-dense and may realize all set-indexed cuts, while still omitting global enumerations by diagonalization.

[merge] *Strengthening (source 12)*. Source 12's Section 5 proves that no uniform family of class well-orders indexed by a class of sets contains every class well-order of $\mathbf{No}$, and that the missing one can be chosen set-like; its corollary 19.39 is theorem 19.37. Its subsection on definable classes explains why enumerating formulas does not evade the theorem.

19.10 Source 12's strengthening: why the collection of all global well-orders is not a class

The formal obstacle is not that individual class relations are illegal. They are legitimate class variables. The obstacle is attempting to make them the elements of an ordinary class of the same theory.

A useful legal replacement is a *uniform family*. For example, $F \subseteq I \times \text{Ord} \times \mathbf{No}$ is a class of triples of sets, and its section F_i can be a class bijection for every $i \in I$. Similarly a class of triples (i, x, y) can code a family of class order relations. The index class I may be proper. These are single classes, not functions whose values are class objects.

Theorem 19.38 (No exhaustive class-coded family). *In GBC, no uniform family of class well-orders of $\mathbf{No}$ indexed by a class of sets contains every class well-order of $\mathbf{No}$. In fact, for every such family there is a set-like class well-order not among its sections.*

Proof. Global choice permits the family to be reindexed by an ordinal or by Ord . Extend an ordinal-indexed family to an Ord -indexed one by repeating an arbitrary default well-order; an empty family is trivial. Write the resulting relations as R_α .

Fix a bijection $B : \text{Ord} \rightarrow \mathbf{No}$ and partition its range into the pairs

$$\{B(2 \cdot \alpha), B(2 \cdot \alpha + 1)\} = \{a_\alpha, b_\alpha\}, \quad a_\alpha < b_\alpha.$$

Here multiplication is ordinal multiplication. The blocks of length two partition Ord . Define a new enumeration E by placing the two labels of pair α in the order *opposite* to their relative order in R_α . Keep the pair blocks in increasing α order.

This is a class bijection $\text{Ord} \rightarrow \mathbf{No}$, defined elementarily from the family and B . Its induced well-order disagrees with R_α on the relative order of a_α, b_α . Thus it is not any section of the family. $\square$

Corollary 19.39 (No uniform set-code solution). *There is no class D of set codes together with a uniform class decoder whose decoded relations are exactly all class well-orders of $\mathbf{No}$.*

This diagonal theorem applies equally to an attempted exhaustive coding of all class subsets of Ord through the pair construction. It does not say that an externally specified set model has too many classes to form an external set. It says that the corresponding exhaustive decoder is not available *internally at the same size level*.

19.10.1 Definable classes and the truth-predicate trap

Formula codes and set parameters are sets, but the semantic map “formula plus parameters $\mapsto$ its extension in the universe” is not a uniformly available truth predicate for the universe. For each fixed formula one may define a class. One may also work externally with all definable classes of a chosen model. Neither maneuver supplies an internal universal decoder defeating [theorem 19.38](#). The same caution applies to quantifying over all possible Lean universe levels: universe polymorphism is not an internal type of all universes.

[merge] Sources 09 and 08 prove the diagonal theorem for families of global enumerations (source 09’s Section 12, “Why all global well-orders cannot be uniformly coded by sets”).

Note 19.40 (Source 09, Theorem 12.1, “Uniform no-coding theorem”). *There is no class of set indices I and no uniform class relation $F \subseteq I \times \text{Ord} \times \mathbf{No}$ whose rows are exactly, or even include, all global enumerations of $\mathbf{No}$.* This is [theorem 19.37](#), by the same diagonal class $D = \{\alpha : (c(\alpha), 2 \cdot \alpha, a_\alpha^-) \in F\}$ along a class surjection $c : \text{Ord} \rightarrow I$, valid with nonfunctional rows. Source 09’s preceding paragraph defines the switches E_A that list the pair $\{a_\alpha^-, a_\alpha^+\} = \{b(2 \cdot \alpha), b(2 \cdot \alpha + 1)\}$ increasingly for $\alpha \notin A$ and decreasingly for $\alpha \in A$; they embed the binary-class comparison into the global one, “the proper-class counterpart of the pair construction” of [note 6.12](#), as in the first half of [theorem 19.32](#).

Note 19.41 (Source 08, Theorem 13.2, “No class-uniform exhaustive list”). *If a class relation $T \subseteq \text{Ord} \times \text{Ord} \times \mathbf{No}$ uniformly gives a global enumeration T_α for each ordinal α , there is a global enumeration e^* different from every T_α ; consequently no uniform family indexed by an ordinary set or class of sets exhausts $\mathfrak{W}_{\text{sl}}$.* This is [theorem 19.37](#), by the same switch at the even coordinate $2 \cdot \alpha$; families indexed by sets or proper classes are reindexed by Ord . Source 08 adds: “the class of set-supported perturbation codes does not parametrize the full collection. The isomorphism $\mathcal{P}_s(b) \cong \mathbf{No}$ is compatible with the much larger size of $\mathfrak{W}_{\text{sl}}$: each orbit is a dense copy of the surreal line, not all global enumerations”, and “an assertion about an embedding of the entire virtual order into a class must first specify a higher-type notion of mapping; it cannot be inferred from ordinary class universality.”

19.10.2 What the theorem does and does not say

The theorem rules out a uniform master list by set parameters, including one indexed by a proper class of such parameters. It does not rule out large useful coded subfamilies such as $\mathcal{P}_s(b)$; indeed the latter is dense and class-universal.

It also does not mean that class functions cannot be discussed in GBC, nor that stronger foundations are inconsistent. They can be discussed one or several at a time, in uniform families, or as elements of a higher-level collection in a suitably stratified theory. What fails is compression of *all* of them into one same-level, set-indexed evaluation relation.

Enumerating formula strings does not evade this issue. A putative uniform evaluation of all formulas over the set universe would require an appropriate truth predicate; it is not obtained from syntax alone. Definitions with additional class parameters also need not remain in the original parameter collection. One must distinguish a metatheoretic list of formulas, individual definable classes, and an internal uniform family of their extensions.

19.11 Omitted ranges and the transition to raw well-orders

Theorem 19.42 (An omitted-range cut). *Fix $r \in \mathbf{No}$ and a class bijection $F : \text{Ord} \rightarrow \mathbf{No} \setminus \{r\}$. There are increasing and decreasing ordinal-indexed families (a_α) and (b_α) in $\mathcal{P}_{\text{bd}}$ such that*

$$a_\alpha <_{\text{lex}} b_\beta \quad (\alpha, \beta \in \text{Ord}),$$

but no member of $\mathfrak{W}_{\text{sl}}$ lies strictly between both entire families. The raw well-order that lists F and then r has type $\text{Ord} + 1$ and is the unique member of $\mathfrak{W}_{\text{all}}$ between these families.

Proof. Use the approximants from Proposition 19.27. They satisfy $a_\alpha <_{\text{lex}} F <_{\text{lex}} b_\beta$ for every pair of indices. If a global bijection E lay between both families and differed from F first at δ , an approximant with index greater than δ would place E on the wrong side. Therefore E would equal F at every ordinal, which contradicts its surjectivity.

Let R be the raw order obtained by appending r after the enumeration F . Its set-like initial part is precisely F : every point listed by F has set-sized predecessors, whereas r has a proper class of predecessors. Its comparisons with the approximants are decided at their prescribed ordinal first differences, so it lies between both families. Conversely, the same first-difference argument forces the core F_S of any raw order S between the families to equal F . By the core decomposition of Theorem 15.1, its residual order must be the unique order on the singleton $\{r\}$. Thus $S = R$. $\square$

The lower and upper families here are indexed by all ordinals, not by sets. Accordingly, the missing interpolant does not contradict Theorem 19.10. It gives a concrete boundary between the set-cut saturation of the set-like comparison and the additional points provided by non-set-like global well-orders.

[merge] *Source 09's route.* Source 09's Section 11 gives a concrete instance of theorem 19.42, with the nonexhaustive word $\xi \mapsto \xi$ and the fresh values -1 and $\alpha + 1$; its remark 19.44 is the general mechanism, which is source 11's theorem for an arbitrary omitted range.

19.12 Source 09's route: a proper-class cut with no interpolant

The interpolation theorem has an exact and visible size limitation. The following construction uses only ordinal-valued prefixes and the single nonordinal value -1 .

Theorem 19.43 (An explicit unfilled (Ord, Ord) configuration). *For any baseline b , there are uniformly class-indexed families $(L_\alpha)_{\alpha \in \text{Ord}}$ and $(U_\alpha)_{\alpha \in \text{Ord}}$ lying in $\mathcal{P}_s(b)$ such that*

$$\alpha < \beta \implies L_\alpha < L_\beta < U_\beta < U_\alpha, \quad L_\alpha < U_\beta \quad \text{for all } \alpha, \beta,$$

but there is no global enumeration E with $L_\alpha < E < U_\beta$ for every α, β .

Proof. Let $g : \text{Ord} \rightarrow \mathbf{No}$ be $g(\xi) = \xi$. This is an injective class word but not an exhaustive enumeration of $\mathbf{No}$. For each α , use [lemma 19.6](#) to choose core completions of the prefixes

$$(g \upharpoonright \alpha) \frown (-1), \quad (g \upharpoonright \alpha) \frown (\alpha + 1),$$

calling them L_α and U_α , respectively. A fixed global well-order on the set support codes makes these choices uniform. The appended letters are fresh. For $\alpha < \beta$, comparison at coordinate α gives $L_\alpha < L_\beta$ and $U_\beta < U_\alpha$. Every lower word is below every upper word, by comparison at the smaller divergence coordinate.

Suppose a global enumeration E lay between all these bounds. If $E(\delta) \neq \delta$ for some ordinal, take the least such δ . If $E(\delta) < \delta$, the lower word $L_{\delta+1}$ agrees with g through coordinate δ and therefore lies above E , a contradiction. If $E(\delta) > \delta$, the upper word $U_{\delta+1}$ lies below E , also a contradiction. Hence $E(\delta) = \delta$ for every ordinal δ , making $E = g$. But g omits, for example, -1 , and is not surjective onto $\mathbf{No}$. $\square$

Here the notation (Ord, Ord) describes an increasing lower and a decreasing upper cofinal presentation, not a pair of set cardinal cofinalities. More explicitly, compare every global enumeration to the nonexhaustive word g . Every enumeration differs from g at a set ordinal and lies on exactly one side. The lower family in the theorem is cofinal on the lower side, and the upper family is cointial on the upper side. Neither side has an endpoint. The missing point would have to be the nonexhaustive branch g itself.

Every set subfamily of these constraints has an interpolant by [note 19.12](#). The failure is therefore genuinely at the proper-class level, not a missed set-sized limit. In particular, the space of all global enumerations is not a class-cut completion of its surreal core.

Remark 19.44 (A general missing-branch mechanism). The same proof works for any injective class word $g : \text{Ord} \rightarrow \mathbf{No}$ which is not surjective. At each set stage choose fresh values below and above $g(\alpha)$ outside the earlier range, and complete the two prefixes. Any interpolant would agree with g at every coordinate and would therefore fail surjectivity. The surreal set-cut property guarantees the required fresh values.

Example 19.45 (Two omitted points produce adjacent raw orders). Let $r < s$ and choose $F : \text{Ord} \rightarrow \mathbf{No} \setminus \{r, s\}$ bijectively. The approximants to F have exactly two raw interpolants: append r, s in that order, or append s, r . Both raw orders have type $\text{Ord} + 2$, and the first is lexicographically smaller. They are adjacent in $\mathfrak{W}_{\text{all}}$. Indeed, a raw order between them cannot have a different core, since an ordinal first difference would put it on the same side of both. With core F fixed, its residual order is a well-order of a two-element set, and these are exactly the two displayed possibilities. Thus the density theorem for $\mathfrak{W}_{\text{sl}}$ does not extend to all raw well-orders.

Corollary 19.46 (Exactly which raw set cuts admit bounded completions). *Let $(A_i)_{i \in I} <_{\text{lex}} (B_j)_{j \in J}$ be a uniformly set-indexed strict cut in $\mathfrak{W}_{\text{all}}$. A member of $\mathcal{P}_{\text{bd}}$ lies between the two sides if and only if*

$$F_{A_i} \neq F_{B_j} \quad (i \in I, j \in J),$$

where F_R is the canonical ordinal enumeration of the set-like initial part of a raw well-order R . The condition is vacuous when a side is empty.

Proof. If the cross cores are different, their comparisons are decided at ordinal first differences and have the same direction as the given raw comparisons. Apply Theorem 19.10 to these uniform families in $\mathfrak{J}$. Every comparison of the resulting completion with a raw input is already decided within its core, giving the required raw inequalities.

Conversely, suppose a cross pair shares a core F . This core cannot be surjective: a surjective core determines the entire raw order, contradicting the strict comparison of the pair. Any $E \in \mathfrak{W}_{\text{sl}}$ therefore differs from F at an ordinal coordinate. That first difference places E on the same side of both raw orders, so E cannot lie between them. This excludes a bounded-support completion in particular. $\square$

The criterion isolates the obstruction without appealing merely to size. Distinct ordinal cores can be separated by one bounded change of the baseline, even for a whole set-indexed cut. Two opposite sides with the same core require information in the residual non-set-like order, beyond every ordinal coordinate available to a global set-like enumeration.

Part V

Foundations, models and formalization

20 Formalizing the subject without changing its meaning

20.1 A bounded development in ordinary set theory

For any ordinal θ , use the set $X = \mathbf{No}_{<\theta}$. The well-order relations on X form a subset of $\mathcal{P}(X \times X)$. Their lexicographic comparison can be defined either by unique ordinal enumeration or by the predecessor formula. This is ordinary ZFC mathematics. No inaccessible cardinal, class theory, or new logical sort is required.

This approach is particularly good for proving the general alphabet theorem, investigating cardinal spectra, and checking how a proposed construction changes birthdays. One should distinguish $\mathcal{W}(X)$, which permits every possible ordinal type of cardinality $|X|$, from its minimal-length stratum. Their cardinalities agree, but their jumps and finer order structure need not.

A family of results for every θ does not automatically give a coherent global map. To take a union, the maps must agree on overlaps. The repository's bounded-surreal nonembedding example makes this restriction concrete.

20.2 Second-order class theory

In GB or GBC, the formula

$$\text{ClassWO}(R) \wedge \text{ClassWO}(S) \implies R = S \vee \text{Lex}(R, S) \vee \text{Lex}(S, R)$$

is meaningful with class variables. One can prove trichotomy, transitivity, existence theorems, and uniform-family embedding theorems in this language. It is unnecessary to form a class whose members are R, S .

When a family is indexed by a class I of sets, one class relation $T \subseteq I \times \mathbf{No} \times \mathbf{No}$ supplies its rows. If the rows are pairwise distinct, substituting them into the elementary predecessor formula produces a class linear order on I . With repeated rows, one instead has the corresponding comparison on codes modulo equal evaluations. This is the correct format for most class-sized examples and applications.

In contrast, an expression such as

$$\{R : R \text{ is a class well-order of } \mathbf{No}\}$$

does not define an ordinary GB class of members. Our fraktur notation abbreviates a domain of class quantification. The diagonal theorem shows why a single uniform set-code replacement cannot recover the full domain either.

20.3 Why KM does not erase the extra level

KM strengthens comprehension by allowing class quantifiers, but its classes still have *sets* as members. It therefore does not turn the totality of class well-orders into an ordinary class. A genuine object containing arbitrary classes as members requires another sort, or a containing universe in which those classes become sets.

This distinction matters when one wants an arbitrary topology of higher-order open collections, a measure on all global well-orders, or a power object of their totality. Such notions require an explicit ambient theory. Hyperclass forcing provides one example of higher-level foundational practice [28]; it is not a reason to silently ignore the level distinction in bare KM.

There is also a separate recursion issue. Extracting the canonical initial Ord-part of one raw well-order is an elementary class construction. Iteratively deleting proper-class tails through an arbitrary transfinite sequence would have class-valued stages and needs its own analysis. General comparability of abstract class well-orders by initial-segment isomorphisms is another stronger topic; ETR implies it [27]. None of that machinery is needed for the labeled comparison proved in this article.

20.4 An explicit containing-universe model

A useful concrete model assumes a strongly inaccessible cardinal κ in an ambient universe and takes

$$U = V_\kappa, \quad \mathcal{C} = \mathcal{P}(V_\kappa).$$

This is a sufficient setting for interpreting a full class universe. The internal surreal class becomes the ambient set

$$\mathbf{No}^U = \mathbf{No}_{<\kappa}, \quad |\mathbf{No}^U| = \kappa.$$

All its ambient well-orders form a set of cardinality 2^κ . Those that are internally set-like have ambient order type κ ; unrestricted well-orders may have any ambient ordinal type of cardinality κ .

To see the set-like assertion, every initial segment of a κ -enumeration has fewer than κ members, with ranks bounded below κ ; regularity puts it in U . Conversely, an ordinal type strictly exceeding κ has an initial segment of cardinality κ , which cannot be a set of U . A smaller type cannot enumerate the entire carrier.

The bounded-support class becomes an ambient order of cardinality κ , while the full κ -permutation order has cardinality 2^κ . Thus a κ -sized dense suborder can coexist with a larger order of all enumerations. The exact ambient ordinal cutoff remains κ^+ .

Every ordinal *of* U is below κ . There is no contradiction between internally embedding all ordinals and having an ambient successor-cardinal barrier. Inaccessibility is an additional assumption for this full universe example; it is not required for an individual birthday fragment.

20.5 A strict nonembedding result

The containing-universe interpretation makes a further comparison decidable by ordinary set-order invariants.

Theorem 20.1 (Nonembedding in a full universe). *In the preceding full interpretation $U = V_\kappa$, $\mathcal{C} = \mathcal{P}(V_\kappa)$, with ambient ZFC and κ strongly inaccessible,*

$$\mathfrak{W}_{\text{sl}}(\mathbf{No}^U) \hookrightarrow \mathfrak{W}_{\text{all}}(\mathbf{No}^U), \quad \mathfrak{W}_{\text{all}}(\mathbf{No}^U) \not\hookrightarrow \mathfrak{W}_{\text{sl}}(\mathbf{No}^U).$$

These are assertions about the actual ambient sets of internal class well-orders and arbitrary ambient order embeddings.

Proof. The first embedding is inclusion. Put $X = \mathbf{No}^U$. The target of the proposed reverse embedding is the dense order of all bijections $\kappa \rightarrow X$. Its bounded-support suborder $D = \mathcal{P}_{\text{bd}}^U$ is dense and has ambient cardinality κ . For the upper bound, codes with domain $\lambda < \kappa$ are subsets of $\lambda \times \lambda$; strong inaccessibility bounds their number below κ , and there are κ possible domains. Transpositions give the matching lower bound.

Fix labels $a < b$ in X . There are 2^κ bijections $F : \kappa \rightarrow X \setminus \{a, b\}$, by the fixed-alphabet cardinality theorem. Each determines a convex adjacent pair

$$R_F = F + (a, b) <_{\text{lex}} S_F = F + (b, a)$$

in the unrestricted order. Distinct F 's determine distinct, disjoint convex fibers; one entire pair precedes the other.

If $\Phi : \mathfrak{W}_{\text{all}}(X) \hookrightarrow \mathfrak{W}_{\text{sl}}(X)$ existed, the intervals

$$(\Phi(R_F), \Phi(S_F))$$

would be 2^κ pairwise disjoint nonempty open intervals in the target. Each contains a point of D . Choosing the least such point under a fixed well-order of D would inject 2^κ into κ , a contradiction. $\square$

Thus equal cardinality 2^κ and equal ordinal embedding cutoff κ^+ do not imply equimorphism. The useful invariant here is the size of a family of disjoint open intervals relative to a dense suborder. The repository already uses related separability and cellularity arguments for the real alphabet [1, Remark `lwo:sp:rem:conditional`]; the argument above supplies the precise surreal-universe version.

There is also a direct class-theoretic obstruction, provided the inverse evaluation relation can be formed by impredicative comprehension.

Theorem 20.2 (No definable uniform embedding in $\text{KM} + \text{GC}$). *For every fixed formula, with fixed set and class parameters, that defines a total transformation $R \mapsto \Phi(R)$ from $\mathfrak{W}_{\text{all}}(\mathbf{No})$ to $\mathfrak{W}_{\text{sl}}(\mathbf{No})$, the theory $\text{KM} + \text{GC}$ proves that this transformation is not an order embedding. The transformation is specified relationally: the defining formula assigns a unique output class graph to each input class well-order. This is a schema for fixed formulas, not quantification over a third sort of arbitrary transformations.*

Proof. Suppose such an order embedding were defined. Fix $a < b$ and a global well-order of the set codes in $\mathcal{P}_{\text{bd}}$. For each class bijection $F : \text{Ord} \rightarrow \mathbf{No} \setminus \{a, b\}$, form the adjacent raw pair R_F, S_F from the preceding proof. Density of $\mathcal{P}_{\text{bd}}$ in $\mathfrak{W}_{\text{sl}}$ supplies codes strictly between $\Phi(R_F)$ and $\Phi(S_F)$. Let q_F be the globally least such code. Comprehension in KM makes this a legitimate least-element

definition, even if the formula specifying Φ contains class quantifiers. Disjointness of the image intervals implies

$$q_F = q_G \implies F = G.$$

Now use KM comprehension to form the ordinary class relation

$$\text{Eval}(q, \alpha, x) \iff \exists \text{ class } F [F : \text{Ord} \xrightarrow{\sim} \mathbf{No} \setminus \{a, b\} \wedge q = q_F \wedge F(\alpha) = x].$$

Its rows are empty or the graphs of unique such bijections, and every such bijection occurs. Compose all rows with one fixed class bijection $\mathbf{No} \setminus \{a, b\} \rightarrow \mathbf{No}$. The resulting single class evaluator, with the ordinary set codes $q \in \mathcal{P}_{\text{bd}}$, contains every global enumeration of $\mathbf{No}$. This contradicts Theorem 19.37. $\square$

The inverse evaluator is the decisive extra comprehension step: its definition quantifies over an arbitrary class F . The preceding proof does not claim that elementary GB comprehension suffices. Nor does it exclude arbitrary external maps between the available-class collections of a nonfull model. In particular, for an externally countable model these collections can both be countable, and its externally dense set-like order can admit an external copy of its unrestricted order. Such a map need not be representable by any uniform definition in the model. This is why the stated interpretation of an embedding matters.

20.6 Internal and external well-foundedness

For a model $(M, \mathcal{X})$, an internally well-founded class relation need not be externally well-founded. An external descending sequence might not be represented by a set in M . Thus comparing the internal theory with an external collection of all relations requires care.

Over GBC, well-foundedness of a class relation can be characterized by the absence of a set-coded descending ω -sequence [27, Section 3]. Keeping the sets and an existing relation fixed while changing only the available classes does not arbitrarily change that relation's internal well-foundedness in suitable models. What can change is which class relations are available at all. External correctness can be ensured by an appropriate well-founded full universe interpretation, as above.

20.7 A practical Lean architecture

The repository already has a universe-indexed sign carrier [7]. Its type structure can be displayed schematically as:

```
structure SignSequence : Type (u + 1) where
  birthday : Ordinal.{u}
  signAt : Ordinal.{u} -> SignType
  signAt_ne_zero_iff : forall i, signAt i != 0 <-> i < birthday
```

This is mathematical pseudocode using ASCII logical notation, not a new executable Lean declaration. The key point is the level separation: the full carrier is one universe above the types indexing its small sign lengths.

The following existing declarations were inspected:

Module or declaration	Role in the proposed development
<code>SignSequence.ordinalOrderEmbedding</code>	The all-plus ordinal embedding
<code>birthdays_bounded</code> , <code>small_bounded</code>	Control of small parameter families and bounded fragments
<code>SignSequence.not_small</code>	The carrier cannot be treated as small at its ordinal level
<code>SignSequenceCutOperation.cut</code> and <code>cut_realizes</code>	Actual separators for small cut data
<code>SizeObstructions.noUnrestrictedCuts</code>	The obstruction to unrestricted cut filling

These are source-level observations, not results of a fresh build. The current repository contains later arithmetic modules as well; historical proposal notes should not be mistaken for the present implementation status.

A sensible formalization proceeds in four parts.

1. **Set carriers.** For $X : \text{Type } u$, represent well-orders as relations on X together with their laws. Prove the predecessor-based comparison and its equivalence with ordinal enumeration. This avoids transporting across ordinal representatives at every comparison.
2. **Ordinal words.** Define words with small ordinal domains, an injective subtype, ordinal concatenation, and the sign code. Formalize prefix-freeness and first disagreement before proving the cardinal spectrum.
3. **Relative set-likeness.** If X lives in $\text{Type } (u+1)$, express the condition that each predecessor subtype is `Small.{u}`. A well-order of this larger X has its unrestricted order type at the larger ordinal universe. A same-level enumeration must not be inferred without the relative smallness hypothesis.
4. **Uniform families.** Keep an explicit index type and an evaluation relation. The bounded-support codes are sets of ordinal pairs at the selected level. Formulate the interpolation theorem only for small index families and the no-code theorem at its correct larger level.

Lean universes and Mathlib ordinals provide precisely the needed separation [29, 30]. Universe polymorphism is a schema of well-typed constructions, not a term denoting the union of all universes. This article supplies no new Lean file and makes no machine-verification claim.

[merge] Source 12's Section 11 identifies both global orders in the full-class model $(V_\kappa, \mathcal{P}(V_\kappa))$ of [theorem 20.1](#). Its clause (iv) gives a second proof of that theorem's non-embedding: the full order $\mathcal{W}(\mathbf{No}_{<\kappa})$ has binary coding length κ^+ ([theorem 6.8](#)), whereas the set-like order $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$ has coding length κ ([theorem 8.3](#), regular case), and coding length cannot decrease under embeddings ([lemma 6.4](#)). Source 11's proof counts 2^κ disjoint intervals against a dense suborder of size κ .

21 A full-class model that connects the two size levels (source 12)

Assume in an ambient ZFC universe that κ is strongly inaccessible. Consider the two-sorted structure

$$\mathcal{M} = (V_\kappa, \mathcal{P}(V_\kappa)),$$

with all external subclasses of V_κ available as its classes. This is a familiar full-class interpretation of strong class set theory: regularity and the strong-limit property ensure the required closure of set operations and class Replacement. It satisfies KM and global choice. We use this as an explicit relative model, not as a claim that inaccessible cardinals are necessary for the earlier GBC theorems.

Theorem 21.1 (Exact external identification). *For the full-class interpretation above:*

- (i) *the internal surreal carrier is externally $\mathbf{No}_{<\kappa}$, and its internal ordinals are the external ordinals below κ ;*
- (ii) *the internal set-like global surreal well-orders, viewed as an external set order, are exactly $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$;*
- (iii) *all internal global surreal well-orders, viewed externally, form exactly $\mathcal{W}(\mathbf{No}_{<\kappa})$;*
- (iv) *both external orders have cardinality 2^κ and ordinal spectrum $\beta < \kappa^+$, but their binary coding ranks are respectively κ and κ^+ . Therefore the full order does not embed into the set-like order.*

Proof. A sign sequence belongs to V_κ precisely when its ordinal domain has length less than κ . Inaccessibility gives $|\mathbf{No}_{<\kappa}| = \kappa$.

Every external relation on $\mathbf{No}_{<\kappa}$ is a permitted internal class: each ordered pair of its labels is an element of V_κ . External well-foundedness implies internal well-foundedness. Conversely, if a relation is externally ill-founded, ambient choice supplies a descending ω -sequence. Its countably many labels have ranks bounded below κ by regularity, so the sequence is an element of V_κ and witnesses internal ill-foundedness. Thus the well-orders agree.

An internal set has external cardinality less than κ . Conversely, a subset of $\mathbf{No}_{<\kappa}$ of cardinality less than κ has bounded ranks and is an element of V_κ . Therefore internal set-likeness means that every proper predecessor segment has cardinality less than κ . An external well-order of κ labels has that property exactly when its order type is κ , proving (ii). Arbitrary external well-orders on those labels have types exactly the ordinals of cardinality κ , proving (iii). Comparisons agree by [theorem 14.7](#). Finally apply [theorems 6.8](#) and [8.3](#) and the monotonicity of coding rank under embeddings. $\square$

21.1 Why this does not contradict surreal universality

Internally, each class linear order has elements that are sets of V_κ and hence externally at most κ elements. The external collection of all class well-orders has 2^κ elements and is not an internal class carrier. Thus it is outside the scope of the internal class-universality theorem.

If $\lambda > \kappa$ is another inaccessible cardinal, the entire external set $\mathcal{W}(\mathbf{No}_{<\kappa})$ can be studied as an ordinary set in V_λ . It then embeds into the larger surreal carrier there by set-order universality. Its labels have moved up one foundational level; no same-level universal object has been created.

The singular cutoff $\beth_\omega$ in [section 8](#) must not be used as though $V_{\beth_\omega}$ were this model. That rank lacks the regularity used in Replacement and in the well-foundedness argument. The singular theorem is an ordinary ZFC theorem about a bounded set, not a full-class model construction at a singular height.

21.2 Henkin classes, countable models, and relativity

If the class collection is some $\mathcal{X} \subsetneq \mathcal{P}(M)$, only those well-orders whose relations lie in $\mathcal{X}$ are admitted. External cardinality and saturation can then differ from the full-class calculation. A countable model may have an externally countable collection of admitted classes and external countable cofinal families, while internally having no *coded set-indexed* cofinal family. The missing external families are simply not internal objects.

For nonstandard or non- β class models, one must additionally separate internal and external well-foundedness. Our exact identification in [theorem 21.1](#) used transitivity, full classes, and regularity; it was not asserted for arbitrary models. The general subtleties of class well-foundedness are discussed in [\[27, §3\]](#).

[merge] Clause (ii) of [theorem 21.1](#) is source 11's statement that the internally set-like well-orders have ambient type κ , and clause (iii) its statement that unrestricted ones have every ambient type of cardinality κ ([section 20](#)). Source 12's remark that the singular cutoff $\beth_\omega$ must not be used as though $V_{\beth_\omega}$ were such a model is the counterpart of source 11's remark that inaccessibility is needed for the full universe example but not for an individual birthday fragment.

[merge] Source 09's inaccessible model is the same setting. Its clause (ii) is the bounded form of [corollary 19.15](#), its clause (iii) of [theorem 19.10](#), and its clause (iv) the bounded form of [theorem 19.43](#); its $\mathcal{E}_\kappa$ is $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$, the set-like order of [theorem 21.1\(ii\)](#).

22 A concrete universe enlargement: all objects become sets (source 09)

A useful consistency-safe visualization of the size jump is available in ordinary set theory with a strongly inaccessible cardinal. This assumption is for the illustration, not for the earlier ZFC or GBC results.

Fix a strongly inaccessible cardinal κ , put $X = \mathbf{No}_{<\kappa}$, and choose a bijection $b : \kappa \rightarrow X$. Externally, X is a set of cardinality κ . In the smaller universe V_κ , it plays the role of the entire surreal class. Every element of X and every set-length prefix of length below κ belong to V_κ .

Define the external set

$$\mathcal{E}_\kappa = \text{Bij}(\kappa, X)$$

and let $\mathcal{P}_s^{<\kappa}(b)$ contain its enumerations eventually equal to b below κ , equivalently the permutations of κ with bounded support. Regularity makes bounded support equivalent here to support of cardinality less than κ .

Theorem 22.1 (Inaccessible set-sized model). *In this setting:*

- (i) $|\mathcal{P}_s^{<\kappa}(b)| = \kappa$ and $|\mathcal{E}_\kappa| = 2^\kappa$.
- (ii) $\mathcal{P}_s^{<\kappa}(b) \cong X$, as linear orders, and the core is order-dense in $\mathcal{E}_\kappa$.
- (iii) Every separated pair of families in $\mathcal{E}_\kappa$ of combined size less than κ has an interpolant in the core.
- (iv) There is a separated (κ, κ) -configuration in the core with no interpolant even in $\mathcal{E}_\kappa$.

Proof. By [proposition 7.3](#), $|X| = 2^{<\kappa} = \kappa$. There are fewer than κ permutations on any fixed $\alpha < \kappa$, because κ is strong limit. Their union over κ stages has size at most κ , and the swaps of position zero with other positions give the reverse bound. The cardinality of $\mathcal{E}_\kappa$ is 2^κ by [note 6.12](#).

For fewer than κ lower and upper constraints, regularity bounds all cross first-disagreement coordinates below κ . The forced-prefix construction in [note 19.12](#) therefore has length below κ . Its chosen separators remain in X : the option families and previously used values have size below κ , their birthdays are bounded below κ , and (1) supplies a separator of birthday below κ . Prefix completion can be carried out at a larger cardinal still below κ . This proves (iii), and in particular density and the $< \kappa$ -cut property of the core.

The same birthday argument gives the $< \kappa$ -cut property of X . A back-and-forth of length κ , with partial maps of size less than κ at every stage, gives the isomorphism in (ii). Finally repeat [theorem 19.43](#) for $\alpha < \kappa$ and the nonexhaustive word $\alpha \mapsto \alpha$ into X . Both -1 and each required successor ordinal belong to X , and the same comparison argument proves (iv). $\square$

This model explains the apparently conflicting assertions that every class-coded portion embeds into the surreal line and yet all enumerations are larger. A class of set codes over V_κ has external

size at most κ ; the set $\mathcal{E}_\kappa$ has external size 2^κ . It is an object of the larger universe, not such a class of codes in the smaller universe. External arbitrary families of κ or more class objects are not automatically small-universe sets.

Even externally, $\mathcal{E}_\kappa$ only represents well-orders of length κ . All well-orders on the external set X also include lengths α with $\kappa < \alpha < \kappa^+$. Thus a universe jump must record not just the carrier but which ordinal lengths are admitted.

[merge] Source 08's Section 15 discusses the same frameworks and the same full-class model; its (S08.13) is [theorem 21.1\(ii\)](#).

23 Universes, class models, and paradox avoidance (source 08)

The preceding proofs can be studied in several foundations, provided the meaning of “all” is kept explicit.

23.1 ZFC with definable classes

In ordinary ZFC, **No** is given by a formula defining sign sequences. Statements about a specified finite list of definable class functions can be translated into statements about their formulas. Set-sized cutoffs and all theorems about $\mathcal{W}(X)$ are ordinary set theorems. Global conclusions can be formulated as schemes relative to specified definable enumerations or in a theory supplying a definable global well-order.

However, “take the set of formulas defining class well-orders and interpret them all in V ” is not a free replacement for a class collection. Formula codes are sets, but uniform unrestricted truth for all formulas of the universe is a separate issue. One cannot silently insert a full satisfaction predicate for V into the base theory. Uniformity and definability claims must state their truth resources.

23.2 Two-sorted GBC and stronger class theories

In a model $(M, \mathcal{C})$ of GBC, the virtual collection $\mathfrak{W}_{\text{sl}}$ means the bijections whose graphs belong to $\mathcal{C}$. The cut families in [note 19.13](#) are represented by one relation in $\mathcal{C}$, indexed by an M -set. The skeleton consists of the M -set support codes, interpreted relative to $b \in \mathcal{C}$. All class unions and selections asserted by the proofs are made inside this model.

Passing from GBC to KM strengthens class comprehension, but it still does not make proper classes eligible as members of ordinary classes. Stronger comprehension is not an additional membership sort. A literal object containing every class enumeration belongs to an explicitly higher-order theory or to an external metatheory.

The literature on class recursion gives useful warnings here: even natural uniform truth or forcing relations may require principles stronger than GBC [\[23, 25\]](#). Our elementary lexicographic operators do not themselves supply such truth relations.

23.3 A concrete full-class model at an inaccessible cardinal

Let κ be strongly inaccessible in an external set universe, and consider the full-class model

$$(V_\kappa, \mathcal{P}(V_\kappa)). \tag{S08.12}$$

The second occurrence of $\mathcal{P}_s$ here is the ordinary power-set notation, not the permutation-code class. Internally, Ord^{V_κ} is a proper class; externally it is the set κ . Likewise the internal surreal class is externally $\mathbf{No}_{<\kappa}$ and has cardinality κ .

The full collection of internal global enumerations is, externally,

$$\{e : \kappa \longrightarrow \mathbf{No}_{<\kappa} : e \text{ is a bijection}\}, \quad (\text{S08.13})$$

a set of cardinality 2^κ . An internal set-supported perturbation has support of size below κ , equivalently bounded support in κ . There are exactly κ canonical codes: at each bound there are fewer than κ possibilities by strong inaccessibility, and there are κ bounds; the first-entry embedding gives the matching lower bound. Thus externally the skeleton has size κ and the full enumeration order size 2^κ .

This is a direct set-theoretic model of the size distinction. The internal isomorphism $\mathcal{P}_s(b) \cong \mathbf{No}$ becomes an external isomorphism between two κ -sized η_κ -orders. Internally “every set cut” means externally every cut specified by fewer than κ elements, not every external set cut.

Also, (S08.13) is the *fixed-length* order $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$. It is not the external order of all well-orderings of the set $\mathbf{No}_{<\kappa}$, which also allows lengths $\kappa+1$, $\kappa+\omega$, and all other ordinals of cardinality κ . Those longer enumerations may have internally proper-class initial segments and are not internal set-like well-orders. This is an important bridge between section 7.2 and the global theory.

Strong inaccessibility is used here to obtain a convenient full model and its size calculation. It is not an additional hypothesis of the GBC theorems: it describes one external way to realize their set/class distinction.

23.4 Restricted classes and external viewpoints

For a general class model, enlarging $\mathcal{C}$ may add new global enumerations and new binary class words while leaving the sets of M unchanged. Thus the virtual ambient orders depend on the chosen class collection. The explicit formulas for pair coding, prefix extension, and diagonalization still work with their stated parameters in each model, but one must not identify an internal “all” with an external one.

Similarly, an internal well-foundedness assertion in a nonstandard or nontransitive model need not mean external well-foundedness. Concrete full-class models such as (S08.12) avoid some of these ambiguities; unrestricted absoluteness claims do not follow from our proofs.

23.5 What prevents the paradoxes

The safeguards are structural. A surreal is a set sign sequence, not a sequence of length Ord. Its birthday is a set ordinal. A support code is a set, while its identity extension is a class function. A uniform family is one class relation with set-valued indices, not a set containing proper classes. The full collection of class enumerations is virtual or lives at a higher type. Finally, “every set cut is realized” never means that a separator is required between two arbitrary proper classes.

For instance, a surreal above every ordinal would contradict proposition 5.4; the left options in that purported cut are a proper class, so theorem 5.6 does not apply. Likewise a list of every set-supported code could be a class enumeration, but an exhaustive class-uniform list of all global enumerations is ruled out by note 19.41. Neither obstruction is bypassed by changing notation.

[merge] The three remaining sections are the other sources’ formalization proposals. The modules they name are proposals and do not exist; every existing declaration they cite exists at the current revision, and none of them states a result of this report (section 1.5).

24 A formalization architecture with explicit size boundaries (source 12)

24.1 Start with a universe-polymorphic set theorem

The most efficient verified starting point is independent of surreal arithmetic. For a type X with a linear order, represent a well-order on X by its binary relation and a proof of well-foundedness and linearity. Define the common prefix by equality of predecessor sets and equality of the restricted relations, exactly as in [section 14.1](#). Prove maximality, distinct next labels, and transitivity of comparison. This gives a generic theorem reusable for reals, surreal cutoffs, and other ordered alphabets.

For a fixed ordinal domain, represent an enumeration as an equivalence between an ordinal's carrier and X . For all well-orders of a small X , using relations on X often avoids unnecessarily quantifying over every ordinal in a larger universe. The ordinal enumeration can be recovered later from the well-order type. This is a representation choice, not an identification of all ordinals with a small type.

24.2 Use the existing sign-sequence foundations selectively

The inspected repository file

```
Algebra/SurrealNumbers/Surreal/Foundations/SignSequenceSimplicity.lean
```

contains, among others,

```
isPrefix_of_minimum_birthday
existsUnique_prefix_of_ordConnected
existsUnique_minimum_birthday_of_ordConnected
existsUnique_simplest_separator
```

[10]. Its comments explicitly distinguish uniqueness of a simplest separator from existence of separators for arbitrary small cuts. The last theorem assumes a separator exists. It must not be imported as though it independently established all cut-filling instances.

A suitable interface for the order-theoretic development needs a birthday map, a numerical linear order, sign concatenation and truncation, the first-difference comparison, and small-cut existence with a birthday bound. Field operations can be omitted except in illustrative examples.

24.3 Suggested proof modules

Module	Principal obligations
CommonPrefix	The elementary definition; initiality; maximality; next-label inequality; generic full lexical linearity.
ExhaustiveLex	Prefix-freeness; equivalence with ordinal first-difference comparison; transport between relations and enumerations.
BinaryCubes	Cylinder fusion; strict hierarchy; characteristic-cut embeddings; precise ordinal concatenation.
PermutationCoding	Pair orientation, product scheduler, transpositions, and exhaustive output.

Module	Principal obligations
BirthdayCutoffs	Size of $\mathbf{No}_{<\kappa}$; prefix-free numerical code; cone partition; regular and singular sums.
ClassFamilies	A formal set/class model, sections of class relations, baseline enumeration, and diagonal non-enumerability.
UniverseModels	A chosen inaccessible-rank interpretation; internal/external well-foundedness and identification of the two orders.

The finite common-prefix theorem should be proved directly for relations, not by first developing all of ordinal arithmetic. The scheduler proof should record both surjectivity and the first-row-appearance invariant; a bare cardinal pairing equivalence is insufficient. The singular theorem must use ordinal sums, never a cardinal cast that silently loses order.

24.4 Universe accounting

Schematically, a sign-sequence type built from small ordinal lengths has one universe level for its ordinal index and potentially a higher level for the total carrier. One should retain explicit levels rather than promise a concrete declaration before checking the chosen ordinal API. A bounded carrier with a specified small birthday bound is the cleanest first implementation.

Three operations must remain visibly different: taking a subtype of one fixed surreal carrier; lifting an existing carrier to a larger universe; and constructing all surreals of the larger universe. The third adds new birthdays and is not merely a type lift. Likewise, a type of relations on a fixed carrier can exist at a higher level without becoming a same-level class of all class relations.

For global theorems, two safe routes are available. One may formalize GB or GBC models and express theorems about their class predicates. Or one may formalize the bounded analogue inside a hierarchy of explicit universes and separately prove its interpretation in a set/class model. Neither route requires an inconsistent type of all types.

24.5 Axioms, kernels, and honest validation

A full formalization should record the uses of classical logic, set choice, global choice, and any ambient model-existence assumptions. The common-prefix comparison should not acquire an ETR assumption merely because abstract class order comparison sometimes uses one. The inaccessible-rank example is conditional on that ambient cardinal, whereas the bounded singular theorem needs no inaccessible cardinal.

The present package includes exact executable tests of finite shadows: sign coding, pair coding, the scheduler, the common-prefix formula, and finite adjacency. These are useful implementation checks and counterexample detectors. They are not kernel-checked proofs of any infinite or class theorem, and no supplied file is advertised as such.

[merge] Source 09's frameworks and Lean dependency plan.

25 Formalization strategies and paradox avoidance (source 09)

25.1 Four suitable frameworks

Birthday-bounded ZFC. For the most immediately formalizable results, fix $\mathbf{No}_{<\theta}$ and a length bound. Everything is then a set: relations, permutations, words, embeddings, and counterexamples.

The exact ordinal spectrum and finite residual adjacency results belong here. No inaccessible cardinal is needed.

Two-sorted GBC. Use set variables for surreal numbers, support codes, ordinals, and bounded prefixes; use class variables for global enumerations. Treat $E <_{\text{lex}} F$ as a formula on class variables. State family theorems with an explicit uniform ternary relation. The core is a genuine class linear order. This framework proves all the global results above without a class-of-classes object.

A larger universe or explicit higher sorts. To make the entire collection of smaller-universe global enumerations an object, put it in a larger universe, or use separate sorts for sets, classes, and collections of classes. Membership and comprehension must be stratified. The inaccessible model in [section 22](#) demonstrates one controlled interpretation. A higher-sort theory requires its own axioms; merely writing a new symbol for all classes does not specify one.

Universe-polymorphic dependent type theory. A type-theoretic formalization can construct a carrier relative to a universe, then form its function types at an appropriate larger level. It must not infer that those functions are indexed by the original small ordinals or that every large subtype is small. Universe polymorphism handles repeated constructions; it does not identify the universe levels.

25.2 Kelley–Morse does not create classes of classes

KM strengthens class comprehension, but its ordinary two-sorted interpretation still has sets as the members of classes. It does not turn proper classes into elements of other classes. Similarly, adding ETR resolves certain class-valued recursion problems but does not invalidate [note 19.40](#). These are separate issues: recursion strength, comprehension strength, and object level.

For non-set-like class well-orders, recursion along predecessors may need class-valued histories. One should therefore not reuse our set-valued recursion justification without checking the new hypotheses. The literature on ETR supplies the appropriate distinctions [\[24, 26\]](#).

25.3 Concrete reuse of the ProveIt sign-sequence foundation

At the pinned snapshot, the foundational module declares

```
structure SignSequence : Type (u + 1) where
  birthday : Ordinal.{u}
  signAt : Ordinal.{u} -> SignType
  signAt_ne_zero_iff : ...
```

The ASCII arrow above is only a typographical transcription. The actual module supplies a linear order via zero-extended signs and contains the following relevant named declarations [\[7\]](#):

Declaration	Role in this project
<code>lt_iff</code>	First-disagreement characterization of sign order.
<code>ordinalOrderEmbedding</code>	Increasing embedding of small ordinals.
<code>birthdays_bounded</code>	Birthday bounds for explicitly small families.
<code>small_bounded</code>	Smallness of bounded sign fragments.
<code>not_small</code>	The entire carrier does not fit in the lower universe.

The companion cut-operation file contains `cut_realizes` and `birthday_cut_le_iSup`, together with the separator theorem `exists_separator_of_small_sets` [9]. Its documentation explicitly distinguishes this order/cut development from supplying field operations. The present proofs need only the order/cut development.

25.4 A proposed Lean dependency plan

A practical implementation should proceed in the following dependency order, keeping the formal claims smaller than the whole manuscript at first.

1. **Fixed-alphabet enumeration order.** Represent an enumeration by its ordinal domain and an equivalence onto a fixed small alphabet. Prove prefix-freeness before defining variable-domain first disagreement. Then prove pair-switch coding, cardinality bounds, and the κ^+ stabilization obstruction.
2. **Small injective prefixes.** Prove the extension lemma for a partial injection into a sufficiently large ordinal. Keep the equal cardinality of both complements as an explicit lemma. This part is independent of surreals.
3. **Abstract small-cut alphabets.** Parameterize the avoidance and forced-prefix interpolation lemmas by a linear order with separators for explicitly small option families. Instantiate with the existing sign carrier only afterwards.
4. **Coded support core.** Use a small support, a permutation on it, and a no-fixed-point condition, or use bounded permutations followed by a proved equivalence relation and canonical normalization. Supply a baseline equivalence as an explicit parameter.
5. **Large-family interface.** Represent rows by a single function or relation, retain its universe levels, and prove bounded reflection for small index types. State separately which large index orders possess an ordinal enumeration with small initial segments.
6. **Back-and-forth and the diagonal.** Formalize the set-valued recursion invariant. Only after its smallness obligations are discharged should the abstract core isomorphism and no-coding statements be instantiated.

In a universe-indexed version, a proposed baseline has the shape $\text{Ordinal}.\{u\} \simeq \text{SignSequence}.\{u\}$. Its existence is not inferred here merely because both types live in $\text{Type}(u + 1)$; equal universe level does not imply equipotence. Likewise, the type of *all* functions between these carriers can be formed in the ambient type theory without becoming a small-universe class of set codes. This is exactly the distinction exhibited by [theorem 22.1](#).

25.5 A checklist of invalid shortcuts

The following shortcuts would change the mathematical problem or introduce an unjustified assumption:

- treating **No** as a set in its original universe, or treating all global enumerations as members of an ordinary class;
- using an arbitrary class option family in the surreal cut constructor, or replacing set-indexed interpolation by class-indexed interpolation;
- assuming every class well-order has type `Ord`, or confusing the well-order represented by an enumeration with the lexicographic order of enumerations;

- completing a partial injection without checking complement cardinalities, or using a set of individually given class witnesses without a uniform family relation;
- transferring a proof of equimorphism to isomorphism, or assuming an order embedding preserves ambient suprema;
- attributing the new global core and diagonal theorems to existing Lean declarations merely because the requisite sign-cut primitives are present.

[merge] Source 08’s Lean plan. The declarations of its table exist; source 08 correctly identifies `exists_cut_separator_of_birthday_lt` as the formal landing point of [theorem 5.6](#) and [corollary 5.7](#) ([section 1.5](#)).

26 A Lean formalization plan tied to ProveIt (source 08)

The repository was inspected at commit

ae7c1e6aae3dc58c9c638b35646865ad6c10387d.

The relevant project is `Algebra/SurrealNumbers`. Its README explicitly distinguishes research drafts from machine-checked results and describes an axiom audit [6]. This manuscript uses that distinction rather than treating all repository prose as a formal theorem.

26.1 Inspected declarations

The source `Surreal/Foundations/SignSequence.lean` defines a universe-indexed carrier of the form

```
structure SignSequence : Type (u + 1) where
  birthday : Ordinal.{u}
  signAt : Ordinal.{u} -> SignType
  signAt_ne_zero_iff :
    forall i, Iff (Not (signAt i = 0)) (i < birthday)
```

This display is an ASCII rendering of the inspected declaration, not a new compiled file. Its zero extension uses the ordered values minus, termination, and plus. The source gives the numerical linear order via Mathlib’s lexicographic function `order` and proves `lt_iff`, `ordinalOrderEmbedding`, `birthdays_bounded`, `small_strict_upper_bounds`, `not_small`, and `small_bounded` [7].

Two declarations in `SignSequenceCut.lean` are particularly relevant: `exists_cut_separator_of_birthday_lt` and `small_cut_fillers`. The construction uses full-length binary words, uses their complete lexicographic order, and produces a separator of exactly a prescribed birthday above all option birthdays [8]. This is the formal landing point for [theorem 5.6](#) and [corollary 5.7](#); it is stronger than merely having a dense linear order.

Inspected ingredient	Role in a formalization
<code>SignSequence.lt_iff</code>	Express every comparison by its least differing coordinate
<code>birthdays_bounded</code>	Bound options indexed by a lower-universe small type
<code>not_small</code>	Exclude collapsing the whole carrier into the birthday universe
<code>small_bounded</code>	Represent a fixed birthday fragment by a genuinely small type
<code>exists_cut_separator_of_birthday_lt</code>	Choose a separator outside a small forbidden family by increasing its birthday
<code>small_cut_fillers</code>	Discharge the ordered-cut interface used by the abstract arguments

26.2 Smallness is not the same as having a type

In Lean, `SignSequence.{u}` and `Ordinal.{u}` live in `Type (u + 1)`. They are not small relative to universe `u`. A `Set SignSequence` is a predicate on the whole larger carrier; it need not be a small family of surreals. Thus a formal cut theorem must carry a condition such as `Small.{u} I` on its index type, not merely declare its options to be a `Set`.

The same applies to permutations. A predicate-defined support in the larger universe is not automatically an admissible set support at the smaller level. The orbit should use explicit small support codes or an equivalent `Small.{u}` hypothesis. Otherwise the crucial prefix bound and the intended distinction between skeleton and full enumeration space are lost.

One safe interface keeps an equivalence

```
b : Equiv Ordinal.{u} SignSequence.{u}
```

as an explicit parameter. The present manuscript does not claim that the repository already constructs this equivalence with all the required relative-smallness properties. An implementation must either prove that baseline theorem in its universe framework or retain the baseline as a stated assumption.

26.3 Suggested module boundaries

The following are proposed modules, not existing files or completed formalizations.

1. `WellOrders/Exhaustive.lean`: exhaustive enumerations of a fixed carrier, no-prefix ambiguity, first-disagreement order, and relations versus enumerations.
2. `WellOrders/PairCoding.lean`: binary pair blocks, small-order universality, cardinality, prefix stabilization, and the exact ordinal spectrum.
3. `WellOrders/FiniteTails.lean`: cylinder extrema, adjacency, and finite-condensation blocks.
4. `WellOrders/PrefixExtension.lean`: small partial injections, choice of a larger support bound, complement bijections, and canonical moved-point codes.
5. `WellOrders/SetCuts.lean`: the active-row invariant, the bound on cross-disagreement coordinates, stopping, and strict separation before completion.
6. `WellOrders/BoundedPermutations.lean`: the code carrier, its small-cut interface, and the relative-universe form of the surreal-skeleton theorem.
7. `WellOrders/ClassBinaryCoding.lean`: terminal block codes, ordinal concatenation, and uniform diagonalization in the chosen higher-type representation.

The first three modules can be developed without surreal arithmetic or class metatheory. The next three isolate the delicate smallness and extension arguments. The final module should be delayed until the meaning of a uniform operator on larger function types is fixed.

26.4 Two complementary implementation routes

A *cutoff-first* implementation proves theorems for $\mathbf{No}_{<\kappa}$ and $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$ with explicit regularity and cardinal hypotheses. All carriers are ordinary types and the formal statements closely match [note 7.6](#). This is the simplest route to substantial checked mathematics.

A *universe-relative* implementation abstracts over a large ordered carrier, a lower-universe small-cut operation, and a baseline equivalence. It formalizes the local orbit argument relative to that notion of smallness. Care is needed when proving back-and-forth: an arbitrary well-order of a large type may have initial segments too large for the small-cut operation. One must use an

enumeration whose proper initial segments are small, rather than invoke an unrestricted large-type choice order.

A third option is a deep formalization of a two-sorted class theory or of a specified model $(M, \mathcal{C})$. This is the best route for comparing axiomatic strengths and internal/external semantics, but is not necessary to certify the set-sized combinatorics.

26.5 Trust boundary

The cited repository files were read, not rebuilt in this work. No new Lean module accompanies this manuscript, and no new kernel-checking claim is made. A future implementation should pin the repository and Mathlib revisions, compile in dependency order, reject unfinished proofs, and inspect the resulting axiom dependencies. Finite tests of codes, even exhaustive ones, cannot certify the class interpolation or back-and-forth arguments.

[merge] The four plans agree on a first milestone that is independent of surreal arithmetic: the predecessor or common-prefix comparison of well-orders on a small carrier, prefix-freeness and the successor-cardinal obstruction, then bounded-support prefix completion and small-family interpolation, reusing the existing small-cut interface. This is a proposal recorded for the collection's formalization work; no such module exists.

Part VI

Set-sized strata: topology, cut spectra and block types

[merge] (3 October 2026.) This Part opens the batch-81 addition of six further manuscripts, sources 13–18 (section 1.8); its notation is fixed in section 1.3.1. It prints everything the six sources prove about set-sized enumeration orders: the density and cellularity of every stratum $\mathcal{W}_\alpha(X)$ of every infinite linear alphabet (section 27), conditional incompleteness and global characters (section 28), the cut spectrum of the alphabet $\mathbf{No}_{<\kappa}$ (section 29), the characters, fusion theorem, exact gap spectrum and gap multiplicities of the minimal slice $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ (section 30), the coding lengths, cube spectra and density of the longer strata $\mathcal{W}_{\mu \cdot \gamma + n}(\mathbf{No}_{<\kappa})$ (section 31), and the six sources' set-sized baselines and warnings (section 32). Section 33 records which questions of this report and of the reals report the Part answers.

The bases are chosen by generality, as in Parts I–V: source 16 for density, cellularity and incompleteness (every infinite alphabet, every stratum), source 17 for the alphabet and the minimal slice (every infinite cardinal cutoff, with no hypothesis on $2^{<\kappa}$), source 15 for the longer strata (the only source treating them). Source 18, written before sources 13–17 had arrived and without knowledge of them, proves most of the minimal-slice results a third time; its density, characters and interpolation proofs differ in detail and are printed as routes, its gap-spectrum proof repeats source 17's and is recorded in notes, and its general permutation principle and exact gap multiplicities are new and printed in full. The whole Part is set theory in ZFC; no class is involved.

Throughout the Part, κ is an infinite cardinal used as a birthday cutoff, $\mu = 2^{<\kappa} = |\mathbf{No}_{<\kappa}|$, $r = \text{cf } \kappa$ and $q = \text{cf } \mu$; a general alphabet X is an infinite linearly ordered set of cardinality μ . Source 16 wrote κ for $|X|$ and θ for the cutoff, and source 18 wrote ν, τ for r, q ; their text is printed in the convention above (section 1.3.1).

27 Density and cellularity at every fixed type (source 16)

[merge] Source 16's Section 5, printed whole. Its κ (the cardinality of X) is printed μ , its cutoff θ is printed κ , and its d, c are printed d, c , cell (section 1.3.1); its equations keep their numbers as (S16.n). Source 18 proves the same theorem independently, with a different lemma and a different lower-bound family (section 27.1); the reals report proves the case $X = \mathbb{R}$ (note 27.5).

For a set linear order L with its *order topology*, let $d(L)$ be the least cardinality of a dense subset. Let $\text{cell}(L)$ be the supremum of the cardinalities of families of pairwise disjoint nonempty open sets. Always $\text{cell}(L) \leq d(L) \leq |L|$. A prefix cylinder is the collection of enumerations extending one fixed injective prefix. Cylinders are convex, but need not themselves be open. Our lower bounds use actual open intervals lying inside cylinders; our upper bounds use a family that meets every nonempty open set.

Lemma 27.1 (No jump between distinct short-prefix branches). *If $e < f$ in $\mathcal{W}_\alpha(X)$ first differ at a coordinate $\delta < \mu$, then they are not adjacent.*

Proof. Write $p = e \restriction \delta = f \restriction \delta$ and $a = e(\delta) < b = f(\delta)$. If there is no point between e and f , then e must be the maximum of its $p \frown a$ cylinder, and f the minimum of its $p \frown b$ cylinder. Any ascent in the remaining suffix of e can be swapped to produce a larger member of its cylinder. Thus that suffix is numerically strictly decreasing. Similarly, the suffix of f must be strictly increasing. Their common remaining label set is $X \setminus (\text{ran}(p) \cup \{a, b\})$, of cardinality μ . Its numerical order would be both a well-order and a reverse well-order. An infinite linear order cannot have both properties. This contradiction proves the claim. $\square$

Theorem 27.2 (Exact density and cellularity). *Let $|X| = \mu$ be infinite and $|\alpha| = \mu$. Then*

$$d(\mathcal{W}_\alpha(X)) = \text{cell}(\mathcal{W}_\alpha(X)) = \begin{cases} \mu^{<\mu}, & \alpha = \mu, \mu + 1, \text{ or } \mu + 2, \\ 2^\mu, & \text{all other } \alpha. \end{cases} \quad (\text{S16.3})$$

Moreover $d(\mathcal{W}(X)) = \text{cell}(\mathcal{W}(X)) = 2^\mu$. No regularity assumption is needed.

Proof. For any $\delta < \mu$ there are $\mu^{|\delta|}$ injective prefixes of length δ , and each extends to an admissible enumeration. One way to see the lower count is to choose, for each coordinate, its label in a separate μ -sized cell of a partition of X ; the choices are automatically injective. Every such cylinder contains at least three points and hence a nonempty ambient open interval. Distinct cylinders at one level are disjoint and convex. Consequently

$$\text{cell}(\mathcal{W}_\alpha(X)) \geq \sup_{\delta < \mu} \mu^{|\delta|} = \mu^{<\mu}.$$

Suppose $\alpha = \mu + n$, $n \leq 2$. Group enumerations by their common prefix of length μ . A nonempty group has exactly $n!$ points, coming from the permutations of the omitted n labels. Thus every group has at most two points. By Lemma 27.1, distinct groups have dense quotient order: if two groups were consecutive, the last point of the first and first point of the second would be adjacent and differ before μ .

It follows that every nonempty open set contains a cylinder specified before μ , except possibly a singleton consisting of an isolated global endpoint. Indeed, select a group strictly between the relevant boundary groups, choose a member of it, and bound its first disagreement with the boundaries by a coordinate below μ . The resulting prefix cylinder lies inside the open set. Select

one completion for each injective prefix of length less than μ and add any isolated global endpoints. This gives a dense set of size at most

$$\sum_{\delta < \mu} \mu^{|\delta|} = \mu^{<\mu}.$$

Together with the lower bound this proves the first line of (S16.3), including singular μ .

If $\alpha = \mu + n$ for finite $n \geq 3$, fix $F \subseteq X$ of size n . There are 2^μ bijections $p : \mu \rightarrow X \setminus F$. The completion cylinder for each is a finite convex chain of $n! \geq 6$ points. An interior point has both immediate neighbors in that chain; the interval between those neighbors is its singleton, an open set in the entire order. Different prefixes give 2^μ disjoint open singletons.

In every remaining case $\alpha \geq \mu + \omega$. Let τ be the order type of the interval $[\mu, \alpha)$, so $\alpha = \mu + \tau$ and $|\tau|$ is infinite. Choose $Y \subseteq X$ with $|Y| = |\tau|$ and $|X \setminus Y| = \mu$. Each of the 2^μ bijections $p : \mu \rightarrow X \setminus Y$ gives a convex cylinder of completions with residual order $\mathcal{W}_\tau(Y)$. Every such cylinder has at least three points, so supplies a nonempty ambient open interval. These intervals are pairwise disjoint. The cardinal upper bound $|\mathcal{W}_\alpha(X)| \leq 2^\mu$ finishes the proof for all strata.

For $\mathcal{W}(X)$ use the same construction with a fixed residual set of three labels. The resulting six-point cylinders are convex even when all enumeration lengths are admitted. Thus $\text{cell}(\mathcal{W}(X)) \geq 2^\mu$, and the cardinality bound gives equality throughout. $\square$

Remark 27.3 (Why two terminal labels differ from three). There can be 2^μ adjacent pairs in $\mathcal{W}_{\mu+2}(X)$ while its density is only $\mu^{<\mu}$. An adjacent pair does not contain an open interval. Three or more terminal labels give a permutation block with an interior point, and that point is isolated. This is why the topological transition occurs at $\mu + 3$, rather than $\mu + 1$ or $\mu + 2$. If $\mu^{<\mu} = 2^\mu$, the two displayed values coincide; the structural distinction still holds.

Corollary 27.4 (Birthday-stage topology). *For every infinite cardinal κ , put $\mu = 2^{<\kappa}$. Formula (S16.3) applies to $\mathbf{No}_{<\kappa}$, whose cardinality is μ . If κ is regular and $2^{<\kappa} = \kappa$, the first three strata have density and cellularity κ , whereas all later strata and the full order have density and cellularity 2^κ .*

Proof. Only the last simplification needs comment. Regularity and $2^{<\kappa} = \kappa$ imply $\kappa^{<\kappa} = \kappa$: for $\lambda < \kappa$, every function $\lambda \rightarrow \kappa$ has bounded range, and for $\rho < \kappa$ one has $\rho^\lambda \leq 2^{\rho \cdot \lambda} \leq \kappa$. Taking the union of κ bounds proves the assertion. $\square$

Note 27.5 ([merge]; the reals report's case). For $X = \mathbb{R}$, the reals report proves $d(\mathcal{W}_\alpha) = \text{cell}(\mathcal{W}_\alpha) = 2^{<\mathfrak{c}}$ for $\alpha \in \{\mathfrak{c}, \mathfrak{c} + 1, \mathfrak{c} + 2\}$ and $2^\mathfrak{c}$ for $\mathfrak{c} + 3 \leq \alpha < \mathfrak{c}^+$ (lwo:prop:all-strata-density), with the adjacent pairs at $\mathfrak{c} + 2$ and the isolated points from $\mathfrak{c} + 3$ on (lwo:prop:finite-topology) and the minimal slice in lwo:prop:fixed-topology. Since $\mathfrak{c}^\lambda = 2^\lambda$ for infinite $\lambda < \mathfrak{c}$, one has $\mathfrak{c}^{<\mathfrak{c}} = 2^{<\mathfrak{c}}$, and theorem 27.2 with $X = \mathbb{R}$ is exactly that result, by the same two mechanisms (short-prefix cylinders, and the finite terminal permutation block). Source 16 does not cite these statements; it cites the reals report only for the set baseline (note 32.1). New in source 16 is the generality: every infinite linear alphabet, of any cardinality including singular ones and $\aleph_0$, with no density or saturation assumption on X .

Note 27.6 ([merge]; the birthday-cutoff instances, and why three formulas agree). For $X = \mathbf{No}_{<\kappa}$, corollary 27.4 gives $d = \text{cell} = \mu^{<\mu}$ on $\mathcal{W}_\mu, \mathcal{W}_{\mu+1}, \mathcal{W}_{\mu+2}$ and 2^μ on every longer stratum. Three other sources compute instances: source 17 for the minimal slice ($d = w = \text{cell} = \mu^{<\mu}$, theorem 30.2), source 18 for every stratum of every alphabet (theorems 27.8 and 27.9, printed next), and source 15 for the block types $\mathcal{W}_{\mu \cdot \gamma + n}$, with the value $2^{<\mu}$ when $\rho_\kappa = \mu$, $\gamma = 1$ and $n \leq 2$, and 2^μ otherwise (corollary 31.7). The formulas agree. If $\mu > \kappa$ then $\mu \leq 2^\kappa$, so for $\lambda < \mu$ one has $\mu^\lambda \leq 2^{\kappa \cdot \lambda} = 2^{\max(\kappa, \lambda)} \leq 2^{<\mu}$, whence $\mu^{<\mu} = 2^{<\mu}$. If $\mu = \kappa$ is regular then $2^{<\kappa} = \kappa$ and $\kappa^{<\kappa} = \kappa$ (proof of

corollary 27.4). If $\mu = \kappa$ is singular, κ is a strong limit, $\rho_\kappa = \kappa \cdot \kappa$ falls under source 15's second case, and $\kappa^{<\kappa} = 2^\kappa$ (corollary 30.26). In every case the four sources give the same cardinal.

27.1 Source 18's route: exact cardinal invariants of bounded enumeration orders

[merge] Source 18's Section 4 up to its residual-alphabet subsection, printed whole as a second route to theorem 27.2. Source 18 presents these theorems as advances; within batch 81 source 16 proves them too, and source 18 could not know it (its pin precedes the arrival of sources 13–17). Source 18's status table says that its proof review “corrected the arbitrary-alphabet density construction to include possible isolated global endpoints”; source 16's proof treats the same endpoints (“except possibly a singleton consisting of an isolated global endpoint”). Its remaining subsections are printed in section 30.8, its residual-alphabet lemmas in notes 29.8 and 29.9.

Throughout this section, X is an infinite linearly ordered set of cardinality μ , identified with its initial ordinal when it is used as an index. If $|\alpha| = \mu$, write

$$\mathcal{W}_\alpha(X) = \{e : \alpha \longrightarrow X : e \text{ is bijective}\},$$

ordered by the first unequal coordinate. A proper-prefix comparison never occurs between two exhaustive enumerations of the same carrier. We use $d(L)$ for the least cardinality of a topologically dense subset of the order topology, and $\text{cell}(L)$ for the supremum of the cardinalities of pairwise disjoint families of nonempty open sets. The lower-bound families constructed below attain the asserted cellularities, so no convention concerning whether the supremum is attained affects the answers. All assertions in this section are theorems of ZFC.

27.1.1 Prefix cylinders and the minimal enumeration type

For an injective prefix $p : \delta \rightarrow X$, $\delta \leq \alpha$, its cylinder is

$$C_p = \{e \in \mathcal{W}_\alpha(X) : e \restriction \delta = p\}.$$

Every cylinder is convex. If $\delta < \mu$, the cylinder is nonempty: both the unused alphabet and the remaining index set have cardinality μ , so a bijection between them completes the prefix. Such a cylinder has at least three elements, obtained, for example, by permuting three unused labels in a fixed completion. It therefore contains a nonempty open interval of the ambient enumeration order. The entire cylinder need not be open; the proof uses only the open interval it contains.

Lemma 27.7 (The dense order of cofinite prefixes). *For $n < \omega$, let*

$$Q_n(X) = \{p : \mu \rightarrow X : p \text{ is injective and } |X \setminus \text{ran } p| = n\}.$$

With first-difference comparison, $Q_n(X)$ is a dense linear order. Consequently $\mathcal{W}_\mu(X) = Q_0(X)$ is dense.

Proof. Take $p < p'$ and let $\delta < \mu$ be their first disagreement. If there are $\delta < \beta < \gamma < \mu$ with $p(\beta) < p(\gamma)$, interchanging these two entries gives $p < s < p'$, with $s \in Q_n(X)$. Similarly, a descent $p'(\beta) > p'(\gamma)$ after δ gives $p < s < p'$ by an interchange in p' . If neither operation is available, the tail of p after δ is strictly decreasing and the tail of p' after δ is strictly increasing. Their ranges have an infinite common subset: they omit from X only their common prefix, of size less than μ , two first-disagreement labels, and two finite sets of omitted labels. On that common subset the numerical order would be both a well-order and a reverse well-order. This is impossible for an infinite set. Indeed, an infinite well-order contains an initial suborder of type ω , which has no largest element. Thus one of the interchanges is available. $\square$

Theorem 27.8 (Exact density and cellularity at the minimal type). *For every infinite linearly ordered alphabet X of cardinality μ ,*

$$d(\mathcal{W}_\mu(X)) = \text{cell}(\mathcal{W}_\mu(X)) = \mu^{<\mu}.$$

No regularity assumption on μ , and no density assumption on X , is needed.

Proof. There are $\mu^{<\mu}$ injective prefixes of length less than μ . For the upper bound, choose one completion for each such prefix. Given $e < f$, density supplies g strictly between them. Choose $\delta < \mu$ larger than both first-disagreement coordinates of (e, g) and (g, f) . Then $C_{g|\delta} \subseteq (e, f)$, so the selected completions are dense. Since a dense linear order with at least two points has no isolated endpoint, meeting all nonempty open intervals also gives topological density. This proves $d \leq \mu^{<\mu}$, and a dense set meets every member of a disjoint nonempty open family, giving $c \leq d$.

For the lower bound, choose distinct labels $a_\delta \in X$ for $\delta < \mu$. For each δ , take every injective prefix of length $1 + \delta$ whose first label is a_δ . There are $\mu^{|\delta|}$ such prefixes. To justify this count for infinite δ , functions $\delta \rightarrow \mu$ can be encoded as injective maps $\delta \rightarrow \delta \times \mu \cong \mu$ by recording the coordinate alongside its value; the reverse inequality is immediate. The finite cases are equally direct. Prefixes in different chosen levels have different first labels, while two distinct prefixes within a level have equal lengths and disagree. Their cylinders are therefore pairwise disjoint. Choose one nonempty open interval in each cylinder. The resulting disjoint family has cardinality

$$\sum_{\delta < \mu} \mu^{|\delta|} = \mu^{<\mu},$$

because $1 + \delta < \mu$ for every $\delta < \mu$. This proves the lower bound and also proves that the stated cellularity is attained. $\square$

27.1.2 A sharp transition at the third terminal coordinate

Theorem 27.9 (Density and cellularity for every fixed type). *If $|X| = \mu$ is infinite and $\mu \leq \alpha < \mu^+$, then*

$$d(\mathcal{W}_\alpha(X)) = \text{cell}(\mathcal{W}_\alpha(X)) = \begin{cases} \mu^{<\mu}, & \alpha \in \{\mu, \mu + 1, \mu + 2\}, \\ 2^\mu, & \mu + 3 \leq \alpha < \mu^+. \end{cases}$$

These formulas are valid for arbitrary numerical orders on X .

Proof. First take $\alpha = \mu + n$ with $n \leq 2$. Restriction to the first μ coordinates exhibits $\mathcal{W}_{\mu+n}(X)$ as a lexicographic sum over $Q_n(X)$ of finite fibers of size $n!$; the omitted n labels are the only possible final labels. Each fiber has at most two points. By [Lemma 27.7](#), the quotient $Q_n(X)$ is dense. Thus every nonempty open interval in $\mathcal{W}_{\mu+n}(X)$ contains a point whose μ -prefix lies strictly between the prefixes of its two bounds: if the bounds had the same prefix, their fiber of at most two points would have empty open interval. A prefix of this intermediate point of length less than μ , taken beyond both relevant first disagreements, has its entire cylinder inside the interval. Choose one completion of every prefix of length less than μ , and also include the global least and greatest points whenever they exist. The resulting family is topologically dense: it meets every nonempty open interval, while the added endpoints handle a possible isolated endpoint in a two-point terminal fiber. Adding at most two points does not change its cardinality bound $\mu^{<\mu}$. The antichain of cylinders from the preceding proof gives the matching lower bound. This argument explicitly deals with the possible jumps when $n = 2$; it does not assume that every cylinder is open, or that every interval contains a point separated from both bounds merely by choosing an arbitrary intermediate point.

Now suppose $\alpha \geq \mu + 3$, and write $\alpha = \mu + \beta$, where β is the order type of its tail after the first μ positions. Choose $T \subseteq X$ with $|T| = |\beta|$ and $|X \setminus T| = \mu$. If $|\beta| = \mu$, use a partition of X into two μ -element sets; otherwise remove a set of size $|\beta|$. There are 2^μ bijections $p : \mu \rightarrow X \setminus T$. This standard count follows either from the cardinality theorem for well-order enumerations or directly by orienting μ disjoint pairs of labels. For each such p , the cylinder C_p consists of all continuations that biject β onto T . Since $\beta \geq 3$, it contains at least three points and hence a nonempty open interval. The cylinders are pairwise disjoint because the prefixes are distinct and have the same length. We obtain 2^μ disjoint nonempty open intervals. On the other hand, $|\mathcal{W}_\alpha(X)| \leq \mu^\mu = 2^\mu$, so no density or cellularity can exceed 2^μ . $\square$

Remark 27.10. For $\mu^{<\mu} < 2^\mu$, the values for $\mu + 2$ and $\mu + 3$ differ. Thus merely adding one terminal coordinate can change both density and cellularity. For singular strong-limit μ , both expressions are already 2^μ , so the geometric distinction remains but the cardinal invariants coincide. The formula is a new consequence beyond the baseline's finite-residual adjacency classification; it is not a new claim about the already-established ordinal embedding spectrum.

28 Countable incompleteness and global characters (source 16)

[merge] Source 16's Section 6 up to its subsection on regular saturated stages, which is printed in [section 30.4](#).

Conditional Dedekind completeness means that every nonempty bounded-above set has a supremum, equivalently every nonempty bounded-below set has an infimum. It should not be confused with filling all small strict cuts, with order density, or with the completeness of a separately adjoined Dedekind completion.

Theorem 28.1 (Every infinite stratum is incomplete). *For every infinite linearly ordered set X and every ordinal α with $|\alpha| = |X|$, $\mathcal{W}_\alpha(X)$ is not conditionally Dedekind complete. A countable bounded monotone family witnesses this failure. The same conclusion holds for $\mathcal{W}(X)$.*

Proof. Every infinite linear order contains an increasing or decreasing ω -sequence. For completeness, choose a countable set of distinct points and color each pair of its indices according to the numerical comparison; the infinite Ramsey theorem gives such a sequence. First assume $a_0 < a_1 < a_2 < \dots$ in X , and put $p(n) = a_{2n}$.

For each $n \geq 0$ choose U_n extending the finite prefix

$$(a_0, a_2, \dots, a_{2n-2}, a_{2n+1}),$$

and for $n \geq 1$ choose L_n extending

$$(a_0, a_2, \dots, a_{2n-2}, a_{2n-1}).$$

All these prefixes extend to the prescribed type α . First disagreement gives $L_n < L_{n+1}$, $U_{n+1} < U_n$, and $L_n < U_m$ for all n, m . Hence (U_n) is bounded below.

Let g be any lower bound. If it first differs from the infinite prefix p at a finite coordinate j , it must differ downward: an upward difference would give $g > U_{j+1}$. Consequently $g < L_n$ for every $n > j$, so it is not the greatest lower bound.

Otherwise g agrees with all of p . Its remaining suffix contains all odd labels a_{2n+1} . That suffix cannot be numerically strictly decreasing: the positions of these increasing labels would form an infinite descending sequence of ordinals. It therefore has an ascent. Swap the two labels of such an

ascent. The new enumeration has the same type, is strictly above g , and still agrees with p at every finite coordinate. It remains below every U_n . Again g is not greatest. If no completion of p with the specified type exists, this second case simply never occurs.

Thus (U_n) has no infimum. Reversing the numerical alphabet treats a decreasing ω -sequence and yields the dual failure of a supremum. The argument for the full order is identical: swapping an ascent preserves whatever ordinal type the candidate bound happens to have. $\square$

Proposition 28.2 (Global characters from the first label). *If X has no greatest element, then*

$$\text{cf}(\mathcal{W}_\alpha(X)) = \text{cf}(\mathcal{W}(X)) = \text{cf}(X)$$

for every admissible infinite α . There is a dual statement for coinitality when X has no least element. In particular, for every infinite cardinal cutoff κ , all the fixed strata and the full order on $\mathbf{No}_{<\kappa}$ have global cofinality and coinitality $\text{cf}(\kappa)$.

Proof. Complete a cofinal family of possible first labels to enumerations. It is cofinal in the lexicographic order. Conversely, the first labels of a cofinal family of enumerations must be cofinal in X , or a larger first label would put an enumeration above the entire family. For $\mathbf{No}_{<\kappa}$, the ordinals below κ are numerically cofinal and the negatives coinital. Any family with bounded birthdays is bounded numerically, so the least possible size is $\text{cf}(\kappa)$. $\square$

Note 28.3 ([merge]; global characters in sources 15, 17 and 18). For the minimal slice of $\mathbf{No}_{<\kappa}$, the global part of [proposition 28.2](#) ($\text{cf} = \text{ci} = r$) is also proved by source 17 ([theorem 30.2](#), with the same value for every cylinder in [lemma 30.1](#)), by source 15 ([note 30.7](#)) and by source 18 ([theorem 30.24](#), also for $\mathcal{W}_{\mu+n}$, and [note 30.13](#) for cylinders), each by the same first-label projection. Source 16's proposition is the general form: it holds for every stratum and for the full order $\mathcal{W}(X)$ of any alphabet without a greatest element. Source 16's subsection "Regular saturated stages", the last part of its Section 6, is printed in [section 30.4](#) with the other gap spectra.

29 The cut spectrum of the alphabet $\mathbf{No}_{<\kappa}$ (source 17)

[merge] Source 17's Section 4, printed whole. Its $\Sigma(L)$ is printed $\text{GapChar}(L)$ (the reals report's symbol), and its displayed equations keep their numbers as (S17. n). At regular uncountable κ its main theorem is already in the repository ([note 29.7](#)). Sources 15 and 18 prove the alphabet lemmas again; their statements follow in [notes 29.8](#) to [29.10](#).

For the remainder of the set-sized analysis, fix an infinite cardinal κ , regarded as an initial ordinal, and write

$$X = \mathbf{No}_{<\kappa}, \quad \mu = 2^{<\kappa}, \quad r = \text{cf } \kappa, \quad q = \text{cf } \mu. \quad (\text{S17.3})$$

Here $\mathbf{No}_{<\kappa}$ consists of all sign sequences of ordinal length below κ , ordered by first difference with $- < \text{termination} < +$. We write Reg for the infinite regular cardinals. No arithmetic closure of X is assumed. In particular, no field hypothesis is being inferred from the mere fact that κ is a cardinal.

Definition 29.1 (Characters and genuine gaps). For a point x of a dense order L without endpoints, its left and right characters are $\text{cf}(L_{<x})$ and $\text{ci}(L_{>x})$. A *gap* is a partition $L = A \sqcup B$ into nonempty initial and final parts, with $A < B$, such that A has no greatest element and B no least element. Its character is $(\text{cf } A, \text{ci } B)$; $\text{GapChar}(L)$ denotes the set of these pairs. Thus a split immediately before or after an actual point is not a gap. Cofinalities of orders with a greatest element would be 1, but do not occur in $\text{GapChar}(L)$.

29.1 Size, intervals, and deletion

Lemma 29.2. *The order X has cardinality μ . Every nonempty open interval and every nonempty open ray in X has cardinality μ . Consequently, deletion of fewer than μ labels leaves an order-dense, cofinal and coinital suborder, with μ labels still in each nonempty interval and ray.*

Proof. There are $2^{|\alpha|}$ sign sequences of length α , so $|X| = \sum_{\alpha < \kappa} 2^{|\alpha|} = 2^{<\kappa}$; the supremum on the right is at least κ and absorbs the number of summands. Given $a < b$, choose a sign sequence z strictly between them, of length below κ . A sufficiently long extension of z fixes its comparisons with a, b before any subsequent signs. Hence an entire sign cone lies inside (a, b) . If its fixed prefix has length $\delta < \kappa$, then all extensions of lengths $\delta + \gamma$, $\gamma < \kappa$, remain below κ : an infinite initial ordinal is closed under sums of two smaller ordinals. This cone has μ elements. The same construction applies to rays. Removing fewer than μ elements cannot empty a set of cardinality μ and leaves its cardinality unchanged, even when μ is singular. $\square$

Lemma 29.3. *The cofinality and coinitality of X are r , and every point of X has left and right character r . The same assertions hold in $Y = X \setminus Z$ whenever $|Z| < \mu$.*

Proof. The ordinal surreals below κ are cofinal, and a cofinal sequence of ordinals of length r gives an upper bound r . A family of fewer than r sign sequences has bounded birthdays below κ and is bounded above and below by ordinal surreals, proving the reverse bound.

At x , the sequences

$$x \frown - \frown +^\gamma \quad \text{and} \quad x \frown + \frown -^\gamma \quad (\gamma < \kappa)$$

are respectively cofinal below and coinital above x ; a cofinal choice of γ 's of length r suffices. Conversely, fewer than r points have bounded birthdays. A longer sign extension of x lies between x and all the points on the appropriate side. Hence the character is not smaller than r .

For Y , replace each member of a cofinal or coinital sequence by a remaining label in an appropriate interval. Such labels exist by Lemma 29.2. Any smaller sequence cofinal in Y would also be cofinal in the relevant interval or ray of X , by density. $\square$

29.2 The alphabet's own gaps

The alphabet is not Dedekind complete. Even at a regular uncountable cutoff, its gaps need not have two equal characters. Distinguishing this fact from its uniform *point* characters is essential.

Theorem 29.4 (Cut spectrum of the birthday alphabet).

$$\text{GapChar}(X) = \{(r, r)\} \cup \{(\theta, r), (r, \theta) : \theta \in \text{Reg}, \theta < \kappa\}.$$

If $|Z| < \mu$, the same formula holds for $X \setminus Z$.

Proof. Let (A, B) be a gap of X . Define a sign sequence s of length κ by set recursion: at stage α , the already defined prefix $s \upharpoonright \alpha$ is in X , so put $s(\alpha) = +$ if that prefix belongs to A , and $s(\alpha) = -$ if it belongs to B . Put

$$I_+ = \{\alpha < \kappa : s(\alpha) = +\}, \quad I_- = \{\alpha < \kappa : s(\alpha) = -\}.$$

The nodes $s \upharpoonright \alpha$ for $\alpha \in I_+$ increase and are cofinal in A ; the nodes for $\alpha \in I_-$ decrease and are coinital in B . To verify cofinality, compare any $x \in A$ with the path s . If x is a path prefix it is one of the displayed lower nodes. Otherwise the first disagreement cannot place x to the right of

the chosen path: at a minus choice the corresponding node is in B . It therefore places x below a lower node. The dual argument handles B .

Neither index set has a greatest element, since the cut is a gap. At least one is unbounded in κ and has cofinality r . An unbounded other side also has cofinality r ; a bounded other side has an infinite regular cofinality $\theta < \kappa$. This proves the upper inclusion. Conversely, the path $+^\theta \frown -^\kappa$, with the evident length κ interpretation, has lower and upper index cofinalities (θ, r) . Reversing signs gives (r, θ) . A path with both signs cofinal in κ gives (r, r) . In each case the induced partition has no endpoint, as the associated node sequences have no last member.

Now let $Y = X \setminus Z$ be as stated. Density of Y implies that a gap of Y corresponds either to a gap of X or to a single removed point of X : there cannot be two X -points between the two sides, because a Y -point would lie between them. At an X -gap, density preserves both cofinalities by choosing intermediate Y -points. At a removed point both characters are r by [Lemma 29.3](#). Conversely, every X -gap survives restriction to Y , with the same characters. This proves the deletion assertion. $\square$

Example 29.5 (A warning about equal-character guesses). For $\kappa > \omega$, the path which first has ω plus signs and then only minus signs determines an (ω, r) gap. It lies at a boundary of a limit sign cone. The cone itself contains many surreal numbers; none is its missing boundary point. Thus uniform point character r does not mean that every gap has character (r, r) .

Lemma 29.6. *One has $\mu \geq \kappa$ and $q \geq r$.*

Proof. The first assertion follows from the count in [Lemma 29.2](#). If the increasing cardinal sequence 2^ν , $\nu < \kappa$, never reaches its supremum μ , a purported cofinal family in μ of size below r is bounded by some 2^ν : choose an index bound for each of its members and then use $\text{cf } \kappa = r$ to bound all those indices. Thus $\text{cf } \mu \geq r$. If the supremum is attained, the powers are constantly μ for all sufficiently large cardinals $\nu < \kappa$. König's inequality $\text{cf}(2^\nu) > \nu$ then gives $\text{cf } \mu \geq \kappa$. For completeness, that inequality follows from $\rho^{\text{cf } \rho} > \rho$ for infinite cardinals ρ , applied to $\rho = 2^\nu$: if $\text{cf } \rho \leq \nu$, then $\rho^{\text{cf } \rho} \leq (2^\nu)^\nu = 2^\nu$, a contradiction. $\square$

Note 29.7 ([merge]; the repository's case of [theorem 29.4](#)). At a regular uncountable cutoff κ , [theorem 29.4](#) is the case $M = N$ of the cutoff gap decomposition of the report `surreal-fields-a-cross-universes` (`univ:gs:lem:boundary`), quoted in `birthday-cutoffs-and-hereditary-sets` (`hset:rem:cutoffgaps`): with no fresh signs, the gap pairs of $\mathbf{No}_{<\kappa}$ are (θ, κ) and (κ, θ) for every infinite regular $\theta \leq \kappa$, realized by the same sign paths $+^\theta \frown -^\kappa$. Source 17 does not cite these reports. New in source 17 are the singular cutoffs, where the bounded side ranges over the regular $\theta < \kappa$ while the unbounded side has character $r = \text{cf } \kappa < \kappa$; the countable cutoff $\kappa = \omega$, where only (ω, ω) occurs; and the deletion clause.

Note 29.8 (Source 18, Lemmas 4.5 and 5.3, and source 15, Lemma 9.1). Source 18's Lemma 4.5 ("Large intervals and unchanged cut sides after deletion"): *every nonempty numerical open interval in $\mathbf{No}_{<\kappa}$ has cardinality μ ; if $U \subseteq \mathbf{No}_{<\kappa}$ has cardinality less than μ , then $\mathbf{No}_{<\kappa} \setminus U$ meets every such interval; in particular (1) $\text{cf}(\mathbf{No}_{<\kappa}) = \text{ci}(\mathbf{No}_{<\kappa}) = r$, (2) immediately below or above any point the cofinality or coinitality is r , and (3) every nonprincipal side of a cut retains its cofinality or coinitality after deletion of U , as do the two open sides at a point, even a deleted one* (an endpoint retained in the residual alphabet still gives a principal side of character 1). Its Lemma 5.3 ("Small deletions preserve order density") repeats this for $R = X \setminus U$ and adds that *every proper gap of R has character in $\text{GapChar}(X) \cup \{(r, r)\}$* . Source 15's Lemma 9.1 ("Full-size numerical intervals"): *every nonempty open interval of $\mathbf{No}_{<\kappa}$, and every nonempty open ray, contains a sign-prefix cone order-isomorphic to $\mathbf{No}_{<\kappa}$; in particular each has cardinality μ* . These are [lemmas 29.2](#) and [29.3](#) and the deletion clause of [theorem 29.4](#), by the same sign-cone argument. Source 15 makes the

cone explicit in three cases (neither endpoint a prefix of the other: take $a \frown +$; a a proper prefix of b : take $b \frown -$; b a proper prefix of a : take $a \frown +$), and source 18 obtains the cone from the birthday-controlled separator theorem (section 5.2) and adds an explicit approach to a point from below by $x \frown - \frown +^\gamma$, γ cofinal in κ , and from above by the sign-reversed words.

Note 29.9 (Source 15, Lemma 9.3, and source 18, Lemma 4.6). Both sources state more than lemma 29.6: $q \geq r$; if the supremum $2^{<\kappa}$ is not attained then $q = r$, and if it is attained, $2^\lambda = \mu$ for some cardinal $\lambda < \kappa$, then $q \geq \kappa$. The proofs are source 17's: cofinal exponents give $q \leq r$ in the unattained case, a shorter cofinal family would have bounded exponent witnesses, and in the attained case König's inequality $\text{cf}(2^\lambda) > \lambda$ for all λ between the attaining exponent and κ gives $q \geq \kappa$.

Note 29.10 (Source 18, Lemma 5.2, "Alphabet gap spectrum"). For $X = \mathbf{No}_{<\kappa}$, $\text{GapChar}(X) = \{(r, r)\} \cup \{(\theta, r), (r, \theta) : \theta < \kappa \text{ an infinite regular cardinal}\}$; every point has left and right character r , and the global cofinality and coinitality are also r . This is theorem 29.4 with lemma 29.3, by the same recursion $s(\alpha) = +$ if and only if $s \restriction \alpha \in A$. Source 18 adds the warning that the lemma "corrects a tempting but false simplification: unfilled alphabet cuts need not all have character (r, r) "; at $\kappa = \omega_1$ the cut below all positive infinite elements, represented by $+^\omega -^{\omega_1}$, has character (ω, ω_1) , and "the asymmetric alphabet gaps will, however, already be covered by the longer-prefix construction in P ". This is the point of example 29.5.

30 The minimal slice $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ (source 17; routes of sources 15, 16 and 18)

[merge] Source 17's Sections 5 and 6 are printed in their order, with two insertions: source 18's general permutation principle (section 30.3), stated before source 17's exact gap spectrum because that theorem and source 16's are instances of it, and source 18's multiplicities (section 30.5). The second and third routes to the characters and the saturation threshold (sources 15 and 18) follow source 17's saturation corollary. Source 17 writes $[p]$ for the cylinder of an injective prefix p ; sources 16 and 18 write C_p for the same set. All three call a cylinder meeting both sides of a cut *straddling*.

30.1 Exact local and topological invariants of the minimal slice (source 17)

Put $P = \mathcal{W}_\mu(X)$. For an injective prefix $p : \alpha \rightarrow X$ with $\alpha < \mu$, let

$$[p] = \{f \in P : f \restriction \alpha = p\}, \quad Y_p = X \setminus \text{ran } p.$$

The set Y_p has cardinality μ . Every such prefix has a completion: the remaining ordinal interval $[\alpha, \mu)$ has order type μ , so a bijection from it to Y_p suffices. This elementary extension fact would be false for arbitrary prescribed prefixes of length μ .

Lemma 30.1 (The cylinder base). *Every $[p]$ is nonempty, convex, and clopen in the order topology of P . The cylinders form a base. No cylinder has an endpoint. Moreover,*

$$\text{cf}[p] = \text{ci}[p] = r.$$

Proof. Convexity is the first-difference property. Given $f \in [p]$, change its entry at coordinate α to a fresh smaller or larger label and then complete. Lemma 29.2 provides such labels after deletion of $f[\alpha]$. This produces bounds on either side of f within $[p]$, proving openness and absence of

endpoints. All length- α cylinders partition P , so the complement is also open. For $a < f < b$, a prefix of f extending beyond its first differences with a, b has its entire cylinder inside (a, b) . This proves the base assertion.

The first unused coordinate maps the cylinder onto Y_p in lexicographic blocks. Choosing one completion for cofinally many first unused labels shows $\text{cf}[p] \leq \text{cf } Y_p = r$. Conversely, the first unused entries of a cofinal family in $[p]$ must be cofinal in Y_p , since otherwise a larger entry starts a cylinder above the entire family. Cointinality is dual. $\square$

Theorem 30.2 (Characters and size invariants). *Every point $f \in P$ has character (q, q) , and*

$$\text{cf } P = \text{ci } P = r, \quad \text{d}(P) = \text{w}(P) = \text{cell}(P) = \mu^{<\mu}. \quad (\text{S17.4})$$

Here d is density, w is topological weight, and cell is cellularity as the supremum of cardinalities of pairwise disjoint families of nonempty open sets. In particular, P is zero-dimensional. The formula for cell does not assert that its supremum is always attained by one family.

Proof. For each coordinate $\delta < \mu$, choose $f_\delta^- < f < f_\delta^+$ whose first difference from f is exactly δ . Such choices are possible by the preceding lemma. Along a cofinal set of coordinates of order type q , the lower choices are increasing and cofinal below f , and the upper choices are decreasing and cointial above it. Conversely, for a family of fewer than q elements below f , the first-difference coordinates are bounded below μ . A lower approximation differing later lies above them all. Thus neither point character is below q . The global cofinalities are [Lemma 30.1](#) at the root.

The number of prefixes is at most $\sum_{\alpha < \mu} \mu^{|\alpha|} = \mu^{<\mu}$, giving that upper bound for weight, density, and cellularity. At each fixed length α , the cylinders are disjoint nonempty open sets. For $1 \leq \lambda < \mu$, the number of injective λ -words over X is μ^λ : a fixed injection $j : \lambda \times X \rightarrow X$ sends any function h to the injection $\xi \mapsto j(\xi, h(\xi))$. Hence both density and cellularity are at least μ^λ for every such λ . Taking suprema gives the reverse bound. Finally $d \leq w$ and $c \leq w$ complete the equalities. The clopen base gives zero-dimensionality. $\square$

Remark 30.3. The point character is controlled by $\text{cf } \mu$, whereas the cofinality of a whole cylinder is controlled by $\text{cf } \kappa$. These can differ. Confusing the two would lose the central feature of the cut classification below. Hereditary density also equals $\mu^{<\mu}$, since every subspace has weight at most $\text{w}(P)$ and P itself attains this density.

Note 30.4 (Source 17, Proposition 5.4, “Set-alphabet background, with a direct proof”). *The order P has cardinality 2^μ . Every linear order of cardinality at most μ embeds into every nonempty open interval of P . An ordinal β embeds in P if and only if $\beta < \mu^+$; the same bound holds for reverse well-ordered suborders.* Source 17 calls these “already part of the source’s theory”. They are the set baseline ([theorem 6.1](#); reals report `lwo:prop:cardinality`, `lwo:thm:ordinal-spectrum`, `lwo:sp:thm:universality`) applied inside a cylinder, whose residual alphabet has size μ , together with [lemma 30.1](#); the proof is the pair-orientation, initial-segment and coordinate-stabilization argument of [section 6](#).

30.1.1 Fusion with an exhaustiveness requirement

Theorem 30.5 (A dense-open fusion theorem). *The intersection of fewer than q open sets is open. The intersection of at most q dense open subsets of P is dense. In particular, when $q = \omega$, P is a Baire space.*

Proof. At a point in fewer than q open sets, choose a cylinder neighborhood for each. Their prefix lengths have supremum below μ , so the longest union prefix gives a cylinder inside the intersection.

Let $(U_i)_{i < q}$ be dense open sets and start with any nonempty cylinder. Write $X = \bigcup_{i < q} X_i$, where the X_i are increasing and each has cardinality below μ . Recursively refine the current prefix to a cylinder contained in U_i , using density and the cylinder base, and then append every member of X_i not yet used, in a set well-order of length below μ . At stages below q , the accumulated length is still below μ , including at limit stages. At the end the union has length exactly μ and range X : its length is at most μ , and it has used μ distinct labels. It is therefore an element of P , belongs to every U_i , and extends the starting prefix. Fewer dense open sets can be padded by copies of P . $\square$

The coverage step is essential. Bare prefix fusion of length μ can produce an injective word omitting a label, which is not a point of P . The theorem does not claim a complete metrizable or a general isomorphism classification.

Note 30.6 ([merge]; repository analogues and the regular size-matched case). For regular κ with $2^{<\kappa} = \kappa$ one has $\mu = r = q = \kappa$, and the character part of [theorem 30.2](#) is source 11's [proposition 12.1](#) (every local and global character κ , and η_κ). For $X = \mathbb{R}$ the reals report proves the analogues: point characters $\text{cf } \mathfrak{c}$ and global characters ω (`lwo:prop:fixed-topology`), weight $2^{<\mathfrak{c}}$ (`lwo:sp:prop:minimalweight`), and that fewer than $\text{cf } \mathfrak{c}$ open sets have open intersection (`lwo:rk:cor:Pspace`). Sources 15, 17 and 18 do not cite these. New in sources 15, 17 and 18 is the alphabet $\mathbf{No}_{<\kappa}$ at every infinite cardinal κ , in particular the separation of the point character $q = \text{cf } \mu$ from the global character $r = \text{cf } \kappa$; new in source 17 is the second half of [theorem 30.5](#), where every fusion sequence must also exhaust the alphabet.

Note 30.7 (Source 15, Theorem 9.2, “Local and global cofinalities”). Source 15 writes $L = \mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ for P , and $\text{cf}^-(f)$, $\text{ci}^+(f)$ for the cofinality of $L_{<f}$ and the coinitality of $L_{>f}$. For every $f \in L$, $\text{cf}^-(f) = \text{ci}^+(f) = \text{cf } \mu$; the global cofinality and coinitality are $\text{cf } \kappa$; consequently the ordinary local topological character of every point of L is $\text{cf } \mu$. This is the character part of [theorem 30.2](#), by the same argument (a swap of the entry at η with a later occurrence of a fresh smaller or larger label gives $f_\eta^\mp$ with first difference exactly η ; cofinally many η suffice, fewer than $\text{cf } \mu$ first differences are bounded). Source 15 adds the local base of size $\text{cf } \mu$ formed by pairs from the two sides, and the examples: “at the cutoff $\kappa = \omega_1$ the alphabet cardinal is $\mu = 2^{\aleph_0}$. The global cofinality is always ω_1 , whereas the local character is $\text{cf}(2^{\aleph_0})$. Under cardinal arithmetic making these cardinals different, the distinction survives at every point of the permutation order. At the singular strong-limit cutoff $\kappa = \beth_\omega$, every point has countable local character, even though the density and cellularity of the minimal slice are 2^κ .”

30.2 The complete nonprincipal cut spectrum of the minimal slice (source 17)

30.2.1 Three ways that a cut can occur

For a gap (A, B) of P , say that a cylinder $[p]$ *straddles* the cut when it meets both A and B . There is at most one straddling cylinder at any fixed length, because the cylinders at that length are disjoint convex blocks.

Lemma 30.8 (Cut trichotomy). *Every gap of P is of exactly one of the following structural forms, using the first stage at which the construction stops.*

1. **A coordinate cut.** Some straddling $[p]$, of length $\alpha < \mu$, has no straddling child. Its children are partitioned by a proper cut of the residual alphabet Y_p .
2. **A limit-prefix boundary.** At a nonzero limit $\delta < \mu$, all shorter prefixes on a coherent path straddle, but the union cylinder $[p]$ lies wholly on one side. The cut is exactly the left or right boundary of $[p]$.

3. An omitted-range branch. *A coherent prefix straddles at every length below μ . Their union is an injection $f : \mu \rightarrow X$ which is not surjective, and the gap is comparison with f .*

Proof. Begin at the empty prefix, whose cylinder straddles. At a successor stage, continue into the unique straddling child when it exists; otherwise every child belongs wholly to one side, giving case 1. At a limit $\delta < \mu$, the union is an injective prefix and has a nonempty cylinder. If that cylinder does not straddle, it is wholly on one side. Every word strictly below it differs at an earlier coordinate, and is below a lower member of an earlier straddling cylinder, so belongs to A . The dual argument holds above the cylinder. This proves exactly the boundary assertion in case 2.

If the process continues through all lengths below μ , let f be the union injection. For any $g \in P$ different from f , its first difference determines its cut side: a later straddling cylinder gives witnesses which force the orientation. If f were surjective it would itself belong to P . Whichever side contains it would then have an endpoint, contrary to the gap hypothesis. Thus f is not surjective, and all words in P are classified by their comparison with it. These alternatives cover the set recursion through μ , and are disjoint by the first stopping stage. $\square$

Lemma 30.9 (Characters in the three cases). *In case 1, the gap character belongs to $\text{GapChar}(X) \cup \{(r, r)\}$. In case 2 it is respectively (cf δ, r) or $(r, \text{cf } \delta)$. In case 3 it is (q, q) .*

Proof. In case 1 partition the residual alphabet into nonempty parts $U < V$, according as a child lies on the lower or upper side. If U has no largest label, the lower character is cf U ; choose one point in cofinally many children. If U has largest label u , the last lower child is cofinal on the lower side and has cofinality r . The upper side has the dual description. When neither side has an endpoint, Theorem 29.4 applies to Y_p . When one has an endpoint, the other has character r by Lemma 29.3; the endpoint side is replaced by an entire child of character r .

At the left boundary of a limit cylinder $[p]$, lower approximations obtained by a smaller fresh entry at coordinates cofinal in δ are cofinal below the cylinder. No family of fewer than cf δ can be cofinal, since its first disagreements with p would be bounded below δ . The upper side has $[p]$ as a coinital subset, and the cylinder has coinitality r . The right boundary is dual.

For an omitted-range branch, approximations from either side at cofinally many coordinates give upper bounds q for both characters. Fewer than q disagreements are bounded, so a later approximation is closer to the branch. This proves equality. $\square$

30.3 A general permutation-order principle (source 18)

The tree argument is not specific to surreal numbers. It separates the contribution of the alphabet from the new cuts introduced by long prefixes and omitted branches.

Theorem 30.10 (Gap spectrum under permutation expansion). *Let D be a dense linear order without endpoints of infinite cardinality μ . Assume that every nonempty open interval has cardinality μ , every point has the same two-sided character ν , and both global cofinalities are also ν . Let $\text{Perm}_\mu(D)$ be the lexicographic order on bijections $\mu \rightarrow D$, and put $q = \text{cf } \mu$. Then*

$$\begin{aligned} \text{GapChar}(\text{Perm}_\mu(D)) = & \text{GapChar}(D) \cup \{(\nu, \nu), (q, q)\} \\ & \cup \{(\theta, \nu), (\nu, \theta) : \theta < \mu \text{ is an infinite regular cardinal}\}. \end{aligned}$$

Proof. Small deletions leave an order dense in D , so the residual-cut analysis and the straddling-cylinder trichotomy apply verbatim. These give the upper bound. All the explicitly added pairs are attained by the same prefix-boundary and omitted-branch constructions. Finally, each gap of D lifts to a gap of the permutation order by placing a permutation on the left or right according to

its first digit. With neither digit side principal, the two characters are unchanged. This gives the reverse inclusion. $\square$

[merge] *The steps the proof leaves to the reader.* Source 18 writes “apply verbatim”; the two facts it uses are written out here. Put $P = \text{Perm}_\mu(D)$.

Small deletions add only (ν, ν) . Let $U \subseteq D$ with $|U| < \mu$ and $R = D \setminus U$. Every interval of D has μ points, so R meets every interval of D ; in particular R is dense, has no endpoints, and has global cofinalities ν and point characters (ν, ν) , because cofinal and cointial families transfer between D and R through points of the other order. Let (A, B) be a gap of R . Two points of D cannot lie between A and B , since a point of R would lie between them. If no point of D lies between them, the downward closure of A and the upward closure of B in D form a gap of D with the same characters. If one point x of D lies between them, then $x \in U$, A is cofinal in $D_{<x}$ and B cointial in $D_{>x}$, and the character is the point character (ν, ν) of x . Hence $\text{GapChar}(R) \subseteq \text{GapChar}(D) \cup \{(\nu, \nu)\}$.

The three cases. Let (A, B) be a gap of P . The prefixes p of length below μ whose cylinders C_p meet both sides form a chain under extension, as in lemma 30.8. If there is a longest one, p , the residual alphabet $R_p = D \setminus \text{ran } p$ (with $|\text{ran } p| < \mu$) is cut into the digits whose child cylinders lie in A and those whose child cylinders lie in B ; both parts are nonempty and, R_p being dense, they do not both have endpoints. If neither has one, the character of (A, B) is the character of this gap of R_p , which lies in $\text{GapChar}(D) \cup \{(\nu, \nu)\}$ by the first step: $A \cap C_p$ is cofinal in A and is the lexicographic sum of the lower child cylinders. If the lower part has a last digit u , A is cofinally the child cylinder $C_{p \frown u}$, whose cofinality is that of its residual alphabet, ν , while B approaches u from above in R_p , with cointiality ν ; the dual case is the same, so the character is (ν, ν) . If the straddling lengths have a limit supremum $\delta < \mu$, the union p has a nonempty cylinder lying on one side, the gap is a boundary of C_p , and its character is $(\text{cf } \delta, \nu)$ or $(\nu, \text{cf } \delta)$: the approximations with a smaller (larger) unused digit at coordinates cofinal in δ exist because rays of D contain intervals of size μ , and C_p has global characters ν . If the lengths are cofinal in μ , the union is a nonsurjective injection $\mu \rightarrow D$ and the character is (q, q) , exactly as in case 3 of lemma 30.9. This proves the upper bound.

Attainment. For a regular $\theta < \mu$, the two boundaries of the cylinder of an injective prefix of length θ have characters (θ, ν) and (ν, θ) ; the lower boundary of a one-digit cylinder $C_{\langle x \rangle}$ has character (ν, ν) , since its lower side approaches the digit x from below and $C_{\langle x \rangle}$ has cointiality ν ; a bijection of μ onto $D \setminus \{x\}$ gives (q, q) ; and a gap of D lifts at the first digit, as source 18 says. $\square$

Theorem 30.11 (Exact gap spectrum). *For every infinite cardinal κ , with the parameters in (S17.3),*

$$\text{GapChar}(\mathcal{W}_\mu(\mathbf{No}_{<\kappa})) = \{(r, r), (q, q)\} \cup \{(\theta, r), (r, \theta) : \theta \in \text{Reg}, \theta < \mu\}. \quad (\text{S17.5})$$

Proof. The upper inclusion follows from the two preceding lemmas and $\kappa \leq \mu$: alphabet gap pairs involving a regular cardinal below κ already belong to the displayed set.

For (r, r) , fix a label $x \in X$ and place on the lower side every word whose first entry is at most x , and on the upper side every word whose first entry exceeds x . The last lower child has cofinality r , and the upper first-entry ray has cointiality r . Neither side has an endpoint.

For any infinite regular $\theta < \mu$, choose an injective prefix of length θ . Its two boundary cuts have characters (θ, r) and (r, θ) by the limit-boundary calculation. The fresh smaller and larger labels used there exist at every coordinate, so both sides of both cuts are nonempty.

Finally choose a bijection $f : \mu \rightarrow X \setminus \{x\}$ for some label x . Each proper prefix is completable in P , but f itself is not in P . Its comparison cut is a gap of character (q, q) . $\square$

Note 30.12 ([merge]; [theorem 30.11](#) as an instance). $D = \mathbf{No}_{<\kappa}$ satisfies the hypotheses of [theorem 30.10](#) with $\nu = r$: [lemmas 29.2](#) and [29.3](#) give the interval size and the uniform point and global characters. With $\text{GapChar}(\mathbf{No}_{<\kappa})$ from [theorem 29.4](#), whose asymmetric pairs $(\theta < \kappa)$ are among those with $\theta < \mu$, the principle returns exactly ([S17.5](#)). Source 17’s proof above is the direct one.

Note 30.13 (Source 18, Theorem 5.1, “Exact gap spectrum”, and Lemma 5.4). Source 18’s Section 5 “settles the cut-spectrum question for the minimal-length permutation order at every infinite cardinal birthday cutoff”. It writes P for the bijections $\mu \rightarrow \mathbf{No}_{<\kappa}$, defines a *proper Dedekind gap* as a partition $P = A \sqcup B$ into nonempty sets with $A < B$, A without greatest and B without least element, of character (cf A , ci B), and writes $\text{GapChar}(P)$ for the set of characters (“this convention excludes principal cuts and the two external ends”). Its Theorem 5.1 is ([S17.5](#)) together with: *every displayed pair is attained; (1) every point of P has left and right character q ; (2) the global cofinality and coinitality are r ; (3) the principal cuts have character $(1, q)$ or $(q, 1)$; (4) if empty sides are assigned cofinality 0, the external ends have character $(0, r)$ and $(r, 0)$* . Its Lemma 5.4: *every cylinder C_p has cofinality and coinitality r and no endpoints; if p has nonzero limit length $\delta < \mu$, the lower boundary of C_p is a proper gap of character (cf δ, r) and the upper boundary one of character $(r, \text{cf } \delta)$* . These are [lemma 30.1](#), case 2 of [lemma 30.9](#) and [theorems 30.2](#) and [30.11](#), and source 18’s proof is source 17’s: the straddling prefixes form a chain under extension, and its three cases (a longest straddling prefix, a limit supremum $\delta < \mu$, lengths cofinal in μ) are the three cases of [lemma 30.8](#), in the same order, with the same character computations. Source 18 attains (r, r) by the boundary of a successor-length cylinder $C_{\langle x \rangle}$ (the exterior side approaches the digit x) rather than by source 17’s first-entry split; “every infinite regular $\theta < \mu$ can be used as δ ” in Lemma 5.4, so the asymmetric pairs are attained “independently of which of them already occur as gaps in the alphabet”.

30.4 Source 16’s route: regular saturated stages over any saturated alphabet

[merge] The last subsection of source 16’s Section 6. It is printed here, after source 17’s spectrum, because it computes the same spectrum for the regular size-matched case and generalizes it to every regular saturated alphabet of the same size ([note 30.15](#)).

Suppose μ is regular infinite and X is an η_μ order of size μ . The earlier report proves that $\mathcal{W}_\mu(X)$ is itself η_μ , with every local and global cofinality and coinitality μ ([proposition 12.1](#)). Here is the mechanism. If one side is empty, choose a fresh first label beyond all first labels on the other side and complete it. For two nonempty families of fewer than μ lower and upper enumerations, bound their cross first disagreements below μ . Follow their forced common coordinates until the active next labels have strictly separated lower and upper sets. Choose a fresh interpolating label using saturation, and complete the resulting prefix. Fewer than μ used labels can be avoided by a further small cut. The procedure must stop before the bound is passed. Small cut filling gives the lower character bound; [Note 32.1](#) gives the upper bound.

Theorem 30.14 (Complete gap-character spectrum). *Under the preceding hypotheses, the cofinality pairs of proper unfilled Dedekind cuts in $\mathcal{W}_\mu(X)$ are exactly*

$$(\mu, \lambda) \quad \text{and} \quad (\lambda, \mu), \quad \omega \leq \lambda \leq \mu, \quad \lambda \text{ regular.} \quad (\text{S16.4})$$

Both sides of a cut here are nonempty, and neither has an endpoint.

Proof. The two characters are infinite regular cardinals. Neither exceeds μ , because a larger character would produce a monotone sequence of type μ^+ , contrary to [Note 32.1](#). They cannot both be below μ : cofinal and coinital sets of those sizes would form a strict small cut, and saturation would insert a point in the supposed unfilled cut. This proves necessity.

For an infinite regular $\lambda < \mu$, fix an injective prefix p of length λ . Its completion cylinder C_p is convex and has no endpoints; its cofinality and coinitality are μ , by the same saturation argument. The cut just above the entire cylinder has lower character μ . Enumerations above the cylinder first differ upward from p at some coordinate below λ . Choose such upward deviations at a cofinal sequence of coordinates in λ ; their completions form a coinital family of size λ on that upper side. Fewer than λ deviation coordinates are bounded, and a later upward deviation lies below all of them. Thus the upper character is exactly λ . The cylinder has no maximum and the upper side has no minimum, so this cut is unfilled. The lower edge gives (λ, μ) .

For (μ, μ) fix $a \in X$ and split the order according to first label $\leq a$ or $> a$. The lower side is cofinally the cylinder with first label a , of character μ . The upper side has coinitality μ , inherited from the native segment above a . Both sides have no endpoint. This realizes the final case. $\square$

For regular κ satisfying $2^{<\kappa} = \kappa$, these conclusions apply to $\mathbf{No}_{<\kappa}$. They answer the earlier report's regular-stage density, cellularity, and cut-character questions under precisely those hypotheses. Singular stages and regular stages of larger cardinality require separate analysis; the theorem does not erase the singular coding phenomenon in (S16.2).

Note 30.15 ([merge]; source 16's theorem as an instance). An η_μ order X of regular cardinality μ satisfies the hypotheses of [theorem 30.10](#) with $\nu = \text{cf } \mu = \mu$: it is dense without endpoints, its intervals have size μ , and every point and both ends have character μ , by saturation and $|X| = \mu$. An η_μ order has no gap with both characters below μ , so the principle gives the pairs (μ, θ) , (θ, μ) with $\theta \leq \mu$ regular, which is (S16.4). For $X = \mathbf{No}_{<\kappa}$ with κ regular and $2^{<\kappa} = \kappa$, [theorem 30.14](#) and [example 30.20](#) coincide. Source 17 is more general in the cutoff (any κ , no hypothesis on $2^{<\kappa}$), source 16 in the alphabet (any regular saturated alphabet). For $X = \mathbb{R}$ the reals report's `lwo:rk:thm:Pgaps` is the corresponding spectrum, $\{(\omega, \theta), (\theta, \omega) : \theta < \mathfrak{c} \text{ regular}\} \cup \{(\text{cf } \mathfrak{c}, \text{cf } \mathfrak{c})\}$; it is source 17's formula with $r = \omega$, $q = \text{cf } \mathfrak{c}$, $\mu = \mathfrak{c}$ and the alphabet spectrum $\text{GapChar}(\mathbb{R}) = \emptyset$ (the formula's (r, r) comes from a principal digit split, and (ω, ω) is the case $\theta = \omega$). Neither source 16 nor source 17 cites it.

30.5 How many gaps have each character? (source 18)

[merge] “The same classification” and “the three cases” below are the straddling-cylinder classification of the gaps of P : a longest straddling prefix, a limit supremum of straddling lengths below μ , and lengths cofinal in μ , which are cases 1, 2 and 3 of [lemma 30.8](#) ([note 30.13](#)). Source 18's d_* is the density $\mu^{<\mu}$ of P ([theorem 30.2](#)); its prefix variable q is printed p' , since $q = \text{cf } \mu$ here. The theorem is the minimal-slice analogue of the reals report's count of $2^{\mathfrak{c}}$ gaps of $\mathcal{W}$ (`lwo:sp:thm:gapcount`), which source 18 does not cite.

The same classification gives exact multiplicities, not just a list of the possible character pairs. Write

$$d_* = \mu^{<\mu} = \sup_{\lambda < \mu} \mu^{|\lambda|}.$$

This is also the cardinality of the tree of injective prefixes of length less than μ .

Theorem 30.16 (Exact multiplicities). *The minimal permutation order P has exactly 2^μ proper Dedekind gaps. For a pair (a, b) in the spectrum of [Note 30.13](#), let $G_{a,b}$ be the set of proper gaps of that character. Then*

$$|G_{a,b}| = \begin{cases} 2^\mu, & (a, b) = (q, q), \\ d_*, & (a, b) \neq (q, q). \end{cases}$$

In particular, the Dedekind completion of P , with or without its two external ends, has cardinality 2^μ .

Proof. First, every full nonsurjective injection $f : \mu \rightarrow X$ gives a proper gap of character (q, q) . Fix $x_* \in X$ and consider injections into $X \setminus \{x_*\}$. There are 2^μ such injections: for a lower bound partition this set into μ disjoint pairs and choose one entry of each pair according to a binary word; the upper bound is $\mu^\mu = 2^\mu$.

Distinct injections give distinct gaps. If $f < g$ first differ at α , choose an unused digit strictly between $f(\alpha)$ and $g(\alpha)$ and extend the common prefix followed by that digit to a bijection. This is possible because the intervening interval has cardinality μ , while fewer than μ digits have been used. The resulting member of P lies strictly between f and g . There are therefore at least 2^μ gaps of character (q, q) .

For an upper bound on all gaps, the tree of short prefixes has cardinality $d_* \leq 2^\mu$, an alphabet of size μ has at most 2^μ cuts, and there are at most $\mu^\mu = 2^\mu$ full injections. The three cases of the straddling-cylinder classification therefore give at most 2^μ gaps altogether. This proves both the total number and the multiplicity of (q, q) .

Next fix an infinite regular $\theta < \mu$. There are exactly d_* prefixes whose lengths have cofinality θ . For the nontrivial lower bound, take any infinite cardinal λ with $\theta \leq \lambda < \mu$ and put

$$\delta = \lambda \cdot \theta.$$

Then $\delta < \mu$, $|\delta| = \lambda$, and $\text{cf } \delta = \theta$. There are μ^λ injective words of this length. One way to see the lower bound is to partition X into λ disjoint sets, each of cardinality μ , and choose the digit at each coordinate from its assigned set. Taking the supremum over λ gives d_* . The opposite bound is the cardinality of the whole prefix tree.

Distinct prefixes give distinct lower-boundary cuts. For incomparable prefixes, an unused digit strictly between their first disagreeing digits yields a permutation between their cylinders; the interval has size μ and fewer than μ digits have been used. If p' properly extends p , choose at the first additional coordinate an unused digit smaller than the corresponding digit of p' . This gives a point of C_p below all of $C_{p'}$; a larger choice gives a point above all of $C_{p'}$. Thus the lower boundaries, and also the upper boundaries, are distinct in this case as well. Consequently there are at least d_* gaps of each character (θ, r) and (r, θ) .

It remains to bound the number of gaps of any character other than (q, q) . Full-length omitted branches contribute none of these. Limit-cylinder boundaries contribute at most d_* gaps. For the maximal-prefix case distinguish two possibilities.

If $q = r$, the character being counted is not (r, r) . In the alphabet sign-path description, a gap not of character (r, r) has one sign set bounded below κ , so its path is eventually constant. There are at most $\sum_{\delta < \kappa} 2^{|\delta|} = \mu$ such paths. Deleting a small set only adds (r, r) gaps, and a principal digit cut likewise lifts to that character. Thus each maximal prefix contributes at most μ gaps of the character under consideration, and all short prefixes together contribute at most $d_* \cdot \mu = d_*$.

If $q \neq r$, then $\mu > \kappa$: equality would give $\text{cf } \mu = \text{cf } \kappa$. Every alphabet cut is represented by a length- κ sign word, so there are at most 2^κ of them. Cuts created by deletions and retained points add at most μ . Since $\kappa < \mu$,

$$2^\kappa \leq \mu^\kappa \leq d_*.$$

Again all maximal prefixes together contribute at most d_* gaps.

Every allowed character other than (q, q) has the form (θ, r) or (r, θ) for a regular $\theta < \mu$. The only apparent exception would be (r, r) with $r = \mu$, but then $\kappa = \mu$ is regular and $q = r$, so that pair is precisely the already excluded (q, q) . The prefix-boundary lower bound therefore applies to every remaining character, completing the multiplicity calculation.

Finally $|P| = 2^\mu$, and its completion adds one point for each proper gap (and optionally its two ends). The completion cardinality is consequently 2^μ . $\square$

30.5.1 Consequences and distinctions

The exact formula makes several distinctions transparent.

1. If $\mu = \kappa$ is regular, then $r = q = \kappa$ and the spectrum is

$$\{(\kappa, \kappa)\} \cup \{(\theta, \kappa), (\kappa, \theta) : \theta < \kappa \text{ infinite regular}\}.$$

In particular, the order's η_κ property does not prohibit asymmetric gaps with one small side.

2. At a singular strong-limit cutoff, $\mu = \kappa$ and $r = q = \text{cf } \kappa$. The asymmetric pairs range over every infinite regular $\theta < \kappa$, including regular cardinals much larger than $\text{cf } \kappa$.
3. If $\kappa = \omega_1$ and $2^{\aleph_0} = \aleph_2$, the proper gap spectrum consists exactly of

$$(\omega, \omega_1), \quad (\omega_1, \omega), \quad (\omega_1, \omega_1), \quad (\omega_2, \omega_2).$$

Every point has character (ω_2, ω_2) , while both global cofinalities are ω_1 . Thus local character, global character, and gap character cannot be conflated.

4. If $\kappa = \omega$, then $\mu = r = q = \omega$ and every proper gap has character (ω, ω) .

The set of possible character pairs depends only on μ , r , and q ; its proof retains substantially more information. Every gap has a precise structural cause: a split between next digits, a boundary of a limit-length cylinder, or a nonsurjective branch of full length. Coinciding character pairs need not have the same structural cause. Nor does equality of these spectra by itself establish order isomorphism.

Natural further problems include classifying the order of the gaps of each character, studying which of the three mechanisms can be recognized intrinsically from the pure linear order, and extending the analysis to strata of nonminimal enumeration length. In the last problem a prefix can leave a small residual alphabet near the end of the domain, so the uniform large-interval argument used here no longer applies without a separate analysis of the ordinal tail.

30.6 Saturation and illustrative cases (source 17)

Definition 30.17. For an infinite regular cardinal τ , an order is an η_τ order when every pair of separated sets $A < B$, each of size below τ , has a strict interpolant. Empty sides are allowed.

Corollary 30.18 (Exact saturation threshold). *The order P is an η_r order, and is not an η_{r+} order.*

Proof. Suppose $A < B$ have sizes below r and no strict interpolant. If one side is empty, the other would be cofinal or coinital in P , contradicting (S17.4). If both are nonempty, let

$$D = \{x : \exists a \in A (x \leq a)\}, \quad U = \{x : \exists b \in B (b \leq x)\}.$$

They are disjoint, and no strict interpolant means that they cover P . If D has a greatest element x , then B is coinital immediately above x , contradicting right point character $q \geq r$. If U has a least element, the dual contradiction uses A . If neither has an endpoint, (D, U) is a gap with both characters below r , forbidden by (S17.5). An (r, r) gap exists; cofinal and coinital witnesses of size r show failure of η_{r+} . $\square$

Example 30.19 (The countable birthday cutoff). At $\kappa = \omega$, X is the order of finite sign sequences, $\mu = r = q = \omega$, and P has countable weight and density, character (ω, ω) at every point, and only (ω, ω) gaps. It is a Baire, zero-dimensional order. These facts do not identify it with the real line, whose order topology has a completely different connectedness structure.

Example 30.20 (Regular size-matched cutoffs). If κ is regular and $2^{<\kappa} = \kappa$, then $\mu = r = q = \kappa$. The gap spectrum is

$$\{(\kappa, \kappa)\} \cup \{(\theta, \kappa), (\kappa, \theta) : \theta \in \text{Reg}, \theta < \kappa\}.$$

The order is η_κ but not $\eta_{\kappa+}$. The small character on one side of a limit-prefix boundary does not violate η_κ , because the other side requires κ witnesses.

Example 30.21 (A singular strong-limit cutoff). At $\kappa = \beth_\omega$, one has $\mu = \kappa$ and $r = q = \omega$. Every point has countable character, but for every infinite regular $\theta < \kappa$ there are gaps of character (θ, ω) and (ω, θ) . Weight, density and cellularity are $\kappa^{<\kappa} = 2^\kappa$. This complements rather than replaces the source's $\kappa \cdot \kappa$ binary-coding-length theorem (theorem 8.3).

[merge] (3 October 2026.) Source 17 wrote “are $\kappa^{<\kappa}$, with no extra cardinal-arithmetic simplification asserted”. At a singular strong limit κ one has $2^\kappa \leq \kappa^{\text{cf } \kappa} \leq \kappa^{<\kappa} \leq 2^\kappa$, so the value is 2^κ ; source 18 proves this identity in corollary 30.26, and source 15 obtains the same value 2^κ from its block-type theorem (corollary 31.7).

Remark 30.22 (What has not been classified). The preceding formulas concern the minimal type μ . They do not extend by replacing μ with an arbitrary ordinal α of cardinality μ : a prefix of length below α can already use μ labels, leaving a radically smaller residual alphabet. Nor do equal values of these invariants imply order-isomorphism. Both issues remain in the research program at the end of this article.

30.7 Source 15's route to the saturation threshold

[merge] Source 15's Section 9, “Exact local characters at every cardinal cutoff”. Its Lemma 9.1 is recorded in note 29.8, its Theorem 9.2 in note 30.7 and its Lemma 9.3 in note 29.9; its saturation theorem is proved directly, by a prefix search, not from the gap spectrum as in corollary 30.18, and is printed in full. Its $\delta = \text{cf } \kappa$ is printed r . Its first sentence describes this report at source 15's pin; the point characters it mentions are now theorem 30.2.

The repository leaves the point characters of $\mathcal{W}_{2^{<\kappa}}(\mathbf{No}_{<\kappa})$ open when $2^{<\kappa}$ need not equal κ . They can be computed exactly. The local invariant is controlled by the enumeration length, whereas the global invariant is controlled by the numerical alphabet.

Throughout this section let

$$X = \mathbf{No}_{<\kappa}, \quad \mu = 2^{<\kappa}, \quad r = \text{cf } \kappa, \quad L = \mathcal{W}_\mu(X),$$

where κ is an infinite initial cardinal. For $f \in L$, write $\text{cf}^-(f)$ for the cofinality of the open initial segment $L_{<f}$, and $\text{ci}^+(f)$ for the coinitality of the open final segment $L_{>f}$.

We use the convention that an order is η_ν if every cut $A < B$ with $|A| + |B| < \nu$ has a strict interpolant, including the cases where one side is empty. This does not assert that an arbitrary cut with larger sides is filled.

Theorem 30.23 (Exact saturation threshold for the minimal slice). *The order $L = \mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ is $\eta_{\text{cf } \kappa}$. For every infinite cardinal $\nu > \text{cf } \kappa$, it is not η_ν . Equivalently, among infinite cardinal cut-size parameters,*

$$L \text{ is } \eta_\nu \iff \nu \leq \text{cf } \kappa.$$

In particular, at every regular κ , the minimal slice is η_κ without the additional assumption $2^{<\kappa} = \kappa$.

Proof. Let $A < B$ be subsets of L with $|A| + |B| < r$. Any family of fewer than r surreal labels has birthdays bounded below κ . Consequently, for any strict numerical cut formed by such labels, there are two points $u < v$ of X strictly between its sides. This follows from the usual surreal cut construction applied twice; all resulting birthdays stay below κ . The same statement holds with an empty side. By [Note 29.8](#), the interval (u, v) has cardinality μ . Thus a numerical cut of this size admits a separator outside any prescribed set of fewer than μ labels.

If one of A, B is empty, choose the first entry strictly above all first entries on the left or below all first entries on the right, and complete to a permutation. Suppose both are nonempty. For each $(a, b) \in A \times B$, let $d(a, b) < \mu$ be its first-difference coordinate. By [Note 29.9](#),

$$\beta = \sup\{d(a, b) + 1 : a \in A, b \in B\} < \mu.$$

Build an injective prefix p , keeping as active those rows of A and B that still agree with p . At a stage η , every active left value at coordinate η is at most every active right value. If a value occurs on both sides, it is unique; append that value to p and keep the rows taking it. Every discarded left row is already below all extensions of the new prefix, and every discarded right row is already above them. The appended value is fresh because it belongs to active injective rows.

If the active value sets have no common value, they form a strict numerical cut. Choose a separator outside the range of p using the preceding full-size-interval argument, append it, and stop. If an active side becomes empty, use the same argument for the resulting one-sided cut. At a limit stage take the union of the prefixes and retain the rows that extend it. The search cannot still have active rows on both sides at stage β : such a pair would agree at every coordinate below β , contradicting the definition of β . Therefore it stops with a prefix of length less than μ .

That prefix has used fewer than μ labels. Its remaining labels and remaining positions both have cardinality μ , and an exhaustive completion has order type μ . Every such completion lies strictly above A and below B , proving η_r .

Finally, [Note 30.7](#) gives a global cofinal subset of size r . Together with an empty right side it is a cut of total size below every cardinal $\nu > r$, but it has no upper interpolant. Hence no such η_ν property holds. $\square$

These results settle the point-character part, and the maximal uniform cut-size threshold, of the earlier minimal-slice question. They do not compute its full two-sided unrealized-cut spectrum. In particular, saturation below $\text{cf } \kappa$ and failure above that threshold do not specify which individual larger pairs of regular cofinalities occur as gaps.

[merge] (3 October 2026.) The “full two-sided unrealized-cut spectrum” that source 15 leaves open is source 17’s [theorem 30.11](#) (and source 18’s [note 30.13](#)); its multiplicities are [theorem 30.16](#).

30.8 Source 18’s route and extension: characters, interpolation and two regimes

[merge] The rest of source 18’s Section 4, printed whole. Its point characters and interpolation theorem are source 17’s [theorem 30.2](#) and [corollary 30.18](#) and source 15’s [theorem 30.23](#), proved again with a different interpolation argument (bounded dropout coordinates); it extends both to the types $\mu + n$ and to the finite-cofinite prefix orders Q_n ([lemma 27.7](#)), and it exhibits a bounded (r, r) obstruction explicitly. Its corollary on two regimes merges with source 17’s examples [examples 30.20](#) and [30.21](#); its second part supplies the value $\kappa^{<\kappa} = 2^\kappa$ used in [example 30.21](#).

30.8.1 Exact point characters and global cofinalities

For a point e in a linear order, its left character means the cofinality of $(-\infty, e)$, and its right character means the coinitality of (e, ∞) . An immediate predecessor or successor therefore gives

character 1 on the respective side.

Theorem 30.24 (Complete local and global characters at the minimal type). *Every point of $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ has left and right character $q = \text{cf}(\mu)$, whereas*

$$\text{cf}(\mathcal{W}_\mu(\mathbf{No}_{<\kappa})) = \text{ci}(\mathcal{W}_\mu(\mathbf{No}_{<\kappa})) = r.$$

More generally, $Q_n(\mathbf{No}_{<\kappa})$ has point characters (q, q) for every finite n . Accordingly, in $\mathcal{W}_{\mu+n}(\mathbf{No}_{<\kappa})$:

- *if $n = 0$ or 1 , every point has characters (q, q) ;*
- *if $n \geq 2$, the first and last points of each finite $n!$ -point fiber have characters $(q, 1)$ and $(1, q)$ respectively;*
- *all interior points of a finite fiber have characters $(1, 1)$.*

Interior points exist exactly when $n \geq 3$. The global cofinality and coinitality remain r for every finite n .

Proof. Fix $p \in Q_n(\mathbf{No}_{<\kappa})$ and $\delta < \mu$. There are μ labels numerically below $p(\delta)$, and equally many above it. The labels used before δ , together with the finitely many labels omitted from p , exclude fewer than μ labels. Choose a smaller available label which therefore occurs later in the μ -prefix, and transpose its occurrence with position δ . This gives $p_\delta^- < p$ with first difference exactly δ . An analogous transposition gives $p_\delta^+ > p$.

As δ runs along an increasing cofinal sequence in μ , the points p_δ^- increase cofinally toward p , while p_δ^+ decrease coinitially toward it. This proves both upper bounds q . Conversely, fewer than q points below p have first-disagreement coordinates bounded below μ . Changing p at a later coordinate gives a point above all of them and below p . They cannot be cofinal. The dual argument gives the right lower bound. This proves the character statement for Q_n .

The map $e \mapsto e(0)$ is onto $\mathbf{No}_{<\kappa}$ and respects comparisons whenever the first coordinates differ. A family cofinal in the enumeration order must therefore have first coordinates cofinal in $\mathbf{No}_{<\kappa}$; conversely, choose one enumeration over every member of a numerical cofinal sequence. This proves the global cofinality formula, and the coinitality formula is dual. Finally, $\mathcal{W}_{\mu+n}(\mathbf{No}_{<\kappa})$ is the lexicographic sum over Q_n of finite ordered fibers of size $n!$. Immediate neighbors inside a fiber give character 1, while approach to the outside of a fiber is controlled by the character q of the quotient point. $\square$

30.8.2 Sharp interpolation at every infinite cardinal cutoff

We say that L is η_r if for all sets $A, B \subseteq L$ with $|A \cup B| < r$ and $a < b$ for every $a \in A, b \in B$, there is $z \in L$ satisfying $A < z < B$. Empty sides are allowed. The sharpness result below also uses a bounded cut with both sides nonempty, so it does not depend on the empty-side convention.

Theorem 30.25 (Exact guaranteed interpolation). *For every infinite initial cardinal κ , with notation as above,*

$$\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \text{ is } \eta_r, \quad r = \text{cf}(\kappa).$$

It has a bounded unfilled cut of character (r, r) , so it is not η_{r+} . The same interpolation and sharpness assertions hold for $\mathcal{W}_{\mu+1}(\mathbf{No}_{<\kappa})$. For $n \geq 2$, adjacent points in the terminal fibers already preclude η_ω .

Proof of interpolation. First $\mathbf{No}_{<\kappa}$ is η_r . A family of fewer than r numerical options has birthdays bounded below κ , so the set-cut theorem supplies a separator with birthday less than κ . Applying it twice supplies a nonempty interval of possible separators, still between the prescribed options. By [Note 29.8](#), such an interval has μ labels.

Let $A < B$ be sets of input enumerations with $|A \cup B| < r$. We construct an injective common prefix p until a completion separating the inputs becomes possible. At a prefix of length $\delta < \mu$, the active inputs are

$$A_p = \{a \in A : a \upharpoonright \delta = p\}, \quad B_p = \{b \in B : b \upharpoonright \delta = p\}.$$

The invariant is that all discarded lower inputs lie below every completion of p , while all discarded upper inputs lie above every completion of p . Let L and U be the sets of next entries of the active lower and upper inputs. Then every $l \in L$ is at most every $u \in U$.

If $L \cap U = \emptyset$, the two next-entry sets are strictly separated (with empty sides permitted). Choose an interval of numerical separators between them. It has μ labels, whereas $\text{ran } p$ has size less than μ . Choose an unused separator z in it. Every completion of $p \frown (z)$ lies strictly between all inputs, and an exhaustive completion of the required type exists by cardinality. The construction stops.

Otherwise, $L \cap U$ consists of exactly one label z : two different common labels would contradict $L \leq U$. This label is unused, since an active input is injective. Append it to p . Inputs with a smaller lower entry or a larger upper entry are discarded in the correct direction, maintaining the invariant. At limits take the union of the prefixes.

Suppose the construction never stopped before μ . The union is an injective word p of length μ . Every input whose μ -prefix differs from p drops out at its first disagreement. There are fewer than $r \leq q = \text{cf}(\mu)$ inputs, so these dropout coordinates are bounded below μ . At most one input can have μ -prefix exactly p : for type μ the whole enumeration is determined, and for type $\mu + 1$ its single omitted label determines the last entry. Since $A < B$, that one input cannot lie on both sides. Beyond the bound on the dropout coordinates, at least one active side is therefore empty, forcing the stopping case. This contradiction proves that the construction stops at a coordinate below μ . $\square$

Proof of the bounded obstruction. Choose a sign word s of length exactly κ such that both its plus positions and minus positions are cofinal in κ , and whose first two signs are $+, -$. For example, use the parity of the finite tail of each ordinal coordinate. Then $0 < s < 1$, while $s \notin \mathbf{No}_{<\kappa}$. Prefixes $s \upharpoonright \gamma$ at cofinally many positions with $s(\gamma) = +$ lie below s and are cofinal below its cut in $\mathbf{No}_{<\kappa}$; prefixes at cofinally many minus positions are coinital above it. Choose such two cofinal collections indexed by r .

Every $x \in \mathbf{No}_{<\kappa}$ has its comparison with s decided before κ . A sufficiently long chosen prefix therefore passes x on the appropriate side. No element of $\mathbf{No}_{<\kappa}$ separates the two prefix families. Neither side can have a smaller cofinal family: fewer than r birthdays are bounded below κ , and a longer appropriate prefix passes every label in that family. Thus this is a bounded numerical gap of character (r, r) .

Pull the cut back under the first-coordinate map $e \mapsto e(0)$. Choose arbitrary completions for the two numerical cofinal families. A hypothetical separating enumeration would need its first entry to lie in the missing numerical cut, which is impossible. Conversely those completions are cofinal and coinital on their respective sides, and the first-coordinate projection proves the matching lower bounds. Enumerations starting with 0 and 1 bound the whole configuration. This supplies the claimed bounded (r, r) obstruction. $\square$

30.8.3 Regular and singular consequences

Corollary 30.26 (Two especially transparent regimes). *1. If κ is regular and $2^{<\kappa} = \kappa$, then*

$$|\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})| = 2^\kappa, \quad \text{d} = \text{cell} = \kappa,$$

and every local character and global cofinality or coinitality is κ . The order is η_κ and has a bounded unfilled (κ, κ) -cut.

2. If κ is a singular strong-limit cardinal, then

$$|\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})| = 2^\kappa, \quad \text{d} = \text{cell} = 2^\kappa,$$

while every point character and both global cofinalities are $\text{cf}(\kappa)$. Its sharp interpolation threshold is $\eta_{\text{cf}(\kappa)}$.

Proof. In the regular case, $\kappa^{<\kappa} = \kappa$. For a fixed $\lambda < \kappa$, every function $\lambda \rightarrow \kappa$ has bounded range, and the possibilities for each range bound are at most $2^{<\kappa} = \kappa$; summing over κ range bounds gives $\kappa^\lambda = \kappa$ for nonzero $\lambda < \kappa$.

In the singular strong-limit case, $2^{<\kappa} = \kappa$. Let $r = \text{cf}(\kappa)$ and take a cofinal sequence $(\kappa_\xi)_{\xi < r}$ below κ . A subset of κ is determined by its intersections with the κ_ξ . Each intersection has fewer than κ possibilities by the strong-limit hypothesis, so

$$2^\kappa \leq \kappa^r \leq \kappa^{<\kappa} \leq \kappa^\kappa = 2^\kappa.$$

Thus $\kappa^{<\kappa} = 2^\kappa$. The other conclusions now follow from the preceding theorems and the baseline cardinality theorem. $\square$

What these results settle, and what they do not. The baseline explicitly left density, cellularity and much of the singular-cutoff interpolation analysis as continuation questions. The preceding theorems determine density and cellularity for every fixed enumeration type and every infinite alphabet, and give exact local/global characters and sharp guaranteed interpolation at every infinite initial birthday cutoff for the minimal slice. They do not identify the order up to isomorphism. In particular, equality of these invariants is not an isomorphism theorem, and the old exact ordinal spectrum remains logically separate from the new topological information. The full spectrum of unfilled cut characters requires an additional analysis of boundaries of transfinite prefix cylinders.

31 Longer strata: every cardinal-block type (source 15)

[merge] Source 15's Section 8, printed whole: the only batch-81 treatment of strata longer than the minimal one. Its r_κ is printed ρ_κ (by [theorem 8.3](#), $\rho_\kappa = \ell_2(\mathcal{W}_\mu(\mathbf{No}_{<\kappa}))$), so that r keeps the meaning $\text{cf} \kappa$; its $D_{\sigma, m}$ is printed $\mathbb{B}_\sigma \times_{\text{lex}} m$, so that no letter clashes with its core $\mathcal{D}$ ([section 1.3.1](#)). The section extends source 12's [theorem 8.3](#) ($\gamma = 1$, $n = 0$) along the targets of [research question 61.11](#). Its entry code ([lemma 31.1](#)) is source 12's [lemma 19.33](#) and source 16's mechanism in (S16.2); its cube clause is [lemma 6.4](#) (reals report `lwo:thm:cube-strict`); its row scheduler ([lemma 31.3](#)) is source 12's [lemma 8.1](#) run blockwise along γ ; its finite-deletion lemma ([lemma 31.4](#)) is [lemma 8.2](#) with a finite set removed; its finite-capacity lemma is, by its own account, the finite-fiber fusion argument of the reals report. Source 15 states its finite-deletion lemma for singular strong-limit κ ; the hypothesis is not used in its proof.

The earlier minimal-slice theorem (section 8) extends to all enumeration types whose remainder on division by the alphabet cardinal is finite. This includes the explicitly proposed next cases $\mu \cdot 2$ and $\mu \cdot \gamma + n$. It does not classify an additional infinite remainder smaller than μ . All products and sums of ordinals in this section are ordinal operations.

Let κ be an infinite initial cardinal, and put

$$X = \mathbf{No}_{<\kappa}, \quad \mu = |X| = 2^{<\kappa}, \quad \rho_\kappa = \begin{cases} \kappa \cdot \kappa, & \kappa \text{ is a singular strong-limit cardinal,} \\ \mu, & \text{otherwise.} \end{cases}$$

For an ordinal α with $|\alpha| = \mu$, write $\mathcal{W}_\alpha(X)$ for the bijections $\alpha \rightarrow X$, compared by their first unequal entry in the numerical order of X . For a positive integer m , write

$$\mathbb{B}_\sigma \times_{\text{lex}} m = \mathbb{B}_\sigma \times_{\text{lex}} \{0, \dots, m-1\}, \quad \mathbb{B}_\sigma = (2^\sigma, <_{\text{lex}}),$$

the finite chain $m = \{0, \dots, m-1\}$ being the last coordinate.

Lemma 31.1 (An explicit entry code and the strict cube hierarchy). *Replace each minus sign by 00, each plus by 11, and terminate with 01. The resulting code $c(s)$ is prefix-free and preserves numerical surreal order. Its length is $2 \cdot \text{bd}(s) + 2$, less than κ for $s \in \mathbf{No}_{<\kappa}$. Furthermore,*

$$\mathbb{B}_\alpha \hookrightarrow \mathbb{B}_\beta \iff \alpha \leq \beta.$$

Proof. The three blocks $00 < 01 < 11$ encode precisely minus, termination, and plus. Distinct completed codes cannot be proper prefixes: at the first termination of one of them the other either terminates too or has a different block. Padding a completed entry therefore does not alter its comparison with another entry.

For the cube assertion, extension by zeros gives the embedding when $\alpha \leq \beta$. Suppose $\alpha > \beta$ and an embedding in the other direction exists. Successively read target bits, retaining at each step a full source child cylinder whose image lies on one side of the next target-bit split. Such a child exists: if the lower child does not lie wholly below the split, every point of the upper child lies above it. At limits take unions of the chosen prefixes. After β steps, all points extending one source prefix of length β have the same whole target word. There are at least two source extensions since $\beta < \alpha$, contradicting injectivity. This is the same cylinder fusion mechanism used in the prior binary-length calculations. $\square$

When $\mu = \kappa$ is singular, κ is a strong limit. Otherwise $2^\rho = \kappa$ for some $\rho < \kappa$ would also give $2^\lambda = \kappa$ at $\lambda = \max(\rho, \text{cf } \kappa) < \kappa$, contradicting König's inequality $\text{cf}(2^\lambda) > \lambda$. Thus the three cases used below are exhaustive.

Theorem 31.2 (Cardinal blocks and a finite terminal permutation). *For every $0 < \gamma < \mu^+$ and every $n < \omega$,*

$$\mathcal{W}_{\mu \cdot \gamma + n}(\mathbf{No}_{<\kappa}) \simeq_e (\mathbb{B}_{\rho_\kappa \cdot \gamma} \times_{\text{lex}} n!).$$

In particular, if κ is a singular strong limit, then

$$\mathcal{W}_{\kappa \cdot \gamma + n}(\mathbf{No}_{<\kappa}) \simeq_e \mathbb{B}_{\kappa \cdot \kappa \cdot \gamma} \times_{\text{lex}} \{0, \dots, n! - 1\}.$$

The statement concerns equimorphism; it does not identify these permutation orders with the generally nondense binary products.

We give the scheduler detail before proving the theorem, since a cardinality-preserving reindexing alone would not preserve lexicographic order.

Lemma 31.3 (Blockwise row scheduling). *Let Y be a linearly ordered set of infinite cardinality κ , let $0 < \gamma < \kappa^+$, and put $\lambda = \kappa \cdot \gamma$. Suppose that Y contains pairwise disjoint subsets $(Y_i)_{i < \lambda}$, each of cardinality κ , with an order embedding $\phi_i : X \rightarrow Y_i$, and that $Z = Y \setminus \bigcup_{i < \lambda} Y_i$ is either empty or has cardinality κ . Then*

$$X_{\text{lex}}^\lambda \hookrightarrow \mathcal{W}_\lambda(Y).$$

Proof. For each $i < \lambda$, fix a bijection $b_i : \kappa \rightarrow Y_i$. For $x \in X$, transpose $\phi_i(x)$ with $b_i(0)$ in that enumeration; denote the result by $b_{i,x}$. Its first value is $\phi_i(x)$, and for equal x the whole row is the same.

Split the variable rows into the consecutive blocks $\{\kappa \cdot \beta + \xi : \xi < \kappa\}$, $\beta < \gamma$. Within one block schedule row-column pairs $(\xi, j) \in \kappa^2$ first by $\max(\xi, j)$ and then lexicographically inside each level. Every pair has fewer than κ predecessors: all earlier pairs lie in a square bounded below κ , and such a square has cardinality less than κ , even when κ is singular. The schedule therefore has order type κ . The first occurrence of row ξ is $(\xi, 0)$, and these first occurrences have increasing row indices.

If $Z \neq \emptyset$, include its fixed κ -long enumeration as a filler row before the variable rows in the first block. The row-index order is $1 + \kappa = \kappa$, so the same construction applies. The filler row has no input parameter. Every later block uses only its κ variable rows.

For $x = (x_i)_{i < \lambda}$, output the entries of b_{i,x_i} by these successive block schedules. Each block has length κ and there are γ blocks, so the output has length exactly λ . The disjoint rows and the filler exhaust Y exactly once.

If inputs $x <_{\text{lex}} x'$ first differ at row $i = \kappa \cdot \beta + \xi$, earlier blocks agree completely. In block β , every output before the first entry of row i comes from a lower row or from the fixed filler. Those outputs agree. The first unequal output is therefore $\phi_i(x_i) < \phi_i(x'_i)$ at the first entry of row i . This proves both injectivity and order preservation. No regularity of γ or κ was used. $\square$

Lemma 31.4 (Finite deletion does not obstruct the singular scheduler). *If κ is a singular strong limit, $X = \mathbf{No}_{<\kappa}$, $F \subseteq X$ is finite, and $\lambda = \kappa \cdot \gamma$ for $0 < \gamma < \kappa^+$, then*

$$X_{\text{lex}}^\lambda \hookrightarrow \mathcal{W}_\lambda(X \setminus F).$$

Proof. For $\xi < \kappa$, let C_ξ be the sign-sequence cone with prefix $+\xi \frown -$. The cones are disjoint, and prefixing gives an order isomorphism $X \rightarrow C_\xi$. Indeed, $\xi + 1 + \eta < \kappa$ for $\xi, \eta < \kappa$, because κ is an initial cardinal; deletion of the prefix gives the inverse. Each cone has cardinality κ . At most finitely many cones meet F . Select κ unaffected cones and index them by λ , which has cardinality κ .

The complement of those selected cones in $X \setminus F$ has cardinality κ : it contains all but finitely many of the all-plus sequences, one for each length below κ , and is a subset of X . The preceding lemma now applies, with that complement as the filler. In particular, no unexplained bijection of a grid with λ is being used in place of a lexicographic scheduler. $\square$

Proof of Theorem 31.2. Write $\lambda = \mu \cdot \gamma$ and $\sigma = \rho_\kappa \cdot \gamma$. We first construct the upper embedding. Use the repository's prefix-free, numerical-order-preserving binary code $c(s)$ for each $s \in X$, with positive length less than κ .

If $\mu > \kappa$, pad each individual entry code to length κ . Prefix-freeness ensures that this preserves all entry comparisons. A block of μ entries then consumes ordinal length $\kappa \cdot \mu = \mu$. Applying this separately to the γ successive blocks gives a prefix code of length $\mu \cdot \gamma = \sigma$.

If $\mu = \kappa$ and κ is regular, concatenate the unpadded codes in each κ -entry block. Each proper partial sum of their lengths is below κ by regularity, while the full sum is κ because all code lengths are positive. Each block therefore consumes exactly κ , again giving length σ .

In the remaining case $\mu = \kappa$ is singular, it is a strong limit. Pad each entry code to length κ . Each block now consumes $\kappa \cdot \kappa$, and the complete λ -entry prefix has code length $\kappa \cdot \kappa \cdot \gamma = \sigma$.

In all cases, a prefix of length λ from a bijection $\lambda + n \rightarrow X$ has a binary code of length exactly σ . The remaining n labels form a finite set determined by that prefix. Append the rank of their terminal permutation in its increasing lexicographic list of $n!$ permutations. Different prefixes compare inside the binary code; equal prefixes have the same remaining labels and compare by the finite rank. This gives $\mathcal{W}_{\lambda+n}(X) \hookrightarrow (\mathbb{B}_\sigma \times_{\text{lex}} n!)$.

For the reverse embedding in either nonexceptional case, reserve a fixed n -element set $F \subseteq X$ and partition $X \setminus F$ into pairs indexed by λ . Such a partition exists because $|\lambda| = \mu$. List each pair in its increasing or decreasing orientation according to the corresponding bit of a member of $\mathbb{B}_\lambda$. Since λ is a nonzero limit ordinal, $2 \cdot \lambda = \lambda$. Append the requested one of the $n!$ permutations of F . Here $\sigma = \lambda$, giving the required embedding.

It remains to obtain the longer cube in the singular strong-limit case. Let $\delta = \text{cf } \kappa < \kappa$, and choose nonzero ordinals $(\tau_j)_{j < \delta}$ cofinal in κ , with $\sum_{j < \delta} \tau_j = \kappa$. The identities

$$\delta \cdot \kappa = \kappa, \quad \delta \cdot \lambda = \delta \cdot (\kappa \cdot \gamma) = \lambda$$

allow the λ input coordinates of X^λ to be grouped into λ consecutive blocks of δ coordinates. In each block, restrict its j th coordinate to surreals whose sign sequences have length exactly τ_j . That coordinate is a copy of $\mathbb{B}_{\tau_j}$, so the entire restricted product is

$$\mathbb{B}_{\sum_{i < \lambda} \sum_{j < \delta} \tau_j} = \mathbb{B}_{\kappa \cdot \lambda} = \mathbb{B}_{\kappa \cdot \kappa \cdot \gamma} = \mathbb{B}_\sigma.$$

By Lemma 31.4, this product embeds into $\mathcal{W}_\lambda(X \setminus F)$. Append any of the $n!$ permutations of F , obtaining $(\mathbb{B}_\sigma \times_{\text{lex}} n!) \hookrightarrow \mathcal{W}_{\lambda+n}(X)$. $\square$

31.1 Exact coding lengths, embedded cubes, and comparisons

The next capacity argument is the finite-fiber fusion argument already used in the real-alphabet report. It is included to make the numerical consequences of the extension explicit.

Lemma 31.5 (Finite fibers over a binary cube). *Let σ, τ be nonzero limit ordinals and let m, n be positive integers. Then*

$$(\mathbb{B}_\sigma \times_{\text{lex}} m) \hookrightarrow (\mathbb{B}_\tau \times_{\text{lex}} n) \iff \sigma < \tau \text{ or } (\sigma = \tau \text{ and } m \leq n).$$

Moreover,

$$\ell_2(\mathbb{B}_\sigma \times_{\text{lex}} m) = \sigma + \lceil \log_2 m \rceil, \quad \mathbb{B}_\theta \hookrightarrow (\mathbb{B}_\sigma \times_{\text{lex}} m) \iff \theta \leq \sigma + \lceil \log_2 m \rceil.$$

Proof. For equal binary lengths, suppose $f : (\mathbb{B}_\sigma \times_{\text{lex}} m) \hookrightarrow (\mathbb{B}_\sigma \times_{\text{lex}} n)$. Recursively choose a source binary prefix s_ξ and target binary prefix t_ξ , both of length ξ , such that

$$f(C_{s_\xi} \times \{0, \dots, m-1\}) \subseteq C_{t_\xi} \times \{0, \dots, n-1\}.$$

At a successor stage the target's next bit cuts the image into two ordered parts. Of the source cylinder's two full children, at least one lies wholly on one side of that cut: if its lower child is not wholly on the lower side, its upper child is wholly on the upper side. Keep that child and the corresponding target bit. At a limit stage take unions of prefixes. The invariant persists, and the source cylinder remains nonempty. At stage σ , all m points over one source binary sequence lie in one target fiber of size n . Thus $m \leq n$. The converse is immediate.

Coding the finite chain of size m by $\lceil \log_2 m \rceil$ bits gives $(\mathbb{B}_\sigma \times_{\text{lex}} m) \hookrightarrow \mathbb{B}_{\sigma + \lceil \log_2 m \rceil}$. No cube shorter than σ can contain $(\mathbb{B}_\sigma \times_{\text{lex}} m)$, since it already contains $\mathbb{B}_\sigma$ and the binary cube hierarchy

is strict. A target $\mathbb{B}_{\sigma+j} = (\mathbb{B}_\sigma \times_{\text{lex}} 2^j)$ with $j < \omega$ can contain it only if $m \leq 2^j$, by the same fusion argument. This proves the coding-length formula.

Likewise, $\mathbb{B}_{\sigma+j}$ embeds into $(\mathbb{B}_\sigma \times_{\text{lex}} m)$ exactly when $2^j \leq m$. Once the first forbidden finite extension has been excluded, every larger cube is excluded because it contains that extension. This proves the complete cube spectrum.

Finally, if $\sigma < \tau$, then $\sigma + \lceil \log_2 m \rceil < \tau$ because τ is a limit. Code into that shorter cube and then into the target. If $\tau < \sigma$, the target codes into a cube of length $\tau + \lceil \log_2 n \rceil < \sigma$, whereas the source contains $\mathbb{B}_\sigma$; strictness excludes an embedding. This proves the comparison criterion. $\square$

Corollary 31.6 (The complete invariant on these block types). *Under the hypotheses of Theorem 31.2, put $\sigma = \rho_\kappa \cdot \gamma$. Then*

$$\ell_2(\mathcal{W}_{\mu \cdot \gamma + n}(\mathbf{No}_{<\kappa})) = \sigma + \lceil \log_2(n!) \rceil,$$

and

$$\mathbb{B}_\theta \hookrightarrow \mathcal{W}_{\mu \cdot \gamma + n}(\mathbf{No}_{<\kappa}) \iff \theta \leq \sigma + \lceil \log_2(n!) \rceil.$$

For $0 < \gamma, \eta < \mu^+$ and $m, n < \omega$,

$$\mathcal{W}_{\mu \cdot \gamma + n}(\mathbf{No}_{<\kappa}) \hookrightarrow \mathcal{W}_{\mu \cdot \eta + m}(\mathbf{No}_{<\kappa}) \iff \gamma < \eta \text{ or } (\gamma = \eta \text{ and } n! \leq m!).$$

Proof. Apply the finite-capacity lemma and use strict monotonicity in the right argument of $\gamma \mapsto \rho_\kappa \cdot \gamma$ for a fixed nonzero left factor. Here each $\rho_\kappa \cdot \gamma$ is a nonzero limit ordinal. $\square$

For an unconditional singular example, take $\kappa = \beth_\omega$. The types $\kappa \cdot 2$, $\kappa \cdot 2 + 3$, and κ^2 have binary coding lengths respectively

$$\kappa^2 \cdot 2, \quad \kappa^2 \cdot 2 + 3, \quad \kappa^3.$$

The type $\kappa \cdot 2 + 3$ contains the cube of length $\kappa^2 \cdot 2 + 2$ but not the cube of its own coding length. These distinctions are invisible to cardinality and to the ordinal embedding spectrum, which remain 2^κ and $\{\beta : \beta < \kappa^+\}$ throughout.

31.2 Density and cellularity, including the singular minimal slice

Write $d(L)$ for the least cardinality of a dense subset in the order topology, and $\text{cell}(L)$ for the supremum of cardinalities of families of pairwise disjoint nonempty open sets. An additional consequence of the coding theorem is an exact answer for these two invariants on the same block types. In particular, the singular minimal slice has the largest possible cellularity, a fact not visible from its unchanged ordinal embedding spectrum.

Corollary 31.7 (Density and cellularity of the block types). *For $L = \mathcal{W}_{\mu \cdot \gamma + n}(\mathbf{No}_{<\kappa})$ as above,*

$$d(L) = \text{cell}(L) = \begin{cases} 2^{<\mu}, & \rho_\kappa = \mu, \gamma = 1, n \leq 2, \\ 2^\mu, & \text{otherwise.} \end{cases}$$

Consequently

$$d(\mathcal{W}_\mu(\mathbf{No}_{<\kappa})) = c(\mathcal{W}_\mu(\mathbf{No}_{<\kappa})) = \begin{cases} 2^\kappa, & \kappa \text{ is a singular strong limit,} \\ 2^{<\mu}, & \text{otherwise.} \end{cases}$$

Proof. First consider $\rho_\kappa = \mu$, $\gamma = 1$, and $n \leq 2$. The finite factor $n!$ has size at most two. There is an embedding $f : L \rightarrow (\mathbb{B}_\mu \times_{\text{lex}} n!)$. No point of L is isolated. Indeed, L has no endpoints because the native surreal alphabet has neither a least nor a greatest element; swapping the first entry with a smaller or larger label moves either way without changing the enumeration type. The adjacency criterion for exhaustive permutations applies inside a fixed type: if the unused remainder at a first difference is infinite, an ascent in the lower tail or a descent in the upper tail gives an intermediate permutation of the same type. Adjacency is therefore confined to the finite terminal permutation component. For $n \leq 2$ each such component has at most two points, so no point has both an immediate predecessor and an immediate successor. Thus no point is isolated, and every nonempty open set contains at least five points.

For every binary prefix s of length less than μ whose cylinder meets the first-coordinate projection of $f[L]$, choose one element $x_s \in L$ whose image has that prefix. There are at most $\sum_{\xi < \mu} 2^{|\xi|} = 2^{<\mu}$ such choices. These choices are dense. To see this, take five increasingly ordered points $x_0 < \dots < x_4$ in any given nonempty open interval. Each binary fiber has at most two points, so the binary first coordinate of $f(x_2)$ is strictly between those of $f(x_0)$ and $f(x_4)$. Choose a prefix long enough to distinguish it from both outer coordinates. Its whole binary cylinder lies strictly between them, and the chosen x_s lies in the original open interval. Hence $d(L) \leq 2^{<\mu}$.

Conversely L contains $\mathbb{B}_\mu$. For each $\xi < \mu$, its $2^{|\xi|}$ cylinders of length ξ each contain three ordered points. The ambient open intervals in L between the images of the outer two points are nonempty, since they contain the image of the middle point, and they are pairwise disjoint. Therefore $\text{cell}(L) \geq 2^{<\mu}$, proving equality in the first case since always $\text{cell}(L) \leq d(L)$.

In every remaining case, put $\sigma = \rho_\kappa \cdot \gamma$. If $\sigma > \mu$, the embedded cube $\mathbb{B}_\sigma$ has 2^μ distinct cylinders of prefix length μ , each with at least three points because σ is a limit ordinal greater than μ . The preceding interval construction gives $\text{cell}(L) \geq 2^\mu$. If $\sigma = \mu$ but $n \geq 3$, the embedded order $(\mathbb{B}_\mu \times_{\text{lex}} n!)$ has a fiber with at least three points over each of its 2^μ binary sequences. The intervals between the images of its first and last fiber points are again pairwise disjoint and nonempty. Thus $\text{cell}(L) \geq 2^\mu$ here as well. Finally $d(L) \leq |L| = 2^\mu$, giving the claimed equalities. $\square$

31.3 The boundary of the result

Write a general admissible type as $\alpha = \mu \cdot \gamma + \zeta$, $\zeta < \mu$. The theorem settles the case that ζ is finite. When ζ is infinite, the preceding long prefix leaves a residual numerical suborder of cardinality $|\zeta|$; that residual order varies with the prefix. Neither the size nor the coding behavior of this smaller alphabet can simply be replaced by those of $\mathbf{No}_{<\kappa}$. An exact treatment must control the possible residual suborders and the code lengths available uniformly for their permutations. Thus the result gives a precise affirmative continuation of the proposed cardinal-block program while leaving the smaller infinite-tail classification explicitly open.

Remark 31.8 ([merge]; the reals report's display at $X = \mathbf{No}_{<\kappa}$). The reals report asks under which hypotheses on X the display $\mathcal{W}_{\lambda+n}(X) \simeq_e \mathbb{B}_\lambda \times_{\text{lex}} n!$ still holds (its question “Other ground orders”, `lwo:sec:research`); its batch-80 note records that [theorem 8.3](#) gives the case $n = 0$, $\lambda = \mu$ for $X = \mathbf{No}_{<\kappa}$, except at a singular strong limit. For $\lambda = \mu \cdot \gamma$, $0 < \gamma < \mu^+$, and every $n < \omega$, [theorem 31.2](#) gives $\mathcal{W}_{\lambda+n}(\mathbf{No}_{<\kappa}) \simeq_e \mathbb{B}_{\rho_\kappa \cdot \gamma} \times_{\text{lex}} n!$, and by [lemma 31.5](#) two such products with nonzero limit lengths are equimorphic only when the lengths are equal. So the display holds whenever κ is not a singular strong limit ($\rho_\kappa = \mu$). At a singular strong limit ($\mu = \kappa$, $\rho_\kappa = \kappa \cdot \kappa$) it holds exactly when $\kappa \cdot \kappa \cdot \gamma = \kappa \cdot \gamma$, that is, when $\gamma = \kappa^\omega \cdot \delta$ for some ordinal δ (then $\kappa \cdot \gamma = \gamma$, by Cantor normal form to base κ); it fails for $\gamma = 1, 2, \kappa, \dots$. The display is therefore not simply false at singular strong limits. Strata λ that are not multiples of μ (the first is $\mu + \omega$), cutoffs that are not cardinals, and other alphabets remain open, as source 15 says in [section 31.3](#).

32 Set-sized baselines and warnings in sources 13–18

[merge] Every batch-81 source restates the set-sized baseline of the reals report and of Part I (section 6 and theorem 6.1): cardinality 2^μ , universality for orders of size at most μ , and the exact ordinal spectrum $\beta < \mu^+$. They are recorded here once, with the sources’ own provenance statements, followed by the set-level warnings of sources 13, 14 and 17. None of these statements is new.

Note 32.1 (Source 16, Theorem 4.1, “Basic set-alphabet results”). Source 16’s Section 4 fixes “an infinite linearly ordered set X of cardinality μ ” and records that exhaustive enumerations are prefix-incomparable and that an injective prefix of length less than μ extends to a bijection of every ordinal type α with $|\alpha| = \mu$. *The orders $\mathcal{W}(X)$ and $\mathcal{W}_\mu(X)$ have cardinality 2^μ . Every linear order of cardinality at most μ embeds into $\mathcal{W}_\mu(X)$. For every ordinal β , $\beta \hookrightarrow \mathcal{W}(X)$ if and only if $\beta < \mu^+$, and the same statement holds with $\mathcal{W}_\mu(X)$ in place of $\mathcal{W}(X)$, and for the reverse of β .* Its proof is the pair-orientation, initial-segment and coordinate-stabilization proof of section 6. Source 16’s status line: “Theorem 4.1 is background from the ProveIt reals and surreal reports; it is reproved here to make the later arguments self-contained. In particular, ‘all ordinals embed’ is false at every fixed set stage.”

Birthday cutoffs and a singular distinction (source 16’s Subsection 4.1). For $\mathbf{No}_{<\kappa}$, κ an infinite cardinal, substitute $\mu = 2^{<\kappa}$ for the alphabet cardinal in Note 32.1. The ordinal embedding cutoff is μ^+ , not automatically κ^+ . The earlier report also proves a sharper binary-coding distinction:

$$\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \simeq_e \begin{cases} 2^{\kappa \cdot \kappa}_{\text{lex}}, & \kappa \text{ is a singular strong-limit cardinal,} \\ 2^\mu_{\text{lex}}, & \text{otherwise.} \end{cases} \quad (\text{S16.2})$$

Here $\kappa \cdot \kappa$ is ordinal multiplication. This statement is equimorphism, meaning embeddings in both directions, not isomorphism.

The upper coding mechanism replaces a negative sign by 00, a positive sign by 11, and termination by 01. These codes are prefix-free and preserve numerical comparison. At a regular stage, a sum of fewer than κ code lengths remains below κ . At a singular strong-limit stage it can already reach κ before all κ coordinates have been processed; padding every entry to length κ gives total length $\kappa \cdot \kappa$. The matching lower bound in the report uses a cofinal sequence of sign lengths and an order-preserving row scheduler (theorem 8.3). We do not reclassify the binary coding ranks of all longer strata here. The topological classification below, by contrast, covers every admissible enumeration length and every infinite alphabet.

Note 32.2 (Source 18, Theorem 3.1, “Set-sized universality and the exact ordinal spectrum”). Source 18’s Section 3 defines $\mathcal{W}(X)$ and $\mathcal{W}_\alpha(X)$ as here, notes that two exhaustive enumerations are never proper prefixes of one another, so that no termination convention is needed, and that “labels are essential: two two-point well-orders can be isomorphic as abstract orders and still have different first numerical labels”. *Let X be an infinite linear order of cardinality μ . Then $|\mathcal{W}(X)| = |\mathcal{W}_\mu(X)| = 2^\mu$. Every linear order of cardinality at most μ embeds into $\mathcal{W}_\mu(X)$. An ordinal β embeds into $\mathcal{W}(X)$ if and only if $\beta < \mu^+$. The same exact spectrum holds for reverse ordinals and for every nonempty stratum $\mathcal{W}_\alpha(X)$.* The proof is that of section 6; source 18 adds that the stabilization argument “uses the regularity of the successor cardinal μ^+ , not regularity of μ ”, and that cardinality 2^μ is a poor guide, since such an order contains no increasing sequence of type μ^+ . Its subsection “What the existing binary-rank result adds” restates theorem 8.3 (binary coding length $\kappa \cdot \kappa$ at a singular strong-limit cutoff, μ otherwise), calls it “background rather than a dependency of our new cut-spectrum proof”, and adds: “Replacing these ordinal sums by cardinalities would erase a

genuine distinction. Our results below add topological information not determined by this coding length.”

Note 32.3 (Source 15, Lemma 3.2 and Theorem 3.3, “Set-sized baseline, from the prior reports”). Source 15’s Lemma 3.2 (“Exhaustiveness removes prefix ambiguity”): *distinct exhaustive bijective enumerations of the same set have a first unequal coordinate in both domains; neither is a proper prefix of the other.* Its Theorem 3.3 is the baseline of [note 32.1](#) for every nonempty fixed-type slice, proved as in [section 6](#), with the fixed type $\alpha = \lambda + n$ handled by reserving n final labels and orienting pairs indexed by λ . Source 15 says: “This proof uses no special property of real or surreal arithmetic. It is reproduced to fix the dependence of the later theorems; its content is already in the audited reports. If X is finite of size n , the whole order is the finite chain of its $n!$ permutations. The empty carrier has one empty well-order.” Its subsections on birthday cutoffs and on set-length words, which close its Section 3, follow. Its Lemma 3.1 (fresh separators) belongs to the class layer and is printed with the other batch-81 restatements of [lemma 5.2](#) ([lemma 51.2](#)).

32.1 Source 15’s birthday cutoffs and set-length words

[merge] The last two subsections of source 15’s Section 3; its X_κ is printed **No**_{< κ} , and its forward reference to its Section 8 points to [section 31](#).

32.1.1 Birthday cutoffs

For an infinite initial cardinal κ , let

$$X_\kappa = \mathbf{No}_{<\kappa}, \quad \mu = |X_\kappa| = 2^{<\kappa}.$$

Counting sign levels proves the cardinality formula: the supremum of their cardinalities is at least κ , and the union of κ levels does not exceed it. The preceding theorem gives

$$|\mathcal{W}(X_\kappa)| = 2^{2^{<\kappa}}, \quad \beta \hookrightarrow \mathcal{W}(X_\kappa) \iff \beta < (2^{<\kappa})^+.$$

At regular κ , a cut using fewer than κ labels has birthdays bounded below κ , and the usual cut construction stays in the cutoff. Singular κ need not have this property. The fixed-type calculation in [Section 31](#) makes the distinction visible even when the cardinalities and ordinal spectra coincide.

32.1.2 Why all set-length words are a different order

If X varies, the class of all pairs (X, R) with X a set of surreals and R a well-order is a legitimate class of sets. Its ordinal enumerations are injective set-length words. Proper prefixes now occur. For example (0) is a prefix of $(0, 1)$. One may put shorter words first, shorter words last, or use a termination symbol placed inside the alphabet. These comparisons agree on exhaustive enumerations of a fixed carrier but need not agree on variable carriers.

The prior report gives an explicit self-delimiting sign code embedding all set-length surreal words, with shorter-first order, into **No**. Its code also exhibits why this order need not be isomorphic to **No**: a word of nonzero limit length is the supremum of its shorter prefixes, whereas a surreal point has no set-sized cofinal family below it. Restricting to successor-length words removes that particular obstruction and yields a class order isomorphic to **No** in **GBC** (this report’s [part II](#), source 11’s Sections 9–11 there). An embedding theorem therefore must not be strengthened to an isomorphism theorem merely because both carriers are proper classes.

32.2 Source 13's fixed-set section

[merge] Source 13's Section 4, "The fixed-set theory and birthday fragments", printed whole; its theorem, a restatement credited by source 13 itself, is printed as a note, and its X_κ is printed $\mathbf{No}_{<\kappa}$.

The set-sized layer is important both in its own right and as a safe formalization starting point. We record its already-established results with their provenance rather than repeating the earlier report in full.

Note 32.4 (Source 13, Theorem 4.1, "Fixed-alphabet results from the repository"). Let X be an infinite linearly ordered set, $|X| = \mu$, and compare exhaustive enumerations using its label order. Then

$$|\mathcal{W}(X)| = 2^\mu, \quad \beta \hookrightarrow \mathcal{W}(X) \iff \beta < \mu^+,$$

and the same spectrum holds for reversed ordinals. Every linear order of cardinality at most μ embeds in $\mathcal{W}(X)$.

Provenance and proof mechanisms (source 13). These are the general-alphabet results of the repository's real well-ordering report, carried into Section 6 of the surreal report ([1]; section 6). The cardinal lower bound codes independent binary choices by pair reversals in an enumeration; the upper bound counts well-order relations on X . Universality uses binary codes for initial segments of a small order. The ordinal upper bound is a prefix-stabilization obstruction, stronger than a cardinality bound. We invoke that established written theorem here; this paragraph is not a replacement proof of its stabilization argument.

For an infinite cardinal κ , let

$$\mathbf{No}_{<\kappa}, \quad \mu = 2^{<\kappa}.$$

Counting signs gives $|\mathbf{No}_{<\kappa}| = \mu$: there are $2^{|\alpha|}$ sign sequences of length α , and the union over $\alpha < \kappa$ has that supremum cardinality. Thus the ordinal spectrum of $\mathcal{W}(\mathbf{No}_{<\kappa})$ is exactly the ordinals below $(2^{<\kappa})^+$. This does not say that only ordinals below the birthday cutoff embed.

32.2.1 A warning about the minimal-length layer

Write $\mathbb{B}_\theta = (\{0, 1\}^\theta, <_{\text{lex}})$, and write $L \simeq_e M$ for mutual order embeddability. The merged surreal report proves the following sharper distinction (theorem 8.3, Theorem 8.3 of the report at its pin):

$$\mathcal{W}_\mu(X_\kappa) \simeq_e \begin{cases} \mathbb{B}_{\kappa \cdot \kappa}, & \kappa \text{ singular strong limit,} \\ \mathbb{B}_\mu, & \text{otherwise,} \end{cases} \quad \mu = 2^{<\kappa}.$$

The multiplication in $\kappa \cdot \kappa$ is *ordinal* multiplication. At a singular strong limit, $\mu = \kappa$, but the lengths κ and $\kappa \cdot \kappa$ need not have the same lexicographic effect. This is an existing result, not an advance of the present manuscript.

There is no automatic passage from equimorphism to isomorphism, from a minimal-length stratum to all strata, or from independently chosen cutoff embeddings to a coherent global map. The last passage requires agreement on overlaps. These distinctions become especially visible in the product obstruction below.

32.2.2 Set-sized approximations to the new cut theorem

For an arbitrary set alphabet, prefix cylinders can terminate by exhaustion and can have extrema. The crossing arguments of source 13's Section 8 (its reconstruction of cuts from prefix cylinders) still provide a useful tree description, but its clean all-ordinal formulation depends on two genuinely global facts: every set prefix has a completion, and after deleting its range there are fresh labels

on both sides of any remaining set cut. A bounded alphabet need not have either property at its boundary. Therefore the new global theorem is not asserted by simply replacing **No** by every birthday fragment in its statement.

Note 32.5 (Source 14, Theorem 2.1, “Set-carrier baseline, inherited”). *In ZFC, if $|X| = \mu$ is infinite, then $|\mathcal{W}(X)| = 2^\mu$ and $\beta \hookrightarrow \mathcal{W}(X)$ if and only if $\beta < \mu^+$, for ordinals β . Every linear order of cardinality at most μ embeds in $\mathcal{W}(X)$. Adjacent pairs are exactly the consecutive permutations of a finite residual alphabet after a common prefix.* This is [section 6](#) with source 11’s adjacency theorem (reals report `two:thm:adjacency`). Source 14 prints it as “inherited from the repository, not a new result of this continuation” and gives no proof: “The complete proof is in the predecessor report; we do not substitute counting for that argument.” Its set-level remark follows.

[merge] The rest of source 14’s Subsection 2.2, “Fixed carriers and the inherited spectra”.

The size assertion is plausible for two different reasons: a relation on X has at most 2^μ possibilities, while independently choosing the orientation of μ disjoint pairs gives 2^μ distinct well-orders. The ordinal upper bound is a lexicographic obstruction, not a cardinality bound: 2^μ can be much larger than μ^+ . The complete proof is in the predecessor report; we do not substitute counting for that argument.

For an infinite cardinal κ , sign sequences give

$$|\mathbf{No}_{<\kappa}| = 2^{<\kappa}.$$

Thus [Note 32.5](#) applies with $\mu = 2^{<\kappa}$. The repository also proves a more refined minimal-slice dichotomy: with $\mathbb{B}_\delta = (\{0, 1\}^\delta, <_{\text{lex}})$,

$$\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \simeq_e \begin{cases} \mathbb{B}_{\kappa \cdot \kappa}, & \kappa \text{ is singular strong limit,} \\ \mathbb{B}_\mu, & \text{otherwise.} \end{cases}$$

Here $\cdot$ is *ordinal* multiplication and $\simeq_e$ means mutual order embeddability, not isomorphism. This inherited result should not be confused with the support-cardinal theorem proved below: birthday bounds restrict the alphabet, whereas support bounds restrict how much of a fixed global enumeration is changed.

32.3 Source 17’s labelled well-orders of a fixed set

[merge] Source 17’s Subsection 2.1, from its map of interpretations. Let $(X, <)$ be a set with a specified linear order. A well-ordering R of X has a unique exhaustive enumeration $e_R : \alpha_R \rightarrow X$, where α_R is its ordinal type. For two distinct well-orders, neither exhaustive enumeration is a proper prefix of the other: after listing every label once there is no fresh label to append. Therefore a first unequal coordinate exists, and we define

$$R <_{\text{lex}} S \iff e_R(\delta) < e_S(\delta), \quad \delta = \min\{\xi : e_R(\xi) \neq e_S(\xi)\}.$$

Write $\mathcal{W}(X)$ for all such well-orders, and $\mathcal{W}_\alpha(X)$ for the suborder with enumeration type exactly α . Equivalently, $\mathcal{W}_\alpha(X)$ consists of bijections $\alpha \rightarrow X$. The notation $\mathcal{W}_\mu(X)$ below always uses the initial ordinal of the cardinal $\mu = |X|$.

The source establishes, for infinite $|X| = \mu$,

$$|\mathcal{W}(X)| = 2^\mu, \quad \beta \text{ embeds in } \mathcal{W}(X) \iff \beta < \mu^+,$$

and universality for linear orders of cardinality at most μ ([theorem 6.1](#); [1]). These are labelled-order results. Quotienting well-orders by abstract order-isomorphism removes exactly the labels on which the comparison depends.

33 What this Part answers

[merge] Dated notes, 3 October 2026. Each item names the question, where it is asked, and the result that answers it. Where the question index of [section 61](#) (dated 2 October 2026) says otherwise, these notes supersede it.

- (a) *Source 11's question on the cut spectrum and point characters of the minimal slice without $2^{<\kappa} = \kappa$* ([research question 61.7](#)): **answered**. Point characters (q, q) , global characters r , density, weight and cellularity $\mu^{<\mu}$ ([theorem 30.2](#)), the exact gap spectrum ([theorem 30.11](#)), its multiplicities ([theorem 30.16](#)), and the saturation threshold η_r ([corollary 30.18](#) and [theorems 30.23](#) and [30.25](#)).
- (b) *Source 12's question on the invariants of $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$* ([research question 61.13](#)): answered except for the isomorphism type and homogeneity, which no source determines (source 17's [remark 30.22](#); source 18 says “equality of these invariants is not an isomorphism theorem”).
- (c) *Source 08's question 2, invariants at regular cutoffs* ([section 61.4](#)): **answered**. Density and cellularity of every stratum ([theorem 27.2](#) and [corollary 27.4](#)), point characters and cut cofinalities ([theorem 30.14](#) and [example 30.20](#)). This completes item (d) of [section 1.7](#).
- (d) *Source 08's question 3, which small cuts fail at singular cutoffs*: answered in character terms by [theorems 29.4](#) and [30.11](#) (source 17 does not claim this question); its coding half was already answered by [theorem 8.3](#) (item (c) of [section 1.7](#)).
- (e) *Source 12's question on all fixed enumeration types* ([research question 61.11](#)) and *source 08's question 4*: answered for $\alpha = \mu \cdot \gamma + n$, $0 < \gamma < \mu^+$, $n < \omega$, the first targets the question suggests ([theorem 31.2](#) and [corollary 31.6](#)); open for an infinite remainder $\zeta < \mu$ ([section 31.3](#)). The density of every stratum is [theorem 27.2](#).
- (f) *Source 09's direction 7, what survives away from inaccessible truncations* ([section 61.3](#)): answered for the minimal slice; it is η_r and not η_{r+} , $r = \text{cf } \kappa$ ([corollary 30.18](#) and [theorem 30.23](#)).
- (g) *The reals report's question “Other ground orders”* ([lwo:sec:research](#); item (e) of [section 1.7](#)): answered further for $X = \mathbf{No}_{<\kappa}$, every stratum $\lambda = \mu \cdot \gamma$ and every n ([remark 31.8](#)); open for strata that are not of this form, for noncardinal cutoffs and for other alphabets.
- (h) *The reals report's questions on other alphabets* ([lwo:rk:q:alphabets](#), [lwo:sp:q:alphabets](#)): the density and cellularity of every stratum and the incompleteness of every stratum are settled for every infinite linear alphabet ([theorems 27.2](#) and [28.1](#)); the gap spectrum is settled for the minimal slice of every alphabet satisfying the hypotheses of [theorem 30.10](#), which bears on the “general cut-spine transfer theorem” that [lwo:sp:q:alphabets](#) asks for; the gap spectrum of the full order $\mathcal{W}(X)$ stays open.

Source 16 also says that its regular-stage results “answer the earlier report's regular-stage density, cellularity, and cut-character questions under precisely those hypotheses” (end of [section 30.4](#)), and source 15 that its block theorem gives “a precise affirmative continuation of the proposed cardinal-block program” ([section 31.3](#)); items (c) and (e) record the same.

Part VII

Words, termination conventions and other comparison orders

[merge] (3 October 2026.) The second Part of the batch-81 addition (section 1.8; notation in section 1.3.1). Part II (part II) compared set-length words with proper prefixes first; the six batch-81 sources ask what changes when the comparison convention changes. This Part prints the six sources' maps of the possible interpretations (section 34); the isomorphism of the unrestricted, injective and numerically increasing shorter-first word orders (section 35); the criterion for every placement of a formal terminator and the list of all isomorphism types it produces (section 36); the exact cut classification of centered increasing words and of the conventions for the entry zero (section 37); and comparison orders other than numerical enumeration lexicography: inverse ranks, relation tables, restrictions and normal forms (section 38). Section 39 records which questions the Part answers.

The bases are chosen by generality, as in Part VI: source 16 for the word isomorphism and the termination classification (every placement of a terminator, unrestricted and injective words), source 18 for centered increasing words and the zero conventions (the only source treating them), source 15 for the other comparison orders (the only source proving theorems about them). Source 13's central termination theorem and source 18's centered theorem are the same special case of source 16's criterion, proved independently, with the same letter code; they are printed as routes. Source 18 was written before sources 13–17 arrived. Re-proofs of Part II are printed as notes. Source 16 uses the macro names of this report for other objects (its `\Words` is the class of *all* words, its `\I` the class injections); here they are printed $\mathcal{T}$ and $\mathfrak{J}$, and $\mathcal{I}$ keeps its meaning of injective words (section 1.3.1). Labels written for the merge in this Part begin `swo:vii:`.

34 The six sources' maps of interpretations

[merge] Each batch-81 source opens with a map of what “all well-orders of the surreal numbers” can mean, as sources 12, 09 and 08 did (section 4). The six maps are printed here, each whole, because their rows on varying carriers, termination, increasing subsets, relation codes and abstract types are the subject of this Part; the subsections on a fixed set carrier that sources 14 and 17 put inside their maps are printed in Part VI (note 32.5 and section 32.3). The six agree with one another and with the batch-80 sources; note 34.2 lists what they share.

34.1 Source 13: the interpretations that must be separated

34.1.1 A taxonomy by carrier and representation

The following table fixes the principal meanings. In every row, numerical comparison of surreal labels is distinct from the auxiliary well-order being represented.

Interpretation	Representation	Logical size and caveat
Fixed set $X \subseteq \mathbf{No}$	Well-order relations on X , equivalently bijections $\alpha \rightarrow X$ from set ordinals	A set; every possible well-order type on X is permitted.
Fixed length on X	Bijections $\lambda \rightarrow X$	A single stratum; $ \lambda = X $ is necessary.

Interpretation	Representation	Logical size and caveat
Arbitrary set subsets	Injective words $p : \alpha \rightarrow \mathbf{No}$	An ordinary class of sets; termination must be specified.
Words with repetitions	Arbitrary $p : \alpha \rightarrow \mathbf{No}$	An ordinary class, but not a class of well-orderings of the ranges.
Inherited numerical order	Subsets on which the restriction of $<$ is a well-order	Not arbitrary relabelings; increasing enumerations only.
Set-like whole-class orders	Class bijections $E : \text{Ord} \rightarrow \mathbf{No}$	The virtual domain $\mathfrak{W}_{\text{sl}}$; its elements are classes.
Unrestricted whole-class orders	Class well-order relations on $\mathbf{No}$	The virtual domain $\mathfrak{W}_{\text{all}}$; proper-class predecessor segments are allowed.
Unlabeled order types	Quotients by order-isomorphism	Label-sensitive lexicographic comparison does not descend.
Relation-code order	Characteristic bits of a relation, read in a chosen order of pairs	A different, coding-dependent lexicographic order.

This covers the standard semantic axes, not every artificial encoding one could invent. Birthday-first comparison, a fixed global well-order of the labels in place of their numerical order, or comparison of unquotiented Conway expressions each changes the problem again.

34.1.2 Fixed carriers have no end-marker ambiguity

For a set X , put

$$\mathcal{W}(X) = \{e : \alpha \xrightarrow{\sim} X : \alpha \text{ an ordinal}\}, \quad \mathcal{W}_\lambda(X) = \{e \in \mathcal{W}(X) : \text{dom}(e) = \lambda\}.$$

These are sets: if $|X| = \mu$ is infinite, the possible domains are among the ordinals below μ^+ of cardinality μ .

Lemma 34.1 (Prefix-freeness of exhaustive enumerations). *Distinct members of $\mathcal{W}(X)$ are never proper prefixes of one another. Their lexicographic comparison is determined at a least common coordinate where their values differ.*

Proof. A prefix that already enumerates all of X cannot be extended injectively by another member of X . If two different enumerations agreed on their common domain, the shorter would be such a prefix. Otherwise their set of disagreement coordinates has a least member. $\square$

An auxiliary well-order R on X has one ordinal type and one increasing enumeration in that order. This is a bijection between relation representations and the displayed exhaustive words. Lexicographic comparison uses the *numerical* order of their values, not the order R itself.

34.1.3 Varying the range introduces a genuine convention

An injective word $p : \alpha \rightarrow \mathbf{No}$ enumerates a well-order of its range. Once ranges may differ, p can be a proper prefix of another word. Under *end-first* dictionary comparison, $p < p \frown x$ for every fresh x . Under end-last comparison the inequality is reversed. A terminal symbol placed inside an enlarged alphabet gives a third possibility.

Deleting repeated letters from arbitrary words is not a harmless change of representation. For example, with $a < b$, the two words (a, a, b) and (a, b) have different dictionary positions but the same first-occurrence enumeration. No quotient order is supplied merely by performing this deletion.

34.1.4 Inherited well-orders and unlabeled quotients

The numerical order of all surreals is not a well-order: it contains, for example, the descending sequence $0, -1, -2, \dots$. Its ordinal subclass, represented by all-plus signs, is a proper-class increasing well-order. Every set ordinal is likewise realized as a numerically well-ordered subset of **No**.

The class of *set-sized* numerically increasing words is an ordinary class of set codes. Under end-first comparison it has an empty minimum and the limit-prefix phenomenon proved in [Section 35.2](#). It therefore differs from the surreal line despite containing a copy of **No** through singleton words. The general class-universality argument below embeds it back into **No**: equimorphism is not isomorphism.

If labels are forgotten altogether, lexicographic comparison is usually lost. On a two-element labeled carrier the two enumerations have the same ordinal type but opposite lexicographic positions. More dramatically, every individual set-like global well-order of **No** has type Ord ; its quotient by abstract order-isomorphism has just one element, although $\mathfrak{M}_{\text{sl}}$ itself is extremely rich.

34.1.5 The Kanovei–Shelah comparison is related, not identical

Kanovei and Shelah’s definable hyperreal construction uses a linearly ordered family of maps from the continuum into $\mathcal{P}(\mathbb{N})$ whose ranges are ultrafilters. The maps are compared lexicographically; injectivity is not the defining requirement [16]. This is an important motivation for ordered collections of enumeration codes. It should not be identified without argument with the exhaustive well-orderings of one fixed labeled carrier. In particular, replacing its index by “all surreal well-orders” would change both the coding and the logical size level.

34.2 Source 14: which interpretation is intended?

34.2.1 A map of the natural meanings

In the table below, “ordinary class” means a collection of sets defined by a class predicate. “Virtual” means that each candidate is itself a class and the theory quantifies over candidates without forming their collection as another class.

Interpretation	Objects and comparison	Size and main warning
Fixed set $X \subseteq \mathbf{No}$	All well-order relations on X , compared by their exhaustive labelled ordinal enumerations.	A set; arbitrary well-order type of cardinality $ X $ is allowed.
Fixed length α	Bijections $\alpha \rightarrow X$, numerical first-disagreement comparison.	A set; one slice of the preceding order.
Birthday fragment	$X = \mathbf{No}_{<\theta}$, with either of the preceding choices.	A set; the ordinal cutoff and cardinality play different roles.
Variable set domain	Pairs (A, R) with $A \subseteq \mathbf{No}$ a set and R a well-order of A , or injective set-length surreal words.	An ordinary proper class; a termination convention is indispensable.

Interpretation	Objects and comparison	Size and main warning
Unrestricted words	All functions $\alpha \rightarrow \mathbf{No}$ for set ordinals α , allowing repeated values.	An ordinary proper class; these are not enumerations of well-ordered labelled subsets.
Increasing subsets	Subsets or subclasses well-ordered by the <i>numerical</i> order inherited from $\mathbf{No}$.	A restriction on labels, not an arbitrary imposed well-order.
Set-like global order	Class bijections $\text{Ord} \rightarrow \mathbf{No}$.	Virtual order $\mathfrak{W}_{\text{sl}}$; each proper initial segment is a set.
Arbitrary global order	Class well-order relations on all of $\mathbf{No}$, including non-set-like ones.	Virtual order $\mathfrak{W}_{\text{all}}$; direct labelled common-prefix comparison is available.
Relation-code lex	Characteristic functions of relations, using a fixed well-ordered schedule of ordered pairs.	A different order; depends on the schedule and on the bit convention.
Order types only	Isomorphism classes after forgetting surreal labels.	Enumeration lex does not descend. Set-like proper class orders all have type Ord .

This list covers the main structural ambiguities, rather than pretending that arbitrary encodings cannot create additional variants. Every subsequent theorem names the interpretation to which it applies.

[merge] Source 14’s Subsection 2.2 (“Fixed carriers and the inherited spectra”) is printed in Part VI (note 32.5); its Subsections 2.3–2.5 follow.

34.2.2 Variable domains and the terminal symbol

An imposed well-order on a set $A \subseteq \mathbf{No}$ is represented uniquely by an injective word $p : \lambda \rightarrow \mathbf{No}$. For varying A , a word may be a proper prefix of another. Three reasonable choices are: *prefix first*, using $\mathbf{end} < x$ for every numerical letter; *prefix last*, using $x < \mathbf{end}$; and a marker inserted at a specified cut in the numerical alphabet. A fourth order is length-first lexicography. They do not agree, and none becomes a well-order simply because the word domains are ordinals: the one-letter words already contain the numerical order of $\mathbf{No}$.

The familiar sign-sequence order of the surreals uses $- < \mathbf{end} < +$. It is not prefix-first dictionary order. A useful self-delimiting numerical code replaces $-$ by 00, $+$ by 11, and appends 01. Comparisons are preserved because

$$00 < 01 < 11.$$

The terminal block makes distinct numerical codewords prefix-incomparable. Concatenating such codes is useful for embeddings, but does not turn arbitrary repeated-letter words into exhaustive enumerations. The predecessor report treats the resulting compression and the special cuts at limit word lengths (sections 9 to 11; Sections 9–11 of this report at the pin).

34.2.3 Numerically increasing well-orders

The numerical order of **No** is not a well-order; for example, $0 > -1 > -2 > \dots$. Hence no numerical order-preserving enumeration can well-order all of **No**. This is different from imposing another well-order on the same carrier.

Proper subclasses can nevertheless be numerically well-ordered and non-set-like. For instance,

$$\left\{ -\frac{1}{\alpha + 1} : \alpha \in \text{Ord} \right\} \cup \{0\}$$

together with numerical order has type $\text{Ord} + 1$. The ordinal arithmetic in the denominator is interpreted by the corresponding surreal ordinal followed by surreal addition; the displayed values are strictly increasing and all precede 0. Thus a numerically bounded subclass can have a point with a proper class of predecessors. Set-likeness is not a consequence of numerical boundedness.

34.2.4 Forgetting labels changes the problem

For a fixed infinite set of cardinality μ , its possible abstract well-order types are the ordinals α satisfying $\mu \leq \alpha < \mu^+$. Ordering these types by ordinal size gives a well-order of type μ^+ , which cannot embed in $\mathcal{W}(X)$ by [Note 32.5](#). Therefore the order on abstract types is not the induced quotient of enumeration lexicography.

More directly, different enumerations of the same type need not occupy a convex subset: choose first labels $x < y < z$ and complete the first and third to type μ , the middle to type $\mu + 1$. All three completions exist, and their lexical positions are decided by those first labels. For a proper class carrier every set-like well-order has abstract type Ord , so quotienting that fragment by abstract type collapses it to a single type. Almost all of the lexicographic information was in the labels, not in the abstract order type.

34.3 Source 15: a map of the interpretations

Every concrete version is determined by six choices: the carrier; whether orders are labelled or identified up to isomorphism; how an order is coded; which order compares the code entries; which well-order scans the entries; and the universe in which all codes are collected. Prefix conventions add a seventh choice when the coded sequences have variable domains and need not exhaust a common carrier.

Interpretation	Objects and comparison	Main distinction
Fixed carrier, all types	All ordinal bijections $e : \alpha \rightarrow X$, compared at first unequal labels in X	A set in ZFC; exhaustive codes are prefix-free.
Fixed enumeration type	Only $e : \alpha \rightarrow X$ for one prescribed α	Same possible ordinal suborders need not mean the same binary coding length.
Variable set carrier	Injective set-length words, or all well-ordered sets of surreal labels	Proper prefixes occur; termination must be ordered explicitly.
Natural well-ordered subsets	Subsets $A \subseteq \mathbf{No}$ with the inherited numerical order well-founded	No arbitrary rearrangement is part of the data.

Interpretation	Objects and comparison	Main distinction
Set-like class orders	Bijections $\text{Ord} \rightarrow \mathbf{No}$; first ordinal difference	All predecessors are sets; a predicate on class functions.
All labelled class orders	Arbitrary class well-orders on $\mathbf{No}$; common labelled prefix	May differ only after a proper class of identical earlier labels.
Fixed class index type	Bijections $J \rightarrow \mathbf{No}$ for one given class well-order J	A chosen type such as $\text{Ord} + \text{Ord}$ restricts the domain.
Inverse-rank lexicography	Rank values of labels scanned in a fixed baseline well-order	Uses ordinal values rather than numerical surreal labels; has different endpoints.
Relation-table lexicography	Characteristic bits of $R \subseteq \mathbf{No}^2$ in a fixed well-ordered scan	Depends on scan and bit orientation; differs already on three points.
Isomorphism types	Quotient well-orders by abstract order isomorphism	Native enumeration lex does not descend to this quotient.
Birthday/simplicity variants	Birthday ranks, sign strings, or simplicity predecessors	Birthday ties need a separate well-order; simplicity is only partial.
Definable or computable codes	A specified syntactic class with an evaluator	Restricts the available objects; syntax alone does not supply universal truth.
Higher-universe totality	All relevant relations become sets in a larger universe	Results depend on the class realization and external well-foundedness.

This table is exhaustive at the level of the structural choices just listed, not a claim to enumerate every artificially possible encoding. Once arbitrary encodings and arbitrary scans are allowed, one can manufacture many further linearizations. The mathematically meaningful task is to say which properties are intrinsic to the chosen order and which depend on its presentation.

34.3.1 The historical Kanovei–Shelah construction

The original construction uses

$$A = \{a : \mathfrak{c} \rightarrow \mathcal{P}(\mathbb{N}) : \text{ran}(a) \text{ is an ultrafilter on } \mathbb{N}\},$$

ordered lexicographically in the initial ordinal $\mathfrak{c}$ coordinates, with $\mathcal{P}(\mathbb{N})$ itself ordered through characteristic functions. Maps may repeat entries, and different maps may represent the same ultrafilter. This is an important precedent for ordering presentations without selecting a preferred presentation. It is not literally an order of all well-orders of $\mathbb{R}$ or of $\mathbf{No}$ [16, Section 1].

34.4 Source 16: a systematic map of the interpretations

Three choices account for most natural readings: the carrier to be well-ordered, whether numerical labels are retained, and the coding used for comparison. A fourth choice, the treatment of termination, becomes necessary when carriers or word lengths vary. There cannot be a literal list of every possible encoding: composing any encoding with an arbitrary permutation can manufacture another order. The following table covers the principal mathematical interpretations and explains how the remaining variants arise.

Interpretation	Objects and comparison	Size and main issue
Fixed set of labels $X \subseteq \mathbf{No}$	Bijections $\alpha \rightarrow X$; first unequal entry compared numerically	A set in ZFC; the order on enumerations is usually far from a well-order.
Fixed enumeration type	Only bijections $\alpha \rightarrow X$, with α fixed	Changing α changes topology and may change embedding invariants.
Variable set carrier	Injective set-length words in $\mathbf{No}$	An ordinary class of set codes; proper-prefix comparison must be specified.
Numerically well-ordered subsets	Increasing enumerations in the numerical order	No arbitrary auxiliary well-order remains on each carrier; proper class subsets can still be non-set-like.
Set-like global well-orders	Class bijections $\text{Ord} \rightarrow \mathbf{No}$	A predicate on class variables; not an ordinary class of such graphs.
Arbitrary global class well-orders	Class relations on all of $\mathbf{No}$; compare after the common labelled prefix	Common prefixes can be proper classes; ordinal-coordinate comparison alone is insufficient.
Relation-table encoding	Binary membership tables for order relations, read along an auxiliary well-order of pairs	Gives a different, coding-dependent lexicographic order.
Abstract order types	Forget surreal labels and identify isomorphic well-orders	Numerical first-disagreement comparison no longer descends to the quotient.
Larger-universe interpretation	Regard internal classes as sets in an explicit containing universe	Makes entire collections available, while changing the meaning of “set”, “class”, and “all”.

34.4.1 Fixed labels versus abstract types

For a linearly ordered set $(X, <)$, write

$$\mathcal{W}_\alpha(X) = \{e : \alpha \xrightarrow{\sim} X\}, \quad \mathcal{W}(X) = \bigcup_{|\alpha|=|X|} \mathcal{W}_\alpha(X).$$

A bijection is identified with the well-order it induces on X . Distinct exhaustive enumerations cannot be proper initial segments of one another: the shorter enumeration has already used every label, leaving no fresh next entry for the longer one. Therefore every distinct pair has a first unequal entry, with no termination convention needed.

If labels are discarded, the order types are simply the ordinals α of cardinality $|X|$. Their ordinary ordinal comparison is meaningful, but it is not numerical lexicography. Even two permutations of a finite set have the same abstract order type and different lexicographic positions. For arbitrary class well-orders, a single class of all abstract “class ordinals” is not supplied by GB or KM. Direct labelled comparison will avoid this issue entirely.

34.4.2 Relation tables genuinely change the answer

Choose an auxiliary well-order $j : \delta \rightarrow X \times X$ and encode a relation R by the binary word $\chi_R(j(\xi))$. The earliest differing bit defines a linear comparison, but its answer depends on j and the bit

convention. For example, on $X = \{0, 1, 2\}$ the enumeration $(0, 2, 1)$ precedes $(1, 0, 2)$ numerically. In row-major relation-table order, their first differing entry is the pair $(0, 1)$: its bits are respectively 1 and 0. The table comparison therefore gives the opposite inequality. A class version needs a specified class ordering of the pairs and appropriately stratified quantification; the word “lexicographic” alone does not choose one.

34.4.3 The Kanovei–Shelah connection

The motivating analogy should not be mistaken for an identification. Kanovei and Shelah use the index

$$A = \{a : \mathfrak{c} \rightarrow \mathcal{P}(\mathbb{N}) : \text{ran}(a) \text{ is an ultrafilter on } \mathbb{N}\},$$

ordered at the first differing coordinate by binary characteristic-function order on $\mathcal{P}(\mathbb{N})$. Repetitions are permitted, and the displayed definition admits ultrafilters generally. The point is a definable order on redundant codes, used in an ultrapower construction. It is not literally the space of all bijective enumerations or all well-orders of $\mathbb{R}$ [16, §2 in the published version; §1 in the preprint].

34.5 Source 17: what can “all well-orders of surreal numbers” mean?

We use the sign-sequence presentation and numerical order of surreal numbers; standard background is given in [11, 12]. A well-ordering relation and the numerical order of its labels are distinct. An enumeration of a well-order need not be numerically increasing. This single distinction already produces two different research programs.

[merge] Source 17’s Subsection 2.1 (“Labelled well-orders of a fixed set”) is printed in Part VI (section 32.3). Its displayed word code keeps its number as (S17.1).

34.5.1 Set-length words and varying supports

Every well-order of a set-sized subset of **No** is represented by an injective word $w : \alpha \rightarrow \mathbf{No}$. Now one word can be a proper prefix of another. We use the *shorter-prefix-first* convention unless specified otherwise. The resulting carrier is an ordinary class of sets. Allowing repeated entries gives a larger class of words, but such a word no longer represents a well-order of its range without an additional occurrence labelling.

There is a useful explicit compression, also present in the source. Represent $x \in \mathbf{No}$ by its sign sequence. Replace a minus sign by $--$, a plus sign by $++$, and append the terminal block $-+$. Call the resulting prefix-free code $c(x)$. The comparison

$$-- < -+ < ++$$

encodes the surreal convention “minus < termination < plus”. For any set-length word put

$$\Phi(w) = \bigcap_{i < \alpha} (+ \cap c(w(i))). \quad (\text{S17.1})$$

At a first differing entry the prefix-free coefficient codes determine the comparison; at a proper word prefix the next sign is $+$, so the longer word is larger. Hence Φ is an order embedding into **No**, with birthday

$$\text{bd } \Phi(w) = \sum_{i < \alpha} (1 + 2 \cdot \text{bd}(w(i)) + 2).$$

Products and sums in this formula are ordinal operations. All lengths are sets. The empty word is least, so the entire word order is not isomorphic to the endpoint-free surreal line, despite embeddings

in both directions. Using $-$ instead of the leading $+$ in each entry block gives the longer-prefix-first convention and makes the empty word greatest.

Length-first comparison, often called shortlex, is another legitimate linear order, but is not this dictionary order. It should not be silently substituted in an embedding or topology theorem.

34.5.2 Numerically well-ordered subsets

A different interpretation asks for subsets $A \subseteq \mathbf{No}$ whose *inherited numerical order* is a well-order. Their set-sized increasing enumerations form a suborder of the injective word order. They include every singleton, so their lexicographic order contains a copy of $(\mathbf{No}, <)$ and is not a well-order. They inherit the embedding (S17.1). Every set-sized increasing word admits a longer increasing extension, because a set of surreal numbers has a strict upper bound.

There is no numerically increasing exhaustive well-order of the whole surreal line: the sequence $0, -1, -2, \dots$ already witnesses the failure of well-foundedness of the numerical order. This does not prohibit an arbitrary labelled class well-order of $\mathbf{No}$.

Proper-class *subsets* with their inherited numerical order must also be distinguished from the entire numerical line. They can be non-set-like well-orders. For example,

$$\left\{ -\frac{1}{\alpha + 1} : \alpha \in \text{Ord} \right\} \cup \{0\}$$

is increasingly well-ordered in type $\text{Ord} + 1$. The reciprocal is surreal field reciprocal, and α denotes its ordinal surreal value. The set of predecessors of 0 in this class order is proper. A collection of all such class subsets again lives at the higher logical level. For varying supports, proper-prefix comparisons require an explicit termination convention, just as for set words. Our main raw comparison keeps the carrier fixed and therefore has no termination ambiguity.

34.5.3 Global enumerations and unrestricted relations

A *set-like* class well-order is one in which each element has a set of predecessors. A set-like well-order of the proper class $\mathbf{No}$ has order type Ord , and is equivalently a class bijection

$$E : \text{Ord} \longrightarrow \mathbf{No}.$$

Write $\mathfrak{W}_{\text{sl}}$ for this predicate on class relations, and $\mathfrak{J}$ for the analogous predicate of injective, not necessarily exhaustive, Ord -words. These symbols are not names of ordinary GB classes whose elements are class functions.

Write $\mathfrak{W}_{\text{all}}$ for arbitrary class well-orders of $\mathbf{No}$. Orders of type $\text{Ord} + 1$, $\text{Ord} + \text{Ord}$, and many other non-set-like types belong here. For example, enumerate $\mathbf{No} \setminus \{a, b\}$ first and append a, b . Interchanging those last two labels creates a first disagreement after a proper-class prefix, not at a set ordinal.

One can also fix a particular class well-order J and compare bijections $J \rightarrow \mathbf{No}$ at their least differing J -position. This is legitimate when least elements of nonempty admitted subclasses of J are part of the hypothesis. It is a fixed-domain alternative, not a universal set ordinal index for every class well-order.

34.5.4 Relation codes, isomorphism types, and ordinal-valued rankings

Fix a class bijection $d : \text{Ord} \rightarrow \mathbf{No} \times \mathbf{No}$. A class relation R has binary code $\xi \mapsto \mathbf{1}_R(d(\xi))$, which can be compared lexicographically. This yields a valid order on the admitted well-order relations,

but depends on d and the bit convention. It is generally not the native labelled-enumeration order. Even on $\{0, 1\}$, listing coordinate $(0, 1)$ first and using $0 < 1$ compares the two well-orders in the reverse of their first-label comparison.

Abstract isomorphism types form yet another subject. For set well-orders they are ordinals, naturally ordered by initial-segment embeddability. The native lexicographic order does not descend to this quotient: the two enumerations (a, b) and (b, a) have the same abstract type but opposite label order when $a < b$. A rank function, a birthday-first code, or a chosen well-order of relation codes introduces additional structure rather than recovering a canonical lexicographic comparison from the abstract types.

34.6 Source 18: interpretations and foundational conventions

34.6.1 A map of the natural interpretations

Object	What a point represents	Size and comparison
$\mathcal{W}(X)$, $X \subset \mathbf{No}$ a set	A labelled well-order of the fixed carrier X , in any possible ordinal type	A set; compare unique exhaustive ordinal enumerations
$\mathcal{W}_\alpha(X)$	A well-order whose type is the specified ordinal α	A stratum of $\mathcal{W}(X)$; nonempty exactly when $ \alpha = X $
Injective set words	A well-order of a variable set-sized subset of $\mathbf{No}$	A genuine proper class of sets; a termination convention is needed
Increasing set words	A subset well-ordered by the numerical surreal order, with its unique increasing enumeration	A subclass of the previous word class; no arbitrary reordering of a fixed subset
$\mathfrak{W}_{\text{sl}}(\mathbf{No})$	A set-like global well-order, equivalently a class bijection $\text{Ord} \rightarrow \mathbf{No}$	A predicate on class relations; distinct enumerations have a set ordinal first difference
$\mathfrak{W}_{\text{all}}(\mathbf{No})$	Any class well-order of the entire surreal carrier	A predicate on class relations; the common labelled prefix can be proper
Inherited class well-orders	A subclass on which numerical $<$ is a well-order	A different restriction: $\mathbf{No}$ itself is not numerically well-ordered
Order types modulo isomorphism	Labels are forgotten	Numerical lexicography does not descend to this quotient
Characteristic-relation codes	A well-order relation encoded by its membership bits in a chosen order of pairs	A valid auxiliary lex order, usually different from enumeration comparison
Containing-universe objects	All orders admitted by a specified smaller set/class universe	Sets in an external universe; “all” is relative to the chosen class realization

There are indefinitely many artificial encodings. The table covers the natural mathematical choices by varying carrier, enumeration type, admissible order, label retention, comparison rule, and size level. An arbitrary encoding cannot be expected to preserve the results for another row.

[merge] Source 18’s Subsections 2.2 and 2.3 (sign sequences with its lemma on surreal order tools, and set and class theories) are foundations, not interpretations, and are not printed in this

Part. Its Subsection 2.4 follows.

34.6.2 Why the Kanovei–Shelah connection needs qualification

Kanovei and Shelah’s published construction [16, §2] orders maps $a : \mathfrak{c} \rightarrow \mathcal{P}(\mathbb{N})$ whose ranges are ultrafilters, using characteristic-function lexicography on $\mathcal{P}(\mathbb{N})$. Repetitions are allowed in those maps and in the ultrafilter presentation. Their index is therefore not literally $\mathcal{W}(\mathbb{R})$ or a permutation order of the reals. The useful connection is the method of linearly ordering many codes without choosing one privileged code. The present article studies different, precisely specified orders inspired by that method. This correction was already explicit in the repository’s reals report [1].

Note 34.2 ([merge]; what the six maps share). All six separate the same carriers as the batch-80 sources (sections 2 and 4.3): a fixed set with all types or one type, varying set carriers (injective words, with a termination convention), unrestricted words, numerically well-ordered subsets, set-like and arbitrary class well-orders, relation codes and abstract types.

- *Termination.* Sources 13, 14, 16 and 17 name the three placements of a terminator (first, last, inside the alphabet) and length-first order as a fourth choice; source 14’s warning that the Conway sign order uses $- < \text{end} < +$ and is not prefix-first dictionary order is the convention printed in section 5.
- *Increasing subclasses.* The example $\{-1/(\alpha + 1) : \alpha \in \text{Ord}\} \cup \{0\}$ of numerical type $\text{Ord} + 1$ is source 12’s (section 18); sources 14, 15, 16, 17 and 18 repeat it.
- *Relation codes.* The two- and three-point examples of sources 14, 16, 17 and 18 make the point of source 12’s shipped example and of proposition 17.1; source 15 proves theorems about these orders (theorem 38.2 and corollary 38.3).
- *Abstract types.* That enumeration lexicography does not descend to abstract order types (sources 13, 14, 15, 16, 17) is printed with source 15’s non-convexity example in section 38; source 14’s argument that the type order μ^+ cannot embed uses note 32.5.
- *Kanovei–Shelah.* The Kanovei–Shelah subsections of sources 13, 15, 16 and 18 printed here (sources 14 and 17 discuss it elsewhere in their texts) make source 08’s point (section 4.4): the index there is a lexicographically ordered family of redundant codes, not a space of well-orders. Source 16 cites Section 2 of the published version and Section 1 of the preprint; source 15 cites “Section 1”.
- *Prefix-freeness.* Source 13’s lemma 34.1 is the prefix-freeness of exhaustive enumerations used in section 6.

35 Shorter-first words: three classes, one prefix order (source 16; routes of sources 13, 14 and 18)

[merge] Source 16’s Section 7, printed whole; its $\mathcal{T}$, $\mathcal{T}^{\text{inj}}$, $\mathcal{T}^{\text{inc}}$ are printed $\mathcal{T}$, $\mathcal{I}$, $\mathcal{T}^{\text{inc}}$ (section 1.3.1), and its equations keep their numbers as (S16. n). Its references to its own fresh-separator lemma, universality proposition, set-cut display and raw-comparison section are printed as references to lemma 5.2, proposition 5.13, and definition 5.1 and theorem 14.2, which state the same facts. The word orders of Part II (part II) are the cases $\mathcal{T}$ and $\mathcal{I}$; what this section adds is the increasing class $\mathcal{T}^{\text{inc}}$ and the isomorphism of the three (theorem 35.1). Source 18 proves the isomorphism independently, with the uniformity of the back-and-forth made explicit (section 35.5); source 13’s and source 18’s word sections and source 14’s explicit codes follow as notes and routes.

Let $\mathcal{T}$ be the class of all set functions $f : \alpha \rightarrow \mathbf{No}$ with α an ordinal. Let $\mathcal{I}$ be its injective subclass, and $\mathcal{T}^{\text{inc}}$ its numerically strictly increasing subclass. Members of all three are sets. In $\mathcal{I}$ the word imposes a well-order on its image; in $\mathcal{T}^{\text{inc}}$ that well-order is the restriction of the numerical order. Initially give all three the convention that a proper prefix is smaller than its extension.

The empty word is then a minimum. More substantially, if f has nonzero limit length λ , its proper prefixes are cofinal below it, giving

$$\text{cf}(\mathcal{T}_{<f}) = \text{cf}(\lambda). \quad (\text{S16.5})$$

At successor-length words there is no set cofinal family below the point, and above any word there is no set coinitial family. For the unrestricted and injective cases these facts appear in the earlier report; they follow by choosing a fresh next letter between the set of active coordinates and the coordinate of the given word. Equation (S16.5) already prevents an isomorphism with $\mathbf{No}$, even if the empty word is removed: a surreal cannot be the supremum of a set lying strictly below it, by the set-cut property (definition 5.1).

Theorem 35.1 (Three interpretations with the same prefix order). *In GBC,*

$$\mathcal{T} \cong \mathcal{I} \cong \mathcal{T}^{\text{inc}}$$

under shorter-prefix-first comparison. There are isomorphisms preserving word length, the prefix relation, unions of set-length prefix chains, and every singleton word. The same assertion holds for the convention that proper prefixes are larger than their extensions.

Proof. For an admissible output prefix q , its possible next letters form, respectively,

$$\mathbf{No}, \quad \mathbf{No} \setminus \text{ran}(q), \quad U(q) = \{x \in \mathbf{No} : \forall y \in \text{ran}(q) (y < x)\}.$$

Each is a proper class with the set-cut property. For the second use the fresh-separator lemma (lemma 5.2). For the third, given sets $A < B$ in $U(q)$, fill the cut $A \cup \text{ran}(q) < B$; this also proves the property with an empty side. Uniform back-and-forth as in class universality (proposition 5.13) gives, for every set parameter q , an order isomorphism h_q from $\mathbf{No}$ onto its available-letter class. Take $h_\emptyset$ to be the identity. A single class relation $H(q, x, y)$ defines all these maps: a pair belongs when it appears in the uniquely specified set-length back-and-forth history. No choice of a proper class for each index is hidden here.

For a free word $f : \alpha \rightarrow \mathbf{No}$, define its output by set recursion:

$$q(i) = h_{q \upharpoonright i}(f(i)) \quad (i < \alpha).$$

The output is injective or increasing according to the target rule. Conversely, an admissible output q uniquely recovers $f(i) = h_{q \upharpoonright i}^{-1}(q(i))$. These operations are inverse and preserve prefixes and lengths. At a first differing coordinate, both inputs use the same strictly increasing local map, so their comparison is preserved. The endpoint termination rule depends only on the prefix relation, which is also preserved. The recursive definition makes compatibility with unions immediate, and the choice at the empty prefix fixes singletons. $\square$

This is an isomorphism theorem, not a literal identification of the objects. Longer labels generally change. In particular, the theorem does not turn an arbitrary chosen well-order of a subset into the numerical order on the *same* subset. It shows that the bare lexicographic order does not recover that conceptual distinction.

35.1 Increasing proper classes are a further case

The preceding theorem concerns set-length words. A numerically well-ordered proper subclass can itself be non-set-like. For example, the increasing class

$$\left\{ -\frac{1}{\xi + 1} : \xi \in \text{Ord} \right\} \cup \{0\}$$

has order type $\text{Ord} + 1$; the point 0 has a proper class of predecessors. The simpler subclass of nonnegative ordinal surreals has type Ord . In GBC, class universality ([proposition 5.13](#)) embeds every class well-order into the numerical surreal line, so every such abstract order can occur on a numerically well-ordered subclass. No increasing enumeration can exhaust **No**: it has a first value and omits every smaller surreal.

If distinct proper class carriers are allowed, comparison must also say what happens when one whole labelled class order is an initial segment of another. The direct common-prefix method of Part III ([theorem 14.2](#)), with an explicitly placed termination symbol, supplies the corresponding variant. It is a different problem from comparing well-orders on the same fixed carrier.

Note 35.2 ([merge]; what transfers from Part II). Display (S16.5) and the two sentences after it are [theorem 10.1](#) (sources 11 and 09: [note 11.4](#)). Because the isomorphisms of [theorem 35.1](#) preserve length and the prefix relation, every statement of Part II about $\mathcal{T}$ or $\mathcal{I}$ that is phrased by lengths and prefixes holds in GBC for $\mathcal{T}^{\text{inc}}$: the exact local characters ([theorem 10.1](#)), the word-cut classification ([theorem 11.1](#)) and the successor-word corollary ([corollary 11.2](#)). Source 18 makes this remark for the cut classification ([section 35.3](#)); its direct proof of the characters of $\mathcal{T}^{\text{inc}}$ needs no global choice ([note 35.5](#)). Part II's explicit code ([theorem 9.1](#)) restricts to $\mathcal{T}^{\text{inc}}$ without choice. The example of source 16's Subsection 7.1 is source 12's ([section 18](#)).

35.2 Source 13: set-length words and their first obstruction

[merge] Source 13's Section 5, printed whole. Its proposition is [theorem 10.1](#) (1) of Part II (and source 09's [note 11.4](#)) and is printed as a note.

Let $\mathcal{T}$ be the ordinary class of all set-length words over **No**, and let $\mathcal{I} \subseteq \mathcal{T}$ be the injective words. The latter is exactly the class of well-orderings of arbitrary set-sized subsets, represented by their increasing enumeration in the auxiliary order. In this section the terminal symbol precedes every label.

Note 35.3 (Source 13, Proposition 5.1, "Limit-prefix character"). *Suppose p has nonzero limit length λ . In either $\mathcal{T}$ or $\mathcal{I}$ when p is injective, the proper prefixes of p are cofinal below p . The cofinality of the strict lower side at p is exactly $\text{cf}(\lambda)$. This is [theorem 10.1](#) (1), with the same proof.*

Source 13's proof. If $q < p$, then either q is a proper prefix of p , or its comparison with p is decided by a smaller letter at some coordinate $\xi < \lambda$. In either case a sufficiently long proper prefix of p lies above q and below p . Restricting to a cofinal sequence of lengths gives the upper bound $\text{cf}(\lambda)$.

For the converse, assign to each $q < p$ a bound $\xi_q < \lambda$ beyond which proper prefixes of p lie above q . A family of cardinality less than $\text{cf}(\lambda)$ has a common such bound. A later proper prefix then witnesses its failure to be cofinal. $\square$

Every point of **No** has no set-sized cofinal family strictly below it: a proposed family together with that point is a set cut and has a strict interpolant. Thus the displayed word orders are not isomorphic to **No**, even after the empty minimum is deleted. They nevertheless embed into **No** by class universality ([proposition 5.13](#)), and **No** embeds back through singleton words. The earlier

report gives explicit sign-coding embeddings and a more detailed end-first analysis (sections 9 to 11; Sections 9–11 of this report at the pin).

The same limit-prefix proof applies to numerically increasing set words, since every proper prefix remains increasing. By reversing the position of termination, one obtains the corresponding small *right*-hand character. Section 36.4 proves that a central termination convention has a fundamentally different outcome.

35.3 Source 18: set-length dictionary words and numerical increase

Let $\mathcal{T}$ be the class of all functions $f : \alpha \rightarrow \mathbf{No}$ with α a set ordinal, and let $\mathcal{I}$ and $\mathcal{T}^{\text{inc}}$ be its injective and strictly numerically increasing subclasses. These are ordinary classes: their members are set graphs. In $\mathcal{I}$ a point represents a well-order of its own range; in $\mathcal{T}^{\text{inc}}$ that well-order is the inherited numerical order. No point exhausts $\mathbf{No}$.

First use ordinary dictionary comparison: compare first unequal entries, and put a proper prefix before every proper extension. This convention is sometimes described by a formal terminator below all surreal letters. Such a terminator is not a surreal number.

Note 35.4 (Source 18, Proposition 6.1, “Explicit order coding; repository result”). *There is an explicit order embedding $\mathcal{T} \hookrightarrow \mathbf{No}$, requiring no Global Choice. Singleton words embed $\mathbf{No}$ into each of $\mathcal{T}, \mathcal{I}, \mathcal{T}^{\text{inc}}$. Thus all three word orders and $\mathbf{No}$ are equimorphic, but the word orders are not isomorphic to $\mathbf{No}$. This is theorem 9.1, with the same letter code $K(x) = C(x) = (+) \cap D(s_x) \cap (+, -)$; source 18 labels it a repository result.*

Source 18’s proof. Duplicate each sign in the sign expansion s_x and write the result as $D(s_x)$. The code

$$K(x) = (+) \cap D(s_x) \cap (+, -)$$

is prefix-free and numerically order-preserving. A terminated source word is correctly compared with a longer one because its delimiter satisfies $(-, -) < (+, -) < (+, +)$. Concatenate the $K(f(i))$ in ordinal domain order. Replacement and ordinal summation give a set sign word. At the first unequal entry prefix-freeness preserves the comparison; at the end of a proper prefix the next block begins with $+$, so the shorter word is smaller. The code length is exactly

$$\sum_{i < \alpha} (1 + 2 \cdot \text{bd}(f(i)) + 2).$$

This proves the embedding. The nonisomorphism follows from the next proposition and holds even after removing the empty word. $\square$

Note 35.5 (Source 18, Proposition 6.2, “The prefix-supremum obstruction”). *The empty word is the minimum of each dictionary word order. For any word f of nonzero limit length α , its proper prefixes are cofinal below it, and the left character of f is $\text{cf } \alpha$. At every successor-length word the left segment has no set cofinal subset. At every word the right segment has no set coinital subset. These assertions hold in $\mathcal{T}, \mathcal{I}$, and $\mathcal{T}^{\text{inc}}$. For $\mathcal{T}$ and $\mathcal{I}$ this is theorem 10.1. The case $\mathcal{T}^{\text{inc}}$ also follows in GBC from theorem 35.1, which preserves lengths and prefixes; source 18’s proof below needs no global choice.*

Source 18’s proof. At a limit length, a smaller word is either a proper prefix or differs at a coordinate below α . A sufficiently long proper prefix exceeds it. A cofinal set of prefix lengths therefore gives the upper bound $\text{cf } \alpha$; fewer than that many comparison coordinates are bounded and cannot be cofinal. At successor length, interpolate at the last entry above all relevant lower next values and

below that entry. In the injective case avoid earlier labels; in the increasing case also stay above all earlier labels. These are set constraints and the surreal cut property satisfies them. To defeat a proposed set coinital family above f , append a new value below all of its relevant next values. In the increasing case impose the additional lower bound $\text{ran } f$. Every family member extending f already satisfies that bound, so a separator still exists. $\square$

In particular, an increasing word of length ω is the supremum of its finite prefixes. No surreal number is the supremum of a set strictly below it, since the surreal set-cut property inserts a smaller upper bound. This invariant distinguishes the word orders from $\mathbf{No}$; mutual embeddings do not preserve suprema in the ambient order.

The repository proves a sharper dictionary cut criterion for arbitrary and injective words (theorem 11.1): a set cut $A < B$ fails to have an interpolant exactly when B contains the empty word, or when B has a minimum of nonzero limit length and A is cofinal below that minimum. A common-prefix search proves this: unequal next-value sets can be separated, while a forced common entry is appended; the search is bounded by the input word lengths. Its only terminal failure is a forced upper minimum reached at a limit. Restricting to nonzero successor-length words removes both failures and gives a copy of $\mathbf{No}$ in GBC. The prefix-tree isomorphism proved below transfers this criterion to increasing dictionary words as well.

The following sections give the genuinely different outcomes obtained by placing termination near zero. They treat zero exclusion, zero padding, and a distinct terminator separately; these choices cannot be conflated.

35.4 Source 14: an explicit word embedding with a termination convention

[merge] The first subsection of source 14's Section 11; it gives a second explicit code for the prefix-first word order (a second route to theorem 9.1) and a code for the prefix-last order. The rest of that section (binary class words, the no-coding theorem and the Kanovei–Shelah remark) is not printed in this Part; its relation-code subsection is in section 38.8. The ordinary class of all set-length surreal words is much smaller in logical sort than the virtual collection $\mathfrak{W}_{\text{sl}}$. An explicit coding makes its embeddability into $\mathbf{No}$ transparent and illustrates why the terminal symbol matters. The underlying compression idea is inherited from the predecessor (section 9; Section 9 of this report at the pin); we spell out a version that also permits repeated letters.

For a surreal x , let $c(x)$ be its binary sign code: replace each minus by 00, each plus by 11, and append 01. The three blocks encode $- < \text{end} < +$. Different complete codes are prefix-incomparable, and their first-difference order is the numerical order of their labels. For a word $p : \lambda \rightarrow \mathbf{No}$, concatenate the codes $c(p(\alpha))$ in ordinal order. Now replace each binary 0 by 01 and each binary 1 by 11, and append 00 to terminate the entire word. Finally read 0, 1 as surreal signs $-, +$.

Proposition 35.6 (Source 14, Proposition 11.1, “Prefix-first word coding”). *This construction is an order embedding of the class of all set-length surreal words, with prefix-first lexicography, into $\mathbf{No}$. Its output birthday is*

$$\sum_{\alpha < \lambda} (4 \cdot \text{bd}(p(\alpha)) + 4) + 2,$$

where every sum and multiplication is ordinal arithmetic.

Proof. All lengths and concatenations are sets. At the first differing label, the prefix-free numerical codes determine the correct binary comparison; the outer replacement preserves it. If one whole word is a proper prefix of the other, its outer terminal block 00 is compared with one of the data

blocks 01, 11, making the shorter word smaller. Complete outer codes are prefix-incomparable, so their comparison as surreal sign sequences never reaches an ambiguity about termination. The length of a label code is $2 \cdot \text{bd}(x) + 2$ and outer replacement doubles that length on the left. Summing those lengths and adding the last two signs gives the displayed formula. $\square$

For prefix-last order, use outer data blocks 00, 10 and terminal block 11 instead. The same proof applies. The embeddings do not claim that these word orders are isomorphic to the entire surreal line: for example the empty word is the least point in the prefix-first order. Nor do they turn an Ord-length class word into a set-length surreal. The birthday formula is explicitly a *set* ordinal expression.

35.5 Source 18's route: shorter-first increasing words are not a new order type in GBC

[merge] The first part of source 18's Section 10: a second proof of [theorem 35.1](#), written without knowledge of source 16. Its proof makes explicit that the family of local isomorphisms is one class given by one elementary formula from a set-like global well-order, and its statement adds that zero may be excluded from any of the alphabets. The corollary of that section, on cones of centered increasing words, is printed with them ([corollary 37.9](#)).

The preceding centered results should not be transferred to the shorter-first convention. For shorter-first words a stronger identification is possible, despite the constraints on coordinates.

Theorem 35.7 (A prefix-preserving tree isomorphism). *In GBC, the classes of arbitrary, injective, and strictly increasing set-length surreal words, all with shorter prefixes first, are pairwise order-isomorphic. Isomorphisms can be chosen to preserve every word length and the prefix relation. The conclusion also holds if zero is excluded from any of the alphabets.*

Proof. For a permitted target prefix p , its admissible next-letter class is respectively

$$\mathbf{No}, \quad \mathbf{No} \setminus \text{ran}(p), \quad D_p = \{x \in \mathbf{No} : \text{ran}(p) < x\}.$$

If zero is excluded, delete it as well. Each class is proper and fills every set cut. For the increasing case, simply add $\text{ran}(p)$ to the lower side of the proposed cut; every upper-side letter already exceeds that range. For the injective case use set avoidance.

Fix a set-like global well-order $\prec$. There is a single uniformly defined family of order isomorphisms

$$\theta_p : \mathbf{No} \longrightarrow D_p,$$

where D_p now denotes the relevant admissible class in any of the cases. Here is a construction that makes the uniformity and the choice strength explicit. Restrict $\prec$ to $\mathbf{No}$ and to D_p ; these give ordinal enumerations, definably in the set parameter p . Run the deterministic back-and-forth that, at each stage, treats the least unmatched source point and then the least unmatched target point, always choosing the $\prec$ -least unused filler of the required cut. Every partial map and every cut at an ordinal stage is a set. For each set ordinal, ordinary set recursion defines the unique initial construction, with p and $\prec$ as parameters. The assertion that a pair occurs in some such initial construction is one elementary formula with these parameters. Elementary class comprehension therefore gives all the θ_p 's as one class relation.

In particular, we have not selected an arbitrary class isomorphism independently for each of a proper class of prefixes. No extra class-choice principle, and no recursion with proper-class-valued stages, has been used.

For an arbitrary source word $f : \alpha \rightarrow \mathbf{No}$, construct a target word by set recursion. Given its prefix p at position ξ , put the next target entry equal to $\theta_p(f(\xi))$; take unions at limits. The output is a permitted word of the same length. Conversely, from any permitted target word g recover

$$f(\xi) = \theta_{g|_\xi}^{-1}(g(\xi)).$$

These procedures are inverse. They preserve prefixes, and the first unequal entries are transformed by the same increasing map θ_p . Thus they preserve the shorter-first lexicographic order. $\square$

This theorem sharpens equimorphism: under the displayed choice assumption the three shorter-first word orders are actually the same order type. It does not identify that type with $\mathbf{No}$. Their positive-length limit prefixes still have the familiar small left cofinalities and nontrivial set suprema.

36 Termination conventions: the criterion and all isomorphism types (source 16; routes of sources 13 and 18)

[merge] Source 16's Section 8, printed whole, as the base of this section: it decides, for every placement of a formal terminator in the alphabet $\mathbf{No}$, whether the unrestricted and the injective word orders fill every set cut ([theorem 36.1](#)), and lists every isomorphism type that occurs ([theorem 36.2](#)). Its termination types $\mathcal{T}_{\min}$, T_κ are printed $\mathbf{T}_{\min}$, $\mathbf{T}_\kappa$; its references to its own universality proposition and set-cut display are printed as references to [proposition 5.13](#) and [definition 5.1](#). Source 13 ([section 36.4](#)) and source 18 ([section 36.5](#)) prove the central case independently, with the same letter code ([note 36.3](#)).

Choose an initial/final partition $\mathbf{No} = L_\varepsilon + R_\varepsilon$ and adjoin a formal symbol ε , distinct from every surreal, with

$$L_\varepsilon < \varepsilon < R_\varepsilon.$$

Compare words at the first differing coordinate, treating a terminated word as having the symbol ε there. This includes prefix-first and prefix-last as the two extreme placements. The Conway convention for sign expansions, $- < \varepsilon < +$, uses a two-letter alphabet; it must not be confused with prefix-first dictionary order over all surreals.

Theorem 36.1 (Termination criterion). *Let $\mathcal{V}$ be either $\mathcal{T}$ or $\mathcal{I}$. Its termination order has a strict interpolant for every separated pair of sets, including empty sides, if and only if:*

1. L_ε and R_ε are nonempty;
2. L_ε has a greatest member or has no set cofinal subset;
3. R_ε has a least member or has no set coinitial subset.

The equivalence uses GB and the surreal set-cut property. In GBC these conditions are also equivalent to this word order being isomorphic to $\mathbf{No}$.

Proof. If either side of the threshold is empty, the empty word is an endpoint. If L_ε has no maximum but a set cofinal subset C , the singleton words (c) , $c \in C$, lie below the empty word and have no strict interpolant below it. Any proposed interpolant has a first letter $x \in L_\varepsilon$, and some $c \in C$ is larger than x , placing (c) above it. The right-side argument is dual. This proves necessity.

For sufficiency, start with a separated set cut $A < B$ of words. Follow a common prefix p , keeping active precisely the input words extending it, including equality. Every discarded lower word is below every extension of p , and every discarded upper word is above every such extension.

At the next coordinate let U, V be the sets of active lower and upper symbols, using ε for a word ending at p . Every $u \in U$ is at most every $v \in V$.

If U and V meet, their common member is unique. It cannot be ε , which would put the same word p on both sides. Append the common surreal letter and continue. In the injective case a witness among the active words ensures freshness. Take unions at limits. The search cannot reach an ordinal larger than every input word length, so eventually $U < V$ are disjoint.

If neither set contains ε , choose a fresh surreal strictly between them and append it. Suppose $\varepsilon \in U$, so $V \subseteq R_\varepsilon$. If a numerical member of U already lies in R_ε , ordinary fresh separation works. Otherwise all numerical lower symbols lie in L_ε . If R_ε has no set coinital subset, choose $r \in R_\varepsilon$ below the whole set V , and then a fresh letter strictly between r and V . If R_ε has a least member r not in V , the same interval above r works.

The remaining case is that $r = \min R_\varepsilon$ belongs to V . Append r , which is fresh in the injective case because an active upper word supplies it. All lower comparisons are now decided. For the upper words continuing through r , collect their next numerical letters and choose a fresh d below this set and below some fixed member of L_ε . The extension $p \frown (r, d)$ is below all those upper words, including any terminating just after r , and above every lower word. Thus two letters handle the possible adjacency between the termination symbol and a numerical letter. The case $\varepsilon \in V$ is the reversed construction using $\max L_\varepsilon$ when needed.

This proves interpolation using only a bounded set recursion. The word carrier is proper, since it contains all singleton surreals. In GBC, class universality (proposition 5.13) gives the claimed isomorphism with **No**. $\square$

36.1 Concrete consequences and explicit encodings

Write 0^- for the formal symbol immediately below numerical zero, and 0^+ for the one immediately above zero. All labels, including zero, remain available. For 0^- the sides are $x < 0$ and $x \geq 0$; for 0^+ they are $x \leq 0$ and $x > 0$. Theorem 36.1 gives

$$\mathcal{T}_{0^-} \cong \mathcal{I}_{0^-} \cong \mathbf{No}, \quad \mathcal{T}_{0^+} \cong \mathcal{I}_{0^+} \cong \mathbf{No} \quad \text{in GBC.} \quad (\text{S16.6})$$

This includes the empty word and all limit-length words.

There are also explicit embeddings that do not use global choice. For a sign sequence s_x , let $D(s_x)$ duplicate each sign, and put

$$C_-(x) = D(s_x) \frown (+, -), \quad C_+(x) = D(s_x) \frown (-, +).$$

Both letter codes are prefix-free and preserve numerical comparison, because either terminal block lies strictly between $(-, -)$ and $(+, +)$. The first sign of $C_-(x)$ is negative exactly when $x < 0$, and that of $C_+(x)$ exactly when $x \leq 0$. Concatenating the codes of the entries therefore embeds the corresponding termination order into **No**. The encoded birthday is

$$\sum_{i < \alpha} (2 \cdot \text{bd}(f(i)) + 2). \quad (\text{S16.7})$$

The operations are ordinal operations; the 2 is on the left. At regular infinite birthday cutoffs, words of length below the cutoff whose entries are below it remain below it under this encoding. Singular cutoffs do not satisfy that closure assertion.

For prefix-first words, add an initial $+$ to each letter code. This makes every continued block begin above termination and again gives an explicit embedding into **No**. Singleton words embed **No** back. Thus equimorphism persists even where local characters prevent isomorphism.

An interior threshold alone does not guarantee saturation. Place it between the surreals bounded above by some natural number and those larger than every natural number. The singleton natural-number words are cofinal below the empty word, so the termination order fails the criterion. Furthermore, (S16.6) does not extend to increasing words: a positive increasing word of length ω still has its finite prefixes cofinal below it. Its permissible continuation letters are all positive. Hence the centered order on $\mathcal{T}^{\text{inc}}$ is not η_{Ord} . The isomorphism of Theorem 35.1 specifically uses an endpoint termination convention.

36.2 Complete isomorphism classification of the termination family

There is a stronger classification for unrestricted and injective words. For an infinite regular cardinal κ , place the formal terminator at the threshold

$$L_\kappa = \{x : \exists \alpha < \kappa (x < \alpha)\}, \quad R_\kappa = \{x : \forall \alpha < \kappa (\alpha < x)\},$$

and call the resulting unrestricted word order $\mathbf{T}_\kappa$. The two classes partition **No**: the regular infinite cardinal κ is a limit ordinal. Its left side has cofinality κ , while its right side has no set coinitial family, since the set of ordinals below κ can be separated from any proposed such family by the set-cut property (definition 5.1).

Theorem 36.2 (All termination types). *In GBC, the isomorphism types obtained from every formal termination cut, for either unrestricted or injective set words, are exactly*

$$\mathbf{T}_{\min}, \quad \mathbf{T}_{\min}^{\text{op}}, \quad \mathbf{No}, \quad \mathbf{T}_\kappa, \quad \mathbf{T}_\kappa^{\text{op}} \quad (\kappa \text{ infinite regular}). \quad (\text{S16.8})$$

There are no repetitions in this list. Here $\mathbf{T}_{\min}$ uses shorter-prefix-first comparison. At every point of $\mathbf{T}_\kappa$ the left cofinality is κ and there is no set coinitial family on the right; its dual has the reversed profile.

Proof. The endpoint placements give the first two types, by Theorem 35.1 and numerical negation. For an interior placement, Theorem 36.1 gives **No** unless one threshold side has no endpoint and a set cofinal or coinitial family. Both sides cannot fail this way: the two sets would contradict the set-cut property (definition 5.1). The one small character is an infinite regular cardinal, say κ .

For a maximum-free left threshold of cofinality κ , extensions $p \frown (c)$ with c in a cofinal subset of the left threshold are cofinal below any word p . In the injective case one may move these letters through intervening nonempty intervals to avoid $\text{ran}(p)$. Conversely, the next letters of a set cofinal below p , among its extending members, must be cofinal in that threshold. This proves the left character κ . The right threshold has no set coinitial family, and the same next-letter argument excludes one above p . If a threshold side has an endpoint instead, two-letter extensions as in Theorem 36.1 exclude small local character on that side. These observations also distinguish the profiles in the list.

Two maximum-free threshold cuts with left cofinality κ are conjugate by a numerical automorphism of **No**. Choose increasing cofinal sequences of length κ on their left sides, match them, and extend this set partial isomorphism by back-and-forth. Each left side is determined as the initial closure of its sequence, so the automorphism carries the whole threshold cut to the other one. Applying it entrywise identifies the unrestricted word orders, and likewise the injective ones. The right-sided case is dual.

To identify unrestricted and injective words in this remaining case, fix the cofinal left sequence. For every set S , choose uniformly a cofinal sequence of the same type in $L_\varepsilon \setminus S$, using fresh points in intervening surreal intervals. Start back-and-forth with the matching of these two set sequences.

It gives a uniform isomorphism $h_S : \mathbf{No} \rightarrow \mathbf{No} \setminus S$ preserving both threshold sides. Taking $h_\emptyset$ as the identity, the recursion

$$q(i) = h_{\text{ran}(q \upharpoonright i)}(f(i))$$

is a length- and prefix-preserving isomorphism from free words to injective words, now preserving comparison with the middle terminator as well. All preparatory sequences and stage histories are sets. The saturated cases were already identified with $\mathbf{No}$.

Finally endpoints distinguish the first two types. The surreal type has no small character on either side of any point, whereas the $\mathbf{T}_\kappa$ types have the displayed one-sided character. Those characters distinguish the different regular cardinals and their duals. This proves exhaustiveness and absence of repetitions. $\square$

Every termination order in this list is nevertheless equimorphic with $\mathbf{No}$, even without global choice. For an arbitrary threshold, the increasing map $\phi(x) = x/(1 + |x|)$ can be rescaled separately on its two sides into $(-1, 0)$ and $(0, 1)$. This preserves all letter comparisons and puts termination at zero. The duplicated-sign codes above then give an explicit word embedding into $\mathbf{No}$; singleton words supply an embedding in the opposite direction. Thus equimorphism leaves a proper class of different isomorphism types undistinguished. This classification excludes numerically increasing words with an interior terminator, whose behavior depends on the prefix as explained above.

36.3 Length-first ordering is another independent choice

In shortlex order one compares the ordinal lengths first, and compares entries only when those lengths agree. It is the ordered sum of the fixed-length layers over Ord , rather than one of the termination orders above. Its empty word is a minimum, and two words can be ordered by their lengths even when they disagree numerically at their first entry. This variant is rigorously definable on set codes but should be named explicitly; none of the global-exhaustive results implicitly uses it.

Note 36.3 ([merge]; one letter code, five spellings). With $D(s)$ the sign sequence s with every sign doubled, the letter codes of the six sources and of Part II are the following; each is prefix-free and order-preserving because its terminal block lies strictly between $(-, -)$ and $(+, +)$.

- $D(s_x) \frown (+, -)$: source 16's C_- (S16.7), source 13's $c(x)$ (proposition 36.5) and source 18's centered $C(x)$ (proposition 36.7); birthday $2 \cdot \text{bd}(x) + 2$ per letter, as in (S16.7). Source 16's C_+ is $D(s_x) \frown (-, +)$.
- $(+) \frown D(s_x) \frown (+, -)$: Part II's prefix-first code (theorem 9.1), source 16's “add an initial +” and source 18's $K(x)$ (note 35.4); birthday $1 + 2 \cdot \text{bd}(x) + 2$ per letter. Source 17's code (S17.1) is $(+) \frown D(s_x) \frown (-, +)$, with the same birthday.
- 00, 11, 01 for $-, +$ and termination, read as signs: source 14's inner code (proposition 35.6 and its Subsection 2.3) and source 15's entry code (lemma 31.1); this is source 16's C_+ . Source 14 doubles it once more for the outer word, which gives $4 \cdot \text{bd}(x) + 4$ per letter and 2 for the end.

Source 16 does not cite Part II for its prefix-first code. No source claims that any of these codes is optimal; source 11's question on optimal birthday bounds (research question 61.6) stays open (section 39).

36.4 Source 13's route: central termination restores a surreal word order

[merge] Source 13's Section 12, printed whole as a route to the central case of theorem 36.1. Its order $\mathcal{T}_0$ is the termination order of the alphabet $\mathbf{No} \setminus \{0\}$ with the terminator in the place of the reserved zero, i.e. with threshold sides the negative and the positive surreals; neither side has an

endpoint or a set cofinal (resp. cointial) subset, so [theorem 36.1](#), transported along an isomorphism $\mathbf{No} \setminus \{0\} \cong \mathbf{No}$ in GBC, gives [theorem 36.4](#). The proofs are the same common-prefix search. Its code ([proposition 36.5](#)) is source 16's code C_- (S16.7), with the same birthday. Source 13's own caveat stands: it does not classify all placements of a marker; source 16 does ([theorem 36.2](#)). Its references to its fresh-separator lemma, class universality and core interpolation are printed as references to [lemma 5.2](#), [proposition 5.13](#), and [theorem 19.10](#).

The end-first convention of [Section 35.2](#) treats every extension as larger than its prefix, irrespective of the sign of the new label. We now keep a genuine middle termination value. To avoid collapsing distinct words, that value must not also be an ordinary letter.

36.4.1 The precise alternative

Let $A = \mathbf{No} \setminus \{0\}$. A word $p : \alpha \rightarrow A$ is padded by the surreal number 0 at every coordinate at or beyond α . Compare two different words at the first difference of these padded functions, using numerical comparison in $\mathbf{No}$. Call this *central termination*, and denote the arbitrary-word and injective-word orders by $\mathcal{T}_0$ and $\mathcal{I}_0$, respectively.

For example,

$$(-1) < () < (1), \quad p \frown (-1) < p < p \frown (1),$$

whenever the displayed appended letters are allowed. The empty word is now an interior point, not an endpoint. For injective words, fresh negative and positive letters can always be used in place of $-1, 1$.

Distinct words really are distinguished by padding: at the end of a proper prefix, the longer word has a nonzero letter. If 0 were also allowed as a letter, p and $p \frown (0)$ would have identical padded functions. The resulting rule would fail even to define an antisymmetric order on raw words. Reserving the marker is essential.

Theorem 36.4 (Central-word isomorphism). *In GBC,*

$$\mathcal{T}_0 \cong \mathbf{No}, \quad \mathcal{I}_0 \cong \mathbf{No}.$$

Both isomorphisms concern ordinary classes of set codes. They do not identify the word structure, concatenation, or birthdays with surreal arithmetic.

Proof. It suffices to prove set-cut saturation. Let a set of lower words be strictly below a set of upper words. Choose an ordinal δ strictly above all their lengths. We use an active-prefix argument similar to the interpolation theorem of the core (source 13's Theorem 7.2, [theorem 19.10](#)), but now an active word that ends exactly at the current prefix contributes the value 0.

At a current nonzero-letter prefix p of length β , an active bound agrees with p and has length at least β . Let L_β and U_β be its next letters, or 0 for a bound ending there. Every cross lower value is at most every upper value. If there is a shared value t , it is unique. It cannot be 0: a lower word and an upper word both ending at p would be identical, contradicting the strict cut. Append t and retain the active rows with next value t . In the injective case t is fresh, because it belongs to an active injective word extending p .

If no value is shared, choose a strict separator x between the two sets, avoiding 0, and also avoiding $\text{ran}(p)$ in the injective case. The fresh-separator lemma ([lemma 5.2](#)) supplies it. The *terminated* word $p \frown x$ now lies strictly between every original bound. Every comparison is already decided at this new coordinate or an earlier one, so its termination afterwards is immaterial.

At limits, take the union of the earlier prefixes. The procedure cannot continue indefinitely: after all input lengths have been passed, no bound can remain active. More specifically, a shared

nonzero letter extends some active word and so cannot occur at a stage $\beta \geq \delta$. The construction stops at a set stage. It also works with an empty side or with both sides empty. This proves set-cut saturation of each order. Their elements are sets; apply class universality ([proposition 5.13](#)) to obtain the two isomorphisms. $\square$

This answers the termination-convention direction for an important interior convention. It is not a classification of all possible placements of a marker in every class alphabet. By translating labels, one may reserve any specified surreal c instead of 0. Since $\mathbf{No} \setminus \{0\} \cong \mathbf{No}$, the construction can also be transported to an abstract alphabet order-isomorphic to the full surreal line. Such a transport is not the identity on numerical labels and should not be hidden in the notation.

36.4.2 An explicit sign-code embedding

The isomorphism proof uses class back-and-forth. There is also a direct order embedding, useful for calculations and finite tests.

Given a surreal letter x with sign sequence s_x of length α , replace each $-$ by $--$ and each $+$ by $++$, then append the pair $+ -$. Write the resulting sign block as

$$c(x) = \text{double}(s_x) \frown (+, -).$$

The family of these blocks is prefix-free. If two original sign sequences first differ by actual signs, their doubled versions decide the comparison in the same direction. If one sequence ends first, its terminal pair $+ -$ lies lexicographically between the two possible continuation pairs $--$ and $++$. Thus the same statement holds when the original comparison used termination.

Proposition 36.5 (Direct central-word coding). *For a set word $p : \lambda \rightarrow \mathbf{No} \setminus \{0\}$, concatenate the blocks $c(p(i))$ in order and regard the result as a surreal sign sequence. This defines an order embedding $C : \mathcal{T}_0 \hookrightarrow \mathbf{No}$, and hence also an embedding of $\mathcal{I}_0$, with*

$$\text{bd}(C(p)) = \sum_{i < \lambda} (2 \cdot \text{bd}(p(i)) + 2).$$

All sums and products in this formula are ordinal operations.

Proof. Every block has positive set-ordinal length and the concatenation has set-ordinal length, so it is a surreal. At a first unequal letter, prefix-freeness and order preservation of the block code decide the same comparison before either block is exhausted.

If one outer word is a proper prefix of the other, its code terminates at the beginning of the next block. Because the next letter is nonzero, the first sign of its block has the sign of that letter. Comparison of termination with this sign has exactly the central termination direction. Thus all comparisons are strict and preserved. The birthday formula is the definition of ordinal concatenation. $\square$

No claim of surjectivity is made for this explicit map. The order isomorphism in [Theorem 36.4](#) is a separate conclusion. The code is a central-termination adaptation of the prefix-free sign-block methods used in the earlier set-word analysis of Part II ([theorem 9.1](#)); the accompanying finite tests check it by independent dyadic evaluation.

36.4.3 Why the obstruction disappears

At a limit-length central word p , the proper prefixes are not uniformly below it: their comparison depends on the next nonzero letter of p . More importantly, a set of lower approximants can be separated from p by a fresh next value at the decisive active stage. The set-cut proof supplies this separator regardless of a particular sign pattern. Hence the small cofinality invariant of [Note 35.3](#) no longer obstructs an isomorphism with **No**.

The two constructions therefore illustrate a general warning: lexicographic comparison of variable-length words is not a minor technical variation of fixed-domain lexicography. A termination convention can change endpoints, point characters, and the entire isomorphism type.

36.5 Source 18's route: centered termination changes the order

[merge] Source 18's Section 7, printed whole as a third proof of the central case. Its classes $\mathcal{C}$, $\mathcal{C}_{\text{inj}}$, $\mathcal{C}_{\text{inc}}$ are printed $\mathcal{T}_c$, $\mathcal{I}_c$, $\mathcal{T}_c^{\text{inc}}$ ([section 1.3.1](#)). Its centered order is source 13's $\mathcal{T}_0$ (the terminator compared as zero with the next entry), and its code $C(x)$ is source 13's $c(x)$ and source 16's $C_-(x)$. Nothing in it is new against sources 13 and 16 except the remark that singleton words also embed **No** into $\mathcal{T}_c^{\text{inc}}$, which the next section uses. Source 18 was written before sources 13–17 arrived.

For a set word p , write $\ell(p)$ for its ordinal length and $p \smallfrown \langle x \rangle$ for ordinal concatenation. The surreal set-cut property used throughout this section is the following: if $L < R$ are sets of surreals, possibly empty, there is a surreal strictly between them. The interpolant can also be required to avoid any prescribed set. Indeed, first find a nonempty surreal interval between the two sides, and then use the set-cut property inside that interval to escape the forbidden set. No global choice is needed for these interpolation statements.

Put $D = \mathbf{No} \setminus \{0\}$. Let $\mathcal{T}_c$ be all set-length words over D , and let $\mathcal{I}_c$ and $\mathcal{T}_c^{\text{inc}}$ be its injective and strictly numerically increasing subclasses. Include the empty word in all three classes. Compare unequal words at their first unequal entry. If one word is a proper prefix of the other, compare its termination symbol $\dagger$ with the next entry of the longer word, using

$$x < \dagger \quad (x < 0), \quad \dagger < x \quad (x > 0).$$

Thus $\dagger$ occupies the missing numerical position of zero. This is a strict linear order, called the centered order below. The definition is not the shorter-first convention.

Theorem 36.6 (Centered arbitrary and injective words). *Both $\mathcal{T}_c$ and $\mathcal{I}_c$ fill every set cut. Consequently, in GBC, both are order-isomorphic to **No**. They have no endpoints; neither side of any point has a set cofinal or a set coinital subset, respectively.*

Proof. For a prefix p , its cone consists of all words extending p , including p itself. A cone is convex. A word outside it is uniformly below the whole cone or uniformly above the whole cone: the comparison has already been decided before p ends.

Given a set cut $A < B$, start with the empty prefix. Maintain the invariant that discarded members of A are below its whole cone, and discarded members of B are above its whole cone. At a current prefix p of length δ , assign next tokens to the active words: an actual next entry to a proper extension, and the numerical token 0 to p itself. Let L_p, R_p be the two sets of tokens. Every member of L_p is at most every member of R_p . Their intersection therefore has at most one member.

If the token sets are disjoint, choose

$$L_p < c < R_p, \quad c \notin \{0\} \cup \text{ran}(p).$$

Then $p \smallfrown \langle c \rangle$ interpolates the original cut. Avoiding the range is needed only for the injective version. If the token sets meet, their common token cannot be zero: zero would mean that the same word p belonged to both sides. Append their common nonzero entry and continue. The new prefix is injective when required, since it is an actual prefix of a witnessing injective word. At a limit stage take the union. The cone invariant persists.

Choose an ordinal strictly above every length occurring in $A \cup B$. The search cannot keep finding common actual entries that long. It therefore terminates with a successor-length interpolant. Empty sides cause no difficulty. Taking one side to be a singleton gives the asserted local character properties.

In GBC, restrict a fixed set-like global well-order to each of the two proper classes and to **No**. An ordinal back-and-forth constructs an isomorphism: at each stage its partial map is a set, and the new point is chosen in the cut imposed by that set. Freshness follows from the set-avoidance version of interpolation. This is a recursion with set-sized partial maps, not an appeal to choice for arbitrary families of proper classes. $\square$

The contrast with shorter-first words is substantive. At a limit word, the centered convention still permits extensions on both sides. These extensions prevent the proper prefixes from creating a one-sided set-cofinal obstruction.

Proposition 36.7 (An explicit centered sign embedding). *There is an explicit order embedding of $\mathcal{T}_c$ into **No**, without global choice. Conversely, **No** embeds into each of $\mathcal{T}_c$, $\mathcal{I}_c$, and $\mathcal{T}_c^{\text{inc}}$.*

Proof. For a nonzero surreal x , let s_x be its nonempty sign expansion. Duplicate each sign, writing $D(s_x)$, and put

$$C(x) = D(s_x) \smallfrown (+, -).$$

This is a prefix-free order-preserving code. When one original sign expansion ends, the delimiter $(+, -)$ is compared with the next doubled block; the inequalities

$$(-, -) < (+, -) < (+, +)$$

give exactly the surreal comparison at termination. The first sign of $C(x)$ is the sign of x . For a word f , concatenate the blocks $C(f(i))$ in ordinal domain order. Replacement makes the resulting sign expansion a set. First disagreements are preserved by prefix freedom; proper-prefix comparisons are preserved by the first sign of the next block. This proves the first assertion. For the reverse embedding send 0 to the empty word and send $x \neq 0$ to $\langle x \rangle$. $\square$

37 Centered increasing words and the entry zero (source 18)

[merge] Source 18's Sections 8, 9 and 11 and the corollary of its Section 10, printed whole; no other source treats these orders. Source 16 observed the first obstruction (end of its Subsection 8.1: a positive increasing word of length ω is the supremum of its finite prefixes, so the centered order on $\mathcal{T}^{\text{inc}}$ is not η_{Ord}) and asked for the classification of increasing words with an interior terminator (its question "Centered increasing words"); source 18 gives it for the threshold at zero ([theorem 37.2](#)), and settles the three ways of treating an actual entry zero ([propositions 37.4](#) and [37.5](#) and [corollary 37.7](#)). Its proofs use the common-prefix search of [theorem 11.1](#) with the termination token evaluated as zero.

37.1 Centered increasing words: an exact cut classification

Strict numerical increase changes which continuations remain available. An all-negative prefix admits fresh negative and positive continuations. Once a positive entry has occurred, every valid continuation is positive. This distinction gives the entire obstruction below.

Theorem 37.1 (Local characters of centered increasing words). *Let $f \in \mathcal{T}_c^{\text{inc}}$ have length α . If α is a nonzero limit and f has a positive entry, then*

$$\text{cf}((\mathcal{T}_c^{\text{inc}})_{<f}) = \text{cf}(\alpha).$$

At every other point the open initial segment has no set cofinal subset. At every point the open final segment has no set coinital subset. The order has no endpoints and is dense.

Proof. If f has a positive entry and has limit length, its sufficiently long proper prefixes form an increasing cofinal family below it. Every smaller word either terminates before f , or first differs at a smaller entry before α ; it cannot properly extend f , since all such extensions are larger. It is consequently below every sufficiently long prefix. A cofinal sequence of prefix lengths proves the upper bound $\text{cf}(\alpha)$. Conversely, for each smaller word choose a prefix exceeding it. Fewer than $\text{cf}(\alpha)$ chosen lengths are bounded, so that many words cannot be cofinal.

Suppose next that every entry of f is negative; this includes the empty word. Given a set $A < f$, collect the next entries of those members of A that properly extend f . They are all negative. Choose a negative d above this set and above $\text{ran}(f)$. The set-cut property permits this even when the range of f has no largest member. Then

$$A < f \frown \langle d \rangle < f.$$

Members of A that do not extend f first differ earlier with a smaller entry. They are also below this new word.

If $f = p \frown \langle x \rangle$ and $x > 0$, a smaller word sharing p either stops there or has next entry below x . For a set of such words, choose $d < x$ above all their next entries, above zero, and above $\text{ran}(p)$. The word $p \frown \langle d \rangle$ is a strictly larger lower bound for f than any member of the given set. This handles the remaining successor case.

Finally, given a set $B > f$, the next entries of members that extend f are positive. Choose a positive d , above the range of f , but below all those entries. Then

$$f < f \frown \langle d \rangle < B.$$

All other members of B are above the whole cone of f . This proves the right-hand assertion and, with empty sets, the absence of endpoints.

For density, if two words first differ, choose a new coordinate strictly between their different entries. It is automatically above the common prefix. If the comparison is a proper-prefix comparison with next entry $x > 0$, choose a coordinate strictly between the set $\{0\} \cup \text{ran}(p)$ and x . For a next entry $x < 0$, choose one strictly between $\{x\} \cup \text{ran}(p)$ and zero. $\square$

Theorem 37.2 (All set-cut obstructions). *A set cut $A < B$ in $\mathcal{T}_c^{\text{inc}}$ has no interpolant if and only if B has a minimum b such that*

1. $\ell(b)$ is a nonzero limit ordinal;
2. b has a positive entry;
3. A is cofinal in the open initial segment below b .

In every other case an interpolant of nonzero successor length can be chosen.

Proof. The stated obstruction plainly prevents interpolation. For the converse use the common-prefix search from [Theorem 36.6](#), evaluating the termination token as zero. Common actual coordinates can still be appended, since each is witnessed by an increasing word.

At a stage where the next-token sets are disjoint, a permissible new coordinate must satisfy

$$L_p \cup \text{ran}(p) < c < R_p, \quad c \neq 0.$$

Every actual token on the right exceeds the whole range of p . Thus these are consistent strict surreal constraints unless zero is a right token and p already has a positive entry. When zero is a right token and all entries of p are negative, the constraints are still consistent and a negative interpolating coordinate can be chosen. Empty right sides are harmless.

The sole exceptional event is therefore reaching $p \in B$ after p has acquired a positive entry. Every proper extension of p is larger, so the active lower side is then empty and $p = \min B$. This event cannot occur at a successor length: the last common coordinate had a witness in A extending that successor prefix, which would be at least p , contradicting $A < B$. It also cannot occur at length zero.

It can occur at a limit length α . For every sufficiently large $\xi < \alpha$, a lower-side witness supplied the coordinate $p(\xi)$. That witness extends $p \restriction (\xi + 1)$, is at least this prefix, and remains below p . The sufficiently long prefixes are cofinal below p by [Theorem 37.1](#); hence A is cofinal there. As before, a bound on all input word lengths ensures that the search terminates using only set recursion. $\square$

Corollary 37.3. *The centered increasing-word order is equimorphic with $\mathbf{No}$, but is not isomorphic to $\mathbf{No}$, even in GBC. Its successor-length suborder fills every set cut and is isomorphic to $\mathbf{No}$ in GBC. The larger suborder obtained by adjoining the empty word and every all-negative limit word also fills every set cut and has the same isomorphism conclusion.*

Proof. Use the explicit embeddings of [Proposition 36.7](#). The increasing word $\langle 1, 2, 3, \dots \rangle$ is the supremum of its finite prefixes in this word order. Such a nontrivial set supremum cannot occur in $\mathbf{No}$, by its set-cut property. For either stated suborder, an upper set cannot contain the exceptional minimum in [Theorem 37.2](#), and the proof returns a successor word in the suborder itself. Apply class back-and-forth in GBC. $\square$

Every nonempty open interval in the centered increasing-word order contains an order copy of $\mathbf{No}$. The density proof can choose a whole surreal interval of permissible new coordinates, rather than one coordinate. This supplies the copy. Thus rich interval universality coexists with the precise limit-word obstructions.

37.2 What if the entry zero is allowed?

Saying only that termination lies “between negatives and positives” does not settle its relation to an actual entry equal to zero. There are at least three different mathematical constructions.

37.2.1 Zero padding and its quotient

One may pad each set word by zeros at every later ordinal coordinate. First-difference comparison then gives a total preorder, not a strict linear order on the original words. For instance the empty word and $\langle 0 \rangle$ become equal. In general put

$$\sigma(f) = \sup\{\xi + 1 : \xi < \ell(f), f(\xi) \neq 0\}, \quad \rho(f) = f \restriction \sigma(f).$$

Two padded words are equivalent exactly when their reduced words $\rho(f)$ agree. A class of canonical set representatives is therefore available: trim the entire terminal zero tail. A limit-length reduced word has nonzero entries cofinal in its length. Reduction preserves injectivity and strict numerical increase.

Proposition 37.4 (The zero-padding quotient). *For all set words, and separately for injective set words, the zero-padding quotient fills every set cut and is isomorphic to $\mathbf{No}$ in GBC. For strictly increasing set words, the quotient has exactly the cut obstruction in Theorem 37.2, applied to the reduced representatives. Its exceptional points are precisely the reduced limit words having a positive entry.*

Proof. Work with reduced representatives. A set of input representatives has lengths bounded by a set ordinal μ . Pad them by zeros through coordinate μ and perform the numerical common-prefix search. This search now allows a common coordinate equal to zero.

In the injective case, the prefix produced by the search remains injective. A common nonzero coordinate is an actual coordinate of every witness. If zero is common, at least one witness has an actual zero at that coordinate: otherwise a witness on each side would have terminated already and, agreeing at all previous coordinates, would represent the same padded word. An actual witness certifies that the enlarged prefix is injective. At a separation step choose a nonzero coordinate outside the range of the prefix, using the surreal set-cut property. This returns a reduced injective word. At coordinate μ , two distinct input classes cannot both remain active, so termination is guaranteed. This proves the first claim.

For increasing words the same witness argument makes every common prefix strictly increasing. At a separating step add its range to the lower numerical constraints. The only potential failure is an upper padded-zero token together with a nonnegative entry in the current prefix.

If the prefix has a positive entry, an increasing extension cannot consist of an extra padded zero. The active upper word must therefore be exactly this reduced prefix. As in the proof of Theorem 37.2, a successor stopping stage contradicts the preceding lower-side witness, while a limit stopping stage gives the stated cofinal-prefix obstruction.

The remaining apparent failure would be a prefix $p = q \smallfrown \langle 0 \rangle$, where q is all-negative, and an active upper representative equal to q . This stage cannot be reached. The common zero that produced it would require a lower witness. If that witness has terminated, it represents q itself; if it has an actual zero, its reduced increasing continuation has a positive entry and lies above q . Both possibilities contradict the cut. There are no further failures. The converse obstruction is again supplied by cofinal proper prefixes of a positive limit word. $\square$

For labeled well-orders the quotient is a change of subject that must be stated explicitly. It identifies distinct codes with different underlying sets: an all-negative increasing word and that word followed by zero are different labeled well-orders, although the padding convention makes them equivalent.

37.2.2 A distinct terminator immediately below or above zero

Alternatively, retain every word as a distinct object and use one of the two enlarged alphabets

$$\mathbf{No}_{<0} < \dagger_- < 0 < \mathbf{No}_{>0}, \quad \mathbf{No}_{<0} < 0 < \dagger_+ < \mathbf{No}_{>0}.$$

The indicated neighboring symbols are adjacent in the enlarged alphabet. These are genuine linear orders on the original words.

Proposition 37.5. *For arbitrary set words, and separately for injective set words, both distinct-terminator conventions fill every set cut. Each resulting class order is isomorphic to **No** in GBC.*

Proof. Use the common-prefix search with tokens in the enlarged alphabet. At a strict token separation an unused surreal can normally be inserted. The only exception is its displayed adjacent pair.

For $\dagger_- < 0$, the exceptional separation has the current prefix p on the lower side, and upper active words with next entry zero. No lower word has next entry zero, since that would have caused a common-coordinate step. Append zero and then append a sufficiently small negative, unused coordinate. The resulting word is above the original lower side and below every upper word in the zero cone, including its root if that root occurs. For $0 < \dagger_+$, use the reverse construction: append zero and then a sufficiently large positive, unused coordinate. The existence of a zero cone certifies that zero has not already been used when injectivity is required. All choices avoid the remaining set of used entries. This resolves the only additional cases, and the length bound still makes the search a set recursion. $\square$

Increasing words behave differently. The following general fact isolates the reason, independently of any surreal coding.

Lemma 37.6 (Adjacency test for increasing words). *Let D be a linearly ordered alphabet with a distinct termination symbol $\dagger$. Suppose every allowed increasing set word has a further admissible entry above $\dagger$. In the corresponding order of all increasing set words, $f < g$ are adjacent if and only if*

$$g = f \smallfrown \langle c \rangle, \quad c = \min\{x \in D : \text{ran}(f) < x, \dagger < x\}.$$

The minimum in this assertion must exist.

Proof. If f is not a prefix of g , append a positive-side admissible entry to f . The new word is above f , and its comparison with g was already settled before the appended entry; hence it remains below g . Such a pair is not adjacent.

If f is a proper prefix of g , its first new entry c must be above $\dagger$. All subsequent entries exceed c , so if g has any further entries, the one-letter extension $f \smallfrown \langle c \rangle$ lies strictly between. For this one-letter extension to be adjacent, no admissible entry strictly between $\dagger$ and c can exist. Conversely, if c is the stated minimum, every word in between would have to extend f ; its first new entry could neither be below c nor above it, and a proper continuation after c would be larger than g . No such word exists. $\square$

Corollary 37.7 (Exactly where the zero jump appears). *For increasing surreal words with $\dagger_-$, the adjacent pairs are exactly*

$$p < p \smallfrown \langle 0 \rangle \quad \text{with every entry of } p \text{ negative.}$$

The empty prefix is allowed. With $\dagger_+$, or with zero excluded and centered termination, there are no adjacent pairs.

Proof. Every set of surreal entries has a strictly larger positive surreal, so the lemma applies. With $\dagger_-$, an all-negative prefix has zero as its least positive-side admissible entry. Once a nonnegative entry is present, the admissible upper class has no least member, by the surreal set-cut property. With $\dagger_+$, its positive-side admissible class consists of positive surreals above the prefix and has no least member. The same argument applies when zero is excluded. $\square$

Thus a convention that differs only in the placement of actual zero can create adjacent pairs even among finite increasing words. The absence of an eligible letter between termination and the zero entry is decisive: increasing words cannot use the negative two-letter detour from [Proposition 37.5](#).

For completeness, the same common-prefix argument gives the full set-cut classification for increasing words with the distinct terminators. With $\dagger_+$, only the positive limit-word obstruction of [Theorem 37.2](#) remains. With $\dagger_-$, one must additionally allow the adjacent-pair obstruction

$$\max A = p, \quad \min B = p \smallfrown \langle 0 \rangle, \quad \text{ran}(p) < 0.$$

Indeed, at the exceptional token gap the word $p \smallfrown \langle 0 \rangle$ itself interpolates unless it is the upper minimum and p is the lower maximum. All other separating-coordinate steps work as before. A positive limit stopping stage is unchanged. For $\dagger_+$, the zero-then-positive detour is increasing and always resolves the adjacent token gap.

Note 37.8 ([merge]; two of the zero conventions are source 16's). For arbitrary and injective words, the conventions $\dagger_-$ and $\dagger_+$ of [proposition 37.5](#) are source 16's 0^- and 0^+ , and that proposition is (S16.6), a case of [theorem 36.1](#); both proofs resolve the adjacent pair by the same two-letter detour. New in source 18 are the zero-padding quotient ([proposition 37.4](#)) and, for increasing words, the exact adjacent pairs ([lemma 37.6](#) and [corollary 37.7](#)).

37.3 Cones of centered increasing words (source 18, Corollary 10.2)

Corollary 37.9 (Two kinds of centered increasing cones). *Work in GBC, with zero excluded and centered termination. The cone above an all-negative increasing prefix p , including its root p , is isomorphic to the entire centered increasing-word order $\mathcal{T}_c^{\text{inc}}$. If p has a positive entry, its cone is isomorphic to the shorter-first arbitrary-word order, including that order's minimum. These isomorphisms can preserve the rooted prefix tree and the lengths of tails after the root.*

Proof. At an all-negative prefix, the two admissible classes

$$\{x : \text{ran}(p) < x < 0\}, \quad \{x : 0 < x\}$$

are both proper and fill every set cut relative to themselves. Once a positive coordinate is chosen, the negative admissible class disappears, and the positive admissible class continues to fill all set cuts. Use the uniform construction in the proof of [Theorem 35.7](#) separately on the negative and positive parts whenever both are present. The source and target therefore pass from the all-negative case to the positive case at the same coordinate, and both entry and termination comparisons are preserved. This proves the first assertion. In the positive case every later entry is above termination, so the rooted cone uses precisely shorter-first comparison. The one-part version of the same construction proves the second assertion. $\square$

37.4 Regular set stages

The cut arguments also give a set-sized version. Let κ be regular infinite, restrict entries to $\mathbf{No}_{<\kappa}$, and restrict word lengths to be below κ . With zero excluded and centered termination, the arbitrary and injective word orders are η_κ . Their increasing-word successor suborders are η_κ as well. The proof of every relevant cut statement uses fewer than κ input words, has its prefix length bounded below κ , and adds fewer than κ coordinate constraints or forbidden entries. Regularity and the η_κ property of $\mathbf{No}_{<\kappa}$ supply the last coordinate. If $2^{<\kappa} = \kappa$, these orders have cardinality κ and are isomorphic to $\mathbf{No}_{<\kappa}$ by ordinary set back-and-forth.

The complete increasing-word order retains every positive limit-length obstruction that is present below the cutoff. When $\kappa = \omega$ no nonzero limit word length is allowed, so this particular obstruction disappears. These statements should not be extrapolated to singular cutoffs or to unrestricted word lengths over a fixed set alphabet: a prefix can then exhaust the available one-sided room. For example, over the nonzero dyadic rationals, an increasing negative word $\langle -1, -1/2, -1/4, \dots \rangle$ has no further negative entry above its whole range, whereas every set of negative surreals does have such a continuation in the full surreal class.

Note 37.10 ([merge]; regular stages in Part II). For shorter-first words the regular-stage statements are Part II's [section 12](#) and [proposition 9.2](#) (1); source 18's paragraph gives the centered analogues. Its warning about singular cutoffs is the one printed after [proposition 9.2](#).

38 Other comparison orders (source 15; remarks of sources 13, 14, 16, 17 and 18)

[merge] Source 15's Section 10, printed whole as the base: three lexicographies on three labels, inverse-rank lexicography ([theorem 38.1](#)), relation tables and the transfer of their endpoints ([theorem 38.2](#) and [corollary 38.3](#)), the failure of restriction-monotone linearizations ([theorem 38.5](#)), the isomorphism quotient, the normal-form embedding ([proposition 38.6](#)) and the birthday and simplicity remarks. Its macros $\mathfrak{W}_{\text{sl}}$, $\mathfrak{W}_{\text{all}}$, $\mathcal{D}$ print as in this report; its rk, bl are this report's rank and ℓ_2 . The relation-code remarks of the other sources follow in [section 38.8](#).

38.1 Three lexicographies disagree on three labels

On $0 < 1 < 2$, an enumeration is a permutation word. Its inverse-rank vector lists the positions of labels 0, 1, 2. For table lexicography scan the directed pairs in this order:

$$(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1),$$

and use bit 1 when the first label precedes the second.

Enumeration	Inverse ranks	Relation table
012	012	110100
021	021	110001
102	102	011100
120	201	001110
201	120	100011
210	210	001011

The increasing orders on the same six objects are respectively

enumeration: 012, 021, 102, 120, 201, 210,
inverse rank: 012, 021, 102, 201, 120, 210,
table: 210, 120, 102, 201, 021, 012.

The table comparison is not just the reverse of enumeration comparison. As abstract finite orders, all three are of course the same six-point chain. The difference here concerns which labelled object precedes which; the next results distinguish abstract infinite order types as well.

38.2 Inverse-rank lexicography

Fix a baseline $q : \text{Ord} \rightarrow \mathbf{No}$. For a set-like well-order R , let $r_R : \mathbf{No} \rightarrow \text{Ord}$ be its rank map. Compare the permutations

$$p_R = r_R \circ q : \text{Ord} \longrightarrow \text{Ord}$$

lexicographically, using the ordinal order on the values. Conversely, every class permutation p corresponds to the enumeration $q \circ p^{-1}$. This comparison uses ordinal ranks of fixed labels, rather than numerical labels at fixed positions. For a non-set-like order an ordinal rank does not exist at every label, so this version needs a further coding choice before it can be applied to all raw orders.

Theorem 38.1 (Endpoint and character of inverse ranks). *The inverse-rank order of set-like global well-orders has a least element, the baseline order, and no greatest element. It is dense. For every nonzero limit ordinal λ , it has a point M_λ with left cofinality $\text{cf}(\lambda)$. No point has a set coinital family strictly above it. In particular this order is not isomorphic to native enumeration lexicography.*

Proof. The identity permutation is least. If p first differs from the identity at α , all smaller ordinals have already appeared, so $p(\alpha) > \alpha$. On every tail of a permutation there is an ascending pair: otherwise a countable subsequence would descend through the ordinals. Swapping that ascending pair makes the permutation lexicographically larger. If $p < q$, perform such a swap strictly after their first disagreement; the result lies strictly between them. This also proves no greatest point. For a set family of points above p , bound all their first disagreements with p and perform an ascending swap later; the result is above p and below the entire family.

Let M_α begin with α and then list the remaining ordinals in increasing order. This is a permutation of type Ord and is the least permutation with first value α , since it always chooses the least unused ordinal thereafter. For limit λ ,

$$M_\lambda = \sup_{\alpha < \lambda} M_\alpha.$$

Indeed, a point below M_λ has first value below λ ; some larger $\alpha < \lambda$ therefore gives a point M_α above it. A cofinal family below M_λ must have first values cofinal in λ , proving the exact character. Native enumeration lex has no least element and no set cofinal family below any point. $\square$

For a set carrier X of cardinality μ , scanning the labels by a bijection $q : \mu \rightarrow X$ gives a least inverse-rank code among all well-orders, again the q order. On a fixed-type slice of type α , a least inverse-rank code exists exactly when $\alpha = \mu$: a least rank vector can have no inversion, since swapping an inverted pair lowers it, and an increasing bijection of its ranks forces precisely that type. Thus even fixed-type endpoint behavior depends on the representation.

38.3 Relation tables and a theorem about all their endpoints

Choose a well-ordered scan of $X \times X$. For a proper class carrier assume the scan is set-like, so every proper initial part of the scan is a set. The table of a strict relation R is its characteristic function, compared with $0 < 1$. The native numerical order on $\mathbf{No}^2$ cannot itself serve as a scan with a universal first-difference principle.

Theorem 38.2 (Endpoint transfer for relation tables). *For a set-like scan, a well-order table is least among all well-order tables if and only if it is least among all strict linear-order tables. The dual statement holds for greatest elements. Both assertions also hold with the class of set-like well-orders in place of all well-orders.*

Proof. Suppose R is a claimed minimum and a linear order T has a smaller table. At the first differing entry (x, y) , one has xRy and yTx . The earlier bits impose a set P of directed order constraints: a true off-diagonal bit gives its indicated direction, and a false one gives the reversed direction. Both R and T satisfy these constraints.

Let D be the downward closure of y under finite P -paths, including y . It is a set, contained in $\{y\}$ and the vertices of the earlier scan. Since T satisfies P and yTx , one has $x \notin D$. Move D to the front of R , preserving the old order within D and within its complement. The resulting relation R' is a well-order. It satisfies every earlier constraint because D is P -downward closed. It puts y before x , so its first changed table bit is smaller, a contradiction. If R was set-like, R' is still set-like: new predecessor classes are contained in an old predecessor set together with D . The proof for a maximum moves the appropriate opposite endpoint at the first larger bit. The reverse implications are immediate. $\square$

Corollary 38.3 (Exactly three possible endpoint patterns). *On an infinite carrier, table lexicography of well-orders has at most one endpoint. By choosing the scan, each of the following can occur: a minimum only, a maximum only, or neither. The same patterns occur for set-like class well-orders of **No**.*

Proof. Reversing a linear order complements all off-diagonal bits, so it reverses the table comparison. If a minimum and maximum well-order both existed, [Theorem 38.2](#) would make them endpoints among all linear orders, and hence reversals of one another. An order and its reversal cannot both be well-orders on an infinite carrier: the first ω points of one form a descending sequence in the other.

For any given linear order T , scan each unordered pair in a two-entry block, putting its T -false direction first and its T -true direction second. Scan the blocks set-likely. Then T is the least linear-order table: a different order changes the first entry of its earliest changed block from 0 to 1. Reversing the two entries makes T greatest. Take T to be a well-order for the one-endpoint cases, and a linear order such that neither it nor its reverse is well-founded for the no-endpoint case. Such an order exists on every infinite carrier; on **No** its native order suffices. Endpoint transfer finishes the argument. $\square$

A scan built around a non-set-like R makes R the minimum among all well-orders but leaves no minimum among set-like well-orders. The greatest linear-order table is then R^{op} , not a well-order, so the set-like subcollection has neither endpoint. Therefore a relation table may erase the natural numerical comparison while retaining or losing particular well-order types according to the chosen scan.

Example 38.4 (Failure of a numerical pair scan). On the integer labels, compare the enumerations

$$0, -1, 1, -2, 2, -3, 3, \dots, \quad 0, 1, -1, 2, -2, 3, -3, \dots,$$

then append the same set-like enumeration of the remaining surreals. The resulting set-like orders disagree on the directed pairs $(-n, n)$ and $(n, -n)$, $n \geq 1$. Their numerical-lexicographic disagreement set in **No**² has no least pair, because its first coordinates include arbitrarily negative integers. A genuine well-ordered scan avoids this problem; the numerical order alone does not.

38.4 Restrictions, extensions, and relabelling

Restriction to a smaller carrier can reverse enumeration lexicography: $021 < 102$ on $0 < 1 < 2$, but on $\{1, 2\}$ the restrictions are $21 > 12$. The obstruction is stronger than a bad choice of lexicographic convention.

Theorem 38.5 (No total linearization respects every restriction). *There is no total order on the six well-orders of a three-element carrier making all three restrictions to two-element carriers monotone, no matter which of the two orders is chosen on each two-element well-order space.*

Proof. Encode each pair restriction by 0 or 1 according to its chosen order. The three restrictions determine the original relation, so the six well-orders map injectively to $\{0, 1\}^3$. If all restrictions were monotone, the six objects in increasing order would give a coordinatewise increasing chain of six distinct Boolean vectors. Every strict step raises the sum of the coordinates, whose possible values are 0, 1, 2, 3. Such a chain has at most four points, a contradiction. $\square$

Thus arbitrary restrictions do not form a monotone system automatically. The theorem does not rule out a carefully chosen nested filtration or restricted family. In the other direction, fixing a well-order Q of $X \setminus Y$ makes $R \mapsto R + Q$ an order embedding from the native well-order space of Y into that of X . First disagreement occurs before the fixed tail. Prepending a common labelled prefix works as well.

No linearization can be invariant under every relabelling of a carrier with at least two labels. A transposition acts as a nontrivial involution on its well-orders, since well-orders are rigid. But an increasing involution of a linear order must be the identity: $x < f(x)$ would imply $f(x) < f(f(x)) = x$. Some auxiliary label structure is indispensable. Increasing automorphisms of the numerical surreal line preserve native enumeration lex ; a table scan or baseline must instead be transported along with the labels.

38.5 The quotient by order isomorphism is a new construction

On the numerical carrier $\mathbb{N}$, let

$$R = (0, 1, 2, \dots), \quad S = (1, 2, 3, \dots, 0), \quad T = (2, 0, 1, 3, 4, \dots).$$

Their types are $\omega, \omega + 1, \omega$, respectively, and their first labels give $R < S < T$. Thus the equivalence class of type ω is not convex. A linear-order quotient requires convex classes, so labelled lexicography does not descend to abstract order types.

One may separately order the possible set types by ordinal comparison. For $|X| = \mu$ infinite they are precisely $\mu \leq \alpha < \mu^+$; in that comparison their order type is μ^+ . The original labelled order embeds no ordinal μ^+ , by [Note 32.3](#). This is a particularly sharp sign that the quotient-type order is a different object. For set-like well-orders of all **No**, quotienting by class-order isomorphism collapses everything to the one type Ord . Arbitrary non-set-like class types require their own collection and comparison framework; raw labelled lexicography provides neither automatically.

38.6 Numerically well-ordered subsets and a normal-form embedding

Let $A, B \subseteq \mathbf{No}$ be sets or classes well-ordered by the inherited numerical order. Their union is well-ordered: a nonempty subclass has its least member among the minima of its intersections with A and B . Consequently $\min(A \triangle B)$ exists whenever $A \neq B$. This makes an intrinsic characteristic comparison possible without a global scan of the entire numerical line:

$$A <_{\chi} B \iff \min(A \triangle B) \in A.$$

It uses membership bits with $1 < 0$. Transitivity is first-difference transitivity on the well-ordered union of three carriers. For increasing enumerations it agrees with numerical first difference and *longer prefixes first*. The empty carrier is greatest. The shorter-first convention agrees on genuine unequal next labels but reverses proper-prefix pairs and makes the empty carrier least.

Proposition 38.6 (Coefficient-one normal forms realize set supports). *For a set-sized naturally well-ordered subset $A \subseteq \mathbf{No}$, the map*

$$H(A) = - \sum_{a \in A} \omega^{-a}$$

is a well-defined surreal, and $A <_{\chi} B$ if and only if $H(A) < H(B)$.

Proof. The exponent support $\{-a : a \in A\}$ is reverse well-ordered and is a set, so the displayed Conway normal form is legitimate. At the least a in the symmetric difference, the largest differing exponent is $-a$. The set containing a has coefficient -1 there rather than 0 , and therefore gives the smaller surreal. Uniqueness of normal form gives injectivity [11, 12]. $\square$

This particular construction does not admit proper-class support: the normal form of one surreal has set support. Naturally well-ordered proper subclasses nevertheless exist with every class well-order type by surreal universality. For example

$$\left\{ -\frac{1}{\alpha + 1} : \alpha \in \text{Ord} \right\} \cup \{0\}$$

has natural type $\text{Ord} + 1$. Its final point has a proper class of earlier points. The reciprocal and addition in this example are surreal operations on the embedded ordinal labels.

At a set limit-length increasing word, shorter-first order makes the word the supremum of its proper prefixes; longer-first order makes it their infimum. The associated one-sided cofinality is the cofinality of the length. These features expose the termination convention in the order itself, even when both conventions admit embeddings into $\mathbf{No}$.

38.7 Birthday and simplicity do not secretly provide a scan

Simplicity, interpreted as initial sign-sequence extension, is a partial tree order. The one-sign strings $+$ and $-$ are incomparable by it. Numerical sign comparison instead uses the middle termination symbol $- < \text{undefined} < +$ and gives the surreal line, not a well-order.

Even birthday-first ordering with numerical tie-breaking fails. The strings

$$s_n = -^n \frown +^\omega \quad (n < \omega)$$

all have birthday ω , since $n + \omega = \omega$, and satisfy $s_{n+1} < s_n$. Thus the birthday- ω level already contains a descending sequence under numerical tie-breaking. A genuine birthday-compatible global well-order requires well-ordering each level with additional coherent choice data. That choice must be stated; it is not supplied by birthdays or simplicity alone.

Note 38.7 ([merge]; repository counterparts of source 15’s Section 10). Relation-table lexicography is source 08’s relation-code order ([proposition 17.1](#): equimorphic with $\mathbb{B}_{\text{Ord}}$ for every coordinate enumeration) and source 12’s remark with its three-point check ([section 18](#)). New in source 15 are the endpoint transfer ([theorem 38.2](#)) and the three endpoint patterns ([corollary 38.3](#)), which show that the endpoints of a relation-code order are artifacts of the scan; inverse-rank lexicography ([theorem 38.1](#)); the restriction theorem ([theorem 38.5](#)); and the normal-form embedding ([proposition 38.6](#)). The birthday tie-break example is source 12’s (“Birthday is not a canonical global well-order”, after [theorem 13.2](#)), and source 18 repeats it ([section 38.8](#)). The non-convexity of type classes is made by sources 13, 14, 16 and 17 in their maps ([section 34](#)).

38.8 Relation-code remarks of sources 13, 14 and 18

[merge] Three further remarks on relation codes, each printed whole: source 13's (from its foundations section), source 14's (from its Section 11) and source 18's (from its section on global well-orders). Source 16's is in its map (section 34.4), source 17's in its map (section 34.5). They make one point, that relation-code lexicography depends on the scan and the bit convention and is not enumeration lexicography; they prove no theorem about it beyond proposition 17.1 and section 38.

38.8.1 Source 13: relation-bit lexicography is a different order

Fix a set-like global enumeration of the ordered pairs of surreal labels. Read the characteristic bits of a raw relation in this order and compare at the first different bit. This is a legitimate code-dependent class lexicography. It is not in general the labeled common-prefix comparison of theorem 14.2: the first edge in a chosen code order need not reveal the first disagreement of the enumerated labels.

The preceding report analyzes this relation-code alternative and its binary-word equimorphism (proposition 17.1; Section 17 of this report at the pin). A choice of pair enumeration is indispensable. Replacing the label order by birthday order or comparing raw Conway construction trees would impose still other auxiliary data. Such choices can be studied rigorously, but no one of them should be advertised as *the* lexicographic order without identifying the coding convention.

38.8.2 Source 14: relation-code lexicography really is another order

One may instead fix a class bijection scheduling $\mathbf{No} \times \mathbf{No}$ by $\mathbf{Ord}$, and compare characteristic functions of well-order relations at their first differing scheduled pair. This defines a linear comparison on the admitted relations but depends on the schedule and on the convention $0 < 1$.

The difference is visible with three labels. Let numerical order be $0 < 1 < 2$, and let the enumerations be $e = (0, 1, 2)$ and $f = (1, 0, 2)$. Enumeration lex gives $e < f$. If the relation-coordinate schedule begins with $(0, 1)$, their strict-relation bits there are 1 and 0, so the relation-code comparison gives $e > f$. Starting instead with $(1, 0)$ reverses those bits and that conclusion. This counterexample does not involve any proper-class complication. Results proved for numerical first-disagreement order cannot simply be transported to relation codes.

38.8.3 Source 18: inherited numerical well-orders and relation codes

Numerical $<$ on $\mathbf{No}$ is not a well-order; the negative integers already descend. Nevertheless, many subclasses are numerically well-ordered. The ordinals give type $\mathbf{Ord}$, and

$$\{-1/(\alpha + 1) : \alpha \in \mathbf{Ord}\} \cup \{0\}$$

has numerical order type $\mathbf{Ord} + 1$. Under Global Choice, class universality gives a numerically ordered copy of every supplied class well-order. This concerns the range of an embedding in the surreal line, not a well-order of every surreal label.

One can instead order well-order *relations* by characteristic bits. Choose a set-like class enumeration of $X \times X$, and compare the bits of two relations at their first unequal pair. The resulting comparison is well-defined, but is a different auxiliary order. Even on $\{0, 1\}$ the enumerations $(0, 1) < (1, 0)$ compare in the opposite direction if the first tested pair is $(0, 1)$ and $0 < 1$ for relation bits. The enumeration rule tests first labels, while the relation rule tests a chosen predicate membership. Its dependence on the chosen pair order should be recorded explicitly.

Finally, ordering birthday first and then ordering sign expansions lexicographically does not produce a global well-order. The fixed birthday- ω slice already contains descending sequences. A genuine birthday-first global well-order must use an auxiliary well-order within each birthday level, not its numerical sign order.

39 What this Part answers

[merge] Dated notes, 3 October 2026, in the form of [section 33](#); where the question index of [section 61](#) (dated 2 October 2026) says otherwise, these notes supersede it.

- (a) *Source 09's direction "Other termination conventions for set words"* ([section 61.3](#)): **answered** for unrestricted and injective words. Exactly the placements of a formal terminator satisfying the three conditions of [theorem 36.1](#) give the set-cut property, hence (in GBC) an order isomorphic to **No**; every placement gives one of the types of [theorem 36.2](#); the distinguished internal value is [theorems 36.4](#) and [36.6](#), and the prefix-free codes are listed in [note 36.3](#). For increasing words with the terminator at zero see (b).
- (b) *Source 08's question 14, increasing-subset and alternate-termination orders* ([section 61.4](#)): **answered** for prefix-small, prefix-large and centered conventions on unrestricted and injective words ([theorems 36.1](#) and [36.2](#)), for shorter-first increasing words (isomorphic to the other two classes in GBC, [theorem 35.1](#)), and for centered increasing words up to their isomorphism type: local characters, all set-cut obstructions, equimorphism without isomorphism ([theorems 37.1](#) and [37.2](#) and [corollary 37.3](#)). Open: the isomorphism type of the centered increasing order (source 18's own question) and increasing words with an interior terminator at a threshold other than zero.
- (c) *Source 13's question "A classification of termination cuts"* (for a class alphabet isomorphic to **No**): answered by source 16 ([theorems 36.1](#) and [36.2](#)), transported along an isomorphism of the alphabet with **No** in GBC.
- (d) *Source 16's question "Centered increasing words"*: answered by source 18 for the threshold at zero (with zero excluded, padded, or adjacent to a distinct terminator: [section 37](#)); open for other interior thresholds.
- (e) *Source 08's question 9 and source 09's direction "Non-set-like well-order relations"* (relation-code orders; [sections 61.3](#) and [61.4](#)): answered in part. Which endpoints a relation-code order of well-orders has is an artifact of the scan: a minimum only, a maximum only, or neither, each by a suitable scan, and never both on an infinite carrier ([theorem 38.2](#) and [corollary 38.3](#)). The isomorphism question for two scans stays open.
- (f) *Source 11's question on optimal birthday bounds* ([research question 61.6](#)): not answered. Sources 13, 14, 16, 17 and 18 give explicit codes with birthdays $\sum(2 \cdot \text{bd} + 2)$, $\sum(4 \cdot \text{bd} + 4) + 2$ and $\sum(1 + 2 \cdot \text{bd} + 2)$ ([note 36.3](#)); none is claimed optimal.

Sources 16 and 18 also ask how much of the word isomorphisms survives without a global well-order; source 18's proof ([section 35.5](#)) shows that one set-like global well-order suffices as a parameter, so the question is whether that parameter can be removed. It is open.

Part VIII

Cuts of the set-coded core: recognition, automorphisms, support and class realizations

[merge] (3 October 2026.) The third Part of the batch-81 addition ([section 1.8](#); notation in [section 1.3.1](#)). Part IV ([part IV](#)) built the core, the ordinary class of set codes for bounded-support perturbations of one baseline enumeration b , proved that it interpolates every uniformly set-indexed cut, and asked which of its class cuts come from global enumerations ([research question 61.1](#); source 09’s third direction in [section 61.3](#)). All six batch-81 sources answer that question, by one argument. This Part prints the answer and what the sources prove around it: source 14’s foundations for the class-level results ([section 40](#)); the core and its cylinders ([section 41](#)), with the five other sources’ re-proofs of Part IV ([section 42](#)); the classification of every proper class cut of the core into a cylinder edge, a numerical gap or a branch ([section 43](#)) and its characters ([section 44](#)); the recognition of the cuts of global enumerations ([section 45](#)) and the five further proofs of it ([section 46](#)); automorphisms of the core ([section 47](#)); class extensions ([section 48](#)); and support-restricted suborders and the exact reserve of a partial injection ([section 49](#)). [Section 50](#) records which questions the Part answers.

The base is source 14 throughout, by generality: it alone classifies *every* proper class cut with unique data, gives the complete orbit invariant, the absoluteness theorem and the support hierarchy. Source 17 is the base for the exact reserve cardinal, the automorphism extension criterion and the clause that every proper class extension adds a represented cut; source 15 adds the finest first-failure partition (a forbidden repeated label) and source 13 the non-rigidity of the cylinder family. The represented-cut theorem itself is printed once as source 14’s [theorems 43.3](#) and [45.1](#), and the other five proofs follow as routes, each whole, because each source builds different examples and corollaries on its own proof ([note 46.1](#)). Source 18 was written before sources 13–17 arrived.

Notation ([section 1.3.1](#)): the core is printed $\mathcal{P}_s(b)$ in the support coding of sources 13, 14, 17 and 18 (their $\mathcal{D}$, $\mathcal{P}_b$, $\mathcal{C}_b$, $\mathcal{C}_b$; source 18’s macro $\backslash\text{Pbd}$), and $\mathcal{P}_{bd}(b)$ in the normalized-permutation coding of sources 15 and 16 (their $\mathcal{D}$ and $\mathcal{P}_b$); the two codings have the same evaluations. Cylinders of codes are printed $\mathcal{C}_p$ (source 13’s $\mathcal{D}_p$ is printed $\mathcal{P}_s(b)_p$, source 17’s $\mathcal{C}_p$ and source 18’s $\mathcal{C}_b(p)$ are printed $\mathcal{C}_p$). Source 14’s $\mathfrak{E}$, $\mathfrak{J}$, $\mathfrak{W}$ are printed $\mathfrak{W}_{sl}$, $\mathfrak{J}$, $\mathfrak{W}_{all}$, its relation macro $\backslash\text{lex}$ as $<_{lex}$; in its support section its $\tau = \text{cf } \kappa$ is printed r , its decreasing sequence (r_i) is printed (u_i) , and the generic regular cardinal of its definition of η_τ -orders is printed θ . A cylinder meeting both sides of a cut *crosses* it in sources 13, 14 and 15 (source 15 also calls such cuts *persistent*) and *straddles* it in sources 16, 17 and 18; the two words mean the same. All six use Ord as a character symbol for “a uniform Ord -indexed cofinal family exists and no set-sized one does”, never as a cardinal. Labels written for the merge in this Part begin `swo:viii:`.

40 Foundations and the audited starting point (source 14)

40.1 Sets, classes, and uniform families

We use sign sequences as canonical set representatives of surreals. Each surreal has a set ordinal birthday. This avoids treating a proper class of equivalent game forms as an individual set.

In GBC, a class $E \subseteq \text{Ord} \times \mathbf{No}$ may be the graph of a global enumeration. It cannot then be an element of a set or an ordinary class. Expressions such as

$$E \in \mathfrak{W}_{sl} = \text{Bij}(\text{Ord}, \mathbf{No})$$

are convenient predicate notation. They mean that the class variable E satisfies the indicated formula. They are not membership assertions in a powerclass object. The same convention applies to the virtual collection $\mathfrak{C}(\mathcal{P}_s(b))$ of class cuts.

A uniformly set-indexed family $(E_i)_{i \in I}$ is one class relation $U \subseteq I \times \text{Ord} \times \mathbf{No}$ whose i -th section is E_i , with I a set. Replacement with the parameter U produces a set of first disagreement ordinals or a set of evaluated prefix values. Merely writing “a set of proper-class functions” does not provide such a family.

A class well-order is a class linear order for which every nonempty subclass has a least member. Under the choice assumptions used here, the standard set-subset well-foundedness formulation is equivalent; see [27, Section 3]. Internal well-foundedness in an arbitrary model is not asserted to be external well-foundedness.

40.2 Why a baseline is an assumption, not a notation

Global Choice supplies a set-like class well-order and hence a class bijection $b : \text{Ord} \rightarrow \mathbf{No}$. Conversely, the predecessor report proves over GB without even assuming set Choice that a class well-order of the sign-sequence surreals implies Global Choice (theorem 13.1; Section 13 of this report at the pin). The proof uniformly well-orders the power sets $V_{\beta+1}$ ([merge] the delivered text says “coherently”; the orders need not extend one another, as corrected in this report on 3 October 2026, commit 0f52c0dca, after source 14’s pin) and constructs rank-by-rank presentations; it is stronger than the observation that each already well-orderable set embeds into $\mathbf{No}$. We use that reported equivalence as background, rather than claim to re-prove its choiceless direction here.

There is no paradox in using Ord as the domain of b : it is a proper class domain. There would be a paradox in calling it a set ordinal or applying a set power-set operation to it inside the same universe.

40.3 Elementary comprehension versus class-valued recursion

The new constructions use formulas with set quantifiers and previously given class parameters. For example, “a set prefix has a core completion on each side of this cut” quantifies over set codes, not over classes. The union of all such coherent prefixes is consequently an ordinary class obtained by elementary comprehension.

Set-valued recursion of length Ord with set histories at each stage can be implemented by coherent set-length recursions in GBC. This must not be conflated with recursions that choose an entire class at each stage, or with recursions along arbitrary non-set-like class well-orders. The literature proves that ETR implies abstract class-well-order comparability [27, Theorem 6]; our labelled comparison and prefix reconstruction are different operations and do not invoke that implication in reverse.

40.4 Repository audit and status of the inputs

The inspected report lives at

`Algebra/SurrealNumbers/docs/foundations-and-computation/surreal-well-orders/article.tex`.

Its README records a 115-page merged report, four source manuscripts, and a pending collection-level review; it explicitly says the report is not formalized (its README at the pin; [merge] the page count and the number of sources have grown with this addition, and the report is still not formalized). We inspected the introduction, set-supported and binary-coding sections, universe

discussion, and research questions relevant to the present work. This is a targeted audit, not a certification of the entire repository.

We also inspected the actual Lean file

`Algebra/SurrealNumbers/Surreal/Foundations/SignSequenceSimplicity.lean`.

It proves uniqueness and prefix properties of simplest separators *assuming existence*. In particular, `existsUnique_simplest_separator` takes an existence hypothesis. It is not a theorem supplying all small-cut separators by itself [10]. This distinction matters for the proposed formalization. [merge] The pinned tree already supplies the existence half and composes the two: `small_cut_fillers` in `SignSequenceCut.lean` [8] proves that small cuts have fillers, and `cut` in `SignSequenceCutOperation.lean` [9] is defined from `existsUnique_simplest_separator` applied to it (section 1.5). Source 14’s recommendation to compose them is therefore already carried out.

The present main theorems have written proofs below. Their novelty claim is only that they extend the inspected predecessor results; historical priority has not been established by the literature check.

41 The set-coded skeleton and its cylinders (source 14)

For the remainder of the class-level development we work in GBC and fix a class bijection $b : \mathbf{Ord} \rightarrow \mathbf{No}$. All occurrences of a class parameter mean a class belonging to the chosen class realization.

41.1 Canonical support codes

A *support code* is a set bijection $\sigma : S \rightarrow S$ with no fixed points, where S is a set of ordinals. The empty code is permitted. Extend it by the identity to a class permutation $\hat{\sigma}$ of $\mathbf{Ord}$, and set

$$E_\sigma = b \circ \hat{\sigma}, \quad \text{supp}(\sigma) = S.$$

Let $\mathcal{P}_s(b)$ be the ordinary class of these codes, ordered by $\sigma < \tau$ if $E_\sigma <_{\text{lex}} E_\tau$. The codes are canonical: two of them have the same extension exactly when they are equal. Every class permutation with set support is represented, because its support is invariant under the permutation and its restriction has no fixed points. A set of ordinals is bounded, so “set support” and “bounded support” are equivalent here. This does not mean finite support.

Let $\mathfrak{W}_{\text{sl}}$ stand for the virtual collection of class bijections $\mathbf{Ord} \rightarrow \mathbf{No}$, and $\mathfrak{J}$ for that of class injections. A statement quantifying over $\mathfrak{W}_{\text{sl}}$ or $\mathfrak{J}$ is a statement with class variables; these symbols do not name additional class objects. Two distinct members of $\mathfrak{J}$ have a least differing ordinal. Their common initial segment has set length, so first-difference comparison is unambiguous.

Lemma 41.1 (Fresh separators). *If A, B, U are sets of surreals and $A < B$, there is a surreal $x \notin U$ such that $A < x < B$. Consequently $\mathbf{No} \setminus U$ has no endpoints and fills all cuts specified by sets of its elements.*

Proof. The ordinary surreal set-cut theorem supplies a separator ([15]; lemma 5.2). In fact there are more than $|U|$ separators: recursively for $|U|^+$ many steps choose a separator above the earlier choices and A , and below B . Every stage uses set options, and the resulting choices are distinct. At least one is outside U . The same argument permits either option set to be empty. This uses set recursion and the usual surreal cut existence theorem, not a purported cut with proper-class options. $\square$

Lemma 41.2 (Completion of a set prefix). *Every injective set word $p : \lambda \rightarrow \mathbf{No}$ extends to some E_σ with $\sigma \in \mathcal{P}_s(b)$.*

Proof. Choose an infinite initial cardinal ν strictly greater than λ and all the ordinals in $b^{-1}[\text{ran } p]$. The map $\alpha \mapsto b^{-1}(p(\alpha))$ is a partial injection from ν to itself with domain of cardinality $< \nu$ and equally small range. Both complements have cardinality ν . Choose a bijection between them to extend the partial map to a permutation of ν , and fix every ordinal outside ν . Restricting the resulting permutation to its moved points gives the required canonical code. $\square$

Definition 41.3 (Cylinder and residual alphabet). For an injective set prefix $p : \lambda \rightarrow \mathbf{No}$, put

$$C_p = \{\sigma \in \mathcal{P}_s(b) : E_\sigma \upharpoonright \lambda = p\}, \quad A_p = \mathbf{No} \setminus \text{ran } p.$$

The empty prefix has cylinder $C_\emptyset = \mathcal{P}_s(b)$.

Proposition 41.4 (Cylinder geometry). *Every C_p is a nonempty convex proper class. It has neither an endpoint nor a set-sized cofinal or cointial subset. Its immediate subdivisions are*

$$C_p = \bigcup_{x \in A_p} C_{p \smallfrown (x)},$$

in increasing order of x . The union is an elementary class statement, not a set-indexed union. Cylinders of two prefixes are either nested or disjoint and strictly ordered.

Proof. Completion proves nonemptiness. If a word between two extensions of p first differed from p before λ , it would lie on the same side of both extensions, proving convexity. The child cylinders partition C_p by the value at coordinate λ , and first difference puts them in numerical order. [Lemma 41.1](#) and completion give a child below or above any specified set of child labels. Thus a set of members of C_p is neither cofinal nor cointial: take their next labels, choose a fresh label above or below all of them, and complete. In particular C_p cannot itself be a set. Finally, incomparable prefixes first differ at a set coordinate, which orders their entire cylinders; comparable prefixes give nested cylinders. $\square$

Lemma 41.5 (Density around injective branches). *If $F, G \in \mathfrak{J}$ and $F <_{\text{lex}} G$, there is a code $\sigma \in \mathcal{P}_s(b)$ with $F <_{\text{lex}} E_\sigma <_{\text{lex}} G$. For every $F \in \mathfrak{J}$ and every set ordinal α , there are codes below and above F agreeing with F before α and differing from it at α .*

Proof. For the first assertion use the first disagreement δ , choose x strictly between $F(\delta)$ and $G(\delta)$ outside their common prefix range, and complete $(F \upharpoonright \delta) \smallfrown (x)$. For the second, choose fresh values below and above $F(\alpha)$ and complete the two prefixes. A fixed global well-order of set codes makes all these choices uniform when a uniform family is needed. $\square$

These elementary cylinder facts, rather than a powerclass completion, are the main engine of the next result. The predecessor proves the stronger statement that $\mathcal{P}_s(b)$ interpolates every uniformly set-indexed strict cut of $\mathfrak{W}_{\text{sl}}$ ([theorem 19.10](#); Section 19 of this report at the pin). We will also recover the ordinary set-cut property of $\mathcal{P}_s(b)$ from our cut classification and its character calculation.

Note 41.6 ([merge]; Part IV counterparts of source 14's Section 4). Source 14's support codes are source 09's coding $\mathcal{P}_s(b)$ ([section 1.3](#)); its [lemma 41.1](#) is [lemma 5.2](#), its [lemma 41.2](#) is [lemma 19.4](#) with the cardinality check of [remark 19.7](#), its [proposition 41.4](#) is the code-level form of [proposition 19.25](#) (sources 11 and 09), and the two-sided approximants of [lemma 41.5](#) are those of [proposition 19.27](#).

Source 14 credits the interpolation theorem ([theorem 19.10](#)) and does not re-prove it; the elementary cylinder facts are what its classification uses. The class T_D of crossing prefixes of a cut, defined in the next section, is the class-scale form of the reals report's cut spine (`lwo:sp:lem:cutspine`, for $\mathcal{W}(\mathbb{R})$); source 14 does not cite it.

42 The core as re-proved in sources 13, 15, 16, 17 and 18

[merge] Each of the other five sources re-proves the core layer of Part IV before its recognition theorem, and each credits it to this report. Their sections are printed here whole, in source order, as re-proofs of the following results of Part IV; none is new against the repository.

- *Fresh separators*: [lemma 5.2](#) (source 13's [lemma 42.1](#), source 17's [lemma 42.12](#)).
- *Class universality, uniqueness and uniform complements* $h_S : \mathbf{No} \cong \mathbf{No} \setminus S$: [proposition 5.13](#) and [theorem 5.17](#) (source 13's [lemma 42.2](#), source 17's [corollary 42.15](#)). Source 16's termination section ([section 36](#)) uses the same family h_S .
- *Prefix completion*: [lemma 19.4](#) with [remark 19.7](#) ([lemmas 42.3, 42.8, 42.9](#) and [42.17](#); source 17 obtains it from its reserve theorem, [corollary 49.10](#)).
- *Interpolation and the core isomorphic to $\mathbf{No}$* : [theorem 19.10](#) and [corollary 19.15](#) ([theorems 42.4, 42.10, 42.13](#) and [42.18](#) and [corollaries 42.5, 42.11](#) and [42.14](#); source 15 cites them).
- *Approximants at a branch*: [proposition 19.27](#) ([lemma 42.6](#)).
- *The diagonal theorem*: [theorems 19.37](#) and [19.38](#) ([theorems 42.7, 42.16](#) and [42.19](#)).
- *Canonical first block and residual fibres* of a raw order: [theorem 15.1](#), [proposition 15.3](#), and [remark 15.4](#) (source 16's Subsection 11.1 below, with displays (S16.12) and (S16.13)).
- *Global choice is an assumption*: [theorem 13.1](#) (end of source 13's Section 3).

The proofs differ only in bookkeeping. Sources 13, 15, 16 and 18 complete a prefix inside an infinite initial cardinal, as Part IV does; source 17 completes it with the least possible reserve ([section 49](#)). Source 13's warning after [lemma 42.3](#) is the point of [remark 19.7](#). Source 16's Subsection 11.1 refers to its block decomposition, printed in Part IX ([section 52.2](#)); source 17's Section 8 and source 18's Section 14 are printed together with their remarks on what density does not give.

42.1 Source 13: surreal cuts and the size of the universe

42.1.1 Signs, birthdays, and small cut filling

We use the sign-sequence presentation of the surreal numbers. A surreal x is a sequence of signs $-$ and $+$ indexed by a set ordinal $\text{bd}(x)$. Extend the sequence by 0 after its domain; comparison is at the first unequal coordinate using $- < 0 < +$. Here 0 in the sign alphabet is a termination value, not an extra sign before the birthday. This is also the representation used by the inspected Lean carrier [7]. General context for the universal ordered-continuum viewpoint is given in [15].

For sets $A, B \subseteq \mathbf{No}$ with $A < B$, a surreal lies strictly between them. More precisely, if an ordinal γ strictly exceeds all their birthdays, there is such a separator of birthday exactly γ . The stronger statement is implemented at the source level by `exists_cut_separator_of_birthday_lt` [8]. We use it in the following form.

Lemma 42.1 (Fresh set-cut separator). *Let A, B, S be sets of surreals and suppose $A < B$. There is an $x \notin S$ with $A < x < B$. Empty sides are allowed.*

Proof. Choose γ strictly larger than every birthday occurring in $A \cup B \cup S$. A separator of birthday γ cannot belong to S . $\square$

The adjective “set” cannot be removed. There is no surreal above every ordinal: any surreal is smaller than a sufficiently long all-plus sign sequence. Likewise, a proper-class sign sequence is not another surreal at a larger birthday.

42.1.2 Set-cut orders and class back-and-forth

Call an ordinary class linear order *set-cut saturated* when every pair of sets $A < B$ has a strict interpolant, including when a side is empty. This is sometimes expressed by an η_{Ord} terminology; we use the explicit condition rather than assigning a set cardinal to it.

Lemma 42.2 (Class universality and uniqueness in GBC). *Every ordinary class linear order embeds into $\mathbf{No}$. Any two nonempty set-cut saturated ordinary class orders are order-isomorphic. In particular, for every set $S \subseteq \mathbf{No}$,*

$$\mathbf{No} \setminus S \cong \mathbf{No}.$$

The complement isomorphisms can be chosen as a single uniform class family $h_S : \mathbf{No} \rightarrow \mathbf{No} \setminus S$, indexed by sets S .

Proof. Global choice supplies a set-like enumeration of any proper class of sets; for a set carrier use a set well-order. At a stage of the usual order-embedding recursion, the images already chosen form a set. Their lower and upper parts determine a set cut in $\mathbf{No}$, and a fresh interpolant gives the next image. The recursion has a set partial map at each set-ordinal stage.

For uniqueness, alternate the two enumerations. A forth step puts the next unused source point into the target cut determined by the current partial isomorphism; a back step does the same for the inverse. Strict interpolation keeps each map injective. Every point on both sides is visited, so the union is an isomorphism. A set-cut saturated nonempty order cannot be a set, since using all its points as a lower side would require a point above itself.

[Lemma 42.1](#) says that $\mathbf{No} \setminus S$ has the same set-cut property. To make its isomorphism uniform, fix the global well-order and always choose its least eligible witness. For each set parameter S and each set stage, the partial history is a set satisfying a fixed recursive specification. Its existence and uniqueness are established by set-length recursion. Elementary class comprehension, quantifying over such set histories, gives one relation $h(S, x, y)$. No choice of a class witness for each S is being assumed. $\square$

This is not an appeal to elementary transfinite recursion with *class-valued* stages. Our stages are sets, even though their total union is a class. Keeping that distinction explicit will matter in [source 13's](#) sections on asymmetric products and on foundational interpretations ([\[merge\]](#) its Sections 10 and 13, printed in Part IX as [sections 55](#) and [57.3](#)).

42.1.3 Global choice is a real assumption

The earlier report proves, over GB without set choice, that the existence of a class well-order of $\mathbf{No}$ is equivalent to global choice ([theorem 13.1](#); Theorem 13.1 at the pin). Thus a global enumeration is not a choice-free consequence of the numerical order of surreals. We work in GBC for the main global constructions and fix a baseline bijection

$$b : \text{Ord} \xrightarrow{\sim} \mathbf{No}.$$

The particular b affects the chosen dense core but not the definition of the full enumeration comparison.

42.2 Source 13: the global enumeration order and its dense core

42.2.1 Uniformity and the code carrier

Let $\mathfrak{J}$ denote the virtual domain of injective class functions $F : \text{Ord} \rightarrow \mathbf{No}$, and let $\mathfrak{W}_{sl} \subseteq \mathfrak{J}$ require surjectivity. Their lexicographic comparison is at the least disagreement ordinal. A nonempty class of ordinals has a least member: find one member and take the least member below it. Thus this comparison is elementary in its class parameters.

A *uniformly set-indexed family* $(F_i)_{i \in I}$ means a set I and a single class relation coding all values $F_i(\alpha)$. A phrase such as “choose a class F_i for each i ” is not a substitute for this uniform family. We consistently use the latter.

Fix $b : \text{Ord} \simeq \mathbf{No}$. A set-supported permutation σ of Ord is the identity except on a set of ordinals. Code it by the set graph of its moved points, omitting all fixed points. Such normalized codes form an ordinary class $\mathcal{P}_s(b)$, and a code d evaluates to

$$E_d = b \circ \widehat{d}.$$

Equivalently, a code is a set bijection $d : S \rightarrow S$, where S is a set of ordinals and $d(\xi) \neq \xi$ for every $\xi \in S$; the empty code is allowed. This is an elementary condition on sets, so defining $\mathcal{P}_s(b)$ does not require quantification over a collection of class permutations. Different normalized codes have different evaluations. Order $\mathcal{P}_s(b)$ by these evaluations. The support of each code is bounded by a set ordinal, but there is no common set bound for all codes.

For an injective set word p , define

$$C_p = \{E \in \mathfrak{W}_{sl} : E \upharpoonright \text{dom}(p) = p\}, \quad \mathcal{P}_s(b)_p = \{d \in \mathcal{P}_s(b) : E_d \upharpoonright \text{dom}(p) = p\}.$$

C_p is virtual notation; $\mathcal{P}_s(b)_p$ is an ordinary class of set codes. Both are convex in their respective orders.

42.2.2 Completion with the missing cardinality check

Lemma 42.3 (Set-prefix completion). *Every injective set prefix $p : \alpha \rightarrow \mathbf{No}$ has a completion in $\mathcal{P}_s(b)$. The completion can be chosen uniformly from p .*

Proof. The class inverse of b , restricted to the set range of p , gives a set injection $f = b^{-1} \circ p : \alpha \rightarrow \text{Ord}$. Choose an infinite initial cardinal δ strictly above α and every member of $\text{ran}(f)$. Both $\delta \setminus \alpha$ and $\delta \setminus \text{ran}(f)$ then have cardinality δ . A bijection between these complements extends f to a permutation of δ . Extend it by the identity outside δ and retain only its moved points as a code. Global choice makes the choices uniform. $\square$

It would be incorrect to assert that every partial injection of a set into itself extends to a permutation without checking the complements. For instance, a bijection from a proper subset onto the entire carrier has no such extension. Enlarging to the cardinal δ is exactly what prevents this obstruction.

42.2.3 The active-prefix interpolation argument

The next theorem is the earlier dense-core interpolation result (theorem 19.10; Section 19 at the pin). We include the algorithm because both cut classification and the central-word theorem depend on its logic.

Theorem 42.4 (Uniform set-cut interpolation). *Let $(A_i)_{i \in I}$ and $(B_j)_{j \in J}$ be uniform set-indexed families in $\mathfrak{J}$, with $A_i < B_j$ for every cross pair. There is an injective set prefix p such that all its global completions lie strictly between the two families. In particular, a code in $\mathcal{P}_s(b)$ interpolates the cut.*

Proof. For a cross pair let $d(i, j)$ be its least disagreement. Class replacement gives a set of these ordinals. Put

$$\theta = \sup\{d(i, j) + 1 : i \in I, j \in J\},$$

with supremum 0 for an empty set. We construct an injective common prefix, keeping as *active* the rows that agree with it so far. Every inactive lower row is already strictly below all completions; every inactive upper row is already strictly above them.

At stage β , let L_β and U_β be the sets of next letters of the active lower and upper rows. Every $l \in L_\beta$ is at most every $u \in U_\beta$, since otherwise the corresponding cross pair would be reversed. There are two cases.

If $L_\beta \cap U_\beta$ is nonempty, it is a singleton $\{t\}$. Append t , which is fresh because the active rows are injective. Keep only the rows whose next letter is t . A dropped lower row has next letter less than t , and a dropped upper row has next letter greater than t , so the invariant is maintained.

If the intersection is empty, then $L_\beta < U_\beta$, allowing empty sides. Choose a fresh strict separator x outside the earlier range by Lemma 42.1, and append it. Every remaining active comparison is now strict. This prefix is the required p .

At a limit stage take the union of the earlier set prefixes and keep exactly the rows agreeing with it. The construction cannot continue with a shared letter through stage θ : surviving active rows on both sides would agree beyond their first disagreement. It therefore stops by that stage. The prefix has length at most $\theta + 1$. Apply Lemma 42.3 to obtain a code. $\square$

The histories in this proof have set length and set entries. Nothing requires collecting a set of class-valued choices, taking a surreal cut with proper-class sides, or invoking a class-valued recursion principle.

Corollary 42.5 (The core and its cylinders). *$\mathcal{P}_s(b) \cong \mathbf{No}$, and every $\mathcal{P}_s(b)_p \cong \mathbf{No}$. The core is order-dense among global enumerations and among distinct injective class words. Every interval specified by a uniform set cut in $\mathfrak{W}_{sl}$ contains a whole cylinder C_p .*

Proof. The theorem says that $\mathcal{P}_s(b)$ fills all its set cuts, so Lemma 42.2 applies. To keep a prescribed prefix p , run the same active-row construction after p , forbidding $\text{ran}(p)$ as well as the later used values; all forbidden values at a stage still form a set. This proves the set-cut property of $\mathcal{P}_s(b)_p$, including empty sides. The other assertions follow directly from the strict cylinder conclusion of the theorem. $\square$

42.2.4 Approximating one branch

Lemma 42.6 (Two-sided ordinal approximants). *For $F \in \mathfrak{J}$ there are uniform sequences of codes a_α, b_α indexed by Ord such that, for $\alpha < \beta$,*

$$E_{a_\alpha} < E_{a_\beta} < F < E_{b_\beta} < E_{b_\alpha}.$$

They are respectively cofinal and coinital among code evaluations strictly below and above F . Neither punctured side has a set-sized cofinal or coinital family.

Proof. Complete $F \upharpoonright \alpha$ followed by a fresh letter below $F(\alpha)$ to obtain a_α , and use a fresh letter above for b_α . The inequalities are decided at the earlier index. A code differing from F at ξ lies on the appropriate far side of every approximant with index greater than ξ . Uniform choices come from the least completion code in the fixed global well-order. Finally, Theorem 42.4, with F on one side, puts a code strictly beyond any proposed set family on the other side. $\square$

42.2.5 The diagonal limitation on all set codes

Theorem 42.7 (No exhaustive uniform evaluator). *No single class relation with set indices can have, among its rows, every class bijection $\text{Ord} \rightarrow \mathbf{No}$.*

Proof. Extend an alleged evaluator to all set indices by giving unused indices empty rows. Fix a class bijection $\alpha \mapsto c_\alpha$ from Ord to the universe of sets. Partition Ord into pairs

$$(2 \cdot \alpha, 2 \cdot \alpha + 1).$$

The multiplication is ordinal multiplication with 2 on the left; these adjacent pairs partition all ordinals.

On pair α , use the two values $b(2 \cdot \alpha), b(2 \cdot \alpha + 1)$. Put the second value first if the row c_α reports $b(2 \cdot \alpha)$ at coordinate $2 \cdot \alpha$; otherwise put the first value first. Put the other value second. This elementary class construction defines a bijection E . Whenever the row c_α is the graph of a function, it differs from E at that even coordinate. Thus every valid row misses E . $\square$

The theorem does not say that $\mathcal{P}_s(b)$ is too small to be dense. It says that density and exhaustive representation are different notions. It also does not prohibit defining some enumerations from set parameters; it prohibits one uniform evaluator from covering *all* of them.

42.3 Source 15: the ordinary class of bounded-support codes

Fix a class bijection $b : \text{Ord} \rightarrow \mathbf{No}$. A set-supported permutation of Ord equals the identity outside some ordinal. Code it by a permutation $\pi : \gamma \rightarrow \gamma$ and extend it by the identity. Normalize $\gamma = \sup\{\xi + 1 : \pi(\xi) \neq \xi\}$, using $\gamma = 0$ and the empty code for the identity. This makes codes unique. Let $\mathcal{P}_{\text{bd}}(b)$ be the ordinary class of these set codes, with evaluation

$$E_d(\xi) = b(\pi_d(\xi)).$$

The values of an evaluation are sets; the whole evaluation is a class function uniformly given by a class relation.

Lemma 42.8 (Every injective set prefix has a core completion). *Every injective set word $p : \alpha \rightarrow \mathbf{No}$ extends to some E_d , $d \in \mathcal{P}_{\text{bd}}(b)$, and a completion can be chosen uniformly.*

Proof. The map $q = b^{-1} \circ p$ is a set injection into Ord . Choose an infinite initial cardinal ν greater than α and every member of $\text{ran}(q)$. The complements of α and of $\text{ran}(q)$ in ν both have cardinality ν . A bijection between them extends q to a permutation of ν . Extend by the identity and normalize. The least permissible ν and a fixed global well-order choose a particular complement bijection, so the construction is uniform. $\square$

We use the following established core theorem from the prior report: $\mathcal{P}_{\text{bd}}(b)$, with the induced native lexicographic order, fills every set cut and is class-order-isomorphic to $\mathbf{No}$ (theorem 19.10 and corollary 19.15; Theorem 19.10 and Corollary 19.15 at the pin, which source 15 cites as “Theorems

19.10 and 19.15”). Its proof combines prefix completion with fresh separators and then applies the ordinary class back-and-forth theorem. This is a statement about a class of sets, so it falls within surreal universality.

For an injective set word $p : \alpha \rightarrow \mathbf{No}$, its cylinder is

$$C_p = \{d \in \mathcal{P}_{\text{bd}}(b) : E_d \upharpoonright \alpha = p\}.$$

Cylinders are nonempty and convex. Those of a fixed length partition $\mathcal{P}_{\text{bd}}(b)$ and are strictly ordered by their different prefixes. Each cylinder has no set cofinal or cointial subset: given a set of its codes, choose a fresh label above or below all their values at coordinate α , append it to p , and complete it using [Lemma 42.8](#).

42.4 Source 16: the bounded-support core of global enumerations

Work in GBC and fix a baseline bijection $b : \text{Ord} \rightarrow \mathbf{No}$. A normalized bounded-support code is a set permutation $\sigma : \lambda \rightarrow \lambda$ of an ordinal satisfying

$$\lambda = \sup\{\xi + 1 : \sigma(\xi) \neq \xi\}.$$

The empty permutation is permitted. Extend σ by the identity above λ , obtaining $\hat{\sigma}$, and evaluate the code as $E_\sigma = b \circ \hat{\sigma}$. Let $\mathcal{P}_{\text{bd}}(b)$ be the ordinary class of these set codes, ordered by comparison of their evaluations. Normalization makes the coding injective. The evaluations are class functions, but their codes are sets; the distinction is essential.

Lemma 42.9 (Uniform completion of every set prefix). *Every injective set function $p : \delta \rightarrow \mathbf{No}$ extends to an evaluation E_q with $q \in \mathcal{P}_{\text{bd}}(b)$. A completion can be selected uniformly in p .*

Proof. The function $b^{-1} \circ p$ is a set injection into Ord . Choose an infinite initial cardinal γ strictly above δ and all the ordinals in its range. The unused positions and unused values below γ both have cardinality γ . A bijection of their complements extends this injection to a permutation of γ . Extend by the identity and normalize. Choose the least suitable set bijection in a fixed global well-order to make the construction uniform. $\square$

Theorem 42.10 (Uniform set-cut interpolation). *Let $(F_i)_{i \in I}$ and $(G_j)_{j \in J}$ be uniformly set-indexed families in $\mathfrak{J}$ satisfying $F_i < G_j$ for all cross pairs. Either side may be empty. Some injective set prefix p has every continuation in $\mathfrak{J}$ strictly between the families. In particular, a member of $\mathcal{P}_{\text{bd}}(b)$ interpolates them.*

Proof. When both sides are nonempty, their cross first-disagreement ordinals form a set by Class Replacement. Bound them strictly below β ; otherwise put $\beta = 0$. Follow a prefix p , calling an input row active if it agrees with the whole prefix. Discarded lower rows are below every continuation and discarded upper rows above every continuation.

At coordinate $\delta = \text{dom}(p)$, let A and B be the sets of next values of the active lower and upper rows. Every value of A is at most every value of B . If they are disjoint, use the fresh-separator lemma ([lemma 5.2](#)) to append an unused value strictly between them and stop. This also handles empty sides. If they meet, their intersection is a singleton; append that forced value and retain only the rows using it. A surviving injective row guarantees freshness. At limits take unions of the set histories.

The process cannot retain both a lower and an upper row through length β , since such rows would agree beyond their bounded first disagreement. Thus it stops at a set stage, with all comparisons decided. [Lemma 42.9](#) provides the bounded-support completion. Surjectivity was not used for any input row. $\square$

Corollary 42.11 (A surreal-sized dense core). *The ordinary class order $\mathcal{P}_{\text{bd}}(b)$ satisfies the set-cut property and is isomorphic to $\mathbf{No}$. It is dense in both $\mathfrak{J}$ and $\mathfrak{W}_{\text{sl}}$ in the sense of their definable comparisons. It has no set cofinal or coinitial subset. The interpolation theorem works for every chosen baseline, including constraints whose rows do not belong to that baseline's core.*

Proof. Apply Theorem 42.10 to evaluations of sets of codes. The core is proper, since transpositions of position zero with arbitrary ordinal positions give distinct codes. Class universality and uniqueness (proposition 5.13 and theorem 5.17) then gives the isomorphism. The remaining assertions are instances of the same interpolation theorem. $\square$

Status. The completion and interpolation theorems, and the core isomorphism, are results of the earlier report (Part IV: lemma 19.4, theorem 19.10, and corollary 19.15). They are proved here because they are the inputs to the new recognition theorem below. Neither $\mathcal{P}_{\text{bd}}(b) \cong \mathbf{No}$ nor its density says that all global enumerations can be coded by sets.

42.4.1 Canonical first cores and residual fibers

For $R \in \mathfrak{W}_{\text{all}}$, the first block of source 16's canonical block decomposition (its Theorem 10.1, [merge] theorem 52.6) is

$$I_R = \{x \in \mathbf{No} : \{y : yRx\} \text{ is a set}\}.$$

It is always a proper class and has a unique enumeration $F_R : \text{Ord} \rightarrow I_R$. If I_R were a set, the least point outside it would have exactly that set of predecessors, a contradiction.

Conversely every $F \in \mathfrak{J}$ occurs as such a first core: enumerate its range by F and then append any class well-order of its complement. The points in the range have set predecessors; every residual point has that whole proper range before it. Restriction and concatenation therefore identify the fiber over F with

$$\mathcal{W}(Y_F), \quad Y_F = \mathbf{No} \setminus \text{ran}(F), \quad (\text{S16.12})$$

where for a proper Y_F the notation denotes the raw class comparison. Each fiber is convex. Distinct fibers are ordered by the first ordinal disagreement of their cores. Symbolically,

$$\mathfrak{W}_{\text{all}} \cong \sum_{F \in \mathfrak{J}} \mathcal{W}(Y_F). \quad (\text{S16.13})$$

This is a summary of uniform relational constructions, not a same-level class sum whose index elements are proper classes. The set-like order $\mathfrak{W}_{\text{sl}}$ consists precisely of the exhaustive, empty-residual fibers. A singleton fiber need not be exhaustive: omitting one label produces a unique raw completion of type $\text{Ord} + 1$.

42.5 Source 17: the ordinary class core inside the global comparison

From now on work in GBC and fix $b : \text{Ord} \rightarrow \mathbf{No}$. Let $\mathfrak{J}$ be the predicate of injective class functions $\text{Ord} \rightarrow \mathbf{No}$, and let $\mathfrak{W}_{\text{sl}}$ be its surjective subpredicate. Comparisons are made at the least ordinal of first disagreement. These are predicates on class variables, not classes whose members are proper classes.

42.5.1 Canonical codes and interpolation

A *support code* is a set bijection $\sigma : S \rightarrow S$, where $S \subseteq \text{Ord}$ is a set and $\sigma(\xi) \neq \xi$ for every $\xi \in S$. The empty code is allowed. Extend it by the identity off S and put $E_\sigma = b \circ \hat{\sigma}$. Let $\mathcal{P}_s(b)$ be the ordinary class of these codes, ordered by the associated enumerations. The fixed-point-free condition makes the coding unique. A global enumeration has such a code exactly when it agrees with b at every sufficiently large ordinal. By [Corollary 49.10](#), every injective set prefix occurs in this core.

We recall the central interpolation theorem of the source, with its proof because the later recognition results depend on its full strength for *injective, possibly nonexhaustive* rows.

Lemma 42.12 (Avoiding a set of labels). *If L, R, Z are sets of surreal numbers with $L < R$, there is $z \notin Z$ with $L < z < R$. Either side of the cut may be empty.*

Proof. The surreal set-cut property supplies a strict separator of any two separated sets. Repeat that construction $|Z|^+$ times, each time adding all earlier chosen separators to the lower side. Every stage has a set history and gives a new separator below all of R . The resulting set has more elements than Z , so one separator is not forbidden. This is set-valued recursion with class parameters, not class-valued recursion. $\square$

Theorem 42.13 (Core interpolation; source result). *Let $(A_i)_{i \in I}$ and $(B_j)_{j \in J}$ be uniformly set-indexed families in $\mathfrak{J}$, with $A_i <_{\text{lex}} B_j$ for every i, j . There is an injective set prefix p such that every row in $\mathfrak{J}$ extending p lies strictly between both families. In particular there is such a row with a code in $\mathcal{P}_s(b)$. Empty index sets are permitted.*

Proof. When both sides are nonempty, Class Replacement collects their cross first-difference coordinates into a set. Choose an ordinal β strictly above all these coordinates. If one side is empty take $\beta = 0$.

Build an injective prefix, retaining as *active* the rows agreeing with it. At stage α form the sets of active next labels

$$L_\alpha = \{A_i(\alpha) : A_i \text{ active}\}, \quad R_\alpha = \{B_j(\alpha) : B_j \text{ active}\}.$$

Every lower label is at most every upper label. Their intersection has at most one member. If it is empty, choose a fresh strict separator by [Lemma 42.12](#), append it, and stop. All active inequalities are now strict. If the intersection is $\{z\}$, append z and keep only rows with next value z . This value is fresh because a surviving row is injective. A discarded lower row is below every later extension; a discarded upper row is above every later extension. This is the invariant throughout the construction. At limits take unions of the set histories.

Both sides cannot still be active after a common prefix of length β , since an active cross pair would disagree only at or beyond β . Thus the procedure stops by the next step. Complete the final prefix in $\mathcal{P}_s(b)$ using [Corollary 49.10](#). All recursions and collections in this proof have set-sized stages and values. $\square$

Corollary 42.14 (A dense surreal order core; source result). *The class order $\mathcal{P}_s(b)$ is isomorphic to the numerical order of $\mathbf{No}$. It is order-dense in the global comparisons of both $\mathfrak{W}_{\text{sl}}$ and $\mathfrak{J}$: between any two distinct rows there is a coded row. Every uniformly set-indexed separated cut of those comparisons has a core interpolant.*

Proof. The last statements are [Theorem 42.13](#). In particular $\mathcal{P}_s(b)$ has the same set-cut interpolation property as $\mathbf{No}$. Under Global Choice enumerate each of these proper classes in a set-like well-order. Construct a back-and-forth partial order-isomorphism along Ord . At each stage its domain and range

are sets, so the required cut is a set cut; interpolation provides a fresh image or preimage. Global choice selects a least permissible choice. Every history at a set ordinal is a set, so elementary class comprehension together with set-valued transfinite recursion gives the full class map. Alternating forth and back ensures exhaustiveness on both sides. $\square$

This is an isomorphism of *linear orders*, not of surreal fields or of structures with the simplicity relation. The additional structure studied in [14] cannot be inferred from numerical order alone. Nor is the core exhaustive in $\mathfrak{W}_{\text{sl}}$.

Corollary 42.15 (Universality for ordinary class linear orders). *Every given linear order on a class of sets embeds by a class map into $\mathcal{P}_s(b)$, hence into $\mathbf{No}$, $\mathfrak{W}_{\text{sl}}$, and $\mathfrak{W}_{\text{all}}$ in their stated senses.*

Proof. Global choice supplies a set-like enumeration of the given class (or a set enumeration if its carrier is a set). At each stage, previous images form a set. Partition them by whether their source labels are below or above the next source label; interpolation in $\mathcal{P}_s(b)$ supplies a fresh image. This forth-only recursion gives the required embedding. The core inclusion and Corollary 42.14 give the other targets. This is the source’s class-universality argument. $\square$

The hypothesis “class of sets” cannot be dropped and replaced with “all class well-orders”. The latter is the higher-order predicate $\mathfrak{W}_{\text{all}}$, not a carrier to which this recursion theorem applies. This distinction is compatible with the no-coding theorem that follows.

42.5.2 Why a dense class of codes is not a universal list

Theorem 42.16 (Uniform diagonal obstruction; source result). *No ordinary class of set indices with one uniform evaluation relation lists all class well-orders of $\mathbf{No}$. A missing order can always be chosen set-like.*

Proof. Global choice permits reindexing a proposed nonempty list by Ord ; a set-sized list can be extended by repetitions. Write its rows as R_α . Partition the baseline labels into the pairs

$$\{b(2 \cdot \alpha), b(2 \cdot \alpha + 1)\} = \{a_\alpha, c_\alpha\}, \quad a_\alpha < c_\alpha.$$

The products here are ordinal products with 2 on the left, so these successive two-element blocks partition Ord . In block α , list the two labels in the order opposite to their relative order under R_α . Keep the blocks in increasing α order. Elementary comprehension defines a class bijection $E : \text{Ord} \rightarrow \mathbf{No}$. Its induced well-order disagrees with R_α on that pair, for every α . The empty-list case is immediate. $\square$

Each input row and the diagonal output are legitimate classes. The forbidden object is a same-level exhaustive decoder, not an individual large relation. Strengthening comprehension does not make a class into a permissible element of another class of the same two-sorted theory. An external set universe, a higher sort, or a stated uniform operator provides a different, legitimate interpretation.

42.6 Source 18: the set-like global order and its ordinary class core

Assume GBC and fix a class bijection $b : \text{Ord} \rightarrow \mathbf{No}$. The set-like global order can be studied through its ordinal enumerations: if $e \neq f$, the nonempty class $\{\alpha : e(\alpha) \neq f(\alpha)\}$ has a least ordinal. Numerical comparison there agrees with source 18’s Theorem 12.3 ([merge] theorem 51.16; theorem 14.7).

42.6.1 Canonical bounded-support codes

A permutation π of $\mathbf{Ord}$ with set-sized support can be encoded by the set graph of its nonfixed pairs; it is the identity elsewhere. This representation is unique. Let $\mathcal{P}_s(b)$ be the genuine class of all such set codes, ordered by their evaluated global enumerations $b \circ \pi$. The notation distinguishes the set code from the class function that it evaluates.

Lemma 42.17 (Completion of every set prefix). *Every injective set word $p : \delta \rightarrow \mathbf{No}$ extends an evaluated member of $\mathcal{P}_s(b)$.*

Proof. The b -indices of the labels in $\text{ran } p$ form a set. Choose an infinite initial cardinal λ greater than δ and all these indices. The first λ values of b contain $\text{ran } p$. List p first, followed by a bijection of the unused values of this initial block in length λ . Since $\delta + \lambda = \lambda$, the result permutes that entire block. Leave all later positions unchanged. This is a bounded-support permutation extending p . $\square$

Theorem 42.18 (Uniform set-cut interpolation). *Let $(e_i)_{i \in I}$ and $(f_j)_{j \in J}$ be uniformly presented set-indexed families of global ordinal enumerations, and suppose $e_i < f_j$ for all i, j . There is $g \in \mathcal{P}_s(b)$ strictly between them. In particular, $\mathcal{P}_s(b)$ fills every set cut and is order-isomorphic to $\mathbf{No}$ in GBC. It is order dense in $\mathfrak{W}_{\text{sl}}$.*

Proof. When both families are nonempty, the first disagreements of all pairs (e_i, f_j) form a set of ordinals by Class Replacement. Take a bound on them. Run a common-prefix search: keep only the family members extending the current prefix; numerical next values on the lower side are at most those on the upper side. If the sets of next values meet, they meet in a unique value, which is appended. If they do not meet, choose a fresh surreal strictly between them, append it, and stop. At a limit stage take the union of the preceding set prefixes.

The search cannot continue through the bound on all pairwise first disagreements while both active sides remain nonempty. When a side becomes empty, the surreal set-cut property also supplies a fresh next value. If one original side is empty, the search stops at the first step. Each inactive lower member is below the whole current cone, and each inactive upper member is above it. Thus the resulting successor prefix puts every completion strictly between the given families. Apply [Lemma 42.17](#).

For the isomorphism, perform set-valued back-and-forth along fixed set-like enumerations of the two proper classes $\mathcal{P}_s(b)$ and $\mathbf{No}$. At each stage only a set of previously matched points must be respected, and interpolation supplies the next point. Taking singleton families in the first assertion proves density in $\mathfrak{W}_{\text{sl}}$. $\square$

The conclusion is stronger than pairwise density and weaker than filling arbitrary proper-class cuts. For example, choose an injective $\mathbf{Ord}$ word that enumerates $\mathbf{No} \setminus \{x\}$. It determines a cut of $\mathcal{P}_s(b)$ but cannot itself be a global enumeration of $\mathbf{No}$. The next section makes the distinction exact.

Class universality of $\mathbf{No}$ now implies that every supplied class linear order embeds into $\mathfrak{W}_{\text{sl}}$, through $\mathcal{P}_s(b)$. The same is true inside any nonempty open interval of $\mathfrak{W}_{\text{sl}}$: a sufficiently long set prefix and a free numerical interval place a copy of the same core inside it. This is a statement about uniform class-indexed subfamilies. It does not say that $\mathfrak{W}_{\text{sl}}$ itself is an ordinary class.

42.6.2 An elementary obstruction to exhaustive class coding

Theorem 42.19 (No ordinary class of codes for all global well-orders). *There is no class-indexed uniform evaluator whose values include every class well-order of $\mathbf{No}$. The same holds if only set-like orders are required as values.*

Proof. Under Global Choice any class of set codes can be listed in an Ord-indexed family, repeating values if the code class is a set. Suppose $(R_\alpha)_{\alpha \in \text{Ord}}$ were exhaustive. Split the fixed baseline b into consecutive pairs

$$\{b(2 \cdot \alpha), b(2 \cdot \alpha + 1)\}, \quad \alpha \in \text{Ord}.$$

In pair α , choose the orientation opposite to the relative order of these two labels in R_α . Keep the pair blocks in their original ordinal order. This is a class bijection $\text{Ord} \rightarrow \mathbf{No}$, hence a set-like well-order. It differs from R_α on the relative order of its designated pair, for every α , contradicting exhaustiveness. The orientation class is defined elementarily from the proposed evaluator. $\square$

This is the relevant Cantor-style diagonal. It blocks a uniform “all codes are sets” shortcut. Individual global well-orders can still be parameters, and any specified uniform family can be studied. Kelley–Morse comprehension strengthens which classes exist; it does not make proper classes into members of other ordinary classes.

43 A trichotomy for class cuts of the skeleton (source 14)

43.1 Normalization and the three constructions

A *proper normalized cut* of $\mathcal{P}_s(b)$ is a class $D \subseteq \mathcal{P}_s(b)$ which is nonempty, proper, downward closed, and has no greatest member. Its complement is allowed to have a least member. Thus the point σ is represented by the lower set $\{\tau : \tau < \sigma\}$, not by the lower set that includes σ . This convention is essential when we calculate characters.

We write D_p^- and D_p^+ for the two edges of a nonempty-prefix cylinder:

$$\begin{aligned} D_p^- &= \{\sigma : (\forall \tau \in C_p) \sigma < \tau\}, \\ D_p^+ &= \{\sigma : (\exists \tau \in C_p) \sigma < \tau\}. \end{aligned}$$

Since C_p has no endpoints, D_p^- consists of the codes below the whole cylinder, whereas D_p^+ consists of those codes together with all of C_p . Both are proper normalized cuts. The empty-prefix edges are the improper endpoint cuts and are excluded.

For a set prefix p and a nonempty proper initial class $H \subset A_p$ with no greatest member and with $A_p \setminus H$ having no least member, define a *numerical-gap cut*

$$D_{p,H} = D_p^- \cup \{\sigma \in C_p : E_\sigma(\lambda) \in H\}, \quad \lambda = \text{dom } p.$$

Here $D_\emptyset^- = \emptyset$ by convention. The nonprincipality conditions prevent this notation from duplicating the two edges of a child cylinder.

Finally, for $F \in \mathfrak{J}$ define the *branch trace*

$$D_F = \{\sigma \in \mathcal{P}_s(b) : E_\sigma <_{\text{lex}} F\}.$$

[Lemma 41.5](#) shows that it is a proper normalized cut. The branch need not be exhaustive. If $F = E_\sigma$, the cut is principal in the normalized sense; otherwise it has no point of the skeleton at its boundary.

43.2 Crossing cylinders and reconstruction

Definition 43.1. A prefix p *crosses* D if its cylinder meets both sides:

$$\text{Cross}_D(p) \iff (\exists \sigma \in C_p \cap D)(\exists \tau \in C_p \setminus D).$$

Let T_D be the class of crossing set prefixes.

All displayed witnesses are sets. Therefore T_D is obtained by *elementary* class comprehension from D and b .

Lemma 43.2 (The crossing-chain lemma). *For every proper normalized cut D , the class T_D contains the empty prefix, is downward closed under prefix restriction, and is a chain under prefix extension. It contains at most one prefix of any given length. Its union is an injective function with domain either a set ordinal or all of Ord.*

Proof. Properness gives the empty prefix, and any ancestor of a crossing cylinder also crosses. If incomparable prefixes p, q both crossed, order their disjoint cylinders as $C_p < C_q$. An excluded point of C_p would then lie below an included point of C_q , contradicting downward closure. Thus they are comparable, which also proves uniqueness at each length. Their union is a function and is injective because every two of its coordinates occur together in a sufficiently long member of the chain. Its domain is an initial subclass of Ord. Such a subclass is all of Ord or a set ordinal. In the latter case, class Replacement on that set domain makes the union a set function. $\square$

Theorem 43.3 (Boundary trichotomy). *Every proper normalized class cut D of $\mathcal{P}_s(b)$ has exactly one of the following forms:*

- (i) a cylinder edge D_p^- or D_p^+ for a nonempty injective set prefix p ;
- (ii) a numerical-gap cut $D_{p,H}$ with the nonprincipality conditions specified above;
- (iii) a branch trace D_F for an injective class function $F : \text{Ord} \rightarrow \mathbf{No}$.

The data within the appropriate form, including the edge direction, are unique. The alternatives and their data can be recovered from the crossing-prefix class by elementary formulas with class parameters.

Proof. We distinguish whether the lengths occurring in T_D are unbounded.

Unbounded case. Let $F = \bigcup T_D$. Its domain is Ord. For a code σ with $E_\sigma \neq F$, let δ be their first disagreement. Take a crossing prefix of length greater than δ . If $E_\sigma <_{\text{lex}} F$, then σ is below that entire cylinder and in particular below one of its included points, so $\sigma \in D$. If $F <_{\text{lex}} E_\sigma$, it is above an excluded point of that cylinder and cannot belong to D . If $E_\sigma = F$, membership in D would make σ its greatest member, since all larger codes have just been excluded. This is prohibited by normalization. Therefore $D = D_F$.

Bounded case with no longest crossing prefix. The union p has nonzero limit length λ . It does not cross, so either $C_p \subseteq D$ or $C_p \cap D = \emptyset$. For every code outside C_p , its first disagreement with p occurs before λ ; a longer crossing prefix then decides its membership exactly as in the unbounded case. Hence $D = D_p^+$ in the first alternative and $D = D_p^-$ in the second.

Bounded case with a longest crossing prefix p . No child $C_{p \frown (x)}$ crosses. Each child is consequently wholly included or wholly excluded. Let

$$H = \{x \in A_p : C_{p \frown (x)} \subseteq D\}.$$

It is an initial class of A_p . Because C_p crosses and the child cylinders partition it, H is nonempty and proper. Points outside C_p are below all of it or above all of it; the crossing witnesses show that

exactly those below belong to D . Thus D has the formula used to define $D_{p,H}$. If H has a greatest member x , that formula is $D_{p\smallfrown(x)}^+$. If $A_p \setminus H$ has a least member x , it is $D_{p\smallfrown(x)}^-$. Both possibilities cannot occur, since A_p is dense. If neither occurs, we have precisely alternative (ii).

For uniqueness, in alternative (iii) the crossing prefixes are exactly all set restrictions of F ; in alternative (ii) they are exactly p and its restrictions; and at an edge they are exactly the *proper* restrictions of its prefix. If the last chain has no longest member, its union recovers the limit prefix and membership of its cylinder recovers the direction. If it has a longest member, the included child labels have a maximum or the excluded child labels have a minimum; that endpoint recovers the additional label and the direction. For a numerical gap neither endpoint exists. These descriptions distinguish the alternatives, and also recover every parameter uniquely. $\square$

Remark 43.4 (What the theorem does not construct). It does not collect all class cuts into an ordinary class, nor insert an Ord-length sign sequence as a surreal number. The theorem is a statement about an arbitrary supplied class D . Its elementary formulas can be used uniformly in models, but the virtual family of every such D still occupies a higher logical level.

43.3 Examples distinguishing the alternatives

For the one-letter prefix (0) , the cut $D_{(0)}^-$ lies just below all core enumerations beginning with 0. There is no numerical “largest first entry below 0.” It is a cylinder boundary, not a branch trace. At the empty prefix, take the numerical gap

$$H = \{x \in \mathbf{No} : (\exists n \in \mathbb{N}) x < n\}.$$

Its complement is the class of surreals at least as large as every natural number. It has no least member: from such a surreal y , the number $y - 1$ is still in the complement and is smaller. The class H has no greatest member. This produces a cut of type (ii), with a countable cofinal left side but no set-sized coinital right side. Finally $F(\alpha) = \alpha$ is an injective branch of type (iii) which omits, for example, -1 . Its trace is coherent at every set length but is not realized by any member of $\mathfrak{W}_{\text{sl}}$. Exhaustiveness is not implied by coherence.

Note 43.5 ([merge]; the three examples in the repository). The first-label cut at 0 (with its upper-edge twin) is printed also by sources 15 and 16 (example 46.10 and source 16’s table after theorem 46.3); the numerical gap $\{x : \exists n x < n\}$ also by sources 15 and 17 (examples 46.11 and 46.23). The branch $F(\alpha) = \alpha$, which omits -1 , is source 09’s theorem 19.43 (printed in Part IV), which source 14 does not cite; source 13 recovers it as example 46.17, crediting this report, and source 15 prints it again (after example 46.11). Its one-label variant, a bijection $\text{Ord} \rightarrow \mathbf{No} \setminus \{r\}$, is source 11’s theorem 19.42, printed again by sources 16 and 17 (source 16’s table, example 46.25), which do not cite it. The limit boundary below the cylinder of $p(n) = n$ ($n < \omega$), which source 14 does not use, is printed by sources 13 (end of its Section 9), 15 (example 46.11), 16 (its table) and 17 (example 46.24, for any injective p of length ω).

44 Characters, saturation, and principal cuts (source 14)

We use Ord as a *character symbol*: it means that a uniform ordinal-indexed cofinal or coinital family exists, while no set-sized one exists. It is not an ordinary cardinal. For any nonempty class linear order with no set-sized cofinal subset, global choice permits an increasing cofinal Ord-sequence: at each set stage dominate the earlier choices and the elements seen so far in a global enumeration. If no such upper choice existed, the already considered set would be cofinal. This is set-valued

recursion. The dual argument handles coinitiality. A set-sized character is the least cardinality of a cofinal or coinitial subset; if there is a corresponding endpoint it is 1.

Theorem 44.1 (Cut-character calculation). *Write the character of a normalized cut as $(\text{cf } D, \text{ci}(\mathcal{P}_s(b) \setminus D))$. For the alternatives of Theorem 43.3, the following formulas hold.*

<i>Cut</i>	<i>Character</i>
D_F , with $F = E_\sigma$ for some core code	(Ord, 1)
D_F , with F not represented by a core code	(Ord, Ord)
D_p^- , $\text{dom } p = \lambda$ a nonzero limit	(cf λ , Ord)
D_p^+ , $\text{dom } p = \lambda$ a nonzero limit	(Ord, cf λ)
D_p^- or D_p^+ , $\text{dom } p$ a successor	(Ord, Ord)
$D_{p,H}$	(cf H , $\text{ci}(A_p \setminus H)$)

For the last row both characters cannot be set-sized. At every actual point of $\mathcal{P}_s(b)$, both strict local characters are Ord.

Proof. For a branch F , complete a prefix agreeing with F before α and taking a fresh smaller entry at α ; call the result a_α . Similarly obtain b_α with a larger entry. If $\alpha < \beta$, first difference gives

$$a_\alpha < a_\beta <_{\text{lex}} F <_{\text{lex}} b_\beta < b_\alpha.$$

Every code below or above F is surpassed on the appropriate side by one of these approximants, once α exceeds its first disagreement. A set of codes differing from F has a set of first disagreements and hence a common ordinal bound; a later approximant defeats its alleged cofinality or coinitiality. If F itself is a core point, the complement of D_F has that point as its least member. This gives (Ord, 1), not (Ord, Ord); only the strict right side has character Ord.

At the lower edge of a limit-length prefix p , take a cofinal sequence of coordinates $\alpha < \lambda$ and complete a word that first differs from p there by a smaller entry. These codes are cofinal below the cylinder. Conversely the first-disagreement coordinates of any cofinal family must be cofinal in λ : a code which follows p longer would otherwise be above the entire family. This proves the character cf λ . On the other side the cylinder has no set-sized coinitial subset by Proposition 41.4, and lies below all other points of the complement. Its character is Ord. The upper-edge calculation is dual.

For a successor-length prefix $p = q \frown (x)$, any set of points below C_p that first differ before the last coordinate can be surpassed by one following q . Among those which follow q , their entries at the last coordinate form a set below x . Lemma 41.1 provides a fresh entry between that set and x . Completion again surpasses the whole family while remaining below C_p . Thus the lower side of the lower edge has no set-sized cofinal subset. The opposite side follows from the cylinder property. The upper edge is the dual case.

For a numerical gap, project included points in C_p to their next-coordinate labels in H . A cofinal set of labels, with one completed word above each chosen label when necessary, produces a cofinal family of codes. Conversely a cofinal family of codes must have cofinal projected labels: otherwise a child cylinder with a larger label in H surpasses it. The absence of a greatest member of H removes any dependence on the internal character of one child. The dual argument proves the right character. If both were sets, cofinal labels on the left and coinitial labels on the right would constitute a set cut of A_p with no separator, contradicting Lemma 41.1. Finally the branch argument at a core point already proves the assertion about strict local characters. $\square$

Corollary 44.2 (Recovery of the core saturation theorem). *Every separated pair of sets of codes $A, B \subseteq \mathcal{P}_s(b)$ has a strict separator in $\mathcal{P}_s(b)$. In particular the core is an η_{Ord} -order: it fills all set-sized strict cuts, including those with an empty side.*

Proof. Empty-side bounds follow by choosing a first label above or below all first labels in the specified set. Suppose both sides are nonempty and no separator exists. If A has a greatest point a , then B would be coinital in the strict right side of a , contradicting its local character. The dual argument applies if B has a least point. Otherwise let $D = \{\sigma : (\exists a \in A) \sigma < a\}$. It is a proper normalized cut with A cofinal. Absence of a separator makes B coinital in its complement: any point there below all of B would be strictly above all of A . The character table has no nonprincipal cut with two set-sized characters, a contradiction. $\square$

Corollary 44.3 (The known surreal order type). *There is a class order isomorphism $\mathcal{P}_s(b) \cong \mathbf{No}$ in GBC.*

Proof. Both are proper class linear orders filling all set cuts. Fix set-like global enumerations and perform back-and-forth along Ord. At a set stage the existing partial map has set domain and range; the next point's lower and upper constraints are sets and admit a fresh image. The union is an elementary definable class bijection preserving order. This is the standard argument, also used by the predecessor; the isomorphism statement itself is inherited, not claimed as a new result. $\square$

The character calculation explains why the core is not Dedekind complete. For instance the increasing core points whose first entries are natural numbers have upper bounds but no supremum. Filling *strict* set cuts supplies another point between all those entries and any proposed upper bound. The countable/non-set-sized numerical-gap character above records this failure rather than contradicting saturation.

Note 44.4 ([merge]; which rows are new). The branch rows of [theorem 44.1](#) are source 11's [proposition 19.27](#) and source 09's [theorem 19.43](#) ((Ord, Ord) at a non-core branch); [corollary 44.2](#) is [theorem 19.10](#) for sets of codes and [corollary 44.3](#) is [corollary 19.15](#), both stated by source 14 as recoveries. New are the characters of the edge and numerical-gap cuts and the statement that no nonprincipal cut has two set-sized characters. The same table is proved, case by case, by source 13 ([proposition 46.19](#)), source 15 ([proposition 46.12](#)), source 16 (the paragraph after its table) and source 17 for non-core branches ([proposition 46.26](#)); those statements are printed with their sources' recognition theorems in [section 46](#).

45 Recognizing exhaustive cuts and preserving old cuts (source 14)

Theorem 45.1 (Exhaustive-cut recognition). *Let D be a proper normalized class cut of $\mathcal{P}_s(b)$. There exists $E \in \mathfrak{W}_{sl}$ with $D = D_E$ if and only if both conditions hold:*

$$\forall \alpha \in \text{Ord} \exists p [\text{dom } p = \alpha \wedge \text{Cross}_D(p)], \quad (\text{C})$$

$$\forall x \in \mathbf{No} \exists p [\text{Cross}_D(p) \wedge x \in \text{ran } p]. \quad (\text{X})$$

Such an enumeration is unique, and its graph is given by

$$E(\alpha) = x \iff \exists p [\text{Cross}_D(p) \wedge \alpha \in \text{dom } p \wedge p(\alpha) = x].$$

All quantifiers in the criterion and reconstruction range over sets; only D and b are class parameters.

Proof. For an exhaustive E , every restriction $E \upharpoonright \alpha$ crosses its trace by the smaller and larger completions in [Lemma 41.5](#). Every surreal occurs in one of these restrictions, giving (C) and (X). Conversely (C) and [Lemma 43.2](#) produce an injective union F on all of Ord. Condition (X) makes

it surjective. The unbounded case of [Theorem 43.3](#) gives $D = D_F$. The displayed graph formula is exactly the union of the crossing chain; its functionality and uniqueness follow from the chain lemma. $\square$

A logical simplification. For this proper-class alphabet, (X) by itself already implies (C). Indeed, the union of the crossing chain then has range all of $\mathbf{No}$; if its domain were a set ordinal, class Replacement would make that range a set. Its domain is therefore Ord, and prefix closure gives (C). We retain the two conditions in the theorem to distinguish coherent nonexhaustive branches from failures that stop at a set stage; we do not claim that the conditions are logically independent.

This answers the predecessor’s question “Which cuts of the core represent exhaustive branches?” ([research question 61.1](#); source 09’s third direction in [section 61.3](#)) *in the prefix-enriched structure*. A bare linear-order isomorphism of the core does not come equipped with its prefix cylinders or the label evaluation. The theorem does not show that those enrichments are definable from the order relation alone. [Corollary 47.3](#) below proves that an automorphism-invariant bare-order criterion cannot recognize exhaustiveness: some core automorphisms carry exhaustive traces to nonbranch cylinder edges.

The criterion also separates two notions of completion. An injective branch always gives a cut coherent through every set stage. It gives a global enumeration only if it eventually visits every label. No condition at a single bounded stage can enforce the latter assertion. For the branch $F(\alpha) = \alpha$, (C) holds and (X) fails at $x = -1$. The two cylinder-edge alternatives fail (C) at a bounded length; a numerical-gap alternative also fails (C), because there is no crossing child after its last crossing prefix.

[merge] The second half of source 14’s Section 7, on preserving old cuts under class extensions (its [theorem 48.1](#), the corollary and the remark after it), is printed in [section 48](#).

46 Five further proofs of the represented-cut theorem (sources 16, 15, 13, 17 and 18)

Note 46.1 ([merge]; one theorem, six proofs). The theorem that a normalized class cut of the core is the cut of an injective class word exactly when a crossing (straddling) prefix exists at every set length, that the word is then unique and elementary in the cut, and that it is exhaustive exactly when the ranges of these prefixes cover $\mathbf{No}$, is proved independently in all six batch-81 sources: source 14’s [theorem 43.3\(iii\)](#) and [theorem 45.1](#), source 16’s [theorem 46.2](#), source 15’s [theorem 46.6](#) with [corollary 46.8](#), source 13’s [theorem 46.15](#), source 17’s [theorem 46.21](#) and source 18’s [theorem 46.28](#) with [corollary 46.31](#). Each source presents it as its own answer to source 11’s [research question 61.1](#); it answers that question ([section 50](#)), and no source has priority over another. The six proofs are the same argument: uniqueness of a crossing prefix at each length, coherence under restriction, the elementary union, and the first disagreement with a code. They are printed as routes because each source builds different examples and corollaries on its own proof.

The analysis of a failing cut is likewise proved five times. Source 14 partitions the proper cuts by their form (edge, numerical gap, branch: [theorem 43.3](#)); sources 13, 15, 16 and 18 partition the failing cuts by their first failed length (successor or limit: [theorems 46.3](#), [46.9](#), [46.18](#) and [46.32](#)). The two partitions are equivalent: a limit failure is an edge of a limit-length cylinder, and a successor failure is an edge of a child cylinder or a numerical gap. Source 15’s successor case is the finest, separating a *forbidden repeated label* (the unused alphabet split at a label already used by the prefix, [example 46.11](#), source 16’s second table row and source 18’s [example 46.33](#)) from a numerical gap

with no surreal separator; in source 14's terms both are numerical-gap cuts $D_{p,H}$, the first with $A_p \setminus H$ and H separated by a point of $\text{ran } p$.

Source 18 states its theorem “in GB with the fixed baseline”. A baseline $b : \text{Ord} \rightarrow \mathbf{No}$ already yields global choice over GB (theorem 13.1), so the hypothesis is not weaker than GBC; the theorem's own content is that no class choice and no ETR are used beyond the baseline. Sources 13, 14, 15, 16 and 17 say GBC.

46.1 Source 16: exact recognition of cuts represented by branches

Fix the core $\mathcal{P}_{\text{bd}}(b)$ and its evaluation. For an injective set prefix $p : \alpha \rightarrow \mathbf{No}$ write

$$C_p = \{q \in \mathcal{P}_{\text{bd}}(b) : E_q \upharpoonright \alpha = p\}.$$

These cylinders are nonempty and convex, and at each fixed length they form an ordered partition. Each cylinder has no endpoint: change the next coordinate of one completion to a fresh smaller or larger surreal and apply Lemma 42.9.

A *lower cut* here means an available class $\emptyset \neq L \subsetneq \mathcal{P}_{\text{bd}}(b)$ that is initial and has no greatest member. Put $U = \mathcal{P}_{\text{bd}}(b) \setminus L$. A prefix *straddles* the cut if $C_p \cap L$ and $C_p \cap U$ are both nonempty. This is a property with only set quantifiers and the class parameters L, b .

Theorem 46.2 (Prefix-cylinder recognition). *The following are equivalent:*

1. *There is an injective class function $F : \text{Ord} \rightarrow \mathbf{No}$ such that*

$$L = \{q \in \mathcal{P}_{\text{bd}}(b) : E_q <_{\text{lex}} F\} =: L_F.$$

2. *At every ordinal length α some injective prefix straddles the cut.*

When these conditions hold, the prefix at each length is unique and is exactly $F \upharpoonright \alpha$. The branch is exhaustive if and only if every surreal occurs in the range of some straddling prefix.

Proof. For a given F , choose fresh values below and above $F(\alpha)$ and complete the corresponding one-letter extensions of $F \upharpoonright \alpha$. They lie on opposite sides of F , so its prefix straddles.

Two different prefixes at the same length cannot both straddle an initial cut. Their cylinders are entirely ordered one before the other, and an upper witness in the earlier cylinder would be below a lower witness in the later one. This contradicts initiality. Restrictions of straddling prefixes still straddle. Hence the prefixes p_α supplied by condition 2 are unique and coherent.

Define the branch directly, by elementary comprehension:

$$F(\alpha) = x \iff \exists p [p \text{ straddles, } \text{dom}(p) = \alpha + 1, p(\alpha) = x]. \quad (\text{S16.14})$$

Here the prefix and the witnesses from both sides are sets. Coherence proves that this is a total injective class function and that $F \upharpoonright \alpha = p_\alpha$. This is not a class-valued recursion.

If $E_q < F$ first differ at δ , every point of $C_{p_{\delta+1}}$ is above q . One such point is in L , so $q \in L$. If $F < E_q$, an upper witness in the same cylinder lies below q , so $q \in U$. Finally, if $E_q = F$, placing q in L would make it the greatest member of L by the preceding two implications. Thus it is in U , and $L = L_F$. The prefixes determine F uniquely, and their range union gives exactly the stated exhaustion criterion. $\square$

The no-greatest convention matters only at a represented core point: without it, both $\{q : E_q < F\}$ and $\{q : E_q \leq F\}$ would straddle every level when $F = E_q$ for a code. The two improper cuts fail already at the empty prefix and represent no injection.

Theorem 46.3 (Complete classification of the first failure). *If a lower cut is not represented, let λ be the least length with no straddling prefix. Then $\lambda > 0$, and exactly one of the following occurs.*

1. **Successor failure.** *If $\lambda = \alpha + 1$, let p be the unique straddling prefix of length α . The unused labels $\mathbf{No} \setminus \text{ran}(p)$ split into nonempty initial and final classes $A + B$, with*

$$C_{p \smallfrown (x)} \subseteq L \ (x \in A), \quad C_{p \smallfrown (x)} \subseteq U \ (x \in B).$$

All codes below C_p are in L and all codes above it in U .

2. **Limit failure.** *If λ is a nonzero limit, the coherent earlier prefixes have an injective set union p of length λ . The cut is exactly the lower or upper edge of its cylinder:*

$$L = C_p^- \quad \text{or} \quad L = C_p^- \cup C_p, \quad C_p^- := \{q : \forall c \in C_p \ (q < c)\}.$$

Proof. Minimize below any ordinal witnessing failure; this uses no ordering of classes. The empty prefix straddles. At a successor, every nonempty child cylinder is wholly on one side, and the children partition the parent cylinder, which straddles. Both parts of the label split are therefore nonempty, and their order follows from that of the children. The outside-code assertion follows by comparison with a lower and an upper witness in the parent.

At a limit, Class Replacement applied to the unique prefixes indexed by the set λ makes their union a set. Coherence ensures injectivity, and completion ensures a nonempty cylinder. Since the length fails, that cylinder is wholly on one side. Any code outside it first differs from p at $\delta < \lambda$; the shorter prefix of length $\delta + 1$ still straddles. Its witnesses force the code to the lower side if it differs downward and to the upper side if upward. Including or excluding the entire limit cylinder gives exactly its two edges. $\square$

A successor failure need not be a numerical gap with both sides lacking endpoints. It can be an edge of a whole child cylinder, or a cut for which the numerically suitable next label has already been used. The theorem distinguishes these from a failure of coherence at a limit length.

46.1.1 Explicit cuts and their meaning

The following table gives concrete, fully defined examples.

Lower side	First failure	Interpretation
$\{q : E_q(0) \leq 0\}$	1	Upper edge of the entire zero-first cylinder.
$\{q : E_q(0) < 0\} \cup \{q : E_q(0) = 0, E_q(1) < 0\}$	2	The apparent second separator would repeat the already used zero.
C_p^- or $C_p^- \cup C_p$, with $p(n) = n$ for $n < \omega$	ω	Every finite prefix straddles, but the whole countable-prefix cylinder lies on one side.
L_F for $F : \text{Ord} \xrightarrow{\sim} \mathbf{No} \setminus \{r\}$	None	A genuine injective branch, failing only the exhaustion test.

All displayed lower sides have no greatest member. For a nonzero limit prefix length λ , its lower cylinder edge has lower cofinality exactly $\text{cf}(\lambda)$ and no set coinital family on its upper side. To see this, make a fresh downward deviation at cofinally many positions of p . The completions increase

cofinally below C_p ; any cofinal family must have unbounded deviation coordinates. On the upper side, the initial portion C_p has no set coinital family, since one can choose a fresh next value below any set of next values. The upper edge has the dual characters.

Corollary 46.4 (Exactly what raw tails add). *For a raw order $R \in \mathfrak{W}_{\text{all}}$, its strict lower cut in $\mathcal{P}_{\text{bd}}(b)$ is L_{F_R} . Thus the cut image of all raw orders consists exactly of the cuts passing the straddling test. Over the cut of F , the fiber is the convex order $\mathcal{W}(\mathbf{No} \setminus \text{ran } F)$. Neither successor nor limit first-failure cuts are filled by any raw class well-order.*

Proof. A bounded-support enumeration unequal to F_R first disagrees at a set ordinal, before any non-set-like tail begins. Its comparison with R is therefore its comparison with F_R . Equality is possible only when F_R is exhaustive, in which case it determines R . Conversely every injection occurs as a first core, by (S16.12). Distinct cores give different cuts by Theorem 46.2, proving exactly the asserted image and fibers. $\square$

This resolves the earlier report’s represented-cut question using the prefix tree with its numerical labels. It does not claim that the tree is recoverable from the bare order on $\mathcal{P}_{\text{bd}}(b)$. Since $\mathcal{P}_{\text{bd}}(b) \cong \mathbf{No}$, forgetting those additional predicates can lose precisely the information used by the recognition test.

Note 46.5 ([merge]; on source 16’s last paragraph). Source 16’s sentence that this “resolves the earlier report’s represented-cut question” is true of the theorem, which all six batch-81 sources prove (note 46.1). Its corollary 46.4 is the statement about raw orders that sources 13, 15, 17 and 18 make in corollary 46.16, corollary 46.8, proposition 46.27 and corollary 46.30; the fibre description in it is source 12’s proposition 15.3, and the image of all raw orders, exactly the cuts passing the straddling test, is the new part. Part IX prints source 14’s trace-fibre theorem (theorem 54.6).

46.2 Source 15: a prefix criterion for all represented core cuts

[merge] Source 15’s Section 7, from its Subsection 7.2 (Subsection 7.1, the class of codes, is printed in section 42.3; Subsection 7.5, the automorphism theorem, in section 47).

46.2.1 The persistent-cylinder theorem

A normalized class cut of $\mathcal{P}_{\text{bd}}(b)$ is a nonempty proper initial subclass L with no greatest member; write $R = \mathcal{P}_{\text{bd}}(b) \setminus L$. The upper side may have a least element. A cylinder *crosses* the cut if it meets both sides. For an injective class word $F : \text{Ord} \rightarrow \mathbf{No}$, put

$$L_F = \{d \in \mathcal{P}_{\text{bd}}(b) : E_d <_{\text{lex}} F\}.$$

The comparison is defined at the first ordinal difference, unless the functions agree, in which case the strict inequality is false.

Theorem 46.6 (Exact criterion for representation). *Extension proved in source 15. In GBC, for a normalized class cut L the following are equivalent:*

1. $L = L_F$ for an injective class word $F : \text{Ord} \rightarrow \mathbf{No}$;
2. for every ordinal α , some cylinder C_p with $\text{dom}(p) = \alpha$ crosses the cut.

The representing word is unique. Its restrictions are exactly the unique crossing prefixes. Its construction uses elementary comprehension, not ETR.

Proof. At most one cylinder at a given level crosses. If $C_p < C_q$ both crossed, a point of R in C_p would precede a point of L in C_q , contradicting initiality. A restriction of a crossing prefix also crosses. Thus the unique prefixes p_α in (2) are coherent.

Define $F(\alpha) = x$ when the crossing prefix of length $\alpha + 1$ has last value x . This is an elementary formula: prefixes and codes are sets, and crossing is expressed by two existential set quantifiers with the cut and evaluation as class parameters. Coherence shows that $F \restriction \alpha = p_\alpha$; injectivity holds on every such restriction and hence globally.

If $E_d \neq F$, let δ be the first difference. The cylinder $C_{p_{\delta+1}}$ crosses the cut. If $E_d(\delta) < F(\delta)$, the whole cylinder is above d , and its lower-side member forces $d \in L$. If $E_d(\delta) > F(\delta)$, its upper-side member lies below d , forcing $d \in R$. If $E_d = F$, the code d cannot lie in L , since it would then be its greatest member. This proves (1).

Conversely, at every α choose fresh labels $a < F(\alpha) < c$, neither in $\text{ran}(F \restriction \alpha)$. Completions of $(F \restriction \alpha) \frown a$ and $(F \restriction \alpha) \frown c$ lie on opposite sides of F . Thus $C_{F \restriction \alpha}$ crosses. These same perturbations show that L_F is nonempty, proper, and has no greatest member. Uniqueness follows from recovery of every prefix. $\square$

Remark 46.7 (The normalization is necessary). Without the no-greatest condition, persistence determines a unique F but allows either side of a principal point:

$$L_F \subseteq L \subseteq L_F \cup \{d : E_d = F\}.$$

Only a word already represented by a core code permits the two choices. Both $\{d : E_d < F\}$ and $\{d : E_d \leq F\}$ cross every level, but the second has a greatest member. Terminal cuts below have neither endpoint.

Corollary 46.8 (Exhaustiveness and the kernel of the raw cut map). *A persistent cut is represented by a set-like global well-order exactly when*

$$\forall x \in \mathbf{No} \ \exists \alpha \quad x \in \text{ran}(p_\alpha).$$

Every persistent cut is represented by an arbitrary raw class well-order. Two raw orders determine the same strict lower cut on $\mathcal{P}_{\text{bd}}(b)$ exactly when their initial words F_R agree. The common fibre is precisely $\mathcal{W}(\mathbf{No} \setminus \text{ran}(F_R))$, interpreted at the appropriate size level.

Proof. The displayed condition says exactly that the recovered word is onto. Every injective word has a raw completion by source 15's Proposition 4.2 (prefix convexity and initial-word fibres; [merge] proposition 51.10; it is proposition 15.3). A raw order and its initial word compare with a core evaluation at the same first ordinal disagreement. If a core evaluation equals that word, its exhaustiveness forces the raw order to have no residual tail. Thus its lower cut is L_{F_R} . Apply uniqueness in the theorem and the fibre description of the same proposition. $\square$

This answers the prior report's represented-cut question by a necessary and sufficient condition on all set-length prefix levels. Missing labels are not a failure of persistence. They distinguish the exhaustive words inside the persistent branches. Any ordering information after the first proper block is invisible to comparisons with the bounded-support core.

46.2.2 An exhaustive description of first failure

Theorem 46.9 (Terminal-cut classification). *Extension proved in source 15. If a normalized class cut fails the criterion, let $\theta > 0$ be the least length without a crossing cylinder. Exactly one of the following descriptions applies.*

1. **Successor failure.** If $\theta = \alpha + 1$, there is a unique crossing prefix $p : \alpha \rightarrow \mathbf{No}$ and a nonempty initial/final partition

$$A + B = \mathbf{No} \setminus \text{ran}(p), \quad A < B,$$

such that each child $C_{p \restriction x}$ lies wholly in L for $x \in A$ and wholly in R for $x \in B$. The four mutually exclusive forms are: an upper boundary of a child cylinder when A has a greatest element; a lower boundary when B has a least element; a unique forbidden repeated label separating A and B when neither endpoint exists but a surreal separator exists; or a numerical class gap with no surreal separator.

2. **Limit failure.** If θ is limit, the coherent union $p = \bigcup_{\alpha < \theta} p_\alpha$ is an injective set word. The cut is exactly the lower or upper boundary of its nonempty cylinder:

$$L = \{d : E_d \restriction \theta < p\} \quad \text{or} \quad L = \{d : E_d \restriction \theta \leq p\}.$$

Proof. The empty cylinder crosses, so $\theta \neq 0$. At a successor, child cylinders partition the crossing parent and none crosses. Each is nonempty, giving the asserted partition of allowed labels. Both parts are nonempty because the parent crosses. Their order follows from the order of the cylinders and initiality of L . All codes below the parent lie in L , and all those above it lie in R .

The numerical residual alphabet is dense by the fresh-separator lemma (source 15's Lemma 3.1, lemma 51.2; lemma 5.2), so A cannot have a greatest element while B has a least one. If neither endpoint exists and s strictly separates them, s cannot be an allowed label: membership in A or B would contradict strict separation. Hence $s \in \text{ran}(p)$. Two different separators would enclose a fresh allowed label, which is impossible. This proves all four subcases.

At a limit, replacement over the set ordinal θ collects the coherent prefixes, and their union is an injective set word. Its cylinder is nonempty by prefix completion but does not cross. A code outside it differs from p at a coordinate $\delta < \theta$; the crossing cylinder of length $\delta + 1$ then fixes its side. The only remaining choice is which side contains all of C_p , giving precisely the two formulas. There is no incoherent-branch case: uniqueness at the earlier levels already ruled it out. $\square$

Example 46.10 (A cut not filled even by arbitrary class well-orders). The initial class $L = \{d : E_d(0) < 0\}$ has no greatest member. Its empty cylinder crosses, but each length-one cylinder lies on one side. It is the lower boundary of $C_{(0)}$. Therefore no injective Ord-word and no raw class well-order represents this cut. The upper boundary $\{d : E_d(0) \leq 0\}$ is likewise unrepresented.

Example 46.11 (Repeated label, numerical gap, and limit boundary). The cut defined by

$$E_d(0) < 0 \quad \text{or} \quad (E_d(0) = 0 \wedge E_d(1) < 0)$$

first fails at length two: the putative continuation of (0) would repeat 0. For a numerical-gap example, use $A = \{x : \exists n \in \omega \ x < n\}$ and its complementary final class of positive infinite surreals, and put $d \in L$ when $E_d(0) \in A$. No surreal strictly separates these label classes; neither relevant endpoint exists. Finally, if $p(n) = n$ for $n < \omega$, the lower and upper boundaries of C_p first fail at length ω even though every finite prefix still crosses.

For a persistent but nonexhaustive example, take $F(\alpha) = \alpha$. Every prefix level crosses, but the range omits the negative surreals. Appending a well-order of the omitted range gives many raw completions with exactly the same lower core cut. If F instead omits just one label, it has one raw completion of type $\text{Ord} + 1$. These are sharply different from the terminal cuts in Example 46.10.

46.2.3 Local characters of terminal cuts

Here a “set cofinal family” is an actual set of codes cofinal in an ordinary class side of a cut. We say explicitly when none exists; Ord is not being used as a set cardinal.

Proposition 46.12 (Cofinality read from first failure). *At a first limit failure θ , the lower-boundary case has $\text{cf}(L) = \text{cf}(\theta)$, while R has no set coinitial family. The upper-boundary case has the dual properties. At a successor failure with label partition $A + B$, if A has no greatest label, L has exactly the same least set cofinality as A , or no set cofinality if A has none. If A has a greatest label, L has no set cofinal family. The dual statements hold for B and R .*

Proof. At a limit, choose cofinally many coordinates $\beta < \theta$ and complete prefixes agreeing with p before β but smaller at β . These form an increasing cofinal family on the lower side. Any cofinal family must have first-divergence coordinates unbounded in θ , since a later lower perturbation dominates every family with bounded divergence coordinates. This proves equality. The upper side begins with the whole cylinder C_p , which has no set coinitial family; all remaining upper-side points are above it.

At a successor with no greatest label in A , a cofinal set of labels can be lifted to completions, and conversely the next labels of any cofinal family eventually inside the parent cylinder must be cofinal in A . If A has a greatest label, its final child cylinder has no set cofinal family and is final on the lower side. The dual argument proves the remaining assertions. $\square$

No gap of $\mathcal{P}_{\text{bd}}(b)$ can have both a set cofinal lower side and a set coinitial upper side: the core’s set-cut filling theorem would put a point between such two sets. The character calculation is compatible with this restriction, while identifying many genuinely different class cuts.

46.2.4 What a completion would have to mean

The normalized class cuts of $\mathcal{P}_{\text{bd}}(b)$, ordered by inclusion, give the natural Dedekind-cut interpretation at the next size level. For a uniform family of such cuts whose union is nonempty and proper, the union is again a normalized class cut: its defining membership formula quantifies over set indices. This is a precise available-family completeness statement, not a same-level class containing all its subclasses.

The map from raw orders to these cuts collapses exactly the residual fibres of [Corollary 46.8](#). Therefore adjoining all raw orders is neither a faithful Dedekind completion of the core nor a realization of every core cut. It both retains multiplicity over persistent cuts and omits all terminal cuts. Any order embedding into a dense completion that fixes the core would have to identify distinct raw orders lying in one fibre, which is impossible. Arbitrary embeddings that move the core are a different question.

46.3 Source 13: exact reconstruction of cuts from prefix cylinders

We now address the earlier report’s question asking for a necessary and sufficient cylinder criterion for represented cuts ([research question 61.1](#)). The criterion is relative to the chosen core and prefix system; [Section 47.4](#) will show why it is not a statement about the pure linear order alone.

46.3.1 Normalization and crossing

Definition 46.13. A *normalized proper cut* of $\mathcal{P}_s(b)$ is a pair of ordinary classes (L, U) such that $L \sqcup U = \mathcal{P}_s(b)$, both sides are nonempty, every member of L precedes every member of U , and L has no greatest member. The upper side may have a least member.

An injective set prefix p crosses this cut when

$$\mathcal{P}_s(b)_p \cap L \neq \emptyset, \quad \mathcal{P}_s(b)_p \cap U \neq \emptyset.$$

The normalization is essential. Immediately below and immediately above a code d , two different cuts would otherwise reconstruct the same branch E_d . We choose the lower trace convention, so the cut at d puts d in the upper side.

Lemma 46.14 (Uniqueness and coherence of crossing prefixes). *At a fixed ordinal length there is at most one crossing prefix. Restrictions of a crossing prefix cross the same cut. Consequently crossing prefixes of different lengths are coherent.*

Proof. Distinct same-length prefixes first differ at a coordinate, placing one entire cylinder below the other. If both cylinders crossed, an upper-side point in the first would precede a lower-side point in the second, impossible. Restriction enlarges a cylinder, so preserves crossing. Apply same-length uniqueness to the restriction of a longer crossing prefix. $\square$

For $F \in \mathfrak{J}$, its lower trace is the class

$$\text{tr}(F) = \{d \in \mathcal{P}_s(b) : E_d < F\}.$$

All comparisons use a least set-ordinal disagreement, except equality, which simply excludes a code from the lower trace.

Theorem 46.15 (Exact branch representation). *In GBC, a normalized proper cut (L, U) of $\mathcal{P}_s(b)$ is the lower trace of a unique $F \in \mathfrak{J}$ if and only if*

$$\text{for every ordinal } \alpha \text{ there exists a crossing prefix of length } \alpha. \quad (\text{S13.1})$$

The represented word is exhaustive, hence belongs to $\mathfrak{W}_{\text{sl}}$, if and only if it also satisfies

$$\text{for every } x \in \mathbf{No} \text{ some crossing prefix has } x \text{ in its range.} \quad (\text{S13.2})$$

Proof. Suppose first that $L = \text{tr}(F)$. For each ordinal α , the prefix $F \upharpoonright \alpha$ has code completions below and above F , by choosing a fresh next value on the desired side. Thus it crosses. It is the unique crossing prefix of that length. Equation (S13.2) is then precisely surjectivity of F .

Conversely, let p_α be the unique crossing prefix of length α . Coherence gives $p_\alpha = p_\beta \upharpoonright \alpha$ whenever $\alpha < \beta$. Define

$$F(\alpha) = p_{\alpha+1}(\alpha).$$

This is an elementary class definition quantifying over the set code $p_{\alpha+1}$. It does not ask for a class-valued choice or a class-valued recursive sequence of prefixes. Every two coordinates belong to one longer injective prefix, so F is injective.

We check the trace equality. If $E_d < F$, let ξ be the first disagreement and choose a crossing prefix of length greater than ξ . Every code in its cylinder is above d . One of those codes is in L , so downward closure gives $d \in L$. If $E_d > F$, the symmetric argument with an upper-side code gives $d \in U$.

The only possible remaining case is $E_d = F$. Every other lower-side code has smaller evaluation, by the two arguments just given. If d were in L , it would therefore be its greatest member, contradicting normalization. Thus $L = \text{tr}(F)$. Distinct words have a code strictly between them by Corollary 42.5, so their traces are distinct. Finally, the union of the prefix ranges is exactly $\text{ran}(F)$, which proves the exhaustion criterion. $\square$

46.3.2 What raw well-orders add, and what they cannot add

Corollary 46.16 (Trace image and fibers of raw orders). *For $R \in \mathfrak{W}_{\text{all}}$, define $L_R = \{d \in \mathcal{P}_s(b) : E_d < R\}$. The normalized proper cuts represented in this way are exactly those satisfying (S13.1). For such a cut the unique reconstructed word is F_R . Its raw realization fiber is precisely the order of all well-orders of $\mathbf{No} \setminus \text{ran}(F_R)$. The fiber contains a set-like realization exactly when the coverage condition holds, and then it is a singleton.*

Proof. Every comparison with a code is decided inside the ordinal head unless the code equals the entire raw order; equality can occur only when the head is exhaustive. Hence $L_R = \text{tr}(F_R)$. Conversely, append a well-order of the omitted range to the branch reconstructed by Theorem 46.15. Source 13's Proposition 6.2 ("Canonical head; prior structural lemma", [merge] proposition 51.8; it is theorem 15.1 and proposition 15.3) gives every realization and proves the fiber statement. $\square$

If the omitted range has n elements, there are exactly $n!$ raw realizations. A single omitted label gives one additional point of type $\text{Ord} + 1$; two omitted labels give two adjacent realizations of type $\text{Ord} + 2$. These examples were already constructed in the earlier report. The new criterion determines the complete trace image: adding arbitrary non-set-like tails still does not fill a cut that fails the all-level crossing test.

Example 46.17 (A branch that fails only exhaustion). Let $F(\alpha) = \alpha$, viewed as the all-plus surreal of length α . Its lower trace crosses every prefix length, but its prefixes never contain -1 . It has raw realizations obtained by well-ordering the nonordinal surreals after its head, and no realization in $\mathfrak{W}_{\text{sl}}$. This recovers the earlier omitted-range example through the exact criterion rather than through a single selected gap.

46.4 Source 13: a complete first-failure classification

The representation theorem reduces the problem to a question about crossing prefixes. The next theorem describes every way the test can fail. In particular, failure at a limit is not an unexplained compactness assumption: it has an exact geometric form.

Theorem 46.18 (Branch, horizontal cut, or limit boundary). *Let (L, U) be a normalized proper cut of $\mathcal{P}_s(b)$. Exactly one of the following descriptions applies.*

- (i) *There is a crossing prefix at every ordinal length. The cut is the trace of a unique injective class word.*
- (ii) *There is a least failed length $\lambda = \beta + 1$. Let p be the crossing prefix of length β . There is a nonempty cut $A < B$ partitioning $Y = \mathbf{No} \setminus \text{ran}(p)$ such that, within $\mathcal{P}_s(b)_p$,*

$$L \cap \mathcal{P}_s(b)_p = \bigcup_{x \in A} \mathcal{P}_s(b)_{p \frown x}, \quad U \cap \mathcal{P}_s(b)_p = \bigcup_{x \in B} \mathcal{P}_s(b)_{p \frown x}.$$

Thus the cut separates whole next-letter cylinders. We call it horizontal at p .

- (iii) *There is a least failed length λ that is a nonzero limit. The union p of the shorter crossing prefixes has length λ , and the cut is exactly the lower or upper boundary of $\mathcal{P}_s(b)_p$.*

All descriptions are expressible with ordinary class parameters in GBC.

Proof. The empty prefix crosses every proper cut. If some length fails, the class of failed ordinals has a least member $\lambda > 0$.

For $\lambda = \beta + 1$, uniqueness gives the crossing p of length β . Every child cylinder is nonempty by prefix completion, but none crosses the cut. Each is therefore entirely on one side. Put $x \in A$ or $x \in B$ according to that side. These are ordinary classes: the predicate “every code in this child is in L ” quantifies only over set codes. Their union is Y , they are disjoint, and the label order of the children shows $A < B$. Both sides are nonempty because the parent crosses. Every code outside the parent lies below all of it or above all of it, so its placement in the cut is forced.

For limit λ , coherence gives an injective set word $p = \bigcup_{\alpha < \lambda} p_\alpha$. Its cylinder is nonempty but cannot cross. If a code does not extend p , it first disagrees at some $\xi < \lambda$. A shorter crossing prefix of length greater than ξ forces its side: a lower disagreement puts it in L , and an upper disagreement puts it in U . Thus every code below $\mathcal{P}_s(b)_p$ is in L , and every code above it is in U . The entire $\mathcal{P}_s(b)_p$ lies on one side, giving precisely one of its two boundaries.

If there is no failed set ordinal, [Theorem 46.15](#) applies. The three cases are disjoint by their first failed length. $\square$

There is no fourth case called “failure at Ord”. An injective branch that omits labels still crosses every set level; its failure is exhaustion, not prefix existence. Treating Ord as one more set stage would conflate the two tests.

46.4.1 One-sided characters of the three cases

For a class order, say that a side has character Ord when it has no set-sized cofinal (respectively coinital) family but does have a uniform increasing (respectively decreasing) Ord-indexed family. This is a notation for a pair of properties, not a set cardinal. When a set-sized family suffices, use the usual least cofinal cardinal; an endpoint has character 1.

Every ordinary class order without a set-cofinal subset has an Ord-indexed increasing cofinal sequence in GBC: enumerate its points by the global well-order and at each stage choose a point above both the earlier set of choices and the next enumerated point. Absence of a set-cofinal subset guarantees this choice. The dual construction handles coinitality.

Proposition 46.19 (Characters at first failure). *The cut descriptions have the following character consequences.*

- (i) *A branch trace has character (Ord, Ord) when its word is not a code evaluation. If $F = E_d$, its normalized trace has character (Ord, 1); the punctured upper side again has character Ord.*
- (ii) *A lower boundary of a limit-length cylinder $\mathcal{P}_s(b)_p$, of length λ , has character (cf λ , Ord). Its upper boundary has character (Ord, cf λ).*
- (iii) *For a horizontal cut associated with $A < B$, the lower character is cf(A) if A has no greatest label, and is Ord if A has a greatest label. The upper character is ci(B) if B has no least label, and is Ord if it has a least label.*

Proof. The branch statement is [Lemma 42.6](#), with the possible equal code placed in the upper side.

At a lower limit boundary, choose prefixes agreeing with p up to ξ and then taking a fresh lower letter. Completions at lengths ξ cofinal in λ are cofinal below the whole cylinder. Conversely, fewer than cf λ many codes below it have their first-disagreement coordinates bounded below λ , so a later such completion lies above them all. The right side has $\mathcal{P}_s(b)_p$ as a coinital subclass, and that cylinder has no set-coinital family by [Corollary 42.5](#). The upper-boundary proof is dual.

For a horizontal cut whose lower label side has no greatest member, choose one code in each child along a cofinal family of labels. This family is cofinal among all lower codes. Conversely, projecting a cofinal family of lower codes in the parent onto its next labels is cofinal in A . Codes

below the parent cannot improve this bound. If A has a greatest label a , the whole terminal child $\mathcal{P}_s(b)_{p \smallfrown a}$ is cofinal on the lower side. Its character is Ord , not 1. The upper assertion is the reversed argument. $\square$

46.4.2 Equal characters do not imply representability

Example 46.20 (A horizontal (Ord, Ord) cut). Let $s : \text{Ord} \rightarrow \{-, +\}$ be the proper-class sign string whose sign is $+$ when the finite remainder of its ordinal index is even and $-$ when it is odd. Both signs occur cofinally. This is *not* a surreal number.

Compare each set-length surreal sign sequence with s at its first difference, regarding its termination as 0. Every surreal differs from s by the time of its termination. Define

$$A = \{x \in \mathbf{No} : x < s\}, \quad B = \{x \in \mathbf{No} : s < x\}.$$

These are nonempty classes partitioning the label order. Neither side has an endpoint or a set-sized cofinal or coinital family. Indeed, the birthdays of any proposed set family are bounded by some γ . Truncating s at a later $+$ position gives a larger lower label than all lower members of the family; truncating at a later $-$ position gives a smaller upper label. The corresponding ordinal sequences of truncations supply cofinal and coinital class families.

Now partition the core by its first label:

$$L = \{d : E_d(0) \in A\}, \quad U = \{d : E_d(0) \in B\}.$$

This cut has character (Ord, Ord) but no crossing prefix of length 1. It is horizontal at the root, and is not the trace of any injective class word or of any raw class well-order.

Thus neither saturation below set size nor the ordered pair of local characters identifies the represented cuts. The labeled prefix criterion contains strictly more information.

46.4.3 An explicit limit-boundary example

Let $p(n) = n$ for $n < \omega$. The cut just below $\mathcal{P}_s(b)_p$ is crossed by every finite initial prefix of p , but not by any prefix of length ω . At each finite level, some lower code leaves the ordinal prefix later, while an upper code extends all of p . The cut has character (ω, Ord) and is not represented by a raw global well-order. Its obstruction is a set-length limit boundary rather than an omitted proper-class range. This is the distinction that an informal “keep taking longer prefixes” argument would overlook.

46.5 Source 17: recognizing exactly which core cuts are global enumerations

The preceding source results place $\mathfrak{W}_{\text{sl}}$ in a larger collection of cuts of $\mathcal{P}_s(b)$, but do not by themselves identify its image. We now give a necessary-and-sufficient test in terms of the displayed prefix cylinders. This addresses source question [swo:n16.1](#) ([research question 61.1](#)) and its parallel in source 09 (its third direction in [section 61.3](#)). It is a *labelled-tree* recognition theorem; a pure-order recognition theorem would be stronger and, as shown below, cannot hold in the naive invariant form.

46.5.1 The straddling test

For an injective set prefix $p : \alpha \rightarrow \mathbf{No}$, define the ordinary class

$$C_p = \{c \in \mathcal{P}_s(b) : E_c \upharpoonright \alpha = p\}.$$

Let L be a proper nonempty lower subclass of $\mathcal{P}_s(b)$ with no greatest element, and put $U = \mathcal{P}_s(b) \setminus L$. We allow U to have a least element. Thus the convention is the *open lower cut*, including the one immediately before an actual core point. Say that p straddles L when C_p meets both L and U .

Theorem 46.21 (Recognition of injective branches). *The following conditions are equivalent.*

1. *There is an injective class word $f : \text{Ord} \rightarrow \mathbf{No}$ such that*

$$L = L_f := \{c \in \mathcal{P}_s(b) : E_c <_{\text{lex}} f\}.$$

2. *For every set ordinal α , there is a straddling injective prefix of length α .*

When these conditions hold, both the straddling prefix at each length and the word f are unique. Moreover f is surjective if and only if

$$\forall x \in \mathbf{No} \exists p (p \text{ straddles } L \wedge x \in \text{ran } p). \quad (\text{S17.8})$$

The reconstruction uses elementary class comprehension over GBC, not ETR.

Proof. Distinct cylinders of the same length are disjoint convex blocks in lexicographic order. At most one can meet both sides of a cut: if an earlier block met U , every later block would be in U .

Suppose first that f is given. For any α , take its prefix $p = f \upharpoonright \alpha$. A fresh smaller next label than $f(\alpha)$, and a fresh larger one, yield core completions respectively below and above f . Thus p straddles. These completions also show that L_f has no greatest element and is proper and nonempty.

Conversely, let p_α denote the unique straddling prefix at length α . Restrictions of straddling prefixes straddle, so uniqueness implies $p_\beta \upharpoonright \alpha = p_\alpha$ for $\alpha < \beta$. Define a class graph by

$$f(\alpha) = x \iff \exists p (p \text{ is a straddling prefix of length } \alpha + 1 \wedge p(\alpha) = x).$$

The quantifier over p ranges over sets. Straddling itself quantifies over the set codes in $\mathcal{P}_s(b)$, with L and b as class parameters. Thus this is elementary comprehension. Coherence makes f a total function; its set restrictions are injective, so it is injective.

If $E_c \neq f$, let δ be their first disagreement. The cylinder of $p_{\delta+1}$ straddles. If $E_c(\delta) < f(\delta)$, then c is below every code in that cylinder, including a code in L , so $c \in L$. The reversed inequality places c in U . If $E_c = f$, putting c in L would make it the greatest member of L , already excluded. Hence $L = L_f$ exactly, including the principal-cut case. Two different words would be separated by a core code, so the representing word is unique. Finally $\text{ran } f$ is the union of the ranges of its straddling prefixes; this proves the coverage criterion. $\square$

Remark 46.22 (Why every ordinal length and the endpoint convention matter). Checking only successor extensions along finite histories is inadequate. A cut can track a coherent prefix through all finite levels and then lie at the boundary of its length- ω cylinder. There is then no straddling cylinder at that limit. In contrast, [Theorem 46.21](#) requires a straddler at *every* set ordinal. Also, a closed principal lower cut would have the same straddling prefixes as the corresponding open one. The hypothesis that L has no greatest element removes that ambiguity.

46.5.2 Three independent failures

Example 46.23 (A coordinate cut that does not follow a branch). Let $A \subseteq \mathbf{No}$ consist of those x with $x < n$ for some natural number n , and let $B = \mathbf{No} \setminus A$. Both are nonempty, A has no largest element, and B has no least element: a positive infinite member of B has a smaller positive infinite half. Partition core codes by whether their first label lies in A or B . The root cylinder straddles, but no length-one cylinder does. This is a genuine cut of the core, but not comparison with an injective Ord-word.

Example 46.24 (A limit boundary that defeats finite-depth testing). Fix an injective prefix $p : \omega \rightarrow \mathbf{No}$. Take the cut immediately before C_p . At every finite length n , the cylinder for $p \upharpoonright n$ meets both cut sides, because a fresh lower label at a later finite coordinate gives a code strictly below C_p . At length ω , the entire cylinder lies on the upper side and no cylinder straddles. This is precisely the limit obstruction in the set-sized cut trichotomy, now at an ordinary class of set codes.

Example 46.25 (A coherent branch which is not exhaustive). Fix $r_0 \in \mathbf{No}$ and a bijection $f : \text{Ord} \rightarrow \mathbf{No} \setminus \{r_0\}$. The cut L_f passes the straddling test at every ordinal length, but fails (S17.8) for r_0 . It is not represented by any member of $\mathfrak{W}_{\text{sl}}$, since the representing injective word is unique.

These examples separate numerical-cut failure, limit coherence, and exhaustiveness. The test is not merely the assertion that a representation exists restated in different notation: its quantifiers over prefixes and core witnesses are over *sets*, and the representing proper-class function is recovered by one elementary formula.

Proposition 46.26 (Characters of branch cuts outside the core). *If $f \in \mathfrak{J}$ has no code in $\mathcal{P}_s(b)$, both sides of its core cut have no set-sized cofinal or coinitial subset, respectively. Both admit uniform Ord-indexed approximation families. This is often abbreviated by saying that the cut has character (Ord, Ord) ; that notation does not make Ord into a cardinal or set.*

Proof. For each α , extend $f \upharpoonright \alpha$ by a smaller or larger fresh label and complete in the core, making the choices uniformly using global choice. As α increases, these codes approach the cut from their respective sides. For any set of codes below f , Class Replacement bounds all their first disagreements with f by a set ordinal. A later lower approximation lies above them all. The upper argument is identical. There is no equality case because f has no core code. $\square$

46.5.3 What the same cut remembers about unrestricted orders

For $f \in \mathfrak{J}$, put $D_f = \mathbf{No} \setminus \text{ran } f$. Append to the well-order listed by f any class well-order of D_f , choosing one using global choice when necessary. The result is a raw class well-order of $\mathbf{No}$. Every appended label has a proper class of predecessors; the initial set-like part is exactly f .

Proposition 46.27 (Raw lifts and loss of residual information). *All raw well-orders with canonical initial enumeration f induce the same comparison cut L_f on $\mathcal{P}_s(b)$. Their convex fibre is the lexicographic order of all well-orders of D_f . Consequently the fibre is a singleton when $|D_f| = 0$ or 1, and has $n!$ members when $|D_f| = n < \omega$. For larger complements it retains the full residual well-order problem.*

Proof. Unless a core enumeration equals f , its comparison is decided at a set ordinal before any appended label. Equality is possible only when f is exhaustive, in which case no tail remains. Restriction to D_f and concatenation with the fixed initial order are inverse order-preserving operations. The source proves this convex-fibre description for all raw orders (proposition 15.3); the argument here identifies its interaction with the reconstructed cut. $\square$

Thus recognition of an injective branch determines the raw order's first Ord-part but not its residual order. The core is dense in $\mathfrak{W}_{\text{sl}}$ and $\mathfrak{J}$, not in the interiors of these nontrivial raw fibres.

46.6 Source 18: exactly which class cuts of a core are represented

Fix a baseline bijection $b : \text{Ord} \rightarrow \mathbf{No}$ and let $\mathcal{P}_s(b)$ be its ordinary class of normalized bounded-support permutation codes. Write E_c for the enumeration evaluated from a code $c \in \mathcal{P}_s(b)$. All such evaluations are supplied by one class relation. For an injective set prefix $p : \delta \rightarrow \mathbf{No}$, put

$$C_p = \{c \in \mathcal{P}_s(b) : E_c \upharpoonright \delta = p\}.$$

The core-completion lemma ensures that every such cylinder is nonempty. It is convex. At a fixed length δ , these cylinders partition $\mathcal{P}_s(b)$ and are linearly ordered without overlap.

Throughout this section, a cut is an ordinary class $L \subseteq \mathcal{P}_s(b)$ that is nonempty, proper, initial, and has no greatest element. This convention uses the strict lower side of a represented point. In particular, it excludes the closed principal cut at a code. Say that p *straddles* L if

$$C_p \cap L \neq \emptyset, \quad C_p \setminus L \neq \emptyset.$$

This is an elementary predicate of the set p , with class parameters b, L . Its witnesses are set codes, not arbitrary classes.

Theorem 46.28 (The prefix criterion for represented class cuts). *In GB with the fixed baseline above, the following are equivalent.*

1. *There is an injective class function $f : \text{Ord} \rightarrow \mathbf{No}$ such that*

$$L = \{c \in \mathcal{P}_s(b) : E_c <_{\text{lex}} f\}.$$

2. *For every ordinal δ there is an injective prefix $p : \delta \rightarrow \mathbf{No}$ that straddles L .*

When they hold, f is unique. At every length δ , the unique straddling prefix is $f \upharpoonright \delta$. The function f is defined by an elementary formula from b, L ; no class choice or ETR is needed to assemble it.

Proof. At most one prefix of a given length can straddle. Indeed, if the cylinder of p lies below that of q , an element outside L in the first cylinder lies below an element of L in the second, contradicting initiality.

Assume the second condition and denote the unique prefix at length δ by p_δ . If $\gamma < \delta$, the restriction $p_\delta \upharpoonright \gamma$ also straddles, since its cylinder contains the straddling cylinder of p_δ . Uniqueness gives

$$p_\delta \upharpoonright \gamma = p_\gamma.$$

Define

$$f(\alpha) = p_{\alpha+1}(\alpha).$$

The graph is elementary because each p_δ is a uniquely specified set. Coherence proves that this is an injective class function and that $f \upharpoonright \delta = p_\delta$.

Take a core code c with $E_c \neq f$, and let α be the first coordinate of disagreement. If $E_c(\alpha) < f(\alpha)$, then c lies below every code in $C_{p_{\alpha+1}}$. That cylinder meets L , so $c \in L$ by initiality. If $E_c(\alpha) > f(\alpha)$, then c lies above every code in the same cylinder. Its member outside L shows that $c \notin L$. Finally, if $E_c = f$ and $c \in L$, the absence of a greatest element of L supplies $d \in L$ with $E_d > f$, contradicting the previous case. Hence the displayed cut identity holds in every case.

Conversely, let L be the strict lower cut of an injective class word f . For any ordinal δ , its prefix $p = f \upharpoonright \delta$ is a set. Choose fresh surreal labels

$$u < f(\delta) < v, \quad u, v \notin \text{ran}(p).$$

Such labels exist because a set cannot exhaust either numerical interval next to a given surreal. Complete p followed by u , and p followed by v , to core enumerations. Both completions belong to C_p , and they lie respectively below and above f . Thus every prefix straddles. Its uniqueness at each length also proves the uniqueness of f . $\square$

For completeness, the strict lower cut of every injective class word is indeed nonempty, proper, and without a greatest element. The first two assertions follow by choosing a smaller or larger first label. If $E_c < f$ first differs at α , complete $f \upharpoonright (\alpha + 1)$ with a fresh next label smaller than $f(\alpha + 1)$. The resulting core row lies strictly between E_c and f .

Remark 46.29 (Why the strict-cut convention matters). If $f = E_c$ is itself a core row, both its strict lower cut and its closed lower cut have a straddling cylinder at every length. The closed cut has greatest element c . The no-greatest-element hypothesis is therefore necessary for the stated strict-cut criterion; it is not an incidental regularity assumption.

46.6.1 Set-like representation and all raw representations

For a raw class well-order R of $\mathbf{No}$, let $F_R : \text{Ord} \rightarrow \mathbf{No}$ be the unique enumeration of its canonical initial set-like part. It is injective, and the remaining labels, if any, occur after its entire ordinal-class prefix. The earlier common-prefix theorem shows that a core row and R are compared at their first ordinal disagreement with F_R , unless they are identical.

Corollary 46.30 (Surjectivity is the exact remaining obstruction). *A cut L is represented by a raw class well-order of $\mathbf{No}$ if and only if it satisfies the prefix criterion of [Theorem 46.28](#). Let f be the unique class word recovered from such a cut. Then:*

1. *L is represented by a set-like global well-order if and only if f is surjective onto $\mathbf{No}$.*
2. *All its raw representatives are exactly the well-orders obtained by putting the sequence f first and then well-ordering $Y_f = \mathbf{No} \setminus \text{ran}(f)$.*
3. *These representatives form the previously described convex fiber isomorphic, in the typed uniform sense, to $\mathcal{W}(Y_f)$.*

Proof. For any raw representative R , its cut on the core is the strict lower cut of F_R . [Theorem 46.28](#) therefore applies and gives $f = F_R$. Conversely, append the restriction of the baseline well-order to Y_f after f . The range of f is proper, and its points have precisely set-many earlier f -values; every appended point has the whole proper range before it. Thus the canonical initial function is exactly f , and the resulting raw order has cut L .

A set-like representative has no residual labels, so its canonical function is a bijection. Conversely a bijective f directly defines a set-like representative. Uniqueness of the recovered class word excludes a different surjective word representing the same cut. The last two statements are the canonical-fiber description. $\square$

Thus allowing arbitrary raw well-orders fills the omitted-range branch cuts, but it still does not fill every class cut of the core. The first-failure theorem below identifies exactly where an unsuccessful prefix criterion breaks down.

Corollary 46.31 (Elementary recognition in uniform families). *Relative to the baseline, raw representability of a cut is an elementary property of its class parameter. Set-like representability is also elementary: in addition to a straddling prefix at every length, require*

$$\forall x \in \mathbf{No} \exists \delta \in \text{Ord} \exists p[p : \delta \rightarrow \mathbf{No} \text{ straddles } L \wedge x \in \text{ran}(p)].$$

Consequently a uniformly class-indexed family of cuts has an elementary class of raw-represented indices, an elementary class of set-like-represented indices, and a uniform relation recovering the injective words of all raw-represented members.

Proof. All displayed variables are sets. The coverage condition is equivalent to surjectivity of the uniquely recovered word. Insert the family index into the same formulas and apply elementary class comprehension. The input cut remains a class parameter; this does not provide set codes for all cuts or all class words. $\square$

46.6.2 The first failure: a numerical split or a limit boundary

Theorem 46.32 (First-failure classification). *Suppose L fails the prefix criterion, and let δ be the least length with no straddling prefix. Then $\delta > 0$ and exactly one of the following descriptions applies.*

1. **Successor failure.** *If $\delta = \eta + 1$, let p be the unique straddling prefix of length η . The unused alphabet $Y = \mathbf{No} \setminus \text{ran}(p)$ has a nontrivial class cut $Y = A \sqcup B$, with $A < B$, such that*

$$C_{p \smallfrown \langle x \rangle} \subseteq \begin{cases} L, & x \in A, \\ \mathcal{P}_s(b) \setminus L, & x \in B. \end{cases}$$

Outside C_p , the codes below that cylinder lie in L and those above it do not.

2. **Limit failure.** *If δ is a limit ordinal, the shorter unique straddling prefixes have a coherent union $p : \delta \rightarrow \mathbf{No}$, which is an injective set prefix. Its nonempty cylinder lies wholly in L or wholly outside L . The cut is respectively the upper or lower boundary of that cylinder: codes strictly below the cylinder belong to L , and codes strictly above it do not.*

Proof. The empty prefix straddles because L and its complement are nonempty, so $\delta > 0$. Unique shorter straddling prefixes are coherent by the proof of [Theorem 46.28](#).

In the successor case, every next-letter cylinder is nonempty and fails to straddle, so it lies wholly on one side. The two sides of the parent cylinder supply a letter of each kind. Initiality orders the resulting classes $A < B$. The straddling parent also forces every code below it into L and every code above it outside L .

In the limit case, Class Replacement on the set ordinal δ collects the uniquely defined shorter prefixes into a set, so their union p is a set. Its cylinder is nonempty by core completion, and does not straddle by the defining failure at δ . A code outside it first differs from p at some $\alpha < \delta$. The straddling cylinder at length $\alpha + 1 < \delta$ then decides its side exactly as in the main proof. All codes below the length- δ cylinder lie in L , all those above it lie outside, and the entire cylinder itself lies on one side. $\square$

The successor split is a class cut in the *unused* numerical alphabet. It can be principal at a remaining label, can occur at a label deleted by the prefix, or can be an unfilled class cut of the surreal numerical order. Surreal set-cut interpolation does not imply that an arbitrary class cut has a surreal boundary. The limit case records the independent possibility of reaching an entire set prefix while the cut lies at one of its cylinder's boundaries. Neither failure type is represented even by a raw class well-order, by [Corollary 46.30](#).

Example 46.33 (A deleted-label cut caused by repetition). Fix $x \in \mathbf{No}$ and put

$$L_x = \{c \in \mathcal{P}_s(b) : E_c(0) < x \text{ or } (E_c(0) = x \text{ and } E_c(1) < x)\}.$$

This is a nonempty proper initial class without a greatest element. The unique straddling prefix of length 1 is $\langle x \rangle$. There is no straddling prefix of length 2: its first entry other than x already decides the side, and when its first entry is x , its second entry must be different from x by injectivity. The first failure is therefore at length 2, where the unused alphabet is split at the deleted label x .

The cut is not represented by any injective ordinal-class word, and hence by no raw class well-order of $\mathbf{No}$. A would-be word would have to begin with x, x . This is an explicit omitted class cut distinct from the omitted-range branch cuts: enlarging the order from set-like to unrestricted raw well-orders does not repair it. If repeated labels are allowed in class words, however, the constant word $x, x, x, \dots$ represents this cut. Every injective core row differs from that word by coordinate 1 at the latest.

Remark 46.34 (Class character does not decide representability). Both sides of L_x have no set-sized cofinal or coinital witness, respectively, although they have uniform witnesses indexed by Ord . To see this, complete the prefixes

$$\left\langle x, x - \frac{1}{\alpha + 1} \right\rangle, \quad \left\langle x, x + \frac{1}{\alpha + 1} \right\rangle \quad (\alpha \in \text{Ord})$$

uniformly to core rows. Since ordinals are cofinal in the numerical surreal order, the first family is cofinal in L_x and the second is coinital in its complement. Any proposed set-sized witness can be improved at its second coordinate by surreal set interpolation. Thus this unrepresented cut has the same (Ord, Ord) class character as omitted-range branch cuts that do have raw representatives. This notation records the stated existence and nonexistence of uniform witnesses; it does not assign a set cardinal to a proper class.

46.6.3 Foundational accounting

All cylinders, cuts, and recovered graphs in this section are ordinary classes of sets. Prefixes are set functions; core rows are accessed through set codes. Uniqueness assembles coherent prefixes without a class-choice principle. Replacement is used only on set-sized domains, including the union at a limit failure. The argument never constructs a class containing all class words, never invokes a truth predicate for arbitrary formulas, and never uses ETR. The fixed baseline already supplies the choice strength needed for the previously proved completion and extension lemmas.

47 Automorphisms of the core (source 14; sources 17, 15 and 13)

[merge] Four sources show that the bare order of the core, or of the full order of global enumerations, does not remember what the recognition theorem uses. Source 14 classifies the automorphism orbits of class cuts by their characters (complete invariant) and derives that exhaustiveness is not invariant; source 17 gives an exact criterion for an automorphism of the core to extend to the global enumerations, the non-extension theorem and the positive family of numerical relabellings; source 15 gives a weaker form of source 14's corollary; source 13 proves a different theorem, that an automorphism of the full order $\mathfrak{W}_{s1}$ can move a cylinder to a non-cylinder. Source 14 is printed first.

47.1 Source 14: automorphism orbits and the limits of bare-order recognition

The prefix classification has a complementary classification in the language of order alone. It is coarser: different prefix-boundary types can belong to the same orbit under class order automorphisms.

Lemma 47.1 (Set homogeneity of the core). *Every increasing bijection between two set-sized suborders of $\mathcal{P}_s(b)$ extends to a class order automorphism of $\mathcal{P}_s(b)$.*

Proof. Begin with the given partial map and carry out the back-and-forth construction used in [Corollary 44.3](#). At any set stage its domain and range are sets, even though they may already contain the entire initial partial map. The lower and upper images of a new point form a set cut. A strict separator automatically avoids all previously used image points, since each of those is on one of the two sides. The inverse step is symmetric. Taking the coherent union gives a surjective class order embedding extending the original map. $\square$

Theorem 47.2 (Complete character invariant for cut orbits). *Two proper normalized class cuts of $\mathcal{P}_s(b)$ lie in the same orbit under class order automorphisms if and only if they have the same character pair. The possible pairs are exactly*

$$(\rho, \text{Ord}), \quad (\text{Ord}, \rho), \quad (\text{Ord}, 1), \quad (\text{Ord}, \text{Ord}),$$

where ρ ranges over the infinite regular set cardinals.

Proof. A class automorphism and its inverse carry sets to sets and preserve cofinality and coinitality, so equality of characters is necessary. A normalized lower cut has no maximum, so any set-sized cofinality is an infinite regular cardinal. The upper complement either has a least point, with character 1, has infinite regular set coinitality, or has character Ord. Both characters cannot be set-sized by the core set-cut property. These observations exhaust the displayed possibilities. They all occur: lower and upper edges of a prefix of length ρ realize the first two, a core point realizes the third, and a successor-prefix cylinder edge realizes the fourth.

Suppose first that D, D' have the same set-sized left cofinality ρ . Choose strictly increasing cofinal sequences of type ρ in them. Matching the sequences gives a set partial isomorphism. [Lemma 47.1](#) extends it to an automorphism. Since a normalized lower set is the strict downward closure of any cofinal sequence with no last member, that automorphism carries D onto D' . For equal right coinitalities use coinital sequences in their complements and the same argument with upward closures. This includes right character 1, where one simply matches the two least points of the complements.

It remains to consider two cuts with character (Ord, Ord) . Each of their four sides is itself a proper class η_{Ord} -order. For example a set cut inside D with a nonempty upper side is filled by core saturation, with the interpolant still below an element of D ; a set with no upper constraints has an upper bound inside D because D has no set-sized cofinal subset. Lower bounds remain available from the ambient core. The corresponding argument for the complement uses its absence of a set-sized coinital subset. None of these sides has an endpoint. Back-and-forth therefore gives one class isomorphism $D \rightarrow D'$ and another between their complements. Their union is an order automorphism of the core, since all points on a lower side precede all points on its upper side. $\square$

Corollary 47.3 (Exhaustiveness is not a bare-order invariant). *There are a global exhaustive enumeration E and a class order automorphism g of $\mathcal{P}_s(b)$ such that $g[D_E]$ is not the trace of any unrestricted class well-order on $\mathbf{No}$. In particular, exhaustive representability and even branch representability cannot be recognized by a criterion invariant under all class automorphisms of the bare core order.*

Proof. Let E swap every consecutive ordinal pair of the baseline:

$$E(2 \cdot \alpha) = b(2 \cdot \alpha + 1), \quad E(2 \cdot \alpha + 1) = b(2 \cdot \alpha).$$

It is exhaustive, but it is not represented by a core code, because it moves a proper class of coordinates. Hence D_E has character (Ord, Ord). The lower edge of the one-letter cylinder $C_{(b(0))}$ has the same character. The orbit theorem supplies g carrying the former cut to the latter. A cylinder edge is not a branch trace by [Theorem 43.3](#), and is not a raw-order trace by source 14's [Theorem 10.2](#) (trace fibres; [\[merge\]](#) [theorem 54.6](#), and [proposition 15.3](#) for the fibre description). This proves the assertion. $\square$

Thus the reference to named cylinders in the recognition theorem is not merely a technical limitation of its proof. A genuine symmetry of the bare linear order can destroy the property being recognized. Principal exhaustive cuts are distinguished by their right character 1, but non-core exhaustive cuts are not distinguished from other (Ord, Ord) boundaries. The stabilizer of the exhaustive-cut collection inside the full automorphism group is therefore a proper subgroup, deserving its own study.

Note 47.4 ([\[merge\]](#); overlaps). [Lemma 47.1](#) is the set-homogeneity of an order filling all set cuts ([theorem 5.17](#), source 09's remark on set-sized partial isomorphisms in its direction 4). [Corollary 47.3](#) and source 17's [theorem 47.7](#) use the same exhaustive enumeration (the baseline with every consecutive pair swapped); source 14 sends its cut to a cylinder edge, source 17 to the cut of a nonexhaustive injective word, and source 15's [theorem 47.10](#) to a first-label cut, so all three show that neither represented cuts nor exhaustive cuts are invariant under automorphisms of the bare core. Source 15's proof that each side of such a cut is a copy of **No** is the argument of source 17's [lemma 47.6](#) and of the last paragraph of the proof of [theorem 47.2](#).

47.2 Source 17: automorphisms, an exact extension test and a negative result

Let $\mathcal{R}_b(L)$ be the predicate saying that L is a proper nonempty open lower cut of $\mathcal{P}_s(b)$ which satisfies both the straddling and coverage conditions of [Theorem 46.21](#). Thus $\mathcal{R}_b$ describes exactly the cuts L_E with $E \in \mathfrak{W}_{sl}$. It is a predicate on individual class parameters, not an ordinary class of cuts.

For a class order-automorphism $h : \mathcal{P}_s(b) \rightarrow \mathcal{P}_s(b)$, its action on cuts is $h_*L = h[L]$. An extension of h to $\mathfrak{W}_{sl}$ means a uniform operator sending each global enumeration to a global enumeration, preserving comparisons and having a uniform inverse, whose action on core enumerations is h . This convention states the higher-order typing of the assertion.

Theorem 47.5 (Extension criterion). *A class order-automorphism h of $\mathcal{P}_s(b)$ extends to an automorphism of $\mathfrak{W}_{sl}$ if and only if*

$$\mathcal{R}_b(L) \iff \mathcal{R}_b(h_*L) \quad \text{for every admissible lower class cut } L. \quad (\text{S17.9})$$

When it exists, the extension is unique and is reconstructed uniformly by [Theorem 46.21](#). No new class-valued recursion is required.

Proof. If H extends h , order preservation gives

$$L_{H(E)} = h_*L_E.$$

Applying the same fact to H^{-1} proves preservation of the represented cut predicate in both directions.

Conversely assume (S17.9). For each E , form L_E , take its image under h , and reconstruct its unique exhaustive word by Theorem 46.21. Call this word $H(E)$. The entire construction is elementary from E, h, b . Distinct rows have distinct cuts, and $E < F$ if and only if $L_E \subsetneq L_F$, because the core is dense between them. Hence H preserves and reflects order. The same recipe using h^{-1} is its inverse. For a code c , the image of the strict lower segment below c is the strict lower segment below $h(c)$, so H agrees with h on the core. Finally an extension's value is determined by its comparisons with every core code, proving uniqueness. $\square$

The criterion gives a precise role to the represented-cut predicate. It is not automatic that an abstract automorphism of the dense core preserves it. In fact it can fail in a strong way.

Lemma 47.6 (Homogeneity of large two-sided core gaps). *Let (L, U) be a gap of $\mathcal{P}_s(b)$ such that L has no set-sized cofinal subset and U no set-sized coinital subset. Each of the class orders L and U is isomorphic to $\mathbf{No}$. Any two such gaps are conjugate by a class order-automorphism of $\mathcal{P}_s(b)$.*

Proof. Consider a separated pair of sets $A < B$ in L . If B is nonempty, Theorem 42.13 gives a core interpolant below every member of B , hence lying in L . If B is empty, the absence of a set-sized cofinal subset of L supplies a member of L strictly above all of A ; this includes the case $A = \emptyset$ after choosing any member. Therefore L has the full set-cut interpolation property. It is proper: a set L would be a set-sized cofinal subset of itself. The dual proof gives the same property and properness for U .

The class back-and-forth proof of Corollary 42.14 now yields $L \cong \mathbf{No} \cong U$. Given two gaps, choose order-isomorphisms between their lower sides and between their upper sides. The union of these two class maps is an order-automorphism of the core carrying the first gap to the second. This uses only two class maps at a time, not an exhaustive choice of maps for a higher collection of gaps. $\square$

Theorem 47.7 (Some core automorphisms do not extend). *In GBC there is a class order-automorphism of $\mathcal{P}_s(b)$ which does not extend to an automorphism of $\mathfrak{W}_{sl}$. More precisely, there is one taking the cut of an exhaustive enumeration to the cut of a nonexhaustive injective word.*

Proof. Construct an exhaustive enumeration E by transposing the two entries in every successive pair of positions of b . It differs from b on a proper class of positions, so it has no core code. Choose also a bijection $F : \text{Ord} \rightarrow \mathbf{No} \setminus \{r_0\}$ for one fixed label r_0 . It cannot have a core code because it is not exhaustive.

By Proposition 46.26, both L_E and L_F are genuine two-sided core gaps with no small cofinal or coinital side. Lemma 47.6 supplies an automorphism h with $h_*L_E = L_F$. The first cut satisfies $\mathcal{R}_b$, whereas the second fails coverage. Theorem 47.5 rules out an extension. Alternatively, any proposed image $H(E) \in \mathfrak{W}_{sl}$ would have cut L_F , whose unique representing injection is the nonexhaustive word F , a contradiction. $\square$

Corollary 47.8 (A limit to pure-order recognition). *Exhaustiveness of a branch cut is not invariant under all automorphisms of the bare class order $\mathcal{P}_s(b)$. In particular it cannot be characterized by a parameter-free, automorphism-invariant property of that order and its cuts alone. The two-sided character (Ord, Ord) does not suffice.*

This does not preclude recognition using the baseline, labelled prefix tree, or suitable additional predicates. Theorem 46.21 provides exactly such a positive result. Nor does it rule out more subtle reconstruction after adding a distinguished represented-cut predicate. It shows why order-isomorphism of dense cores, by itself, cannot be used to extend maps of the entire higher-order comparison.

47.2.1 A natural positive family

Proposition 47.9 (Relabelling by numerical automorphisms). *If $\varphi : \mathbf{No} \rightarrow \mathbf{No}$ is an increasing class bijection, then $E \mapsto \varphi \circ E$ is an automorphism of $\mathfrak{W}_{\text{sl}}$, and transporting relations by φ is an automorphism of the unrestricted comparison $\mathfrak{W}_{\text{all}}$. It carries the core associated with b onto the core associated with $\varphi \circ b$.*

Proof. The first differing labels are transformed by an increasing bijection, so their comparison is unchanged. Surjectivity, injectivity, and the class well-order property are preserved under relabelling. Finally $\varphi \circ (b \circ \hat{\sigma}) = (\varphi \circ b) \circ \hat{\sigma}$, with the same set support code. $\square$

This is a sufficient family of extendible maps, not a classification of all of them. Arbitrary permutations of the label class need not preserve lexicographic order. A general isomorphism between two cores whose baselines differ on a proper class of positions need not extend either; [Theorem 47.5](#) identifies the additional condition that must be checked.

47.3 Source 15: the bare order does not determine represented cuts

Theorem 47.10 (An automorphism destroys prefix representability). *Extension proved in source 15. In GBC, there is a class order automorphism of $\mathcal{P}_{\text{bd}}(b)$ taking a cut represented by an injective Ord-word to a cut represented by no such word, and therefore by no raw class well-order.*

Proof. Take a nonexhaustive word F . Neither side of (L_F, R_F) has a set cofinality toward the cut: a set of first-disagreement coordinates can be bounded, and a perturbation agreeing with F beyond that bound improves every proposed bound. Each side consequently fills all its own set cuts. For a cut with constraints on both sides, use the core interpolation theorem between bounding points in that side. For an empty bounding side, use the absence of a set cofinal or cointial family toward the removed boundary, together with the unboundedness away from it. Thus each side is a copy of $\mathbf{No}$.

The class order $\mathbf{No} + \mathbf{No}$ also fills every set cut. If constraints straddle the summands, choose a point in the first summand above the set of lower constraints; cases with both bounds in one summand reduce to that summand. The cases of constraints extending into the second summand are handled there. Class uniqueness gives $\mathbf{No} + \mathbf{No} \cong \mathbf{No}$. Hence there is a numerical partition $\mathbf{No} = A + B$ with both sides isomorphic to $\mathbf{No}$.

Let $L' = \{d : E_d(0) \in A\}$ and $R' = \{d : E_d(0) \in B\}$. This cut has no crossing cylinder at length one, so it is unrepresented. Neither side has a set cofinality toward the cut, as is seen already in its first labels. The preceding interpolation argument shows that L' and R' are also copies of $\mathbf{No}$. Choose class isomorphisms $L_F \rightarrow L'$ and $R_F \rightarrow R'$ and take their union. It is the required automorphism. All these maps have sets as inputs and outputs. $\square$

Thus even knowing the abstract class order $\mathcal{P}_{\text{bd}}(b) \cong \mathbf{No}$ is insufficient to identify the enumeration cuts. One must retain predicates for the cylinders, the evaluation map, or equivalent structure. This supplies a negative answer to a natural reconstruction question raised in the prior report. The theorem does not claim that every automorphism of the core extends to the higher collection of all global enumerations.

[merge] Source 15 states [theorem 47.10](#) without naming its foundation; its proof chooses class isomorphisms of the sides of two cuts with $\mathbf{No}$, so it is a theorem of GBC, and the words “In GBC” are added at the merge. Its Proposition 4.2 ([proposition 51.10](#)) needs GBC for the same reason. Its closing sentence, a “negative answer to a natural reconstruction question raised in the prior report”, answers [research question 61.4](#) in part only ([section 50](#)).

47.4 Source 13: the pure order does not recover the prefix cylinders

The earlier report asks whether prefix structure can be recovered from the pure order ([research question 61.4](#)). We give a negative answer for the family of all set-prefix cylinders, using only a finite change in how information is grouped.

Lemma 47.11 (Finite tuples as a surreal class order). *For each positive integer n , the class P_n of injective n -tuples of surreals, ordered lexicographically, is isomorphic to $\mathbf{No}$.*

Proof. First, finite lexicographic powers $\mathbf{No}^n$ fill every set cut. At the first coordinate, the lower and upper projected values satisfy all cross weak inequalities. If no cross value is shared, choose a strict separator for that coordinate and fill the remaining coordinates arbitrarily. If a value is shared, it is the unique common one; keep that coordinate and use the active rows in the remaining coordinates. Continuing this way cannot reach two identical opposing tuples, because the original cut was strict. This proves the set-cut property. [Lemma 42.2](#) gives $\mathbf{No}^n \cong \mathbf{No}$.

There is a uniform order isomorphism $\mathbf{No}^n \rightarrow P_n$: at coordinate i , map the input value increasingly into the complement of the previously output finite set, using h_S . Recover the input by the inverse complement map coordinate by coordinate. First disagreements are preserved. Compose the two isomorphisms. $\square$

Fix an order isomorphism $k : \mathbf{No} \rightarrow P_2$. Write $k(x) = (u_x, v_x)$, where $u_x \neq v_x$. There is a uniform increasing bijection

$$g_x : \mathbf{No} \setminus \{x\} \longrightarrow \mathbf{No} \setminus \{u_x, v_x\}, \quad g_x = h_{\{u_x, v_x\}} \circ h_{\{x\}}^{-1}.$$

Theorem 47.12 (Cylinder non-rigidity). *In GBC there is a uniform order automorphism of $\mathfrak{W}_{\text{sl}}$ that maps some prefix cylinder to a class of enumerations that is not any prefix cylinder. Thus the family of all prefix cylinders is not invariant under automorphisms of the pure linear order.*

Proof. Define $H(E)$ by replacing the first label $x = E(0)$ with the pair (u_x, v_x) , and translating the remaining labels by g_x :

$$H(E)(0) = u_x, \quad H(E)(1) = v_x, \quad H(E)(2 + \alpha) = g_x(E(1 + \alpha)).$$

This is a class bijection onto $\mathbf{No}$. The first two output labels determine x by k^{-1} ; the inverse of g_x recovers the tail. Therefore H is a bijection on the virtual domain. Different first input labels are compared by the order isomorphism k , and equal first labels leave an increasing tail transformation. Hence H is an order automorphism, with

$$H(C_{(x)}) = C_{k(x)}.$$

Fix a label u . Then

$$H^{-1}(C_{(u)}) = \bigcup_{x \in A_u} C_{(x)}, \quad A_u = \{x : \text{the first coordinate of } k(x) \text{ is } u\}.$$

The class A_u contains more than one label, because P_2 contains (u, v) for every $v \neq u$. It is nonempty and proper, since pairs with another first coordinate also exist. The displayed inverse image therefore contains enumerations with different first labels, but is not the whole domain. It cannot be a nonempty-prefix cylinder, all of whose members have the same first label, and it cannot be the empty-prefix cylinder, which is the whole domain. Apply the automorphism H^{-1} to $C_{(u)}$. $\square$

47.4.1 What the counterexample does not say

The theorem rules out a parameter-free order-theoretic characterization whose members are exactly the prefix cylinders. It also rules out recovering their original tree canonically from the pure linear order. It does not say that no useful interval structure is recoverable, that every order automorphism arises by block recoding, or that the chosen core $\mathcal{P}_s(b)$ is preserved by H .

In particular, the expanded structure containing the core, its set-support filtration, birthday data, or an explicit prefix predicate may have a much smaller automorphism group. The cut criterion of [Theorem 46.15](#) is correctly a criterion in such an expanded prefix presentation. Its non-invariance under forgetting that presentation is a substantive feature, not a foundational flaw.

47.4.2 Finite block depth is not intrinsic either

Using P_n instead of P_2 replaces one first label by any fixed positive finite number of distinct labels, with a complement isomorphism applied to the tail. The same construction gives an order automorphism that sends first-level cylinders to selected n th-level cylinders. Thus even finite prefix depth has no absolute meaning in the pure order, despite being indispensable to a particular lexicographic presentation.

[merge] Source 13's opening sentence refers to [research question 61.4](#), which its theorem answers in part. [Lemma 47.11](#) uses the family h_S of [lemma 42.2](#).

48 Class extensions (source 14; source 17)

[merge] Sources 14 and 17 prove the same theorem: for two class realizations with the same sets and a common baseline, an old core cut is the cut of a global enumeration in the larger realization exactly when it already is in the smaller one, and the enumeration is then old. The proof is the elementarity of the recognition criterion. Source 17 adds that every proper class extension adds a new global enumeration and a new represented cut ([theorem 48.4](#)). Source 14's text, from the second half of its Section 7, is printed first.

Theorem 48.1 (No new realization of an old trace cut). *Suppose $(M, \mathcal{C})$ and $(M, \mathcal{D})$ are models of GBC with the same set structure, $\mathcal{C} \subseteq \mathcal{D}$, and a common baseline enumeration $b \in \mathcal{C}$. If a proper normalized core cut D belongs to $\mathcal{C}$, then*

$$(M, \mathcal{C}) \models \exists E \in \mathfrak{W}_{\text{sl}} (D = D_E) \iff (M, \mathcal{D}) \models \exists E \in \mathfrak{W}_{\text{sl}} (D = D_E).$$

Moreover, when it exists, the realizing enumeration in the larger model is already a class of the smaller model.

Proof. The canonical support codes, the cylinders, and their membership relations are defined from the same sets and b . Every formula in (C), (X), and the reconstruction formula has set quantifiers and the old class parameters D, b only. Its truth is therefore the same in the two structures. If the larger model contains a realizing enumeration, the recognition theorem gives (C) and (X) there and hence in the smaller model. Elementary comprehension in the smaller model constructs the displayed graph from D, b . It is the same graph as the larger model's enumeration by uniqueness. The converse is immediate. $\square$

Corollary 48.2. *An enumeration newly available after such a class extension has a trace cut not belonging to the old class collection. An old nonexhaustive branch cut, cylinder edge, or numerical-gap cut is not turned into the trace of a global enumeration by enlarging only the class part.*

Remark 48.3 (Scope of absoluteness). The theorem holds because realizability of this particular kind of cut has an elementary, uniquely reconstructing criterion. It is not an assertion that all class-existence questions are absolute, that the virtual enumeration order is unchanged, or that arbitrary internal well-foundedness is externally absolute. The set structure and the baseline must remain fixed. New classes can, and in suitable realizations do, supply new enumerations and new cuts simultaneously.

48.1 Source 17: what changes when the available classes change?

Let $(M, \mathcal{X}) \subseteq (M, \mathcal{Y})$ be two transitive two-sorted models of GBC with the same sets and $\mathcal{X} \subseteq \mathcal{Y}$. Assume the baseline b belongs to $\mathcal{X}$. The assertions below can also be stated internally as preservation statements for a specified same-set extension; the transitive formulation avoids irrelevant ambiguities about external well-foundedness.

Theorem 48.4 (New enumerations are exactly new represented core cuts). *The set-code carrier $\mathcal{P}_s(b)$, its order relation, and its labelled set-prefix cylinders are unchanged between the two models. For any $E \in \mathfrak{W}_{sl}^{\mathcal{Y}}$,*

$$E \in \mathcal{X} \iff L_E \in \mathcal{X}.$$

Consequently every genuinely larger class realization $\mathcal{X} \subsetneq \mathcal{Y}$ adds a global enumeration and a represented cut of the unchanged core.

Proof. Support codes and prefixes are sets, and the two models have exactly the same sets. Their evaluations use the same baseline class. All relations in the definition of a cylinder are therefore the same.

If $E \in \mathcal{X}$, elementary comprehension in the smaller model gives $L_E \in \mathcal{X}$. Conversely, suppose $L_E \in \mathcal{X}$. At every ordinal length, the assertion that a set prefix straddles this cut has the same truth value in the two models: its witnesses are set codes and its quantifiers range over their common sets. Thus the smaller model can reconstruct the unique branch by the elementary formula in [Theorem 46.21](#). Its graph must be E , so $E \in \mathcal{X}$. Coverage is also an elementary assertion with the same set domain.

For the final assertion, choose $A \in \mathcal{Y} \setminus \mathcal{X}$. A global well-order already in $\mathcal{X}$ gives a class bijection $g : \text{Ord}^M \rightarrow M$. The pullback $D = g^{-1}[A]$ is a subclass of Ord^M in $\mathcal{Y}$ and not in $\mathcal{X}$, since A is recovered from it by the same old g . Encode D by listing each sorted baseline pair in increasing or decreasing order according as its index is absent from or present in D . This yields $E_D \in \mathfrak{W}_{sl}^{\mathcal{Y}}$. Pair orientation recovers D , so E_D is not in $\mathcal{X}$. Its core cut is new by the first part of the theorem. $\square$

For pre-existing global enumerations, their comparisons and their surjectivity are unchanged. A bijection from the common ordinal class still induces a well-order: any nonempty subclass of its index class has a least ordinal, witnessed below any one of its members. Thus the old set-like comparison sits inside the new one with its order intact, even though the represented-cut collection strictly expands.

This result answers a precise part of the source's class-realization question. It does not reconstruct the entire class collection from the *unlabelled external order type* of its admitted enumerations. The proof uses the baseline and its fixed decoder. That stronger reconstruction problem remains open here.

49 Support-restricted suborders and the exact reserve (source 14; source 17)

[merge] Source 14 measures how much support an interpolant needs; source 17 measures how much support beyond the forced positions a completion of a partial injection needs. Source 14's Section 9 is printed first, then source 17's Section 7.

49.1 Source 14: a sharp hierarchy of support-restricted orders

Fix an infinite set cardinal κ and define

$$\mathcal{P}_{<\kappa}(b) = \{\sigma \in \mathcal{P}_s(b) : |\text{supp}(\sigma)| < \kappa\}.$$

For an infinite regular cardinal θ , an η_θ -order is a linear order in which every separated pair $A < B$ with $|A \cup B| < \theta$ has a strict separator, including when one side is empty. This convention makes η_ω mean dense without endpoints. The class-sized orders here need not have cardinality θ .

The predecessor distinguishes finite support from arbitrary set support ([proposition 19.31](#); Section 19 of this report at the pin). We now give an exact intermediate hierarchy, including singular thresholds. The key is to compress the support of a separator without preserving its irrelevant later coordinates.

Lemma 49.1 (Small-support extension of a partial injection). *Let $f : D \rightarrow \text{Ord}$ be a set injection. It extends to a class permutation of Ord with finite support if D is finite, and with support of cardinality at most $|D|$ if D is infinite. One may require that any specified set of already prescribed fixed positions, disjoint from $D \cup f[D]$, stay fixed.*

Proof. Put $T_0 = D \cup f[D]$. For finite D , the complements $T_0 \setminus D$ and $T_0 \setminus f[D]$ have the same finite cardinality. Choose a bijection between them and combine it with f ; fix everything outside T_0 . For infinite D of cardinality ν , adjoin a fresh set B of ν ordinals disjoint from T_0 and from the specified fixed positions. With $T = T_0 \cup B$, both $T \setminus D$ and $T \setminus f[D]$ have cardinality ν . A bijection between them extends f to a permutation of T , again extended by the identity. The resulting support has cardinality at most ν . $\square$

Lemma 49.2 (Support compression). *Let A, B be separated sets of core codes and let*

$$S = \bigcup \{\text{supp}(\sigma) : \sigma \in A \cup B\}.$$

There is a strict separator whose support is finite if S is finite, and has cardinality at most $|S|$ if S is infinite.

Proof. By [Corollary 44.2](#), choose a separator $Q \in \mathcal{P}_s(b)$. If its support is contained in S , nothing remains to prove. Otherwise let

$$\gamma = \min(\text{supp}(Q) \setminus S).$$

Every input permutation fixes γ , while $\hat{Q}$ moves it. Thus the first disagreement between E_Q and every input enumeration occurs no later than γ . Any permutation agreeing with $\hat{Q}$ through γ will preserve all those comparisons.

Let $D = \{\alpha \leq \gamma : \hat{Q}(\alpha) \neq \alpha\}$. By minimality of γ , $D \subseteq S \cup \{\gamma\}$. Keep the partial injection $\hat{Q} \upharpoonright D$. None of its images is a prescribed fixed position $\beta \leq \gamma$ outside D : otherwise $\hat{Q}(\beta) = \beta = \hat{Q}(\alpha)$ for a distinct $\alpha \in D$, contradicting injectivity. Apply [Lemma 49.1](#) while retaining those fixed positions. For infinite D its buffer can explicitly be chosen above γ . The resulting permutation agrees with $\hat{Q}$ through γ , hence is still a strict separator. Its support is finite when S is finite, and is at most $|S|$ when S is infinite, since $|D| \leq |S \cup \{\gamma\}|$. $\square$

Theorem 49.3 (Exact support saturation and an attained obstruction). *For every infinite set cardinal κ , put $r = \text{cf } \kappa$. Then $\mathcal{P}_{<\kappa}(b)$ is an η_r -order but not an η_{r+} -order. More precisely, there are increasing $(l_i)_{i<r}$ and decreasing $(u_i)_{i<r}$ in $\mathcal{P}_{<\kappa}(b)$ such that $l_i < u_j$ for all i, j , and every core separator moves at least κ coordinates. A separator of support cardinality exactly κ exists in $\mathcal{P}_s(b)$.*

Proof. For the positive assertion take $A, B \subseteq \mathcal{P}_{<\kappa}(b)$ with $|A \cup B| < r$. Fewer than $\text{cf } \kappa$ sets, each of cardinality $< \kappa$, have union of cardinality $< \kappa$. Their union of supports therefore satisfies $|S| < \kappa$. Support compression gives an interpolant with support $< \kappa$. For $\kappa = \omega$ both the family and all its supports are finite, so the finite clause applies.

For sharpness choose strictly increasing ordinals $(\lambda_i)_{i<r}$ cofinal in κ . Consider the injective prefix

$$p : \kappa \rightarrow \mathbf{No}, \quad p(\alpha) = b(\alpha + 1).$$

Every infinite initial cardinal is a limit ordinal, so this formula uses coordinates below κ . It moves every coordinate of its domain relative to b . For each i , choose fresh labels

$$a_i < p(\lambda_i) < c_i, \quad a_i, c_i \notin \text{ran}(p \upharpoonright \lambda_i),$$

and extend the two prefixes

$$(p \upharpoonright \lambda_i) \frown (a_i), \quad (p \upharpoonright \lambda_i) \frown (c_i)$$

to enumerations with codes l_i, u_i . Their partial ordinal maps have domains of cardinality $|\lambda_i + 1| < \kappa$; by Lemma 49.1 they can be completed with support $< \kappa$, even when a chosen label has a very large baseline index. Set choice suffices to make the r choices simultaneously.

For $i < j$, comparison at λ_i gives $l_i < l_j$ and $u_j < u_i$. The same first-coordinate argument gives $l_i < u_j$ for every i, j . Suppose any global enumeration E separated the two families and first differed from p at some $\alpha < \kappa$. Choose i with $\lambda_i > \alpha$. Both E_{l_i} and E_{u_i} agree with p through α . According to its differing value there, E would lie below both or above both, a contradiction. Hence any separator satisfies $E \upharpoonright \kappa = p$. Relative to b it must move all κ coordinates of that prefix. In particular no member of $\mathcal{P}_{<\kappa}(b)$ separates the families.

Finally extend p itself by Lemma 49.1. Its domain has cardinality κ , so the extension has support at most κ ; the forced moves show the support is exactly κ . It separates the two families by comparison at λ_i . The obstruction has combined cardinality r , hence violates η_{r+} , and its minimum support cost is attained. $\square$

49.1.1 Consequences and examples

For a regular κ the answer is exactly η_κ but not $\eta_{\kappa+}$. In particular finite support is dense without endpoints but can fail a countable cut; countable support fills every countable strict cut but can fail a cut of size ω_1 . For $\kappa = \aleph_\omega$, the cofinality is ω , so support of every cardinality below $\aleph_\omega$ still does not guarantee countable-cut filling. The same construction takes supports cofinal in the threshold and forces their full union at the limit.

This sharpness statement is stronger than merely exhibiting some gap: it quantifies how much extra support any solution must use. It also explains why replacing a singular support bound by its cardinal size in a saturation argument is invalid. The closure property used in the positive direction is closure under unions of fewer than $\text{cf } \kappa$ supports, not fewer than κ supports.

Proposition 49.4 (Universality without high saturation). *Every $\mathcal{P}_{<\kappa}(b)$ contains an order-isomorphic copy of $\mathbf{No}$, already among the permutations with support at most two. Thus, in GBC, it contains every ordinary class linear order up to embedding. Nevertheless no fixed $\mathcal{P}_{<\kappa}(b)$ is isomorphic to $\mathbf{No}$, and it is not dense in $\mathfrak{W}_{\text{sl}}$.*

Proof. For $x \in \mathbf{No}$, transpose the baseline positions 0 and $b^{-1}(x)$, using the identity if these coincide. Call the code t_x . Then $E_{t_x}(0) = x$, so $x < y$ implies $t_x < t_y$. The familiar class universality of $\mathbf{No}$ follows by inserting points of any ordinary class linear order, in a fixed set-like global enumeration, between the images of their earlier lower and upper neighbors; every stage has only set many constraints. Composing gives the asserted embeddings.

Theorem 49.3 supplies a set cut not filled by $\mathcal{P}_{<\kappa}(b)$, whereas $\mathbf{No}$ fills all set cuts, so the two orders are not isomorphic. For ambient nondensity, use the prefix p of length κ in that theorem. Choose two distinct fresh next labels and complete both prefixes to core enumerations. Every enumeration between them shares p and hence moves at least κ baseline coordinates. The nonempty open interval they determine misses $\mathcal{P}_{<\kappa}(b)$ entirely. $\square$

Remark 49.5 (Cardinality versus location of support). The theorem restricts $|\text{supp } \sigma|$, not the ordinal height of the support. A transposition with an arbitrarily late baseline coordinate still has support two. Replacing the cardinality condition by $\text{supp } \sigma \subseteq \gamma$ for one fixed ordinal γ gives a set-sized permutation space with quite different universality. These two filtrations should never share an undifferentiated “bounded” label.

Note 49.6 ([merge]; the finite case is source 08’s). At $\kappa = \omega$, **theorem 49.3** is source 08’s **proposition 19.31** (printed in Part IV), with the same witness prefix $p(n) = b(n+1)$ and the same fresh labels; source 14 cites the finite-support subsection but not the proposition. New in source 14 are the positive half ($\mathcal{P}_{<\kappa}(b)$ is η_r), the threshold at every infinite κ , including singular κ , and the attained support cost κ of a separator. **Proposition 49.4** extends source 08’s remark that the finite-support codes contain a copy of $\mathbf{No}$ (**note 19.9**).

49.2 Source 17: the optimal extra support needed to extend a partial injection

The source extends a prescribed prefix inside a sufficiently large cardinal block. That argument is safe but leaves a more elementary optimization question: how many *additional moved positions* are actually necessary? The following answer applies to arbitrary set-sized partial injections, not just prefixes.

Theorem 49.7 (Exact reserve cardinal). *Let $A \subseteq \text{Ord}$ be a set and let $h : A \rightarrow \text{Ord}$ be injective. Define*

$$D = \{\xi \in A : h(\xi) \neq \xi\}, \quad R = h[D], \quad S = D \cup R, \quad u = |R \setminus D|, \quad v = |D \setminus R|.$$

Among class permutations $\pi : \text{Ord} \rightarrow \text{Ord}$ with set support which extend h , the minimum possible value of $|\text{supp}(\pi) \setminus S|$ is

$$\tau(h) = \begin{cases} 0, & u = v, \\ \max(\aleph_0, u, v), & u \neq v. \end{cases} \quad (\text{S17.6})$$

Moreover every point of S must move. An extension attaining the bound can be constructed by a bijection between two explicitly specified sets. The theorem uses only set choice and elementary operations with the class of ordinals; with a fixed global choice parameter the construction can be made uniform.

Proof. The fixed points $A \setminus D$ are disjoint from R , by injectivity of h . Every $d \in D$ must move. If $r \in R \setminus D$, it already has a preimage $d \neq r$ under the prescribed map, so an extending permutation cannot fix r . Thus every point of S must move.

Suppose π is an extension and put $T = \text{supp}(\pi) \setminus S$. No point of T is in $A \setminus D$, whose values are prescribed to be fixed. The support of a permutation is invariant under the permutation and

its inverse. After the prescribed bijection $h : D \rightarrow R$ has been removed, π must therefore give a bijection

$$(R \setminus D) \cup T \longrightarrow (D \setminus R) \cup T. \quad (\text{S17.7})$$

In particular $u + |T| = v + |T|$. If $u \neq v$ are finite, finite cancellation rules out finite T , and $\aleph_0$ suffices. If at least one of u, v is infinite, unequal cardinals can be equalized by addition of $|T|$ only when $|T| \geq \max(u, v)$. This follows from the formula for addition of infinite cardinals as their maximum. These observations prove the lower bound in (S17.6).

Conversely, choose a fresh set of ordinals T of cardinality $\tau(h)$, disjoint from $A \cup R$, and choose a bijection in (S17.7). Combine it with $h : D \rightarrow R$, and take the identity outside $S \cup T$. The result is a permutation extending h , with support contained in $S \cup T$. All points of S move, as proved above. If $\tau(h) > 0$, the lower bound forces exactly $\tau(h)$ further points to move; if $\tau(h) = 0$, no reserve was used. This proves attainment, including cases where the chosen bijection happens to fix some points of T . $\square$

Example 49.8 (Why a finite reserve can never repair a one-sided shift). Prescribe $h(n) = n + 1$ for every $n < \omega$. Then $D = \omega$, $R = \omega \setminus \{0\}$, and $(u, v) = (0, 1)$. No permutation on ω extends h . More strongly, no extension can use only finitely many additional moved positions outside ω . Exactly countably many are necessary and sufficient. One concrete completion chooses fresh $t_0, t_1, \dots$, sends t_0 to 0, and sends t_{n+1} to t_n ; all other ordinals are fixed.

Example 49.9 (Finite partial injections require no reserve). When D is finite, $|D| = |R|$ implies $|R \setminus D| = |D \setminus R|$. Hence every finite partial injection extends to a permutation supported exactly on $D \cup R$. The distinction between this fact and the preceding infinite example is precisely the failure of finite cancellation for infinite sets; finite testing alone cannot find the infinite obstruction.

Corollary 49.10 (Application to prescribed surreal prefixes). *Fix a class bijection $b : \text{Ord} \rightarrow \mathbf{No}$. If $p : \alpha \rightarrow \mathbf{No}$ is an injective set prefix, apply Theorem 49.7 to $h = b^{-1} \circ p$. The resulting $b \circ \pi$ extends p and agrees with b outside a set. Formula (S17.6) gives the exact minimum number of moved coordinates beyond the necessarily moved set $D \cup h[D]$.*

This is a cardinal optimum relative to a given partial injection and baseline. It does not assert a minimum *ordinal* support bound, a minimum birthday of a code, or a canonical extension independent of choice. Those are different optimization problems. The source’s larger-cardinal completion remains a useful simple implementation; the reserve formula says exactly when its extra room is genuinely necessary.

Note 49.11 ([merge]; the reserve theorem in this report). Theorem 49.7 sharpens source 09’s remark 19.7: the example $n \mapsto n + 1$ there is source 17’s example 49.8, and the theorem says exactly when the larger cardinal of lemma 19.4 is needed and how many extra positions suffice. The two support theorems are consistent: lemma 49.1 bounds the support of an extension of a set injection $f : D \rightarrow \text{Ord}$ by $|D|$ when D is infinite and keeps it finite when D is finite, while theorem 49.7 computes the least number of moved positions outside the forced set: 0 for finite partial injections (example 49.9), and otherwise 0 or $\max(\aleph_0, u, v) \leq |D|$. Source 17’s question table maps the reserve theorem to source 11’s research question 61.6; that question asks for birthday bounds, and the theorem bears instead on source 09’s direction “Sharp birthday and support bounds” and source 08’s question 13 (section 50).

50 What this Part answers

[merge] Dated notes, 3 October 2026, in the form of [section 33](#); where the question index of [section 61](#) (dated 2 October 2026) says otherwise, these notes supersede it.

- (a) *Source 11's research question 61.1 and source 09's direction "Which cuts of the core represent exhaustive branches?"* ([section 61.3](#)): **answered**, in the labelled-tree form the question asks for. A normalized class cut of the core is the cut of an injective class word exactly when a crossing prefix exists at every set length; the word is unique and elementary in the cut; it is exhaustive exactly when the crossing prefixes cover **No** ([theorem 45.1](#); six proofs, [note 46.1](#)). Every other proper cut is a cylinder edge or a numerical gap after a set prefix, with unique data ([theorem 43.3](#)); the missing labels, the forbidden repeated labels and the cuts escaping every coherent branch that [research question 61.1's](#) commentary asks to separate are exactly the coverage failure, source 15's repeated-label case and the edge and gap cuts. A recognition from the bare order is impossible (item (c)).
- (b) *Source 08's question 8, higher-size cuts and completion* ([section 61.4](#)): answered for the class cuts of the core relative to its prefix presentation: the trichotomy and characters ([theorems 43.3](#) and [44.1](#)), the image of the raw orders ([corollary 46.4](#)), and source 15's remark that adjoining the raw orders is neither a Dedekind completion of the core nor a realization of every core cut ([section 46.2](#)). Cuts given by higher collections of classes stay open.
- (c) *Source 11's research question 61.4, source 09's direction "Automorphisms beyond set-sized partial maps" and source 08's question 6*: answered in part, negatively. The automorphism orbits of class cuts of the core are exactly the four character classes ([theorem 47.2](#)); neither represented nor exhaustive cuts are invariant under automorphisms of the bare core ([corollary 47.3](#) and [theorems 47.7](#) and [47.10](#)); the prefix cylinders of $\mathfrak{W}_{\text{sl}}$ are not invariant under automorphisms of $\mathfrak{W}_{\text{sl}}$, nor is finite prefix depth ([theorem 47.12](#)). An automorphism of the core extends to $\mathfrak{W}_{\text{sl}}$ exactly when it preserves represented cuts ([theorem 47.5](#)). Open: the stabilizer of the exhaustive cuts, and which extra predicates make the reconstruction unique (source 14's and source 17's own questions).
- (d) *Source 11's research question 61.5, baselines*: its suggested restricted first case is settled: a baseline change by an increasing class bijection of **No** is an automorphism of $\mathfrak{W}_{\text{sl}}$ and of $\mathfrak{W}_{\text{all}}$ carrying one core onto the other ([proposition 47.9](#), source 08's remark in its question 6). The general case is open.
- (e) *Source 11's research question 61.8, source 12's research question 61.18 and source 08's question 12, class realizations*: answered in part. Over a fixed set structure and baseline, the represented-cut status of an old core cut is absolute, its enumeration is old, and every proper extension of the class part adds a global enumeration and a represented cut ([theorems 48.1](#) and [48.4](#)). What the unlabelled external order type remembers is open here (Part IX prints source 18's countable models, [section 58](#), and [section 60](#) (e) records what they answer).
- (f) *Source 09's direction "Sharp birthday and support bounds" and source 08's question 13*: answered in part for support. The least support cardinal of an interpolant in $\mathcal{P}_{<\kappa}(b)$ is governed by $\text{cf } \kappa$ ([theorem 49.3](#)), and the least reserve of a completion of a partial injection is 0 or $\max(\aleph_0, u, v)$ ([theorem 49.7](#)). Ordinal support bounds, birthday bounds and the forced-prefix length remain open, as source 17 says.
- (g) *Source 11's research question 61.6* is not affected by the reserve theorem ([note 49.11](#)).

Part IX

Unrestricted class well-orders: canonical blocks, condensation, products, comprehension and models

[merge] (3 October 2026.) The fourth and last Part of the batch-81 addition ([section 1.8](#); notation in [section 1.3.1](#)). Part III ([part III](#)) compared arbitrary class well-orders of **No** directly, extracted the canonical initial set-like part of each, and asked how far that extraction can be iterated ([research question 61.3](#)); Part V ([part V](#)) compared the two global orders in larger universes. This Part prints what sources 13–18 prove beyond those Parts, and every remaining section of theirs on foundations, universes and formalization:

- their restatements of the order facts and of the direct comparison of class well-orders ([section 51](#)), all re-proofs of Parts I and III;
- the canonical decomposition of a class well-order into proper blocks of type Ord and one optional final set block, and the theorem that every class well-order is a block-index order ([section 52](#)), proved by sources 15, 16 and 18;
- explicit condensation histories of every class length and what they say about recursion strength ([section 53](#), source 15);
- the maximal finite components of the unrestricted order and the trace fibres of the core ([section 54](#), sources 17 and 14);
- asymmetric products ([section 55](#), source 13);
- binary class words, the diagonal theorem and a Σ_1^1 comprehension bound for uniform nonembedding ([section 56](#), sources 16 and 14);
- larger universes and class realizations ([section 57](#), sources 16, 14, 13, 15 and 18);
- the external classification of the full order in countable ω -standard models ([section 58](#), source 18);
- the six formalization plans ([section 59](#)).

[Section 60](#) records which questions the Part answers. The sources' foundations sections and Lean plans are printed here, at the end of the addition, rather than in Part V, so that no existing section is renumbered.

Notation in this Part (in addition to [section 1.3.1](#)). The block notations are the sources' own and differ: source 15 writes B_W for the class of least representatives of the blocks of a class well-order (X, W) and $C_b = \pi_W^{-1}(b)$ for the block with least representative b ; source 16 writes J_R for the least representatives of the proper blocks of R and α_R for the type of the final set block; source 18 writes B_R and θ_R . *False readings*: in source 15's block section b is a point of the order, not the baseline enumeration, and C_b is a block, not a prefix cylinder C_p of the core; source 14's J_R ([section 54](#)) is the canonical initial branch $\text{Ord} \rightarrow \mathbf{No}$ of R (source 15's F_R , source 13's F_R), not source 16's class of block representatives. Sources 16 and 18 write $\rho_R(x)$ for the rank of x inside its block; it is unrelated to source 15's coding length ρ_κ . In the sections on larger universes κ denotes a strongly inaccessible cardinal, as in Part V (source 14 writes Ω , sources 17 and 18 write λ); it is not a birthday cutoff there. Source 18 writes H_{sl} , H_{all} with a plain H in its later sections; they are printed $\mathfrak{W}_{\text{sl}}$, $\mathfrak{W}_{\text{all}}$, as its macros are. Source 15's finite ordinal polynomials $P(\omega)$ and source 18's

class $\mathcal{P}$ of finite-support functions $\omega \rightarrow \text{Ord}$ are the same order (note 53.6). Labels written for the merge in this Part begin `swo:ix:`.

51 Order facts and direct comparison as re-proved in sources 15, 16, 18, 17 and 13

Note 51.1 ([merge]; Part I and Part III counterparts). Before their class-level results, sources 13, 15, 16, 17 and 18 re-prove the surreal order facts and the direct comparison of class well-orders that Parts I and III already contain; each credits this report (source 18 labels its comparison theorems “repository result”). Their sections are printed whole below, in the order 15, 16, 18, 17, 13 for the order facts and 13, 15, 16, 18 for the comparison; none is new against the repository.

- *Fresh separators*: lemma 5.2 (source 15’s lemma 51.2, source 16’s lemma 51.3, parts (1)–(2) of source 18’s lemma 51.5). Parts (3)–(4) of source 18’s lemma are facts about the alphabet $\mathbf{No}_{<\kappa}$ proved in Part VI (theorem 29.4).
- *Universality and uniqueness of $\mathbf{No}$* : proposition 5.13 and theorem 5.17 (source 16’s proposition 51.4; the paragraph after source 15’s lemma 51.2).
- *Choice*: theorem 13.1 (source 16’s proposition 51.11, source 18’s theorem 51.14, the first subsection of source 15’s section 51.7; source 17 cites it in its foundations). Sources 15, 16 and 18 prove it over GB without set choice, as theorem 13.1 does; source 16 says that the recursion well-orders the hierarchy “uniformly” and source 18 that its orders “need not extend one another”, and none of the three repeats the “coherent” overstatement corrected in this report on 3 October 2026.
- *Direct comparison by the largest common labelled prefix*: theorems 14.2 and 14.7 (source 13’s proposition 51.7, source 15’s theorem 51.9, source 16’s lemma 51.12 with (S16.9), source 17’s proposition 51.6 with (S17.2), source 18’s lemma 51.15 and theorem 51.16 with (S18.4); source 14’s version opens section 54).
- *Canonical initial part and convex fibres*: theorem 15.1, proposition 15.3, and remark 15.4 (source 13’s proposition 51.8, source 15’s proposition 51.10; source 14’s lemma 54.5 in section 54).
- *Adjacency*: theorems 16.2 and 16.6 (source 16’s proposition 51.13, source 18’s theorem 51.17; source 17’s theorem 54.2 in section 54).

Source 15 states proposition 51.10 without naming its foundation. Its convexity half is a theorem of GB, but the realization of every injective word as an initial word appends “any fixed well-order” of the omitted range, which needs global choice; the words “In GBC” are added at the merge (source 13’s proposition 51.8 likewise uses “global choice” for this step). Source 15’s remark after it (“If F omits m labels, its fibre has $m!$ elements”) is remark 15.4 and proposition 15.3, as it says. Source 13’s Section 6 and source 16’s Section 9 continue in the cut sections of Part VIII, which cite them as proposition 51.8 and Theorem 10.1 of source 16 (theorem 52.6).

51.1 Source 15: sign sequences, properness, and cut filling

We realize a surreal as a sign sequence $s : \alpha \rightarrow \{-, +\}$ of ordinal length α . At a first difference the comparison is $- < \text{undefined} < +$; the birthday is the length. The all-plus sequences give an increasing copy of Ord , so $\mathbf{No}$ is a proper class. Every set of surreals has a set bound on its birthdays, by replacement.

The numerical surreal order fills every set cut: if L, R are sets and $L < R$, some x satisfies $L < x < R$, with either side allowed to be empty. This is the familiar Conway construction. It implies density and no endpoints, but it does not assert filling for arbitrary class cuts. The numerical order and the simplicity relation are distinct: extending a sign string makes it more complicated in simplicity, but may increase or decrease its numerical value [11, 12, 14].

Lemma 51.2 (Fresh separators). *If $L < R$ are sets of surreals and S is a set, there is $x \notin S$ with $L < x < R$. In particular, deleting a set from $\mathbf{No}$ leaves a dense order without endpoints.*

Proof. If every separator belonged to S , the class $I = \{x : L < x < R\}$ would be a set, by separation from S . It is nonempty. Apply set-cut filling to $L \cup I < R$: every member of I is less than every member of R . A resulting separator belongs to I and exceeds all of I , a contradiction. Thus the interval has a point outside S . The same proof allows empty bounding sides. $\square$

In GBC, a forth construction embeds every ordinary class linear order in $\mathbf{No}$: enumerate its underlying class by Ord , and at each stage separate the sets of earlier lower and upper images. A back-and-forth construction shows uniqueness, up to class order isomorphism, of a proper class order with this set-cut property. Each stage has a set of earlier values; these are ordinary set-valued ordinal recursions. This is the classical universality and absolute-linear-continuum phenomenon [15, 20]. It does not apply to a collection whose elements are themselves class functions.

51.2 Source 16: foundations and surreal order facts

51.2.1 Conventions about sets, classes, and choice

Here GB has ZF as its set theory and no choice principle built into its notation. We write $\text{GBC} = \text{GB} + \text{GC}$; global choice implies ordinary set choice. The set-sized theorems are proved in ZFC. A *class well-order* is a strict class linear order for which every available nonempty subclass has a least element. A relation is *set-like* if each predecessor class is a set. A proper set-like class well-order is uniquely isomorphic to Ord : send each point to the ordinal type of its predecessor set.

An individual class function $F : \text{Ord} \rightarrow \mathbf{No}$ is legitimate. The notation “ $F \in \mathfrak{W}_{\text{sl}}$ ” below abbreviates that this class function is a bijection; it does not make F an element of a GB class. Likewise, a uniform family $(F_i)_{i \in I}$ means one class of triples (i, α, x) whose rows are the graphs in question. Its index I may be a set or a class of sets. One cannot silently choose a class for each set index and assume that the rows assemble into an available uniform family.

51.2.2 Sign sequences, small cuts, and bounded stages

We use the sign presentation

$$\mathbf{No} = \bigcup_{\alpha \in \text{Ord}} \{-, +\}^\alpha,$$

with numerical comparison at the first different or missing sign, using $- < \text{missing} < +$. The birthday $\text{bd}(x)$ is the length of the sign sequence. Every set of surreals has bounded birthdays, whereas all birthdays occur. Thus $\mathbf{No}$ is proper. The basic separation property is:

$$A, B \text{ sets, } A < B \implies \exists x \in \mathbf{No} (A < x < B), \quad (\text{S16.1})$$

with either side allowed to be empty. These are standard facts about the surreal order [11, 12, 14]; the corresponding smallness-sensitive primitives in ProveIt are audited in [Section 59.4](#).

Lemma 51.3 (Fresh separators). *If $A < B$ are sets of surreals and S is any set, some $x \notin S$ satisfies $A < x < B$. Every nonempty numerical interval is a proper class. Deleting a set from **No** preserves the set-cut property.*

Proof. First choose u, v with $A < u < v < B$ by two applications of (S16.1). The interval (u, v) cannot be a set: if it were, choose an element above all of (u, v) but below v , a contradiction. Hence the set S cannot exhaust this interval. Any remaining point works. The same argument gives the assertions about intervals and deletion. $\square$

For infinite cardinal κ , put

$$\mathbf{No}_{<\kappa} = \{x : \text{bd}(x) < \kappa\}, \quad \mu = 2^{<\kappa}.$$

Then $\mathbf{No}_{<\kappa}$ is a set of size μ . For regular infinite κ it is an η_κ order: every separated pair of sets with total cardinality less than κ has an interpolant in the same birthday stage. Bound the option birthdays below κ using regularity, and use the usual birthday-controlled separator. At singular κ this argument fails because a small set can have birthdays cofinal in κ .

51.2.3 Which uses of recursion are elementary?

A recursion producing a set at each ordinal stage, with a set history before each stage and fixed class parameters, is available in GB. One constructs each set-length approximation by ordinary recursion and joins the compatible approximations by elementary comprehension. This applies to the forth and back-and-forth constructions used below.

By contrast, a recursion whose stages are classes is an instance of elementary transfinite recursion when its defining formula has only set quantifiers. Such a recursion may exceed GBC even along ω ; truth recursion is a standard example. ETR supplies abstract class well-order comparability by initial-segment isomorphisms, but we neither identify comparability with ETR nor assert a necessity result for our particular constructions [27, 22]. Size of the stage and history, rather than the mere appearance of the symbol **Ord**, is decisive.

Proposition 51.4 (Universality and uniqueness). *In GBC every class linear order embeds into **No**. Any two proper class linear orders satisfying (S16.1) are isomorphic, provided they have set-like global enumerations. The construction can be made uniform in set parameters when the carrier and order predicates are uniform.*

Proof. Enumerate the source by an ordinal or by **Ord** using global choice. At each stage insert its next point between the images of its earlier lower and upper points. These are sets. For an isomorphism alternate forth and back steps through fixed global enumerations of the two orders, inserting the next unused point on the appropriate side at each step. Each partial map is a set. Choose the least permissible point under fixed global well-orders to make every stage unique. Its graph is the union of the set-length approximations. With an extra set parameter carried throughout, the same defining formula gives a uniform class relation for the whole parameterized family. This is the classical mechanism of surreal universality and homogeneity [15, 33]. $\square$

51.3 Source 18: surreals as sign sequences

Represent a surreal number by a function $s_x : \text{bd}(x) \rightarrow \{-, +\}$, where $\text{bd}(x)$ is an ordinal. Numerical order is comparison at the first unequal sign, counting an absent sign as strictly between $-$ and $+$. Sign extension is the simplicity relation, which is a tree order, not a total order. This presentation and its substructures are developed in the standard surreal literature [12, 14].

We shall repeatedly use the following elementary consequences.

- Lemma 51.5** (Surreal order tools). *1. If L, R are sets of surreals and $L < R$, some surreal lies strictly between them. Either set may be empty.*
- 2. The separator can avoid any prescribed set of forbidden values.*
- 3. For infinite initial κ , every nonempty open interval of $\mathbf{No}_{<\kappa}$ contains a sign-prefix cone isomorphic to $\mathbf{No}_{<\kappa}$; hence it has cardinality $\mu = 2^{<\kappa}$.*
- 4. $\mathbf{No}_{<\kappa}$ fills every cut with fewer than $r = \text{cf } \kappa$ input values. Its cofinality, coinitality, and both characters at every point are r .*

Proof. For (1), choose an ordinal strictly above all birthdays in $L \cup R$. Comparison in a sufficiently long full binary level supplies a sign sequence separating the two sets; equivalently use the usual simplest surreal cut construction. To get (2), the separator interval contains a proper class of values: an interval is order-isomorphic to $\mathbf{No}$, or one can extend a sign prefix by arbitrary ordinal lengths. A set of forbidden values cannot exhaust it.

For (3), first choose a sufficiently long but still $< \kappa$ prefix whose whole cone lies inside the given interval. Prefix concatenation preserves numerical order. If $\beta, \gamma < \kappa$, then $\beta + \gamma < \kappa$, since this sum still has cardinality below the initial cardinal κ . This works at singular cardinals too. Thus concatenating the chosen prefix with every sign word of length $< \kappa$ gives the required copy. Counting levels gives $|\mathbf{No}_{<\kappa}| = \sup_{\alpha < \kappa} 2^{|\alpha|} = 2^{<\kappa}$.

For (4), fewer than $\text{cf } \kappa$ birthdays are bounded below κ , so the construction in (1) stays below κ . Ordinal labels along a cofinal sequence in κ give a cofinal sequence; their negatives give a coinital one. No smaller family is cofinal, since its birthdays are bounded. At a fixed point, append a negative sign and then arbitrarily long positive tails to approach from below; use the dual construction above. These give character at most r . The small-cut property gives the reverse inequality. $\square$

Part (2) is also available in the bounded setting when the forbidden set has size $< \mu$ and a whole interval of separators is available. This distinction matters: a used prefix in a minimal permutation can have length much greater than κ , but it still uses fewer than μ labels.

51.4 Source 18: set and class theories

Set-sized results are proved in ZFC. In class arguments, GB denotes Gödel–Bernays theory over ZF with elementary class comprehension and Class Replacement. GBC adds Global Choice. A class is a collection of sets, never an element of another ordinary class. We regard a class well-order R as a strict class linear order for which every nonempty available subclass has an R -least element. This definition is internal to the specified two-sorted model.

A class order is *set-like* if every predecessor class is a set. A proper set-like class well-order is isomorphic to Ord : map a point to the ordinal type of its predecessors. The range is an initial subclass of Ord , and if it were a set, Class Replacement would make the whole carrier a set. Conversely, an Ord enumeration is set-like.

All uses of class back-and-forth below choose one new set label at each ordinal stage. The preceding history at a stage is a set, and the next choice is determined using a fixed global well-order. This is ordinary set-valued recursion with class parameters. We do not infer that arbitrary class-valued recursions follow from GBC. Elementary transfinite recursion has substantially greater strength in general [24].

An expression such as $\mathfrak{M}_{\text{all}}$ is initially notation for a domain of class variables, not a class whose members are class relations. A uniform family $(R_i)_{i \in I}$ means one class relation $F(i, x, y)$ with set labels i, x, y . A set-indexed family of class relations is not silently replaced by a set of classes.

51.5 Source 17: foundations and direct comparison without a class of classes

51.5.1 Four safe levels

Our set theorems use ZFC. Our global theorems use GBC, meaning Gödel–Bernays set theory with Global Choice; GB alone includes no choice principle. This agrees with the conventions in the source and in Hamkins’s account of Global Choice [19].

A class well-order here is a linear class relation for which every nonempty admitted subclass has a least element. Set-likeness is an additional property, not part of that definition. Which subclasses are admitted is specified by the surrounding class theory or model.

There are four distinct safe ways to interpret a higher-order-looking statement. One may work with ordinary sets such as $\mathcal{W}_\mu(X)$; with an ordinary class of *set codes* such as $\mathcal{P}_s(b)$; with finitely many class variables or a uniform class relation coding a family; or externally in a larger universe in which the objects from the smaller universe are sets. None permits forming an unrestricted same-level set or class of all proper classes.

A uniform family indexed by a class I of sets is a single relation $F \subseteq I \times \text{Ord} \times \mathbf{No}$, with each row F_i an enumeration. A uniform transformation of enumerations is a formula which defines an output class from an input class and for which output existence and uniqueness are proved. It is not a GB function with proper classes as arguments stored in its graph.

51.5.2 The direct common-prefix formula

For a strict class well-order R let $P_R(x) = \{y : yRx\}$. Given two class well-orders R, S of the same labelled carrier X , define

$$C(R, S) = \{x : P_R(x) = P_S(x) \text{ and } R \upharpoonright P_R(x) = S \upharpoonright P_S(x)\}. \quad (\text{S17.2})$$

Equality and restriction here expand to quantifiers over elements of X ; no class quantifier is hidden in the defining formula.

Proposition 51.6 (Direct labelled comparison; background with proof). *The class $C(R, S)$ is the largest common labelled initial segment of R, S . If $R \neq S$, its complement is nonempty and the two least residual labels $a = \min_R(X \setminus C)$ and $b = \min_S(X \setminus C)$ are different. Declaring $R <_{\text{lex}} S$ exactly when $a < b$ gives a strict linear comparison of all admitted class well-orders of X .*

Proof. If $x \in C$ and yRx , the equality of predecessor classes puts y before x in both orders; agreement below x then gives equal predecessor classes and equal predecessor orders for y . Thus $y \in C$. The same argument works with S , and proves agreement on C . Every common labelled initial segment satisfies the defining condition, so is contained in C . If $C = X$, the relations agree. Otherwise well-foundedness provides the least residual elements. If $a = b$, both its predecessor classes are C and the orders there agree, contradicting $a \notin C$.

For transitivity compare $C(R, S)$ and $C(S, T)$ as initial segments of S . Initial segments of a linear order are comparable by inclusion. If one is properly shorter, its first disagreement is the first disagreement relevant to comparing R, T , and has the same orientation. If the two prefixes are equal, the three next labels $a < b < c$ give $a < c$. Trichotomy and asymmetry are immediate. This is an elementary proof; it does not compare arbitrary abstract class order types by a recursive isomorphism construction. $\square$

This result is already in the merged report (theorems 14.2 and 14.7). Its inclusion is important: it eliminates a false dilemma between restricting to Ord-enumerations and introducing a stronger class recursion theory merely to define the comparison.

51.5.3 Choice, recursion, and external semantics

Global Choice supplies a baseline bijection $b : \text{Ord} \rightarrow \mathbf{No}$. The source also proves the converse strength statement: existence of any class well-order of the sign-sequence surreals implies Global Choice over choiceless GB (theorem 13.1). That converse is not needed in our proofs; we explicitly assume GBC for the global results.

Recursion whose value at each ordinal stage is a set is available with class parameters in GBC: set recursion constructs compatible histories on each set ordinal, and elementary comprehension collects them. This is not the assertion that arbitrary class-valued recursions have solutions. The latter territory is governed by principles such as ETR, whose strength is substantially more than elementary comprehension [22]. We will reconstruct branches by a uniqueness formula, so that even the appearance of such a class-valued recursion is avoided.

In a transitive two-sorted model $(M, \mathcal{X})$, “all classes” means $\mathcal{X}$. An external observer may have more subclasses of M . With a strongly inaccessible cardinal λ in an ambient universe, $(V_\lambda, \mathcal{P}(V_\lambda))$ provides a useful full-class interpretation. Externally, its set-like surreal enumerations are the λ -length permutations of $\mathbf{No}_{<\lambda}$, while its unrestricted well-orders allow all external ordinal types of cardinality λ (theorem 21.1). This is a conditional model construction, not an assertion that ZFC proves an inaccessible cardinal exists.

Finally, neither passing to KM nor enumerating formula strings gives an internal universal class decoder. The latter would need a uniform truth predicate, not merely syntax. Section 42.5 gives an independent diagonal explanation for the failure of exhaustive same-level coding.

51.6 Source 13: direct comparison of unrestricted class well-orders

The class $\mathbf{No}$ itself is fixed throughout this section. A class well-order R is a strict linear order for which every nonempty subclass has an R -least member. A set-like class order additionally has a set of predecessors at every point. The totality $\mathfrak{W}_{\text{all}}$ is notation for the domain of such class parameters, not an ordinary class whose members are relations.

51.6.1 The largest common labeled prefix

The following construction is from the earlier report (theorem 14.2; Section 14 at the pin); its explicit formula is useful because it does not demand an enumeration of a non-set-like order by Ord.

Proposition 51.7 (Elementary labeled comparison). *Two distinct class well-orders R, S of $\mathbf{No}$ have a largest common labeled initial part G . There are distinct labels r, s that are the respective first points after G . Defining*

$$R <_{\text{lex}} S \iff r < s$$

gives a strict linear comparison on $\mathfrak{W}_{\text{all}}$ by an elementary formula with R, S as class parameters. No class-well-order comparability principle is required.

Proof. Define G to consist of those x such that $\text{Pred}_R(x) = \text{Pred}_S(x)$ and the two relations agree on that common predecessor class. Equality and agreement here abbreviate formulas quantifying over set elements. They do not form the predecessor classes as elements of a larger collection.

If $x \in G$, all its predecessors belong to G . The restrictions of R and S to G agree, and any common labeled initial part is contained in G . If G were the whole carrier, the relations would agree. Otherwise take the least points r, s of the complements in the two orders. Their predecessor classes are both G . They cannot coincide, since their common value would then be a member of G .

Trichotomy follows from numerical trichotomy of r, s . For transitivity, consider $R < S < T$. The common prefixes of (R, S) and (S, T) are both initial parts of S , hence one contains the other. If one is properly shorter, its first disagreement also decides the comparison of R, T . If they are equal, the three next labels satisfy $r < s < t$, which gives $R < T$. This proves transitivity without assigning an ordinal length to a proper-class prefix. $\square$

This is not comparison of abstract class order types by initial-segment isomorphisms. For that different problem, elementary transfinite recursion is a known sufficient principle [27, Theorem 6]. Our formula compares identical labels on one carrier, rather than constructing an isomorphism between unrelated carriers.

51.6.2 The ordinal head and the residual fiber

Proposition 51.8 (Canonical head; prior structural lemma). *For $R \in \mathfrak{W}_{\text{all}}$, the points with a set of R -predecessors form a proper-class initial part H_R , uniquely enumerated by a class injection*

$$F_R : \text{Ord} \longrightarrow \mathbf{No}.$$

Conversely, for every injective class word $F : \text{Ord} \rightarrow \mathbf{No}$, appending any class well-order of $Y = \mathbf{No} \setminus \text{ran}(F)$ produces a raw order whose head is F . All raw orders with this head form a convex fiber, with their internal comparison equal to the labeled lexicographic comparison of well-orders of Y .

Proof. H_R is initial. Were it a set, the least point outside it would have exactly H_R as its predecessor class and would belong to it, a contradiction. Ranking each point by the ordinal type of its set of predecessors gives the unique enumeration: its ranks form a proper initial subclass of Ord only if they are a set, so they are all of Ord .

After an Ord -head is listed, every residual point has a proper class of predecessors, whereas each head point has a set of predecessors. This proves the converse. Distinct heads disagree at a set ordinal, and that disagreement places one entire fiber below the other. Equal heads leave only the tail comparison, proving convexity. A well-order on Y is available from global choice, also when Y is proper. $\square$

This is the earlier report's decomposition (theorem 15.1 and proposition 15.3; Section 15 at the pin). We shall identify exactly which cuts of a fixed dense core these fibers represent, thereby supplying an additional characterization not contained in the decomposition alone.

51.7 Source 15: native lexicography on class well-orders

51.7.1 Existence, set-likeness, and the first proper block

A class relation R on $\mathbf{No}$ is *set-like* when $\text{Pred}_R(x) = \{y : yRx\}$ is a set for every x . An already-given set-like class well-order of a proper class has a unique order isomorphism with Ord : assign to each point the ordinal type of its predecessor set. The image is initial in Ord and cannot be a set, since the inverse map would make the whole carrier a set. This collapse needs GB, not a separate global choice assumption [19].

Existence is different. Over GB, the existence of any class well-order of $\mathbf{No}$, even a non-set-like one, is equivalent to global choice (theorem 13.1; Theorem 13.1 at the pin). One direction restricts a set-like global well-order of the universe. For the other, a well-order of $\mathbf{No}$ orders the sign codes for every $\mathcal{P}(\alpha)$. Starting with an ordinal enumeration of a rank stage, transport this order to its power

set; at limits combine the already constructed orders by rank. This set-valued ordinal recursion produces well-orders of all rank stages and hence a global well-order. This explains why merely assuming a convenient global baseline already carries substantive choice information.

An arbitrary class well-order can have type $\text{Ord} + 1$, $\text{Ord} + \text{Ord}$, or much more complicated class types. The final element of an $\text{Ord} + 1$ order has a proper class of predecessors. It is still an ordinary surreal label; the new size occurs in its position in the chosen order.

51.7.2 A first disagreement after a proper class of agreement

If R, S are class well-orders of the same labelled carrier X , put $\text{Pred}_R(x) = \{y \in X : yRx\}$. A common labelled prefix is initial in both orders and carries the same restricted relation. The comparison below is the prior report's elementary construction ([theorem 14.2](#); Section 14 at the pin); we include its proof because it makes the foundational advantage precise.

Theorem 51.9 (First unequal labels without class ordinals). *If $R \neq S$, there are unique distinct a, b such that*

$$\text{Pred}_R(a) = \text{Pred}_S(b) = C, \quad R \restriction C = S \restriction C.$$

The class C is their maximal common labelled prefix. If the labels carry a fixed linear order $<$, declaring $R <_{\text{lex}} S$ when $a < b$ defines a strict linear comparison on the class-well-order predicate. Its formula has only set quantifiers with R, S as class parameters.

Proof. Let a be the R -least point whose predecessor classes in R and S differ, and put $C = \text{Pred}_R(a)$. For $y \in C$, these predecessor classes agree. Such y must also precede a in S : otherwise aSy would imply aRy . If zSy for $y \in C$, agreement of predecessor classes gives zRy , so C is also S -initial. The restricted orders agree, and $C \subsetneq \text{Pred}_S(a)$. Let b be the S -least point outside C . Then bSa , $b \neq a$, and $\text{Pred}_S(b) = C$. Any other first-disagreement pair must have its first member equal to the R -least point with changed predecessors, proving uniqueness.

For transitivity, suppose $R < S < T$. Their two common prefixes with the middle order S are initial segments of that same order, hence comparable. If one is shorter, its first difference also decides R versus T ; if they are equal, the next labels satisfy $a < b < c$, which decides the comparison. Totality and asymmetry follow from unique first disagreement. Finally the formula can be written directly as

$$\exists a, b [a < b \wedge \forall z(zRa \leftrightarrow zSb) \wedge \forall u, v((uRa \wedge vRa) \Rightarrow (uRv \leftrightarrow uSv))].$$

No quantification over a collection of prefix classes is required. □

For example, enumerate $\mathbf{No} \setminus \{a, b\}$ in type Ord , where $a < b$, and append either a, b or b, a . Their first unequal labels occur after every ordinal stage of the common enumeration. The displayed formula still compares them. It is not necessary to invent a set ordinal “beyond Ord ” to name their first difference.

For one given class index order J , bijections $J \rightarrow \mathbf{No}$ may instead be compared at the J -least unequal coordinate. This agrees with the raw comparison, but restricts the allowed order type. Two different index orders should be compared through the induced labelled relations unless an explicit alignment has been supplied.

Proposition 51.10 (Prefix convexity and initial-word fibres). *In GBC, orders extending a common labelled prefix form a convex family. For an arbitrary class well-order R of $\mathbf{No}$, let*

$$I_R = \{x : \text{Pred}_R(x) \text{ is a set}\}, \quad F_R : \text{Ord} \longrightarrow I_R$$

be its unique increasing enumeration. Then I_R is a proper class. Every injective class word $F : \text{Ord} \rightarrow \mathbf{No}$ occurs as F_R , and its fibre is a convex copy of the well-order space of $Y_F = \mathbf{No} \setminus \text{ran}(F)$. Distinct fibres compare at the first unequal coordinate of their words.

Proof. A comparison that first diverges inside a common prefix puts the competing order on the same side of every extension, proving convexity. The class I_R is initial and set-like. If it were a set, the least point outside it would have precisely I_R as its predecessors, a contradiction. Its enumeration is therefore by all ordinals.

Given F , order its range by its ordinal indices and append any fixed well-order of Y_F . Every point of the range has set predecessors, and every residual point has the proper range of F among its predecessors. Thus the resulting initial word is exactly F . Restriction to Y_F and concatenation with F give inverse order-preserving maps on the fibre. For unequal words, first ordinal difference precedes both residual tails. $\square$

If F omits m labels, its fibre has $m!$ elements. An exhaustive word and a word omitting just one label both have singleton fibres; only the first is set-like. The unrestricted space has adjacent finite residual permutation blocks, whereas the set-like space is dense and has no endpoints. These distinctions are already established in the prior report.

51.8 Source 16: global choice and direct comparison of arbitrary class well-orders

Write $\mathfrak{W}_{\text{sl}}$ for the predicate of class bijections $\text{Ord} \rightarrow \mathbf{No}$, and $\mathfrak{W}_{\text{all}}$ for that of all class well-order relations on $\mathbf{No}$. These are notations for assertions about class variables. We will also use $\mathfrak{J}$ for class injections $\text{Ord} \rightarrow \mathbf{No}$, allowing nonexhaustive ranges.

Proposition 51.11 (Existence has a precise choice strength). *Over GB, the following are equivalent: global choice; a class bijection $\text{Ord} \rightarrow \mathbf{No}$; a set-like class well-order of $\mathbf{No}$; and an arbitrary class well-order of $\mathbf{No}$.*

Proof. Global choice gives a set-like global well-order of the universe. Its restriction to the proper class $\mathbf{No}$ has order type Ord , yielding the bijection. A bijection induces a set-like well-order, which is an arbitrary class well-order as well.

For the reverse implication suppose R well-orders $\mathbf{No}$. Every $\mathcal{P}(\delta)$ injects into $\mathbf{No}$ by sign sequences of length δ , so R gives it a well-order. One can now well-order the hierarchy V_β uniformly by set-valued recursion, starting with $w_0 = \emptyset$. At a successor, use the existing well-order w_β of V_β to identify it with an ordinal δ , transfer subsets to $\mathcal{P}(\delta)$, and use the restriction of R on their sign codes. At a nonzero limit λ , order the disjoint rank layers by their ranks, using $w_{\gamma+1} \upharpoonright (V_{\gamma+1} \setminus V_\gamma)$ on layer γ , where $\gamma + 1 < \lambda$. Each history is a set; no choice of an unspecified family of well-orders is used. Finally order all sets first by rank and then by the resulting well-order on that rank layer. This is a set-like global well-order of the universe and supplies global choice. This expands the choice-equivalence argument recorded in the prior report ([theorem 13.1](#)). $\square$

For distinct $F, G \in \mathfrak{J}$, their first unequal ordinal coordinate exists: minimize the disagreement class below any one witness. Compare $F(\delta)$ and $G(\delta)$ numerically. This is a total, transitive lexicographic comparison. For general $R, S \in \mathfrak{W}_{\text{all}}$, however, a first disagreement may occur after a proper class of common labels.

Lemma 51.12 (A common labelled prefix without class ordinals). *Let $R \neq S$ be class well-orders of the same class X . There are unique distinct labels a, b such that*

$$\{x : xRa\} = \{x : xSb\} = C, \quad R \upharpoonright C = S \upharpoonright C.$$

The class C is their largest common labelled initial segment.

Proof. Let a be R -least among the labels whose predecessor classes in R and S differ. For every xRa the two predecessor classes of x agree. Consequently $C = \{x : xRa\}$ is also S -initial and the relations agree on it. Moreover C is a proper subclass of the S -predecessors of a . Let b be the S -least label outside C . Then bSa , $b \neq a$, and its S -predecessor class is C . In any pair satisfying the displayed conditions, every earlier label has identical predecessor classes, while the next R -label does not. It must therefore be a , and then both C and b are determined. $\square$

Define $R <_{\text{lex}} S$ if the corresponding next labels satisfy $a < b$ in the numerical or other fixed native order of X . In a triple R, S, T , the common prefixes of (R, S) and (S, T) are initial segments of S and are inclusion-comparable. Whichever ends first decides the comparison of R and T ; if they end together, transitivity of the next numerical labels decides. Thus the definition is a strict linear comparison.

The first-pair condition can be expressed using only set quantifiers:

$$\begin{aligned} a \in X \wedge b \in X \wedge a \neq b \quad \wedge \quad & \forall z (zRa \leftrightarrow zSb) \\ & \wedge \quad \forall x \forall y ((xRa \wedge yRa) \Rightarrow (xRy \leftrightarrow xSy)). \end{aligned} \tag{S16.9}$$

This is why no abstract comparison of class order types and no ETR is required. On an injectively indexed uniform family of class well-orders, it gives an ordinary class linear order on the set-valued indices.

Proposition 51.13 (Convexity and finite residual jumps). *Fix any labelled common prefix, possibly a proper class. Its completions form a convex suborder of $\mathfrak{W}_{\text{all}}$. Two distinct raw orders are adjacent exactly when the labels remaining after their largest common prefix form a finite set and their residual permutations are consecutive.*

Proof. An order that disagrees before a fixed common prefix ends compares in the same direction with all its completions, proving convexity. If $R < S$ were adjacent and the remaining set or class of labels were infinite, R would have to be maximal in its next-letter cylinder and S minimal in its next-letter cylinder. As in Lemma 27.1, swapping an ascent or descent forces the respective suffixes to be numerically decreasing and increasing. If Y is the residual carrier and a, b the first unequal labels, the numerical order on $Y \setminus \{a, b\}$ is consequently both a well-order and a reverse well-order. Such a class is finite: otherwise repeatedly taking its least remaining member produces a set-coded increasing ω -sequence, contradicting reverse well-foundedness. Thus Y is finite, and convexity reduces adjacency exactly to that of its finite permutation list. Conversely consecutive residual permutations have no intervening completion, and convexity excludes an intervening order outside the prefix cylinder. $\square$

51.9 Source 18: global well-orders: existence and elementary comparison

51.9.1 Existence is exactly a global choice issue

No set-length function exhausts **No**, since its range is a set. A well-order of the whole surreal carrier must therefore be a class relation. It need not be set-like: $\text{Ord} + 1$, $\text{Ord} \cdot 2$, and more complicated class concatenations can be transported to the carrier under Global Choice. Here these expressions denote explicitly constructed class orders, not set ordinals.

Theorem 51.14 (Existence of a global well-order; repository result). *Over GB with ZF as its set theory, the following are equivalent: Global Choice; the existence of a set-like global well-order of **No**; the existence of any class well-order of **No**.*

Proof. Global Choice gives a set-like global well-order of the set universe and hence of **No**. The second statement implies the third. For the reverse implication, let R be a class well-order of the sign-sequence surreals. For every ordinal θ , the length- θ binary sign sequences form a set, and the restriction of R well-orders that set. It thereby well-orders $\mathcal{P}(\theta)$, uniformly in θ .

Construct a well-order of each V_α by set-valued recursion with parameter R . Start with the empty set. Given a well-order of V_α , use its unique ordinal enumeration to identify subsets of V_α with subsets of an ordinal; pull back the already specified well-order of that power set. At a limit stage, order the disjoint rank layers first by their ordinal rank and then by the relevant previously constructed well-order. These orders need not extend one another; uniform existence is what the argument requires. Finally order the whole universe first by rank and then by the chosen well-order of each rank layer. Its predecessor classes are sets, so it is a set-like global well-order and supplies Global Choice. This is the choiceless argument used in the repository ([theorem 13.1](#)). $\square$

The theorem is distinct from asking whether universality of the *numerical* surreal order for all class linear orders implies Global Choice. Our arguments use Global Choice for that universality; they do not establish its converse. Under Global Choice, the usual forth construction embeds a class linear order in **No**: enumerate its carrier in type Ord, and at each stage place the next point between the already embedded lower and upper sets. Hamkins gives a research-paper account of this class universality [[33](#), §5].

51.9.2 Comparing orders after a proper-class common prefix

Let R, S be class well-orders on the same linearly ordered class X . The numerical order on X need not coincide with either well-order. Write $\text{Pred}_R(x) = \{y : yRx\}$. Define $C(R, S)$ to consist of the labels x such that

$$\text{Pred}_R(x) = \text{Pred}_S(x) \quad \text{and} \quad R \upharpoonright \text{Pred}_R(x) = S \upharpoonright \text{Pred}_R(x). \quad (\text{S18.4})$$

Both equalities expand to formulas quantifying only over set labels, with the two relations as class parameters.

Lemma 51.15 (Largest common labelled initial segment). *$C(R, S)$ is an initial segment of both orders, on which they agree. It contains every common labelled initial segment. If $R \neq S$, it is proper, and the first remaining labels a in R and b in S satisfy*

$$\text{Pred}_R(a) = C(R, S) = \text{Pred}_S(b), \quad a \neq b.$$

Proof. If x satisfies (S18.4), each of its predecessors does too: the two orders agree below x , and therefore assign identical predecessor classes below any such label. Thus C is initial in both, and agreement follows by testing a pair below its larger label. Every common labelled initial segment satisfies the displayed conditions. If $C = X$, the two relations agree everywhere. Otherwise their well-order property supplies least labels outside C . Their predecessor classes are exactly C . If those labels were equal, that label would satisfy (S18.4), a contradiction. $\square$

Theorem 51.16 (Elementary full lexicographic order). *For distinct R, S , declare $R <_{\text{lex}} S$ if $a < b$ in the numerical order, with a, b as in [Lemma 51.15](#). This is a strict linear order on the available well-orders of X . It is elementary in the two class parameters. It requires neither a system of abstract class ordinals nor a principle comparing arbitrary class order types.*

Proof. Trichotomy and asymmetry are immediate. For transitivity, compare the common prefixes for R, S and for S, T . They are initial segments of the same order S , hence nested. If the first prefix ends earlier, S and T still agree at its next label, so the comparison of R, T is the comparison of R, S . The opposite strict containment gives the comparison of S, T . If the prefixes agree, the three next labels are numerically ordered and transitivity follows in X .

For explicit elementary definability, use the formula defining C , assert that the predecessors of a under R and of b under S are exactly the labels satisfying that formula, and assert $a < b$. No existential quantification over an additional class is needed. $\square$

The theorem extends the usual first-ordinal-difference rule. In the case of distinct set-like orders, C is a set and has an ordinal length. In the full case, it may already have type Ord , $\text{Ord} \cdot 2$, or a more complicated type. The comparison still exists because the two orders use the same labels. Abstractly isomorphic but differently labelled prefixes are not declared equal.

51.9.3 Residual cylinders and finite adjacency

Fix a labelled well-order Q of $C \subsetneq X$. Its cylinder is the family of all global orders beginning with Q . Put $Y = X \setminus C$. Restriction to Y and concatenation with Q are inverse, order-preserving operations. The cylinder is convex: any order that diverges before finishing Q lies on the same side of every extension of Q .

This gives the following useful size-sensitive reading of a cylinder. If $|Y| = n < \omega$, its tails form the $n!$ permutations of Y . If Y is an infinite set of size λ , its tails form $\mathcal{W}(Y)$, of cardinality 2^λ and ordinal spectrum $\beta < \lambda^+$. If Y is proper, its tails form a predicate on class relations. Under GBC this last family contains a uniformly presented copy of every class linear order. Indeed, characteristic functions of initial segments embed any class linear order into binary Ord words; independent pair swaps in a fixed enumeration of Y then embed those words into set-like well-orders of Y .

Even when Y is a set, the corresponding full orders of X are class relations. The statement is a set-parameterized isomorphism with their tails, not the assertion that those class relations themselves form an ordinary set.

Theorem 51.17 (Adjacency is finite residual adjacency). *Let $R < S$, with common prefix C , next labels $a < b$, and residual carrier $Y = X \setminus C$. Then R, S are adjacent if and only if Y is finite and their residual permutations are consecutive.*

Proof. If R, S are adjacent, no remaining numerical label lies strictly between a, b , or it could be put next to produce an intermediate order. The tail of R after a must be lexicographically greatest: otherwise a greater replacement remains below S . A greatest well-order tail must list its labels in decreasing numerical order, since any numerical ascent can be transposed to increase the code. Dually, the tail of S after b must be numerically increasing. On $Y \setminus \{a, b\}$, numerical order and its reverse would therefore both be well-orders. Such a carrier is finite: otherwise successively taking the least remaining element produces an increasing ω sequence with no greatest member, contradicting reverse well-ordering. Conversely, convexity of the cylinder reduces a finite residual carrier exactly to finite permutation adjacency. $\square$

For example, choose $a < b$ in $\mathbf{No}$ and give $\mathbf{No} \setminus \{a, b\}$ a set-like order Q . The two orders $Q + (a, b)$ and $Q + (b, a)$ are adjacent and have type $\text{Ord} + 2$. Neither is set-like. The set-like global order has no adjacent pairs, because a set prefix leaves an entire proper class of unused numerical separators. It is consequently not dense in the full global order: intervals inside a six-point terminal permutation fiber contain only non-set-like orders.

52 The canonical block decomposition (source 15; routes of sources 16 and 18)

Note 52.1 ([merge]; one theorem, three proofs). Three batch-81 sources decompose an arbitrary class well-order at once by the same relation: $x \sim y$ exactly when the closed interval between x and y is a set. Each proves that $\sim$ is an elementary convex equivalence, that every block is set-like of type Ord or a set ordinal, that at most one block is a set and it is final, and that the whole order is canonically the lexicographic sum of Ord over the class of proper-block representatives, followed by the optional final set block; each realizes every nonempty class well-order as a block-index order in GBC. The statements are source 15's [theorems 52.3](#) and [52.5](#) and [corollary 52.4](#), source 16's [theorems 52.6](#) and [52.8](#) with (S16.11), and source 18's [lemma 52.10](#), [theorem 52.11](#) with (S18.3), and [theorem 52.14](#). Source 15 is printed first because it also writes out the coordinate map $x \mapsto (\pi_W(x), r_W(x))$ and the realization of every block profile; the other two follow as routes, whole, because each adds statements of its own: source 16 the largest set final segment ([corollary 52.7](#)) and the uniform family of residual tails; source 18 the functoriality of the decomposition ([proposition 52.12](#)), the cofinality of R read off its blocks ([corollary 52.13](#)), uniform removal of the first α blocks by one closed definition, and a class well-order isomorphic to its own block quotient ([proposition 52.15](#)). The first block is source 11's canonical initial part ([theorem 15.1](#)), as all three say. The sources do not cite one another (source 18 was written before sources 15 and 16 arrived), and no source has priority. All three state that the decomposition is elementary in GB, without global choice or ETR, and that GBC is used only to transport an abstract order to **No**. The block decomposition answers the block-index clause of [research question 61.3](#) ([section 60](#)).

52.1 Source 15: canonical condensation into proper-class blocks

The preceding extraction of the initial set-like part can be extended to *all* blocks at once. This resolves an ambiguity in the question of how much recursion is required: constructing the canonical first-level decomposition is elementary; enumerating its blocks by a prescribed index order, or iterating the decomposition on an arbitrary input, is a different request.

Interval condensation is a standard construction in order theory [34]. Here the size ideal consists of sets, not of countable subsets. Thus the construction belongs to class theory and depends on which collections are sets in the underlying model. Its comparison with familiar countable-interval condensation is structural, not an identification of the two notions.

Work first in GB, with no choice assumption, and let (X, W) be a class well-order. We use the formulation that every nonempty available subclass of X has a least element. Put

$$[x, y]_W = \{z \in X : x \leq_W z \leq_W y\} \quad (x \leq_W y).$$

Definition 52.2 (Condensation by set intervals). For $x, y \in X$, write $x \sim_W y$ when the closed interval between them is a set. Let

$$B_W = \{b \in X : \text{there is no } aWb \text{ with } a \sim_W b\}.$$

The *condensation* of W is the induced order on B_W .

This definition is elementary. For example, “ $[x, y]_W$ is a set” can be written

$$\exists u \forall z (z \in u \longleftrightarrow (z \in X \wedge x \leq_W z \wedge z \leq_W y)),$$

where u, z are set variables and X, W are class parameters. Neither the equivalence classes nor a collection of those classes is being used as an element of another class. Their least representatives are sets and form the legitimate class B_W .

Theorem 52.3 (Elementary canonical block decomposition). *In GB, the following hold.*

1. *The relation $\sim_W$ is a convex equivalence relation. Every equivalence class has a least element, and the class function $\pi_W : X \rightarrow B_W$ sending an element to that least representative is elementary and unique.*
2. *The order (B_W, W) is a class well-order. Each fiber $C_b = \pi_W^{-1}(b)$ is set-like in its induced order. It therefore has the order type of a nonzero set ordinal if it is a set, and has order type Ord if it is a proper class.*
3. *Every nonfinal fiber is a proper class. Consequently at most one fiber is a set, and such a fiber must be final.*
4. *There is a uniform class order isomorphism*

$$x \mapsto (\pi_W(x), r_W(x)), \quad r_W(x) = \text{otp}([\pi_W(x), x)_W),$$

onto a lexicographically ordered subclass of $B_W \times \text{Ord}$. Every nonfinal vertical section is all of Ord ; a final section is either all of Ord or a nonzero set ordinal.

No elementary-transfinite-recursion principle is required.

Proof. Reflexivity and symmetry are immediate. In a linear order the interval between x and z is contained in the union of the intervals between x, y and between y, z . A subclass of a set is a set by Separation with class parameters; hence $x \sim_W y$ and $y \sim_W z$ imply $x \sim_W z$. If $x \leq_W z \leq_W y$ and $x \sim_W y$, the intervals from x to z and from z to y are subsets of $[x, y]_W$, proving convexity.

For any x , its equivalence class is nonempty and elementary in the parameters, so it has a W -least element b . Such b belongs to B_W , and conversely every member of B_W is the least member of its equivalence class. The graph of π_W is therefore given by one elementary formula. Every nonempty available subclass of B_W is also a subclass of X , and its W -least member belongs to it. This proves the quotient's well-order property without any theorem comparing abstract class well-orders.

Fix $b \in B_W$ and $x \in C_b$. The predecessors of x in C_b are exactly $[b, x)_W$, a set. Thus C_b is set-like. Its set-sized predecessor ranks form an initial subclass of Ord . If this initial subclass were a set ordinal while C_b were proper, the inverse rank map and class Replacement would make C_b a set. Therefore the rank range is all of Ord precisely in the proper-class case. In the set case it is the ordinary ordinal order type of C_b .

Suppose now that C_b is a set and is not final. Let d be the least point after this fiber. More explicitly, choose the least member of the nonempty class

$$\{z \in X : bWz \text{ and } z \not\sim_W b\}.$$

Convexity and minimality give $[b, d)_W = C_b$. Hence $[b, d]_W = C_b \cup \{d\}$ is a set, contradicting $d \not\sim_W b$. This proves the final-fiber assertion.

For each x , the set well-order $[\pi_W(x), x)_W$ has a unique ordinal order type. Its order-isomorphism witnesses are sets, so r_W is defined by elementary comprehension. Elements in different fibers are compared by their representatives; elements in one fiber are compared by their ranks. The displayed map is therefore a class order isomorphism. Its image is defined directly as the class of pairs that occur, so no symbol denoting a “class ordinal” has to be treated as a set-valued section length. $\square$

The first fiber is exactly the familiar initial part consisting of elements with set-sized predecessor classes. If X is proper, the following equivalence is particularly useful:

$$W \text{ is set-like} \iff |B_W| = 1.$$

This is an invariant of the individual class well-order. It does not claim that set-likeness is definable in the *bare lexicographic order of all labeled well-orders*; that reconstruction question remains separate.

Corollary 52.4 (Allowed block profiles). *Every proper-class well-order is canonically a sum of Ord-blocks indexed by a class well-order, with the sole possible exception of a final block of nonzero set-ordinal type. The index order need not be set-like, or even have type Ord.*

The word “canonically” refers to the least-representative quotient and the rank map above. It does not assert that arbitrary class index orders have canonical set codes or admit an ordinal classification inside GB.

Theorem 52.5 (Every class well-order occurs as a block index). *Assume GBC. Let J be any nonempty class well-order, with a set or proper-class carrier.*

1. *There is a class well-order of **No** whose canonical condensation is order-isomorphic to J and whose every block has type Ord.*
2. *For every nonzero set ordinal α , there is one whose block index is $J + 1$, whose blocks indexed by J have type Ord, and whose final block has type α .*

More generally these realize every profile allowed by Theorem 52.3, subject only to the whole carrier being a proper class.

Proof. Order $J \times \text{Ord}$ lexicographically, comparing the J coordinate first. This is a class well-order: a nonempty subclass has a least first coordinate and then a least ordinal coordinate. It is proper because J is nonempty. The interval between two points of the same row is a set. Between points of different rows lies a final segment of the earlier Ord-row, which is proper. Consequently the rows are exactly the equivalence classes, with representatives $(j, 0)$.

Global Choice supplies a class bijection between this proper carrier and **No**. Transfer the relation along it. A class bijection takes sets to sets by class Replacement, and the same is true of its inverse; it therefore preserves the set/proper-class distinction used in the condensation definition. This proves the first assertion.

For the second, append the set ordinal α to $J \times_{\text{lex}} \text{Ord}$. This final tail is one new equivalence class. It does not merge with any preceding row because an interval crossing into it contains a proper final segment of an Ord-row. Transfer again to **No**.

For a desired profile with final set block, remove the final index from its proposed index order and apply the second construction. The remaining order must be nonempty: otherwise the whole carrier would be a set. Profiles with no set block use the first construction. $\square$

For instance, the formal class-order expression $\text{Ord} \cdot (\text{Ord} + 1)$ has an $\text{Ord} + 1$ block index. Its final block occurs after a proper class of complete Ord-blocks. No procedure that lists only set-ordinally many initial blocks reaches it. Here the product denotes the explicitly described ordered sum; it is not an ordinal object larger than every ordinal.

52.2 Source 16: simultaneous canonical decomposition into ordinal-class blocks

The earlier report extracts the first set-like part of a class well-order and asks how to organize the successive residual parts. A direct interval definition answers this question without carrying out a class-valued recursion. In this section GB suffices; assume that the given relation already is a class well-order in the sense of Section 51.2.

For $x, y \in X$, let $[x, y]_R$ mean the closed interval between them, regardless of which comes first. Put

$$x \sim_R y \iff [x, y]_R \text{ is a set.} \quad (\text{S16.10})$$

Formally the right side is $\exists s \forall z (z \in s \leftrightarrow z \in [x, y]_R)$, with s a set variable.

Theorem 52.6 (Canonical block decomposition). *The relation $\sim_R$ is a convex equivalence relation. Every block has a least member and is set-like in its inherited order. Each block is therefore either a set well-order of ordinal type or a proper class well-order of type Ord. There is at most one set block, and it is terminal.*

Let J_R be the class of least representatives of the proper blocks, ordered by R , and let α_R be the ordinal type of the terminal set block, or zero when it is absent. There is a canonical class order isomorphism

$$(X, R) \cong \left(\sum_{j \in (J_R, R)} \text{Ord} \right) + \alpha_R. \quad (\text{S16.11})$$

All the data and the isomorphism are definable by elementary comprehension. The sum is the ordinary class $J_R \times \text{Ord}$ ordered lexicographically, not a class whose members are classes.

Proof. The interval between x and z is contained in the union of the intervals between x, y and between y, z . Finite union and separation therefore prove transitivity; the other equivalence properties and convexity follow immediately.

Each block is an available nonempty subclass and has an R -least point $m(x)$. The formula saying that $j \sim_R x$ and no R -earlier point is equivalent to x defines m elementarily. Thus representatives form a subclass of X , avoiding a quotient carrier made of proper classes. If $j = m(x)$, the predecessors of x within its block are exactly the set $[j, x)_R$. Define

$$\rho_R(x) = \text{otp}([j, x)_R, R).$$

This uses only a set well-order and its unique ordinal type. The range of ρ_R on the block is an initial subclass of Ord. If that range were a set, Class Replacement applied to its inverse would make the block a set. Hence a proper block has rank range all of Ord; a set block has an ordinal rank range.

Suppose a set block B were not terminal. Let c be the least point strictly above all of B . There is nothing between B and c . For $b \in B$, the interval $[b, c]_R$ is contained in $B \cup \{c\}$, a set. This contradicts maximality of the equivalence block. Thus a set block is terminal, and there is at most one.

On proper blocks, $x \mapsto (m(x), \rho_R(x))$ is the desired isomorphism; on the optional final block use its ordinal rank. All assertions defining these graphs quantify over sets with R as a fixed class parameter, so elementary comprehension suffices. $\square$

52.2.1 Every residual tail is now available uniformly

For $j \in J_R$ put

$$T_j = \{x : j \leq_R x\}, \quad B_j = \{x : m(x) = j\}.$$

The family of tails is one class relation of pairs (j, x) . Within T_j , a point x has a set of predecessors exactly when $[j, x)_R$ is a set, equivalently when $x \in B_j$. Hence B_j is precisely the canonical initial set-like part of T_j . The tail T_j is what remains after all earlier blocks have been deleted. This supplies the whole successive residual family even when J_R is non-set-like.

Corollary 52.7 (Largest set final segment). *The class*

$$A_R = \{x \in X : \{y : x \leq_R y\} \text{ is a set}\}$$

is always a set. It is the largest set-sized final segment of R , and is exactly its optional terminal block.

Proof. If A_R is nonempty, let a be its least point. Every later point has a tail contained in the set tail above a , so A_R is precisely that tail, a set. Every set final segment is contained in A_R by definition. Conversely, a block containing a point of A_R is the union of a set of block predecessors and a subset of its set tail, and so is a set block. [Theorem 52.6](#) identifies it. $\square$

Deleting A_R leaves a labelled prefix consisting of all the proper blocks and having no nonempty set-sized final segment. If that prefix is fixed, its completions form a convex copy of $\mathcal{W}(A_R)$. Appending any well-order on A_R leaves it the largest set final segment, since a tail meeting an earlier Ord block would already be proper. This refines the earlier first-block decomposition: after every proper block is removed, the residual carrier is always a set.

Theorem 52.8 (All block-index orders occur). *Every nonempty class well-order J , whether its carrier is a set or a proper class, occurs as the index of the proper blocks of a class well-order. An arbitrary set ordinal can be prescribed as its final block. In GBC the resulting order can be carried by surreal labels.*

Proof. Order $J \times \text{Ord}$ with the J coordinate primary. A nonempty subclass has a least occupied J coordinate and then a least ordinal in that section, so this is a class well-order. Intervals inside one section are sets; an interval crossing two sections contains a proper tail of Ord. Its blocks are exactly the sections. Appending a set ordinal gives the stipulated final block. The product is proper, and global choice supplies a class bijection of its carrier with **No**, along which the relation can be transported. $\square$

If J_R is a set of type γ , formula (S16.11) means $\text{Ord} \cdot \gamma + \alpha_R$. If J_R is proper and set-like, it means $\text{Ord} \cdot \text{Ord} + \alpha_R$. The example $\text{Ord} \cdot (\text{Ord} + 1) + \alpha$ has block index $\text{Ord} + 1$: an Ord-indexed list of its initial blocks would miss its last full block, whereas (S16.10) finds it directly.

52.2.2 What this does and does not require from ETR

The construction resolves successive residual extraction for one original order. Repeatedly condensing the *quotient order itself* is another operation. If $C_R(D)$ denotes the least representatives of set-interval blocks in the induced order on $D \subseteq X$, a tower such as

$$D_0 = X, \quad D_{\beta+1} = C_R(D_\beta), \quad D_\lambda = \bigcap_{\beta < \lambda} D_\beta$$

is an elementary class-valued recursion. Each specified finite iteration is elementary; appropriate ETR supplies a uniform transfinite tower. No necessity or exact reversal is established here. Also, since the block-index order can be an arbitrary class well-order, [Theorem 52.6](#) does not solve abstract class well-order comparability or give a universal class ordinal.

52.3 Source 18: a simultaneous decomposition into ordinal-class blocks

The repository asks how much class recursion is needed to iterate the extraction of the initial set-like part of a class well-order. A direct definition organizes all of these blocks simultaneously. It avoids a recursion whose successive states are proper classes.

Work in GB and let R be a class well-order on a proper class X . Here “class well-order” means that every nonempty available subclass has an R -least element. For $x \leq_R y$, write

$$[x, y]_R = \{z \in X : x \leq_R z \leq_R y\}.$$

The assertion that this interval is a set is the elementary formula

$$\exists a \forall z [z \in a \longleftrightarrow (z \in X \wedge x \leq_R z \leq_R y)].$$

In particular, no quantification over candidate classes is needed.

Definition 52.9 (Set-interval equivalence). Put $x \sim_R y$ when the closed R -interval with endpoints x, y is a set. Let $m_R(x)$ be the R -least element of the equivalence class of x . A block is represented by this least label, never by placing a proper class inside another class.

Lemma 52.10 (Block structure). *The relation $\sim_R$ is an elementary, convex equivalence relation. Every one of its blocks is set-like in its induced order. Every nonfinal block is a proper class. Consequently there is at most one set-sized block, and if present it is final.*

Proof. Reflexivity and symmetry are immediate. For transitivity, the interval between two of three points is contained in the union of the intervals from those points to the third; separation and the union of two sets give the result. Convexity follows by taking subsets of an interval.

Fix a block C . Its least element b exists by the assumed class well-ordering. If $x \in C$, its predecessors within C form the set $[b, x]_R \setminus \{x\}$. Thus the block is set-like. Suppose now that C is a set and is not final. There is a least point d after all of C . Convexity and this minimality imply $[b, d]_R = C \cup \{d\}$. This is a set, so $d \sim_R b$, contradicting $d \notin C$. The final assertion follows because two distinct final blocks are impossible. $\square$

Let B_R be the class of least labels of the proper-class blocks, ordered by R . If there is a final set block, let θ_R be its ordinal order type; otherwise put $\theta_R = 0$. Since X is proper, B_R is nonempty. It may be a set or a proper class.

Theorem 52.11 (Canonical division by the ordinal class). *In GB, every class well-order R on a proper class has a canonical class isomorphism*

$$(X, R) \cong (B_R \times \text{Ord})_{\text{lex}} + \theta_R,$$

where the product compares the B_R coordinate first. All proper blocks have type Ord. The block map, the block index order, and the isomorphism are elementary and uniform in R . No global choice or elementary transfinite recursion principle is used.

Proof. A proper, set-like class well-order has type Ord: assign to each point the ordinal type of its set of predecessors. The ranks form an initial subclass of the ordinals; if that subclass were a set, the inverse rank map and class replacement would make the domain a set. Hence the rank range is all of Ord.

Apply this observation to each proper block. More explicitly, define

$$\rho_R(x) = \text{otp}(\{z : m_R(x) \leq_R z <_R x\}, R).$$

The displayed predecessor interval is a set. Its restricted relation and its unique ordinal order type are sets as well. Thus the graph of ρ_R is elementary with class parameter R , and $(m_R(x), \rho_R(x))$ gives the desired coordinate in a proper block. The same rank map identifies a final set block with θ_R . Convexity of the blocks proves order preservation and exhaustiveness.

The class B_R consists of actual elements of X , characterized by an elementary formula. Class well-foundedness restricts to it. None of the construction forms a collection having the blocks themselves as elements. $\square$

Proposition 52.12 (Uniqueness and functoriality). *A class order isomorphism preserves set-interval equivalence, induces an isomorphism of the proper-block index orders, and preserves the final ordinal θ_R . Hence the pair (B_R, θ_R) is unique up to class order isomorphism of its first coordinate. The construction works simultaneously for any uniformly coded set- or class-indexed family of input relations.*

Proof. A class isomorphism and its inverse send sets to sets by class replacement. They therefore preserve whether an interval is a set, and send block minima to block minima. Ordinal order types of set tails are unique. Conversely an isomorphism of block indices and equality of the final ordinals combine, using the displayed coordinates, to produce an isomorphism of the original orders. Uniformity follows by inserting the family index into the same elementary formulas. $\square$

The first proper block is exactly the repository's canonical initial set-like part I_R . Thus the theorem is a simultaneous form of the successive residual decomposition. It does not prove comparability of arbitrary abstract class well-orders: the remaining index order B_R can be as complicated as any class well-order.

Corollary 52.13 (Set-sized cofinality from the blocks). *If $\theta_R > 0$, the least size of a set cofinal in R is $\text{cf}(\theta_R)$. If $\theta_R = 0$ and B_R has a greatest element, no set is cofinal in R . If $\theta_R = 0$ and B_R has no greatest element, R has a set cofinal subset exactly when B_R does, and the least sizes agree.*

Proof. A nonempty final set tail determines cofinality. A final proper block has type Ord , so every set of points in that block is bounded, while earlier blocks cannot help with cofinality. In the remaining case, projecting a set cofinal subset to block minima gives a set cofinal in B_R . Conversely a set cofinal in B_R provides a cofinal set of least block labels in R : because B_R has no maximum, a cofinal family includes a block strictly beyond the block of any prescribed point. Class replacement preserves the relevant size inequalities. $\square$

In a full containing-universe interpretation based on an inaccessible κ , the theorem agrees with ordinary ordinal division. The internal ordinal class is externally κ . An internal class interval is an internal set exactly when its external order type is less than κ : regularity bounds the ranks of fewer than κ labels, and internal sets have size less than κ . Thus a well-order of external ordinal type γ , with $|\gamma| = \kappa$, decomposes as

$$\gamma = \kappa \cdot \beta + \rho, \quad \rho < \kappa.$$

The elementary internal argument does not require the existence of such a containing universe.

Theorem 52.14 (Exact spectrum of block index orders). *In GBC, every nonempty class well-order Γ on a class of set labels occurs as B_R , up to isomorphism, for a class well-order R of No. Any prescribed set ordinal can simultaneously be realized as the final remainder θ_R .*

Proof. Form $(\Gamma \times \text{Ord})_{\text{lex}} + \theta$. It is a class well-order: a nonempty subclass either meets the product, in which case choose its least first coordinate and then its least ordinal coordinate, or lies in the

final ordinal. Within one product fiber, closed intervals are sets. Between distinct fibers, an interval contains a proper final part of an Ord fiber and is proper. The proper blocks are exactly the fibers, and the final θ , when nonzero, is one set block. Global choice supplies a class bijection from this proper carrier to No; transport the order. $\square$

52.3.1 Uniform removal through ordinary ordinal stages

The direct decomposition also answers a narrower recursion question. Let K_R be the initial set-like part of B_R (all of B_R if it is a set), and let $r_R(b)$ be the ordinal type of the predecessors of b in B_R , for $b \in K_R$. For every set ordinal α , put

$$U_\alpha = \{x \in X : m_R(x) \in K_R \wedge r_R(m_R(x)) < \alpha\}.$$

The family $(U_\alpha)_{\alpha \in \text{Ord}}$ is one elementary class relation. Its complement carries the residual order after removal of the first α proper blocks. At a successor stage, if a proper residual carrier remains, its initial Ord block is removed; at a limit stage the removed domain is the union of the earlier removed domains. If B_R is a set, removal stops after its ordinal type, leaving θ_R .

Thus even this uniform ordinal-indexed instance of apparently class-valued recursion is already available in GB through a closed definition. This is a statement about this particular operation, not a claim that arbitrary class-valued recursions are provable in GB.

52.3.2 A class well-order isomorphic to its own block quotient

Let $\mathcal{P}$ consist of all functions $f : \omega \rightarrow \text{Ord}$ with finite support. For distinct f, g , put

$$f \triangleleft g \iff f(n) < g(n), \quad n = \max\{i : f(i) \neq g(i)\}.$$

Proposition 52.15 (Self-similar block quotient). *The order $(\mathcal{P}, \triangleleft)$ is a class well-order. Its set-interval blocks consist precisely of functions that agree at every coordinate $i \geq 1$. Every block has type Ord, and its block index order is isomorphic to $(\mathcal{P}, \triangleleft)$ itself. In GBC the example can be transported to a class well-order of No.*

Proof. For a nonempty class of functions, the zero function is least if present. Otherwise choose its least possible largest nonzero support index d . Among functions with that index, minimize the coefficient at d , then at $d - 1$, and so on through 0. A nonempty class of ordinals has a least element, and there are finitely many steps. The result is least among functions of degree d , and every larger degree is greater. This proves class well-foundedness.

If f, g agree above coordinate 0, every function between them has that same tail, and its zeroth coordinate lies in the set ordinal interval between $f(0)$ and $g(0)$. Their interval is a set. If their tails differ and $f \triangleleft g$, keep the tail of f fixed and let the zeroth coefficient run through all ordinals greater than $f(0)$. The resulting proper class lies strictly between f and g . This proves the description of blocks.

The least label in a block has zeroth coefficient 0. Deleting that coordinate identifies the class of block minima, with its inherited order, with all of $\mathcal{P}$. Varying the zeroth coefficient identifies each block with Ord. Finally apply global choice to transfer the proper carrier to No. $\square$

This example is a rigorous class analogue of the finite-polynomial presentation informally denoted by Ord^ω . The notation is not needed in the construction. It shows that passage to the block quotient need not reduce the order to a simpler isomorphism type. It does not assert nontermination of every possible limit-stage condensation convention; a limit operation must be defined separately.

52.3.3 What remains open after the direct decomposition

The repository's question about uniformly organizing successive initial Ord blocks has an elementary answer above, including an exact unrestricted spectrum of block index orders. A distinct question remains: what is the strength of uniformly iterating the *quotient operation itself* through arbitrary class well-orders, with an explicitly specified limit convention? General ETR is a sufficient framework for suitable elementary class recursions, but the present arguments establish neither its necessity for this specific operation nor the strength of alternative limit conventions. The self-similar example rules out a proof based merely on a claimed strict decrease of quotient isomorphism types.

53 Condensation histories and the strength of iterated condensation (source 15; sources 16 and 18)

[merge] Source 15's Section 6 constructs, for every class well-order J , a class well-order whose iterated condensation runs through exactly J proper stages before reaching a singleton, by one elementary formula. Sources 16 and 18 discuss the same recursion question in the last subsections of their block sections (printed in [section 52](#)); the three discussions agree: finitely many condensations, fixed externally, are elementary; a uniform transfinite tower of condensations is a class-valued recursion for which a suitable ETR suffices; no source claims that ETR is necessary.

53.1 Source 15: explicit condensation histories of arbitrary class length

The block theorem admits an additional extension. Every class well-order can be realized as the order of the nontrivial stages of an explicit condensation history. The construction has a direct elementary formula and therefore does not require a general recursion axiom, even when some stages have proper-class many predecessors.

Definition 53.1 (Finite ordinal polynomials over a class order). For a nonempty class well-order $(J, <_J)$, let $P(J)$ consist of the *set codes* for finite partial functions

$$f : J \rightarrow \text{Ord} \setminus \{0\}.$$

Values off the finite domain are interpreted as zero. The empty function is denoted 0. For distinct f, g , let j be the greatest member of their finite disagreement set and put

$$f <_{P(J)} g \iff f(j) < g(j).$$

For completeness, put $P(\emptyset) = \{0\}$. These are finite set codes, not proper-class functions used as elements of a class. The greatest disagreement exists because its domain is finite; there is no last-coordinate assumption on J itself.

Lemma 53.2 (The polynomial order is a class well-order). *In GBC, $P(J)$ is a proper class well-order.*

Proof. The order laws reduce to ordinary finite lexicographic comparison on the union of the finitely many supports of the polynomials under comparison. If j_0 is the least point of J , varying the positive ordinal coefficient at j_0 shows that $P(J)$ is proper.

We first rule out a descending ω -sequence (f_n) . Its union of supports is a set $A \subseteq J$; the restriction of $<_J$ to A is a set well-order. Let $r : A \rightarrow \eta$ be its ordinal rank map. Choose an

additively indecomposable ordinal Λ strictly above every coefficient of every f_n , for example a sufficiently large power of ω . For each f_n , form the ordinary ordinal

$$v(f_n) = \sum_{j \in \text{supp}(f_n)}^{\text{decreasing } <_J} \Lambda^{r(j)} \cdot f_n(j).$$

This is a finite sum. If two polynomials first differ from the top at j , all their lower terms are less than $\Lambda^{r(j)}$. Indeed, each lower exponent is $\sigma < r(j)$, each coefficient is less than Λ , and $\Lambda^\sigma c < \Lambda^{\sigma+1} \leq \Lambda^{r(j)}$; finite sums remain below the latter additive principal ordinal. When $r(j) = 0$ there are no lower terms. Thus the different coefficients at j decide the ordinal comparison, and v preserves it strictly. The sequence $(v(f_n))$ would be descending in Ord, a contradiction.

For completeness, Global Choice converts a nonempty class with no least member into a descending ω -sequence. Given f in such a class, consider the nonempty class of its smaller members. Its members of least set-theoretic rank form a nonempty set. Apply the fixed global choice function to that set to choose the next member. This is a class function, and ordinary set-valued recursion through ω constructs the descending sequence. Hence every nonempty available subclass of $P(J)$ has a least member. $\square$

Lemma 53.3 (One condensation removes the least exponent). *Let $j_0 = \min J$. In $P(J)$, two polynomials lie in the same set-interval block exactly when they agree at every $j >_J j_0$. The least representative of that block has coefficient zero at j_0 . Consequently the canonical condensation is the evident copy of $P(J \setminus \{j_0\})$.*

Proof. If the polynomials agree above j_0 , every polynomial between them has those same upper coefficients. Its coefficient at j_0 lies in a bounded ordinal interval, so the closed interval is a set.

Conversely let $f < g$ disagree at a coordinate above j_0 . Keep all coefficients of f above j_0 fixed and replace its coefficient at j_0 by $f(j_0) + \beta + 1$, for $\beta \in \text{Ord}$. The resulting polynomials lie strictly above f and strictly below g : the greatest disagreement with g is unchanged. They form a proper class, so $[f, g]$ is not a set. Finally zero is the least ordinal coefficient, which identifies the least block representative. $\square$

Theorem 53.4 (Every class well-order is an explicit condensation length). *In GBC, for every nonempty class well-order J there is a class well-order of **No** with an elementary, uniformly coded condensation history indexed by $J + 1$, such that every stage indexed by J is proper and the appended final stage is a singleton.*

Proof. It suffices to use $P(J)$ and transfer the result to **No** by a class bijection. For $j \in J$ define the retained representatives

$$H_j = \{f \in P(J) : \text{supp}(f) \subseteq J_{\geq j}\}, \quad H_* = \{0\},$$

where $*$ is one new final index. Code the indices with disjoint tags if necessary. The class of pairs (j, f) satisfying this condition is defined by an elementary formula, including when $J_{< j}$ is proper.

At the least index, $H_j = P(J)$. If k is the immediate successor of j , Lemma 53.3 applied to the tail $J_{\geq j}$ shows that the condensation of the induced order on H_j is exactly H_k . If j is last, that same argument gives H_* .

At a nonzero limit index j , a polynomial belongs to every H_k for $k <_J j$ if and only if none of its support lies below j . For if $t <_J j$ lies in the support, choose k with $t <_J k <_J j$; then the polynomial is absent from H_k . Thus

$$H_j = \bigcap_{k <_J j} H_k.$$

This intersection is legitimate elementary notation: membership is the set-quantifier statement $\forall k (k <_J j \Rightarrow (k, f) \in H)$. If J has no greatest element, the same argument shows that the intersection over all $j \in J$ is $\{0\} = H_*$.

For every $j \in J$, arbitrary ordinal coefficients supported at j belong to H_j , so H_j is proper. If $j <_J k$, the polynomial with coefficient one at j belongs to $H_j \setminus H_k$. The history is therefore strictly decreasing at distinct indices and first becomes a singleton at its appended final stage. All formulas, including the full class of stages, have been supplied explicitly. No invocation of elementary transfinite recursion has occurred. $\square$

Examples include finite histories, a history of length ω , one of length Ord , and one of length $\text{Ord} + \text{Ord}$. These refer to the order of the nontrivial stages, rather than to additional set ordinals. In particular there is no universal set-ordinal bound on the depth of the proper-class block hierarchy.

Remark 53.5 (A proper quotient can have the same abstract order type). For $J = \omega$, deleting the least exponent leaves a tail isomorphic to ω . Hence $P(\omega)$ is order-isomorphic to its own canonical condensation, even though the retained representatives form a proper subclass. The explicit history nevertheless reaches the singleton at stage ω . Likewise every set-ordinal tail of Ord is isomorphic to Ord , while the history for $P(\text{Ord})$ becomes a singleton only at its appended class-limit stage. Consequently one cannot prove termination by asserting that each condensation strictly decreases the abstract order type.

53.1.1 Exactly what this says about recursion strength

For an arbitrary input W , every *externally fixed finite* number of successive condensations can be constructed in GB by composing elementary definitions. This finite theorem scheme should not silently be replaced by the assertion that one class uniformly codes all finite iterations. The standard principle ETR_ω is a sufficient hypothesis for that uniform history; its final intersection can then be formed by elementary comprehension. Similarly, an appropriate elementary-recursion principle is a sufficient hypothesis for an arbitrary prescribed class-length history on a general input.

No lower bound or equivalence with ETR_ω is claimed for this particular condensation operator. Proving that such a history exists for every W in GB, or calibrating its additional strength, remains a sharper reverse-mathematical question. The explicit polynomial histories show why a general warning about class-valued recursion must not be converted into an assertion that every individual long construction needs ETR. Here one elementary formula already defines every stage, even along arbitrary given class well-orders.

53.1.2 A full-universe consistency check

Suppose an ambient universe contains a strongly inaccessible cardinal κ , and use the full two-sorted model $(V_\kappa, \mathcal{P}(V_\kappa))$. For a subclass of V_κ , being an internal set is equivalent to having ambient cardinality less than κ . One direction uses the strong-limit property; the other uses regularity to bound the ranks of fewer than κ many elements. Thus the preceding condensation becomes condensation by intervals of cardinality less than κ .

If the ambient ordinal type of an internal class well-order is θ , write its ordinary left ordinal division by κ as

$$\theta = \kappa \cdot \gamma + \delta, \quad \delta < \kappa.$$

Its canonical blocks are the successive κ -segments, followed by one δ -segment if $\delta > 0$. The quotient has type γ if $\delta = 0$ and $\gamma + 1$ otherwise. This is exactly the canonical block theorem as seen from the

containing universe. An internal Ord-block is an ambient κ -block, rather than an ambient proper class.

This check also identifies the essential relativity: changing which collections count as sets changes the condensation ideal. The construction is intrinsic to (X, W) together with the chosen set/class boundary; it is not absolute between arbitrary universes.

Note 53.6 ([merge]; the self-similar example of sources 15 and 18). Source 15's remark [remark 53.5](#) and source 18's [proposition 52.15](#) describe the same class well-order: source 15's $P(\omega)$, the finite partial functions $\omega \rightarrow \text{Ord} \setminus \{0\}$ compared at the greatest coordinate of disagreement, is source 18's $(\mathcal{P}, \triangleleft)$, the finite-support functions $\omega \rightarrow \text{Ord}$ compared at the largest coordinate of disagreement. Both show that it is isomorphic to its own block quotient, so that one condensation step need not lower the isomorphism type; source 18 proves it in full, source 15 deduces it from [lemma 53.3](#). Their full-universe checks agree as well: source 15's subsection above, source 18's remark after [corollary 52.13](#), and source 18's [section 57.5](#) all identify the internal block decomposition in $(V_\kappa, \mathcal{P}(V_\kappa))$, κ inaccessible, with left ordinal division by κ .

54 Finite components and trace fibres (source 17; sources 18 and 14)

Note 54.1 ([merge]; what is repeated and what is new). Source 17's [theorem 54.2](#) is source 11's [theorem 16.2](#) and source 12's [theorem 16.6](#), which it calls a “source result”. Its [theorem 54.4](#) identifies the maximal finite-adjacency components of the unrestricted order: they are the $n!$ permutations of the maximal finite final segment. Finite permutation blocks after a common prefix are already source 12's [corollary 16.7](#) and, for set alphabets, source 08's [note 6.26](#); new in [theorem 54.4](#) are the maximality of these blocks on the class carrier, the uniqueness of the factorization $R = P + \tau$, and the density of the quotient. Source 18 proves the same component theorem independently ([lemmas 58.1](#) and [58.2](#) and [proposition 58.3](#), printed with its countable models in [section 58](#)), and adds that every factorial block size occurs densely ([lemma 58.5](#)).

Source 14's Section 10, printed second, recalls the direct comparison and the canonical initial branch ([lemma 54.5](#); source 11's [theorem 15.1](#), source 13's [proposition 51.8](#), source 15's [proposition 51.10](#)) and proves that the trace of an unrestricted order on the core is the cut of its initial branch ([theorem 54.6](#)). Its [example 54.7](#) (singleton fibres need not be set-like; $m!$ elements over m omitted labels) is source 11's [remark 15.4](#) and [proposition 15.3](#), which source 14 does not cite, and its [corollary 54.8](#) is the statement about raw orders printed in Part VIII by sources 13, 15, 16, 17 and 18 ([note 46.5](#)). New in source 14's section is the exact description of the trace fibres over the core as the convex families $F + Q$.

54.1 Source 17: unrestricted orders and their finite condensation

The restriction to set-like orders is not cosmetic. After a class prefix has exhausted almost every label, the residual alphabet can be finite, and numerical density of **No** is no help. We recall the source's exact adjacency theorem and then identify the maximal components generated by adjacency.

Theorem 54.2 (Adjacency after a common prefix; source result). *Let $R <_{\text{lex}} S$ be well-orders of the same labelled class X , and let C be their largest common labelled prefix. Write $a < b$ for the next labels and $Y = X \setminus C$. The orders are adjacent if and only if Y is finite and their permutations of Y after C are consecutive in the finite lexicographic order.*

Proof. A remaining label strictly between a and b would start an intermediate order, followed by a restriction of one given well-order. Also, the tail of R after a must be a lexicographically greatest well-order of its remaining labels; otherwise replacing that tail gives an intermediate order. Such a greatest order exists exactly when the native decreasing order is itself a well-order: any pair listed in native increasing order can be transposed to make the enumeration larger. Dually, the S -tail after b must be the native increasing order.

Thus $Y \setminus \{a, b\}$ is well-ordered both in native increasing and native decreasing order. It is finite. Indeed, otherwise successively choosing increasing elements by their well-order creates a strictly increasing ω -sequence, a set with no greatest element, contradicting reverse well-ordering. Hence Y is finite, and no intermediate residual permutation exists precisely when the two permutations are consecutive. The converse follows directly from convexity of the common-prefix cylinder. These are the source's arguments ([theorems 16.2 and 16.6](#)). $\square$

Definition 54.3 (Finite-adjacency components). Two orders are in the same component when a finite chain of adjacent pairs connects them. This is a uniform relation on the well-order predicates, or an ordinary equivalence relation in an explicitly larger set-universe interpretation. We do not form a same-level class of proper-class equivalence classes.

Theorem 54.4 (Exact maximal finite components). *Every well-order R of the whole surreal carrier has a unique factorization*

$$R = P + \tau,$$

where τ is a finite final segment of length n , and P has no greatest element. The integer n may be zero. The finite-adjacency component of R consists exactly of all $n!$ permutations of these same n final labels following the fixed prefix P .

After identifying each such component, the resulting higher-order linear comparison is dense and has no endpoints. A nonsingleton component intrinsically determines its terminal length $n \geq 2$ from its size $n!$. Singleton components do not distinguish terminal lengths 0 and 1.

Proof. Repeatedly delete the greatest point for as long as one exists. This process must stop after finitely many deletions: otherwise set-valued recursion on ω produces a strictly descending ω -sequence in R , contradicting the well-order property. The remaining prefix is still a proper class, since only finitely many labels were removed, and has no greatest element. This establishes existence and uniqueness of the maximal finite final segment.

All its $n!$ permutations form a finite convex block, by common-prefix convexity. Consecutive members are adjacent by [Theorem 54.2](#), so they are in one component. Suppose an order in this block has an adjacent neighbor. Their common prefix is cofinite. It must contain all of P : if it stopped strictly inside P , its nonempty residual final part of P would be finite and would have a greatest element, making P have a greatest element. That is impossible. Thus an adjacent neighbor can change only the finite final labels and remains in the displayed block. No adjacency chain leaves it.

All components are finite and consequently have first and last elements. If two distinct components were adjacent in the quotient, the last member of the earlier one and first member of the later one would be adjacent in the original order, joining the components. Hence the quotient is dense. The original comparison has no endpoints: replace its first label by any smaller or larger label, then append the restriction of its well-order to the remaining labels. Since each component is finite, the quotient has no endpoints either. Strict increase of $n!$ for $n \geq 2$ proves the intrinsic size assertion. $\square$

This is a description of the *maximal adjacency components*, not just of some finite cylinders. It does not identify the quotient with $\mathfrak{W}_{\text{sl}}$ or with **No**. The many non-set-like convex fibres from [Proposition 46.27](#) can retain substantial structure after all finite components have been collapsed.

54.2 Source 14: unrestricted class well-orders and the trace quotient

The enumeration picture only covers *set-like* well-orders of the whole surreal class. For arbitrary class well-orders, some elements may have proper-class predecessor collections, so an ordinal-valued position need not exist for every label. We first recall how to compare such orders without inventing those positions.

54.2.1 Elementary comparison of labelled relations

Let R, S be strict class well-orders on the same labelled carrier **No**. For a label x , write $\text{Pred}_R(x) = \{y : yRx\}$. Define C by

$$x \in C \iff \text{Pred}_R(x) = \text{Pred}_S(x) \text{ and } R \upharpoonright \text{Pred}_R(x) = S \upharpoonright \text{Pred}_S(x).$$

Restrictions in this formula concern binary relations on the indicated predecessor classes. Equality expands into quantifiers over set labels; no class variable is bound. It follows directly that C is initial in both orders and their restrictions to C agree. If they differ, C is proper. Let a be the R -least label outside C and c the S -least such label. Their existence uses that the relations are class well-orders. They are distinct: if they coincided, that label too would have identical predecessors and identical predecessor relations, and would belong to C . Set

$$R <_{\text{lex}} S \iff a < c \text{ in numerical surreal order.}$$

This is the direct comparison developed in the predecessor ([theorems 14.2, 15.1 and 16.2](#); Sections 14–16 at the pin). For completeness, its transitivity has a particularly small proof. If $R <_{\text{lex}} S <_{\text{lex}} T$, the common prefixes for (R, S) and (S, T) are initial segments of the same well-order S , hence nested. If one is shorter, its first discrepancy already compares R with T in the asserted direction; the third order still follows S there. If the two prefixes agree, the three next labels satisfy $a < b < c$ in numerical order. This gives $R <_{\text{lex}} T$. Totality and asymmetry follow from the distinct next labels. When both orders are set-like, this definition agrees with enumeration comparison.

There is no step comparing two abstract class order types by a newly constructed class isomorphism. The labels allow comparison by literal agreement of predecessors. That is the reason no appeal to ETR is needed here.

54.2.2 The canonical initial branch

Lemma 54.5 (Initial set-like branch). *Every $R \in \mathfrak{W}_{\text{all}}$ has a canonical injective class function $J_R : \text{Ord} \rightarrow \mathbf{No}$ whose range is an initial segment of R . It is characterized by*

$$J_R(\alpha) = x \iff \text{Pred}_R(x) \text{ is a set and } \text{otp}(\text{Pred}_R(x), R) = \alpha.$$

Its range is exactly the labels with set-sized predecessor collections. The relation R is set-like if and only if J_R is exhaustive.

Proof. For each set ordinal α , recursively take the least unused label for $\alpha + 1$ stages. At every intermediate stage the used labels are a set, whereas the carrier is proper, so the remaining class is nonempty and has an R -least element. Its predecessor set is precisely the labels already chosen. These set-length recursions are unique and compatible. Their elementary characterization in the display gives a class function on Ord . Conversely any label with set predecessors occurs at the order type of that predecessor set. Initiality and the remaining assertions follow immediately. $\square$

For $F \in \mathfrak{J}$, put $X_F = \mathbf{No} \setminus \text{ran } F$. Concatenate the order given by F with any class well-order Q of X_F , writing $F + Q$ for the resulting labelled relation. Such a Q exists, for example by restricting the baseline well-order to X_F . The concatenation is a class well-order: a nonempty subclass meeting the first block has a least ordinal position there, and otherwise has a least element in the second block. Its initial branch is exactly F , because every label in the second block has the proper class $\text{ran } F$ among its predecessors. Thus every injective branch, exhaustive or not, arises as J_R .

54.2.3 What the skeleton can and cannot distinguish

Define the trace of an unrestricted relation on the core by

$$\text{Tr}_b(R) = \{\sigma \in \mathcal{P}_s(b) : E_\sigma <_{\text{lex}} R\},$$

where a core enumeration is identified with its induced set-like well-order relation.

Theorem 54.6 (Trace fibers and their exact loss of information). *For every unrestricted labelled class well-order R ,*

$$\text{Tr}_b(R) = D_{J_R}.$$

Consequently

$$\text{Tr}_b(R) = \text{Tr}_b(S) \iff J_R = J_S.$$

For a fixed $F \in \mathfrak{J}$, its trace fiber is the convex family

$$\{F + Q : Q \text{ is a class well-order of } X_F\}.$$

The quotient by equality of core traces is therefore identified, in the virtual sense, with the lexicographic order $\mathfrak{J}$. Its image in the normalized cuts is exactly the branch alternative of Theorem 43.3, not the entire cut collection.

Proof. If E_σ differs from J_R at α , the labelled orders agree before that ordinal and their next labels are $E_\sigma(\alpha)$ and $J_R(\alpha)$. Both comparisons therefore have the same sign. If they agree everywhere, $J_R = E_\sigma$ is exhaustive, so R is exactly that set-like order and neither strict comparison holds. This proves the trace formula.

If two initial branches differ, Lemma 41.5 gives a core code between them, which belongs to the larger trace but not the smaller one. Thus trace equality forces branch equality. The concatenation description above proves the asserted fiber and its surjectivity over all F . Distinct branches are compared at a set coordinate before either tail, so every fiber is convex and the order of fibers agrees with the lexicographic order of $\mathfrak{J}$. Finally the trichotomy separates branch traces from cylinder and numerical edges. $\square$

Example 54.7 (Singleton fibers need not be set-like). If X_F is empty, the fiber is the singleton exhaustive order F . If $X_F = \{x\}$, it is again a singleton, but its order is the non-set-like concatenation of an Ord -long initial block with the last label x . Its trace is not realized by any exhaustive enumeration. Thus “singleton trace fiber” is not a criterion for being set-like. For finite $|X_F| = m$, the fiber has exactly $m!$ elements, ordered as the finite permutations of that residual alphabet. Larger residual alphabets retain their full well-ordering problem inside the fiber.

This gives a useful diagram of interpretations:

$$\mathfrak{W}_{\text{sl}} \subsetneq \mathfrak{J} \longrightarrow \{\text{normalized core cuts}\}, \quad \mathfrak{W}_{\text{all}} \twoheadrightarrow \mathfrak{J},$$

where the first arrow into cuts is injective and the last map has the fibers just described. The braces and arrows here abbreviate virtual relations on classes. They do not form a same-sort diagram of class objects in GBC.

The quotient is not the usual finite condensation of a linear order. It can collapse an infinite, even higher-order, family of tails sharing a proper-class initial branch. Conversely finite adjacent blocks elsewhere in the full order are governed by the finite residual-alphabet analysis of the predecessor. Equality of traces is precisely the observational equivalence given by *all set-supported test enumerations*.

Corollary 54.8 (Raw orders do not fill every core boundary). *A cylinder-edge or numerical-gap cut of $\mathcal{P}_s(b)$ is not the trace of any unrestricted class well-order of the full surreal carrier. Allowing non-set-like tails adds realizations of omitted-range branch cuts, but cannot supply all missing core cuts.*

The distinction is sharper than saying that the set-like order is incomplete. Some missing boundaries correspond to an honest Ord-long initial branch with unvisited labels and are supplied by an unrestricted tail. Other boundaries stop at a set stage and are not supplied by any such tail.

55 Asymmetric products and the role of semantics (source 13)

55.1 Absorbing a class-sized left factor

An isomorphism involving $\mathfrak{W}_{\text{sl}}$ is understood first as a pair of inverse uniform transformations of class parameters. For example, a map on $\mathbf{No} \times \mathfrak{W}_{\text{sl}}$ takes a set label and a class enumeration and returns a class enumeration by a fixed definition. It is not asserted to be an ordinary class of pairs whose entries include arbitrary classes.

Theorem 55.1 (Left-factor absorption). *In GBC there are uniform inverse order isomorphisms*

$$\mathfrak{W}_{\text{sl}} \cong \mathbf{No} \times_{\text{lex}} \mathfrak{W}_{\text{sl}}.$$

Consequently, for every ordinary class linear order K , there is a uniform embedding

$$K \times_{\text{lex}} \mathfrak{W}_{\text{sl}} \hookrightarrow \mathfrak{W}_{\text{sl}}.$$

Proof. Use the uniform complement isomorphisms of [Lemma 42.2](#). Given (x, E) , define $\Phi(x, E)$ by

$$\Phi(x, E)(0) = x, \quad \Phi(x, E)(1 + \alpha) = h_{\{x\}}^{-1}(E(\alpha)) \quad (\alpha \in \text{Ord}).$$

The map $\alpha \mapsto 1 + \alpha$ is the increasing parametrization of the ordinal tail after its first point. The displayed function is a bijection, since its tail covers exactly $\mathbf{No} \setminus \{x\}$.

Recover x as the first value and apply $h_{\{x\}}^{-1}$ to the tail to obtain the inverse. Different first labels are compared immediately; with the same first label, the increasing complement isomorphism preserves the first tail disagreement. Thus Φ is an order isomorphism. Embed K into $\mathbf{No}$ by [Lemma 42.2](#) and compose with Φ . $\square$

This includes every ordinary class-indexed lexicographic sum of copies of $\mathfrak{W}_{\text{sl}}$, including index orders such as Ord, Ord* and Ord + Ord. It does *not* include the full $\mathfrak{W}_{\text{sl}}$ as the index order: its members are classes, not set codes.

55.2 A dense-core obstruction to splitting every point

Lemma 55.2 (Split-order density bound). *Let M be a dense set-sized linear order and let $D \subseteq M$ be order-dense. If $K \times_{\text{lex}} 2$ embeds into M , then $|K| \leq |D|$ in ZFC.*

Proof. For each k , the images of $(k, 0)$ and $(k, 1)$ enclose a nonempty open interval of M . The intervals for different k are disjoint. Choose a point of D in each, using the least one in a fixed well-order of D . This injects K into D . $\square$

The source pairs may be adjacent; that does not invalidate the argument. An order embedding need not preserve adjacency, and the target’s density puts a point between their images. The proof is an interval counting argument rather than a count of all points in the target.

Theorem 55.3 (No full split in an inaccessible universe). *Work in ambient ZFC and let κ be strongly inaccessible. Interpret internal sets as V_κ and classes as all ambient subsets of V_κ . For the resulting global enumeration order,*

$$\mathfrak{W}_{\text{sl}} \times_{\text{lex}} 2 \not\hookrightarrow \mathfrak{W}_{\text{sl}}.$$

In particular, $\mathfrak{W}_{\text{sl}}^2 \not\hookrightarrow \mathfrak{W}_{\text{sl}}$ and $\mathfrak{W}_{\text{sl}} \times_{\text{lex}} \mathbf{No}^{V_\kappa} \not\hookrightarrow \mathfrak{W}_{\text{sl}}$, although $\mathbf{No}^{V_\kappa} \times_{\text{lex}} \mathfrak{W}_{\text{sl}} \cong \mathfrak{W}_{\text{sl}}$.

Proof. Externally, the internal surreal carrier is $X = \mathbf{No}_{<\kappa}$ of cardinality κ , and the available set-like global enumerations are exactly all bijections $\kappa \rightarrow X$. Every restriction to an internal set has ranks bounded below κ by regularity, so these functions are legitimate full-model classes. The cardinality of this order is 2^κ : permutations give the upper bound, and independent pair reversals the lower bound.

The core D of internal set-support codes has ambient cardinality κ . For the upper bound, all its codes belong to V_κ , whose ambient cardinality is κ ; transpositions give the lower bound. The internal interpolation theorem makes D order-dense in the external enumeration order. This order is dense. Applying Lemma 55.2 would force $2^\kappa \leq \kappa$, contrary to Cantor’s theorem.

Choose two distinct points in the second factor of either proposed larger product. This gives a suborder isomorphic to $\mathfrak{W}_{\text{sl}} \times_{\text{lex}} 2$, so both nonembeddings follow. The reverse-order placement of $\mathbf{No}^{V_\kappa}$ is covered by Theorem 55.1. $\square$

The earlier report uses the same dense-core mechanism to rule out an embedding of all raw well-orders into the set-like fragment (theorem 20.1). The new point is that merely splitting *each existing set-like point into two* already creates the obstruction. It is unnecessary to introduce the entire raw well-order domain. [merge] That proof already uses only the adjacent raw pairs $F + (a, b) < F + (b, a)$, with F ranging over the bijections of κ onto $X \setminus \{a, b\}$; these pairs form a copy of $\mathfrak{W}_{\text{sl}} \times_{\text{lex}} 2$ (the set-like orders of $X \setminus \{a, b\}$ form a copy of $\mathfrak{W}_{\text{sl}}$). So the proof of theorem 20.1 already establishes theorem 55.3; source 13’s theorem isolates that conclusion and states it for $\mathfrak{W}_{\text{sl}}$ itself, and its “new point” is a new statement, not a new mechanism.

55.3 The uniform Kelley–Morse version

Theorem 55.4 (No definable uniform split embedding). *For each fixed formula, with fixed set and class parameters, that defines a total transformation from $\mathfrak{W}_{\text{sl}} \times 2$ to $\mathfrak{W}_{\text{sl}}$, the theory $\text{KM} + \text{GC}$ proves that the transformation is not an order embedding. The transformation may be specified relationally by a unique output class for each input. This is a schema about formulas, not quantification over arbitrary third-sort functions.*

Proof. Suppose a formula defines an embedding Ψ . For every $E \in \mathfrak{W}_{\text{sl}}$, choose the globally least code $q_E \in \mathcal{P}_s(b)$ satisfying

$$\Psi(E, 0) < E_{q_E} < \Psi(E, 1).$$

The interval is nonempty and meets the core. The definition of its eligible code class can use the formula for Ψ and its output classes; KM permits the necessary comprehension. For distinct E, F , the image intervals are disjoint. Hence

$$q_E = q_F \implies E = F.$$

Now form the ordinary class evaluator

$$T(q, \alpha, x) \iff \exists \text{ class } E [E \in \mathfrak{W}_{\text{sl}} \wedge q = q_E \wedge E(\alpha) = x].$$

This is the decisive impredicative comprehension step. Each nonempty row is the graph of a unique enumeration, and every enumeration occurs in some row. It contradicts [Theorem 42.7](#). $\square$

The same conclusion applies to uniform embeddings of $\mathfrak{W}_{\text{sl}}^2$ or $\mathfrak{W}_{\text{sl}} \times_{\text{lex}} \mathbf{No}$ into $\mathfrak{W}_{\text{sl}}$, by restricting the second coordinate to two fixed choices. The argument adapts the prior report's impredicative inverse-evaluator proof for raw orders ([theorem 20.2](#)). Elementary GB comprehension alone does not justify the displayed existential class quantifier in T , and no claim to that effect is made. [merge] Applied to the same pairs $F + (a, b) < F + (b, a)$, the proof of [theorem 20.2](#) is the proof of [theorem 55.4](#).

55.4 Externally countable models reverse the conclusion

Proposition 55.5 (External countable collapse of the order). *Let $(M, \mathcal{C})$ be an externally countable model of GBC. The external order of its available global enumerations is isomorphic to $(\mathbb{Q}, <)$. Consequently, with arbitrary external maps,*

$$\mathfrak{W}_{\text{sl}}^2 \cong \mathfrak{W}_{\text{sl}}, \quad \mathfrak{W}_{\text{sl}} \times_{\text{lex}} 2 \hookrightarrow \mathfrak{W}_{\text{sl}}.$$

Proof. The collection of available class objects is externally countable. The internal theorems asserting density and absence of endpoints provide actual witnesses among those objects, so its induced external order is a countable dense linear order without endpoints. The classical back-and-forth for such orders identifies it with $\mathbb{Q}$. The lexicographic square of a countable dense order without endpoints is again countable, dense, and without endpoints. The two-point split is a countable linear order, which embeds into $\mathbb{Q}$ by the elementary countable embedding construction. $\square$

Even if the model also satisfies KM + GC, such an external embedding need not be represented by an internally uniform definition. Likewise, an external countable cofinal sequence need not be an internal set-coded sequence. This explains exactly why the proposition and [Theorem 55.4](#) do not conflict.

The question “does the global order contain its lexicographic square?” therefore has no unqualified model-independent answer until the meaning of the collection and of an allowed map has been fixed.

Note 55.6 ([merge]; credits, and source 08's question on products). [Theorem 55.1](#) is proved with the uniform complement isomorphisms h_S of [lemma 42.2](#) (source 11's [proposition 5.13](#), source 09's [theorem 5.17](#)). Its map $\Phi(x, \cdot)$ is the isomorphism of $\mathfrak{W}_{\text{sl}}$ onto the first-label cylinder $C_{(x)}$ constructed in [proposition 19.25](#) and [note 19.26](#), which source 13 does not cite; the absorption theorem assembles these cylinder isomorphisms over all first labels x . As the merge notes inside the section say, the

proofs of [theorems 55.3](#) and [55.4](#) are the proofs of source 11's [theorems 20.1](#) and [20.2](#) applied to the pairs they already use, and [proposition 55.5](#) is the observation source 11 makes after [theorem 20.2](#) ("for an externally countable model these collections can both be countable") turned into an isomorphism with $\mathbb{Q}$; source 18's [theorem 58.6](#) reproves its second clause and classifies the full order ([section 58](#)). Together they answer source 08's question 7 on full virtual lexicographic products ([section 60](#)); the index of [section 61](#) had it open on 2 October 2026, although the negative half already followed from source 11's proofs. Source 13's critical scope conditions apply: the product nonembedding is not claimed as a theorem of bare GBC about a third sort of maps (its status file, item 3).

56 Coding, diagonalization, and a sharper comprehension bound (source 16; source 14)

56.1 Binary class words and representable suborders

Let $\mathbb{B}_{\text{Ord}}$ denote binary class functions on Ord , with first-disagreement comparison. In the sense of uniform transformations,

$$\mathfrak{W}_{\text{sl}} \simeq_e \mathfrak{J} \simeq_e \mathbb{B}_{\text{Ord}}. \quad (\text{S16.15})$$

For one direction, partition the baseline values into ordinal-indexed pairs and list each pair increasingly or decreasingly according to the input bit. Here the coordinate pairs are $\{2 \cdot \xi, 2 \cdot \xi + 1\}$, with ordinal multiplication. This embeds all binary classes into exhaustive enumerations. For the other, globally enumerate the set codes of $\mathcal{P}_{\text{bd}}(b)$ as $(q_\xi)_{\xi \in \text{Ord}}$ and send $F \in \mathfrak{J}$ to

$$D_F = \{\xi : E_{q_\xi} < F\}.$$

If $F < G$, density gives a core point strictly between them. Thus $D_F \subsetneq D_G$, whose characteristic words have the same first difference direction. Inclusion $\mathfrak{W}_{\text{sl}} \subseteq \mathfrak{J}$ finishes (S16.15). This extends the earlier report's coding argument to the injective comparison as well.

Equimorphism is not isomorphism. Binary class words have least and greatest words and adjacent pairs, while $\mathfrak{W}_{\text{sl}}$ and $\mathfrak{J}$ are dense and have no endpoints. Likewise, any one injectively and uniformly indexed class family in $\mathfrak{W}_{\text{sl}}$ is an ordinary class linear order on its indices, and hence embeds into **No** in GBC. This does not supply an embedding of the entire virtual collection of global enumerations into **No**. There is no exhaustive uniform family at the same level.

Theorem 56.1 (No exhaustive uniform class of set codes). *In GBC, no class of set indices with one uniform evaluation relation contains every class well-order of **No** among its rows. In fact a set-like well-order is missing from every such family.*

Proof. Use global choice to reindex the rows by Ord , repeating a default row if the original family was a nonempty set. For each ordinal ξ , sort the two labels $b(2 \cdot \xi), b(2 \cdot \xi + 1)$ as $a_\xi < c_\xi$. Define a new enumeration by keeping the pair blocks in ordinal order. Within pair ξ , place c_ξ before a_ξ if (a_ξ, c_ξ) belongs to relation row ξ , and otherwise place a_ξ before c_ξ . This local rule is elementary even for empty or malformed rows. The new enumeration is exhaustive and set-like; it differs from every well-order row ξ on that pair. The empty-family case is immediate. $\square$

Formula strings do not evade this theorem. Each fixed formula defines its own class when comprehension applies, but uniformly evaluating all formulas over the entire universe would require a suitable truth predicate. Neither a metatheoretic list of formulas nor a proper class of their finite syntactic codes supplies such an evaluator.

56.2 A sufficient bound below full Kelley–Morse

The prior report proves in KM plus global choice that no fixed definable total transformation from $\mathfrak{W}_{\text{all}}$ to $\mathfrak{W}_{\text{sl}}$ is an order embedding, allowing arbitrary formulas defining the transformation. For elementary formulas the following sharper sufficient bound holds.

Theorem 56.2 (Elementary transformations and Σ_1^1 comprehension). *Work in $\text{GBC} + \Sigma_1^1\text{-CA}$. Let $\vartheta(R, E)$ be a fixed formula with only set quantifiers, with fixed set and class parameters allowed. Suppose it specifies a unique class output $E \in \mathfrak{W}_{\text{sl}}$ for every $R \in \mathfrak{W}_{\text{all}}$. The specified transformation is not an order embedding.*

Proof. Assume an embedding is specified. Fix numerical labels $a < c$. For each class bijection $F : \text{Ord} \rightarrow \mathbf{No} \setminus \{a, c\}$ form

$$R_F = F + (a, c), \quad S_F = F + (c, a).$$

These are adjacent raw orders, and different F give disjoint, ordered pairs. Their alleged images $E_0 < E_1$ have a bounded-support code between them. Let q_F be the least such code in a fixed global well-order of $\mathcal{P}_{\text{bd}}(b)$. The image intervals are disjoint, so $q_F = q_G$ implies $F = G$.

The required inverse-evaluation predicate has precisely the form

$$\begin{aligned} \text{Eval}(q, \xi, x) \iff \exists F \exists E_0 \exists E_1 [\text{Bij}(F) \wedge \vartheta(R_F, E_0) \wedge \vartheta(S_F, E_1) \\ \wedge \text{LeastBetween}(q, E_0, E_1) \wedge F(\xi) = x]. \end{aligned} \quad (\text{S16.16})$$

The three displayed existential variables range over classes; the matrix has only set quantifiers. The constructed raw orders can be substituted directly into the elementary formula ϑ , and “least code between” quantifies only over set codes. Thus Σ_1^1 class comprehension forms this evaluator.

Its rows are empty or unique bijection graphs, and every such F occurs as a row. Compose each nonempty row with a fixed bijection $\mathbf{No} \setminus \{a, c\} \rightarrow \mathbf{No}$, and fill empty rows with a default. This is an exhaustive uniformly class-indexed family of global enumerations. Replace each bijection row by its induced well-order relation, an elementary conversion, and apply [Theorem 56.1](#) to obtain the contradiction. $\square$

This proves a sufficient comprehension bound, not a reversal. It does not show that Σ_1^1 comprehension is necessary, nor that the same proof works for arbitrary ϑ with unrestricted class quantifier alternations. The syntax of the claimed transformation is part of the theorem. The distinction between elementary and impredicative class definitions is consistent with the known hierarchy around ETR and KM [\[27, 22, 24\]](#).

56.3 Source 14: binary class words and global permutations

A characteristic function $d : \text{Ord} \rightarrow \{0, 1\}$ can be turned into a global enumeration by pair orientations. Partition the ordinals into pairs $2 \cdot \alpha, 2 \cdot \alpha + 1$; ordinal multiplication has 2 on the left. Sort the corresponding baseline labels as $u_\alpha < v_\alpha$. Put the pair in order (u_α, v_α) when $d(\alpha) = 0$ and (v_α, u_α) when $d(\alpha) = 1$. Every baseline label is used exactly once. The first differing bit produces the first differing pair in the correct lexical direction.

Conversely, enumerate the core codes by a fixed class bijection $\text{Ord} \rightarrow \mathcal{P}_s(b)$. The characteristic function of D_E in this schedule is an order embedding of $\mathfrak{W}_{\text{sl}}$ into binary class words: if $E < F$, then $D_E \subsetneq D_F$ by core density, and the first changed bit is 0 versus 1. Thus the known equimorphism

$$\mathfrak{W}_{\text{sl}} \simeq_e (\{0, 1\}^{\text{Ord}}, <_{\text{lex}})$$

is entirely uniform. It is not an isomorphism. The binary class-word order has endpoints and adjacent pairs of the form $p \frown (0) \frown (1, 1, \dots)$ and $p \frown (1) \frown (0, 0, \dots)$; $\mathfrak{W}_{\text{sl}}$ is dense without endpoints. These are class-word constructions, not sets of every binary class.

Theorem 56.3 (No exhaustive uniform class of codes, inherited). *There is no uniformly Ord-indexed family containing every member of $\mathfrak{W}_{\text{sl}}$. More generally no ordinary class of set codes, with one uniform evaluation relation, exhausts all global enumerations.*

Proof. Suppose $(E_\alpha)_{\alpha \in \text{Ord}}$ were such a family. In the pair at $2 \cdot \alpha, 2 \cdot \alpha + 1$, choose the orientation whose first entry differs from $E_\alpha(2 \cdot \alpha)$. At least one of the two orientations works, since the pair labels are distinct. The resulting class function is a bijection, as it only reorders disjoint baseline pairs, and it differs from every row E_α at the designated coordinate. This is an elementary diagonal construction.

An ordinary proper class of set codes admits a set-like global enumeration in GBC; a set of codes can be padded to an Ord-indexed family. Either kind of exhaustive uniform evaluation would therefore contradict the first assertion. This is the predecessor's diagonal obstruction in an explicit pair form. $\square$

A skeleton is therefore a dense *test family*, not a list of all objects. The new cut recognition theorem explains what extra information a class cut of that test family can encode; it does not defeat the diagonal theorem by relabelling cuts as sets.

Note 56.4 ([merge]; repetitions, and the new comprehension bound). The binary-class equimorphism (S16.15) and source 14's display are source 11's [theorem 19.32](#) (with source 08's [note 19.35](#) and source 12's [theorem 19.34](#)); source 16 adds the injective words $\mathfrak{J}$ between the two ends, by the same core-density argument. The diagonal theorems [theorem 56.1](#), [theorem 56.3](#) and source 15's [theorem 57.4](#) (printed with source 15's foundations in [section 57](#)) are source 11's [theorem 19.37](#) and source 12's [theorem 19.38](#), as are source 13's [theorem 42.7](#), source 17's [theorem 42.16](#) and source 18's [theorem 42.19](#) printed in Part VIII; all credit this report. New is source 16's [theorem 56.2](#): for transformations defined by a formula with only set quantifiers, GBC with Σ_1^1 class comprehension suffices to refute an order embedding of $\mathfrak{W}_{\text{all}}$ into $\mathfrak{W}_{\text{sl}}$, where source 11's [theorem 20.2](#) used KM + GC and arbitrary formulas. It is a sufficiency result; source 16 asks for the reversal ([research question 61.63](#)). Its proof uses the adjacent raw pairs only through the disjointness of their image intervals; with the pairs $(E, 0) < (E, 1)$ of $\mathfrak{W}_{\text{sl}} \times_{\text{lex}} 2$ in place of $R_F < S_F$ the same Σ_1^1 evaluator can be formed, which bears on source 13's [research question 61.24](#). (This adaptation is an observation of the merge, not a theorem of either source.)

57 Larger universes, class realizations and paradox avoidance (sources 16, 14, 13, 15 and 18)

Note 57.1 ([merge]; Part V counterparts). Five sources discuss the full-class model $(V_\kappa, \mathcal{P}(V_\kappa))$ at an inaccessible κ and the relativity of class notions. Source 16's [theorem 57.2](#) is source 12's exact external identification [theorem 21.1](#) (both identify $\mathfrak{W}_{\text{sl}}$ with $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$ and $\mathfrak{W}_{\text{all}}$ with $\mathcal{W}(\mathbf{No}_{<\kappa})$), and its [theorem 57.3](#) is source 11's [theorem 20.1](#) with one sharpening: already $\mathcal{W}_{\kappa+2}(X)$ does not embed into $\mathcal{W}_\kappa(X)$, although by [theorem 27.2](#) both have density and cellularity κ . Source 18 proves the same nonembedding by counting cellularity, display (S18.5). Source 14's model subsection recalls source 12's and source 09's calculations ([theorems 21.1](#) and [22.1](#)), and source 15's full-universe paragraph repeats the coding-length contrast of [theorem 21.1](#). The rest of these sections are foundational discussions in the manner of Part V ([section 23](#)); they are printed whole, in the order

16, 14, 13, 15, 18, because each states its own list of invalid inferences and its own boundary of comprehension.

57.1 Source 16: an explicit larger-universe interpretation

To treat the entire collections as actual sets, choose an ambient ZFC universe and assume that κ is strongly inaccessible. Consider the two-sorted structure

$$\mathcal{M} = (V_\kappa, \mathcal{P}(V_\kappa)).$$

Its sets are the elements of V_κ , and its available classes are *all* external subsets of V_κ . It satisfies KM plus global choice. Inaccessibility controls replacement: the image of an internal set under any available class function has size less than κ and bounded rank. Full class comprehension is immediate from the external power set. Such full models are a standard way to make the next logical level explicit [28].

Theorem 57.2 (Exact external identification). *In this full model the internal surreal class is externally $X = \mathbf{No}_{<\kappa}$, of cardinality κ . Internally set-like well-orders are exactly the externally well-ordered relations of type κ on X . All internal class well-orders correspond exactly to the full external set $\mathcal{W}(X)$. Thus the two comparisons become*

$$\mathfrak{W}_{\text{sl}}^{\mathcal{M}} = \mathcal{W}_\kappa(X), \quad \mathfrak{W}_{\text{all}}^{\mathcal{M}} = \mathcal{W}(X). \quad (\text{S16.17})$$

Proof. All sign sequences of length below κ belong to V_κ , and these are exactly the internal sign sequences. Strong inaccessibility gives $|X| = 2^{<\kappa} = \kappa$. Every external binary relation on X is an available internal class. Well-foundedness agrees in this full interpretation; in particular any external descending ω -sequence has bounded rank below κ and is an internal set witnessing failure.

A subset of X is an internal set exactly when it has external size less than κ . One direction follows from the strong-limit property of V_κ ; the other follows from regularity bounding the ranks of fewer than κ elements. Consequently an external well-order of cardinality κ is internally set-like exactly when all its proper initial segments have cardinality less than κ , which is exactly external order type κ . All other external well-orders are still admitted classes, yielding the unrestricted identification. $\square$

Theorem 57.3 (Strict embedding separation in the full model). *In the preceding external interpretation,*

$$\mathfrak{W}_{\text{sl}}^{\mathcal{M}} \hookrightarrow \mathfrak{W}_{\text{all}}^{\mathcal{M}}, \quad \mathfrak{W}_{\text{all}}^{\mathcal{M}} \not\hookrightarrow \mathfrak{W}_{\text{sl}}^{\mathcal{M}}.$$

Indeed even $\mathcal{W}_{\kappa+2}(X)$ does not embed into $\mathcal{W}_\kappa(X)$, although both have density and cellularity κ .

Proof. Inclusion gives the forward embedding. The target $\mathcal{W}_\kappa(X)$ is dense and has a dense subset of size κ . For fixed $a < c$ there are 2^κ prefixes $F : \kappa \rightarrow X \setminus \{a, c\}$. Their two completions $F + (a, c) < F + (c, a)$ are disjoint adjacent pairs in $\mathcal{W}_{\kappa+2}(X)$. An order embedding into the dense target would turn these pairs into 2^κ disjoint nonempty open intervals. Each interval must meet the κ -sized dense subset, contradicting Cantor's theorem. $\square$

This rederives the earlier report's full-model nonembedding, while identifying why equal cardinality, ordinal spectra, density, and cellularity can still fail to imply equimorphism. The adjacent pairs provide an additional obstruction when the target is dense.

57.1.1 Henkin models and internal well-foundedness

For a general model $(M, \mathcal{C})$, one has only the admitted classes $\mathcal{C} \subseteq \mathcal{P}(M)$, not automatically every external subset. External families of these classes need not be coded by any one internal family relation. If the model is externally countable, both virtual collections can be externally countable, and the external dense set-like order can admit an external copy of the raw order. Such an embedding need not be internally definable or uniformly represented, and so does not contradict [Theorem 56.2](#) or the internal diagonal theorem.

There is also a precise correction to a common ambiguity. Over GBC, for a fixed class relation, well-foundedness on nonempty sets, well-foundedness on all available nonempty subclasses, and absence of a set-coded descending ω -sequence are equivalent. Therefore two class expansions with the same sets agree about internal well-foundedness of any relation they share. Expanding the class collection can add relations; it does not change that property of a shared relation. Nevertheless, even a transitive class model can contain a relation it regards as well-founded that is externally ill-founded; one must distinguish a general model from a β -model [27, §3].

The full inaccessible model removes this ambiguity by its explicit fullness and rank closure. A singular birthday cutoff is a legitimate set alphabet for [Note 32.1](#); it is not an automatic replacement for V_κ in this model-theoretic construction.

57.2 Source 14: universe-relative models and paradox avoidance

57.2.1 A full-class model with externally set-sized objects

For a concrete interpretation, assume in an external set universe a strongly inaccessible cardinal Ω and consider

$$(V_\Omega, \mathcal{P}(V_\Omega)).$$

Internally the ordinal and surreal collections are proper classes; externally they are the sets Ω and $X = \mathbf{No}_{<\Omega}$. Strong inaccessibility gives $|X| = \Omega$. A baseline is externally a bijection $b : \Omega \rightarrow X$. The core consists of permutations with support of size $< \Omega$, equivalently bounded support in Ω , and has external cardinality Ω . The external global-enumeration order is

$$\text{Bij}(\Omega, X), \quad |\text{Bij}(\Omega, X)| = 2^\Omega.$$

These are the predecessor's full-model calculations ([theorems 20.1, 21.1 and 22.1](#); the universe-model sections at the pin). The lower bound for the second cardinality is pair orientation; the upper bound counts graphs. For the core, each bounded stage has fewer than Ω permutations by the strong-limit property, while first-entry transpositions supply Ω distinct codes.

Internally “every set cut” means externally cuts with fewer than Ω parameters. Thus the core and X are externally Ω -sized η_Ω -orders. The full enumeration order has strictly more points while the core remains dense in it. This is an ordinary set-theoretic example of a dense order and a larger order of its selected cuts; no paradox occurs.

The new trichotomy becomes an external classification of cuts using prefixes of length $< \Omega$. A branch extending through all internal ordinals has external length Ω . It is not an internally set prefix to which one may apply a small-cut theorem. The coverage condition is precisely surjectivity onto X .

All external well-orders on X also allow order types α with $\Omega < \alpha < \Omega^+$. They correspond to internal non-set-like orders, not to additional bijections $\Omega \rightarrow X$. Consequently “increase the universe so everything is a set” does not license replacing one interpretation by another without recording the allowed ordinal lengths.

57.2.2 ZFC schemes, GBC, KM, and higher sorts

In ordinary ZFC, sign sequences and their set-sized fragments can be handled directly; global statements can be treated as schemes for specified definable class parameters. A definable baseline must be supplied where needed. Formula codes alone do not solve the uniformity problem: evaluating every possible class definition in the whole universe requires truth resources not furnished by a list of formulas.

In two-sorted GBC, the class parameters and set-coded skeleton are literal objects of their respective sorts. The classification theorem has the form “for every class cut D , one of the stated alternatives holds.” It is not a third-sort object containing all those cuts. Stronger comprehension in KM does not change the rule that proper classes cannot be members of ordinary classes. A literal collection of class orders or cuts requires a specified additional sort or an external universe. Relative consistency assumptions for any such enlargement must be stated separately.

In a universe-polymorphic type theory, one can form a type of relations on a fixed small carrier. A larger-universe type can hold all such relations without identifying it with the original carrier. Lifting a carrier changes its ambient type level, not its mathematical elements; constructing surreals with larger allowed ordinal birthdays adds new elements. These are different operations. A proof assistant should not hide the distinction under coercions or ambiguous names.

57.2.3 Which assertions depend on the class realization?

Changing the available classes while retaining the same sets leaves each existing support code and each existing prefix unchanged. It can add new global enumerations, binary class words, and cut classes. [Theorem 48.1](#) is a more precise invariant than merely saying that first disagreements are unchanged: it prohibits a new enumeration from realizing an *old* cut. At the same time the external cardinality and order type of the entire admitted enumeration collection can change. The results are compatible, because a new point brings a new trace parameter.

For unrestricted orders, having an initial branch reconstructible from an old trace does not similarly force an arbitrary new *tail* to be old. The trace deliberately forgets that tail. This illustrates why one must not generalize the absoluteness theorem from set-like realizations to uniqueness of raw realizations.

57.2.4 A short paradox audit

There is no set ordinal containing all ordinals, no surreal whose sign sequence has domain Ord, and no application here of the surreal cut constructor to arbitrary proper-class option sets. All recursive histories used for the main constructions at a set stage are sets. Class predicates such as “is a global enumeration” are never turned into same-sort membership objects. Equal abstract class types are not silently represented by a canonical set ordinal. The baseline choice is explicit. Finally, class-model absoluteness is proved from formulas with the stated old parameters, not assumed for all higher-order properties. These restrictions are what allow the positive theorems, rather than obstacles to developing the subject.

57.3 Source 13: foundational interpretations without paradoxes

57.3.1 Ordinary set theory: the cutoff-first route

For any set ordinal θ , the sign sequences of birthday below θ form a set. Relations on this set form another set, and the subset satisfying well-order axioms is available by separation. All fixed-carrier order theory therefore lives in ordinary ZFC; no inaccessible cardinal or extra class sort is needed.

When a proof uses a family of all birthday cutoffs, the family is a class description rather than a set containing the entire surreal line. A compatible directed system requires explicitly commuting transition maps. Merely proving existence of an embedding at each cutoff does not make the union of independently chosen maps a function.

For set-length words, a word is itself a set of ordered pairs. The class of all such words is a legitimate class of sets. It is not the same logical object as the totality of class-length words.

57.3.2 Two-sorted class theory: quantify, do not collect

In GB or GBC, class variables can stand for relations on **No**. One may assert

$$\forall R \forall S [\text{ClassWO}(R) \wedge \text{ClassWO}(S) \implies R = S \vee R <_{\text{lex}} S \vee S <_{\text{lex}} R].$$

There is no need to form an object whose members are R and S . The fraktur notations $\mathfrak{W}_{\text{sl}}, \mathfrak{J}, \mathfrak{W}_{\text{all}}$ used here abbreviate precisely such restricted class quantification.

A family indexed by an ordinary class K of sets is supplied by one class relation $T(k, \alpha, x)$. First-disagreement comparison of two rows is elementary in T . If repeated rows occur, a fixed set-like global well-order selects the least set code in each equal-row class; row equality itself quantifies only over set coordinates and values. The distinct-row representatives then form an ordinary class linear order, which embeds into **No**.

This explains how every uniformly class-coded portion of $\mathfrak{W}_{\text{sl}}$ can embed into **No** while an exhaustive uniform code family is impossible. The local statement is about a particular class evaluator. It does not quantify a class of all evaluators or identify all class functions with their set codes.

57.3.3 The exact comprehension boundary of the new proofs

The crossing predicate

$$\text{Cross}(L, p) \iff (\exists d \in \mathcal{P}_s(b)_p \ d \in L) \wedge (\exists d \in \mathcal{P}_s(b)_p \ d \notin L)$$

quantifies over set codes. “For every ordinal length a crossing prefix exists” is therefore elementary with L as a class parameter. The uniqueness lemma makes $F(\alpha) = p_{\alpha+1}(\alpha)$ an elementary class definition. This is the reason cut reconstruction does not need ETR or impredicative comprehension.

The uniform complement isomorphisms, support completions, and active prefixes similarly use set histories. In contrast, the inverse evaluator in [Theorem 55.4](#) asks whether *some class enumeration* produced a specified code. That is a genuine existential class quantifier in a class comprehension formula. The proof uses KM at exactly this point.

General elementary transfinite recursion allows class-valued stages along a class well-order. It is distinct from the set-valued recursion used above. Its relevance to abstract class-well-order comparison is discussed in [\[27\]](#); none of our direct labeled comparison arguments relies on the stronger comparison theorem.

57.3.4 Kelley–Morse does not turn classes into members

KM permits class quantifiers in comprehension formulas, but its classes still have sets as members. Thus even KM does not justify an ordinary power class whose elements are all class well-orders of **No**. A third sort, a containing universe, or a fixed syntactic schema is necessary for arbitrary maps, topologies, or families whose elements are such classes.

Hyperclass forcing is an example of work that explicitly addresses this higher level, with additional model and coding hypotheses [28]. Its existence as a theory should not be read as permission to suppress the level distinction in an argument stated only over GB or KM.

57.3.5 Full universes versus selected class realizations

The full interpretation $(V_\kappa, \mathcal{P}(V_\kappa))$ at a strongly inaccessible κ is a convenient sufficient model for our external cardinal arguments. The available internal surreals are externally the set $\mathbf{No}_{<\kappa}$, and internal set-like enumerations have external length κ . A raw well-order may have any external ordinal type of cardinality κ ; types beyond κ create proper-class predecessor segments internally.

For a nonfull model $(M, \mathcal{C})$, only classes in $\mathcal{C}$ are available. Changing $\mathcal{C}$ can change the available branches, global enumerations, cuts, and external maps without changing the underlying set universe. The cut theorem remains an internal statement about the cuts actually present. If an outer class realization adds a normalized cut satisfying the crossing criterion, its unique branch becomes available there by elementary comprehension from that cut. Thus missing classes, not merely missing set elements, are part of the geometry.

Internal well-foundedness must also be kept separate from external well-foundedness. Over the relevant global-choice class theories, class well-foundedness can be tested by the absence of a set-coded descending ω -sequence; an external descending sequence need not be so coded in the model [27, Section 3]. This is why an arbitrary internally well-ordered class is not automatically a well-order in a surrounding metatheory. Our full inaccessible model avoids that particular ambiguity, while [Proposition 55.5](#) deliberately studies an external order of available objects.

[merge] Source 13’s subsection “Relation-bit lexicography is a different order” comes here in its text; it is printed in [section 38](#).

57.3.6 A concise list of invalid inferences

The most tempting false steps are logically different. A proper-class ordinal word is not a surreal; a predicate on surreals is not necessarily a small set; a set of individual class witnesses is not automatically a uniform family; a class well-order need not be set-like; an order embedding need not preserve suprema, cuts, or adjacency; and equal cardinality, equimorphism, and isomorphism are three distinct claims. Finally, the isomorphism type of an external available-object order need not describe the internally definable transformations on that order. Each of the main theorems above uses at least one of these distinctions essentially.

57.4 Source 15: formalization without paradoxes

57.4.1 Four sound frameworks and their different meanings

Framework	What it directly represents	Important boundary
ZFC, fixed sets	Relations on a set X , all ordinal enumeration types, binary cubes, birthday cutoffs	No set of all surreals; a definable-class claim is a scheme or a formula.
Two-sorted GB/GBC	Individual class orders, uniform families, elementary raw comparison and set-interval condensation	Classes contain sets, not other proper classes.

Framework	What it directly represents	Important boundary
KM and recursion extensions	Stronger class comprehension; appropriate class-valued recursions	Stronger comprehension does not add a third sort automatically.
Containing universe or stratified types	A lower universe's classes become higher-universe objects	Size and well-foundedness are relative; lifting does not identify all levels.

57.4.2 Why Burali–Forti and unrestricted cuts do not arise

In the sign presentation, each birthday is a set ordinal, while birthdays are unbounded in the class of ordinals. The order type of an arbitrary class well-order is not asserted to be a set ordinal. An expression such as $\text{Ord} + \text{Ord}$ abbreviates a relation on a class of tagged sets. A collection of all these relations is treated either by class-variable predicates or in a genuinely larger universe.

Likewise, the cut principle permits sets $L < R$. Substituting all of **No** on the left and the empty class on the right would demand a surreal greater than every surreal and is outside that principle. The represented cut theorem explicitly admits unfilled class cuts, so it does not smuggle in such an unrestricted principle.

Theorem 57.4 (No exhaustive uniform class of set codes). *In GBC no class of set codes with one uniform decoder can list every class well-order of **No**. An omitted order can always be chosen set-like.*

Proof. Reindex any proposed family by Ord , repeating a default row if the indexing class is a set. Let its rows be R_α . Partition a baseline enumeration of **No** into consecutive two-label blocks, using positions $2 \cdot \alpha, 2 \cdot \alpha + 1$. In block α , place its two labels in the order opposite to their relative order in R_α . Listing these blocks constructs a class bijection $\text{Ord} \rightarrow \text{No}$, and its well-order differs from each row on that row's designated pair. Its definition uses only the one family relation and the baseline as class parameters. $\square$

This is the prior report's diagonal obstruction, restated to locate the size boundary (theorems 19.37 and 19.38; Theorems 19.37–19.38 at the pin). An ordinary class of codes may be dense and realize every set cut without listing every global enumeration. Formula strings do not evade the theorem: a uniform semantic evaluation of all formulas of the universe would require an appropriate truth predicate. Individual formulas, an external collection of definable classes of a model, and an internal universal evaluator are different resources.

57.4.3 Exactly where recursion principles enter

In GBC, for a class strict linear order, the following internal conditions are equivalent: every nonempty subclass has a least element; every nonempty set subset has a least element; and there is no descending ω -sequence in the model. If a subclass has no least element, choose below each of its members a witness of least possible set rank, then use global choice within that rank. Ordinary recursion with this class parameter constructs a descending ω -sequence, whose range is a set by class replacement. The other implications are the usual set well-foundedness arguments. This explains the role of choice in the equivalence; the article's primary definition remains the subclass minimum condition [27, Section 2].

Set-valued recursion along Ord can be assembled from its unique set initial solutions using elementary comprehension. This covers the ordinary surreal forth construction and prefix com-

pletions. The raw comparison formula, simultaneous canonical block quotient, and persistent-cut reconstruction do not even need a class-valued recursive construction.

A general recursion whose stage value is itself a proper class is a stronger request. The principle ETR supplies elementary transfinite recursion along class well-orders. It suffices for comparison of arbitrary abstract class well-orders by initial-segment isomorphisms [27, Theorem 6]. That is a different problem from comparing identical labelled prefixes. The cited work poses converse-strength questions; we do not turn those questions into a claimed independence theorem or assert that their historical open status is settled as of the date of this article. For general class recursion and truth-predicate strength see [22, 35].

For condensation, a fixed finite number of iterations is available by substituting elementary formulas. A uniform history on every arbitrary input is a separate statement; ETR of the appropriate length is sufficient, but this article proves no necessity. In the polynomial examples the entire history already has an elementary formula, including at stages with proper-class many predecessors. A warning about general recursion must not be used to reject an explicitly defined construction.

57.4.4 Full-universe and Henkin interpretations

Let a containing universe have an inaccessible κ , and consider $(V_\kappa, \mathcal{P}(V_\kappa))$. Internally, the surreals are $\mathbf{No}_{<\kappa}$ and the ordinals are the ambient κ . Every external subclass of the set universe is available as a class. Internal class well-foundedness therefore agrees with external well-foundedness for these relations.

The internal set-like enumeration space becomes the ambient $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$, while the full raw space becomes all ambient well-orders of the carrier. Both have cardinality 2^κ , and both embed exactly the ordinals below κ^+ , but their finer structure differs. The set-like space is equimorphic with $\mathbb{B}_\kappa$; the unrestricted one has unbounded binary coding lengths below κ^+ and cannot embed into that one cube. This is a genuine nonembedding theorem in a specified external model, not a same-level cardinality argument about a class of classes (theorems 20.1, 21.1 and 22.1; Sections 20–22 at the pin).

For a general two-sorted model $(M, \mathcal{C})$, only the classes in $\mathcal{C}$ may occur as relations or witnesses. A relation can be internally well-founded while an external descending sequence is absent from $\mathcal{C}$. A countable model may have externally countably many classes without possessing an internal exhaustive decoder. Adding classes can add new global enumerations and can destroy the internal well-foundedness of an old relation by adding a descending witness. The elementary first-disagreement formula on previously available valid orders remains the same, but the ambient totality and existence assertions may change.

57.5 Source 18: containing universes, class realizations, and paradox avoidance

The mathematical distinctions above are also distinctions between formal objects. A type, a set, a definable class, an available class in a two-sorted model, and a collection of available classes have different membership and quantifier rules. A rigorous development should choose one of the following settings for each theorem.

57.5.1 Ordinary ZFC and bounded birthday stages

All results about $\mathcal{W}_\alpha(\mathbf{No}_{<\kappa})$ are ordinary set theorems. There is no need for class theory, an inaccessible cardinal, or a global well-order to construct these spaces. Choice well-orders the set alphabet, and all enumerations and cuts live in ordinary power sets. In particular, the exact gap spectrum does not rest on the existence of an external model or on an unformalized “class ordinal.”

One may also work in ZFC with definable classes as metanotation. A particular formula defines a particular global relation, but one must not silently quantify in the object theory over all formulas and their semantic truth sets. A uniform truth predicate for the whole set universe is unavailable. A scheme covering each fixed defining formula is different from a single theorem about all available classes.

57.5.2 GBC and named class parameters

GBC is a natural setting for supplied global orders, uniform families, the elementary common-prefix comparison, the block decomposition, and the represented-cut criterion. These constructions use actual set labels to represent their quotients or prefixes. A block minimum is a set; the proper block it represents is not an element.

The status of a well-order is relative to the class realization. In a model $\mathcal{M} = (M, \mathcal{X})$, “every nonempty subclass” means every appropriate member of $\mathcal{X}$. An external descending sequence might not be represented in M or $\mathcal{X}$. Consequently external countability or external ill-foundedness cannot simply be substituted for internal set size or internal well-foundedness. The factorial-shuffle theorem carefully uses external countability and, for its finite blocks, ω -standardness; it does not assume that all internal well-orders are externally well-founded.

57.5.3 A full-class model at an inaccessible cardinal

For a concrete comparison with the external countable setting, let λ be strongly inaccessible in an ambient ZFC universe and consider the full class model

$$(V_\lambda, \mathcal{P}(V_\lambda)).$$

Its internal surreal carrier is the ambient set of sign sequences of length below λ , of ambient cardinality λ . Every ambient subset of the carrier is available as a class. Class well-foundedness therefore means actual external well-foundedness. An internal predecessor class is a set precisely when its ambient cardinality is below λ : regularity bounds the ranks of a small set of elements of V_λ , and inaccessibility gives the converse cardinal bound.

It follows that, from the ambient viewpoint,

$$\mathfrak{W}_{\text{sl}} = \mathcal{W}_\lambda(\mathbf{No}_{<\lambda}), \quad \mathfrak{W}_{\text{all}} = \mathcal{W}(\mathbf{No}_{<\lambda}).$$

The former has cellularity λ , since $\lambda^{<\lambda} = \lambda$. The latter has cellularity 2^λ : fix a residual set of three labels and vary the λ -long prefix on its complement. Each of 2^λ disjoint cylinders contains a nonempty open interval. Hence

$$\mathfrak{W}_{\text{all}} \not\rightarrow \mathfrak{W}_{\text{sl}} \quad \text{in this full-class interpretation.} \tag{S18.5}$$

An order embedding would send the endpoints and an interior point of each such interval to 2^λ disjoint nonempty intervals in the target, contradicting its cellularity.

There is no conflict with the external countable-model embedding in the preceding section. The two class realizations have different external sizes and admit different external families. There is also no conflict with surreal universality: $\mathfrak{W}_{\text{all}}$ is an external set of size 2^λ , not an internal class of set labels to which the internal universality theorem applies.

The canonical block theorem supplies a further consistency check. An ambient order of the carrier has ordinal type γ with $|\gamma| = \lambda$. Its division

$$\gamma = \lambda \cdot \beta + \rho, \quad \rho < \lambda,$$

is exactly the external interpretation of the internal decomposition into Ord blocks and a final set tail. Thus the class theorem recovers ordinary ordinal division in this model without treating Ord as an internal set.

57.5.4 KM, stronger recursions, and genuine higher sorts

KM permits impredicative class comprehension. It remains a two-sorted theory: classes are not members of classes. To make the whole domain of global well-orders an object one needs, for example, an explicitly specified higher sort, an external containing universe, or a carefully defined unrolling. These are substantive changes of framework, not notation that erases the diagonal in [Theorem 42.19](#).

The use of ETR should be equally explicit. General class-valued recursions, including iterated truth constructions, exceed GBC [\[24\]](#). ETR is a sufficient framework for suitable abstract class well-order comparisons and recursive quotient constructions [\[35\]](#). In contrast, three procedures in this article have elementary closed definitions or set-valued histories: comparison of shared labels, simultaneous set-interval decomposition, and reconstruction of a branch from unique set prefixes. None requires ETR. We do not infer an exact reverse-mathematical equivalence for an unspecified iteration of the quotient operator.

57.5.5 A concise list of invalid identifications

The following substitutions would change a theorem’s meaning or invalidate its proof.

1. Replacing Ord by a set ordinal while still claiming to enumerate all surreals of the same universe.
2. Treating all class relations as members of a class merely because each individual relation is available.
3. Replacing an arbitrary class well-order by an Ord enumeration; this discards its later proper blocks and set tail.
4. Replacing a proper-class cut by a set cut when applying surreal interpolation.
5. Using cardinal multiplication where the proof concerns ordinal lexicographic coding length.
6. Assuming that an order embedding preserves suprema or density in the ambient order.
7. Treating a set-indexed uniform family of classes as an ordinary set of class objects.
8. Extending numerical lexicography to isomorphism classes after discarding the labels on which it depends.

These are mathematical type distinctions, not additional consistency axioms imposed on the bounded results.

Note 57.5 ([merge]; internal well-foundedness under class extensions). The last paragraph of source 15’s subsection on full-universe and Henkin interpretations (in [section 57.4](#)) says that adding classes “can destroy the internal well-foundedness of an old relation by adding a descending witness”. For class realizations that are models of GBC with the same sets this is not so, and source 16’s subsection on Henkin models proves the opposite: over GBC a class relation is well-founded on all admitted subclasses exactly when no set-coded descending ω -sequence exists (source 15 proves the same equivalence in its subsection “Exactly where recursion principles enter”), and a descending ω -sequence admitted by a larger realization is a set of the common set part, by class Replacement, so it was already present. Hence two GBC realizations with the same sets agree on the internal well-foundedness of every relation they share. Source 15’s sentence can hold only

for a larger class collection in which that equivalence fails, for example one that is not a model of global choice; as a statement about GBC realizations it is withdrawn here. The same correction was made to source 08's foundations after the Hilbert's-tenth review of batch 80 (section 5.6 and the note before it). Source 15's [research question 61.56](#) asks which statements “require well-foundedness preservation”; by the argument above, none does among GBC realizations with the same sets.

58 An exact external classification in countable models (source 18)

The distinction between an internal class and an external collection of classes leads to a surprisingly uniform external order type. The statement below concerns the raw labelled comparison of all admitted class well-orders, including those that are not set-like. It does not concern their abstract order types. It strengthens the observation that an external countable collection of admitted orders can embed into the external rational line.

For a linear order L , write $x \sim_{\text{fin}} y$ when the closed interval between x and y is finite. These equivalence classes are the *finite-condensation blocks*. For a nonempty countable collection $\mathcal{A}$ of linear order types, its *shuffle* $\sigma(\mathcal{A})$ is obtained by densely colouring $\mathbb{Q}$ with one colour for each member of $\mathcal{A}$, then replacing a point by the order type of its colour. A colour-preserving back-and-forth proves that the resulting order type is independent of the chosen dense colouring. These are standard order-theoretic constructions; see, for example, Dimitrov–Harizanov–Morozov–Shafer–Soskova–Vatev, *On cohesive powers of linear orders*, Definitions 3.3 and 6.2, [36]. The identification with the well-order space below is derived here.

58.1 Terminal tails and finite-condensation blocks

The next arguments can first be carried out internally in GBC. Let X be an infinite linearly ordered class, and consider the predicate of all class well-orders of X , with the raw labelled lexicographic comparison. It is enough that the required well-orders and constructions exist; for $X = \mathbf{No}$, global choice supplies them.

Lemma 58.1 (The maximal finite terminal tail). *Every class well-order R on an infinite carrier has a unique finite terminal segment T_R such that R restricted to $X \setminus T_R$ has no greatest element. Write $n(R) = |T_R|$. Every finite terminal segment of R is contained in T_R .*

Proof. If R has no greatest element, take $T_R = \emptyset$. Otherwise start at the greatest element and repeatedly take its immediate predecessor, for as long as one exists. If this process did not terminate at a finite stage, ordinary set-valued recursion with parameter R would produce a descending ω -sequence. Class Replacement makes its range a set, contradicting well-foundedness. At termination the selected finite chain is terminal and the remaining prefix has no greatest element. Any finite terminal segment is obtained by finitely many of the same uniquely determined predecessor steps, proving maximality and uniqueness. $\square$

Lemma 58.2 (Adjacency forces a finite residual carrier). *Suppose $R < S$ are adjacent in the lexicographic order of all class well-orders of X . Let C be their common labelled prefix, and let $a = \min_R(X \setminus C) < \min_S(X \setminus C) = b$ in the ambient label order. Then $Y = X \setminus C$ is finite.*

Proof. Keep the prefix C , then the next label a , fixed. The R -tail must be the greatest available well-order in that prefix cylinder: any greater tail gives an order strictly between R and S . At each subsequent position, therefore, its label must be the greatest remaining label. Otherwise choose a

larger remaining label and well-order the remaining carrier to obtain a greater tail. Consequently the reverse ambient label order on $Y \setminus \{a\}$ is a class well-order.

The symmetric argument, fixing C then b , shows that the ambient label order on $Y \setminus \{b\}$ is a class well-order. On $Z = Y \setminus \{a, b\}$, both an order and its reverse are therefore well-orders. This forces Z to be finite. Indeed, if Z were infinite, repeatedly choosing the least unused point in one of those well-orders would produce an ascending ω -sequence in that order, hence a descending ω -sequence in the other. Thus Y is finite. $\square$

Proposition 58.3 (Factorial blocks). *The finite-condensation block containing R consists exactly of the orders obtained by keeping R fixed on $X \setminus T_R$ and permuting the $n(R)$ labels in T_R . It is a finite chain of size $n(R)!$.*

Proof. The common-prefix cylinder with residual carrier T_R is convex and contains exactly its $n(R)!$ permutations. In lexicographic order they form a finite chain, so they lie in one finite-condensation block.

Conversely, any adjacent order S has a finite residual carrier at the first disagreement, by the preceding lemma. That residual is a finite terminal segment in R , hence lies in T_R . Thus an adjacent step cannot leave the displayed permutation cylinder. A finite interval is a finite chain of adjacent steps; no member outside this cylinder can be at finite distance from R . This proves the assertion. $\square$

The cases $n = 0$ and $n = 1$ both give a singleton block. The maximal finite-tail length distinguishes these as labelled well-orders, but their block sizes do not distinguish them in the pure linear order.

58.2 Every factorial block size occurs densely

Lemma 58.4 (An interval across different blocks contains an infinite residual cylinder). *If $R < S$ belong to different finite-condensation blocks, there is a labelled prefix cylinder lying entirely inside (R, S) whose residual carrier is infinite. The prefix may be a proper class.*

Proof. Let C be the common prefix, $Y = X \setminus C$, and let $a < b$ be the first unequal labels. The class Y is infinite, since a finite residual would put the two orders in one finite permutation cylinder.

Consider the following two possible changes. In R , at a position $x \in Y \setminus \{a\}$ outside T_R , keep the whole earlier prefix and replace x by a larger unused label. In S , at a position $x \in Y \setminus \{b\}$ outside T_S , keep the whole earlier prefix and replace x by a smaller unused label. A change of the first kind gives a prefix cylinder above R and below S , because the earlier first label a remains fixed; a change of the second kind gives the same conclusion with the first label b fixed. The unused carrier before such a change is infinite, since the changed position lies outside the maximal finite terminal tail. Removing one label leaves the new cylinder with infinite residual carrier.

At least one such change must be possible. Otherwise let

$$Z = Y \setminus (\{a, b\} \cup T_R \cup T_S).$$

This class is infinite. In R , every $x \in Z$ must be numerically greater than each of its R -successors in Z , since any larger unused successor would permit the first change. Thus $R \upharpoonright Z$ is the reverse ambient label order. In S , every $x \in Z$ must be numerically smaller than each of its S -successors in Z , since a smaller one would permit the second change. Thus $S \upharpoonright Z$ is the ambient order. Both restrictions are class well-orders, contradicting the infinitude of Z as in the preceding proof. The orders in this cylinder also lie in blocks different from both endpoint blocks. For an upward change,

its first disagreement with R occurs while the residual is infinite, and its first disagreement with S occurs at the original infinite residual Y ; the downward case is symmetric. $\square$

Lemma 58.5 (Prescribing the finite tail). *For every $n \in \omega$, every interval joining different finite-condensation blocks contains an order W with $n(W) = n$.*

Proof. Take the cylinder from Lemma 58.4, with prefix P and infinite residual carrier Y . Choose n distinct labels $F \subseteq Y$. Well-order $Y \setminus F$ without a greatest element: if it is a set, use its infinite initial cardinal as order type; if it is a proper class, use a set-like well-order of type Ord . Both are available in GBC. Append these labels after P , and then append the n labels of F in any order. The resulting well-order belongs to the cylinder, and its maximal finite terminal tail is exactly F . $\square$

These arguments are parameterwise constructions of individual classes. They do not assert that the collection of all orders, or its quotient, is an internal class whose elements are classes.

58.3 The external classification

Let $\mathcal{M} = (M, \mathcal{X})$ be an externally countable two-sorted model of GBC. Write $\mathfrak{W}_{\text{sl}}^{\mathcal{M}}$ and $\mathfrak{W}_{\text{all}}^{\mathcal{M}}$ for the external sets of its admitted set-like and arbitrary class well-orders of its internal surreal carrier. Comparisons are those computed in $\mathcal{M}$.

Theorem 58.6 (Countable-model factorial-shuffle theorem). *If $\mathcal{M}$ is ω -standard, then, externally,*

$$\mathfrak{W}_{\text{all}}^{\mathcal{M}} \cong \sigma(\{1, 2, 6, 24, 120, \dots\}) = \sigma(\{n! : n \in \omega\}).$$

Its finite-condensation quotient is a copy of $\mathbb{Q}$, and each factorial block size occurs densely in that quotient. In particular, the pure external order type is independent of the particular countable ω -standard model and its admitted collection of classes.

For every externally countable model of GBC, even without ω -standardness,

$$\mathfrak{W}_{\text{sl}}^{\mathcal{M}} \cong \mathbb{Q}.$$

Proof. Internal linearity gives an actual external linear order on each admitted collection. The collections are countable because the entire class sort is externally countable. Internal nonemptiness and variation of the first surreal label make them infinite and give no endpoints.

In an ω -standard model, the number $n(R)$ of Lemma 58.1 is a standard natural number. Its $n(R)!$ -element permutation cylinder is therefore genuinely externally finite. Proposition 58.3 identifies all external finite-condensation blocks. Its converse is also external: an external finite interval is a standard finite chain of adjacent steps, and adjacency has its literal external meaning because the class sort ranges over precisely the admitted orders.

The quotient is countable, has no endpoints, and is dense by Lemma 58.5. The same lemma makes every block-size colour dense. The countable colour-preserving back-and-forth identifies it with the densely coloured rational order used in the definition of the shuffle. Each finite block is a rigid chain, so this quotient isomorphism lifts uniquely within each block to the claimed order isomorphism.

For the set-like orders, their internal lexicographic order is dense: at the first unequal ordinal position, the common prefix is a set; choose an unused surreal strictly between the next two labels and complete the prefix to a global enumeration. The surreal cut-filling property permits avoidance of the set of used labels. There are no endpoints by varying the first label. Thus its external order is a countable dense linear order without endpoints, hence isomorphic to $\mathbb{Q}$. This part does not use ω -standardness. $\square$

Remark 58.7 (An explicit computable presentation). For $q \in \mathbb{Q}$, let $d(q) > 0$ be its denominator in lowest terms, and set $c(q) = (1 + v_2(d(q)))!$. Then the lexicographically ordered set

$$\{(q, i) : q \in \mathbb{Q}, 0 \leq i < c(q)\}$$

is an explicit computable copy of the factorial shuffle. For each fixed $k \geq 0$, reduced rationals with denominator of 2-adic valuation k are dense: choose a sufficiently large odd denominator factor and an odd numerator inside the desired interval. Cancelling common odd factors does not change that valuation. This computable presentation does not supply a computable decoder for the model's class relations.

Remark 58.8 (The hypothesis cannot simply be dropped). If $\mathcal{M}$ is ω -nonstandard, choose a nonstandard $N \in \omega^{\mathcal{M}}$ and append an internally N -element tail to a global well-order without a last element. The corresponding internal $N!$ -element permutation cylinder is externally an infinite discrete interval with endpoints. Starting at its first element and taking the immediate successor any standard finite number of times produces infinitely many points in one external finite-condensation block. Consequently the external blocks need not be finite, and the displayed factorial-shuffle classification fails. The proof requires standardness of ω , not external well-foundedness of every admitted class relation; no β -model assumption is needed.

Corollary 58.9 (External embeddings and intrinsic automorphism orbits). *In an externally countable ω -standard model, the unrestricted order is not externally isomorphic to its set-like suborder: it has adjacent pairs. Nevertheless it externally embeds into that suborder, since every countable linear order embeds into $\mathbb{Q}$.*

The automorphism group of the unrestricted external order is the automorphism group of the rational quotient coloured by factorial block size. Its point orbits are indexed by a factorial block size and a position within that block.

Proof. Only the automorphism assertion needs comment. Finite-distance blocks and their sizes are preserved by every order automorphism. Every colour-preserving quotient automorphism lifts uniquely because finite chains are rigid. Conversely, the colour-preserving back-and-forth is transitive on the quotient points of each fixed colour. Hence two points lie in the same orbit exactly when their block sizes and within-block positions agree. $\square$

Corollary 58.10 (Set-likeness is not definable from the external order). *In the countable ω -standard setting, the distinguished subset $\mathfrak{W}_{\text{sl}}^{\mathcal{M}}$ is not first-order definable, even with a finite tuple of parameters, in the pure linear order $\mathfrak{W}_{\text{all}}^{\mathcal{M}}$.*

Proof. Set-like orders have no last element and hence lie in singleton blocks. Orders of type $\text{Ord} + 1$ also lie in singleton blocks, since their maximal finite terminal tail has size one, but they are not set-like. The preceding automorphism description can exchange any two singleton-block points.

To allow finitely many parameters, take a surreal label smaller than the first labels of all the parameter orders. There are both a set-like order and an order of type $\text{Ord} + 1$ starting with that label. They lie below all the parameters and occupy singleton blocks. A colour-preserving quotient back-and-forth sends one to the other while fixing each parameter block, and its lift fixes the parameter points. No formula with those parameters can distinguish the two points. $\square$

The natural inclusion of $\mathfrak{W}_{\text{sl}}^{\mathcal{M}}$ in $\mathfrak{W}_{\text{all}}^{\mathcal{M}}$ is not dense. For example, fix a proper class prefix of type Ord leaving three labels, and vary the final three labels. The resulting six-point cylinder has a nonempty open interval containing no set-like order. More substantially, leaving an infinite set of labels gives whole intervals between different finite-condensation blocks that contain no set-like

order. Thus the pure order classification does not classify the expanded pair $(\mathfrak{M}_{\text{all}}^{\mathcal{M}}, \mathfrak{M}_{\text{sl}}^{\mathcal{M}})$, or the external evaluation relation connecting these orders to their surreal labels. Those richer structures remain appropriate further research targets.

Note 58.11 ([merge]; relation to sources 13 and 17 and to Part V). The second clause of [theorem 58.6](#) ($\mathfrak{M}_{\text{sl}} \cong \mathbb{Q}$ externally in every externally countable model) is source 13’s [proposition 55.5](#), with the same proof; source 11 already observed, after [theorem 20.2](#), that in such a model the external set-like order can admit an external copy of the unrestricted order, which source 18’s [corollary 58.9](#) makes precise. [Lemmas 58.1](#) and [58.2](#) and [proposition 58.3](#) are a second proof of source 17’s [theorem 54.4](#). New in this section are the factorial-shuffle classification of the full order in countable ω -standard models, its computable presentation, the failure without ω -standardness, the description of the automorphism orbits, and the non-definability of set-likeness in the pure order ([corollary 58.10](#)).

59 Formalization plans of sources 13–18

Note 59.1 ([merge]; status of the six plans). None of sources 13–18 ships Lean code, and no statement of theirs is formalized ([section 1.5](#)). Their plans are printed whole, in source order. Every existing declaration they cite exists at the current revision, including `SignSequence`, `SignSequence.lt_iff`, `birthdays_bounded`, `small_bounded`, `not_small`, `ordinalOrderEmbedding`, `small_cut_fillers`, `exists_cut_separator_of_birthday_lt`, `cut`, `cut_realizes`, `birthday_cut_le_iSup`, `exists_separator_of_small_sets`, `existsUnique_simplest_separator`, `SmallCutData`, `noUnrestrictedCuts`, and, cited by source 18, `le_iff_options_lt` and `lt_iff_exists_option` in `SignSequenceComparison.lean`. The modules they propose do not exist: source 14’s nine (`LabelledCommonPrefix ... ClassRealizations`), source 17’s eight files under `Well0Orders/`, `ClassOrders/` and `Surreal/`, and source 16’s `Well0OrderOn X`; sources 13, 15 and 18 describe layers without file names. Source 14’s remark that the existence half must be imported separately is already carried out in the tree (the merge note inside its plan).

59.1 Source 13: a formalization plan tied to the inspected Lean code

59.1.1 What already exists at the pinned snapshot

The inspected source `Surreal/Foundations/SignSequence.lean` defines [7]:

```
structure SignSequence : Type (u + 1) where
  birthday : Ordinal.{u}
  signAt : Ordinal.{u} -> SignType
  signAt_ne_zero_iff :
    forall i, signAt i != 0 <-> i < birthday
```

This display is an ASCII schematic rendering of the actual declaration, not a new compiled Lean file. The source uses Lean’s native logical notation. Its numerical linear order is lifted from the lexicographic order of zero-extended sign functions.

Inspected declaration	Role in this development
<code>SignSequence.lt_iff</code>	First-disagreement characterization of numerical comparison.
<code>ordinalOrderEmbedding</code>	The all-plus embedding of the ordinal carrier into the sign carrier.

Inspected declaration	Role in this development
<code>birthdays_bounded</code>	A strict birthday bound for a family with an explicit lower-universe smallness instance.
<code>not_small</code>	The whole sign carrier is not small in the universe indexing its birthdays.
<code>small_bounded</code>	Each fixed birthday-bounded fragment is small relative to that lower universe.
<code>exists_cut_separator_of_birthday_lt</code>	A separator of exactly a prescribed birthday above all option birthdays; the fresh separator lemma’s direct input.
<code>small_cut_fillers</code>	The constructed carrier realizes every small strict cut.

The last two are in `SignSequenceCut.lean` [8]. They are not merely postulated field axioms: the inspected source constructs them from fixed-length Boolean lexicographic words. We inspected the files, not a fresh build of the project. No theorem in this manuscript is thereby automatically a kernel-checked theorem.

59.1.2 Relative smallness must stay visible

`SignSequence.{u}` and `Ordinal.{u}` both live in `Type (u + 1)`, while bounded sign sequences and initial segments are small relative to `u`. A `Set SignSequence` is merely a predicate on the larger carrier. It need not be an admissible small family of options. The formal counterpart of “set cut” therefore requires an index type with `Small.{u}`, not just the word `Set` in a declaration.

The same applies to supports and prefix lengths. A support that is a predicate on a larger type is not automatically a small support. A code should carry a small graph or an explicit smallness proof. Otherwise the completion bound, the dense-core distinction, and the diagonal interpretation can silently change meaning.

A universe-relative approach may take a baseline equivalence

$$b : \text{Ordinal}.\{u\} \simeq \text{SignSequence}.\{u\}$$

as an explicit parameter. Its existence is not inferred merely because the types occupy the same universe. A formal project must either construct the required equivalence, with the intended small-initial- segment properties, or retain it as a stated assumption.

59.1.3 Suggested modules and their proof obligations

The following names are proposals, not claims about existing files.

Exhaustive orders and finite predicates. A first module should define exhaustive ordinal enumerations on a small carrier, prove prefix-freeness, and compare the relation-based and enumeration-based first-disagreement definitions. These proofs do not require surreal arithmetic or class theory. The finite checker included here provides examples, not a substitute for this module.

Prefix codes and completion. Represent a moved-point graph canonically and prove evaluation injective on codes. Prove the complement-cardinality extension lemma before using any prefix-completion operation. Formulate cylinders by restriction equality and prove convexity and same-level disjointness.

Small-family interpolation. Use an explicit small index type for the two families. Establish a bound on cross first-disagreement coordinates, then implement the active-row invariant with a set-sized history. The stopping lemma, not a generic unbounded recursion tactic, is the critical finite-length formal obligation at this stage.

Cut traces and failure. Define crossing by two existential code witnesses. Same-level uniqueness and restriction coherence are purely order-theoretic. Under all-level existence, reconstruct the value at α from the unique prefix of length $\alpha + 1$. Prove trace equality with the no-greatest-lower normalization stated explicitly. For first failure, split successor from limit ordinals and prove the boundary formula rather than assuming that crossing commutes with limit intersection.

Products and recoding. First formalize [Lemma 55.2](#) for ordinary set types and an order-dense subset. It has no class-theoretic ambiguity and should be a small, reusable result. The uniform complement-map interface then supports prefix insertion and finite-block automorphisms. The KM inverse evaluator belongs in a model or syntax layer, not in an unqualified theorem about arbitrary Lean functions on a larger type.

Central words. Represent termination by an option type or by a reserved symbol with an explicit proof that it is not a letter. For injective words, retain injectivity as an invariant when selecting the shared next letter. The explicit sign-block embedding can be formalized independently of the class back-and-forth isomorphism.

59.1.4 Three complementary formalization routes

The *cutoff-first route* proves the fixed-set and density-bound lemmas inside ordinary types. It requires the least foundational infrastructure and makes cardinal hypotheses explicit.

The *universe-relative route* abstracts a large ordered carrier with a lower-universe small-cut operation, a set-like enumeration, and support codes. It can certify the local combinatorics and many of the new constructions relative to those interfaces. It should not be described as a proof of every GBC or KM metatheorem merely because it uses two universes.

The *deep model route* formalizes a two-sorted class theory or a specified model $(M, \mathcal{C})$. This is the route for the exact comprehension schema, the external countable contrast, and the full-model interpretation. Taking all predicates on an arbitrary carrier as “classes” does not by itself establish the intended replacement or universe axioms; the model assumptions must be proved or supplied.

59.1.5 Verification boundary

A completed formalization should pin both the repository revision and its dependency revisions, compile the new modules, and audit theorem axioms and unfinished proofs. None of those actions is claimed here. The deliverable is a proof manuscript, a reproducible LaTeX build, and finite consistency checks. Existing Lean inputs are identified by actual declarations and source hashes in the audit appendix, so that a future formalization has concrete starting points.

59.2 Source 14: a realistic formalization plan

The new proofs admit a modular formalization in which most of the work is ordinary order theory. A full class-theoretic metamodel should not be the first dependency. The finite checks shipped with this article are useful regression tests, but do not discharge any infinite or class-level theorem.

59.2.1 A generic set-level kernel

Start with a type X carrying a linear order and a chosen ordinal or well-ordered index type. An exhaustive presentation is a bijection from that index type to X . For arbitrary well-order types on one fixed carrier, relations are often easier than a dependent sum over every ordinal: a relation together with linearity and well-foundedness proofs is already a set-sized object at the carrier's type level.

Prove the labelled common-prefix comparison before importing surreal arithmetic. Its main obligations are initiality, agreement of the restricted relations, existence and inequality of the two next labels, and transitivity via nested initial segments. For fixed-domain words, prove the equivalent least-disagreement formulation. Finite relation matrices and permutation words give independently checkable examples.

A second module handles partial-injection extension. Its specification must retain *agreement on a prescribed prefix* and the *support cardinality bound*; bare existence of an extension is not enough for the compression theorem. The finite case and the infinite buffer case should be separate lemmas, since equality of infinite cardinalities of domain and range alone does not imply equality of their complement cardinalities.

59.2.2 An abstract saturated alphabet interface

The core theory needs an ordered alphabet with set-cut filling and fresh separators, a global baseline in the class version, and set-prefix completion. Numerical surreal field operations are not required. A bounded preliminary development may use a regular-cardinal saturated set order and prefix lengths below that cardinal, with all relevant smallness hypotheses explicit. It proves the crossing-chain and bounded versions of the trichotomy without formalizing surreal addition.

For actual sign sequences, the pinned repository already supplies simplicity lemmas. The inspected declaration `existsUnique_simplest_separator` requires a proof that a separator exists. The appropriate architecture imports an independent small-cut existence theorem and then applies this uniqueness result; it does not erase the hypothesis to manufacture saturation. [merge] The pinned tree already has this architecture: `small_cut_fillers` in `SignSequenceCut.lean` proves that small cuts have fillers, and `cut` in `SignSequenceCutOperation.lean` is defined from `existsUnique_simplest_separator` applied to it (section 1.5; the same remark in source 14's audit section, section 40).

59.2.3 Proposed modules and theorem dependencies

Proposed module	Principal obligations
<code>LabelledCommonPrefix</code>	Initiality, next labels, lexical linearity; comparison with ordinal enumerations.
<code>SupportExtension</code>	Finite extension; infinite buffer; preservation of fixed positions; cardinal bounds.
<code>CoreCylinders</code>	Canonical support codes, prefix completion, convex cylinders, ordered children.
<code>CrossingPrefixes</code>	Elementary crossing predicate; chain and uniqueness by length; union graph.
<code>CutTrichotomy</code>	Unbounded branch; bounded limit edge; terminal numerical gap; uniqueness.

Proposed module	Principal obligations
ExhaustiveTrace	Coherence and coverage; graph reconstruction; trace injectivity.
SupportSaturation	Support compression; union bound at cf κ ; sharp forced-prefix configuration.
RawTraceFibers	Canonical set-like initial branch; tail concatenation; equality and convexity of trace fibers.
ClassRealizations	Same-set structures and elementary absoluteness with old parameters.

These are proposed modules, not files claimed to exist or compile. A dependency path is

completion $\longrightarrow$ cylinders $\longrightarrow$ crossing chain $\longrightarrow$ trichotomy $\longrightarrow$ recognition.

Support compression can instead import the predecessor’s verified-in-writing core interpolation theorem while that path is being formalized. The character theorem and support threshold then provide two largely independent audits of the prefix machinery.

59.2.4 Class-level interfaces and avoiding a hidden third sort

A formal metatheory of GBC can express cuts and enumerations as class variables and use elementary comprehension explicitly. An alternative is to formalize the external inaccessible model, where all relevant collections are sets in a larger type universe. These routes have different assumptions and should not be conflated.

The central reconstruction specification can be written schematically as

```

Crosses(D, p) :=
  exists a b, InCylinder(a,p) and InCylinder(b,p)
    and D(a) and not D(b)

RecoveredGraph(D, alpha, x) :=
  exists p, Crosses(D,p) and alpha in domain(p) and p(alpha)=x

Coherent(D) := forall alpha, exists p,
  domain(p)=alpha and Crosses(D,p)

Covers(D) := forall x, exists p, Crosses(D,p) and x in range(p)

```

This is logical pseudocode, not Lean syntax and not compiled code. Its purpose is to display the quantifiers: a, b, p, α, x are set objects. No line chooses a class as an element of a class family. The proof of functionality precedes any use of the recovered graph as a function, and the coverage theorem is a separate proof of surjectivity.

59.2.5 What must be checked before a kernel-verification claim

A completed formalization must pin its Lean and Mathlib revisions, compile all files, expose any axioms beyond the chosen foundations, and connect the generic ordered alphabet to the actual surreal carrier. It must also verify that internal ordinals, universe lifts, and prefix domains match the statements. None of those steps has been completed for the new theorems in this package; the package contains conventional written proofs and executable finite diagnostics only.

59.3 Source 15: a concrete Lean architecture

The inspected ProveIt declaration is

$$\text{SignSequence.}\{u\} : \text{Type } (u+1).$$

Its birthday belongs to $\text{Ordinal.}\{u\}$. The file provides `ordinalOrderEmbedding`, `birthdays_bounded`, `small_bounded`, `not_small`, and a well-founded simplicity relation, among other declarations [7]. The essential statement is relative smallness: the entire sign carrier is not u -small, even though it is a legitimate type one universe higher. The official Lean universe rules support this distinction [29]. This article has inspected those declarations but has not rebuilt the repository or certified any new theorem in Lean.

A practical development should use the following dependencies.

1. **Generic set layer.** Define exhaustive ordinal enumerations of an arbitrary linearly ordered carrier; prove prefix-freeness, first-difference comparison, pair orientation, and the successor-cardinal obstruction. Surreal arithmetic is unnecessary.
2. **Bounded surreal layer.** Reuse sign order and birthdays. Formalize the 00/11 code with terminal 01, ordinal concatenation, the max-coordinate scheduler, and the three coding-length cases. Track ordinal products explicitly instead of replacing them by cardinalities.
3. **Relative-smallness layer.** For a relation on a higher-universe carrier, express the assertion that an interval or predecessor type is u -small. Implement the least-representative condensation and its rank map. This models the chosen set/class boundary without a type of all types.
4. **Prefix and cut layer.** Define one uniform evaluation relation on small support codes. State crossing using quantification over small prefixes and codes, and reconstruct a persistent branch. Quantify over the cut as a parameter at the permitted universe level.
5. **Model layer.** To claim a theorem about GB, GBC, or KM itself, formalize a model or an axiomatic interface for that theory. Universe polymorphism alone is not a proof that a chosen type-theoretic encoding satisfies precisely those set-theoretic axioms.

Each layer should record its universe parameters, smallness assumptions, choice use, and available recursion principle. A relation on the entire higher-universe sign type with ordinary Lean well-foundedness may encode external well-foundedness stronger than a Henkin model's internal class condition. These should be separate definitions, linked only by proved translation theorems.

59.4 Source 16: a concrete formalization plan tied to ProveIt

59.4.1 Inspected primitives and actual status

The following declarations were inspected at the pinned repository revision. They are existing foundations, not formal proofs of the new results of this article. The earlier well-order report explicitly states that no theorem with its `swo:` labels has a Lean mapping.

Existing file	Relevant declarations or interface	Reuse here
<code>SignSequence.lean</code>	<code>SignSequence</code> , <code>birthday</code> , <code>birthdays_bounded</code> , <code>small_bounded</code> , <code>not_small</code>	Carrier, numerical order, bounded birthday stages, and properness relative to a universe.

Existing file	Relevant declarations or interface	Reuse here
<code>SignSequenceCut.lean</code>	<code>small_cut_fillers</code> , <code>exists_cut_separator_of_birthday_lt</code>	Small-cut realization and prescribed upper birthday control.
<code>SignSequenceCutOperation.lean</code>	<code>cut</code> , <code>cut_realizes</code> , <code>birthday_cut_le_iSup</code> , <code>exists_separator_of_small_sets</code>	Canonical separators with explicit smallness assumptions.
<code>SignSequenceSimplicity.lean</code>	<code>existsUnique_simplest_separator</code> and minimum-birthday results	Deterministic choices and control of sign-prefix cuts.
<code>SizeObstructions.lean</code>	<code>SmallCutData</code> , <code>noUnrestrictedCuts</code>	Keep small inputs distinct from an unrestricted self-cut.

All paths in the table lie below `Algebra/SurrealNumbers/Surreal/Foundations/`; the pinned sources are recorded in the bibliography [7, 8, 9, 31]. No Lean build was performed for this article, and no proposed module name below should be read as an existing theorem declaration.

59.4.2 Two formalization routes

The most economical first route is ordinary type-theoretic order theory. Use a bounded birthday carrier known to be small, transport it with `Shrink` if convenient, and prove the set-alphabet theorems. Store the numerical label order separately from the well-order being encoded. A proposed `WellOrderOn X` can store the binary relation, strict linearity, and `WellFounded` evidence; an enumeration stores an ordinal domain with a bijection to X . Prove the relation/enumeration correspondence, prefix-incomparability, and first-disagreement comparison before taking any quotient. Quotienting first by abstract isomorphism would destroy the required labels.

The second route explicitly formalizes a model of class theory. Introduce separate carriers of set and admitted-class codes, membership, formula semantics, and internally admitted family relations. State elementary comprehension, replacement, and any GC or ETR assumptions at that level. This route is appropriate for exact axiom-strength claims and Henkin models. It is heavier than proving the bounded combinatorial theorems, but it prevents an accidental use of all ambient predicates as internal classes.

59.4.3 Universe levels and logical levels are different

`ProveIt`'s actual carrier is

$$\text{SignSequence}.\{u\} : \text{Type } (u + 1), \quad \text{birthday} : \text{Ordinal}.\{u\}.$$

Its small-cut interfaces explicitly require `Small.{u}` for the index families. A bounded birthday subtype is small at that level, while the full sign carrier is not.

One must not say that every power set automatically forces the next syntactic Lean universe. If $S : \text{Type } (u + 1)$, then both `Set S` and `S → S → Prop` also inhabit `Type (u + 1)`. Being well-typed does not make them u -small, nor does it identify them with the class collection of an internal GBC model. Native quantification over all such predicates is an ambient full-type interpretation. Likewise, native `WellFounded R` asserts ambient well-foundedness, which is not automatically a model's internal assertion [29, 38, 30].

For a direct universe-relative global development, keep a baseline $b : \text{Ord}^u \simeq \mathbf{No}^u$ as an explicit parameter until its existence is separately established. Encode supported permutations by genuinely small data, and represent families of their evaluated graphs uniformly. This gives meaningful theorems without pretending that every ambient type is an internal set or every function on class predicates is a same-level class function.

59.4.4 Suggested milestones

1. **Finite and set-alphabet layer.** Formalize first disagreement, prefix convexity, pair coding, and the stabilization bound. This isolates the ordinal-spectrum theorem from all surreal-specific facts.
2. **Topological layer.** Formalize the no-cross-fiber-jump lemma, prefix families meeting every open set, and disjoint finite-tail cylinders. Then prove the countable incompleteness construction and the regular-stage gap spectrum.
3. **Small-support layer.** Formalize normalization, completion of a small prefix, and the bounded common-prefix interpolation search. Reuse the existing small-cut separator rather than postulating a new unrestricted cut operation.
4. **Cut recognition.** Define straddling on set codes, prove uniqueness and coherence at each length, and implement the direct formula (S16.14). Only then package the exhaustive branch criterion and first-failure classification.
5. **Class semantics layer.** In an explicit model interface, formalize set-interval equivalence, least representatives, local rank, and the terminal set. These are elementary constructions; a general class-recursion engine is unnecessary for them.
6. **Higher comprehension.** State the inverse evaluator with its exact syntactic complexity before invoking a comprehension axiom. Prove the elementary diagonal theorem independently, and use it as the final contradiction in the nonembedding theorem.

These milestones deliberately separate ordinary mathematics about a large type from metatheorems about a two-sorted model. No new consistency axiom is needed for the bounded results, and the inaccessible model is an optional explicit interpretation rather than a prerequisite for the whole subject.

59.5 Source 17: formalization routes and a practical verification plan

59.5.1 Start with the alphabet interface, not surreal arithmetic

The new set-sized results depend on a small interface. A formal development can begin with a linear order X of infinite cardinality μ , with μ elements in every nonempty open interval and ray. It should record its ray and point character r and the gap spectra of deletions of fewer than μ elements. Prefix completion, clopen cylinders, divergence bounds, the cut trichotomy, and fusion can then be proved abstractly. The birthday sign alphabet is one implementation of this interface. This organization prevents arithmetic lemmas about surreal addition or multiplication from obscuring a proof that uses only order and signs.

The exact gap theorem has four independently testable mathematical interfaces: a gap is a two-sided partition without endpoints; a straddling prefix is a set witness; the first stopping stage is a successor, a proper limit, or full length; and each resulting character is computed separately. Separating them makes it difficult to accidentally substitute point character q for cylinder character r .

The support theorem is another independent target. Its statement needs only a set partial injection, the necessarily moved domain and range, cardinal addition, and a complement bijection. No surreal construction appears until the final composition with a baseline.

59.5.2 Proposed modules and proof obligations

The following names describe a proposed architecture, not files claimed to exist in the repository or in this package.

Proposed module	Principal obligation
<code>WellOrders/CommonPrefix.lean</code>	Elementary predecessor formula, maximality, first residual labels, transitivity
<code>WellOrders/PartialSupport.lean</code>	Mandatory support, deficiency balance, exact reserve cardinal and extension
<code>WellOrders/PrefixTopology.lean</code>	Completion, convex clopen cylinders, point versus cylinder character, base cardinality
<code>WellOrders/CutTrichotomy.lean</code>	Unique straddling path, first failure, limit boundary, nonexhaustive branch
<code>Surreal/CutoffGapSpectrum.lean</code>	Sign-path classification of alphabet gaps and stability under small deletion
<code>WellOrders/Fusion.lean</code>	Dense-open refinement with explicit coverage of the alphabet
<code>ClassOrders/BranchRecognition.lean</code>	A chosen model or size interface; elementary reconstruction from unique set witnesses
<code>ClassOrders/CoreAutomorphisms.lean</code>	Represented-cut preservation, uniqueness of extension, nonextension witness

For the topology modules, one should formalize the definition of cellularity as a supremum before claiming an exact cardinal value. A supremum of disjoint-family sizes need not be attained. Likewise the gap spectrum should not include point cuts of character 1, and the recognition module should not quietly switch between open and closed principal lower cuts.

59.5.3 Universe-relative type theory

Lean’s universes offer a practical way to keep a fixed smaller universe separate from a larger carrier. A type, a type of relations on it, and a universe of all smaller types must have their levels checked explicitly. Function types do not automatically jump one level merely because they represent a mathematically large collection: their levels are computed from the domain and codomain. Universe polymorphism supplies a family of declarations, not an internal type of all universes; and universe lifting is explicit rather than an automatic cumulative coercion [29].

For this project, choose a lower notion of “small” and an ordinal index which well-orders the intended large carrier with small proper initial segments. State the cut operation only for lower-small families. A back-and-forth argument along an arbitrary choice well-order of the large type is unsafe: its proper initial segments may be too large for the assumed cut operation. The correct interface is the small-history property used in the proof of [Corollary 42.14](#).

A set-sized cutoff implementation avoids this difficulty entirely. Every object is an ordinary type with explicit cardinal bounds, and a formal proof can be instantiated at many cardinals. A universe-relative version then transports the argument, without equating an ordinal type from one universe with all ordinals from every universe.

59.5.4 A deep model-theoretic alternative

To study exactly which comprehension principles suffice, formalize a specified two-sorted model $(M, \mathcal{K})$ and its elementary comprehension and replacement axioms. Each theorem about global rows is then a theorem about admissible class relations of that model. This is the most direct route for [Theorem 48.4](#) and for questions comparing GBC, ETR, and KM.

The two routes should not be conflated. A successful theorem about types in a fixed universe does not, without an interpretation theorem, prove a conservativity or axiomatic-strength statement about class theory. Conversely, a GBC proof with explicit class variables is not automatically a compiled theorem about the repository’s current sign-sequence carrier.

59.5.5 Verification boundary

No Lean code accompanies this article and no kernel-verification claim is made. The source’s foundation modules were not rebuilt. A future formalization should pin both the repository and Mathlib revisions, compile the modules, reject unfinished proofs, and audit theorem axiom dependencies. The companion Python checks test only finite mechanisms. They cannot decide the existence of class maps, prove cardinal cofinality formulas, or certify the infinite support obstruction.

59.6 Source 18: a concrete formalization plan for ProveIt

59.6.1 What the inspected source actually provides

The source audit used the pinned revision 83befe707f2840c2b53e0606701d8a2b28598e47. The relevant files are in `Algebra/SurrealNumbers/Surreal/Foundations/`. The following declarations were inspected directly in the source files [\[32\]](#), rather than inferred from a report’s title.

Source file	Useful declarations and their role
<code>SignSequence.lean</code>	<code>SignSequence</code> : <code>Type (u + 1)</code> , numerical linear order, <code>lt_iff</code> , <code>birthdays_bounded</code> , <code>small_bounded</code> , <code>not_small</code> , prefixes, truncations, and the ordinal order embedding
<code>SignSequenceCut.lean</code>	<code>exists_cut_separator_of_birthday_lt</code> and <code>small_cut_fillers</code> ; the bounded and small-family separator interface
<code>SignSequenceSimplicity.lean</code>	<code>existsUnique_simplest_separator</code> and minimum-birthday uniqueness; deterministic choice of separators where appropriate
<code>SignSequenceComparison.lean</code>	<code>le_iff_options_lt</code> , <code>lt_iff_exists_option</code> , and canonical-cut comparison

The declaration `SignSequence.{u}` living in `Type (u+1)` is crucial. It contains the surreals with u -small birthdays; the theorem `not_small` says that this whole type is not itself u -small. The existence of a Lean type is therefore not the same as being a set at the intended smaller size level. Lean’s universe hierarchy enforces these distinctions [\[29\]](#). A Tarski–Grothendieck implementation is another well-developed option; Pąk and Kaliszky’s Mizar formalization illustrates a concrete universe-based route [\[37\]](#).

This article does not assert that the new spectrum, condensation, or shuffle theorems are already formalized in these files. No Lean build or complete imported-axiom audit was performed for this article. Reading a proof file is not equivalent to running its elaborator and kernel in the pinned environment.

59.6.2 The first set-level modules

The most reusable initial target is independent of surreal arithmetic. Represent a labelled well-order by a relation on a fixed carrier and its well-order proof, or by a bijection from an ordinal domain. Prove the equivalence of these representations and the absence of proper prefixes for exhaustive enumerations. Keep “increasing by numerical order” as an extra predicate on words, not as part of the definition of a well-order code.

The suggested sequence is:

1. **Ordinal words.** First disagreement, concatenation, prefix cones, alternative terminators, and the finite permutation adjacency criterion.
2. **Set spectra.** Pair coding, eventual coordinate stabilization, and the μ^+ obstruction. The central induction is on a regular successor cardinal, with the domain and range cardinalities stated separately.
3. **Permutation topology.** Order topology, dense sets, disjoint interval families, $Q_n(X)$, and the density/cellularity transition. Include global endpoints in dense-set constructions for arbitrary alphabets.
4. **Surreal cutoff interface.** Large sign cones, preservation of cut characters after deleting fewer than μ labels, and $\text{cf}(2^{<\kappa}) \geq \text{cf } \kappa$.
5. **Cut classification.** Encode the chain of straddling prefixes and prove its three cases. The multiplicity theorem then uses explicit injections and the cardinalities of the prefix tree and full branch set.

This order gives a checkable sequence of finite dependency boundaries. The proof of the gap spectrum depends only on order, ordinal words, cardinal arithmetic, and the cutoff interface. It does not require surreal addition, multiplication, or real closedness.

59.6.3 The class-level modules

There are two sensible implementations.

Semantic set/class models. Define a first-order set universe and a collection of admitted class predicates, with the GB axioms made explicit. A class well-order is a relation in that collection with the internal well-order property. Then prove the elementary common-prefix theorem, set-interval decomposition, uniform branch criterion, and factorial-block facts as statements about that model. An external countability hypothesis subsequently gives the shuffle classification. This route preserves the exact meaning of internal set size and available classes.

Universe-relative carriers. Fix the smallness level u , work with a larger carrier type, and formulate set-likeness using u -small predecessor types. Relations on that carrier live in an appropriate larger universe. This is convenient for a concrete containing-universe development, but theorems must say which subsets are admitted and which smallness level “set” means. Merely forming the type of all relations in the next universe does not identify it with a particular GBC class realization.

In either route, give quotient blocks actual representatives. For set-interval condensation use the least label of a block; for the external finite-condensation quotient use the usual set quotient in the ambient universe. Their types and their purposes differ.

59.6.4 Suggested proof obligations and trust reporting

For a mechanized milestone, require the following explicit outputs: the precise theorem statement with universe levels; a compiling source file against the pinned toolchain; an imported-axiom report;

and a short explanation of the semantic correspondence with the article. Finite examples can test a comparator implementation, but they do not verify the μ^+ argument, Global Choice equivalence, or any claim about proper classes. The current deliverable supplies conventional proofs and source provenance, rather than presenting finite computations as certificates for infinitary assertions.

60 What this Part answers

[merge] Dated notes, 3 October 2026, in the form of [section 33](#); where the question index of [section 61](#) (dated 2 October 2026) says otherwise, these notes supersede it.

- (a) *Source 11's research question 61.3 and source 12's research question 61.14, iterated residual decomposition and proper-class blocks*: the block-index clause is **answered**. Every class well-order decomposes at once, elementarily in GB, into proper blocks of type Ord indexed by a class well-order, followed by at most one final set block; every nonempty class well-order occurs as the index, with any final set ordinal ([note 52.1](#)); explicit condensation histories of every class length exist ([theorem 53.4](#)). In particular all the subpredicates $\text{Ord} \cdot \gamma + \alpha$ of [research question 61.14](#) are nonempty and are recognized by their block profile. Open: the strength needed to iterate the condensation of an arbitrary input uniformly ([research question 61.48](#), [research question 61.64](#), [research question 61.86](#)), and, for [research question 61.14](#), the embedding and nonembedding theorems between those subpredicates and the set-like order, of which only the full-model separation $\mathcal{W}_{\kappa+2} \not\leftrightarrow \mathcal{W}_{\kappa}$ ([theorem 57.3](#)) is proved.
- (b) *Source 11's research question 61.2, weakest comprehension for the uniform nonembedding*: answered in part. For transformations defined by formulas with only set quantifiers, GBC with Σ_1^1 class comprehension suffices ([theorem 56.2](#)); whether it is necessary, and whether GBC alone suffices, is open ([research question 61.63](#)). The question's bounded variant, comparing density and cellularity of all well-orders of $\mathbf{No}_{<\kappa}$ with the minimal stratum, is answered by [theorem 27.2](#) (Part VI).
- (c) *Source 08's question 7, full virtual lexicographic products* ([section 61.4](#)): **answered** for the interpretations it names. In GBC, $\mathfrak{W}_{\text{sl}} \cong \mathbf{No} \times_{\text{lex}} \mathfrak{W}_{\text{sl}}$ and every ordinary class-indexed lexicographic sum of copies of $\mathfrak{W}_{\text{sl}}$ embeds into $\mathfrak{W}_{\text{sl}}$ ([theorem 55.1](#)); in the full inaccessible universe $\mathfrak{W}_{\text{sl}} \times_{\text{lex}} 2 \not\leftrightarrow \mathfrak{W}_{\text{sl}}$, hence $\mathfrak{W}_{\text{sl}} \times_{\text{lex}} \mathfrak{W}_{\text{sl}} \not\leftrightarrow \mathfrak{W}_{\text{sl}}$ ([theorem 55.3](#)); no fixed definable uniform embedding exists in $\text{KM} + \text{GC}$ ([theorem 55.4](#)); in externally countable models $\mathfrak{W}_{\text{sl}} \times_{\text{lex}} \mathfrak{W}_{\text{sl}} \cong \mathfrak{W}_{\text{sl}}$ externally ([proposition 55.5](#)). The negative half already followed from the proofs of [theorems 20.1](#) and [20.2](#) ([note 55.6](#)). A finer absorption spectrum (source 13's [research question 61.25](#)) is open.
- (d) *Source 12's research question 61.17, automorphisms and finite condensation*: answered in part. The maximal finite components of the unrestricted order are the $n!$ permutations of the maximal finite final segment, and the condensed order is dense without endpoints ([theorem 54.4](#), [proposition 58.3](#)); in every countable ω -standard model the full order is externally the factorial shuffle and its automorphism orbits are known ([theorem 58.6](#), [corollary 58.9](#)). The internal classification of the condensed order (source 17's [research question 61.77](#)) is open.
- (e) *Source 11's research question 61.8, source 12's research question 61.18 and source 08's question 12, class realizations*: answered further. In addition to [section 50](#) (e): the pure external order of all admitted class well-orders is the same factorial shuffle in every countable ω -standard model, so it records nothing about the class realization there, and set-likeness is not definable in it even with parameters ([theorem 58.6](#) and [corollary 58.10](#)); among GBC realizations with the same sets, internal well-foundedness of a shared relation never changes

(note 57.5). The expanded structures with the set-like predicate or the label evaluation are open (research question 61.88).

- (f) *Source 11's research question 61.4, recovering prefix structure from the bare order*: a further negative part. Set-likeness is not first-order definable, even with finitely many parameters, from the pure order of all admitted class well-orders of a countable ω -standard model (corollary 58.10); within one class well-order it is recognized by the number of blocks (source 15: W is set-like exactly when it has one block). With section 50 (c), the question's negative half is settled; the positive question, which extra predicates suffice, stays open.
- (g) *Comprehension questions* (source 12's research question 61.16, source 09's direction 2, source 08's question 10): bear. The direct comparison, the block decomposition, the represented-cut reconstruction and the uniform removal of blocks all have elementary definitions in GB or GBC (sources 15, 16 and 18); no reversal is proved.
- (h) *Source 11's research question 61.10, source 09's direction 9 and source 08's question 15 (Kanovei–Shelah)*: not affected; sources 14 and 17 restate the size boundary (section 4).

The questions of sources 13–18 that a sibling source answers are marked in section 61.

Questions, conclusions and status

61 Research questions

[merge] The four sources' questions are printed per source in the following subsections, as proposed continuations, not as a census of open problems. The index below groups the duplicates, credits every source that asks them, and records the status after the merge (dated 2 October 2026). Questions of source 09 and source 08 are numbered by the order of their subsections and paragraphs in [sections 61.3](#) and [61.4](#). [merge] (3 October 2026.) The status column below carries dated updates for the batch-81 addition; the questions of sources 13–18 have their own index in [section 61.5](#).

Topic	Asked by	Status
Weak-choice strength of a well-order of No	12: remark 61.15	answered by theorem 13.1
Comparison of non-set-like class well-orders	08: question 11	answered by theorems 14.2 and 14.7
Coding lengths at singular cutoffs	08: question 3	answered in part by theorem 8.3 ; <i>3 Oct.</i> : cut half answered (section 33)
Invariants, cut spectra and isomorphism type at cutoffs	11: research question 61.7 ; 12: research question 61.13 ; 08: questions 1, 2	open; regular case partly by proposition 12.1 ; <i>3 Oct.</i> : research question 61.7 and question 2 answered, research question 61.13 except the isomorphism type, question 1 open (section 33)
Fixed noninitial enumeration types	12: research question 61.11 ; 08: question 4	open (reals report: lwo:sec:research); <i>3 Oct.</i> : answered for $\mu \cdot \gamma + n$ (section 33)
Noncardinal cutoffs; away from inaccessible truncations	12: research question 61.12 ; 09: direction 7	open; <i>3 Oct.</i> : direction 7 answered for the minimal slice (section 33)
Represented cuts, exhaustive branches and higher-size cuts	11: research question 61.1 ; 09: direction 3; 08: question 8	open; <i>3 Oct.</i> : answered ; question 8 for cuts of the core (section 50)
Full virtual lexicographic products	08: question 7	open; <i>3 Oct.</i> : answered (section 60)
Weakest comprehension and minimal axioms	11: research question 61.2 ; 12: research question 61.16 ; 09: direction 2; 08: question 10	open; <i>3 Oct.</i> : Σ_1^1 sufficiency for research question 61.2 (section 60)
Iterated residual decomposition; proper-class blocks	11: research question 61.3 ; 12: research question 61.14	open; <i>3 Oct.</i> : block-index orders answered, iteration strength open (section 60)

Topic	Asked by	Status
Automorphisms; recovering prefix structure; baselines	11: research questions 61.4 and 61.5 ; 12: research question 61.17 ; 09: direction 4; 08: questions 5, 6	open; <i>3 Oct.</i> : in part, mostly negative (sections 50 and 60)
Birthday and support bounds	11: research question 61.6 ; 09: direction 6; 08: question 13	open; <i>3 Oct.</i> : support in part (section 50); birthday bounds open
Dependence on the class realization	11: research question 61.8 ; 12: research question 61.18 ; 08: question 12	open; <i>3 Oct.</i> : in part (sections 50 and 60)
A first verified milestone	11: research question 61.9 ; 09: direction 1; 08: question 13	proposals
Uniform constructions indexed by codes (Kanovei–Shelah)	11: research question 61.10 ; 09: direction 9; 08: question 15	open
Relation-code comparisons	09: direction 5; 08: question 9	open; equimorphism type by proposition 17.1 ; <i>3 Oct.</i> : endpoint patterns (section 39)
Termination conventions for set words	09: direction 8; 08: question 14	open; <i>3 Oct.</i> : direction 8 answered , question 14 except one type (section 39)
Compatibility between birthday cutoffs	12: research question 61.19	open

61.1 Source 11’s research questions and topics

The questions below are proposed extensions of the present development. Except where a source is explicitly named, they are not asserted to be established open problems in the literature. Several ask for a stronger classification than the proved results supply.

61.1.1 A precise description of the represented cuts

Fix the dense ordinary class $\mathcal{P}_{\text{bd}} \cong \mathbf{No}$. Every set-like global enumeration determines a lower cut of $\mathcal{P}_{\text{bd}}$, and so does every injective nonexhaustive Ord-word. The omitted-range example shows that the first collection does not fill every cut generated by the second.

Research question 61.1. Characterize, using the prefix tree and its numerical labels, exactly which class cuts of $\mathcal{P}_{\text{bd}}$ arise from injective Ord-words, and which of these words are exhaustive.

A useful first target is a necessary-and-sufficient compatibility condition on the intersections of a cut with prefix cylinders. The challenge is to separate missing labels, impossible repeated labels, and cuts that escape every coherent branch. This should first be stated for a fixed uniform class cut, before quantifying over an entire hyperclass of cuts.

61.1.2 The embedding relationship between the two global orders

The set-like fragment and binary class words are equimorphic, while the unrestricted order has additional convex fibers. [Theorem 20.1](#) rules out a reverse embedding in the full universe interpretation. [Theorem 20.2](#) also rules out a definable uniform reverse embedding in $\mathbf{KM} + \mathbf{GC}$.

Research question 61.2. What is the weakest comprehension scheme over GBC that proves the uniform nonembedding result? Can the inverse-evaluation step be replaced by an elementary argument, or is a genuine strengthening necessary?

A precise project should specify the syntactic complexity of the proposed transformation and of the comprehension used to invert its code assignment. In the bounded setting one can instead compare the density and cellularity of all well-orders of $\mathbf{No}_{<\kappa}$ with their minimum-length stratum, especially when κ is not inaccessible. The full-universe case is already settled above; the remaining questions concern strength and wider hypotheses.

61.1.3 Iterated residual decomposition

One extraction of the canonical initial Ord-part requires only elementary class comprehension. A residual proper class has another such part, suggesting a hierarchy of successive blocks.

Research question 61.3. What is the exact recursion strength needed to organize iterated residual decompositions uniformly, and what class order types can occur as their block-index orders?

An initial project could analyze finitely many blocks and specified set-length constructions, recording where the recursion state becomes a class. The general case should be compared with known ETR results on class well-order comparability, rather than assumed to follow from the one-step theorem.

61.1.4 Recovering the prefix structure from the bare order

The definitions exhibit a labeled tree of prefixes. An automorphism of the bare lexicographic order need not obviously preserve its displayed labels or cylinders.

Research question 61.4. Which prefix cones, canonical-core fibers, and finite permutation blocks are definable from the linear order alone? What extra predicates make the reconstruction unique?

Adjacency directly detects finite stretches, but a singleton fiber can be exhaustive or can omit one label. Thus local block size alone cannot recover set-likeness. An analysis of the automorphism group may clarify what the order remembers about the original enumeration.

61.1.5 The bounded-support choice of baseline

Every baseline enumeration gives an ordinary class $\mathcal{P}_{\text{bd}}$ abstractly isomorphic to $\mathbf{No}$. If two baselines differ on a proper class of positions, their bounded-support subfamilies can be disjoint.

Research question 61.5. How do the two dense classes sit inside the same $\mathfrak{W}_{\text{sl}}(\mathbf{No})$, and when does an automorphism of that higher-order order carry one onto the other?

The abstract isomorphism of the two dense orders does not automatically extend to all global enumerations. The represented-cut problem is likely relevant. A restricted first case is a baseline change obtained by an order automorphism of the numerical surreal line.

61.1.6 Sharper birthday bounds and minimal codes

The explicit code gives a usable bound

$$\text{bd}(E(f)) = \sum_{i < \text{dom } f} (1 + 2 \cdot \text{bd}(f(i)) + 2).$$

It is designed for a transparent proof, not asserted to be optimal.

Research question 61.6. Which birthday bounds are optimal for uniform embeddings of injective surreal words? Can shorter prefix-free coefficient codes preserve useful simplicity relations?

Regular cutoffs permit same-stage compression; singular cutoffs show why a cardinal bound alone is insufficient. A sharpened result should specify whether it preserves only numerical order, prefix extension, simplicity, or all of these.

61.1.7 Saturation of more general bounded alphabets

If X is an η_κ order of cardinality κ , its minimal-length permutation order inherits η_κ . Birthday fragments often have cardinality $2^{<\kappa}$, which may exceed κ .

Research question 61.7. Determine the exact cut spectrum and point characters of $\mathcal{W}_\lambda(\mathbf{No}_{<\kappa})$, where $\lambda = 2^{<\kappa}$, without assuming $2^{<\kappa} = \kappa$.

The issue combines the size of a used prefix with the number and birthdays of possible numerical separators. The real-alphabet formulas in the precursor report cannot settle this case by substitution.

61.1.8 Class extensions and available global orders

Two models may have the same sets and hence the same individual surreal numbers, but different available classes and different global well-orders.

Research question 61.8. Which features of the lexicographic theory are invariant under extensions of the class part, and how does the represented-cut collection change?

Uniform formulas such as first divergence remain the same for the existing relations. Existence of additional enumerations changes the larger totality. This offers a way to distinguish properties of the labeled surreal order from properties of the surrounding class theory.

61.1.9 A verified first milestone for ProveIt

Research question 61.9. Can the explicit word compression and the direct predecessor comparison be formalized first, with a small, reusable interface independent of surreal arithmetic?

These are concrete, finite-dependency targets: ordinal concatenation, numerical first difference, injectivity, and well-order predecessor sets. The next milestone is the general successor-cardinal obstruction, followed by bounded-support completion and the small-family interpolation theorem. This ordering keeps the central algorithms separate from a formalization of an entire higher-order set theory.

61.1.10 Uniform mathematical constructions indexed by well-orders

The Kanovei–Shelah method suggests using a definable linear order of codes to index constructions without selecting one privileged code. The surreal setting raises additional smallness conditions.

Research question 61.10. Which finite-support or bounded-support constructions can be indexed by a uniform class family of global well-orders, and which require a genuine higher-order index object?

A first investigation should specify its support sets, evaluation maps, and quantifier domains before adding algebra or model theory. The diagonal theorem prevents an exhaustive class-code shortcut, but leaves many useful uniform families available.

[merge] Status of source 11’s questions after the merge: all remain open. Source 12’s dichotomy (theorem 8.3) computes the binary coding length of $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ in every case, but not the cut spectrum or the point characters that research question 61.7 asks for.

[merge] Dated note, 3 October 2026 (batch 81). Research questions 61.1 and 61.7 are answered (sections 33 and 50); research questions 61.2 to 61.5 and 61.8 are answered in part (sections 50 and 60); research questions 61.6, 61.9 and 61.10 stay open, the explicit codes and support bounds of sources 13–17 bearing on research question 61.6 (sections 39 and 50).

61.2 Source 12’s research questions and prioritized program

The following are proposed continuations. Except where an older question is explicitly attributed, they are not claimed to be recognized open problems in the literature.

Research question 61.11 (All fixed enumeration types). Determine $\ell_2(\mathcal{W}_\alpha(\mathbf{No}_{<\kappa}))$ for every ordinal α of cardinality $2^{<\kappa}$, not only the minimal type. Can the answer be expressed by a precise ordinal-sum invariant of admissible entry-code lengths, together with the finite terminal permutation factor?

The singular theorem identifies the first obstruction that any such formula must capture. A useful initial target is $\alpha = \mu \cdot 2$ and then $\mu \cdot \gamma + n$ for finite n .

Research question 61.12 (Noncardinal birthday bounds). For an arbitrary infinite ordinal θ , classify the minimal permutation order of $\mathbf{No}_{<\theta}$. Which features of the ordinal tail of θ , beyond $|\theta|$ and $|\mathbf{No}_{<\theta}|$, survive in its coding rank and local character?

The cone argument for an initial cardinal relies on closure under $i + 1 + \alpha$ below the cutoff. At a noncardinal cutoff that closure can fail and should be replaced by a stratified family of cones.

Research question 61.13 (Isomorphism, not merely equimorphism). Classify the order topology and isomorphism invariants of $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$. At regular cutoffs, and at singular strong limits, what are its density, cellularity, cut-character spectra, and degrees of homogeneity? Which distinguish it from other orders of the same binary rank?

The binary cube itself cannot be the literal answer: it has endpoints and adjacent pairs, while the surreal permutation order has different local behavior. A dense suborder equimorphic with a cube can retain substantial information not measured by rank.

Research question 61.14 (The hierarchy of proper-class blocks). Study the subpredicates of $\mathfrak{W}_{\text{all}}(\mathbf{No})$ with types $\text{Ord} \cdot \gamma + \alpha$ for fixed set ordinals γ, α . Give explicit, size-safe embedding and nonembedding theorems between these subpredicates and the set-like order.

The inaccessible-model calculation provides concrete bounded targets and shows that the full order is strictly more complex than the set-like order there. Extending binary fusion to uniform transformations on class predicates may require stronger comprehension; that dependence should be proved, not assumed away.

Remark 61.15 (Source 12’s question “The exact weak-choice strength”, answered). *Source 12’s text:* Remove set-level AC from the base theory in theorem 13.2. What principle is equivalent to a class

well-order of the sign-sequence surreals then? Compare it with coherent well-ordering of power sets of ordinals and with uniform ordinal presentations of transitive closures.

[merge] *Answered (2 October 2026)*. Source 11’s [theorem 13.1](#) proves the equivalence over GB with no set-level choice: a class well-order of the sign-sequence surreals is equivalent to global choice. Its proof is precisely a uniform well-ordering of the power sets $V_{\beta+1}$, obtained by set-valued recursion from the given well-order on the sign sequences of each length θ ; see [remark 13.4](#). (Corrected on 3 October 2026 after the Hilbert’s-tenth review `review_surreal_synthesis_0be9b9134.md`: “uniform” replaces “coherent”, since these orders need not extend one another; [remark 13.4](#) gives a finite example.)

This is separate from the question whether surreal universality for class linear orders itself implies global choice, raised by Hamkins [21]. The present equivalence concerns a well-order of all surreal labels and does not settle that universality question.

Research question 61.16 (Elementary common-prefix theory). Formalize [theorem 14.7](#) over a precisely chosen weak class theory. How little comprehension and class well-foundedness are needed? Can one characterize the same relation directly by a formula avoiding explicit mention of the common-prefix class?

The definition already expands to elementary quantifiers. A mechanized proof would make the separation from abstract class comparability particularly transparent and yield a reusable theorem for other carriers.

Research question 61.17 (Automorphisms and finite condensation). Determine natural automorphism groups of the full lexicographic order, and of the order after collapsing finite residual blocks. Which operations arise from numerical automorphisms or permutations of the underlying surreal carrier, and which genuinely act on enumeration structure?

Finite-tail blocks are now characterized intrinsically by adjacency. That makes condensation a plausible first invariant, but no complete classification of its quotient is claimed here.

Research question 61.18 (Dependence on the class realization). Fix a transitive set universe M and vary its admissible class collection $\mathcal{X}$. Which external order invariants of the admitted global well-orders detect this change? Can the order reconstruct useful information about available truth predicates or class choice functions?

The diagonal obstruction persists internally, while external cardinality can vary. This makes the distinction between full and Henkin semantics a source of mathematical questions rather than merely a formal warning.

Research question 61.19 (Compatibility between birthday cutoffs). Construct and analyze coherent embeddings from the well-order spaces at smaller birthday cutoffs into those at larger cutoffs. Which choices of reserve labels and tail enumerations are functorial, and which preserve finite condensation, coding ranks, or canonical cylinders?

Naively extending every enumeration by an arbitrary tail can change minimal length, and fixed choices need not be compatible across three cutoffs. A coherence theorem would make universe-lifting arguments much more useful in a proof assistant.

[merge] Status of source 12’s questions after the merge: [remark 61.15](#) is answered (it is printed above as a remark). [Research question 61.11](#) continues the reals report’s question “Other ground orders” for the surreal cutoffs, whose minimal type [theorem 8.3](#) settles. The others remain open.

[merge] Dated note, 3 October 2026 (batch 81). [Research question 61.11](#) is answered for the types $\mu \cdot \gamma + n$, and [research question 61.13](#) except for the isomorphism type and homogeneity ([section 33](#)); [research questions 61.14, 61.17 and 61.18](#) are answered in part ([sections 50 and 60](#)); [research questions 61.12, 61.16 and 61.19](#) stay open, and sources 13–18 ask them again ([section 61.5](#)).

61.3 Source 09’s research directions

The following are proposed problems emerging from the development, not a claim that all are already catalogued open problems. Each is stated with a foundational level so that its intended meaning is clear.

61.3.1 Formal proof of the surreal core theorem

Framework: universe-indexed Lean with explicit smallness and a baseline parameter. Complete the prefix-extension and interpolation modules, then prove the back-and-forth isomorphism. The difficult part is expected to be the bookkeeping of small histories and variable ordinal prefixes, not the finite first-disagreement arguments. A useful first milestone is the inaccessible or regular-cardinal set-sized version, parameterized by an abstract saturated alphabet.

61.3.2 Exact axiomatic strength of the global conclusions

Framework: GB with ordinary set choice but without global choice. [Proposition 13.7](#) settles the baseline question: a set-like global well-order of **No** is equivalent to global choice. Determine the exact strength of alternative universality and homogeneity assertions that do not already assume such a baseline. In particular, which conclusions can be stated and proved for individually definable class orders without supplying a single uniform global enumeration? Distinguish uniform existence assertions from a schema treating one fixed definition at a time; the latter is not automatically a uniform class construction.

61.3.3 Which cuts of the core represent exhaustive branches?

Framework: class cut parameters over $\mathcal{P}_s(b)$ or an explicitly higher-sorted completion. Every global enumeration determines a cut in the dense core, and density makes this representation injective. Characterize its image intrinsically. The missing-branch construction shows that injectivity of a class word is insufficient: exhaustiveness must be detected. A promising formulation separates coherence of set prefixes, realizability of each next-coordinate cut, and eventual coverage of every surreal value.

61.3.4 Automorphisms beyond set-sized partial maps

Framework: GBC for the core; higher-sort or universe-bounded versions for the full comparison object. Set-sized partial isomorphisms extend in the core, but proper-class partial maps need not. Classify which preserve enough prefix or support structure to extend. Also distinguish automorphisms of the bare order from those preserving the preferred filtration by support bounds. These are different classification problems even though the bare core is always **No**.

61.3.5 Non-set-like well-order relations

Framework: GBC or GBC + ETR, with a fixed relation-code coordinate order. Study the alternative binary comparison of all global well-order relations. Which order properties depend on the coordinate

enumeration? Does it contain a natural set-coded dense core, and when are its cuts represented by actual class well-orders? No conclusion about that different comparison follows merely from the enumeration results proved here.

61.3.6 Sharp birthday and support bounds

Framework: ZFC on bounded fragments. Given bounds on the birthdays and first-disagreement coordinates of a set family, determine optimal bounds for an interpolating prefix and its supported completion. The present proof bounds the forced prefix length by $\sup\{\Delta(E_i, F_j) + 1 : i \in I, j \in J\} + 1$ and uses an oversized cardinal for support. That bound is safe but not generally minimal. This direction could yield concrete complexity estimates for symbolic surreal order calculations.

61.3.7 What survives away from inaccessible truncations?

Framework: ZFC with specified cardinal arithmetic. Replace the inaccessible κ by a regular cardinal and a set alphabet with appropriate cut saturation. Separate the roles of regularity, $\kappa^{<\kappa} = \kappa$, and closure under the required cut constructor. At singular cardinals, the first-disagreement bound can reach the ambient length even for relatively small families; identify the resulting interpolation spectrum and construct sharp counterexamples.

61.3.8 Other termination conventions for set words

Framework: GBC or bounded ZFC. Our dictionary order makes proper prefixes smaller and therefore creates the limit-prefix suprema of [note 11.4](#). Study a distinguished internal termination value, restricted alphabets, or prefix-free codes of set well-orders. Determine exactly which variants have the set-cut property and hence are isomorphic to **No**. The convention must be part of the statement, not an implicit analogy with surreal signs.

61.3.9 Uniform infinitary constructions indexed by redundant codes

Framework: bounded universes first. Kanovei–Shelah-style redundant indexing suggests avoiding a choice of one representative for each well-order or auxiliary object. Investigate constructions using the support core or bounded strata as indices, while checking that the resulting products, functions, and quotient relations remain at the declared level. The diagonal theorem forbids a uniform same-level list of all global enumerations, but does not preclude purpose-built redundant families adequate for a particular theorem.

[merge] Status of source 09's directions after the merge. Direction 2: the baseline question it already settles over GB+AC needs no set choice ([theorem 13.1](#)); the strength of the other universality and homogeneity assertions remains open. Direction 3 is source 11's [research question 61.1](#). Direction 5: source 08's [proposition 17.1](#) gives the equimorphism type of a relation-code order; the dense core and the representation of its cuts remain open. Direction 7: source 12's dichotomy treats coding lengths at singular cutoffs, not the interpolation spectrum the direction asks for. The others remain open.

[merge] Dated note, 3 October 2026 (batch 81). Direction 3 is answered ([section 50](#)), direction 7 for the minimal slice and direction 8 for unrestricted and injective words ([sections 33](#) and [39](#)); directions 4, 5 and 6 are answered in part ([sections 39](#) and [50](#)); directions 1, 2 and 9 stay open, with source 16's Σ_1^1 bound bearing on direction 2 ([section 60](#)).

61.4 Source 08's research questions and topics

The following are directions not resolved by the present manuscript. They are not blanket assertions that no relevant answer exists in the literature. Each should begin with a targeted literature search and a precise choice of size level.

61.4.1 Set-sized enumeration orders

1. Isomorphism classification beyond the common spectrum. The ordinal spectrum and cardinality of $\mathcal{W}(X)$ depend only on $|X|$, while its local cylinder structure remembers the numerical order of residual alphabets. Which additional invariants of X can be recovered from $\mathcal{W}(X)$? Determine sufficient and necessary conditions for $\mathcal{W}(X) \cong \mathcal{W}(Y)$, rather than merely embeddings in both directions. Dense alphabets, well-orders, and mixed sums provide natural test cases.

2. Exact invariants at regular cutoffs ([merge] partly answered; answered 3 October 2026; see the notes after this list). For regular κ with $2^{<\kappa} = \kappa$, compute the density, cellularity, point characters, and cut cofinalities of $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$. The binary equimorphism in (S08.6) does not by itself determine its order-topological isomorphism type. Investigate whether the support skeleton construction has a natural κ -sized dense realization inside this set order under weaker assumptions than strong inaccessibility.

3. Singular birthdays and code lengths ([merge] answered in part; see the notes after this list). At singular κ , exactly which cuts of size below κ fail to be realized in $\mathbf{No}_{<\kappa}$ or in its fixed-length enumeration order? Determine the least binary coding lengths for these orders and how they depend on cofinality and cardinal arithmetic. The three specific uses of regularity in note 7.6 identify where a new argument is needed.

4. Prescribed noninitial lengths. For α of cardinality κ , classify $\mathcal{W}_\alpha(\mathbf{No}_{<\lambda})$. Finite tails are understood through note 6.25, but noninitial limit lengths may preserve finer lexicographic complexity. Compare this problem with the fixed-type analysis proposed in the continuum manuscript [2], checking its hypotheses and proofs independently for nonseparable alphabets.

61.4.2 The skeleton and the virtual ambient order

5. Definable choices of the surreal isomorphism. The back-and-forth isomorphism $\mathcal{P}_s(b) \cong \mathbf{No}$ depends on witness choices. With a specified global well-order, can it be chosen to respect useful support or birthday bounds? What additional normalization gives uniqueness, if any? Distinguish definability from parameters, parameter-free definability, and invariance under a change of baseline.

6. Automorphisms and extension from a skeleton. An order automorphism φ of $\mathbf{No}$ acts on $\mathfrak{W}_{\text{sl}}$ by $e \mapsto \varphi \circ e$ and sends O_b onto $O_{\varphi \circ b}$. Which automorphisms of one skeleton extend to uniform automorphisms of the full virtual order? Can order alone recognize eventual-agreement orbits, or are these additional structure supplied by the coordinate presentation?

7. Full virtual lexicographic products ([merge] answered 3 October 2026; see the notes after this list). For an ordinary class carrier, every set-indexed lexicographic power of $\mathbf{No}$ embeds into a skeleton. The *full virtual* product $\mathfrak{W}_{\text{sl}} \times_{\text{lex}} \mathfrak{W}_{\text{sl}}$ is a different object. Does it admit a uniform

order embedding into $\mathfrak{M}_{\text{sl}}$? More generally, determine an absorption spectrum for uniform virtual powers. Simply concatenating two Ord-long words has domain $\text{Ord} + \text{Ord}$, not a set-like Ord domain, so the set-sized concatenation argument is unavailable.

8. Higher-size cuts and completion ([merge] answered 3 October 2026 for cuts of the core; see the notes after this list). The set-cut theorem deliberately leaves out cuts specified by proper class families of enumeration codes or by higher collections of classes. Classify which such cuts are realized once a precise higher-type language is chosen. A proposed Dedekind completion must identify the type of its cut objects and cannot be treated as another ordinary class without a size proof.

9. Classification of relation-code orders. For two global coordinate enumerations q, q' of $\mathbf{No}^2$, the relation-code orders have the same binary equimorphism type. Are there useful hypotheses yielding an isomorphism? Determine which invariants are artifacts of the coordinate order and which reflect genuine properties of class well-orders.

61.4.3 Foundations and formalization

10. Minimal axioms, stated theorem by theorem. Separate the strength of a baseline $\text{Ord} \cong \mathbf{No}$, uniform prefix completion, orbit cut filling, and the global back-and-forth theorem. Over variants of GB with set Choice, which global selection principles are necessary for their uniform formulations? The present proofs establish sufficiency of GBC, not an exact reverse-mathematical classification.

11. Comparison of non-set-like class well-orders ([merge] answered; see the note after this list). Formulate an explicit alignment or comparison principle for arbitrary class well-orders and locate its recursion requirements. Which restricted domains, such as finite sums of Ord and set ordinals, admit direct elementary algorithms? A theorem about all class domains should not be extrapolated from these examples without proof.

12. Dependence on the class collection. For fixed M and $\mathcal{C} \subseteq \mathcal{D}$, compare the virtual orders of global enumerations in $(M, \mathcal{C})$ and $(M, \mathcal{D})$. The same set-coded skeleton relative to a shared baseline remains available, but new class rows may appear. Study which higher-size cut properties, operator embeddings, and definability properties change.

13. Quantitative and certified prefix extension. Formalize the support bound in [note 19.8](#), the forced-prefix length in [note 19.13](#), and the birthday of its stopping entry. This would make the central theorem a sequence of explicit bounded lemmas rather than one monolithic existence proof. It is a strong first target for the repository's existing small-cut interface.

14. Increasing-subset and alternate-termination orders ([merge] answered 3 October 2026 except one type; see the notes after this list). Study lexicographic orders of numerically increasing set sequences, as opposed to arbitrary enumerations, and compare prefix-small, prefix-large, and centered-termination conventions. The limit-prefix counterexample shows that these choices affect local cofinalities even when the alphabet is $\mathbf{No}$.

15. An order-theoretic interface for class nonstandard models. Determine which part of a Kanovei–Shelah-style indexing construction can use a set-coded skeleton and which genuinely needs a larger collection. Keep the order embedding problem separate from the existence of a class ultrafilter, satisfaction relation, or transfer principle. A successful construction should include an explicit uniform-family and smallness audit at every step.

[merge] Status of source 08’s questions after the merge, dated 2 October 2026. Question 11 is answered: the labelled comparisons of [theorem 14.2](#) and [theorem 14.7](#) compare arbitrary class well-orders of a labelled class with elementary comprehension, without ETR (see the note after [proposition 17.1](#)). Question 3 is answered in part: the least binary coding length of the fixed-length enumeration order at a singular cutoff is $\kappa \cdot \kappa$ at a strong limit and $2^{<\kappa}$ otherwise ([theorem 8.3](#)); which cuts of size below κ fail to be realized remains open. Question 2 is answered in part by [proposition 12.1](#) (the order is η_κ , and all its global and local cofinalities and coinitalities equal κ). Question 4 is source 12’s [research question 61.11](#). The others remain open.

[merge] Dated note, 3 October 2026 (batch 81). Questions 2 and 7 are answered ([sections 33](#) and [60](#)); question 3 also in its cut half, in character terms ([section 33](#)); question 8 for class cuts of the core ([section 50](#)); question 14 except the isomorphism type of the centered increasing order ([section 39](#)); questions 4, 6, 9, 12 and 13 in part ([sections 33, 39, 50](#) and [60](#)); questions 1, 5, 10 and 15 stay open.

61.5 The questions of sources 13–18: index and status (3 October 2026)

[merge] Sources 13–18 ask 74 questions (13, 13, 12, 11, 13 and 12), printed per source in the following subsections as proposed continuations, not as a census of open problems. Many repeat each other or a question above; the table groups them, names the earlier question each continues, and records the status after the whole batch-81 addition. A question answered by a sibling source is marked in its source’s subsection. The status of the questions of sources 11, 12, 09 and 08 after the addition is recorded in the dated notes after their subsections and in [sections 33, 39, 50](#) and [60](#).

Topic	Asked by	Status, 3 October 2026
Local geometry and gap spectrum of the minimal slice	14: research question 61.41 ; 16: research question 61.59	answered by source 17 (theorem 30.11 ; Part VI)
Isomorphism type, arrangement of gaps, finer topology of set strata	15: research question 61.52 ; 17: research questions 61.71 and 61.72 ; 18: research question 61.82 (continuing 12: research question 61.13)	open; characters, density, gap spectrum and multiplicities of the minimal slice are known (section 33)
Longer and nonminimal enumeration types	15: research question 61.46 ; 16: research question 61.58 ; 17: research question 61.69 ; 18: research question 61.83 (continuing 12: research question 61.11)	coding length and cube spectrum for $\mu \cdot \gamma + n$ (theorem 31.2), density of every stratum (theorem 27.2); open for infinite remainders and for gap spectra beyond the minimal slice
Noncardinal birthday cutoffs	15: research question 61.47 ; 17: research question 61.70 ; 18: research question 61.84 (= 12: research question 61.12)	open

Topic	Asked by	Status, 3 October 2026
Coherent embeddings across cutoffs; quantitative compatibility	14: research question 61.42; 15: research question 61.54; 16: research question 61.68; 17: research question 61.79; 18: research question 61.85 (= 12: research question 61.19)	open
Normal forms for class cuts; cutoff analogues of first failure	13: research questions 61.22 and 61.23	in part: unique-data trichotomy (theorem 43.3); minimal-slice cuts (theorem 30.11)
Recovering prefix structure, exhaustiveness or set-likeness from the bare order	13: research question 61.20; 14: research question 61.34; 15: research question 61.49; 16: research question 61.61 (continuing 11: research question 61.4)	negative half answered (sections 50 and 60); which enrichments suffice is open
Automorphisms preserving a core or its represented cuts; changes of baseline	13: research question 61.21; 14: research question 61.33; 15: research question 61.51; 16: research question 61.62; 17: research questions 61.73 and 61.74; 18: research question 61.87 (continuing 11: research question 61.5)	extension criterion (theorem 47.5), non-extension (theorem 47.7), relabelling (proposition 47.9); stabilizers and general baselines open
Completion with residual data	15: research question 61.50	open
Products and splits	13: research questions 61.24 and 61.25	in part: KM + GC suffices (theorem 55.4); hierarchy open
Class realizations	13: research questions 61.26 and 61.27; 14: research question 61.43; 15: research question 61.56; 16: research question 61.66; 17: research question 61.75; 18: research questions 61.88 and 61.89 (continuing 11: research question 61.8)	in part (section 50 (e), section 60 (e))
Termination conventions	13: research question 61.28; 16: research question 61.60; 18: research question 61.91	13's answered by source 16; 16's answered by source 18 at the zero threshold; 18's open (section 39)
Birthday, support and code bounds	13: research question 61.29; 14: research questions 61.35 to 61.37; 17: research question 61.78; 18: research question 61.90 (continuing 11: research question 61.6)	support: hierarchy and exact reserve (theorems 49.3 and 49.7); the rest open
Iterated condensation, residual blocks, raw fibres	13: research question 61.30; 14: research questions 61.38 and 61.39; 15: research question 61.48; 16: research question 61.64; 17: research questions 61.76 and 61.77; 18: research question 61.86 (continuing 11: research question 61.3)	block indices answered (note 52.1); 17's research question 61.77 answered externally for countable ω -standard models (theorem 58.6); iteration strength open
Relation codes and other scans	13: research question 61.31; 14: research question 61.40; 15: research questions 61.53 and 61.55	in part: endpoint patterns (theorem 38.2)

Topic	Asked by	Status, 3 October 2026
Comprehension and choice strength	14: research question 61.44 ; 16: research questions 61.63 and 61.65 ; 18: research question 61.92 (continuing 11: research question 61.2)	in part: Σ_1^1 sufficiency (theorem 56.2); one set-like parameter suffices for the word isomorphisms (section 39)
First formal milestones	13: research question 61.32 ; 14: research question 61.45 ; 15: research question 61.57 ; 16: research question 61.67 ; 17: research question 61.80 ; 18: research question 61.93	proposals; nothing is formalized
Typed constructions indexed by presentations	17: research question 61.81 (continuing 11: research question 61.10)	open

61.6 Source 13: further research questions and proposed projects

These questions are proposed continuations of the present development, not a claim that each is a previously published open problem. The prefix-trace criterion, the first-failure classification, and the negative pure-order cylinder result settle the versions stated here; the questions below ask for stronger or different conclusions.

61.6.1 Cuts, trees, and enriched structure

Research question 61.20 (Minimal extra structure for recognizing branches). What is the weakest expansion of the pure core order $\mathcal{P}_s(b) \cong \mathbf{No}$ that makes its branch traces recognizable? Is the family of all cylinders enough as an unranked family, or must one retain the prefix-length relation? Can a support filtration recover the necessary levels without explicitly naming prefixes?

[Theorem 47.12](#) shows that the pure order alone cannot canonically recover the original cylinder family. The positive target is therefore an enriched reconstruction theorem, not a repetition of the disproved pure-order assertion.

Research question 61.21 (Automorphisms preserving a chosen core). Classify the uniform order automorphisms of $\mathfrak{W}_{sl}$ that preserve the specific eventual-agreement core $\mathcal{P}_s(b)$. Which preserve its support filtration, and which preserve the set of exhaustive traces after an isomorphism between two cores? Does adding birthday or simplicity information force a substantial rigidity theorem?

The finite-block automorphisms need not preserve the baseline orbit. A useful initial project is to characterize tail alphabet transformations that do preserve it.

Research question 61.22 (A normal form for class cuts). Can the branch/horizontal/limit-boundary trichotomy be organized into a canonical recursive notation for every available normalized cut? For horizontal cuts, how much of the next-label cut can be encoded by surreal simplicity data? What uniqueness survives when changes of prefix presentation are allowed?

A complete class-cut collection is not an ordinary class object in GBC. The first implementation should concern one supplied class cut or one uniform class-indexed family, before introducing a higher sort.

Research question 61.23 (Sharp cutoff analogues). For a birthday fragment X_κ , determine which part of the first-failure classification survives when prefix extension can fail through exhaustion,

complement-cardinality mismatch, or a missing label separator. Can these three obstructions be combined with the singular strong-limit transition into a practical classification of cut characters in fixed-length strata?

This asks for more than substituting X_κ into the global theorem; the hypotheses that must be replaced are explicitly identified in [Section 32.2](#).

61.6.2 Products, comprehension, and model dependence

Research question 61.24 (The weakest uniform split obstruction). Over GBC, what additional comprehension principle suffices to rule out a fixed-formula uniform embedding $\mathfrak{W}_{\text{sl}} \times_{\text{lex}} 2 \rightarrow \mathfrak{W}_{\text{sl}}$? Can the inverse-evaluator argument be reduced to a restricted existential-class comprehension schema? For more restrictive formulas defining the proposed embedding, is there an elementary diagonal argument avoiding that comprehension altogether?

[Theorem 55.4](#) gives a sufficient theory and identifies the point to optimize. It is not a proof that full KM is necessary.

Research question 61.25 (A hierarchy beyond one split). In a full inaccessible universe, classify the mutual embeddability of $\mathfrak{W}_{\text{sl}}^n$ for finite $n \geq 2$, of $\mathfrak{W}_{\text{sl}} \times_{\text{lex}} K$ for various right factors K , and of iterated point-splittings. Which invariants distinguish these orders after the first dense-core obstruction has saturated at cardinality 2^κ ?

All these orders can have the same cardinality, so a useful invariant must record more than the number of elements. Families of disjoint intervals, patterns of adjacent pairs, and iterated condensation are natural candidates.

Research question 61.26 (Class realizations with intermediate behavior). For a fixed set model M , how do different collections of available classes $\mathcal{C}$ affect the external density, cellularity, and product-embedding behavior of its enumeration order? Which intermediate behaviors lie between the externally countable rational case and the full inaccessible-universe obstruction?

A precise answer should state whether a map is arbitrary external, definable over the two-sorted model, or coded in an additional sort. Those are distinct classification problems.

Research question 61.27 (Uniformity versus external isomorphism). For an externally countable model, can the complexity of an external isomorphism $\mathfrak{W}_{\text{sl}} \cong \mathbb{Q}$ be measured relative to presentations of its set and class sorts? What information about the model is necessary to compute, define, or select such an isomorphism?

This connects the order-theoretic collapse to computability and model presentation without mistaking an external construction for a class available internally.

61.6.3 Termination, bounds, and residual orders

Research question 61.28 (A classification of termination cuts). For a class alphabet order-isomorphic to **No**, characterize the placements of an extra terminal marker for which arbitrary set-word or injective set-word lexicography is set-cut saturated. What happens when the marker is placed immediately on one side of an existing label, rather than reserving that label? Can the answer be expressed in terms of the two characters of the marker cut?

The end-first and end-last failures and the reserved-central success are settled here. They need not exhaust all intermediate behavior.

Research question 61.29 (Optimal birthday and support bounds). For prescribed bounds on input birthdays, word length, and baseline indices, sharpen the prefix-length and support-cardinality bounds in [Theorem 42.4](#). Is the explicit central-code birthday sum optimal within a useful class of prefix-free local codes? Which bounds are compatible with embeddings across birthday cutoffs?

The current support completion deliberately chooses a large cardinal to equalize complements. It proves existence, not optimality.

Research question 61.30 (Iterating raw residual decomposition). The ordinal-head fiber theorem removes one set-like initial part. How far can residual decomposition be iterated with set-valued codes, and what class-recursion principles are needed when an iteration has class-valued stages? Is there a canonical rank or notation for a substantial family of raw class well-orders beyond the first head?

This is where ETR or a stronger scheme could become genuinely relevant. The elementary one-head construction alone does not justify an arbitrary iteration.

Research question 61.31 (Relation-code dependence). For two set-like enumerations of pairs, determine which transformations relate their relation-bit orders on raw well-orders. When do the two induced orders become isomorphic, merely equimorphic, or distinct under definability restrictions? Which code orders can be characterized by simple invariants of the pair enumeration?

61.6.4 A concrete formalization milestone

Research question 61.32 (First checked extension). Formalize the split-order density lemma and the crossing-prefix uniqueness/reconstruction argument in a size-explicit abstract setting. Then instantiate the small-cut and fresh-value interfaces using `SignSequenceCut.lean`. Can the central-word saturation theorem be completed without introducing surreal field arithmetic?

The answer to the last design question should be positive at the mathematical level: the proof uses only order, the reserved marker, smallness, and fresh set-cut separators. Establishing this dependency boundary in Lean would be a useful substantial first checked result.

[merge] Status of source 13's questions, 3 October 2026. [Research question 61.28](#) is **answered** by source 16 ([theorems 36.1](#) and [36.2](#), transported along an isomorphism of the alphabet with **No**; [section 39](#) (c)). [Research question 61.22](#) is answered in part by source 14's unique-data trichotomy ([theorem 43.3](#)) and source 15's finer successor cases; [research question 61.23](#) in part, for the minimal slice, by source 17's gap spectrum ([theorem 30.11](#)). [Research question 61.21](#) bears on source 17's extension criterion ([theorem 47.5](#)), [research question 61.26](#) on source 18's countable models ([theorem 58.6](#)), [research question 61.29](#) on source 17's exact reserve ([theorem 49.7](#)), [research question 61.30](#) on the block decomposition ([note 52.1](#)), [research question 61.31](#) on source 15's endpoint transfer ([theorem 38.2](#)), and [research question 61.24](#) on source 16's Σ_1^1 bound ([note 56.4](#)). The others remain open.

61.7 Source 14: further research questions and priorities

The following are proposed continuations of the present work. They are not all established named open problems in the literature. In particular, the prefix-structured exhaustive-cut question and the support-cardinal threshold are now answered by the theorems above; the questions here state stronger or different targets rather than restating those results.

61.7.1 Cut recognition beyond the named prefix structure

Research question 61.33 (The exhaustive-trace stabilizer). Characterize the subgroup of class order automorphisms of $\mathcal{P}_s(b)$ that preserves the exhaustive-trace collection. How does it compare with the groups preserving all prefix cylinders, or all label incidences? Can it be generated by explicit transformations of enumeration structure?

[Theorem 47.2](#) classifies the bare-order cut orbits and [Corollary 47.3](#) proves that this stabilizer is proper. What remains is its structure, not the now-answered yes/no question whether every core automorphism preserves exhaustive representability.

Research question 61.34 (Minimal enrichment for exhaustiveness). Can the coverage predicate be recovered from cylinder inclusion and numerical order of children without naming the label evaluation itself? What additional incidence relation records that the same surreal label appears at different possible coordinates?

Coherence is essentially a branch property of the cylinder tree, whereas exhaustiveness compares labels along that branch with the entire alphabet. Separating these logical resources should clarify exactly what the recognition theorem uses.

61.7.2 The support hierarchy beyond one saturation invariant

Research question 61.35 (Isomorphism of support layers). For infinite cardinals κ, λ with $\text{cf } \kappa = \text{cf } \lambda$, when are $\mathcal{P}_{<\kappa}(b)$ and $\mathcal{P}_{<\lambda}(b)$ isomorphic as bare class linear orders? Does some cut invariant recover more than the cofinality of the support threshold?

[Theorem 49.3](#) separates layers with different cofinalities but leaves this equal-cofinality case untouched. Their shared embedded copy of **No** is too weak to decide it. The attained minimum support cost is an invariant of the *support-enriched* structure, not yet an invariant of its bare order.

Research question 61.36 (Support ideals). Replace the cardinality condition by an ideal of permitted set supports. Under what closure and permutation-invariance hypotheses is saturation controlled by the ideal's additivity? Can a failure of union closure always be converted into a coherent prefix configuration forcing a support outside the ideal?

The compression lemma bounds cardinality, not membership in an arbitrary ideal: it can adjoin images of moved coordinates and a buffer. Therefore a general ideal theorem needs explicit closure under those operations. It would be incorrect simply to replace $< \kappa$ by “belongs to the ideal” in the existing proof.

Research question 61.37 (Optimal simultaneous support cost). Given a specific separated set family, characterize the least support cardinality of a separator by its forced-prefix tree, rather than only by the union of all input supports. Can the finite version be computed without enumerating all possible separator permutations?

The sharp examples attain the threshold, but many input supports contain coordinates irrelevant to their comparisons. The first-outside-support argument suggests an algorithmic pruning principle. Finite optimization and the transfinite cardinal minimum should be studied separately.

61.7.3 Raw tails, condensation, and alternative comparison orders

Research question 61.38 (Iterated initial-branch quotients). After quotienting raw well-orders by their first Ord-long branch, can one iterate a similarly explicit trace construction to recover successive proper-class blocks? What can be proved for prescribed block patterns without assuming a canonical class ordinal for every well-order?

The trace-fiber theorem completely describes the first loss of information. Iterating it along class-valued histories may require stronger recursion principles. The exact point at which elementary comprehension ceases to suffice should be isolated as a theorem.

Research question 61.39 (Automorphisms of raw trace fibers). Which automorphisms of the branch order extend to the unrestricted well-order order, and what compatibility conditions do the residual alphabets impose? Which automorphisms preserve finite residual blocks but fail to preserve the core-trace quotient?

The singleton omitted-label fiber shows that being a singleton is not enough to recognize the exhaustive suborder. Any extension theorem must also account for residual alphabets with more than one label.

Research question 61.40 (Relation-code comparison). For a specified coordinate schedule on pairs of surreal labels, classify the class cuts represented by genuine well-order relations. Is there a useful analogue of crossing cylinders that detects transitivity, linearity, and well-foundedness from a coherent branch of relation bits?

Coherent binary bits always define a relation, but well-foundedness is a global constraint. The strong reconstruction phenomenon for exhaustive enumerations may not carry over. A schedule-independent theorem must explicitly quantify which coordinate changes it tolerates.

61.7.4 Birthday cutoffs and universe coherence

Research question 61.41 (Exact local geometry of set slices). For $\mu = 2^{<\kappa}$, determine the complete cut-character spectrum of $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$, especially when $\mu > \kappa$ and at singular cutoffs. Which parts of the crossing-cylinder proof survive when deleting a long used prefix can exhaust intervals of the alphabet?

This continues an unfilled target of the predecessor. The global proof uses the fact that deleting *any set* leaves plentiful fresh surreals; a fixed birthday fragment lacks that unrestricted property. The inherited binary equimorphism does not decide the local characters.

Research question 61.42 (Coherent embeddings across cutoffs). Construct explicit embeddings between enumeration spaces for increasing birthday bounds that compose coherently. Which preserve the minimal allowed enumeration type, the prefix cylinders, or the relevant support filtration?

Appending an arbitrary tail preserves a first disagreement but may change the ordinal type. A functorial construction should record both the alphabet inclusion and the position schedule, rather than silently identify cardinally equivalent domains.

61.7.5 Class realizations and proof strength

Research question 61.43 (Which new cuts can a class extension add?). For a fixed set model and old baseline, characterize the additional exhaustive trace cuts admitted by specified class extensions. Can the new-cut collection distinguish natural classes of realizations even when their old-cut realizability relations agree?

[Theorem 48.1](#) supplies a rigidity constraint: no old cut changes status. It does not classify the new parameters. The tail fibers also suggest studying extensions that add raw orders with an old trace but a new tail.

Research question 61.44 (Axiomatic economy). What is the weakest class theory proving the crossing-chain trichotomy and its reconstruction theorem once a baseline and fresh-separator property are supplied? Can the proof of the support threshold be formalized in a set-only regular-cardinal model before passing to classes?

The existing proofs visibly use elementary comprehension and class Replacement, plus set Choice for extensions. Identifying which are essential is more informative than merely observing that a stronger class theory also proves the theorem.

Research question 61.45 (First kernel-verified milestone). Can the common-prefix comparison and the support-compression lemma be formalized as small reusable Lean modules, then independently connected to the pinned surreal sign-sequence library?

These targets avoid the largest foundational infrastructure while capturing the most error-prone finite and cardinal bookkeeping. A useful deliverable would include separate tests for fixed-position preservation, complement cardinalities, and the distinction between strict right character and the character of a normalized cut complement.

[merge] Status of source 14's questions, 3 October 2026. [Research question 61.41](#) is **answered** by source 17 ([theorems 30.2](#) and [30.11](#), with source 18's multiplicities, [theorem 30.16](#)), at every cutoff, including $\mu > \kappa$ and singular κ . [Research question 61.43](#) is answered in part by source 17's [theorem 48.4](#) (every proper same-set extension adds new represented cuts), [research question 61.38](#) in part by the block decomposition ([note 52.1](#)), and [research question 61.33](#) bears on source 17's extension criterion ([theorem 47.5](#)), which identifies the stabilizer of the represented-cut predicate abstractly. [Research question 61.42](#) is source 12's [research question 61.19](#). The others remain open.

61.8 Source 15: further research questions and a prioritized program

Research question 61.46 (All fixed enumeration types). Write $\alpha = \mu \cdot \gamma + \zeta$ with $\zeta < \mu$ infinite. Classify the possible residual numerical suborders left after the long prefix, and use them to compute $\mathcal{W}_\alpha(\mathbf{No}_{<\kappa})$. The theorem here handles exactly finite ζ . A useful first family is $\zeta = \omega$, followed by regular and singular infinite remainders.

Research question 61.47 (Noncardinal birthday bounds). For an arbitrary ordinal θ , determine the order structure of $\mathbf{No}_{<\theta}$ and its permutation strata. The cone and ordinal absorption arguments for an initial cardinal need not survive unchanged. Identify which ordinal invariants of θ replace the present regular/singular dichotomy.

Research question 61.48 (The exact strength of arbitrary-input condensation). Does GB prove a uniform class coding all finite condensations of every class well-order? If not, determine a sharp sufficient and necessary principle. Then study longer histories with specified limit intersections. The explicit polynomial examples show that existence for particular long histories alone is insufficient to calibrate the general scheme.

Research question 61.49 (Intrinsic information in the full raw order). Which prefix cones, first-block fibres, and finite residual blocks are definable from the bare lexicographic order of raw well-orders? [Theorem 47.10](#) shows that the abstract core does not determine its represented cuts. Determine how much extra structure is needed for reconstruction, and whether the full raw order restores any of the lost distinctions.

Research question 61.50 (Completion with residual data). Construct and characterize a stratified completion that contains all terminal cuts while recording the residual well-order fibres over persistent branches. Determine its universal property and its minimum required size level. Ordinary Dedekind completion alone loses the residual data.

Research question 61.51 (Automorphism extension). Characterize those automorphisms of a bounded-support core that preserve persistent cuts, preserve exhaustive cuts, or extend to native global enumerations. Compare cores built from baselines differing on a proper class of positions. The abstract isomorphism of two cores does not decide their placement inside the surrounding order.

Research question 61.52 (Finer topology of bounded slices). After the density, cellularity, and point-character calculations, classify the exact gap spectra, local bases, and isomorphism types of the bounded permutation spaces. Equimorphism with a binary product does not determine all these invariants. Study which distinctions survive when cardinal arithmetic makes displayed density values coincide.

Research question 61.53 (Inverse ranks and prescribed relation scans). Classify the cuts and point characters of ordinal-rank lexicography beyond the points M_λ . For table lexicography, use endpoint transfer to characterize scans whose extremal linear-order table is well-founded or set-like. Then seek invariants beyond the three possible endpoint patterns.

Research question 61.54 (Coherent restricted filtrations). The three-label theorem rules out monotonicity for every restriction. Which nested chains of carriers, or which restricted families of orders, admit a useful monotone restriction system? Can birthday cutoffs be linked by canonical order embeddings rather than by unrestricted deletion?

Research question 61.55 (Naturally well-ordered supports). Characterize the image and omitted cuts of $A \mapsto -\sum_{a \in A} \omega^{-a}$. Compare shorter-first and longer-first orders on increasing set words, and develop a higher-level analogue that treats proper-class supports without calling the result a single surreal normal form.

Research question 61.56 (Class-model dependence). For a fixed set universe M , study changes in the lexicographic theory as the class realization grows from $\mathcal{C}$ to $\mathcal{C}'$. Which statements about existing orders are absolute, which require well-foundedness preservation, and which detect newly available cuts or enumerations? Keep full-class and Henkin interpretations separate.

Research question 61.57 (First formal milestones). Formalize the set-prefix common-disagreement formula and the elementary set-interval quotient before the entire class theory. Next verify the binary scheduler and finite-fibre fusion; then the persistent-cut reconstruction with explicit smallness. These have concise interfaces and clear failure cases, making them useful independent additions to ProveIt.

The most concrete next mathematical projects are the smaller infinite-tail classification, exact bounded gap spectra, and the automorphism-extension criterion. The most concrete foundational project is the arbitrary-input condensation scheme. Each now has a precise point where the proved construction stops, rather than an undifferentiated request to study “all well-orders.”

[merge] Status of source 15’s questions, 3 October 2026. [Research question 61.51](#) is answered in its extension clause by source 17: a core automorphism extends to the global enumerations exactly when it preserves the represented cuts ([theorem 47.5](#)), and some do not ([theorem 47.7](#)); the comparison of cores from baselines differing on a proper class stays open. [Research question 61.52](#) is answered in part for the minimal slice (gap spectrum and multiplicities, [theorems 30.11](#) and [30.16](#));

local bases and isomorphism types stay open. [Research question 61.49](#) and [research question 61.56](#) are answered in part as recorded in [section 60](#) (e)–(f); for [research question 61.56](#) see also [note 57.5](#). [Research questions 61.46](#) and [61.47](#) are source 12’s [research questions 61.11](#) and [61.12](#) beyond the cases answered in Part VI. The others remain open.

61.9 Source 16: further research questions, with the resolved parts removed

The following are proposed continuation problems, not a census of currently open problems in the literature. In particular, the exact termination family, represented-cut recognition with the labelled prefix tree, simultaneous residual blocks, and the stated regular-stage topological invariants have been settled under the assumptions of this article. They should not be relisted as wholly open.

61.9.1 Highest-priority structural problems

Research question 61.58 (Longer-stratum binary coding ranks). For $\mathbf{No}_{<\kappa}$ and $\mu = 2^{<\kappa}$, determine the least ordinal ρ for which $\mathcal{W}_\alpha(\mathbf{No}_{<\kappa}) \hookrightarrow 2_{\text{lex}}^\rho$ for every $|\alpha| = \mu$, and determine the reverse cube embeddings.

The topology is now known for all these strata, but it does not determine binary coding rank. The singular strong-limit transition ([S16.2](#)) and the finite residual permutation blocks give concrete starting cases. The first targets should be $\alpha = \mu + 1$, $\mu + 2$, and $\mu + \omega$, with separate treatment of singular strong-limit κ .

Research question 61.59 (Singular birthday gaps). Determine the unfilled cut-character spectrum of $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ when κ is singular, and when κ is regular but $\mu > \kappa$.

The global characters $\text{cf}(\kappa)$ and the exact density $\mu^{<\mu}$ are known here. The missing information is local: which numerical gaps in the truncated alphabet combine with exhaustion and prefix edges to produce each pair of cofinalities? The bounded first-failure analysis should be formulated before attempting a general classification by cardinal arithmetic alone.

Research question 61.60 (Centered increasing words). Classify the order of numerically increasing set-length words when the termination symbol is placed at a fixed interior numerical threshold.

[Theorem 36.2](#) covers unrestricted and injective words. The increasing case has prefix-dependent available alphabets and already fails the centered saturation theorem. A useful first problem is to determine all point characters for the 0^- and 0^+ placements, including words that cross the threshold only at a limit stage.

Research question 61.61 (Recovering prefix structure from order alone). Which features of the labelled prefix tree can be reconstructed from the bare order $\mathcal{P}_{\text{bd}}(b)$, from $\mathfrak{W}_{\text{sl}}$ as a definable comparison, or from $\mathfrak{W}_{\text{all}}$?

Our cut recognition uses the family of cylinders and numerical labels as extra predicates. A bare core is isomorphic to $\mathbf{No}$, and singleton raw fibers can arise either from an exhaustive branch or from a branch omitting one label. Thus neither the core order nor the size of a fiber immediately identifies exhaustion. One can ask for the least additional predicates that make the tree and exhaustion relation definable.

Research question 61.62 (Changing baselines and extending automorphisms). For two baselines b, c , characterize when an automorphism of the set-like global comparison carries $\mathcal{P}_{\text{bd}}(b)$ onto $\mathcal{P}_{\text{bd}}(c)$.

An increasing numerical automorphism h gives the explicit example $E \mapsto h \circ E$, carrying $\mathcal{P}_{\text{bd}}(b)$ onto $\mathcal{P}_{\text{bd}}(h \circ b)$. General abstract isomorphisms between the two cores need not preserve the distinguished set of exhaustive branch cuts. [Theorem 46.2](#) suggests an extension criterion stated in terms of precisely those cuts.

61.9.2 Foundational problems with explicit targets

Research question 61.63 (Exact strength of uniform nonembedding). Can the conclusion of [Theorem 56.2](#) for elementary transformation formulas be proved in GBC alone? If not, what is its reverse-mathematical strength?

The missing step is sharply localized: forming the inverse evaluator ([S16.16](#)). A lower bound would need a model in which a specified elementary transformation exists or a reduction from a comprehension principle. The mere presence of an existential class quantifier in the present proof does not establish necessity.

Research question 61.64 (Repeated condensation of the quotient). Determine the strength and possible stabilization behavior of the transfinite tower obtained by repeatedly taking least representatives of set-interval blocks and intersecting at limits.

[Theorem 52.6](#) has removed the need for ETR to extract all blocks of one order. The new question concerns repeated condensation of the resulting index orders. Suitable ETR is a sufficient bound; whether this particular operator needs it, and which class orders can remain nontrivial at prescribed stages, are separate questions.

Research question 61.65 (Choice needed for the word isomorphisms). How far can [Theorems 35.1](#) and [36.2](#) be proved without a global well-order?

Their explicit embeddings and the termination saturation criterion use less than the GBC back-and-forth isomorphisms. Investigate whether set choice, a restricted class-enumeration principle, or a canonical sign construction suffices for particular cases. This does not settle the broader historical question about the choice strength of universal embeddability into **No**.

Research question 61.66 (Class extensions and canonical data). For a fixed set universe with two admitted-class expansions, describe which new branch cuts, global well-orders, and uniform transformations become available.

Shared relations retain the same internal well-foundedness under GBC, but the available class graphs can change. It is useful to distinguish absoluteness of the elementary formulas for blocks and recognition from existence of their class parameters or of the transformations being considered.

61.9.3 Formalization and quantitative coding

Research question 61.67 (A first verified theorem package). Implement the set-alphabet ordinal spectrum and the $\mu, \mu + 1, \mu + 2$ versus later-strata density theorem, then instantiate them at the small birthday carriers already in ProveIt.

This package would verify both the combinatorial mechanism and its smallness boundaries without first formalizing arbitrary class models. The two-point-fiber case is a particularly useful regression target: counting jumps as disjoint open intervals would produce a false proof.

Research question 61.68 (Quantitative compatibility of the constructions). Find sharp birthday and support bounds for prefix completion, termination encodings, and the tree isomorphisms; determine which maps commute with increasing birthday cutoffs.

Equation (S16.7) is explicit, while the global back-and-forth isomorphisms generally are not birthday-minimal. Compatibility should mean equality or a specified coherent commuting diagram of maps, not only the existence of unrelated embeddings at each stage.

[merge] Status of source 16's questions, 3 October 2026. [Research question 61.59](#) is **answered** by source 17 ([theorem 30.11](#); [section 33](#)). [Research question 61.60](#) is answered by source 18 for the threshold at zero ([section 37](#); [section 39 \(d\)](#)); other interior thresholds are open. [Research question 61.58](#) is answered for $\alpha = \mu \cdot \gamma + n$, which includes the first targets $\mu + 1$ and $\mu + 2$, by source 15's [theorem 31.2](#); $\alpha = \mu + \omega$ and the reverse cube embeddings beyond those types are open. [Research question 61.62](#) is answered in part by source 17's extension criterion and relabelling ([theorem 47.5](#) and [proposition 47.9](#)), which is the criterion the commentary anticipates. [Research question 61.65](#): source 18's proof of the word isomorphisms uses one set-like global well-order as a parameter ([section 39](#)). [Research question 61.61](#) and [research question 61.66](#) are answered in part as recorded in [sections 50](#) and [60](#). The others remain open.

61.10 Source 17: further research questions and a prioritized program

These questions are continuations of this article, not claims that each is a historically catalogued open problem. The results above narrow their scope: questions about minimal-slice characters, the displayed gap spectrum, or mere recognition with the full labelled prefix structure are no longer left open within this development.

61.10.1 Set-sized order geometry

Research question 61.69 (Nonminimal enumeration types). For $|\alpha| = \mu = 2^{<\kappa}$, classify the characters, gap spectrum, density, and binary coding length of $\mathcal{W}_\alpha(\mathbf{No}_{<\kappa})$. Begin with $\alpha = \mu \cdot 2$ and $\mu + n$.

A prefix can now use μ labels before exhausting its coordinate domain, so residual alphabets can be small, have endpoints, or be finite. The proof of [Theorem 30.11](#) suggests replacing the single residual-alphabet interface by a tree whose nodes carry their actual residual order types. A theorem expressed only in $|\alpha|$ cannot capture all such effects.

Research question 61.70 (Noncardinal birthday cutoffs). Replace the initial cardinal κ by an arbitrary infinite ordinal θ . Which local invariants depend on the ordinal tail of θ , not merely on $|\theta|$ and $|\mathbf{No}_{<\theta}|$?

Our cone argument uses $\delta + \gamma < \kappa$ for smaller ordinals. That property fails at many noncardinal cutoffs. A useful first target is a cutoff of the form $\kappa + \omega$, followed by $\kappa \cdot 2$, with the contributions of each birthday layer kept separate.

Research question 61.71 (Isomorphism after the invariant calculation). When do two minimal permutation orders with the same values of $r, q, \mu^{<\mu}$ and the same gap spectrum have the same order type? Which additional invariants distinguish them?

The explicit gap classification provides more information than binary equimorphism, but still need not determine an order. Candidate invariants include the arrangement of cylinder-boundary cuts, densities of particular convex suborders, and the interaction of branch cuts with a chosen prefix filtration. No general isomorphism conclusion is asserted here.

Research question 61.72 (Beyond the proved fusion length). Determine the least size of a family of dense open sets whose intersection fails to be dense. When $q = \omega$, determine complete metrizability and more precise descriptive topological properties directly from the permutation presentation.

[Theorem 30.5](#) supplies a lower bound: every family of size at most q succeeds. For longer constructions, intermediate prefixes can already have length μ , so the simple small-history argument no longer applies. This is an obstruction to that proof, not a proof that longer fusion always fails.

61.10.2 Global branches, automorphisms, and class realizations

Research question 61.73 (The extendible automorphism group). Describe the stabilizer of the represented-cut predicate in $\text{Aut}(\mathcal{P}_s(b))$. Can one characterize it by preservation of fewer predicates than the entire labelled cylinder and coverage structure?

[Theorem 47.5](#) identifies this stabilizer abstractly, and [Theorem 47.7](#) shows it is proper. The remaining task is a useful structural description. Numerical relabellings give a natural family, but need not exhaust it.

Research question 61.74 (Changing baselines). For two baselines b, b' , when does an automorphism of $\mathfrak{W}_{\text{sl}}$ carry $\mathcal{P}_s(b)$ onto $\mathcal{P}_s(b')$? How does the answer depend on their eventual-agreement classes and on their represented-cut predicates?

Relabelling by an increasing numerical bijection is sufficient when it carries one baseline to the other. Abstract order-isomorphism of the two cores is not sufficient by itself; the obstruction proved here explains why the represented cuts must enter the analysis.

Research question 61.75 (Recovering a class realization). What information about $\mathcal{X}$ can be reconstructed from the labelled core together with its admitted represented cuts? What is lost if the baseline and cylinder predicates are forgotten?

[Theorem 48.4](#) detects every proper same-set extension through new represented cuts. Its pair encoding recovers an individual ordinal subclass from a suitable enumeration. A stronger reconstruction should specify the decoding language and distinguish existence of individual codes from a universal truth or satisfaction predicate.

Research question 61.76 (Iterated proper-class blocks). Determine the precise recursion strength needed to organize successive canonical Ord-blocks of a raw well-order along a prescribed index order. Compare finite iteration, set-length iteration, and class-length iteration.

One extraction is elementary; iterating a procedure whose state is a class can require more. The distinctions involving ETR in [22] provide the appropriate setting for a reverse- mathematical analysis. Our direct comparison and branch recognition do not by themselves solve this recursion problem.

Research question 61.77 (The condensed unrestricted order). Classify the dense comparison obtained by collapsing the finite components of [Theorem 54.4](#). Which residual fibres remain detectable from its bare order, and what embeds into it?

Finite adjacency is now fully described. The quotient can still contain nontrivial fibres with infinite or proper-class residual carrier. It would be incorrect to conclude from density alone that it is the set-like order or an ordinary surreal-sized class.

61.10.3 Effective bounds, coherence, and verification

Research question 61.78 (Ordinal and birthday optima for supported completion). Refine the cardinal reserve formula to optimal ordinal support bounds and optimal birthdays of canonical support codes. Which optima are uniform without an additional global choice parameter?

Two fresh reserves of the same cardinality can occupy very different ordinal intervals. [Theorem 49.7](#) isolates the unavoidable cardinality; ordinal placement and coding complexity remain separate. A constructive version should specify the representation of partial injections and the operation used to choose a complement bijection.

Research question 61.79 (Coherent cutoff embeddings). Construct embeddings between the orders at different birthday cutoffs which compose exactly and preserve a useful prefix filtration. Determine which can also preserve cut types or finite condensation.

Appending an arbitrary tail preserves some comparisons but can change the required enumeration type. A coherent reserve mechanism should be chosen once for the whole system rather than separately at each pair of cutoffs. The support theorem may help isolate its minimal size costs.

Research question 61.80 (First checked milestones). Formalize the reserve theorem, elementary common-prefix comparison, and set-sized cut trichotomy before attempting the class automorphism theorem. Can the abstract alphabet interface support other saturated or non-Archimedean ordered alphabets without changes to the proofs?

These milestones have explicit input and output types and no dependency on a formal universe of all class well-orders. Their value is not merely in reproducing finite tests, but in certifying the limit and cardinal arguments that the tests cannot inspect.

Research question 61.81 (Typed constructions indexed by presentations). Which algebraic or model-theoretic constructions can use the set orders $\mathcal{W}_\alpha(X)$ or uniformly coded global subfamilies as index orders, while keeping every support and evaluation map within a specified size level?

A promising first investigation should use a bounded index set, verify that the output is independent of the chosen presentation in the intended sense, and only then consider a uniform class version. The analogy with Kanovei–Shelah is a guide to construction, not permission to use an uncoded higher-order totality as a set.

[merge] Status of source 17's questions, 3 October 2026. [Research question 61.77](#) is answered externally for countable ω -standard models by source 18: there the collapsed order is the rational order, densely coloured by the factorial block sizes ([theorem 58.6](#)); its internal classification is open. [Research question 61.69](#) is answered for the binary coding length and the density of the strata $\mu \cdot \gamma + n$ by sources 15 and 16 ([theorems 27.2](#) and [31.2](#)); characters and gap spectra beyond the minimal slice are open. [Research question 61.76](#) is answered in its block clause by the block decomposition ([note 52.1](#)). [Research questions 61.70](#) and [61.79](#) are source 12's [research questions 61.12](#) and [61.19](#). Source 17's own table of the earlier questions it addresses (in [section 4](#)) is corrected there. The others remain open.

61.11 Source 18: revised research questions and priorities

The following are proposed directions left after the theorems above. They are not presented as a list of previously catalogued open problems. Each states the extra information that is still needed.

61.11.1 Bounded order types and their completions

Research question 61.82 (The arrangement of gaps). For fixed κ , classify $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ up to order isomorphism, or classify its completion together with the subset of realized points. Which features go beyond the complete character spectrum and the multiplicities proved here?

The new spectrum and counting formulas turn this into an arrangement problem. In particular, a large family of gaps of one character may have a highly nontrivial induced order. A first target is $\kappa = \omega$, followed by ω_1 under specified continuum arithmetic. Equimorphism with a binary cube does not settle this.

Research question 61.83 (Longer fixed enumeration types). Determine gap spectra and binary coding lengths for $\mathcal{W}_\alpha(\mathbf{No}_{<\kappa})$ when $\alpha > \mu + 1$, especially $\alpha = \mu \cdot 2$ and $\mu \cdot \gamma + n$.

Density and cellularity are now known for every α of cardinality μ , and finite-tail point characters are explicit. Later infinite blocks can use up an entire μ -sized residual region, so the minimal-type proof cannot simply be repeated.

Research question 61.84 (Noncardinal birthday bounds). For a noninitial ordinal θ , determine the spectrum of $\mathcal{W}_{|\mathbf{No}_{<\theta}|}(\mathbf{No}_{<\theta})$. Which ordinal features of θ survive in interval sizes and cut characters?

The present proof uses the closure $\beta + \gamma < \kappa$ for $\beta, \gamma < \kappa$, which may fail for θ . The appropriate replacement is a stratified description of sign cones, not a substitution of $|\theta|$ for θ .

Research question 61.85 (Coherent enlargement of cutoffs). Construct embeddings between order spaces for increasing birthday cutoffs that compose exactly, and specify whether they preserve minimal enumeration length, prefix cylinders, finite condensation, or selected cut characters.

Appending a fixed tail gives an embedding of full well-order spaces, but need not preserve minimum-length strata or compose with a second independently chosen tail. A global rank-layer schedule gives a candidate for coherent full spaces; the sharper minimal-stratum requirements remain separate.

61.11.2 Class cuts, blocks, and automorphisms

Research question 61.86 (Iterating the quotient rather than removing blocks). Choose a precise limit rule and determine the strength of iterating $R \mapsto B_R$ along specified class well-orders. Which fixed points, periodic behaviors, or invariants occur?

The single decomposition and ordinary ordinal-indexed block removal are elementary. The finite-support example is already a fixed point up to isomorphism. Any claim that repeated condensation automatically reduces complexity must confront that example. The axiomatic strength of the genuinely iterated operation is not determined here.

Research question 61.87 (Order structure on represented class cuts). Study the class cuts satisfying the straddling-prefix criterion, with predicates marking exhaustive branches and residual tail types. Which natural automorphisms of the bounded-support core preserve that distinguished collection of cuts?

The criterion gives a recognition theorem. It does not classify the automorphism action on all class cuts, or show that every automorphism of the core extends to the global order. Baseline changes differing on a proper class of positions are a concrete first case.

Research question 61.88 (The expanded countable model). Classify $(\mathfrak{W}_{\text{all}}^{\mathcal{M}}, \mathfrak{W}_{\text{sl}}^{\mathcal{M}})$, and then add the evaluation relation to the surreal labels. Which features depend on the countable class realization once the pure order has become the same factorial shuffle?

The pure order cannot define set-likeness even with finite parameters. An added predicate can therefore retain information that the shuffle classification necessarily discards.

Research question 61.89 (ω -nonstandard class models). Describe the external finite-condensation blocks and the elementary theory of $\mathfrak{W}_{\text{all}}$ for countable ω -nonstandard GBC models. Do these orders share the first-order theory of the standard factorial shuffle, while realizing additional types?

Internally finite permutation cylinders become externally infinite discrete intervals. Standard finite adjacency chains inside them already invalidate the standard classification. Determining the possible global assembly of these intervals is a concrete model-theoretic problem; no positive answer to the question is claimed here.

61.11.3 Words, encoding, and formalization

Research question 61.90 (Sharp word-code bounds). Optimize the birthday of explicit order embeddings of injective and increasing words, recording separately preservation of numerical order, prefix extension, and simplicity. Can the centered encodings be made comparably efficient at singular stages?

The duplicated-sign delimiters provide transparent proofs, rather than optimal codes. Uniform tree isomorphisms obtained by back-and-forth supply little direct birthday control.

Research question 61.91 (The centered increasing order itself). Classify the entire centered increasing-word order up to isomorphism and determine the intrinsic definability of its all-negative cones and its positive-limit points.

Its local characters, its complete set-cut obstruction, and the two types of rooted cones are known here. The contrast between that intrinsic information and the displayed entry signs makes the automorphism problem meaningful.

Research question 61.92 (Weak choice and uniform constructions). Which word-tree isomorphisms and class-core identifications can be proved without Global Choice? For a specified operation, distinguish existence, a uniform definable implementation, and an effective implementation on concrete codes.

The explicit sign embeddings and many cut arguments need no Global Choice. Our class isomorphisms use deterministic global back-and-forth. These are different strength claims; a uniform family of class maps should not be selected without proof.

Research question 61.93 (A first machine-checked milestone). Formalize the elementary common-prefix relation and the set-interval decomposition for a supplied class-model relation, alongside the set-level ordinal spectrum and minimal permutation cut trichotomy. Can one order-theoretic interface serve all four?

This is a realistic modular target because it avoids almost all surreal arithmetic. The initial deliverables should expose their size assumptions in the theorem signatures and make the conventional proof-to-formal-statement correspondence reviewable.

[merge] Status of source 18's questions, 3 October 2026. Source 18 was written before sources 13–17 arrived. [Research question 61.83](#) is answered in part by source 15 (binary coding lengths of $\mu \cdot \gamma + n$, [theorem 31.2](#)); [research question 61.87](#) in part by source 17's extension criterion ([theorem 47.5](#)), which shows that a core automorphism extends to the global order exactly when it preserves the represented cuts; [research question 61.92](#) bears on source 16's [research question 61.65](#). [Research questions 61.84](#) and [61.85](#) are source 12's [research questions 61.12](#) and [61.19](#); [research question 61.90](#) is source 11's [research question 61.6](#). The others remain open.

62 The sources' conclusions

[merge] Source 11 has no conclusion section; its summary is [section 63](#). The conclusions of sources 12, 09 and 08 are printed whole.

62.1 Source 12's conclusion

There is a rigorous full lexicographic order of all global surreal well-orders, provided it is understood as a relation on class objects, not as an ordinary class containing those objects. Its comparison needs no universal enumeration of abstract class order types: the largest common *labelled* prefix is elementary and sufficient.

The set-like suborder has a particularly uniform geometry. It fills set-indexed cuts, contains every class linear order even locally, and is equimorphic with binary classes on Ord. The unrestricted order adds new proper-class prefix phenomena, including finite adjacent blocks. The diagonal theorem prevents either construction from being exhausted by a same-level class of set codes.

At set-sized birthday bounds the theory becomes quantitative. The full well-order space of a carrier of cardinality μ has exact ordinal spectrum below μ^+ and binary rank μ^+ . The minimal surreal slice has rank μ , except at singular strong-limit cutoffs, where it has rank $\kappa \cdot \kappa$. This is an order-theoretic effect of cofinal blocks of code lengths, invisible to cardinal counting alone.

The inaccessible-rank interpretation shows exactly how these facts can coexist: internal classes and the external collection of all such classes occupy different levels. Respecting those levels removes the apparent paradox without weakening the embedding and universality questions into trivialities.

62.2 Source 09's conclusions

The lexicographic well-ordering problem has three different answers, depending on its carrier. On a fixed set of surreals of size κ , the exact ordinal spectrum stops at κ^+ . Allowing arbitrary set-length words gives a class-universal dictionary order, but prefix limits give it local order properties different from those of **No**. For set-like global well-orders, first disagreement again defines a linear comparison. Existence of a global enumeration is equivalent to global choice over GB with set choice, and all global enumerations cannot be collected into an ordinary class of set codes.

The eventual-agreement core resolves the constructive side of the global problem: it is a genuine class, it is isomorphic to the surreal line, it realizes every uniformly set-sized cut of global enumerations, and it is dense in every global interval. Bounded reflection explains why set-indexed configurations can always be studied in a suitable bounded permutation stage. The explicit missing branch and binary diagonal identify two independent boundaries: arbitrary class-sized cuts need not have solutions, and no single uniform class-indexed coding covers all global enumerations.

These boundaries are not paradoxes. They are the information needed to state theorems at the correct logical level.

62.3 Source 08’s conclusion

The global enumeration problem has a particularly clean answer once the types of objects are separated. Set-like well-orders of **No** are represented by class bijections from **Ord**, and numerical first-disagreement gives a rigorous virtual linear order. That order is large enough to encode every class binary word, yet its set-cut geometry is already witnessed by a single class of set-supported perturbations. This code class is another copy of the surreal line.

At a fixed set carrier, the structure is different: its exact ordinal spectrum stops at the successor of the carrier cardinal, and finite permutation tails account for every jump. At varying set carriers, termination introduces new local cofinalities. For non-set-like well-orders, a further index convention or relation code is necessary. None of these distinctions is optional bookkeeping: each changes a mathematical conclusion.

The principal proofs are elementary in their stated foundations. Their formalization should begin with bounded objects and the repository’s existing sign-sequence cut theorem, and only then pass to the uniform class formulation. The dense-skeleton theorem provides a concrete target: it combines first-disagreement combinatorics, controlled set support, small-cut filling, and universe-aware back-and-forth without requiring a paradoxical collection of all classes.

[merge] (3 October 2026.) The conclusions of sources 13, 14 and 17 follow; source 13’s includes its result ledger. Sources 15, 16 and 18 end with status tables instead, printed with the ledgers in the next section. In source 13’s ledger, the row “No full two-point split embedding” is a further statement whose proof is source 11’s ([note 55.6](#)).

62.4 Source 13: conclusions and result ledger

The strongest organizing principle is not that a single enormous well-order has been discovered. It is that different notions of “all” and different representations produce distinct linear orders, while a small collection of explicit tools connects them safely.

The fixed-set layer has its own cardinal and ordinal spectra. The ordinary class of set words depends decisively on termination. The full set-like enumeration domain has a surreal-sized dense core but no exhaustive uniform class of set codes. The raw order domain expands individual branch fibers, not the entire cut space of that core. A higher-level product can exceed the dense core’s capacity even though ordinary class-indexed sums are absorbed. Finally, pure order isomorphisms can change the prefix presentation itself.

Result	Status in this article	Stated setting
Fixed-alphabet spectra and singular cutoff dichotomy	Credited prior written results	ZFC
Direct raw comparison and head fibers	Prior mechanism, re-proved for use	GBC suffices here
Core completion, set interpolation, diagonal	Prior foundation, self-contained proofs	GBC
Crossing criterion and label coverage	Further result, Theorem 46.15	GBC
First-failure trichotomy and character formulas	Further result, Theorem 46.18 and Proposition 46.19	GBC

Result	Status in this article	Stated setting
No-left-factor absorption	Further formulation and proof, Theorem 55.1	Uniform GBC maps
No full two-point split embedding	Further result, Theorem 55.3–55.4	Full inaccessible universe; uniform KM + GC schema
External rational presentation	Semantic contrast, Proposition 55.5	Externally countable model
Cylinder non-rigidity	Further result, Theorem 47.12	Uniform GBC automorphism
Central-word surreal isomorphism	Further result, Theorem 36.4	GBC, reserved zero marker
New Lean proofs	Not supplied	Formalization plan only

The follow-up therefore gives explicit answers to several concrete questions raised by the repository while leaving stronger classification and minimal-axiom problems clearly identified. The proofs and their stated size hypotheses are the basis on which these conclusions should be evaluated.

62.5 Source 14: conclusion

The strongest organizing distinction is not between large and small cardinalities alone. It is between set-coded presentations, class-valued branches, and raw class relations with further tails. Prefix cylinders make that distinction mathematically visible.

Every normalized class cut of the set-supported core either stops at a cylinder edge, stops at a genuine numerical gap, or follows an injective branch through all ordinals. Exhaustive well-orders are exactly the branches that cover the alphabet. Because this criterion reconstructs a graph using only set quantifiers and old parameters, it yields a no-new-realization theorem for old cuts. Raw well-orders have the same trace precisely when their canonical initial branches agree, leaving an explicit residual well-ordering problem in each fiber.

Support restrictions provide a different axis of complexity. Allowing support of cardinality $< \kappa$ gives exactly the cut-filling threshold $\text{cf } \kappa$, and the failure beyond it has an attained support cost. All these layers still contain a copy of the full surreal order. Thus class universality, density in an ambient comparison order, and saturation are genuinely different properties.

The resulting theory remains paradox-free because each “all” has a specified sort, each prefix has a specified size, and each use of surreal cut filling has set option families. The automorphism-orbit theorem sharpens this distinction: bare-order symmetries can exchange exhaustive traces with nonbranch boundaries. The principal open targets are now the exhaustive-trace stabilizer, finer support invariants, and the interaction between raw tail structure and higher class realizations.

62.6 Source 17: conclusion

The lexicographic ordering problem separates into several mathematically substantial structures. On the minimal birthday-cutoff slice, the relevant parameters are the cardinality $\mu = 2^{<\kappa}$ and two different cofinalities $r = \text{cf } \kappa$, $q = \text{cf } \mu$. Point characters are (q, q) , cylinder cofinalities are r , and three distinct cut mechanisms give the exact gap spectrum. Exhaustiveness is also the key issue in fusion: a coherent full-length injection need not be a permutation.

The same distinction persists at class scale. A single lower class cut of a supported core determines an injective global word exactly when its labelled cylinders straddle at every ordinal

length. A separate coverage condition distinguishes exhaustive words. Reconstruction is elementary, but exhaustion is invisible to the bare core order: a core automorphism can carry an exhaustive branch gap to a nonexhaustive one. The represented- cut predicate is therefore genuine additional structure, and same-set class extensions change it while leaving the code carrier unchanged.

For unrestricted class well-orders, a proper-class prefix can leave a finite residual alphabet. The resulting maximal adjacency components are exactly the factorial blocks of a finite final segment; their condensation is dense without becoming an ordinary class of all class orders. Across all these conclusions, typing the objects and bounding the recursion histories are part of the mathematics, not merely notational cautions.

63 Results, assumptions, and verification status

Result	Foundation used here
Fixed-alphabet cardinality and ordinal spectrum	ZFC; existing repository theorem
Explicit set-length word compression	ZF, as a definable class construction
Local word characters and set-cut arguments	Surreal set cuts; ZFC for cardinal-valued characters
Any class well-order of No implies global choice	GB with ZF set part
Raw comparison, canonical initial part, finite-jump criterion	GB, with the given class orders
Bounded-support completion and interpolation	GBC, with a fixed global enumeration
$\mathcal{P}_{\text{bd}} \cong \mathbf{No}$ and class-order universality	GBC, by set-valued back-and-forth
Binary class coding and no exhaustive uniform evaluator	GBC, interpreted with class parameters
Unrestricted order does not embed into its set-like fragment	Full inaccessible universe; or KM + GC for definable uniform maps
All higher-order collections treated as actual objects	An explicitly chosen higher-order or containing-universe setting

The proof dependencies are short. Set cuts yield fresh coefficients after deleting a set. Fresh coefficients and bounded-support completion yield strong interpolation. Strong interpolation yields density, local cofinality statements, the **No**-isomorphism, and cut coding. Independently, the predecessor comparison yields convex cones, the canonical initial-part decomposition, and the finite-residual jump theorem. Pair swaps supply both positive binary embeddings and the diagonal obstruction.

The bundled Python script checks finite instances of the two comparison rules and of the sign encoding. It checks 465 coefficient-code comparisons, 79,800 word-code comparisons, and 266,272 pairs of finite permutations, with the predecessor rule checked in both directions and the adjacency criterion compared with the finite lexicographic order. These checks help detect a reversed delimiter or an incorrect finite comparison. They do not verify an ordinal induction, a class comprehension argument, or a cardinal nonembedding theorem.

The article contains full mathematical proofs for the claims it promotes to theorems. It does not claim that the new results have been checked in Lean, that the whole repository has been audited, or that every research question listed here is a previously published open problem.

[merge] The other sources' ledgers follow; each records the hypotheses its source used, not that each is necessary.

63.1 Source 12's proof-dependency and assumption ledger

Result	Stated setting	Essential input
Common labelled prefix and full lexical comparison	GBC; comparison is elementary	Existing well-orders; class minima; ambient linear order.
Global-choice equivalence	GB + set AC	Ordinal coding of transitive closures; restriction of a surreal well-order to a set of codes.
No exhaustive class-coded family	GBC	A baseline Ord-bijection; elementary diagonal pair orientation.
Set-like cut interpolation	GBC	Set-sized discrepancy bound; surreal small-cut filling; prefix extension.
Local class universality	GBC	Set-valued forth and back-and-forth recursions; global enumeration.
Finite-tail adjacency	GBC	Label transpositions; finite residual cylinders; no infinite order well-ordered in both directions.
Set-sized full spectra	ZFC	Pair coding; regularity of μ^+ ; cylinder-fusion nonembedding.
Minimal-slice singular transition	ZFC	Cone partition; scheduler; ordinal block sums; regularity or singular cofinality as appropriate.
Full-class rank model	Ambient ZFC + an inaccessible κ	Closure of V_κ ; all external classes; regularity for descending sequences.

63.2 Source 09's theorem-status ledger

Result	Logical setting and proof input	Status here
Surreal set cuts	Set options; standard sign-sequence simplicity	Classical; repository primitives inspected.
Class universality and back-and-forth	GBC; set-valued recursion	Classical mechanism, proved in full.
Global baseline $\Leftrightarrow$ GC	GB plus set choice; well-founded ordinal codes	Standard coding argument, proved here.
Fixed-set cardinality and ordinal spectrum	ZFC; pair switches and regular-cardinal stabilization	Generalization/reproof of the set-sized route.
Limit-prefix character of $\mathcal{I}$	ZFC schema or GBC class statement	Written proof.
Bounded-support extension	GBC with fixed b ; set cardinal arithmetic	Written proof; finite mechanism tested.
Core interpolation and $\mathcal{P}_s(b) \cong \mathbf{No}$	GBC; uniform families, small cuts, extension	Main construction; written proof only.
Bounded reflection	GBC; Replacement for a uniform set family	Written proof.
Missing (Ord, Ord) branch	GBC; ordinal prefixes and uniform completions	Explicit counterexample with proof.
Uniform no-coding	GBC; pair switches and diagonal class	Written proof; finite diagonal tested.
Inaccessible model	ZFC plus a strongly inaccessible cardinal	Conditional set-sized theorem, proved here.
Proposed Lean modules	Explicit universes, smallness, and baseline parameter	Plan only; not compiled or verified.

63.3 Source 08’s assumption and proof-dependency ledger

Result	Assumptions	Main dependencies
Fixed-carrier cardinality and universality	ZFC, $ X = \kappa$ infinite	Pair blocks and lower-cut bits
Exact ordinal spectrum	Same	Regularity of κ^+ , not of κ
Adjacency classification	Same	Extreme residual enumerations, two-sided well-foundedness
Regular cutoff theorem	κ regular, $2^{<\kappa} = \kappa$	Birthday bounds, terminal coding, active-row cuts
Pairwise global comparison	Class parameters, elementary comprehension	A least disagreement below one witness
Orbit prefix extension	Baseline b , class replacement, set Choice	Enlargement to a bigger set cardinal
Orbit interpolation	Baseline and uniform set families	Cross-disagreement bound, fresh entry, prefix extension
Skeleton isomorphism and class universality	GBC	Global enumerations, uniform set-valued recursion
Virtual binary equimorphism	Baseline, elementary class operations	Pair blocks and ordinal concatenation
No exhaustive class-uniform list	GBC for arbitrary class indices	Pairwise diagonalization
Full-class model example	External strongly inaccessible κ	Relative sets versus external sets

The ledger records the hypotheses actually used in the supplied proofs. It does not claim that every hypothesis is logically necessary. In particular, the foundational optimization questions in [section 61.4](#) remain distinct from the sufficiency results.

[merge] (3 October 2026.) The ledgers and status tables of sources 14, 15, 16, 17 and 18 follow (source 13’s is with its conclusions in [section 62](#)). Each records what its source proved in writing and what it inherited; “new” there means new relative to this report at the source’s pin, not new relative to the sibling sources, which proved several of the same theorems independently ([section 4.5](#)). Source 18’s table is printed whole, as is its remark that its review corrected the arbitrary-alphabet density construction to include isolated global endpoints (the correction is in the printed theorem, [theorem 27.9](#)). Source 16’s last subsection describes its delivered package; the shipped names are listed in [section B](#).

63.4 Source 14: result ledger and verification status

Result	Status in this package
Fixed-set cardinality, ordinal spectrum, adjacency	Inherited from the pinned ProvelT report; stated with source, not newly established here.
Birthday minimal-slice dichotomy	Inherited; ordinal multiplication and mutual embeddability explicitly retained.
Core cylinders and prefix completion	Reproved as prerequisites; basic constructions already present in the predecessor.
Boundary trichotomy, Theorem 43.3	Main continuation result; complete written proof, relative to named cylinders.

Result	Status in this package
Character table, Theorem 44.1	Written derivation from the trichotomy, including the principal-cut correction.
Exhaustive recognition, Theorem 45.1	Answers the prefix-enriched version of a predecessor research question.
Class-model absoluteness, Theorem 48.1	Written proof using the elementary reconstruction formula and fixed old parameters.
Cut automorphism orbits and nonrecognizability, Theorem 47.2	Written proof by set homogeneity and two-sided back-and-forth; distinguishes order-invariant classification from prefix classification.
Support compression and exact threshold, Theorem 49.3	Main continuation result; finite kernel checked computationally, infinite/cardinal proof supplied in text.
Core isomorphism with No ; diagonal non-coding	Inherited statements; proofs or recovery arguments included for context.
Direct raw comparison and canonical branch	Inherited constructions; short proofs given to support the trace analysis.
Trace fibers, Theorem 54.6	New synthesis of the canonical branch with cut recognition; written proof.
Historical priority of continuation results	Not established; no claim of a first result in the published literature.
Lean verification of continuation results	Not performed. Proposed module names and pseudocode are not compiled artifacts.

The entire manuscript is an unrefereed research contribution. The ledger separates what is proved in writing, what is inherited, and what remains unverified; a finite diagnostic is never substituted for a class theorem.

63.5 Source 15: what is settled here and what remains

Topic	Result and boundary	Foundation
Set-sized spectra	Cardinality and ordinal universality are inherited background, with proof reproduced	ZFC
Native class comparison	Direct first unequal labels; no abstract class-type comparison required	GB with given class well-orders
Canonical block quotient	All first-level blocks defined simultaneously; any class index can occur	GB for quotient; GBC for realization
Long condensation histories	Explicit polynomial histories for every given class well-order; no arbitrary-input equivalence to ETR	GBC
Represented cuts	Persistence at every set prefix level is necessary and sufficient; all first failures classified	GBC with fixed core
Abstract reconstruction	An automorphism of the core can destroy representability	GBC

Topic	Result and boundary	Foundation
Bounded local geometry	Local character $\text{cf}(2^{<\kappa})$, global cofinality $\text{cf } \kappa$, and exact $\eta_{\text{cf } \kappa}$ threshold	ZFC
Longer fixed types	Exact cube and finite-factor classification for $\mu \cdot \gamma + n$; not all smaller infinite residual tails	ZFC
Alternative codes	Rank endpoints, table endpoint classification, quotient and restriction obstructions	ZFC or GBC as stated
Formal verification	Concrete interfaces proposed; finite examples checked; new transfinite proofs not machine-verified	Explicit status, not an added axiom

The research questions below are proposed extensions of this development. Except for a cited historical question, they are not claims that a named problem has an independently verified current open status in the literature.

63.6 Source 16: proof status, assumptions, and reproducibility

Result	Assumptions	Status relative to the inspected report
Set-alphabet cardinality and ordinal spectrum, Note 32.1	ZFC	Existing result, reproved.
All-stratum density and cellularity; countable incompleteness	ZFC	New extension here, with full proofs.
Exact regular-stage gap spectrum	ZFC; regular μ ; η_μ alphabet of size μ	New extension here.
Endpoint word-tree isomorphisms; complete termination type classification	GBC	New extension here; explicit embeddings need less choice.
Direct raw comparison and GC equivalence	GB; given class well-orders	Existing result, explained and reproved.
Simultaneous set-interval block decomposition and largest set tail	GB	New extension here; no ETR.
Bounded-support completion and interpolation	GBC; uniform families	Existing result, reproved.
Represented-cut recognition and first-failure classification	GBC; one available lower cut	New extension here; elementary formulas.
Elementary uniform nonembedding	GBC plus Σ_1^1 class comprehension	Sharper sufficient bound; no reversal claimed.
Full-model identification and nonembedding	Ambient ZFC plus an inaccessible cardinal	Existing result, rederived with additional stratum comparison.
Proposed Lean architecture	Explicit universe and model choices	Design only; no new Lean verification claimed.

63.6.1 What was checked

The source audit read the merged surreal well-order article and README, the reals report README, the project agent instructions, and the specific sign-sequence and size-obstruction declarations listed above, at revision 491ed8411cccf7bd57916eb7666bbba762e92f96. The audit was targeted, rather than a proof that no overlapping statement occurs anywhere else in the repository. The new arguments were developed and checked independently within this preparation, including the set-theoretic scopes of the cut, block, topology, and termination proofs. This is mathematical review, not a kernel proof or external peer review.

The accompanying Python script checks finite relation-prefix comparison, the centered and prefix-first sign encodings, and the distinction between relation-table and enumeration comparison. It evaluates finite surreal sign strings as exact dyadic rationals, so the numerical side of the encoding check is independent of the encoded lexicographic comparator. Its successful finite tests do not establish class comprehension, transfinite well-foundedness, cardinal arithmetic, or saturation.

The PDF was compiled from the included standalone LaTeX source and checked for missing references, compilation errors, and layout defects. The archive includes the source, PDF, finite diagnostic script and its results, a short build guide, and a source/status note. A standard TeX installation with the packages named in the preamble is sufficient:

```
latexmk -pdf -interaction=nonstopmode surreal_well_orders_further.tex
python3 finite_checks.py
```

No network access is needed to compile the article or run the finite checks. The bibliography gives pinned repository locations and primary literature links for independent investigation.

63.7 Source 17: theorem and dependency ledger

Result	Main dependency	Foundation	Status here
Alphabet gaps, Theorem 29.4	Sign paths, set recursion, small deletion	ZFC	Proved
Minimal-slice invariants, Theorem 30.2	Cylinder base, divergence bounds	ZFC	Proved
Fusion, Theorem 30.5	Dense refinement, coverage schedule	ZFC	Proved
Gap spectrum, Theorem 30.11	Alphabet gaps, cut trichotomy, character calculation	ZFC	Proved
Exact reserve, Theorem 49.7	Cardinal deficiency balance, complement bijection	Set choice; class notation elementary	Proved
Core interpolation and isomorphism	Surreal set cuts, supported completion, set histories	GBC	Source results re-proved
Recognition, Theorem 46.21	Unique straddlers, elementary class comprehension	GBC	Proved
Automorphism extension and failure	Recognition, large-gap homogeneity	GBC	Proved
Class-extension detection	Recognition, pair coding, common baseline	Same-set GBC models	Proved
Finite condensation	Source adjacency, no descending ω -chain	GBC; comparison-level statement	Proved

Result	Main dependency	Foundation	Status here
All-type or noncardinal classification	Residual alphabets not controlled by current interface	Not fixed	Proposed questions

“Proved” means that a written argument is supplied in this article. It does not mean external proof review, historical priority, or formal verification. The targeted source audit read the merged report, its README, its open-question sections, and the relevant source proofs at the pinned revision. It was not an exhaustive audit of the repository or the literature.

63.8 Source 18: assumptions, provenance, and verification status

Result family	Framework	Status in this deliverable
Set ordinal spectra	ZFC	Repository result, proof supplied
Bounded cardinal invariants and exact gaps	ZFC	Proved here; independent mathematical audit
Elementary labelled class comparison	GB with supplied well-orders	Repository method, proof supplied
Canonical set-interval decomposition	GB with supplied well-order	Proved here; no ETR used
Block-index realizations on No	GBC	Proved by explicit construction and transport
Core interpolation and exhaustive-code obstruction	GBC	Repository methods, proofs supplied
Represented-cut criterion	GB with a fixed baseline	Proved here; surjectivity distinguishes the set-like case
Countable factorial shuffle	Externally countable, ω -standard GBC model	Proved here; external classification
Centered word cut theorems	GB/definable-class formulation	Proved here; isomorphisms additionally use GBC
Full-class nonembedding model	Ambient ZFC with inaccessible cardinal	External interpretation and cardinal proof supplied
Lean realization of new results	Future work	Not claimed or built

The repository audit was targeted to the surreal well-order report, the reals precursor, their source-status documents, and four relevant sign-sequence foundation files. It was not an audit of the whole repository. We fixed a revision so that subsequent repository changes cannot silently change the basis of this article. The mathematical proof review checked the cutoff hypotheses, singular-cardinal steps, class replacement and comprehension uses, finite terminal fibers, and external-model assumptions. It also corrected the arbitrary-alphabet density construction to include possible isolated global endpoints.

The complete source is self-contained LaTeX with an embedded bibliography. The PDF was compiled from that source and its rendered pages inspected. Those checks establish document consistency and layout; the mathematical validation remains the conventional proofs and their independent review. The attached package makes no claim of a formal kernel certificate or a literature-priority result.

A Source crosswalk and reproducibility

The immutable sources used in the targeted audit are recorded in the accompanying source audit. For convenient verification, the most relevant mappings are:

Source	Relevant content
[1]	Labels <code>lwo:lem:prefix-free</code> , <code>lwo:lem:cylinder</code> , <code>lwo:prop:cardinality</code> , <code>lwo:thm:no-long</code> , <code>lwo:thm:ordinal-spectrum</code> ; bounded sign orders in <code>lwo:sp:thm:surreals</code> .
[3]	Set-coded surreals and bounded birthdays; unrestricted cut obstruction; class-quantification and universe distinctions; set-valued recursion.
[7]	Actual sign carrier, ordinal embedding, smallness obstruction, and bounded fragments.
[9]	Small cut operation, separator properties, and birthday bounds.
[5]	A nearby use of a set-like global well-order and ordinary class-parameter recursion.

To compile the delivered standalone source, run `latexmk -pdf surreal_well_orders.tex` from its directory. All article sections and the bibliography are included in that single file; no separate bibliography processor is needed. Run `python3 finite_checks.py` for the finite checks. The supplied PDF was compiled from the included sources and inspected after rendering.

[merge] *Delivery names.* The paragraph above describes source 11’s delivered package. Its `surreal_well_orders.tex` is shipped as `article.tex`, which this merged text replaces; its PDF is not shipped (`article.pdf` is a build of this text); its `finite_checks.py` and `finite_checks_results.txt` are `code/11-raw-orders-finite_checks.py` and `data/11-raw-orders-finite_checks_results.txt`; the “accompanying source audit” is `11-raw-orders-repository_audit.md`. Build this report with `latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex`, and rerun the checks as in [section B](#). The same applies to the reproducibility records of the other sources that follow.

A.1 Source 12: reproducibility and inspection record

The package contains `article.tex`, the compiled `article.pdf`, a build script, an exact finite-check program, its recorded JSON output, and a research-status and source-audit file. The Python program uses only the standard library. The recorded run passed all 180,905 finite comparison cases. Its comparisons are exact; there are no floating-point tolerances or inferred transfinite results. The PDF was compiled and rendered for visual inspection.

The repository audit directly inspected the merged real-well-order README and the sign-sequence simplicity Lean source at the stated commit. It did not independently compile the repository or verify the predecessor report’s proofs. The needed predecessor patterns have therefore been reproved in the present manuscript. Literature checking was focused on surreal foundations, Kanovei–Shelah indexing, and class recursion and well-orders; it was not sufficient to certify historical originality.

A.2 Source 09: repository and source audit

The repository was inspected through its GitHub connector. All cited repository files are pinned to

63bb8b1d99359e7188abf0a33fb152e121858150.

A later search surfaced a newer commit; the cited foundational files were fetched at the earlier pinned snapshot rather than mixing versions. No repository file was changed.

The targeted inspection covered the merged real well-ordering report’s README and the relevant returned source of the sign-sequence and cut-operation modules. The real report identifies itself as unrefereed and not formalized. Its README describes a merged 54-page report and separates its results from nearby Lean developments. That report is used as research context, not as a kernel-certified dependency.

The sign-sequence file provides actual Lean declarations for the universe-indexed carrier, order, ordinal embedding, and smallness properties; the cut file provides the small-cut operation and its birthday bound. Their source statements were read, but this project did not rebuild the repository or audit all transitive dependencies. No claim of independent kernel verification is made.

The primary literature checked includes Kanovei–Shelah’s original index definition, Rangel–Mariano’s explicit set/class framework, and the class-recursion distinctions in Gitman–Hamkins–Holy–Schlicht–Williams. The Kanovei–Shelah index page was also visually inspected. The literature search was targeted rather than exhaustive. General universality and back-and-forth arguments are classical; the support-core organization and the accompanying obstruction statements are proved here without a priority claim.

A.3 Source 09: verification and reproducibility

The accompanying `code/finite_checks.py` uses only Python’s standard library. It checks finite versions of the pair-switch lexicographic embedding, partial-prefix permutation completion, the forced-prefix separator algorithm, support extension preserving first disagreements, and the diagonal switch construction. Exact rational arithmetic is used for the separator tests. A deterministic random seed makes the sampled checks reproducible; other suites exhaust their specified finite ranges.

These checks do not verify any transfinite theorem. Their role is to detect indexing, orientation, and extension mistakes in the finite combinatorial mechanisms. Proofs of the infinite statements are the written arguments in the article. The recorded run passed 164,560 assertions, including 1,101 separator-family cases. The test report is `data/finite_checks.json`. The archive also contains a build script and a theorem-status ledger.

To compile the article, run `bash build.sh` in the archive’s root directory. It makes three passes with `pdflatex`, resolves the internal bibliography and cross-references, and writes build products into `build/` before copying the finished PDF to the root.

A.4 Source 08: finite regression checks

The package includes `verify_finite.py`, requiring Python 3.10 or later and only its standard library. Running it writes `verification_results.json`. The supplied run passed:

Check	Instances
Consecutive permutation pairs, sizes 1 through 7	5,906
All increasing pairs of distinct permutations, sizes through 5	7,432
Pair-code order comparisons, bit lengths 0 through 7	21,845
Sign words of lengths 0 through 6	127
All sign-code order and prefix-free comparisons	16,129

The permutation test checks both directions of the adjacency rule on all manageable pairs, not just a few positive examples. The sign test compares the numerical convention $- < 0 < +$ with the binary blocks $00 < 01 < 11$, including pairs in which one sign word is a proper prefix. The pair-code test checks that every output is a permutation of the correct finite alphabet and that comparisons agree with the input binary order.

These tests can detect a reversed pair orientation, a wrong terminal block, or a mistaken finite-tail rule. They cannot prove regularity arguments, transfinite stabilization, set/class separation, or the existence of a global isomorphism. Those depend on the mathematical proofs in the text.

A.5 Source 08: source audit and reproducibility

Two mathematical source files were inspected. One is `SignSequence.lean`; the other is `SignSequenceCut.lean`. Both were read at the pinned ProveIt commit. The project’s README was used only for its stated organization and audit policy. No build was run, and no claim was imported merely because a research report’s title suggested it. The source files have blob hashes

```
SignSequence      e2c093860f29b0b9ea312d50099fc4aa89a76a99
SignSequenceCut   c5c35afd98a5ad293687367ee9fb0244f94cc91c
```

The Kanovei–Shelah preprint was inspected directly, including the page defining its index. The class-recursion references were consulted through the author’s research summaries. Classical book and journal references provide background attribution; no claim of an exhaustive literature or priority search is made. The prior continuum manuscript was identified in the user’s file library and used as a contextual predecessor, not as a formally certified dependency.

The package contains a self-contained L^AT_EX source with an embedded bibliography, the compiled PDF, a build script, the finite verification script and its output, and a README. No external figure or private font file is needed. Compile with a standard TeX Live installation using `latexmk -pdf surreal_well_orders.tex`, or run `pdflatex` sufficiently many times to resolve cross-references.

[merge] (3 October 2026.) The audit and reproducibility records of sources 13, 14, 15 and 17 follow; source 16’s is the last subsection of its status section and source 18’s is part of its status table (both printed with the ledgers). They describe the delivered packages, whose layout differs from this directory: the shipped names of every delivered file are in [section B](#), and none of the six manuscripts or their PDFs is shipped. Their checksum manifests (sources 13, 14, 16, 17) and source 13’s build logs are not shipped either. Source 15 ships no checksums; source 18 ships no code or data.

A.6 Source 13: repository and literature audit

A.6.1 Pinned source and scope of inspection

The current main branch was resolved through the connected GitHub interface to

```
0ecd158fd163bc4aca0ee40ccf6479ca6cf5f2a8
```


with commit timestamp 3 October 2026, 17:15:49 UTC. The working tree was not modified. The audit was targeted, not an exhaustive review of every proof in the repository or a rebuild of the Lean project.

The merged surreal report’s introduction, guide, selected dense-core and cylinder proofs, raw omitted-range and model sections, formalization plan, and research-question index were inspected. Retrieval targets included source ranges around 5000–5600, 5730–6205, and 6800–7120; these describe the targeted portions, not a claim that the entire 115-page report was independently certified. The real well-ordering report’s guide was inspected for provenance and conventions.

Source	Path or Git blob identity
Merged surreal report	<code>Algebra/SurrealNumbers/docs/foundations-and-computation/surreal-well-orders/article.tex</code>
Report blob	<code>f20837b76f8d981cb1dce3da79f633e280edad41</code>
Sign carrier	<code>Algebra/SurrealNumbers/Surreal/Foundations/SignSequence.lean</code>
Carrier blob	<code>e2c093860f29b0b9ea312d50099fc4aa89a76a99</code>
Cut constructor	<code>Algebra/SurrealNumbers/Surreal/Foundations/SignSequenceCut.lean</code>
Cut blob	<code>c5c35afd98a5ad293687367ee9fb0244f94cc91c</code>
Real report guide	<code>SetTheory/Cardinals/docs/reports/ordinals-and-order-types/lexicographic-well-orderings-of-reals/README.md</code>

The mathematical claims newly developed here were checked by reviewing their proofs and by finite tests of the code-level constructions. No independent referee report or machine verification is implied. The literature search was targeted to primary sources on the surreal continuum, class well-orders and recursion, higher-class foundations, and the Kanovei–Shelah index. It was not a complete priority search for every abstract order-theoretic lemma.

A.6.2 Specific dependency choices

The fresh-value lemma uses an already constructed small-cut interface. The dense-core interpolation argument is written out rather than attributed only to a theorem name. The global product obstruction explicitly reuses the earlier report’s density and inverse-evaluator method. The cut and automorphism theorems do not require its singular-cardinal classification, and the central-word theorem does not require surreal arithmetic. Thus a later correction in a nonessential prior theorem would not automatically affect the main new proofs.

A.7 Source 13: reproducibility and finite checks

The archive contains the standalone `article.tex`, its compiled PDF, a build script, a standard-library Python checker, its recorded JSON output, and a status note. No external image or private font file is required. The bibliography is internal to the TeX source.

The finite checks cover the following distinct mechanisms. First, they build well-order relations on finite alphabets and compare the common-predecessor formula with direct permutation lexicography. Second, they test convexity of every finite prefix cylinder and the uniqueness and coherence of

crossing prefixes at all finite cut positions. Third, they test finite prefix completion. Fourth, they compare the central-word sign code with independent rational values of finite surreal sign sequences.

The test alphabets for the last group are $-2, -1, -\frac{1}{2}, \frac{1}{2}, 1, 2$, with all words of length at most three, including repeated-letter words. Finite signs are evaluated as dyadic rationals, not merely re-compared by a second copy of the same lexicographic function. The reserved-marker condition is also checked by exhibiting the collision that would result from admitting a trailing zero as a letter.

These tests do not establish class comprehension, ordinal-limit coherence, global choice, the back-and-forth theorem, or a transfinite birthday bound. They are reproducible checks of finite instances and independent representations, with the logical trust boundary stated explicitly.

```
# Run from the extracted archive directory.
python3 code/finite_checks.py --output data/finite_results.json
bash build.sh
```

The build script runs three LaTeX passes to settle cross-references and the table of contents. A PDF inspection record and checksums are included in the delivery's status files.

A.8 Source 14: finite diagnostics and reproducibility

The accompanying `verification.py` uses only Python's standard library. It checks finite instances of the following kernels: relation-based common prefixes versus enumeration prefixes; convexity of prefix cylinders; the chain property of crossing prefixes at every finite cut; extension of partial injections on a finite support; preservation of all comparisons under support compression; finite self-delimiting sign and word codes; and pair-orientation coding. The exact tested ranges, case counts, and pass/fail results are written to `verification_results.json` by the delivered execution. The delivered run passed every diagnostic. Its principal counts are:

Kernel	Tested cases or comparisons
Labelled common-prefix comparison	15,018
Prefix cylinders / proper finite cuts	2,371 / 867
Finite partial injections	15,126
Compressed candidates / retained comparisons	3,868 / 66,873
Numerical sign-code pairs	16,129
Outer word-code pairs, two termination conventions	320,000
Pair-orientation comparisons	21,844

Finite alphabets have endpoints and adjacent labels, unlike the residual surreal alphabet. Therefore the finite crossing-prefix test checks the chain lemma, not the full boundary trichotomy. The buffer argument for infinite partial injections, cofinality at singular cardinals, Global Choice, and class-model absoluteness are not experimentally verified. Those assertions stand on their written proofs and remain appropriate subjects for independent mathematical and formal review.

To reproduce the package locally:

```
python3 verification.py
bash build.sh
```

The build script compiles the standalone source twice with `pdflatex`; the bibliography is internal, so no external database or network access is needed for compilation. Standard TeX packages named in the preamble must be installed. `SHA256SUMS.txt` records the delivered source, PDF, scripts, status files, and diagnostic output. The archive does not include fonts, repository clones, or LaTeX auxiliary files. The repository itself has not been modified.

A.9 Source 14: source audit and bibliographic scope

The primary repository inputs were read through the GitHub connector at the pinned revision below. The targeted mathematical ranges were the opening status and abstract, the core/interpolation and binary-coding sections, the universe and formalization discussion, and the continuation questions. The actual Lean simplicity file was separately inspected. The metadata in the README distinguishes four source manuscripts from their merged report. We did not treat their statements as kernel-verified results or as established published literature.

External sources were used to check the Kanovei–Shelah indexing construction, the distinction between elementary comparison and class-valued recursion, and standard surreal background. The consulted version of the class-recursion paper is explicitly dated in the bibliography. No unresolved question quoted from it is asserted here to remain open as of the present date. The new arguments in this article are self-contained relative to the stated surreal set-cut theorem, GBC, and explicitly marked inherited set-spectrum results.

A.10 Source 15: source provenance and reproducibility

The repository inputs were read through the GitHub connection at the pinned revision stated in [Section 4.8](#). The main paths are the surreal well-orders report, the real well-orderings report, and `Algebra/SurrealNumbers/Surreal/Foundations/SignSequence.lean`. The new results are deductions in this article, not changes committed to the repository. No claim of source-wide novelty follows from this targeted audit.

The primary literature was checked for the precise Kanovei–Shelah index, surreal order versus simplicity, global choice, class well-foundedness and ETR, and the standard interval-condensation mechanism. In particular the countable condensation studied by Brown and Suárez is related methodology, not the same as the model-relative set-interval condensation used here [\[34\]](#). The final journal and arXiv versions of a source may have different pagination; theorem references specify the version where needed.

The accompanying `examples.py` regenerates the three-label table, checks the explicit restriction reversal and the Boolean restriction bound, and checks the finite predecessor-comparison formula. These calculations are reproducibility aids for finite examples only. They do not validate cardinal arithmetic, class comprehension, transfinite recursion, or the new proper-class theorems.

The source is a standalone L^AT_EX document with an internal bibliography and no external figures. Build it with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error
surreal_lexicographic_orders.tex
```

The PDF was compiled and inspected for unresolved references, overflowing formulas, and page-layout defects. Mathematical review and layout review are separate from formal proof checking.

A.11 Source 17: finite checks and reproducibility

The package contains `finite_checks.py`, which requires only the Python standard library. Its deterministic recorded run is in `finite_checks.json`. The run passed all of the following checks:

Finite mechanism checked	Instances
Predecessor-signature comparison against tuple lexicography	533,418
Exact adjacency condition against consecutive finite permutations	266,272
Convexity of prefix cylinders, carriers of size at most 7	16,072
Uniqueness and coherence of straddlers at finite cut levels	852
Finite partial-injection extension on its necessary support	15,126
Prefix-free coefficient-code comparison	961
Variable-length word-code comparison (400 words, repetitions allowed)	160,000

The empty carrier and empty word are included where appropriate. The coefficient tests use all sign sequences of length at most four. Word tests use an alphabet of seven sign sequences of length at most two, and words of length at most three. Finite partial injections are tested on carriers of size at most six, including partial maps with fixed points.

These tests do not verify singular cardinals, limits of prefix chains, class comprehension, large-gap homogeneity, or the necessity of an infinite reserve. They are consistency checks on the finite mechanisms used by the proofs, not experimental evidence for an unproved theorem.

To reproduce the checks and build the PDF from a TeX Live installation, run the supplied `build.sh`. The bibliography is internal to the LaTeX source, so no external bibliography database is required. The ZIP includes a README and a research-status note, with SHA-256 checksums of its payload. It does not bundle fonts or claim to reproduce the separate source report.

B The finite checks as shipped

[merge] All four programs use only the Python standard library and check finite instances; they verify no transfinite, class-theoretic or cardinal statement. Rerun on 2 October 2026 on copies, with Python 3.14.4 on Windows, each reproduced its recorded output exactly apart from line endings: on Windows all four write CRLF, while the records have LF, so compare with `diff -strip-trailing-cr`.

Source	Program and record	Checks	Output
11	<code>code/11-raw-orders-finite_checks.py</code> ; <code>data/11-raw-orders-finite_checks_results.txt</code>	465 sign-code pairs; 79,800 word-code pairs; predecessor comparison and adjacency on all 266,272 pairs of permutations of at most 6 points (867 adjacent)	prints to standard output
12	<code>code/12-singular-finite_checks.py</code> ; <code>data/12-singular-finite_checks.json</code>	180,905 cases: sign code, pair orientation, common prefix and adjacency, larger adjacent pairs, row scheduler	writes the file named by <code>-output</code>
09	<code>code/09-core-finite_checks.py</code> ; <code>data/09-core-finite_checks.json</code>	164,560 assertions, 1,101 separator families, seed 20261002: pair switch, prefix completion, forced separator, bounded reflection, diagonal switch	default <code>-output</code> is <code>data/finite_checks.json</code> beside <code>code/</code> , an unprefixed new file; always pass <code>-output</code>

Source	Program and record	Checks	Output
08	<code>code/08-skeleton-verify_finite.py</code> ; <code>data/08-skeleton-verification_results.json</code>	5,906 consecutive and 7,432 ordered pairs of permutations; 21,845 pair-code comparisons; 16,129 sign-code comparisons on 127 words	writes <code>verification_results.json</code> beside itself
<i>[merge] Batch 81 (sources 13–17; source 18 ships no code), added 3 October 2026</i>			
13	<code>code/13-products-finite_checks.py</code> ; <code>data/13-products-finite_results.json</code>	533,417 relation-versus-lexicographic pairs of permutations of at most 6 points; 2,371 convex cylinders and prefix completions; 5,885 cut-level uniqueness checks; 867 successor-failure partitions; 2,044 transitivity triples; 67,081 central-word pairs on 259 words	default <code>-output</code> is <code>data/finite_results.json</code> relative to the working directory, an unprefixed new file; always pass <code>-output</code>
14	<code>code/14-trichotomy-verification.py</code> ; <code>data/14-trichotomy-verification_results.json</code>	15,018 labelled common-prefix pairs; 2,371 cylinders and 867 finite cuts; 15,126 partial injections; 3,868 compressed candidates with 66,873 comparisons; 16,129 sign-code and 320,000 word-code pairs; 21,844 pair orientations	writes <code>verification_results.json</code> beside itself (no option); run a copy
15	<code>code/15-blocks-examples.py</code> ; <code>data/15-blocks-finite_examples.txt</code>	the three-label table, restriction reversal and relabelling, all eight orientations of the pair-restriction spaces, and the raw predecessor formula against enumeration lexicography for all permutations of 0–5 points	prints to standard output
16	<code>code/16-strata-finite_checks.py</code> ; <code>data/16-strata-finite_check_results.json</code>	four sign-code schemes on 241 words (232,324 word comparisons, exact dyadic values); the raw predecessor formula on 15,018 relation pairs of at most 5 points; the relation-table counterexample	default <code>-output</code> is <code>finite_check_results.json</code> beside the script, an unprefixed new file in <code>code/</code> ; always pass <code>-output</code>

Source	Program and record	Checks	Output
17	code/17-gaps-finite_checks.py; data/17-gaps-finite_checks.j son	533,418 predecessor comparisons; 266,272 adjacency checks; 16,072 cylinders; 852 straddling checks; 15,126 partial injections; 961 coefficient-code and 160,000 word-code comparisons	default -output is finite_checks.json in the working directory; always pass -output

Run them on copies, never in the report directory. From the report directory, with an empty scratch directory `$S`:

```
cp code/08-skeleton-verify_finite.py "$S/"
python "$S/08-skeleton-verify_finite.py"          # writes $S/verification_results.json
python code/09-core-finite_checks.py --output "$S/09.json"
python code/11-raw-orders-finite_checks.py > "$S/11.txt"
python code/12-singular-finite_checks.py --output "$S/12.json"
```

and compare with the four records in `data/`. The delivered build scripts build nothing in this layout: `code/08-skeleton-build.sh` runs `verify_finite.py` and compiles `surreal_well_orders.tex` beside itself, `code/09-core-build.sh` compiles an `article.tex` in `code/`, and `code/12-singular-build.sh` changes into `code/` and then calls `code/finite_checks.py`. They are kept as delivered.

[merge] (3 October 2026, batch 81.) The five programs of sources 13–17 also use only the Python standard library and check finite instances only. Rerun on 3 October 2026 on copies, with Python 3.14.4 on Windows, each reproduced its recorded output apart from line endings and, for sources 13 and 14, the recorded `python_version` (3.13.5). From the report directory, with an empty scratch directory `$S`:

```
python code/13-products-finite_checks.py --output "$S/13.json"
cp code/14-trichotomy-verification.py "$S/"
python "$S/14-trichotomy-verification.py"          # writes $S/verification_results.json
python code/15-blocks-examples.py > "$S/15.txt"
python code/16-strata-finite_checks.py --output "$S/16.json"
python code/17-gaps-finite_checks.py --output "$S/17.json"
```

and compare with the records in `data/` using `diff -strip-trailing-cr`. The delivered build scripts build nothing in this layout: `code/13-products-build.sh` and `code/14-trichotomy-build.sh` compile an `article.tex` in `code/` (the first writes `build-pass-1.log` there before failing), `code/15-blocks-build.sh` and `code/18-shuffle-build.sh` run `latexmk` on the unshipped manuscripts `surreal_lexicographic_orders.tex` and `surreal_lexicographic_orders_further_study.tex`, and `code/17-gaps-build.sh` first runs `python3 finite_checks.py`, absent under that name. They are kept as delivered.

Delivery names. The reproducibility records printed in [section A](#) and the ledgers name the delivered files. Source 13's `code/finite_checks.py`, `data/finite_results.json`, `data/pdf_inspection.json`, `build.sh` and `RESEARCH_STATUS.md` are `code/13-products-finite_checks.py`, `data/13-products-finite_results.json`, `data/13-products-pdf_inspection.json` (it records the layout check of source 13's 33-page PDF, which is not shipped), `code/13-products-build.sh` and `13-products-RESEARCH_STATUS.md`; source 14's `verification.py`, `verification_results.json`, `build.sh` and `RESEARCH_STATUS.md` are `code/14-trichotomy-verification.py`, `data/14-trichotomy-verification_results.json`, `code/14-trichotomy-build.sh`

and 14-trichotomy-RESEARCH_STATUS.md; source 15's examples.py, finite_examples.txt and build.sh are code/15-blocks-examples.py, data/15-blocks-finite_examples.txt and code/15-blocks-build.sh; source 16's finite_checks.py, finite_check_results.json and SOURCE_STATUS.txt are code/16-strata-finite_checks.py, data/16-strata-finite_check_results.json and 16-strata-SOURCE_STATUS.txt; source 17's finite_checks.py, finite_checks.json, build.sh and research_status.md are code/17-gaps-finite_checks.py, data/17-gaps-finite_checks.json, code/17-gaps-build.sh and 17-gaps-research_status.md; source 18's build.sh is code/18-shuffle-build.sh. The six manuscripts, their PDFs, their delivery READMEs and the checksum manifests are not shipped; they remain in the arrival commits (section 1.8).

C The other sources' title pages and abstracts

[merge] The results these summaries announce are printed in the body; the summaries are kept for the record.

Source 12

Title-page summary. Main distinction. An arbitrary class well-order of **No** need not have order type Ord. Nevertheless, two such well-orders have an *elementarily definable largest common labelled initial segment*, possibly a proper class. This supplies a lexicographic comparison in GBC without invoking general class well-order comparability or ETR.

Principal quantitative result. Put $\mathbf{No}_{<\kappa} = \{s : \alpha \rightarrow \{-, +\} : \alpha < \kappa\}$ and $\mu = |\mathbf{No}_{<\kappa}| = 2^{<\kappa}$, for an infinite initial cardinal κ . The minimal-length permutation order satisfies

$$\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \simeq_e \begin{cases} \mathbb{B}_{\kappa \cdot \kappa}, & \mu = \kappa \text{ and } \text{cf } \kappa < \kappa, \\ \mathbb{B}_\mu, & \text{otherwise,} \end{cases}$$

where $\mathbb{B}_\gamma = (2^\gamma, <_{\text{lex}})$ and $\simeq_e$ means mutual order-embeddability, not isomorphism.

Status. This is an AI-assisted, unrefereed research manuscript. Proofs are supplied for the stated results; historical priority is not claimed. The accompanying exact finite tests do not verify the transfinite arguments. No new Lean formalization is claimed.

Abstract. We study the lexicographic order of well-orderings of the surreal numbers, carefully distinguishing bounded sets of surreals, set-like global well-orders, and arbitrary global class well-orders. For the full class problem, ordinal-indexed enumerations are insufficient. We construct the largest common labelled initial segment of two well-orders directly from their relations, prove that comparison of their next elements defines a linear order, and show that this construction needs only elementary class comprehension once the well-orders exist. Over GB with set choice, the existence of any class well-order of the surreals is equivalent to global choice. The collection of all these well-orders is not itself an ordinary class of GBC or Kelley–Morse theory; a diagonal theorem rules out any uniform exhaustive coding by a class of sets.

For set-like global well-orders we prove density, interpolation for set-indexed cuts, local universality for class linear orders, and mutual embeddability with the lexicographic comparison of binary classes on Ord. For unrestricted global well-orders we classify adjacent pairs: they arise precisely from finite residual permutations after a common, necessarily proper-class, prefix. At a birthday cutoff κ , we obtain an exact regular/singular dichotomy for the binary coding rank

of minimal-length enumerations. We also establish the exact ordinal and binary-cube embedding spectra for the full set-sized order. Finally, a full-class model based on an inaccessible rank explains exactly how the bounded and global theories fit together, and provides a rigorous example in which the full order and the set-like order have the same cardinality and ordinal spectrum but different coding ranks. Explicit proof-assistant interfaces and further research questions are included.

Source 09

Abstract. The phrase “the lexicographic order of all well-orders of the surreal numbers” has several mathematically different interpretations. We develop three: exhaustive enumerations of a fixed set of surreals; all set-length injective surreal words; and class bijections from the ordinals onto the surreal numbers, representing set-like global well-orders. For an infinite set X of cardinality κ , we give a self-contained proof that its lexicographic well-ordering space has cardinality 2^κ , contains every linear order of size at most κ , and contains an ordinal precisely when that ordinal is below κ^+ .

For the global problem, we work in Gödel–Bernays set theory with global choice and do not pretend that a collection of proper classes is itself an ordinary class. Fix a global enumeration b . Its eventually-equal perturbations admit canonical set codes. We prove that their lexicographic order is isomorphic to the surreal line, meets every set-prefix cylinder, and interpolates every uniformly set-indexed separated pair of families of arbitrary global enumerations. Thus every class linear order embeds into this genuine class core. We also prove bounded reflection for set-indexed families, construct an explicit unfilled (Ord, Ord) -configuration, and diagonalize against every uniform class-indexed list of all global enumerations.

A strongly inaccessible truncation gives a fully set-sized model of the size distinction: the core has cardinality κ , whereas all length κ enumerations have cardinality 2^κ . The article includes foundational and Lean formalization plans, a repository audit, finite verification code, and a focused research agenda.

Title-page status. Status: AI-assisted research manuscript; complete written arguments are provided for the stated results. Not peer-reviewed or Lean-verified. Classical ingredients and repository results are distinguished from the constructions developed here. No claim of historical priority is made.

Source 08

Abstract. There is no single unqualified object called “the lexicographic order of all well-orders of the surreal numbers.” We distinguish well-orderings of a fixed set of surreals, well-orderings of varying set-sized subsets, set-like well-orderings of the whole surreal class, and arbitrary class well-orderings. For an infinite ordered set X of cardinality κ , the exhaustive-enumeration order $\mathcal{W}(X)$ has size 2^κ , contains every linear order of size at most κ , and contains an ordinal β exactly when $\beta < \kappa^+$. Its adjacent pairs arise precisely from finite permutation tails.

For the whole surreal class, work in Gödel–Bernays set theory with Global Choice. The numerical first-disagreement comparison of class bijections $\text{Ord} \rightarrow \mathbf{No}$ is meaningful without treating classes as elements of classes. Fix one such bijection b . We prove that every set-sized separated family of global enumerations has a strict separator obtained from b by a permutation of set support. The class of canonical codes for these perturbations is therefore order-isomorphic to $\mathbf{No}$. It is a dense, set-cut-realizing skeleton inside the much larger virtual enumeration order, and every class linear order embeds into it. The full virtual order is equimorphic with the lexicographic order of class binary sequences, but admits no class-uniform exhaustive parametrization. We supply proofs, cutoff

versions, counterexamples to tempting identifications, a universe-sensitive Lean implementation plan tied to inspected ProveIt sources, and a research agenda.

Status box.

Status and scope. The results below are mathematical derivations with explicit proofs, not claims of historical priority. The manuscript has not been independently refereed, and its new results have not been checked by a proof assistant. Repository declarations are identified separately from proposed extensions. The accompanying executable checks cover finite coding and adjacency examples only. All uses of “class” are relative to the stated foundational model.

[merge] (3 October 2026.) The title pages of sources 13–18 follow, with their abstracts and status statements; layout commands are omitted.

Source 13

Title page. “Further research report”. *Lexicographic Orders of Surreal Well-Orderings: Cut reconstruction, asymmetric products, prefix non-rigidity, and size-safe foundations* (PDF metadata: “Further Structure”). Research manuscript prepared for Vladimir Reshetnikov, 3 October 2026.

Abstract. We further investigate the lexicographic comparison of well-orderings of the surreal numbers, separating set-sized alphabets, set-length words, set-like global enumerations, unrestricted class well-orders, and code-dependent alternatives. The central new result is an exact prefix-cylinder criterion for a normalized class cut of a surreal-sized dense core to be represented by an injective ordinal-class word. Exhaustive words are characterized by an additional label-coverage condition. Cuts that fail the criterion have a first failure of one of two precisely described kinds: a next-label cut or a limit-prefix boundary.

We prove an asymmetric product phenomenon. The global enumeration order absorbs a surreal left factor, and more generally admits every ordinary class-indexed lexicographic sum of copies of itself. Its two-point split, however, does not embed back in the full inaccessible-universe interpretation, nor by a definable uniform transformation in Kelley–Morse theory with global choice. Externally countable models give the opposite answer for unrestricted external maps. A finite-block recoding automorphism shows that the pure linear order does not recover its prefix cylinders. Finally, replacing end-first dictionary comparison by a central termination convention on the nonzero alphabet makes both arbitrary and injective set-length word orders isomorphic to the surreal line.

The proofs specify their logical size levels. An implementation plan uses the repository’s actual universe-indexed sign-sequence and small-cut interfaces, without claiming a new Lean formalization.

Title-page status. Research status. AI-assisted, unrefereed mathematical manuscript. The proofs are presented for scrutiny; historical priority and kernel verification are not claimed. This is a follow-up to the repository’s merged report, not a replacement or a certification of all its results. The inspected repository snapshot is pinned in [Section A.6](#). Finite checks accompany the article, but do not establish transfinite or class-theoretic theorems.

Source 14

Title page. “Research continuation”. *Lexicographic Well-Orders of the Surreals II: Cut classification, support thresholds, and class-model absoluteness*. Prepared for Vladimir Reshetnikov,

3 October 2026.

Abstract. The earlier ProveIt reports identify a set-coded dense skeleton of the lexicographic order of global surreal enumerations, but leave open which cuts of that skeleton come from exhaustive enumerations. We answer the prefix-structured version of that question. Every proper normalized class cut of the skeleton has exactly one of three forms: an edge of a set-prefix cylinder, a nonprincipal numerical gap after a set prefix, or a coherent injective branch of length Ord . The last form is realized by a global well-order of set-like type exactly when its branch covers every surreal. Reconstruction is elementary; a separate orbit theorem shows that bare-order automorphisms can destroy exhaustive representability.

This yields an absoluteness theorem: enlarging the class part of a model while retaining its sets and a fixed baseline enumeration cannot newly realize an old skeleton cut by a global enumeration. New enumerations have genuinely new trace cuts. For unrestricted labelled class well-orders, the trace map forgets exactly the tail after the canonical Ord -long initial segment; its fibers are described explicitly.

A separate support analysis gives a sharp hierarchy. If κ is an infinite set cardinal, the permutations of a baseline with support of cardinality $< \kappa$ form an $\eta_{\text{cf } \kappa}$ -order, but not an $\eta_{(\text{cf } \kappa)^+}$ -order. A separated pair of $\text{cf } \kappa$ -long families forces at least κ moved coordinates, and this support cost is attained in the full skeleton. Thus universality for ordinary class linear orders can coexist with sharply limited saturation.

We distinguish fixed carriers, variable set domains, increasing subclasses, global set-like orders, unrestricted class relations, relation-code lexicography, and order-type quotients. The exposition includes proofs, exact quantifier conventions, a repository audit, universe-relative and Lean formalization plans, finite regression tests, and a focused research agenda.

Title-page status. Research status. The new results below are proved in this manuscript, not machine-verified. They are unrefereed and no claim of historical priority is made. The cut classification is relative to the named prefix cylinders; it is not a classification in the language of bare linear order alone. The repository is unchanged.

Source 15

Title page. *Lexicographic Orders of Surreal Well-Orders: Interpretations, canonical blocks, represented cuts, and fixed-type spectra.* “A continuation of the ProveIt research report”. Research article prepared for Vladimir Reshetnikov, 3 October 2026.

Abstract. There is no single unqualified object called the lexicographic order of all well-orders of the surreal numbers. The carrier, representation, comparison alphabet, indexing order, prefix convention, and foundational size level all affect the answer. We give a systematic account of the principal interpretations and develop three extensions of the existing ProveIt report. First, the relation that a closed interval in a class well-order is a set produces its entire canonical block decomposition by an elementary formula; every class well-order can occur as the block-index order. Explicit ordinal polynomials realize arbitrarily long class-indexed condensation histories. Second, a cut of the bounded-support surreal core is represented by an injective ordinal-class word exactly when every set-length level has a cylinder meeting both sides. First failure gives an exhaustive description of terminal cuts, including repeated-label obstructions and limit-prefix boundaries. The lower-core-cut map forgets exactly the residual well-order after the first proper block. Third, for birthday cutoffs we extend the minimal-slice dichotomy to enumeration types consisting of any

admissible set number of full cardinal blocks followed by a finite tail, with exact binary coding lengths. Exact local characters and the saturation threshold are computed for every cardinal birthday cutoff. Complete proofs, counterexamples, foundational assumptions, source provenance, and a prioritized research program are included.

Title-page status. Status. This is an AI-assisted, unrefereed research exposition. “Extension” means a deduction beyond the audited repository report, not a claim of historical priority. Its new theorems have written proofs and independent mathematical review within this task. They have not been checked by Lean or another proof assistant. Finite calculations check only the explicit finite examples.

Repository revision inspected: 0ecd158fd163bc4aca0ee40ccf6479ca6cf5f2a8.

Source 16

Title page. “Order theory and foundations”. *Lexicographic Orders of Surreal Well-Orders: Cuts, Canonical Blocks, and the Limits of Set-Sized Coding*. “A further study of the ProveIt research program”. Research article prepared for Vladimir Reshetnikov, 3 October 2026.

Abstract. The phrase “the lexicographic ordering of all well-orders of the surreal numbers” describes several substantially different objects. We separate well-orders on a fixed set, ordinal-length words, numerically increasing subsets, set-like global well-orders, arbitrary class well-orders, and relation-table encodings. Building on the ProveIt report, we establish several extensions. For every infinite alphabet we compute the density and cellularity of every fixed enumeration stratum, including a transition at three terminal positions, and prove that every stratum fails conditional Dedekind completeness. At regular saturated stages we determine the complete cofinality spectrum of unfilled cuts. For global enumerations we give an exact prefix-cylinder criterion for the cuts represented by injective ordinal-class branches, with an additional criterion for exhaustiveness. Every failure has a first coordinate or limit-prefix obstruction. For an arbitrary class well-order, the elementary relation “the interval between these points is a set” simultaneously extracts all its canonical ordinal-class blocks, without elementary transfinite recursion. We also classify alternative termination conventions, sharpen a comprehension bound for uniform nonembedding, and give a formalization plan that distinguishes smallness, Lean universes, and internal class semantics. All new results are presented with ordinary mathematical proofs; no claim of historical priority or machine-checked verification is made.

Title-page status. Repository snapshot. 491ed8411cccf7bd57916eb7666bbba762e92f96.

The source audit is targeted. The article is independent of later changes to the repository, and its new theorems are not additions to the Lean library.

Source 17

Title page. *Lexicographic Well-Orders of the Surreal Numbers, II: Exact cut spectra, branch recognition, optimal support reserves, and automorphism-extension obstructions*. Research manuscript prepared for Vladimir Reshetnikov, 3 October 2026.

Abstract. We continue the ProveIt report on lexicographic well-orders of surreal numbers, separating set-sized enumeration spaces, set-length words, set-like global well-orders, unrestricted class well-orders, and relation-code orders. For every infinite cardinal κ , put $X = \mathbf{No}_{<\kappa}$, $\mu = 2^{<\kappa}$,

$r = \text{cf } \kappa$, and $q = \text{cf } \mu$. For the minimal-length enumeration order $P = \mathcal{W}_\mu(X)$ we prove that every point has two-sided character (q, q) , that $\text{cf } P = \text{ci } P = r$, and that density, weight, and cellularity are $\mu^{<\mu}$. Its nonprincipal cut-character spectrum is exactly

$$\{(r, r), (q, q)\} \cup \{(\theta, r), (r, \theta) : \theta \in \text{Reg}, \theta < \mu\}.$$

Here Reg denotes the infinite regular cardinals. These statements require neither regularity of κ nor $2^{<\kappa} = \kappa$. We also establish an exact saturation threshold and a fusion theorem for dense open sets.

In **GBC**, we give a necessary-and-sufficient cylinder criterion for a class cut of a bounded-support surreal core to represent an injective **Ord**-word; a separate coverage condition characterizes exhaustive words. The reconstruction is elementary and needs no **ETR**. This yields an exact criterion for extending core automorphisms to the full global enumeration order and an obstruction: some core automorphisms do not extend. Proper extensions of the available classes necessarily add new represented cuts while leaving the set-coded core unchanged. A cardinal deficiency formula gives the optimal number of additional moved positions needed to extend a set partial injection to a supported permutation. We also identify maximal finite adjacency components in the unrestricted order and show that their quotient is dense. All higher-order statements are formulated without a same-level class of classes.

Title-page status. Status. The new assertions have ordinary mathematical proofs in this manuscript. They are not claimed to be historically first, peer-reviewed, or Lean-verified. The repository was read at the pinned revision stated in [Section 4.10](#); it was not modified or built. Finite checks supplied with this article test finite mechanisms, not the transfinite theorems. The remaining questions are explicitly distinguished from the results proved here.

Source 18

Title page. *Lexicographic Well-Orders of the Surreal Numbers: Cuts, canonical class blocks, and changes of interpretation.* Research article prepared for Vladimir Reshetnikov, 3 October 2026.

Abstract. The phrase “the lexicographic ordering of all well-orders of surreal numbers” describes several different objects. We give a common framework for well-orders of fixed sets, fixed enumeration types, set-length words, numerically increasing words, ordinal-indexed global enumerations, and arbitrary class well-orders. We retain the repository’s elementary comparison by common labelled prefixes and develop its unresolved topological and foundational questions.

For $X = \mathbf{No}_{<\kappa}$ and $\mu = 2^{<\kappa}$ we compute the density, cellularity, point characters, saturation threshold, and complete Dedekind-gap character spectrum, including exact multiplicities, of the minimal permutation order $\mathcal{W}_\mu(X)$. A third terminal position causes its density and cellularity to reach 2^μ . For arbitrary class well-orders, an elementary set-interval equivalence yields a canonical decomposition into **Ord** blocks and a final set ordinal, without **ETR**. The quotient can be isomorphic to the original order. An exact prefix criterion recognizes which class cuts of a dense core are represented by global well-orders; a repeated-label cut gives an explicit obstruction. We also classify the external full lexicographic order of every countable ω -standard **GBC** model as a dense shuffle of finite chains of factorial sizes, and show that its pure order does not define the set-like fragment. Finally, termination conventions distinguish word orders isomorphic to **No** from orders with limit-prefix obstructions and even adjacent pairs.

All new claims are supplied with conventional proofs. Their status is relative to a pinned repository snapshot, without a claim of literature priority or machine verification. Explicit size boundaries and a staged formalization plan accompany a revised research agenda.

Title-page status. Repository basis. <https://github.com/VladimirReshetnikov/ProveIt> Snapshot: 83befe707f2840c2b53e0606701d8a2b28598e47. This is an AI-assisted research manuscript with mathematical proof review; it is not a refereed paper or a Lean certificate.

D Crosswalk of the sources' numbered statements

[merge] Each row gives a numbered statement of a source (numbered as in that source's own PDF), and where it is printed or recorded here. A note records a statement proved here by the same argument; a section labelled "route" prints a different argument. Labels are the source's own with its prefix, or `swo:sub:nnumber` for unlabelled statements.

D.1 Source 11 (base)

No.	Statement in source 11	Printed here as
3.1	Definition	Definition 5.1, p. 33
3.2	Lemma (Deleting a set)	Lemma 5.2, p. 33
3.3	Proposition	Proposition 5.13, p. 40
4.1	Theorem (Cardinality and exact ordinal spectrum)	Theorem 6.1, p. 42
4.2	Proposition (Exactly where jumps occur)	Proposition 6.2, p. 43
5.1	Corollary	Corollary 7.1, p. 49
5.2	Lemma	Lemma 7.2, p. 49
6.1	Theorem (Explicit sign coding)	Theorem 9.1, p. 54
6.2	Proposition (Birthday bounds)	Proposition 9.2, p. 55
7.1	Theorem (Exact local character)	Theorem 10.1, p. 56
7.2	Corollary	Corollary 10.2, p. 56
8.1	Theorem (Word-cut classification)	Theorem 11.1, p. 57
8.2	Corollary (Successor words)	Corollary 11.2, p. 57
9.1	Proposition	Proposition 12.1, p. 59
10.1	Theorem (Existence and global choice)	Theorem 13.1, p. 60
11.1	Lemma (Unique first disagreement)	Lemma 14.1, p. 64
11.2	Theorem (Raw lexicographic order)	Theorem 14.2, p. 65
11.3	Proposition (Convex prefix cones)	Proposition 14.3, p. 65
12.1	Theorem (Canonical initial part)	Theorem 15.1, p. 67
12.2	Example (Two proper blocks)	Example 15.2, p. 68
12.3	Proposition (Convex fibers)	Proposition 15.3, p. 68
12.4	Remark (Singleton fibers and iteration)	Remark 15.4, p. 68
13.1	Lemma (Extreme enumerations)	Lemma 16.1, p. 69
13.2	Theorem (Exact adjacency criterion)	Theorem 16.2, p. 69
13.3	Example (An adjacent pair of type $\text{Ord} + 2$)	Example 16.3, p. 70
14.1	Definition	Definition 19.1, p. 74
14.2	Definition	Definition 19.3, p. 75
14.3	Lemma (Completion of an arbitrary set prefix)	Lemma 19.4, p. 75
14.4	Theorem (Prefix interpolation)	Theorem 19.10, p. 78
14.5	Corollary	Corollary 19.15, p. 79
14.6	Corollary (The class of realizations of a set cut)	Corollary 19.18, p. 80
14.7	Proposition	Proposition 19.20, p. 80
14.8	Proposition (Prefix cylinders)	Proposition 19.25, p. 82
14.9	Proposition (Approximants at every point)	Proposition 19.27, p. 82
14.10	Corollary (Set saturation is not Dedekind completeness)	Corollary 19.29, p. 83

No.	Statement in source 11	Printed here as
14.11	Theorem	Theorem 19.32, p. 85
14.12	Corollary	Corollary 19.36, p. 86
14.13	Theorem (No exhaustive uniform class of codes)	Theorem 19.37, p. 87
14.14	Theorem (An omitted-range cut)	Theorem 19.42, p. 89
14.15	Example (Two omitted points produce adjacent raw orders)	Example 19.45, p. 90
14.16	Corollary (Exactly which raw set cuts admit bounded completions)	Corollary 19.46, p. 90
15.1	Theorem (Nonembedding in a full universe)	Theorem 20.1, p. 93
15.2	Theorem	Theorem 20.2, p. 93
16.1	Research question	Research question 61.1, p. 284
16.2	Research question	Research question 61.2, p. 284
16.3	Research question	Research question 61.3, p. 285
16.4	Research question	Research question 61.4, p. 285
16.5	Research question	Research question 61.5, p. 285
16.6	Research question	Research question 61.6, p. 286
16.7	Research question	Research question 61.7, p. 286
16.8	Research question	Research question 61.8, p. 286
16.9	Research question	Research question 61.9, p. 286
16.10	Research question	Research question 61.10, p. 286

D.2 Source 12

No.	Statement in source 12	Printed here as
3.1	Definition (Common labelled prefix)	Definition 14.4, p. 66
3.2	Lemma (Maximal-prefix lemma)	Lemma 14.5, p. 66
3.3	Definition (Full lexicographic comparison)	Definition 14.6, p. 66
3.4	Theorem (Full class lexicographic order)	Theorem 14.7, p. 66
3.5	Remark (Why this does not prove class order-type comparability)	Remark 14.8, p. 67
3.6	Example (A disagreement after every ordinary ordinal)	Example 14.9, p. 67
3.7	Corollary (Agreement with enumeration lexicography)	Corollary 14.10, p. 67
4.1	Theorem (Choice equivalence)	Theorem 13.2, p. 61
4.2	Remark (The explicit set-choice hypothesis)	Remark 13.3, p. 62
5.1	Theorem (No exhaustive class-coded family)	Theorem 19.38, p. 87
5.2	Corollary (No uniform set-code solution)	Corollary 19.39, p. 88
6.1	Lemma (Extending a set prefix)	Lemma 19.5, p. 76
6.2	Lemma (Avoiding a forbidden set)	Lemma 5.3, p. 33
6.3	Theorem (Set-indexed interpolation)	Note 19.11, p. 78
6.4	Proposition (Cofinality and local character)	Note 19.28, p. 83
6.5	Lemma (Surreal universality, with the size of the recursion exposed)	Lemma 5.14, p. 40
6.6	Lemma (Deleting a set does not change the numerical class order)	Lemma 5.15, p. 40
6.7	Theorem (Local class universality and cylinders)	Note 19.21, p. 81
7.1	Lemma (A self-delimiting numerical code)	Lemma 19.33, p. 85
7.2	Theorem (Binary-class equimorphism)	Theorem 19.34, p. 85
8.1	Definition (A prefix cylinder in the full order)	Definition 16.4, p. 70

No.	Statement in source 12	Printed here as
8.2	Proposition (Residual-order decomposition)	Proposition 16.5, p. 70
8.3	Theorem (Exact adjacency criterion)	Theorem 16.6, p. 71
8.4	Corollary (Proper-class prefixes are responsible)	Corollary 16.7, p. 71
9.1	Lemma (Prefix-freeness of exhaustive enumerations)	Lemma 6.3, p. 45
9.2	Lemma (Strict binary hierarchy)	Lemma 6.4, p. 45
9.3	Lemma (Characteristic-cut coding)	Lemma 6.5, p. 45
9.4	Lemma (Pair orientation coding)	Lemma 6.6, p. 45
9.5	Lemma (No μ^+ -long monotone chain)	Lemma 6.7, p. 45
9.6	Theorem (Cardinality and exact spectra)	Theorem 6.8, p. 46
9.7	Remark (What the rank does not say)	Remark 6.9, p. 46
10.1	Lemma (Row scheduler)	Lemma 8.1, p. 52
10.2	Lemma (Surreal cone decomposition)	Lemma 8.2, p. 52
10.3	Theorem (Exact minimal-slice dichotomy)	Theorem 8.3, p. 53
10.4	Corollary (A complete cube test for the minimal slice)	Corollary 8.4, p. 53
10.5	Example (An unconditional singular example)	Example 8.5, p. 53
10.6	Example (Regular cutoffs)	Example 8.6, p. 53
11.1	Theorem (Exact external identification)	Theorem 21.1, p. 96
14.1	Research question (All fixed enumeration types)	Research question 61.11, p. 287
14.2	Research question (Noncardinal birthday bounds)	Research question 61.12, p. 287
14.3	Research question (Isomorphism, not merely equimorphism)	Research question 61.13, p. 287
14.4	Research question (The hierarchy of proper-class blocks)	Research question 61.14, p. 287
14.5	Research question (The exact weak-choice strength)	Remark 61.15, p. 287
14.6	Research question (Elementary common-prefix theory)	Research question 61.16, p. 288
14.7	Research question (Automorphisms and finite condensation)	Research question 61.17, p. 288
14.8	Research question (Dependence on the class realization)	Research question 61.18, p. 288
14.9	Research question (Compatibility between birthday cutoffs)	Research question 61.19, p. 288

D.3 Source 09

No.	Statement in source 09	Printed here as
2.1	Definition	Definition 5.9, p. 38
2.2	Lemma (Avoiding a prescribed set)	Lemma 5.10, p. 38
3.1	Theorem (Class universality)	Theorem 5.16, p. 41
3.2	Theorem (Uniqueness and set-homogeneity)	Theorem 5.17, p. 41
3.3	Corollary	Corollary 5.18, p. 41
4.1	Definition	Note 6.10, p. 46
4.2	Lemma (Exhaustiveness removes the prefix ambiguity)	Note 6.11, p. 47
4.3	Theorem (Cardinality and universality)	Note 6.12, p. 47
4.4	Lemma (Eventual stabilization)	Lemma 6.13, p. 47
4.5	Theorem (Exact ordinal spectrum)	Theorem 6.14, p. 47
4.6	Remark	Remark 6.15, p. 47
4.7	Proposition	Note 6.16, p. 47

No.	Statement in source 09	Printed here as
4.8	Proposition (Finite residuals are exactly the adjacency mechanism)	Note 6.17, p. 48
5.1	Proposition	Proposition 7.3, p. 50
6.1	Proposition	Proposition 11.3, p. 58
6.2	Proposition (Limit prefixes create small one-sided character)	Note 11.4, p. 58
7.1	Definition	Definition 13.5, p. 63
7.2	Proposition	Proposition 13.6, p. 63
7.3	Proposition (The baseline already entails global choice)	Proposition 13.7, p. 63
8.1	Lemma (Bounded-support prefix completion)	Lemma 19.6, p. 77
8.2	Remark (Why extra unused positions matter)	Remark 19.7, p. 77
9.1	Theorem (Uniform set-indexed interpolation)	Note 19.12, p. 79
9.2	Corollary (The surreal core theorem)	Note 19.16, p. 79
9.3	Corollary (Dense copies in every interval)	Note 19.19, p. 80
10.1	Theorem (Bounded reflection)	Theorem 19.30, p. 83
10.2	Theorem	Note 19.22, p. 81
10.3	Proposition (Representable suborders do not exceed surreal universality)	Proposition 19.24, p. 81
10.4	Proposition (Cylinder self-similarity)	Note 19.26, p. 82
11.1	Theorem (An explicit unfilled (Ord, Ord) configuration)	Theorem 19.43, p. 90
11.2	Remark (A general missing-branch mechanism)	Remark 19.44, p. 90
12.1	Theorem (Uniform no-coding theorem)	Note 19.40, p. 88
13.1	Theorem (Inaccessible set-sized model)	Theorem 22.1, p. 97

D.4 Source 08

No.	Statement in source 08	Printed here as
2.1	Proposition	Proposition 5.4, p. 34
2.2	Lemma	Lemma 5.5, p. 34
2.3	Theorem (Birthday-controlled cut filling)	Theorem 5.6, p. 34
2.4	Corollary (Fresh separators)	Corollary 5.7, p. 35
2.5	Definition	Definition 5.8, p. 35
3.1	Lemma (No prefix ambiguity)	Note 6.18, p. 48
3.2	Lemma (Pair coding)	Note 6.19, p. 48
3.3	Lemma	Note 6.20, p. 48
3.4	Theorem	Note 6.21, p. 48
4.1	Theorem (Prefix stabilization)	Note 6.22, p. 48
4.2	Corollary (Exact spectra)	Note 6.23, p. 48
4.3	Proposition (A uniform binary upper coding)	Note 6.24, p. 49
5.1	Theorem (Adjacency classification)	Note 6.25, p. 49
5.2	Corollary (Finite condensation blocks)	Note 6.26, p. 49
6.1	Proposition	Note 7.4, p. 51
6.2	Lemma	Note 7.5, p. 51
6.3	Theorem (The initial-length cutoff order)	Note 7.6, p. 51
6.4	Remark	Note 7.7, p. 51
7.1	Proposition	Note 11.5, p. 59
8.1	Definition	Definition 5.11, p. 39
8.2	Theorem	Note 5.12, p. 39

No.	Statement in source 08	Printed here as
9.1	Proposition	Note 19.2 , p. 74
10.1	Lemma (Prefix extension inside one orbit)	Note 19.8 , p. 77
10.2	Corollary	Note 19.9 , p. 77
11.1	Theorem (Orbit interpolation)	Note 19.13 , p. 79
11.2	Corollary	Note 19.14 , p. 79
12.1	Theorem (Uniqueness of a class saturated line)	Note 5.19 , p. 41
12.2	Theorem (Surreal-sized skeleton)	Note 19.17 , p. 79
12.3	Theorem (Universality for class linear orders)	Note 5.20 , p. 42
12.4	Corollary	Note 19.23 , p. 81
12.5	Proposition	Proposition 19.31 , p. 84
13.1	Theorem (Virtual binary equimorphism)	Note 19.35 , p. 86
13.2	Theorem (No class-uniform exhaustive list)	Note 19.41 , p. 88
14.1	Proposition	Proposition 17.1 , p. 72

[merge] (3 October 2026.) Sources 13–18 follow, numbered as in their own PDFs. Each is printed in full where it is first proved, or recorded in a numbered note where it repeats a printed result (the batch-80 practice).

D.5 Source 13

No.	Statement in source 13	Printed here as
2.1	Lemma (Prefix-freeness of exhaustive enumerations)	Lemma 34.1 , p. 140
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3.2	Lemma (Class universality and uniqueness in GBC)	Lemma 42.2 , p. 180
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5.1	Proposition (Limit-prefix character)	Note 35.3 , p. 151
6.1	Proposition (Elementary labeled comparison)	Proposition 51.7 , p. 229
6.2	Proposition (Canonical head; prior structural lemma)	Proposition 51.8 , p. 230
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7.3	Corollary (The core and its cylinders)	Corollary 42.5 , p. 182
7.4	Lemma (Two-sided ordinal approximants)	Lemma 42.6 , p. 182
7.5	Theorem (No exhaustive uniform evaluator)	Theorem 42.7 , p. 183
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8.2	Lemma (Uniqueness and coherence of crossing prefixes)	Lemma 46.14 , p. 201
8.3	Theorem (Exact branch representation)	Theorem 46.15 , p. 201
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15.2	Research question (Automorphisms preserving a chosen core)	Research question 61.21, p. 295
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15.4	Research question (Sharp cutoff analogues)	Research question 61.23, p. 295
15.5	Research question (The weakest uniform split obstruction)	Research question 61.24, p. 296
15.6	Research question (A hierarchy beyond one split)	Research question 61.25, p. 296
15.7	Research question (Class realizations with intermediate behavior)	Research question 61.26, p. 296
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15.9	Research question (A classification of termination cuts)	Research question 61.28, p. 296
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15.11	Research question (Iterating raw residual decomposition)	Research question 61.30, p. 297
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15.13	Research question (First checked extension)	Research question 61.32, p. 297

D.6 Source 14

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4.3	Definition (Cylinder and residual alphabet)	Definition 41.3, p. 178
4.4	Proposition (Cylinder geometry)	Proposition 41.4, p. 178
4.5	Lemma (Density around injective branches)	Lemma 41.5, p. 178
5.1	Definition	Definition 43.1, p. 190
5.2	Lemma (The crossing-chain lemma)	Lemma 43.2, p. 190
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14.6	Research question (Iterated initial-branch quotients)	Research question 61.38 , p. 299
14.7	Research question (Automorphisms of raw trace fibers)	Research question 61.39 , p. 299
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D.7 Source 15

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3.2	Lemma (Exhaustiveness removes prefix ambiguity)	Note 32.3 , p. 135
3.3	Theorem (Set-sized baseline, from the prior reports)	Note 32.3 , p. 135
4.1	Theorem (First unequal labels without class ordinals)	Theorem 51.9 , p. 231

No.	Statement in source 15	Printed here as
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5.1	Definition (Condensation by set intervals)	Definition 52.2, p. 236
5.2	Theorem (Elementary canonical block decomposition)	Theorem 52.3, p. 237
5.3	Corollary (Allowed block profiles)	Corollary 52.4, p. 238
5.4	Theorem (Every class well-order occurs as a block index)	Theorem 52.5, p. 238
6.1	Definition (Finite ordinal polynomials over a class order)	Definition 53.1, p. 244
6.2	Lemma (The polynomial order is a class well-order)	Lemma 53.2, p. 244
6.3	Lemma (One condensation removes the least exponent)	Lemma 53.3, p. 245
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6.5	Remark (A proper quotient can have the same abstract order type)	Remark 53.5, p. 246
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7.2	Theorem (Exact criterion for representation)	Theorem 46.6, p. 197
7.3	Remark (The normalization is necessary)	Remark 46.7, p. 198
7.4	Corollary (Exhaustiveness and the kernel of the raw cut map)	Corollary 46.8, p. 198
7.5	Theorem (Terminal-cut classification)	Theorem 46.9, p. 198
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7.7	Example (Repeated label, numerical gap, and limit boundary)	Example 46.11, p. 199
7.8	Proposition (Cofinality read from first failure)	Proposition 46.12, p. 200
7.9	Theorem (An automorphism destroys prefix representability)	Theorem 47.10, p. 214
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10.2	Theorem (Endpoint transfer for relation tables)	Theorem 38.2, p. 169
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13.2	Research question (Noncardinal birthday bounds)	Research question 61.47, p. 300
13.3	Research question (The exact strength of arbitrary-input condensation)	Research question 61.48, p. 300
13.4	Research question (Intrinsic information in the full raw order)	Research question 61.49, p. 300
13.5	Research question (Completion with residual data)	Research question 61.50, p. 301
13.6	Research question (Automorphism extension)	Research question 61.51, p. 301
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13.8	Research question (Inverse ranks and prescribed relation scans)	Research question 61.53, p. 301
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13.11	Research question (Class-model dependence)	Research question 61.56, p. 301
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D.8 Source 16

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5.3	Remark (Why two terminal labels differ from three)	Remark 27.3, p. 108
5.4	Corollary (Birthday-stage topology)	Corollary 27.4, p. 108
6.1	Theorem (Every infinite stratum is incomplete)	Theorem 28.1, p. 111
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8.2	Theorem (All termination types)	Theorem 36.2, p. 157
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10.2	Corollary (Largest set final segment)	Corollary 52.7, p. 240
10.3	Theorem (All block-index orders occur)	Theorem 52.8, p. 240
11.1	Lemma (Uniform completion of every set prefix)	Lemma 42.9, p. 184
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11.3	Corollary (A surreal-sized dense core)	Corollary 42.11, p. 185

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12.3	Corollary (Exactly what raw tails add)	Corollary 46.4, p. 197
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14.1	Theorem (Exact external identification)	Theorem 57.2, p. 257
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16.3	Research question (Centered increasing words)	Research question 61.60, p. 302
16.4	Research question (Recovering prefix structure from order alone)	Research question 61.61, p. 302
16.5	Research question (Changing baselines and extending automorphisms)	Research question 61.62, p. 302
16.6	Research question (Exact strength of uniform nonembedding)	Research question 61.63, p. 303
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16.8	Research question (Choice needed for the word isomorphisms)	Research question 61.65, p. 303
16.9	Research question (Class extensions and canonical data)	Research question 61.66, p. 303
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4.4	Theorem (Cut spectrum of the birthday alphabet)	Theorem 29.4, p. 113
4.5	Example (A warning about equal-character guesses)	Example 29.5, p. 114
4.6	Lemma	Lemma 29.6, p. 114
5.1	Lemma (The cylinder base)	Lemma 30.1, p. 115
5.2	Theorem (Characters and size invariants)	Theorem 30.2, p. 116
5.3	Remark	Remark 30.3, p. 116
5.4	Proposition (Set-alphabet background, with a direct proof)	Note 30.4, p. 116
5.5	Theorem (A dense-open fusion theorem)	Theorem 30.5, p. 116
6.1	Lemma (Cut trichotomy)	Lemma 30.8, p. 117
6.2	Lemma (Characters in the three cases)	Lemma 30.9, p. 118
6.3	Theorem (Exact gap spectrum)	Theorem 30.11, p. 119
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No.	Statement in source 17	Printed here as
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15.2	Research question (Noncardinal birthday cutoffs)	Research question 61.70, p. 304
15.3	Research question (Isomorphism after the invariant calculation)	Research question 61.71, p. 304
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15.10	Research question (Ordinal and birthday optima for supported completion)	Research question 61.78, p. 306
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15.12	Research question (First checked milestones)	Research question 61.80, p. 306
15.13	Research question (Typed constructions indexed by presentations)	Research question 61.81, p. 306

D.10 Source 18

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4.3	Theorem (Density and cellularity for every fixed type)	Theorem 27.9, p. 110
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4.5	Lemma (Large intervals and unchanged cut sides after deletion)	Note 29.8, p. 114
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5.2	Lemma (Alphabet gap spectrum)	Note 29.10, p. 115
5.3	Lemma (Small deletions preserve order density)	Note 29.8, p. 114
5.4	Lemma (Characters of cylinders and their limit boundaries)	Note 30.13, p. 120
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5.6	Theorem (Exact multiplicities)	Theorem 30.16, p. 121
6.1	Proposition (Explicit order coding; repository result)	Note 35.4, p. 152
6.2	Proposition (The prefix-supremum obstruction)	Note 35.5, p. 152
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13.5	Corollary (Set-sized cofinality from the blocks)	Corollary 52.13, p. 242
13.6	Theorem (Exact spectrum of block index orders)	Theorem 52.14, p. 242
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14.2	Theorem (Uniform set-cut interpolation)	Theorem 42.18, p. 188
14.3	Theorem (No ordinary class of codes for all global well-orders)	Theorem 42.19, p. 188
15.1	Theorem (The prefix criterion for represented class cuts)	Theorem 46.28, p. 207
15.2	Remark (Why the strict-cut convention matters)	Remark 46.29, p. 208
15.3	Corollary (Surjectivity is the exact remaining obstruction)	Corollary 46.30, p. 208
15.4	Corollary (Elementary recognition in uniform families)	Corollary 46.31, p. 209
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19.12	Research question (A first machine-checked milestone)	Research question 61.93 , p. 308

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